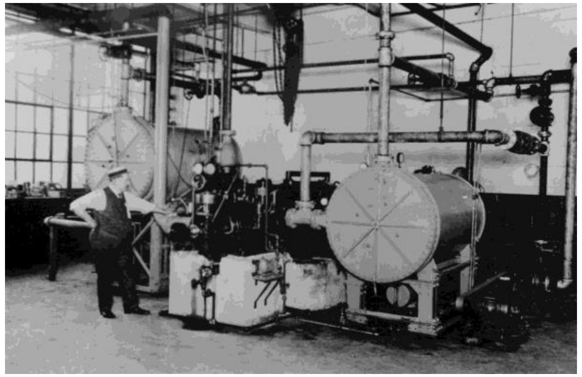
STS Training Series

DVM S Refrigeration

Why Attend Training?

Air Conditioning through the ages

- As much as things change, they stay the same?



1902

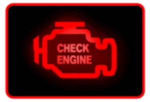
1950

2024

Are there differences in manufacturers? Are There Similarities?

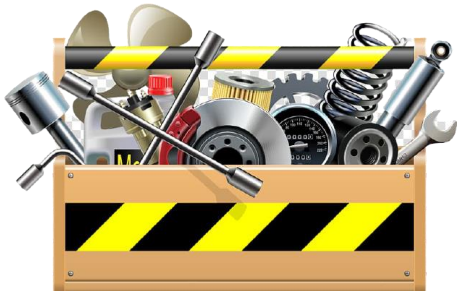
Are there differences in operation, performance, service tools and techniques between manufacturers?

Do you have the right tools and knowledge to work on the equipment?



What is your expectation for service?

OR



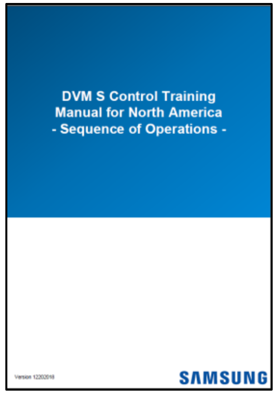
What is your customers expectation for service?

OR

And

Tools

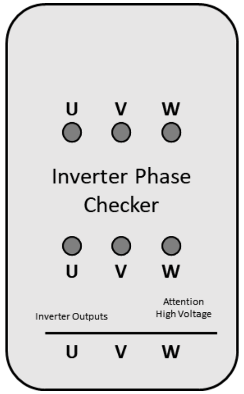
- Are these specialty tools?
- What value do these tool bring?
- How proficient can you be with out them?



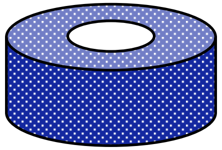
Sequence of Operation Guide

HAVC App.
Phone with internet

S-Net Pro Converter



Inverter Tester



EEV & Solenoid Magnet

Laptop with manufacturers service and design software

Review

- Given the following system determine the fan speed settings for heat and cooling .
  - 70,000 BTU Input furnace?
  - 2 tons air 20 SEER air conditioning?
- Heat Speed.
- Cooling Speed
- More information
  - 95% 70,000 BTU Input furnace with a design TD of 55 degrees
  - 2 tons air high efficiency air conditioning requiring 350 cfm/ton?
- The CFM requirement for A/C is 700 CFM. The heating CFM can be determined on the following page

Calculating CFM

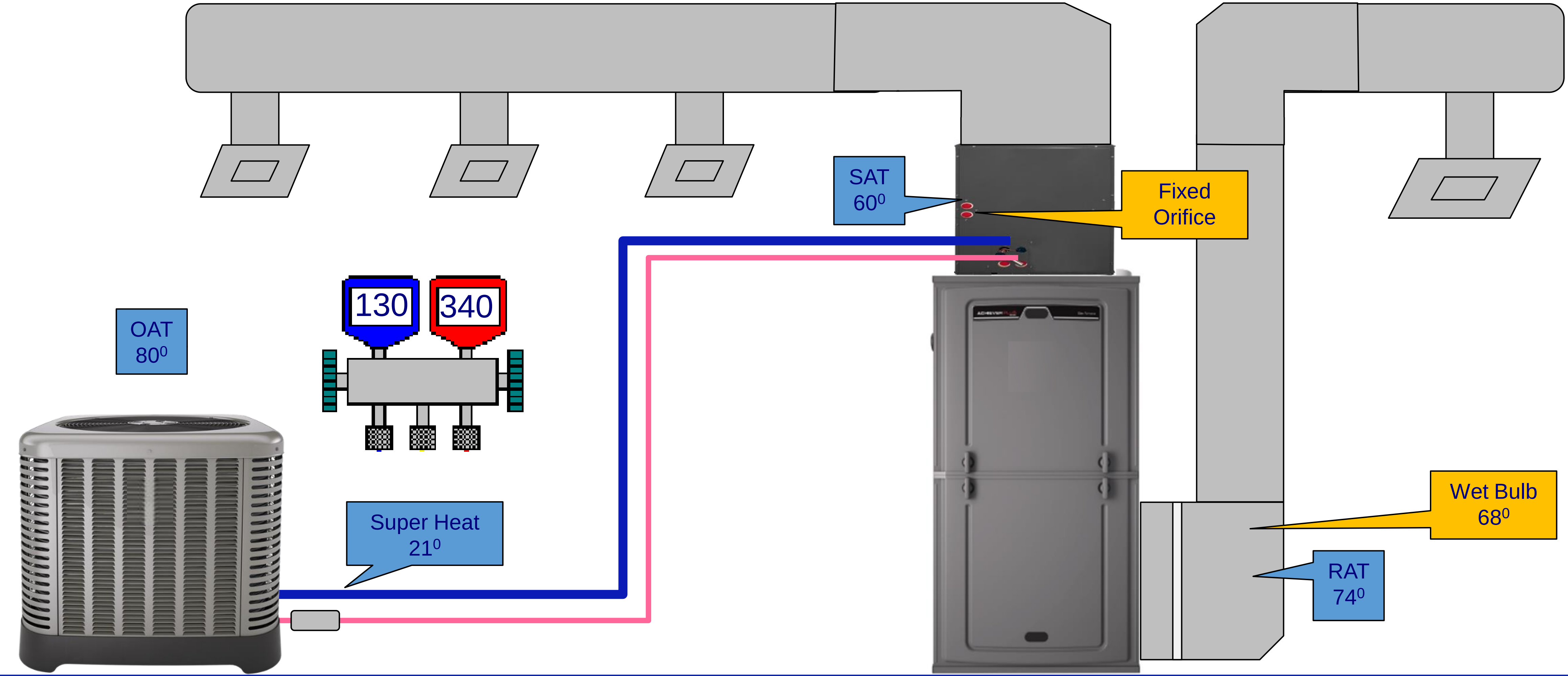
Gas Furnace

- When installing a gas burning piece of equipment you must calculate the required CFM needed for proper operation.
- To calculate the required CFM you need to know the systems Btu Output and the design TD.
- The formula is:
  - **CFM = BTU Output / TD x 1.08** (the specific heat of a cu’ of air).
- Calculate the CFM requirement for a 95% 70,000 BTU Input furnace with a design TD of 55 degrees?
- CFM = 66,500 / 55 x 1.08
- CFM = 1,103

Conclusion

- The heating system requires more air than the air conditioning system.
- To get the correct performance out of the air conditioning system it requires 700 CFM not 800.
- Why is this important?
- Does it really make a difference?
- Do your contractors know this?

Evaluate the System Operation



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Fixed Orifice Charging Charts

Condenser Entering Air Temperature

Indoor Wet Bulb Temperature

	55	60	65	70	75	80	85	90	95
52	12	10	6	-	-	-	-	-	-
54	14	12	10	7	-	-	-	-	-
56	17	15	13	10	6	-	-	-	-
58	20	18	16	13	9	5	-	-	-
60	23	21	19	16	12	8	-	-	-
62	26	24	21	19	15	12	8	5	-
64	29	27	24	21	18	15	11	9	6
66	32	30	27	24	21	18	15	13	10
68	35	33	30	27	24	21	19	16	14
70	37	35	33	30	28	25	22	18	15
72	40	38	36	33	31	28	26	24	22

Required Super Heat at Service Valve +/- 5^0 F

Super Heat Calculator

Indoor Entering Wet Bulb Temperature

Indoor Dry Bulb Temperature

	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72
70	51	51	52	52	53	53	54	55	55	56	57	59	60	-	-	-
72	52	52	53	53	54	55	55	56	57	57	58	59	60	61	62	63
74	53	53	53	54	55	55	56	57	58	58	59	60	61	62	63	64
76	54	54	54	55	55	56	57	57	58	59	60	61	62	63	64	65
78	55	55	55	56	56	57	57	58	59	60	61	62	63	64	65	66
80	56	56	56	56	57	58	58	59	60	61	62	63	64	65	66	67
82	57	57	57	57	58	59	60	60	61	62	63	64	65	66	67	68
84	-	58	59	59	60	60	61	61	62	63	63	64	65	66	67	68

Proper Evaporator Leaving Dry Bulb Temperature +/- 3^0 F

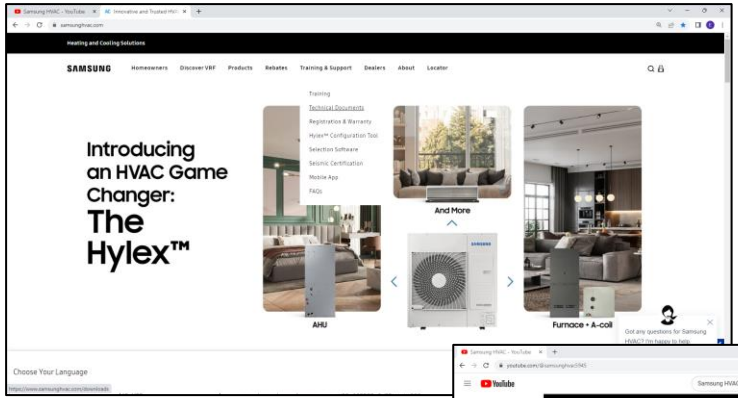
Air Flow Calculator

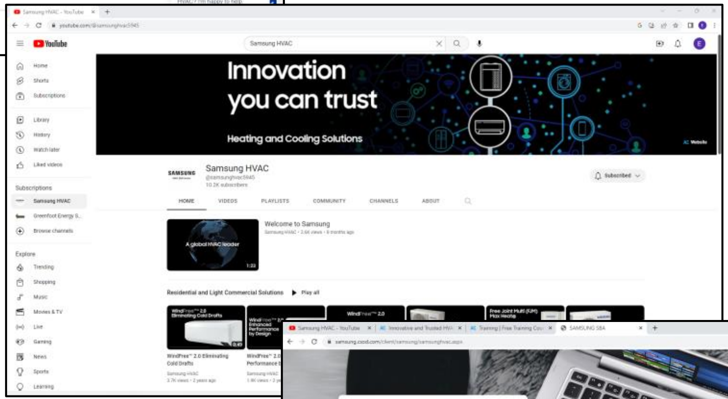
What is "Normal"

- When troubleshooting it is important know what each manufacturer's normal or targets values of operation are.
- There are no more "rules of thumbs".
- Make sure you ask the right questions and use the manufacturer's information when troubleshooting.

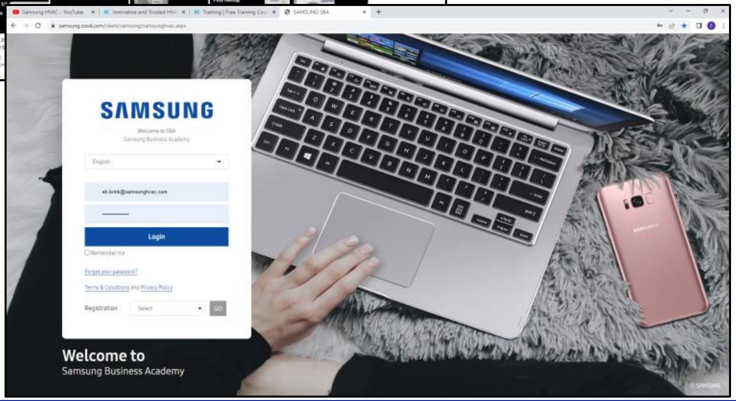
There is none!!!!
It depends on what Outdoor unit the indoor is connected to and the outdoor air temperature

Tools - Technical Resources



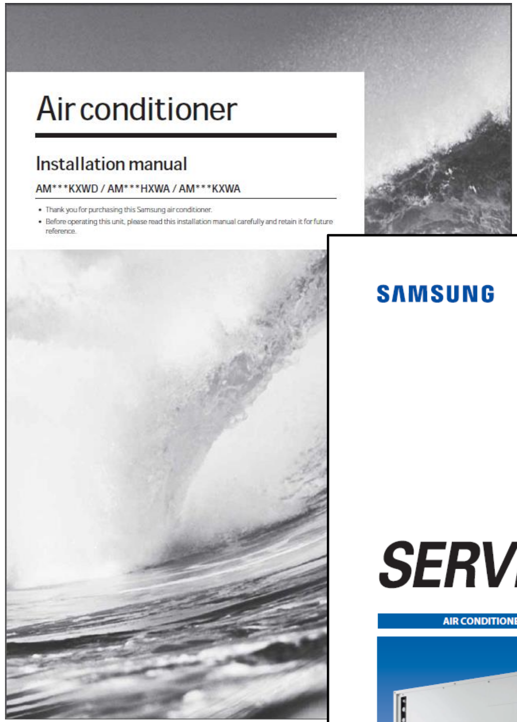


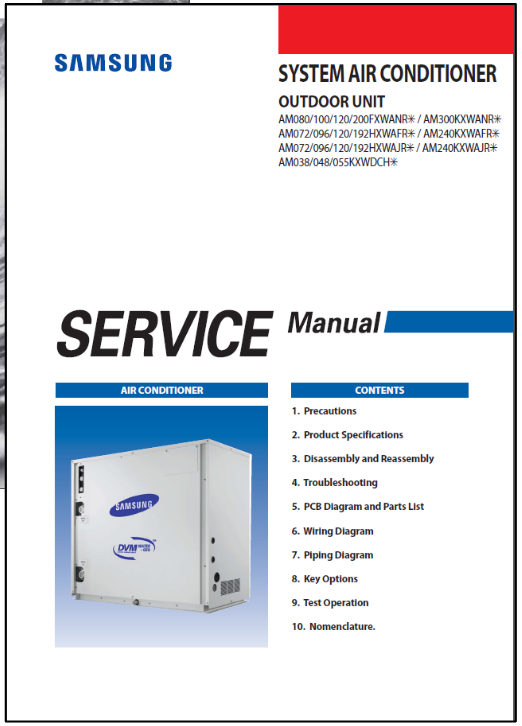
Web Sites



Mobile App.

Are you using the correct procedures and service information to diagnose the system?





Manuals

What do these have in common?

Ductless

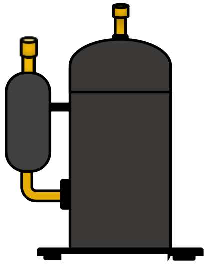
VRF

Unitary A/C

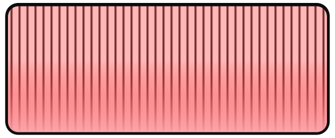
Refrigeration

Basic Refrigeration

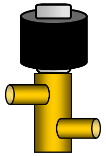
- All manufacturers are bound by the same Laws of thermodynamics and refrigeration theories when designing their equipment.
- A compressor, condenser, evaporator and metering devices functions do not change once applied to a VRF system.
- Understanding the basics of a refrigeration cycle is key to understanding a VRF system.
- Understanding the function of each component individually makes understanding the system as a whole easier.



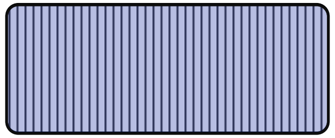
Compressor



Condenser



Metering Device



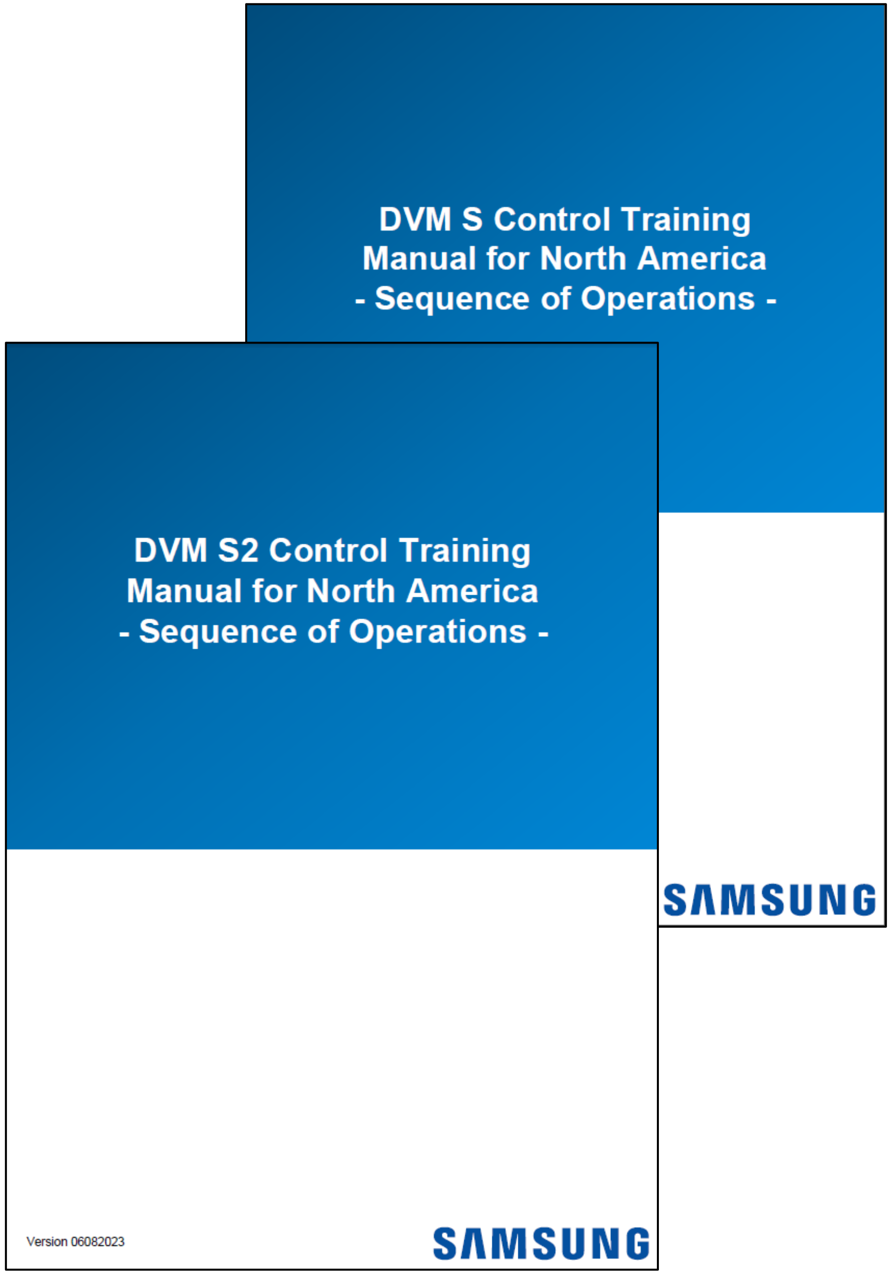
Evaporator

Overview

- This class will cover the basics of refrigeration and apply those concepts to troubleshooting VRF refrigeration circuits.
- We will review the components found in the VRF system individually and then will discuss how they interact within the system as a whole.
- We will discuss the sequence of operation for the DVM S product, review its target operation parameters and discuss the control parameter for system components.
- We will use the DVM S sequence of operation packet as a reference and resource throughout the class.

Have you ever wondered?

- The DVM S Sequence of operation book is a great tool for understanding the VRF system.
- It provides the control parameters for most of the components within the VRF system.
- It is a key component in understanding what is happening within the system.


DVM S Control Training
Manual for North America
- Sequence of Operations -

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DVM S2 Control Training
Manual for North America
- Sequence of Operations -

Version 06082023

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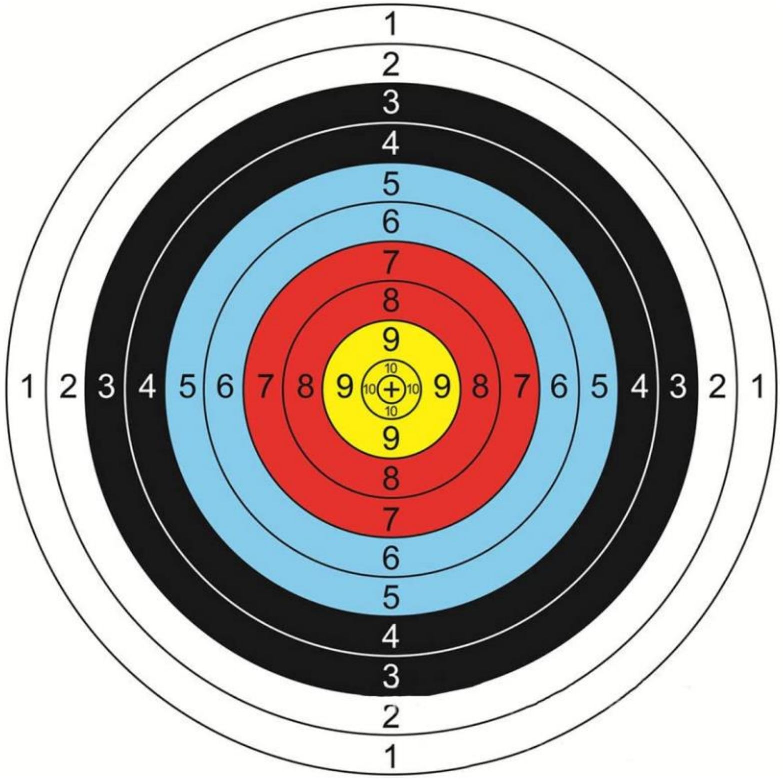
VRF Troubleshooting

- Due to the dynamic nature of VRF systems troubleshooting is more difficult than unitary systems.
- The system is constantly adjusting to changes within the system.
- This causes system reading to constantly fluctuate.
- Sometimes it feels like you're hitting a moving target.

Increasing your accuracy for troubleshooting.

- You must have a good understanding of refrigeration basics.
- You must understand the cause and effect of the system components.
- Look at the system over an extended run time.
- The more information you gather will increase your troubleshooting accuracy.
- Using the right tools and diagnostic information is the key to success.



How can you increase your accuracy in making the correct diagnosis?.

The Importance of Multiple Data

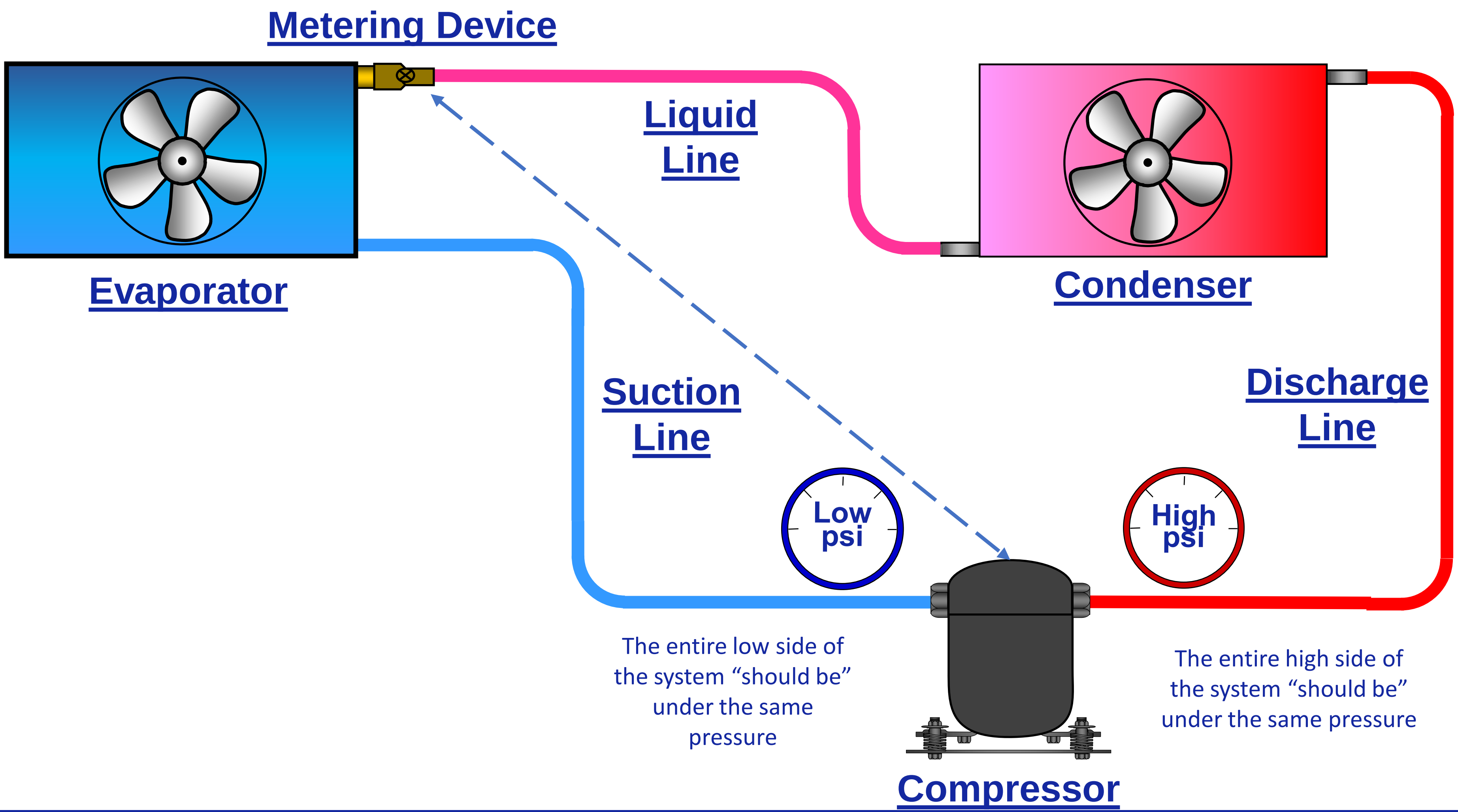
The System is Experiencing Low, Low Side Pressure in Cooling. What's the problem?

Air Conditioning Troubleshooting Chart				
System Problem	**INDICATORS**			
	Discharge Pressure	Suction Pressure	Superheat	Sub-Cooling
Over Charge	High	High	Low	High
Undercharge	Low	Low	High	Low
Liquid Line Restriction (in outdoor)	High	Low	High	High
Liquid Line Restriction (at indoor)	Low	Low	High	High
Low Evaporator Air Flow	Low	Low	Low	Low
Dirty Condenser	High	High	Low	Low
Low Outdoor Temperature	Low	Low	High	High
Inefficient Compressor	Low	High	High	High
Metering Device Not Opening	Low	Low	High	High
Un-insulated Sensing Bulb	High	High	Low	Low
Low Load	Low	Low	Low	Low

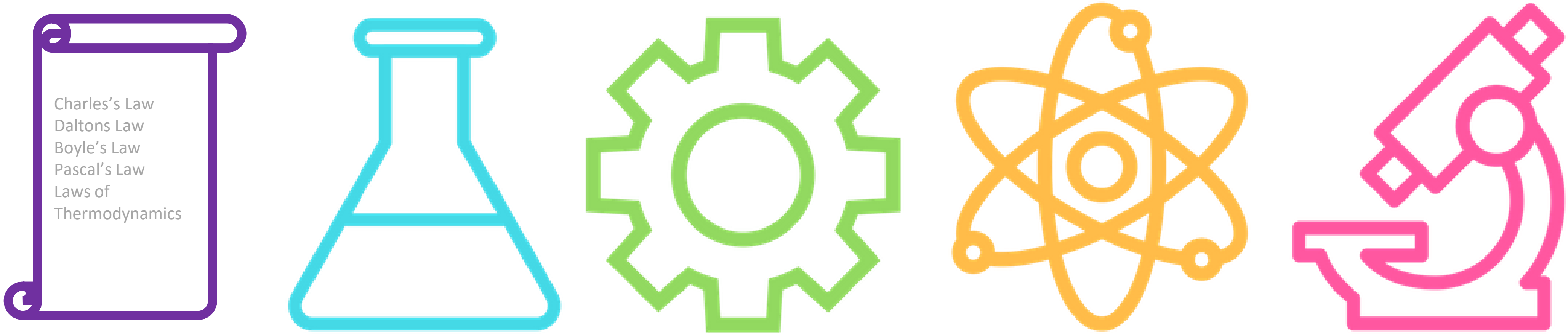
Refrigeration Basics Review

- In this section we will review the basics of the refrigeration cycle.
- Rember that the foundation of VRF refrigeration systems are grounded in the core fundamentals of thermodynamics and basic refrigeration theory.
- Don’t forget all manufacturers are bound to the same laws.

Basic Refrigeration System



The Science of Refrigeration



Theory | Chemicals | Mechanical | Science | Troubleshooting

A + B = C

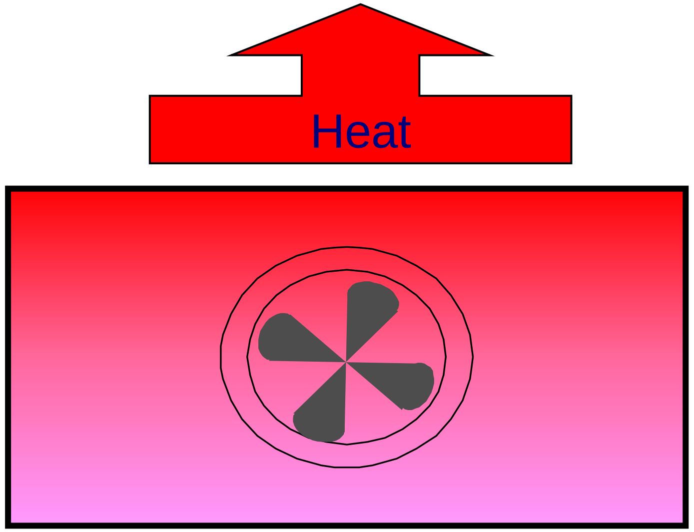
Thermodynamics

- Refrigeration is based on the principle of thermodynamics.
- The laws of thermodynamics state:
  - Heat travels from hot to cold.
  - In order of heat to be transferred that there must be a difference in temperature.
  - The rate at which heat can be transferred depends on surface area, contact time and temperature difference.

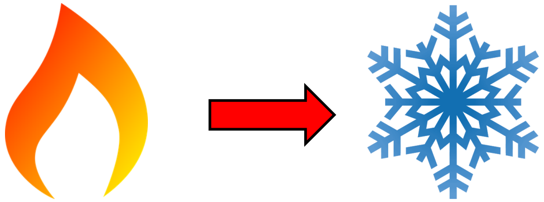
Always remember that refrigeration is based on temperatures NOT pressures

A + B = C

Heat Transfer of a Fluid

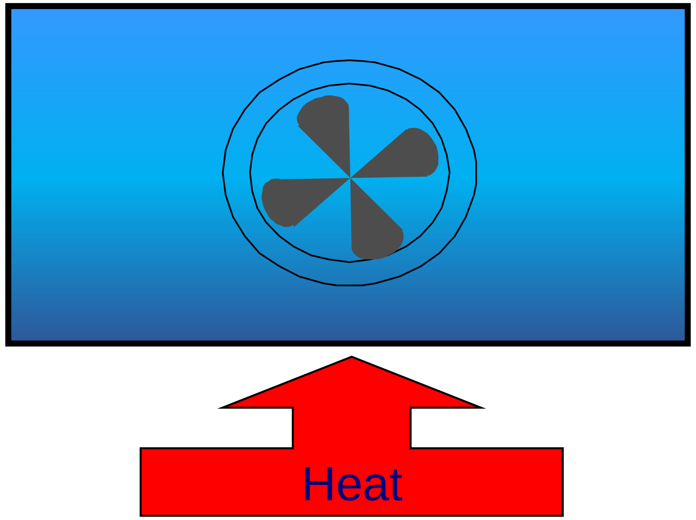


If you remove enough heat from a vapor, it will condense into a liquid

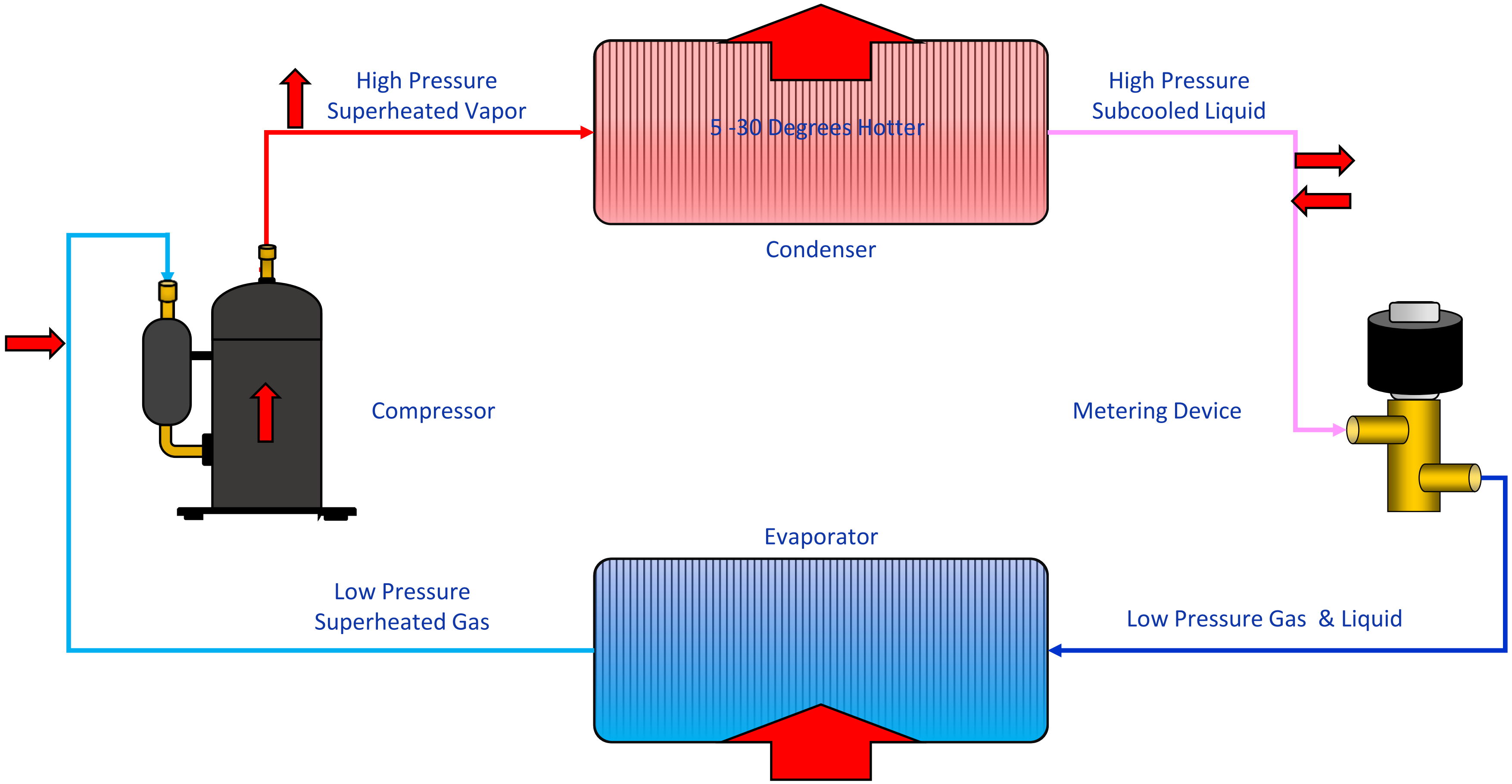


Heat Travels for hot to cold

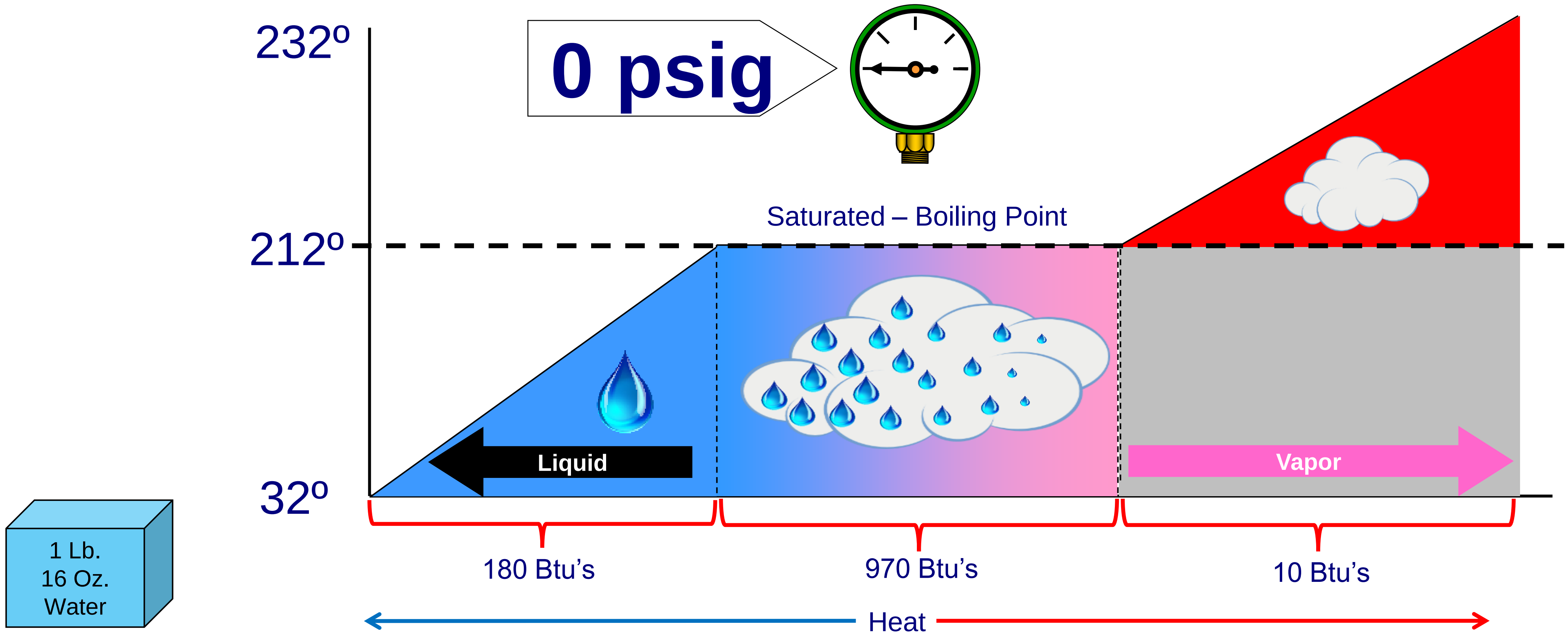
If you add enough heat to a liquid, it will evaporate into a vapor



Heat Transfer & Basic Refrigeration Cycle



Heat Transfer Rates for a States of a Refrigerant

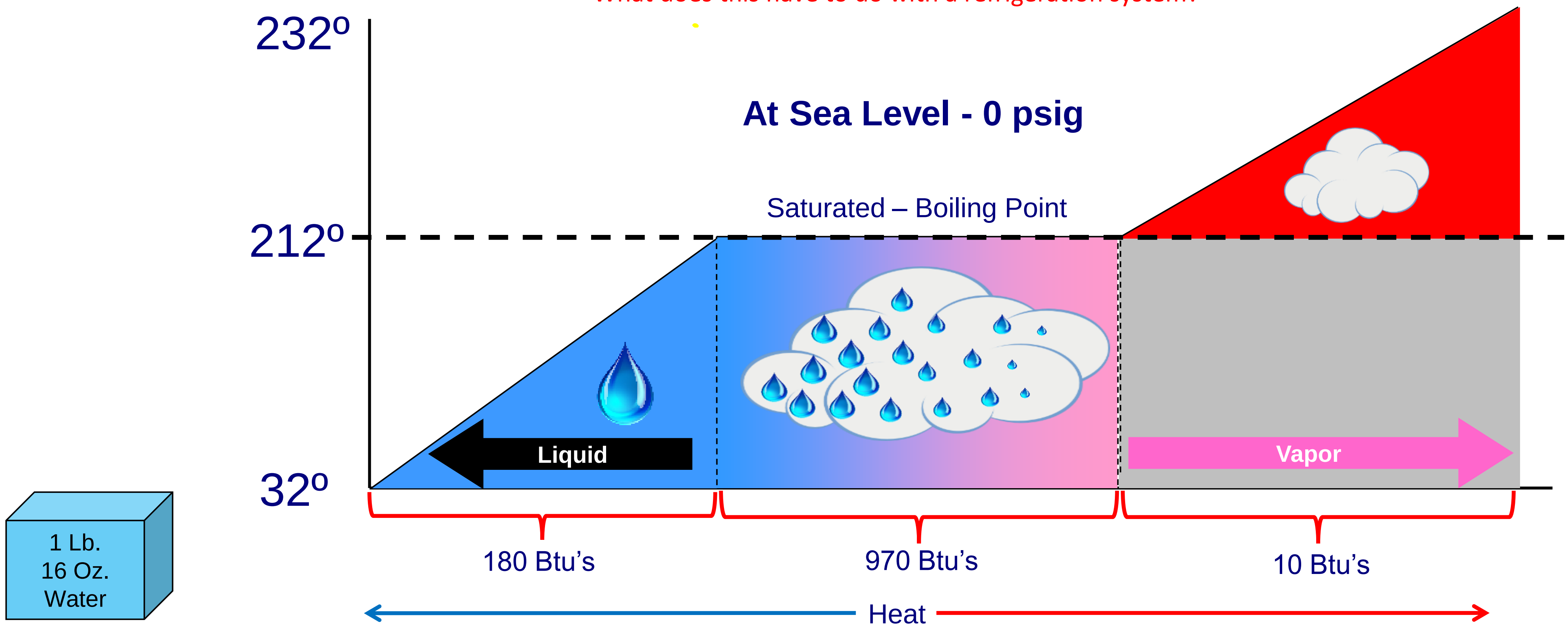


The more liquid refrigerant in the evaporator the more heat that can be removed from the space

This is why high efficiency units have lower evaporator superheats and high subcooling?

Heat Transfer Rates for a States of a Refrigerant

What does this have to do with a refrigeration system?

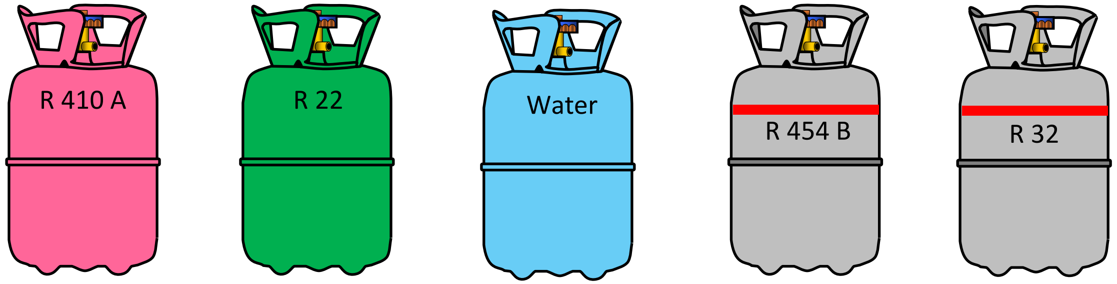


The more liquid refrigerant in the evaporator the more heat that can be removed from the space

This is why high efficiency units have lower evaporator superheats and high subcooling?

Refrigerant

- A fluid that picks up heat by evaporating at a low temperature and pressure and gives up heat by condensing at higher temperature and pressure
- Saturated Refrigerants have a specific temperature / pressure relationship.
- Knowing one you can find the other using a P/T chart



Refrigeration is based on TEMPERATURE not pressures

States of a Refrigerant

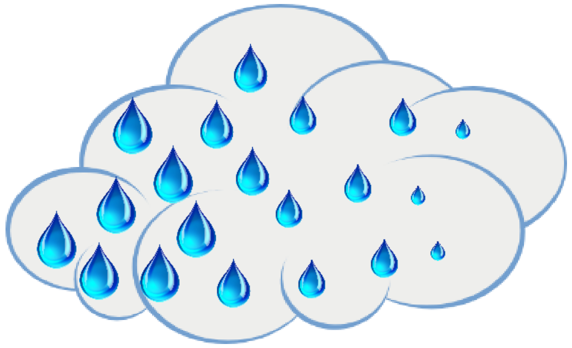
- Refrigerants can exist in one of three states :
  - Saturated (mixture of vapor and liquid)
  - Superheated (vapor only)
  - Subcooled (liquid only)
- Determining the state of a refrigerant through out a given system is the foundation of refrigeration service.
- Understanding what effect operating conditions have on a refrigerant will aid in having an understanding on how the system is operating.

States of a refrigerant

(Water as an example at atmospheric pressure)

Liquid
Sub-Cooled

<212º



Mixture of Liquid and Vapor
Saturated
(it's boiling point)

212º

Vapor/Gas
Super Heated

>212º

Saturated

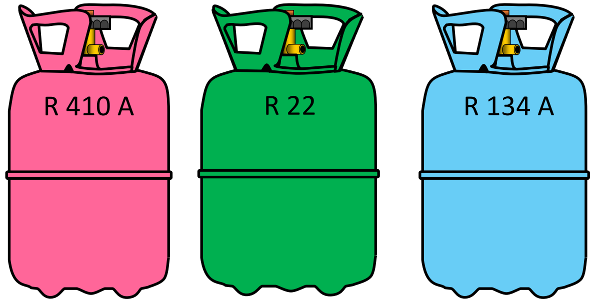
- Saturation Temperature is the temperature where liquid and vapor can exist at a given pressure.
- Also referred to as the boiling point of a refrigerant.
- May be called the condensing temperature or evaporator temperature.
- The saturation point is directly related to the pressure. The lower the pressure the lower the boiling point.
- Saturated refrigerant normally exist in the condenser and evaporator.
- How can you change the boiling point of water?

Pressure	Boiling Point
3O psig	271^0
O psig	212^0
25,000 Microns	80^0
500 Microns	0^0

Pressure / Temperature (PT) Charts for Saturated Refrigerants

Pressure Temperature (PT) Charts show the relationship between the pressure and saturation temperature (boiling point) of a refrigerant.

Remember that refrigeration is based on Temperature NOT pressure.



125^0

PSIG	134a	22	410A
180	123	94	63
185	125	96	65
190	127	98	66
195	128	100	68
200	130	101	70
210	134	105	73
220	137	108	76
230	140	111	79
240	143	114	82
250	146	117	84
260	149	120	87
270	152	123	89
280	155	126	92
290		128	94
300		131	96
325		137	102
350		143	107
375		149	112
400		155	117
425			121
450			126
475			130

Saturation Temperatures

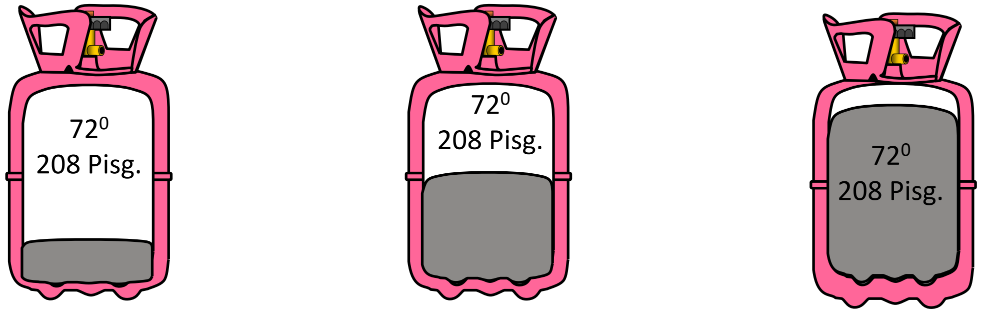
Example:
The condenser coil is designed to transfer the proper amount of heat if the refrigerant temperature is 125 degrees.

What would the high side pressure be if the following refrigerants were used:

R 410A 445 psig
R 22 275 psig
R 134A 185 psig

Saturation

- As long as there is a mixture of liquid and vapor the refrigerant is in a saturated condition.
- If the mixture is 99% vapor and 1% liquid it is considered saturated
- Same as if the mixture in 99% vapor and 1 % liquid it is considered saturated.
- This is why each cylinder has the same pressure for its given temperature .
- Because we can not look inside the piping system to determine the state of the refrigerant, we as technician need to determine the state of the refrigerant throughout the system by calculating superheat and subcooling.



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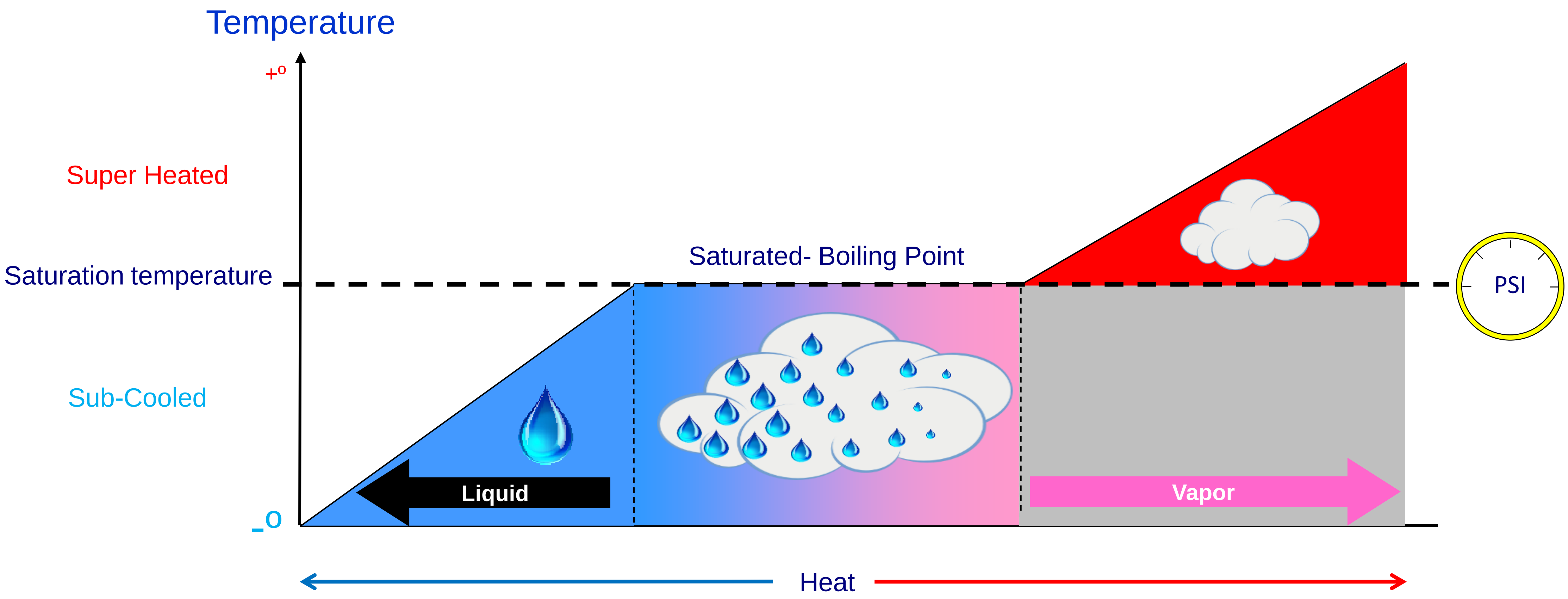
Superheat

- Superheat refers to the temperature of the refrigerants above its saturation temperature for the pressure it is under.
- Is an indicator of how well the evaporator is working.
- If a refrigerant is superheated no liquid is present.
  - Only a gas can be superheated.
- Superheated gas normally exist in:
  - The outlet of the evaporator.
  - The suction line.
  - The discharge line.
  - The Inlet of the condenser.

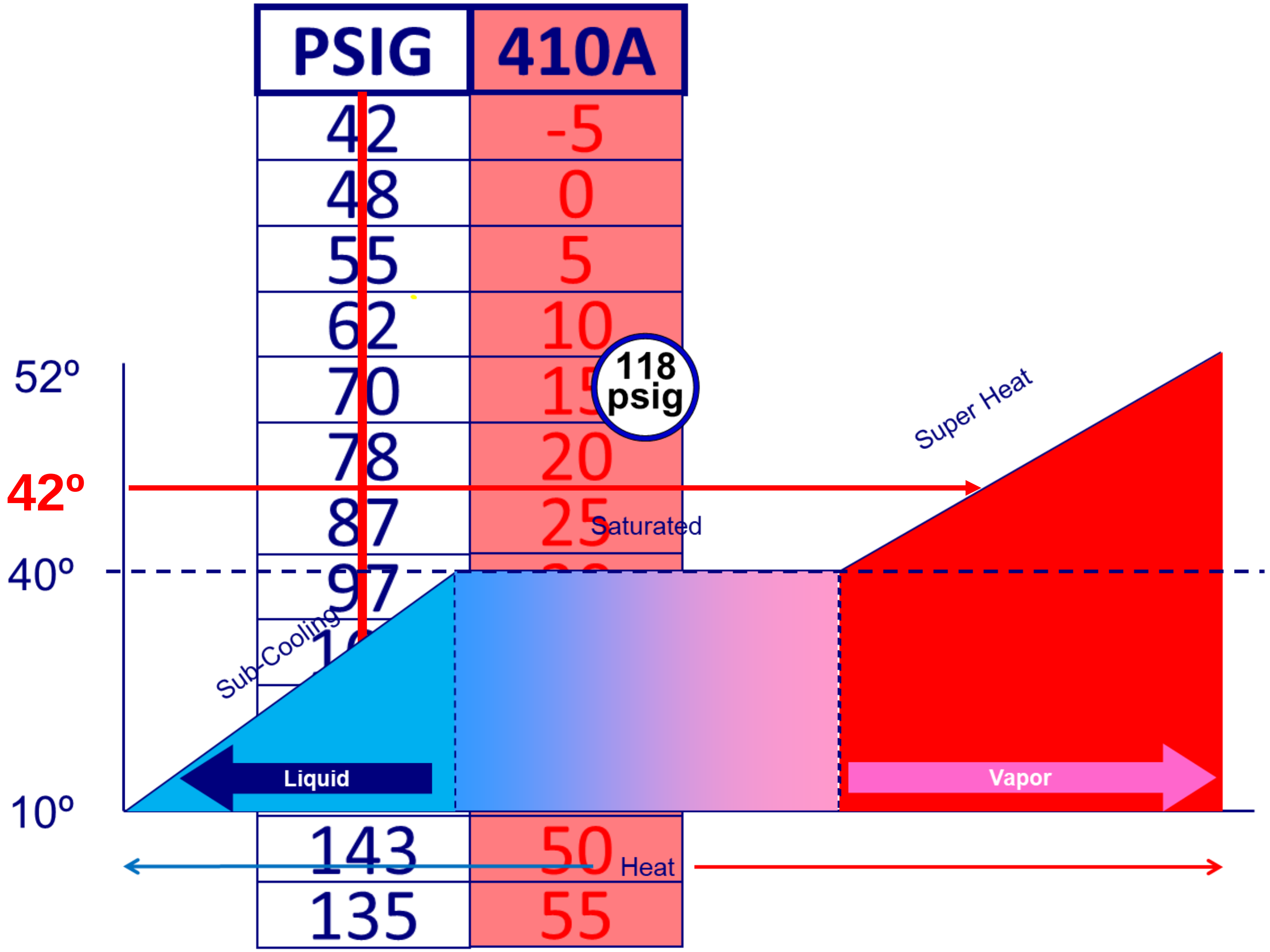
Subcooling

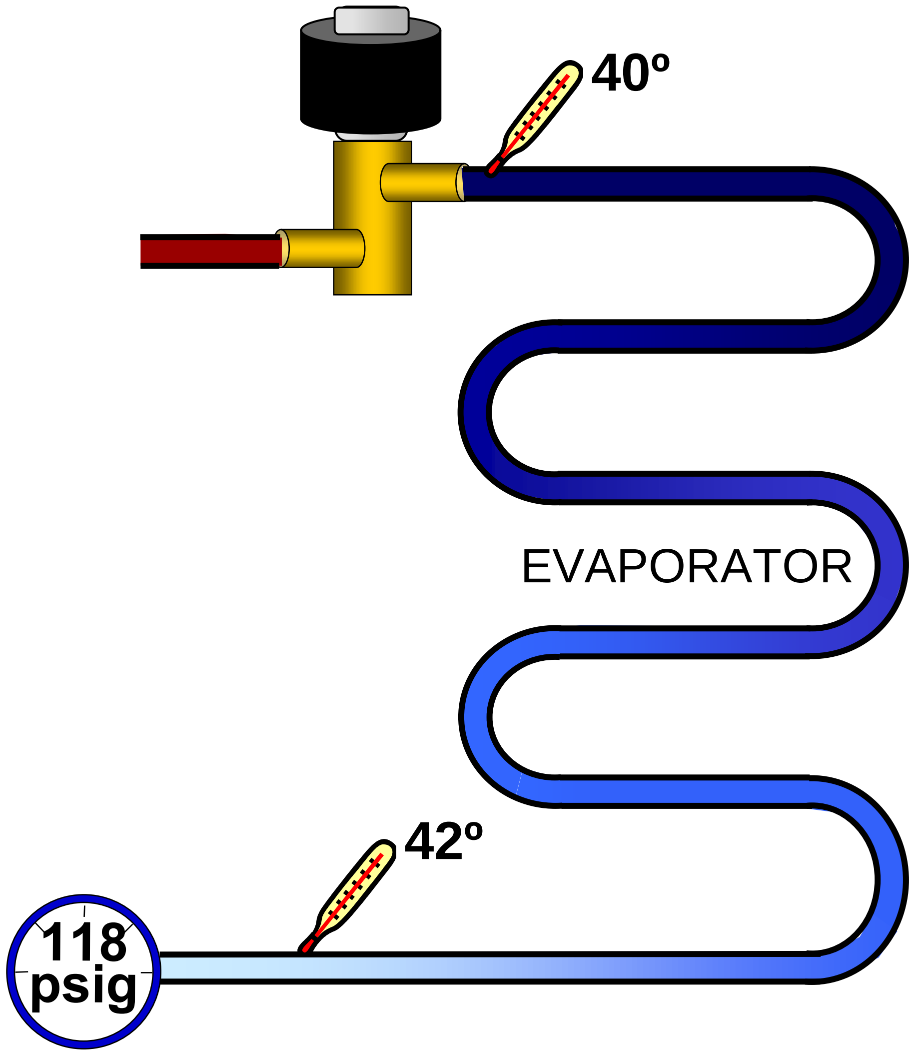
- Sub-cooled refers to the temperature of the refrigerant below its saturation temperature for the pressure it is under.
- If a refrigerant is sub-cooled no vapor is present.
  - Only a liquid can be sub-cooled.
- Is and indicator of the amount of charge in the system.
- Sub-cooled liquid normally exist in the liquid line and outlet of the condenser.
- Why do ductless units require greater subcooling?
- For a metering device to work correctly it must be supplied with sub-cooled liquid. (minimum of 5.4^{0} verified 10/24)

States of Refrigerants Review



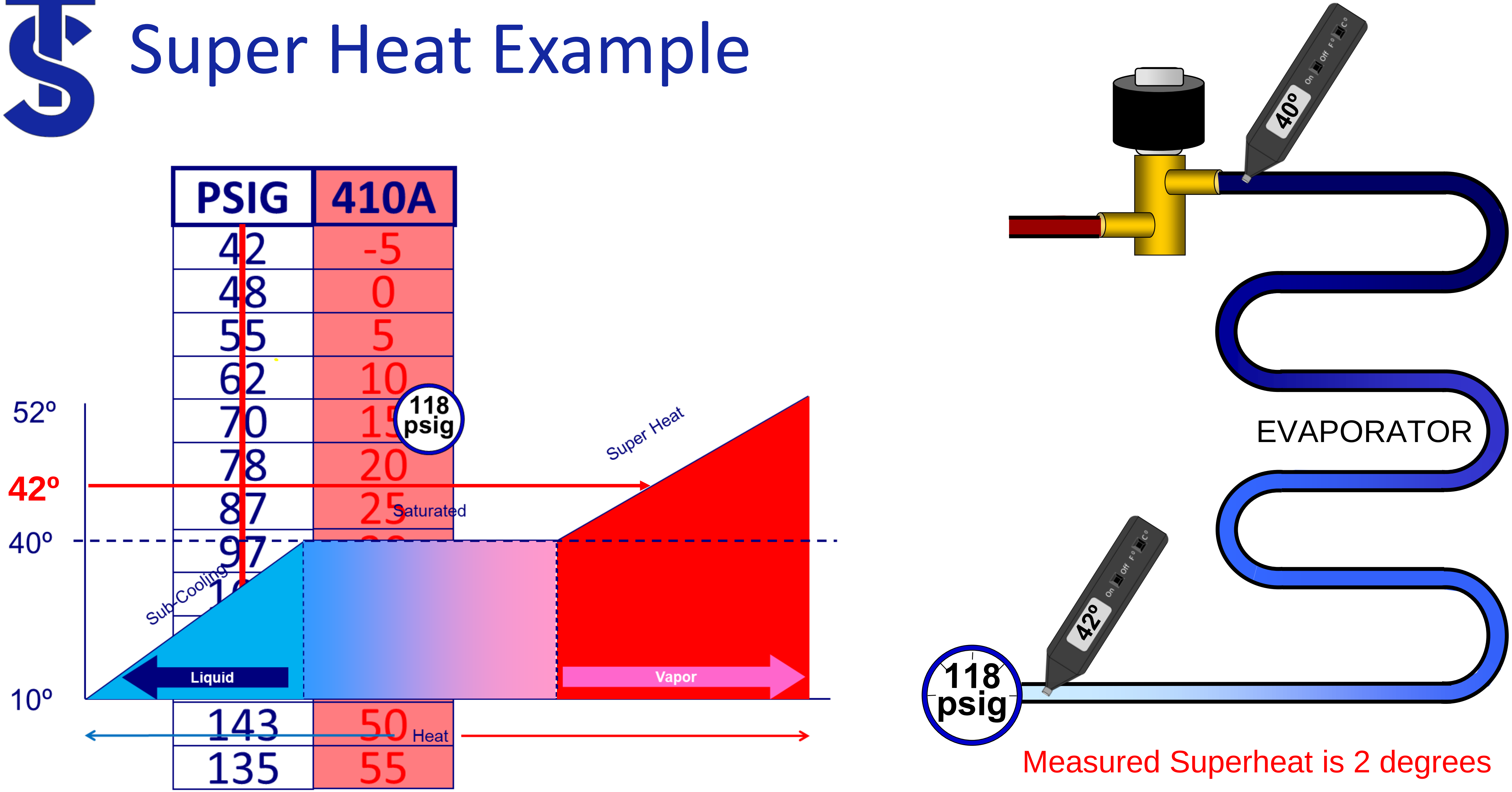
Super Heat Example





Measured Superheat is 2 degrees

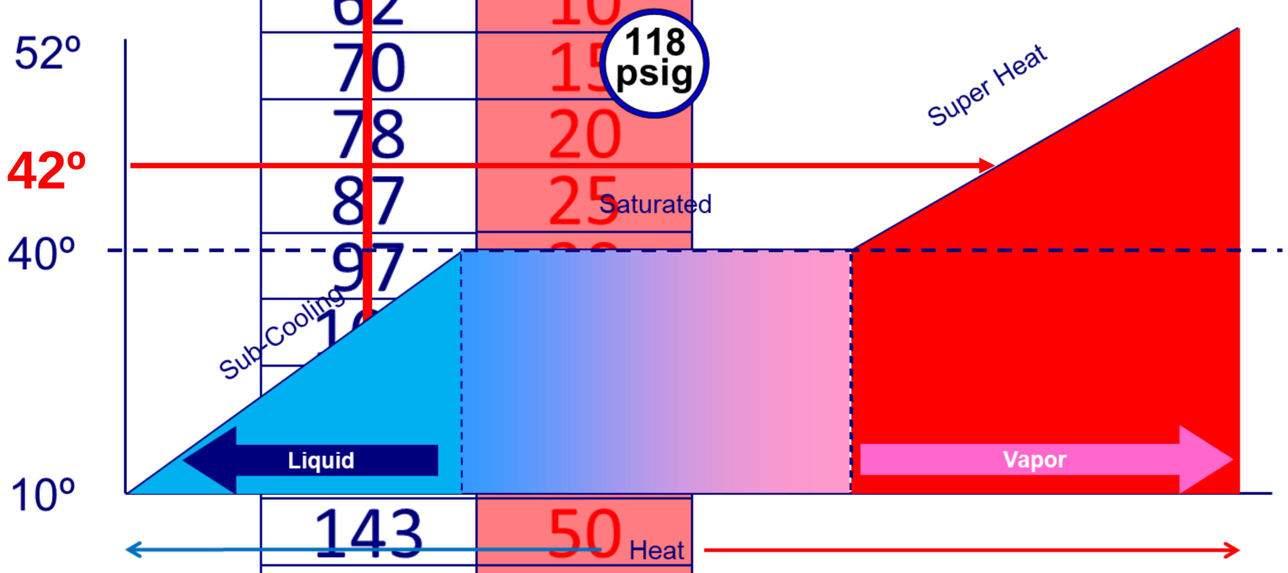
Super Heat Example

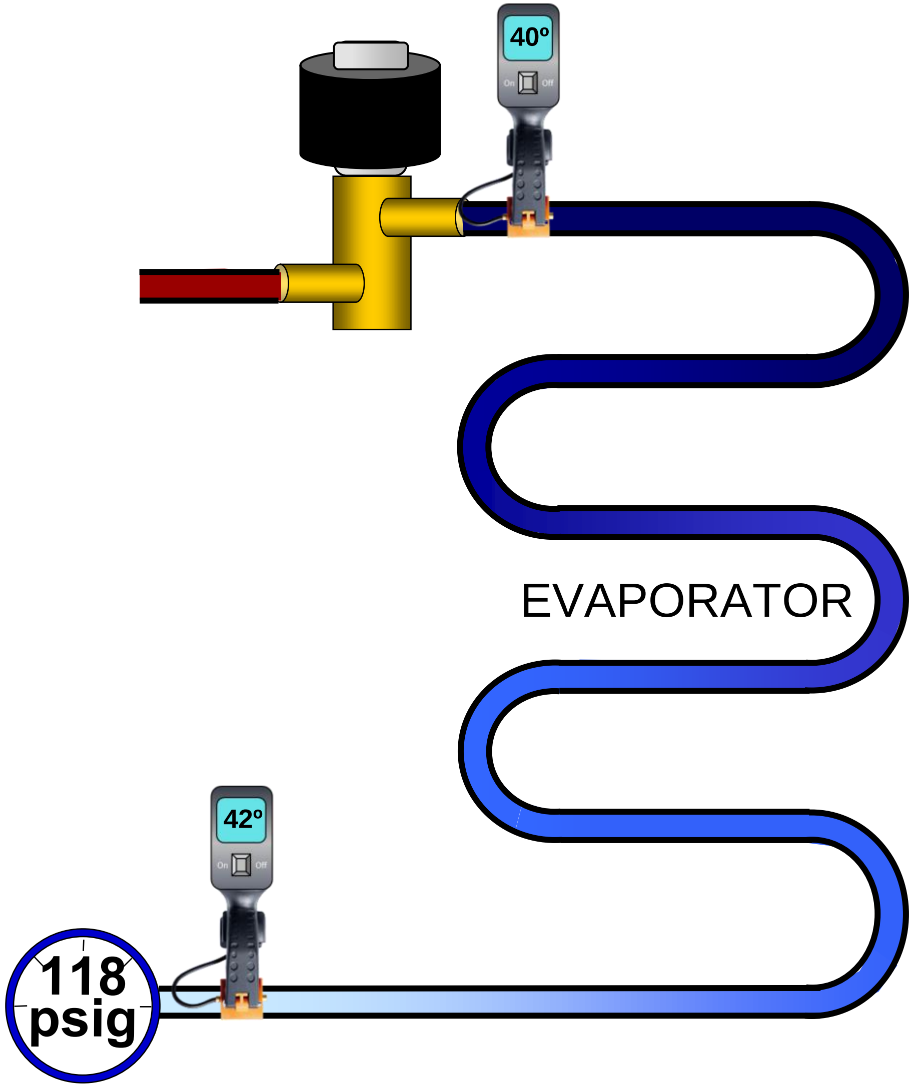


Measured Superheat is 2 degrees

Super Heat Example

PSIG	410A
42	-5
48	0
55	5
62	10
70	15
78	20
87	25
97	[illegible]
[illegible]	[illegible]
143	50
135	55





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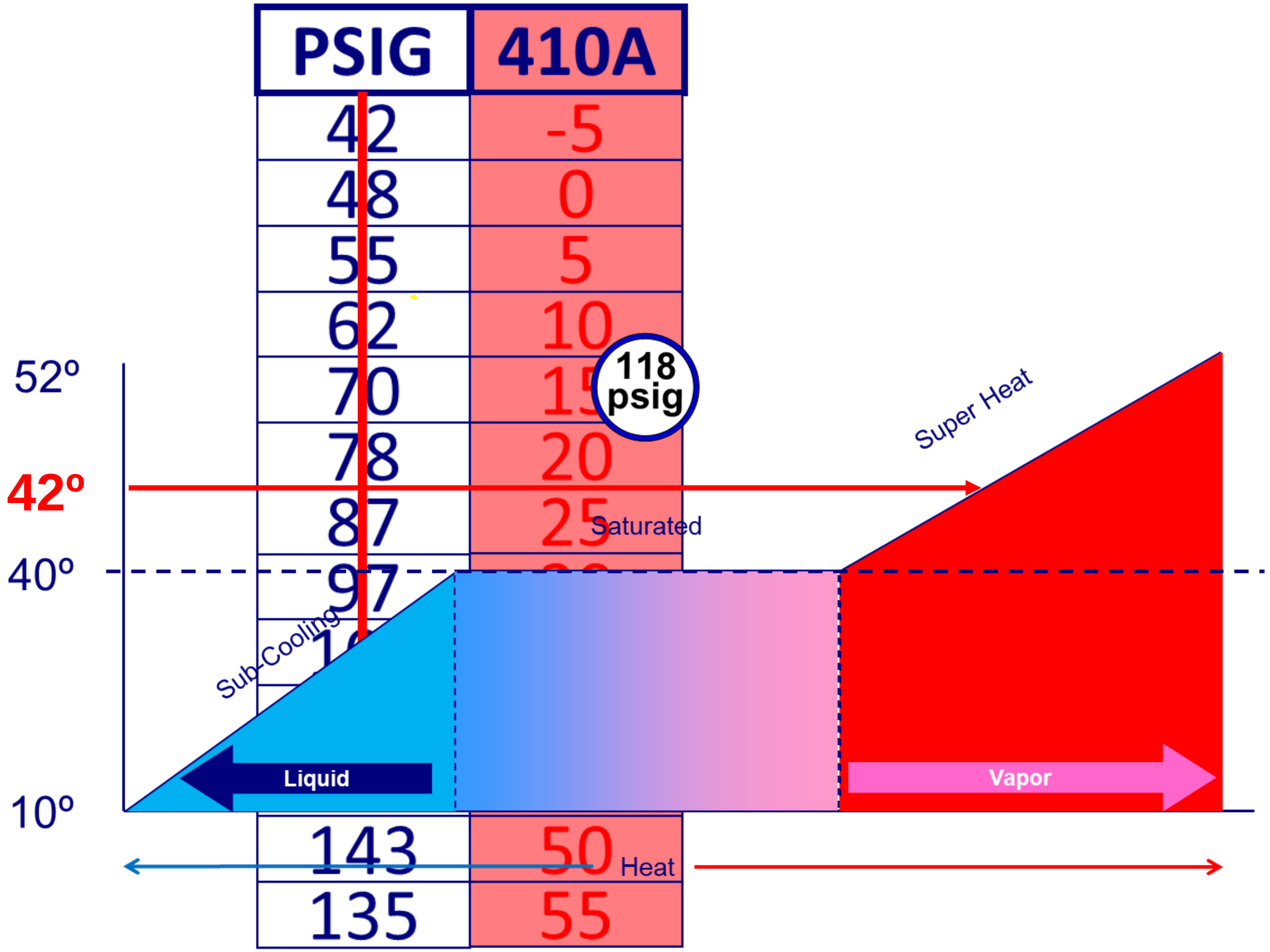
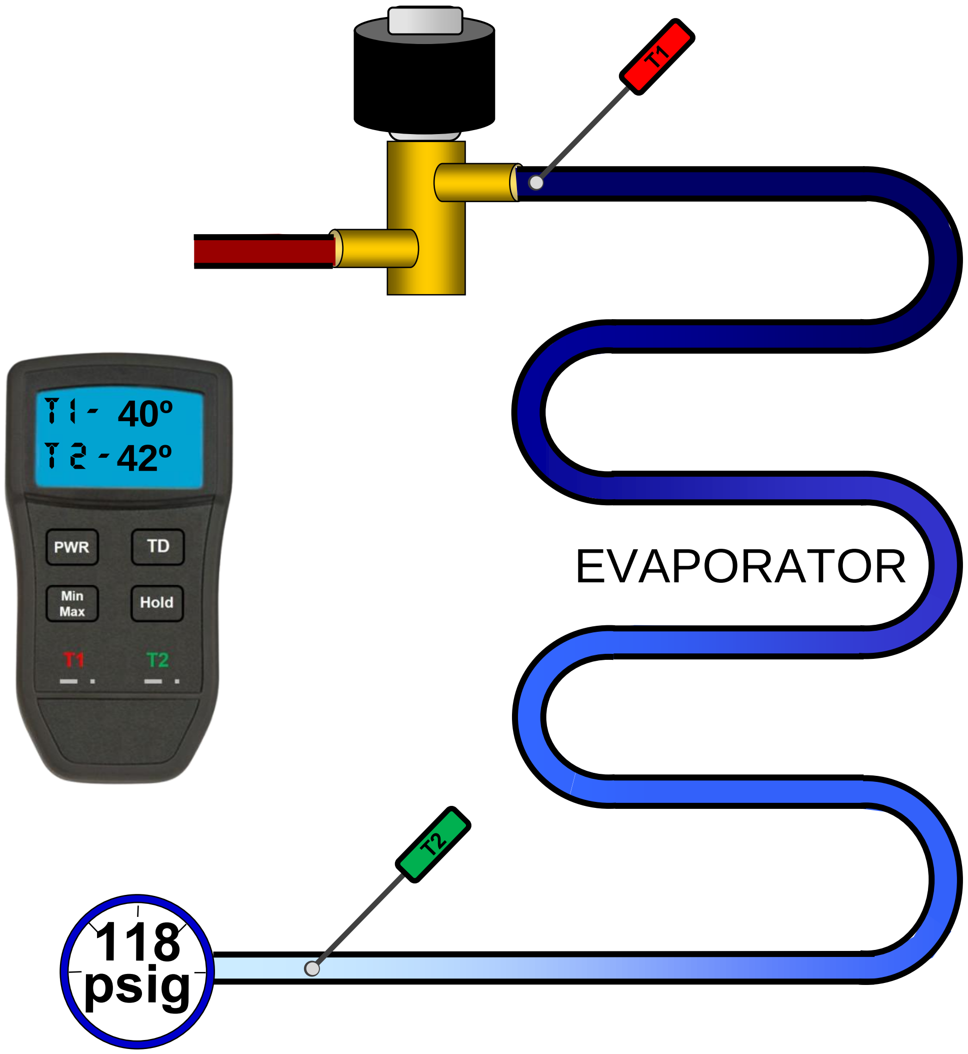
Super Heat Example





Measured Superheat is 2 degrees

Super Heat Example

PSIG	410A
42	-5
48	0
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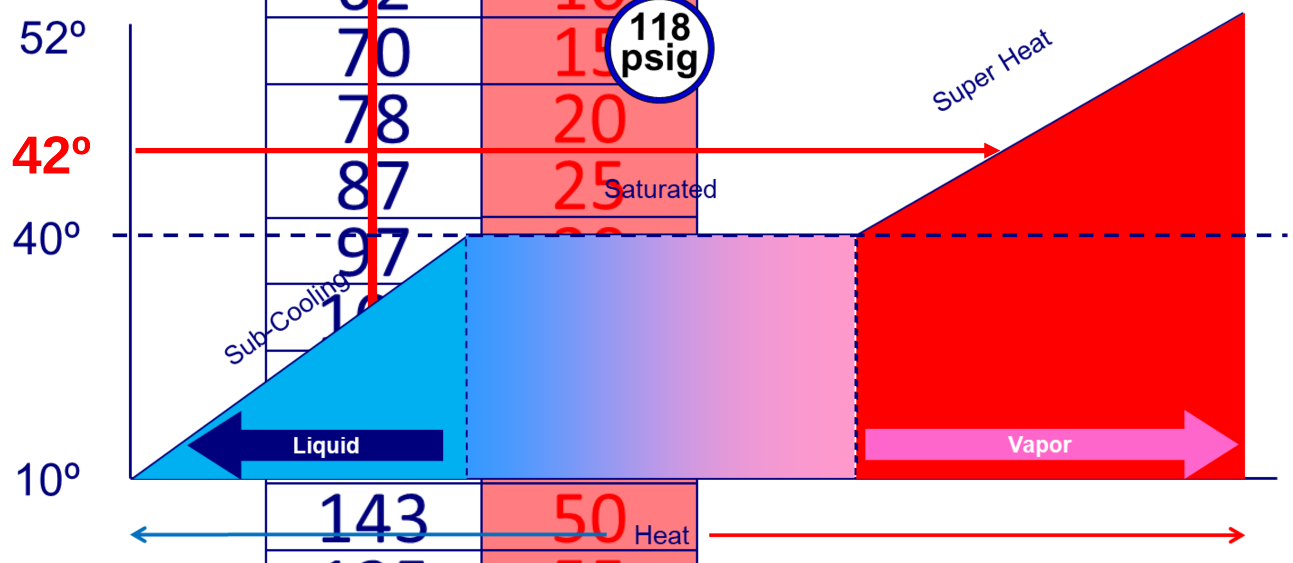
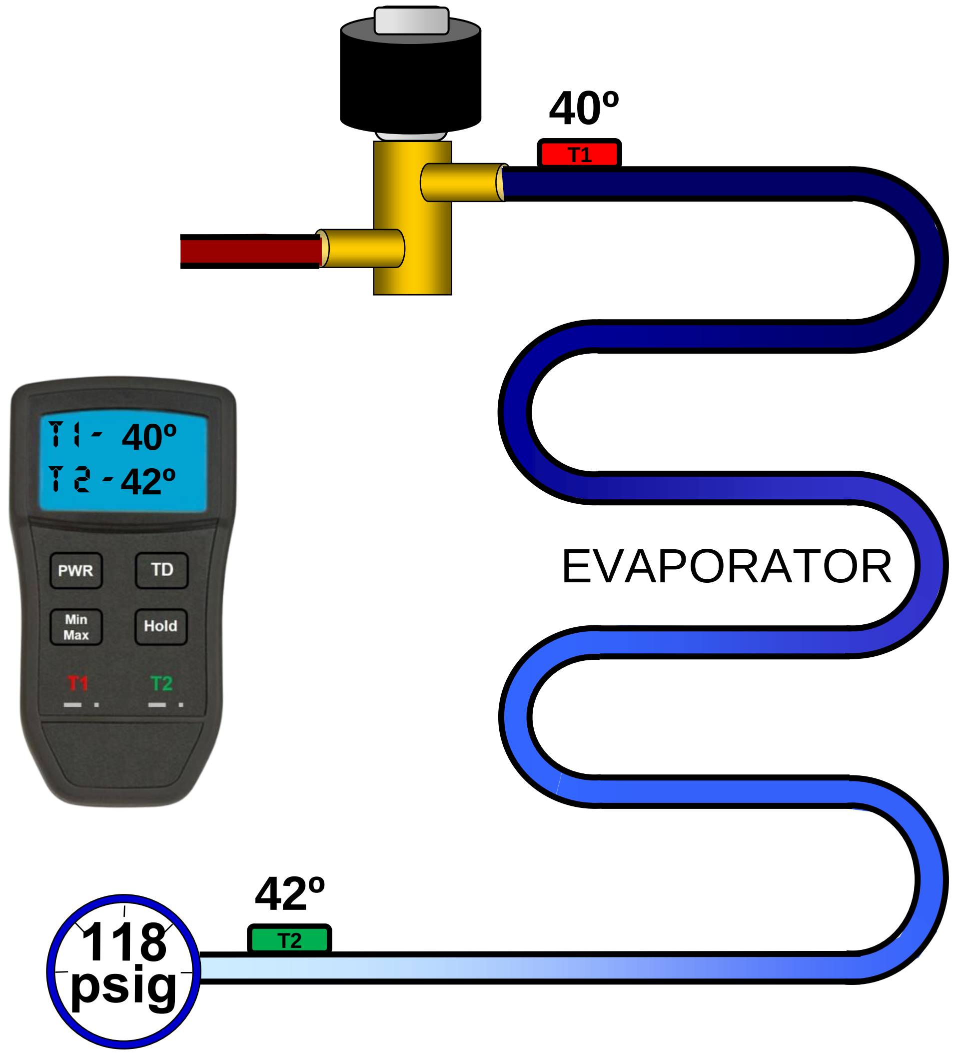
70	15
78	20
87	25
97	[illegible]
[illegible]	[illegible]
143	50
135	55





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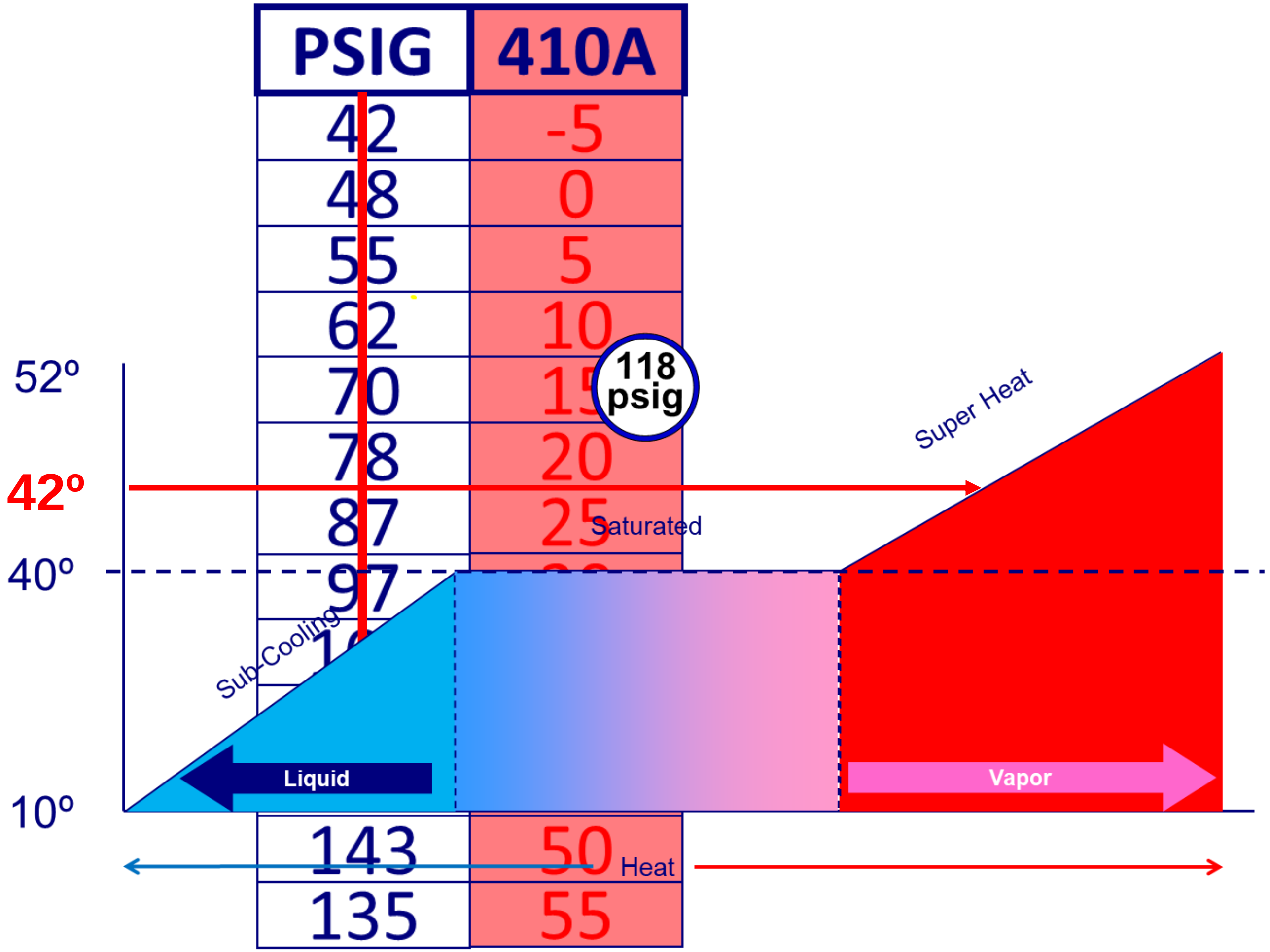
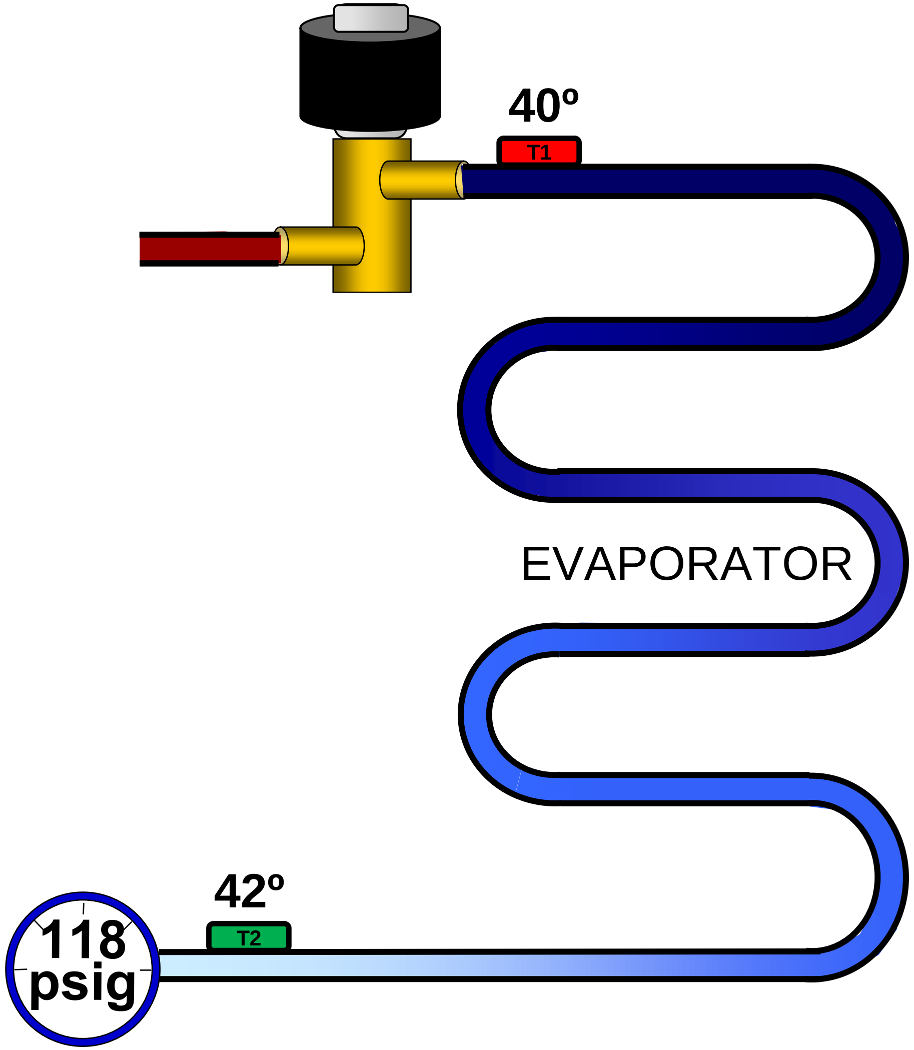
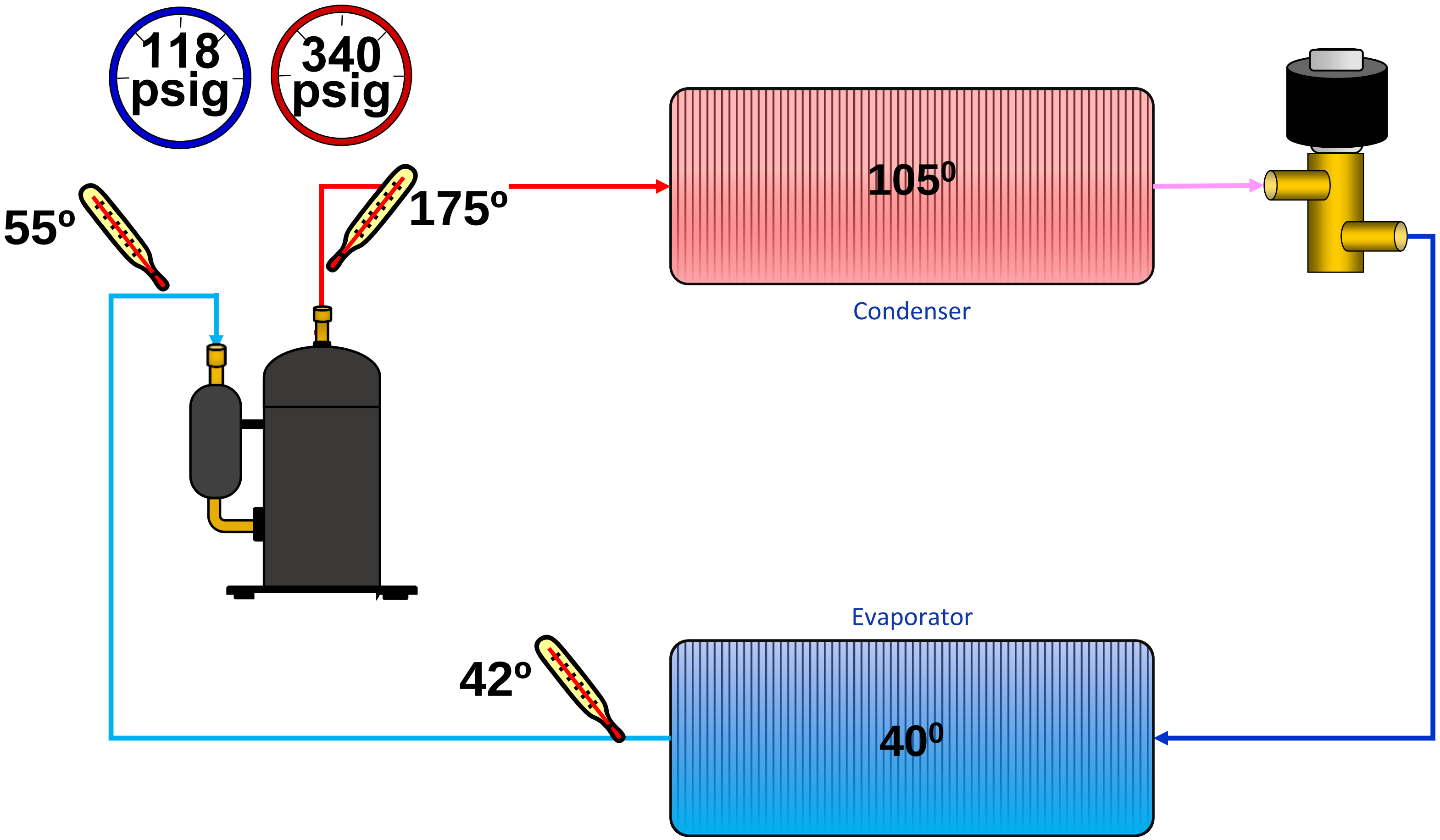
Super Heat Example

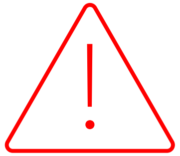




Measured Superheat is 2 degrees

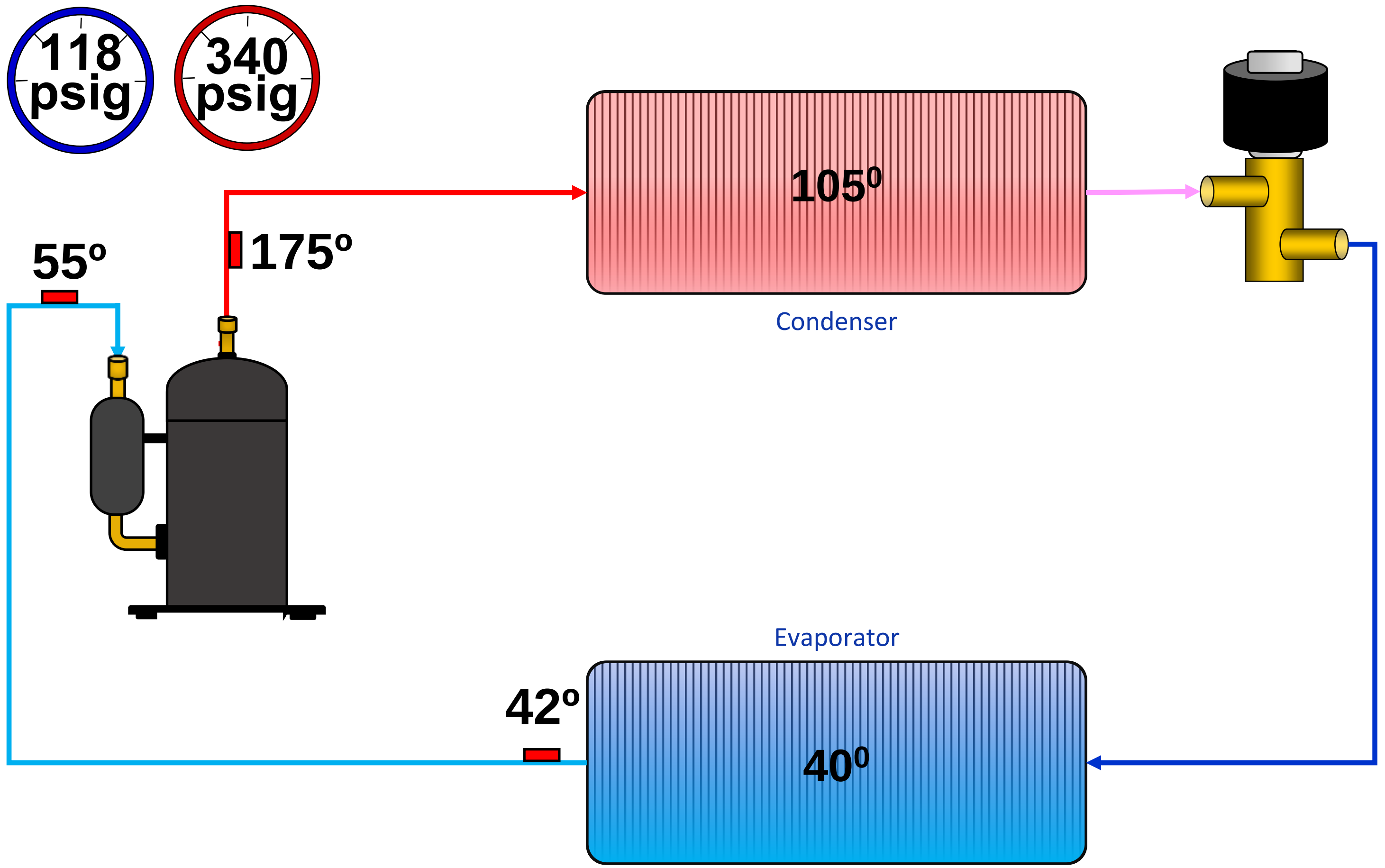
Measuring Superheat





Its good practice to measure the superheat and sub-cooling at the component due to pressure drops in the system and long pipe lengths.

Measuring Superheat



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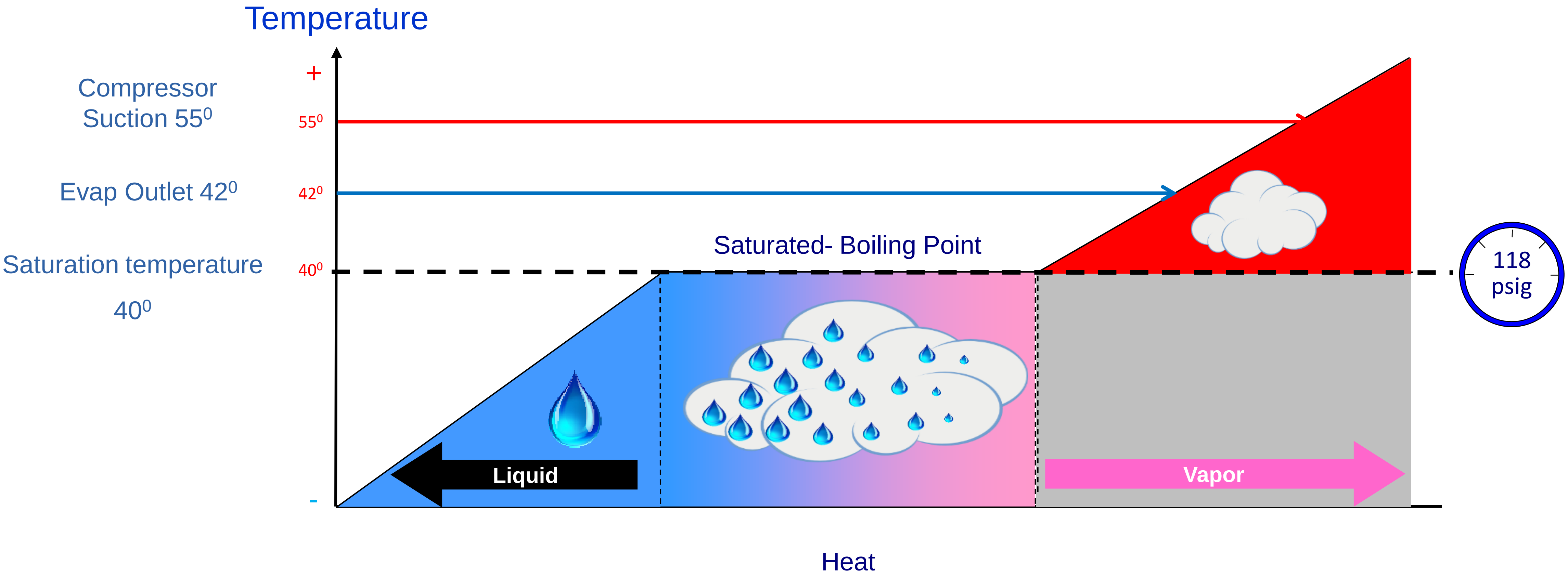
Example-



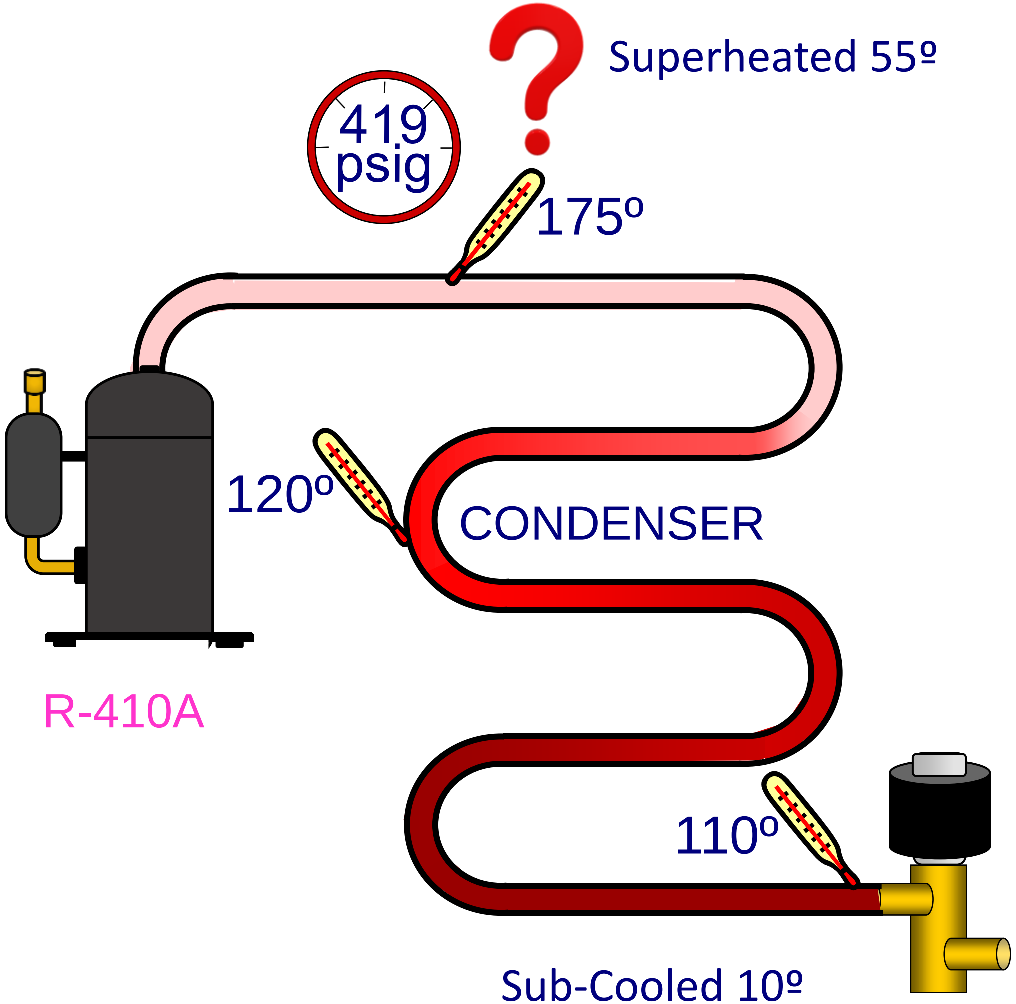
Temperature
Compressor
Suction 55⁰
Evap Outlet 42⁰
Saturation temperature
40⁰
+
55⁰
42⁰
40⁰
-
Saturated- Boiling Point
118
psig
Liquid
Vapor
Heat

How to Calculate Sub-Cooling

- Measure high side pressure.
  - Best practice are to measure the pressure at the location where you are calculating sub-cooling. (this is not always possible)
- Use this pressure to determine the saturation temperature from the PT chart.
- Measure the liquid line temperature.
- Subtract the line temperature from the saturation temperature.
- The difference is the degrees of sub-Cooling.

Sub-Cooling Example

PSIG	134a	22	410A
180	123	94	63
185	125	96	65
190	127	98	66
195	128	100	68
200	130	101	70
210	134	105	73
220	137	108	76
230	140	111	79
240	143	114	82
250	146	117	84
260	149	120	87
270	152	123	89
280	155	126	92
290		128	94
300		131	96
325		137	102
350		143	107
375		149	112
400		155	117
425			121
450			126
475			130



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250	146	117	84
260	149	120	87
270	152	123	89
280	155	126	92
290		128	94
300		131	96
325		137	102
350		143	107
375		149	112
400		155	117
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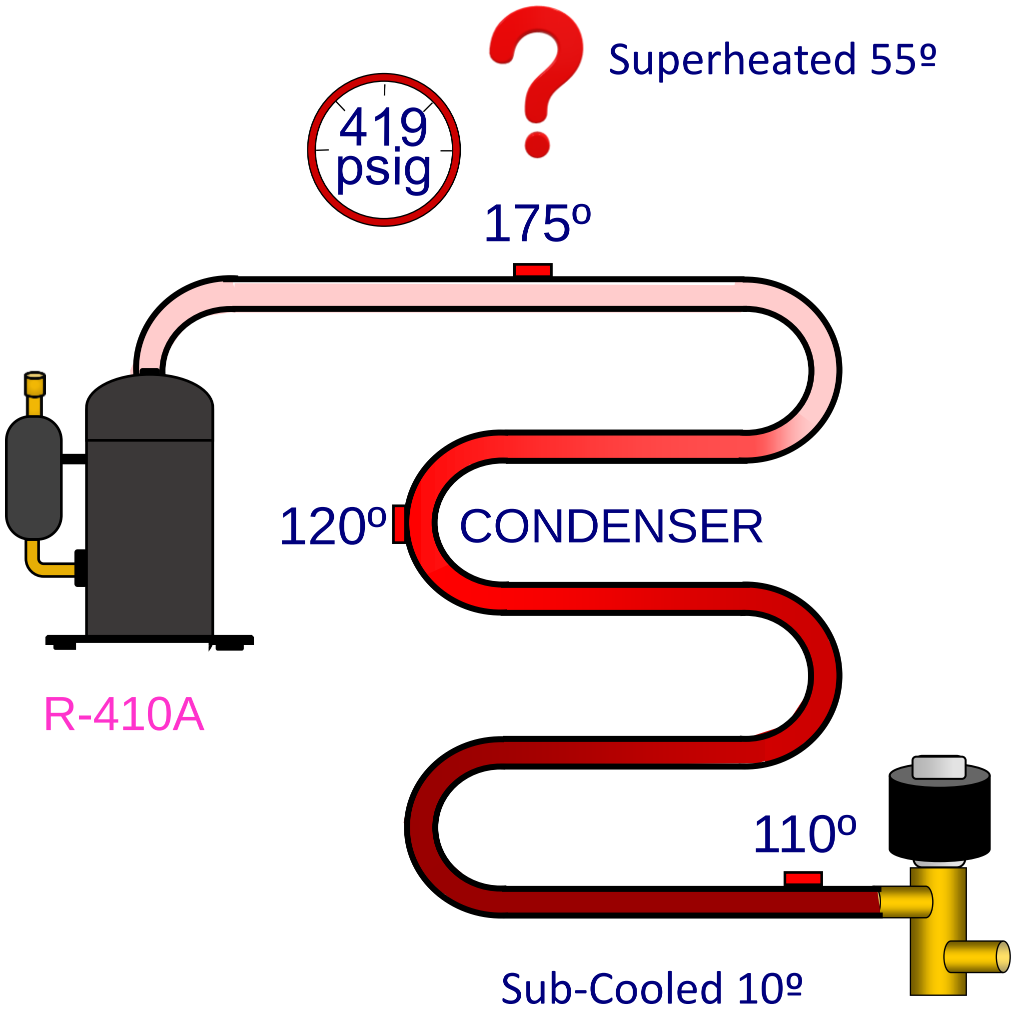
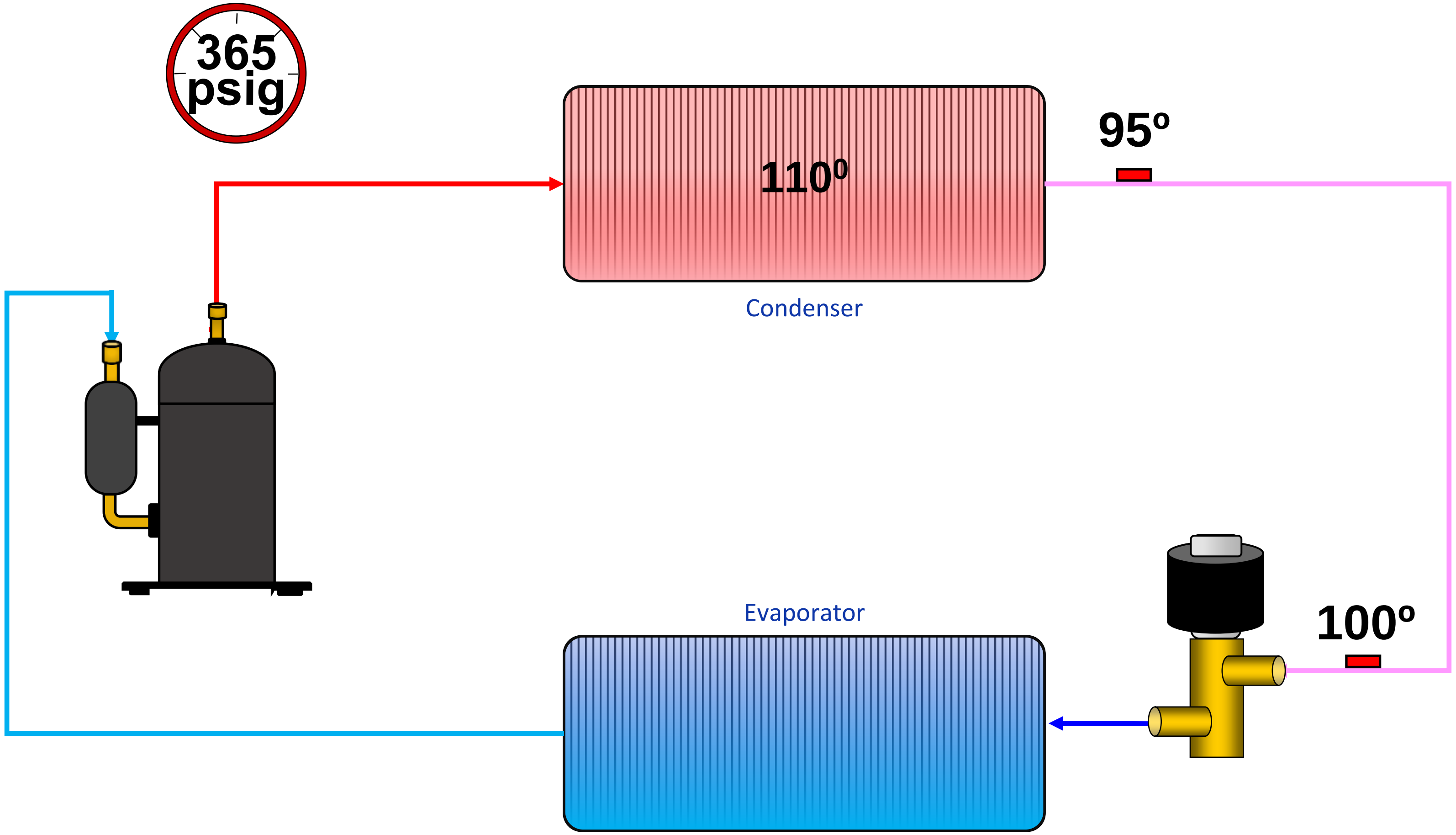
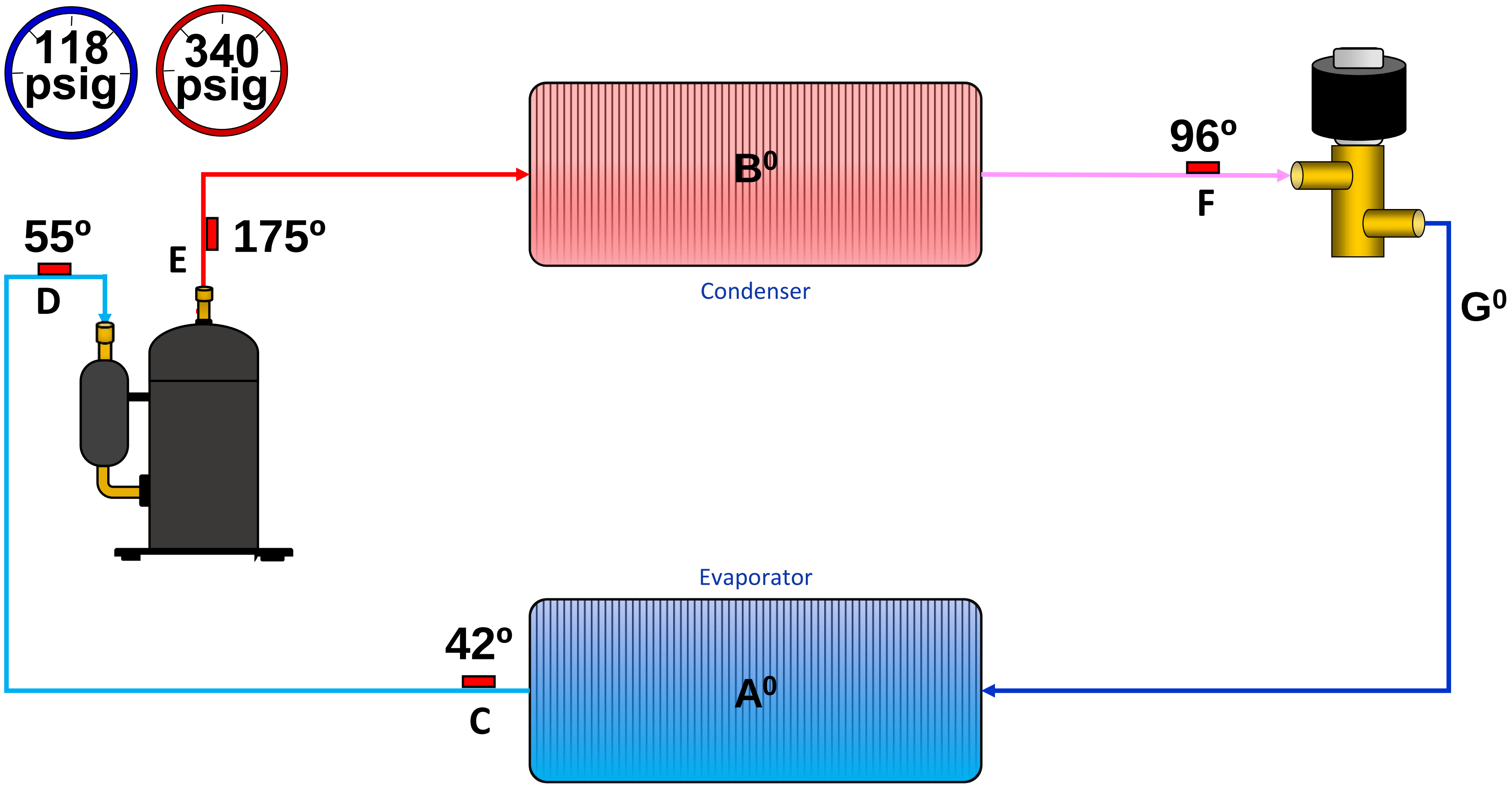


Measuring Sub-Cooling



What happens to the sub-cooling if the piping Is exposed to hot or cold ambient conditions ?

Calculate the state of the refrigerant or temperature at each location



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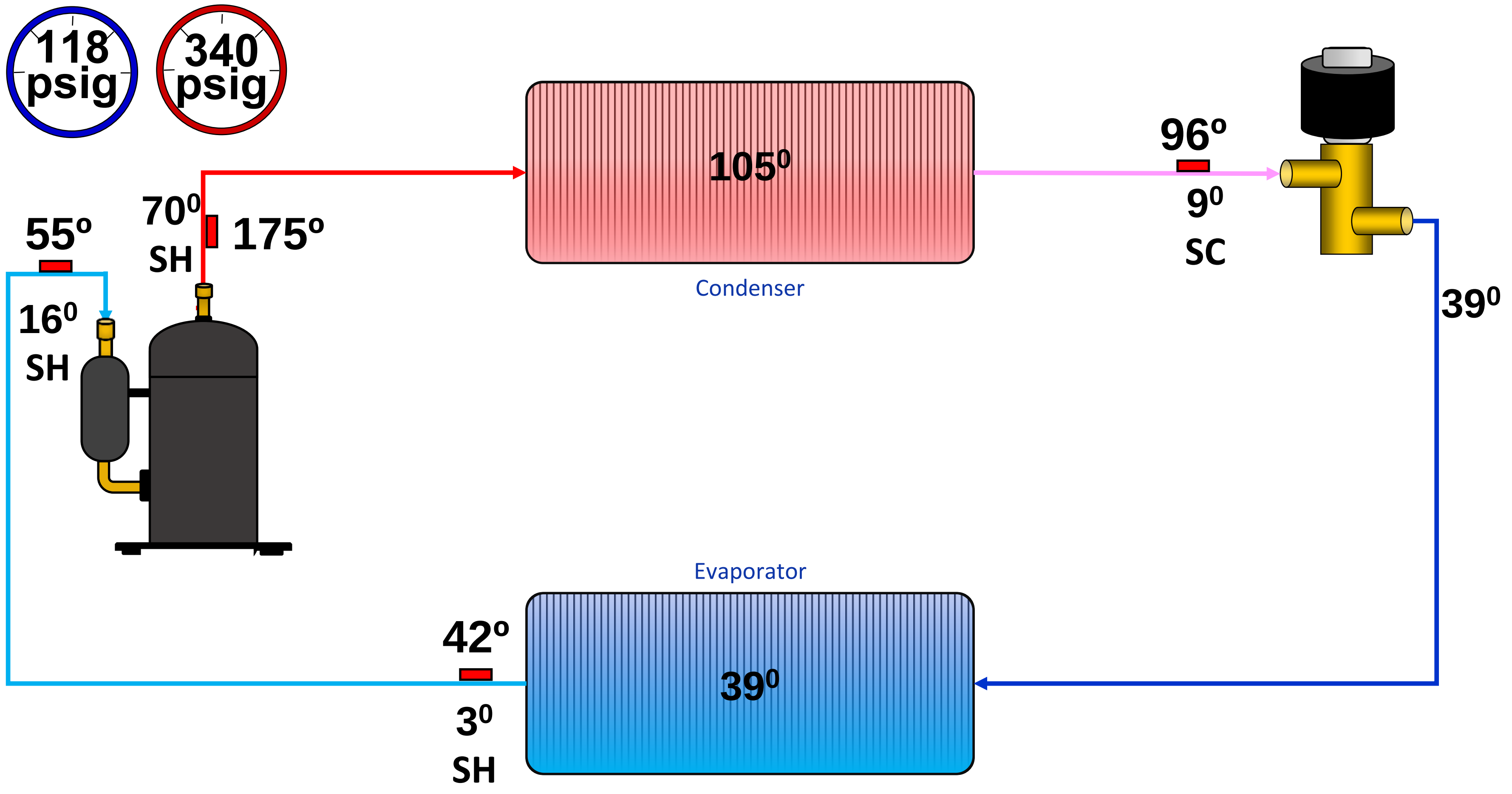
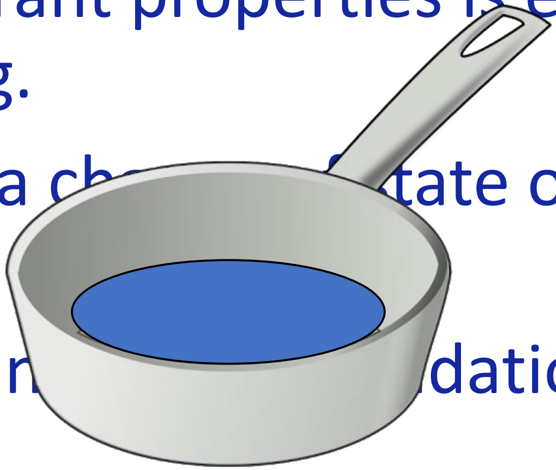


What does this mean?

- Having a good understanding of refrigerant properties is essential to understanding what the system is doing.
- More hea erred when there is a ch tate of the refrige
- Unders rheat and sub-coolin dation of refrigeration servicing.



Which pot requires more heat to boil all the water?

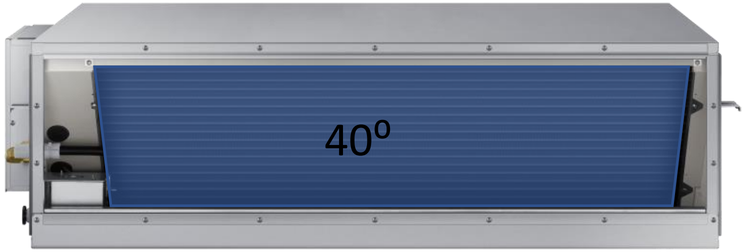
Why?

What does this have to do with refrigeration?

The more liquid refrigerant in the evaporator the more heat that can be removed from the space
This is why high efficiency units have lower evaporator superheats and high subcooling?

R 410A to R32

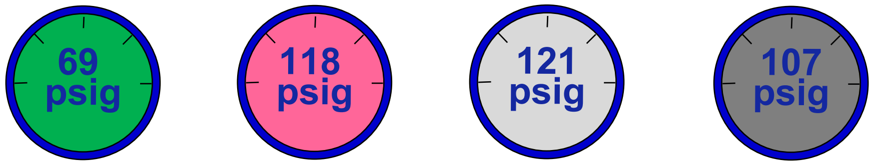
What Changes?



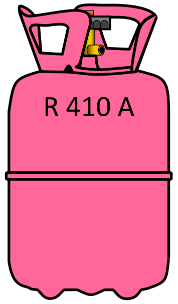
- Refrigeration is based on the transfer of heat.
- A quantity of heat is temperature.
- The design temperatures for air conditioning have remained the same since the beginning.
- A 40^0 evaporator is a common for air condition
- The only thing that changes is the operating pressure because that is dictated by the refrigerant in the system

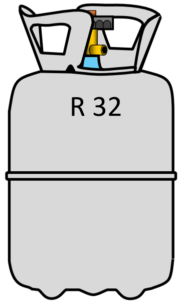
R 410 A Pressure Temperature Chart			
°F	°C	PSI	KPA
13.8	-10.1	68	468.8
17.5	-8.1	74	510.2
21	-6.1	80	551.6
24.3	-4.3	86	592.9
28.5	-1.9	94	648.1
32.2	0.1	102	703.3
35.5	1.9	110	758.4
39.5	4.2	118	813.6
40.5	4.7	120	827.4
43	6.1	126	868.7
46.3	7.9	134	923.9
50.3	10.2	144	992.8

R-32 Pressure Temperature Chart			
°F	°C	PSI	KPA
14	-10	69.79	481.2
17.6	-8	75.81	522.7
21.2	-6	82.15	566.4
24.8	-4	88.82	612.4
28.4	-2	95.84	660.8
32	0	103.21	711.6
35.6	2	110.95	765
39.2	4	119.07	821
42.8	6	127.58	879.6
46.4	8	136.49	941.1
50	10	145.81	1005.3



R 22 R 410 A R 32 R 454 B



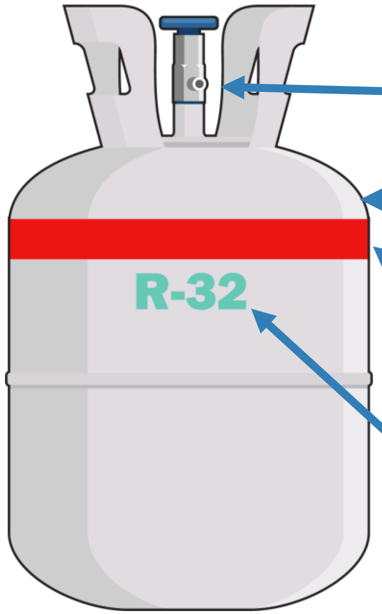


R – 32

- Is an A2L refrigerant
- Requires service tools rated for A2L refrigerants.
- System service valves will require Left-Handed Thread adapters or Hoses.
- Samsung will continue to use 5/16 connectors.
- The troubleshooting process for system with R-32 does not change!
- It is the responsibility of the contractor to get educated on the new refrigerants

A2L connectors will have a double knurled rings on the fitting

A New Look



- Reverse threads on port
- No tank color designation
- Added red band to identify A2L refrigerant
- Refrigerant type may follow traditional color code

Working A2L Refrigerant

- Vacuum pumps, recovery machines and Leak detectors must be rated for A2L Refrigerants
- Leak detectors should be a ultra sonic or infrared type.
- Existing Gauge sets can be used
- Nitrogen MUST be purged anytime a torch is applied during the installation and service process.
- System must be leak checked after installation
- System must hold a vacuum during installation.

New A2L Service Tools

Vacuum Pump

Recover Machine

Leak Detector

UL Service Requirements

- A2L refrigerants must be recovered.
- Tubing may be un-brazed.
- Ventilation required during work. It is assumed that the fan should be UL507 listed.
- Pressure tests must be conducted after repair
- It is also required to perform a leak test on the system after repairs have been made

Charging By Sub-Cooling or Superheat

- There is a specific amount of refrigerant is the system to transfer the correct amount .
- If you feel there is a refrigeration charge issue, verify the piping lengths of the system.
- Remove the charge and weigh it in!
- Do not:
  - "Gas and Go"
  - Do not charge to subcooling or superheat.
  - Do not charge to beer can cold.
- Why?

Because of the dynamic nature of a VRF system charging by sub-cooling is not allowed.

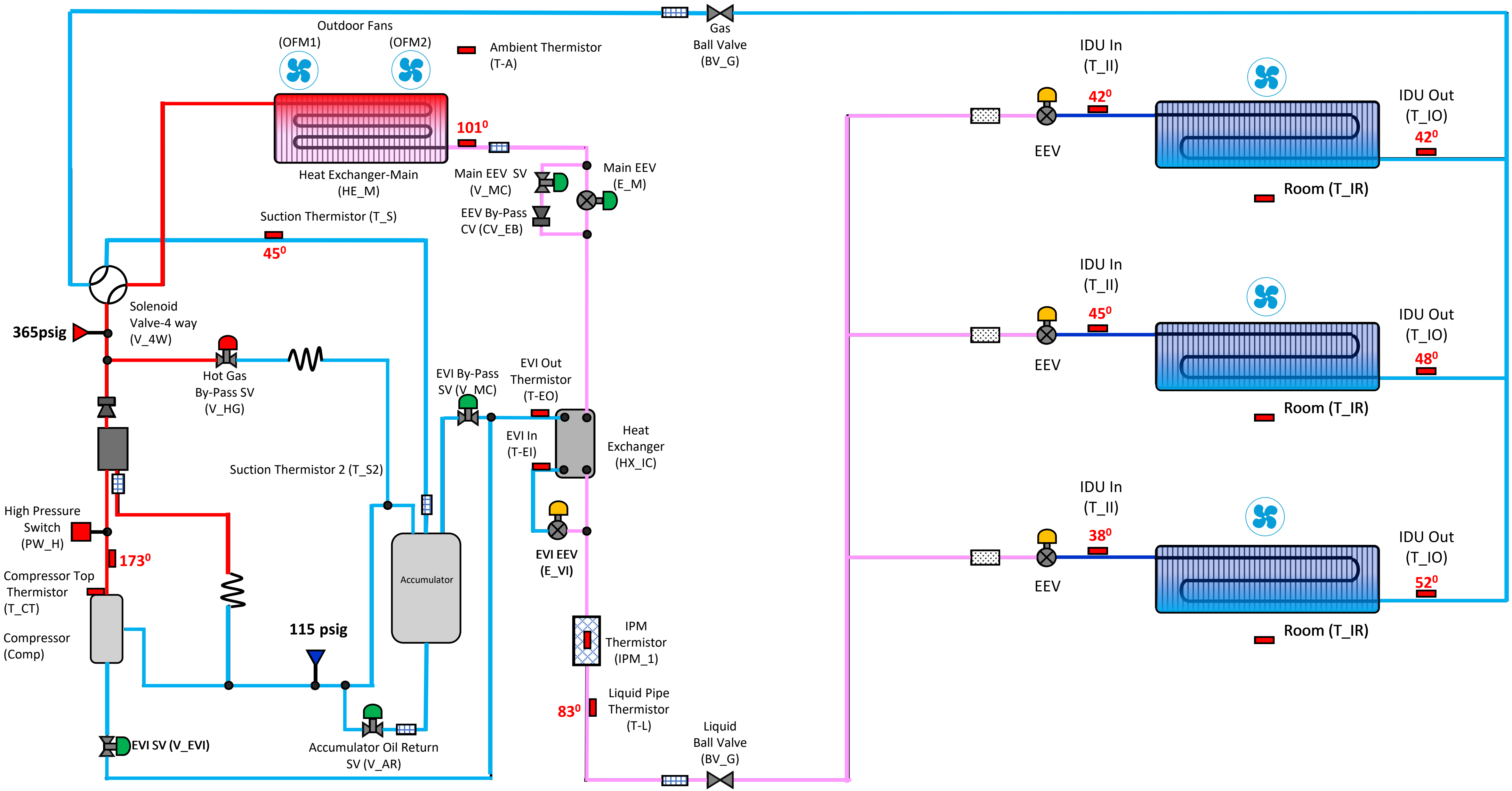
There is a charge imbalance between the heating and cooling cycles

Exercise

Use the following diagram and:

- Calculate the following
  - High Side Saturation Temperature
  - Low Side Saturation Temperature
  - Evap. 1 Saturation Pressure
  - Evap. 2 Saturation Pressure
  - Evap. 3 Saturation Pressure
  - Discharge Superheat
  - Evap. 1 Superheat
  - Evap. 2 Superheat
  - Evap. 3 Superheat
  - System Superheat
  - Condenser Sub-Cooling
  - ODU Sub-Cooling

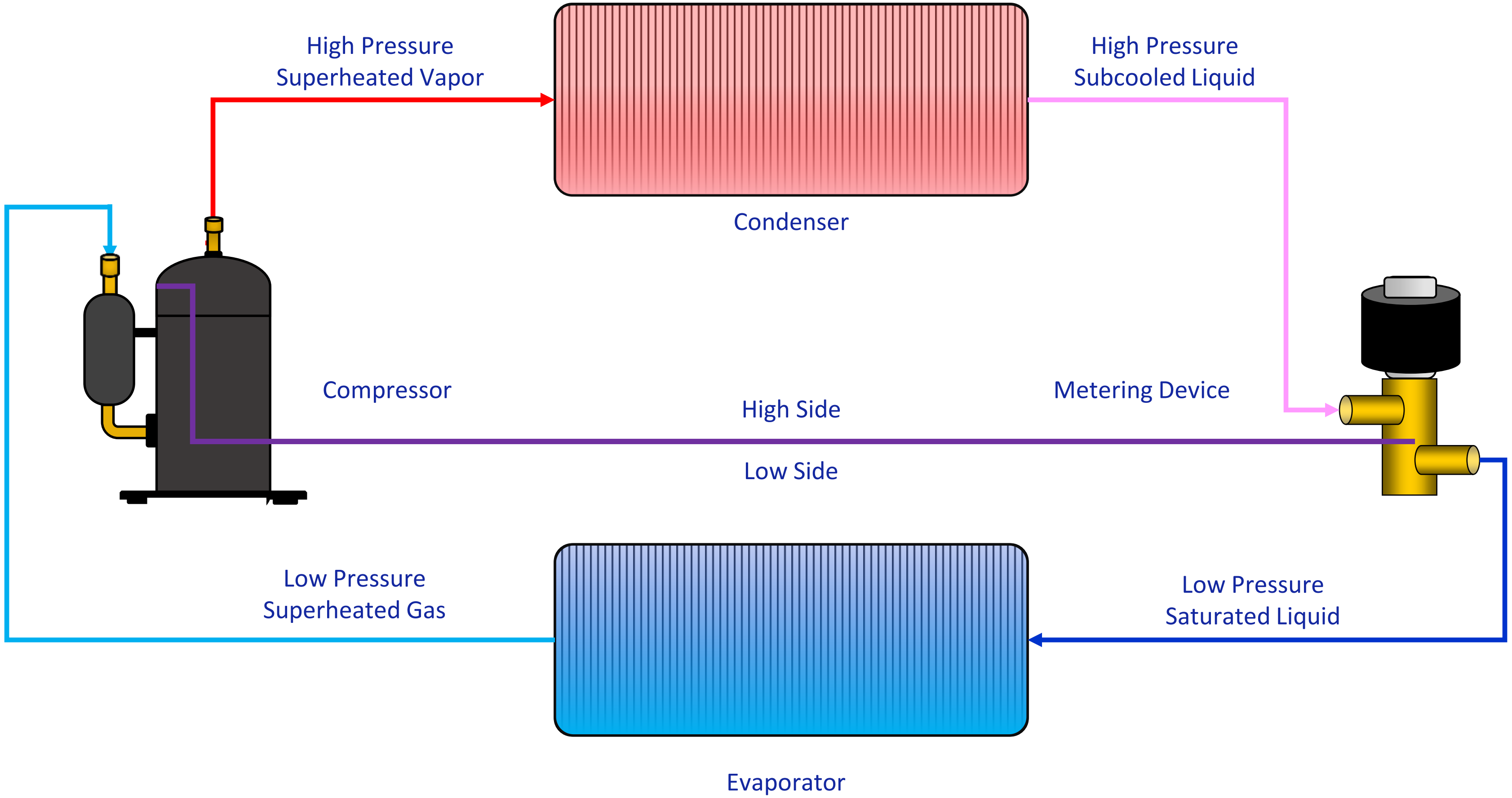
Calculate the state of the refrigerant at the given locations



Components of a refrigeration circuit

- All refrigeration circuits share the same basic components.
- Their functions do not change when used in different application.
- Having a basic understanding of the components individually will give you a better understand as to how the system as whole operates.
- Let’s look at the basic operation of each of these components.

Basic Refrigeration Cycle Components

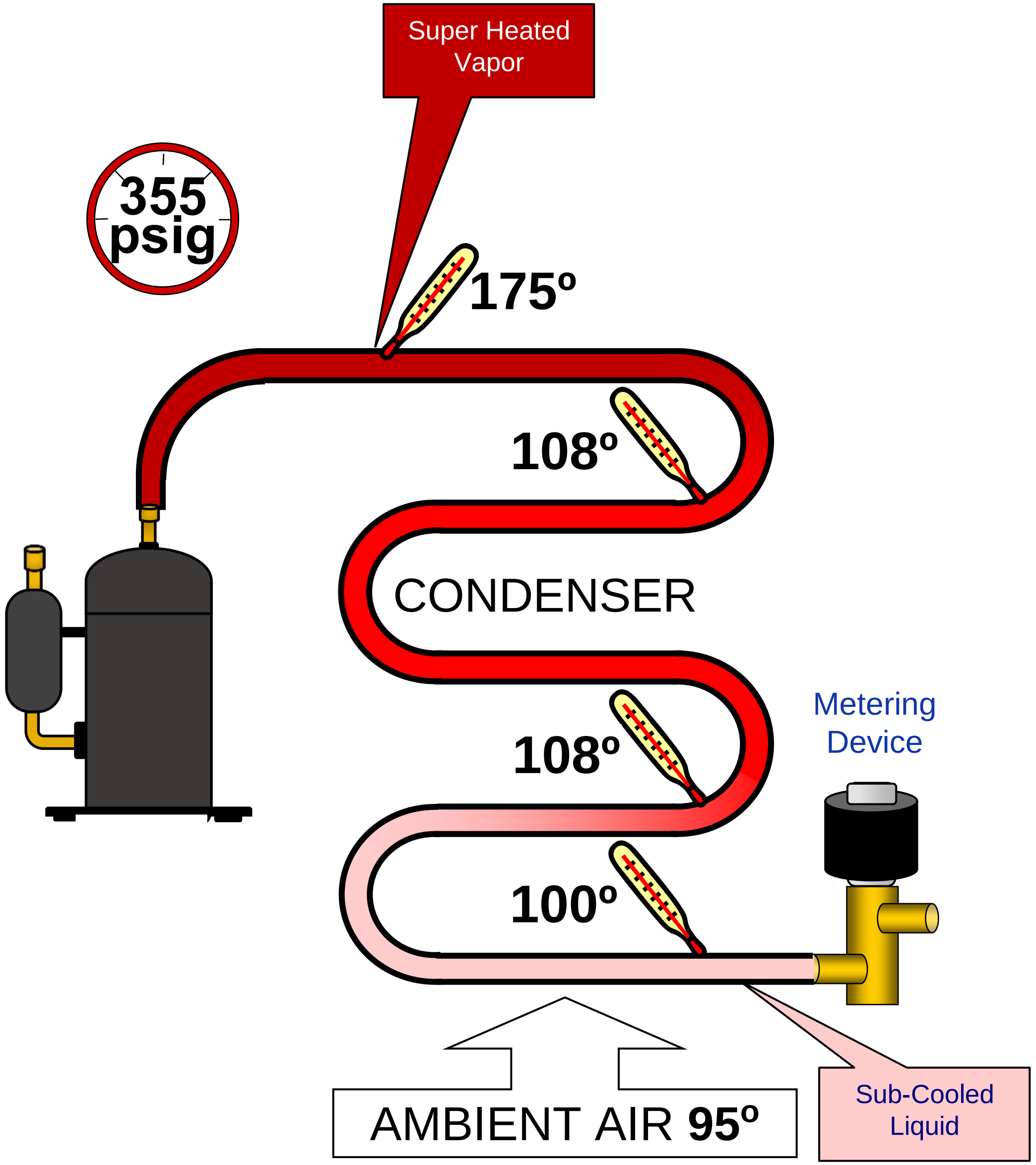


The Compressor

- It circulates refrigerant through the system.
- It compress low pressure gas to high pressure gas.
- Gas enters through the suction side of the compressor and leaves through the discharge line.
- The discharge temperature of the compressor is directly related the temperature of the suction gas.
- The compression ratio is the difference between the low side and high pressures
- Compressors are more efficient when it operates as low compression ratios.

The Condenser

- Its function is to remove the heat from the superheated refrigerant and in doing so changes the vapor refrigerant into a sub-cooled liquid
- The Condenser HAS TO BE HOTTER than the air going across it for heat to be transferred
- Does its function change in a heat pumps system?



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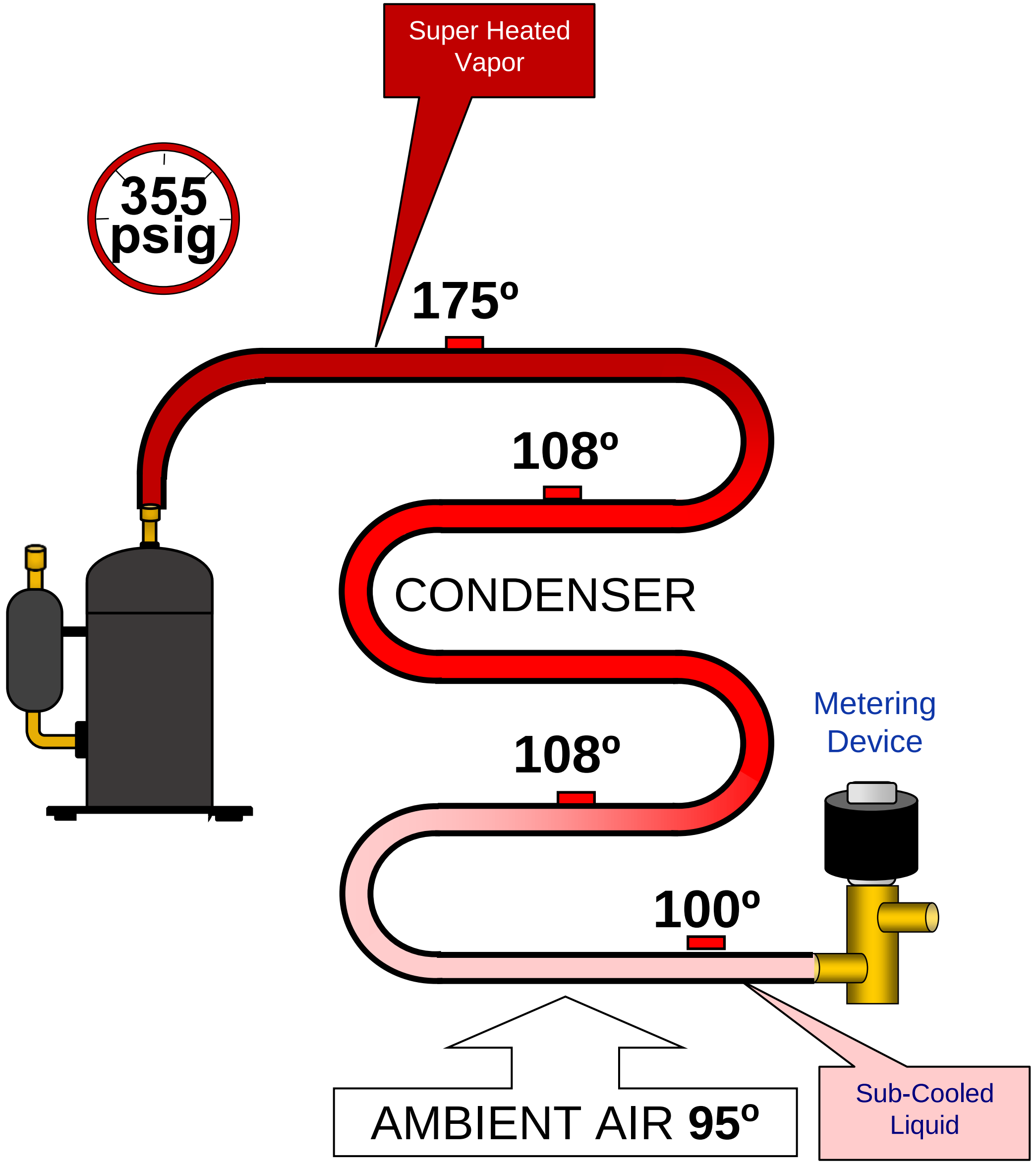
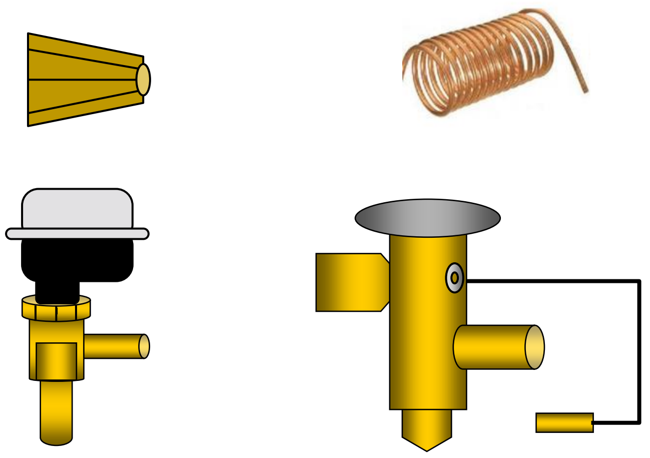
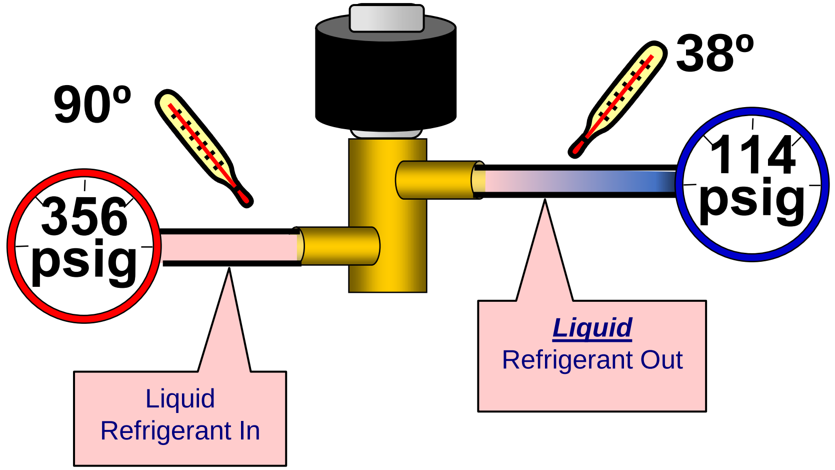


Metering Device

- It creates a pressure drop.
- It is designed to take condensing pressure (temperature) and lower it to a design evaporator pressure so that its corresponding temperature is below the design air temperature.
- It regulates flow to maintain a design temperature differential through the evaporator.

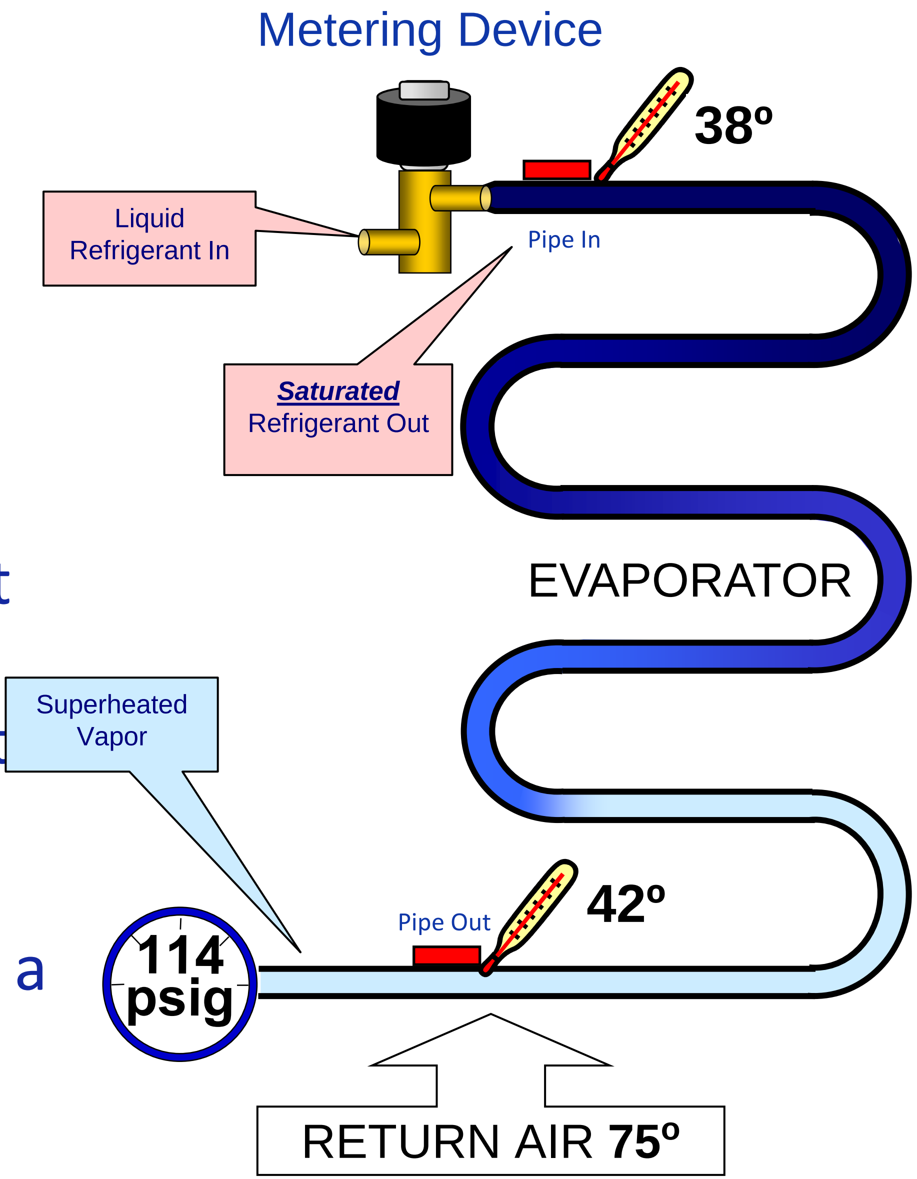




The refrigerant leaving the metering device is MOSTLY liquid.

The Evaporator

- Its function is to absorbed heat into the refrigerant.
- In doing so it changes the saturated liquid refrigerant into a superheated vapor
- For heat to be absorbed into the evaporator it MUST BE COLDER than the air going across it.
- The evaporator capacity is directly related to it airflow, coil temperature and the space humidity and temperature.
- Does the function of the evaporator change in a heat pump?



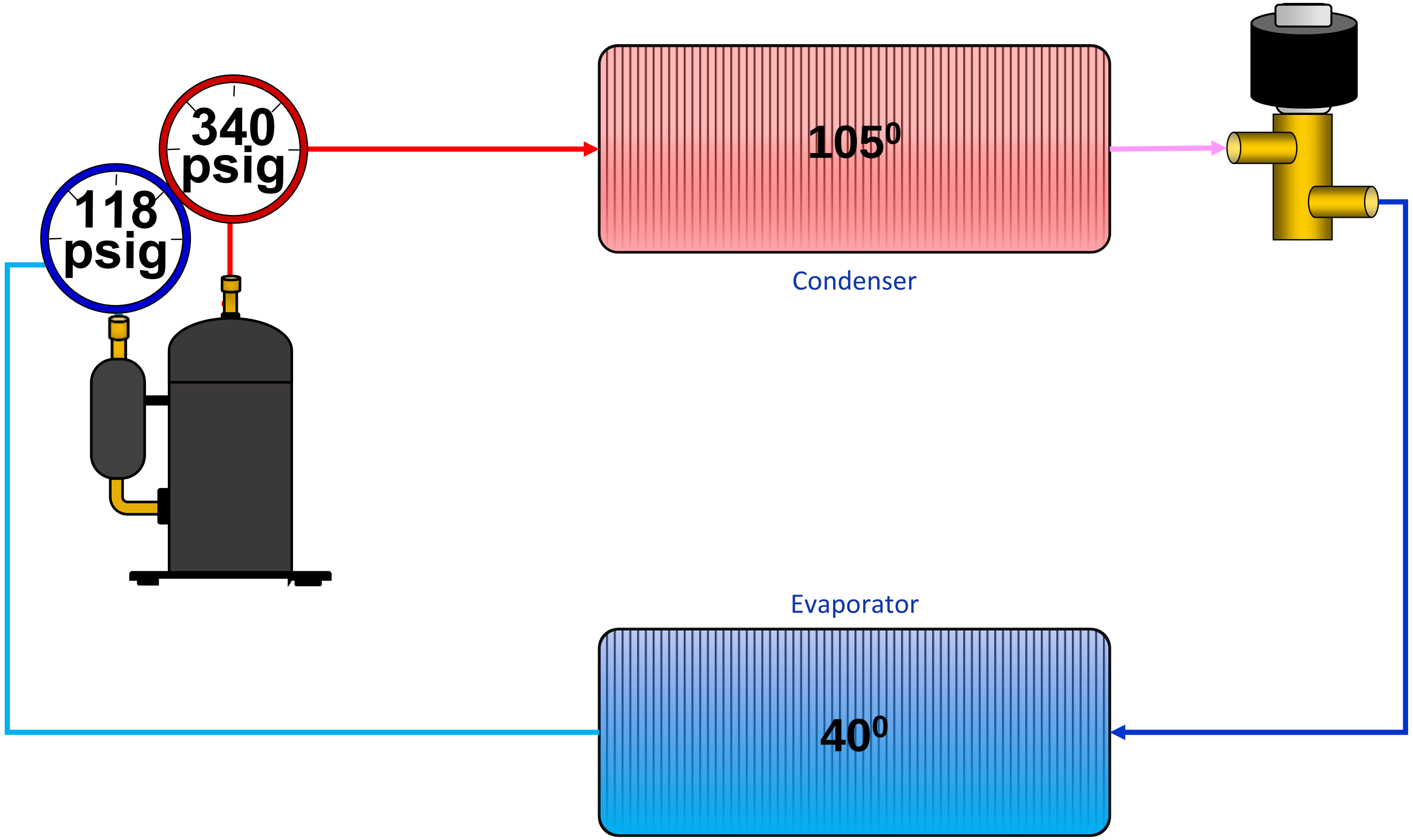
Piping systems

- The piping system is commonly overlooked when it comes to refrigeration system.
- A basic refrigeration system will contain:
  - Discharge line / Hot gas line
    - Connects the discharge of the compressor to the inlet of the condenser
    - Carries high pressure, high temperature superheated vapor.
  - Liquid line
    - Connects the outlet of the condenser to the inlet of the metering device.
    - Carries high pressure high temperature sub-cooled liquid to the metering device.
  - Suction Line
    - Connects the outlet of the evaporator to the inlet of the compressor.
    - Carries low pressure low temperature superheated vapor.

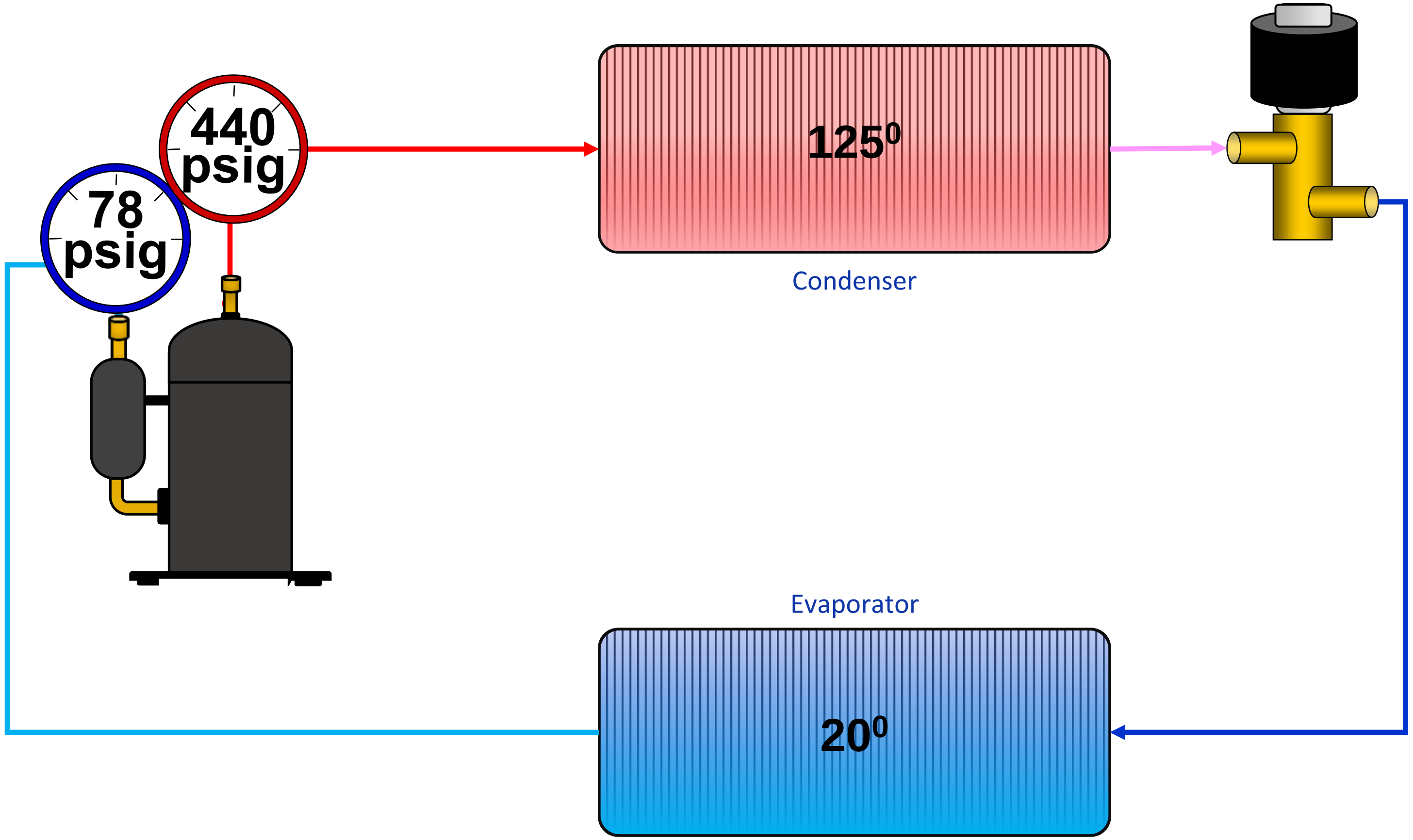
Review

- Understanding superheat and sub-cooling are the foundations for troubleshooting a refrigeration system.
- Being able to measure and calculate superheat and sub-cooling are the fist steps in the troubleshooting process.
- If the Evaporator superheat is high it can indicate specific issues.
- If the Evaporator superheat is low that can indicate a different set of issues.
- Knowing the sub-cooling can help determine what the issues are present in the evaporator.

Why are gauges used during service?

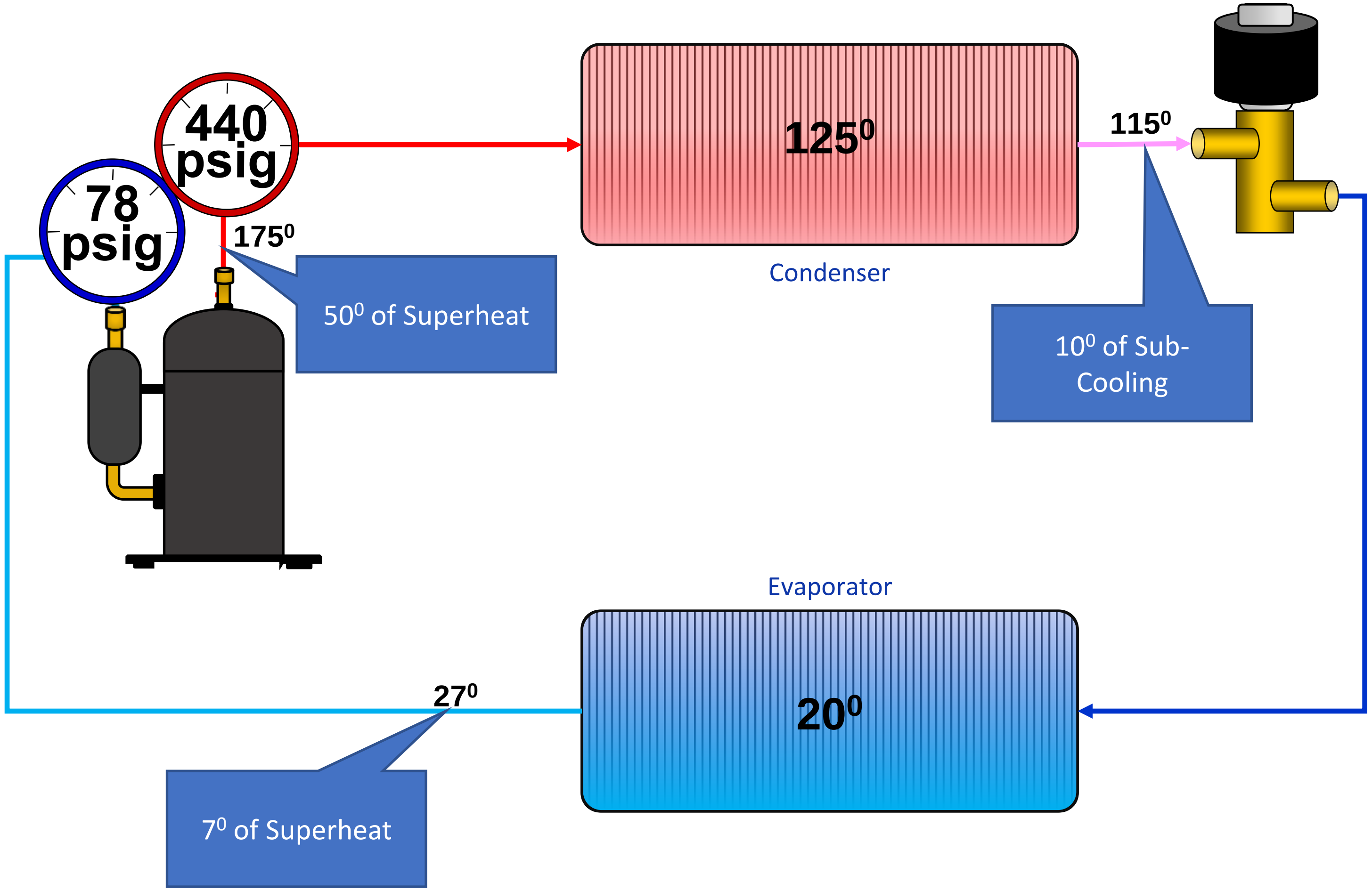


If the coil temperatures are known, can you determine the system pressures ?

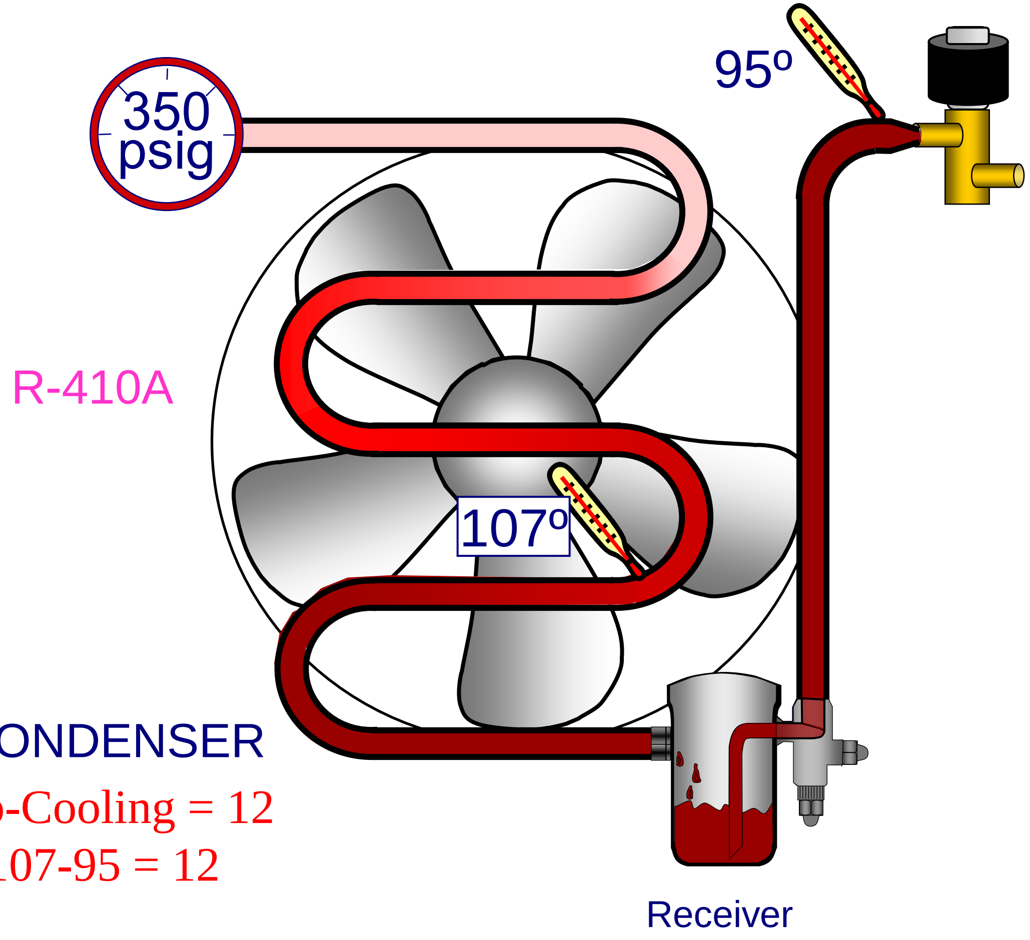


What should the pipe temperatures be?

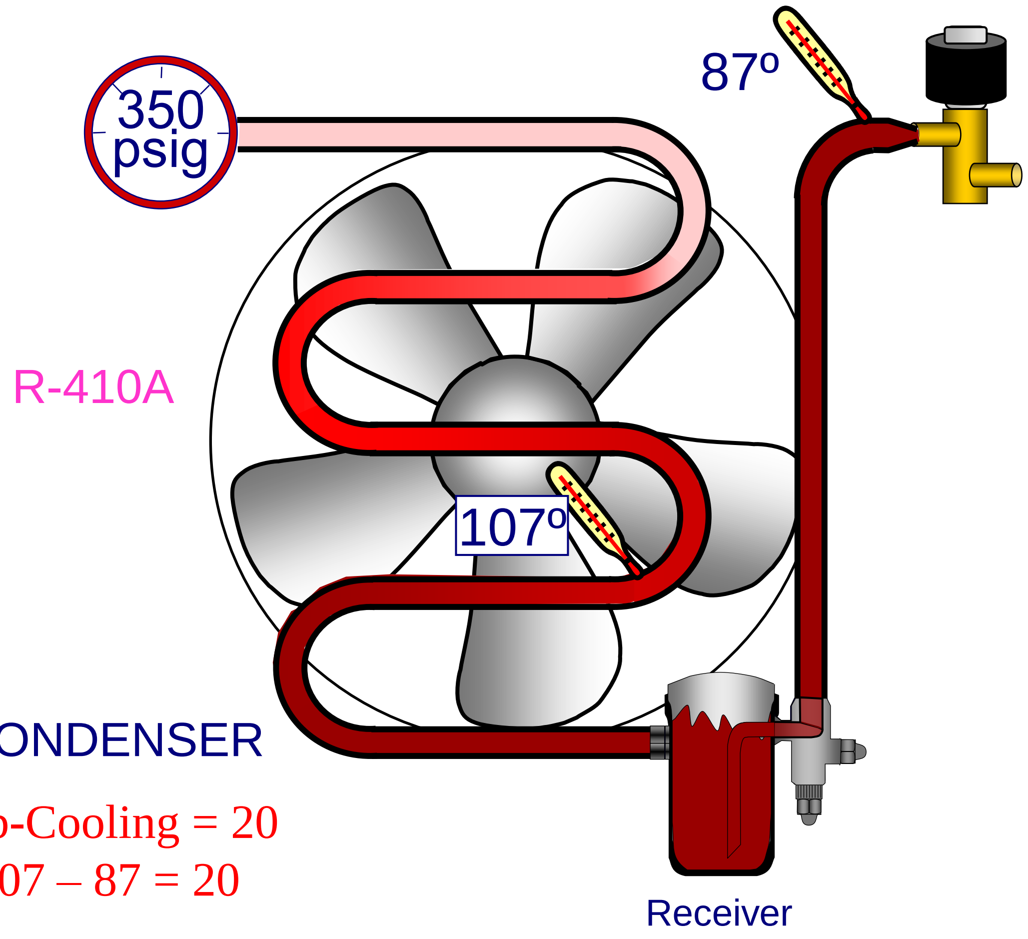
A+B=C



Which System Has More Refrigerant?

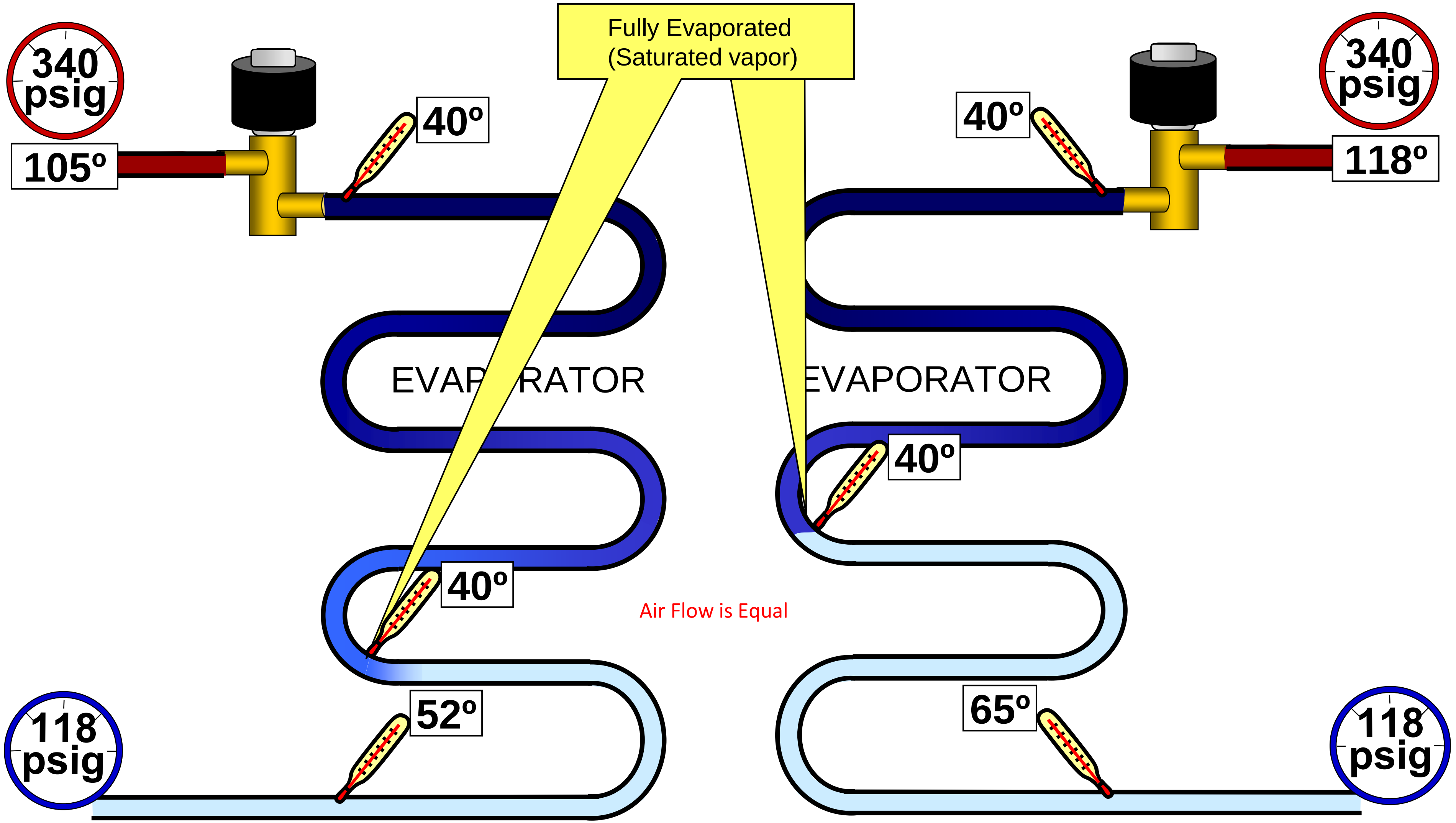


Sub-Cooling = 12
107-95 = 12



Sub-Cooling = 20
107 – 87 = 20

Which evaporator absorbs more heat?



Thermodynamics
Contact Time

- The amount of time that the air is contact with the refrigerant is regulated by:
  - The size of the coil.
  - The fan
    - CFM of the Evaporator and Condenser
  - The compressor speed
    - Frequency of the compressor

Air Flow – Compressor Speed Relationship

- There is a direct relationship between the speed of the compressor and air flow.
- The compressor speed is selected to match the air flow provided.
- If the compressor speed is too high the operating suction pressure will be low, and the system superheat will be low.
- If the compressor speed is too low, the operating suction pressure, evaporator saturation temperature and superheat will be high.
- If air flow is low, it will cause the operating suction and temperature to be low cooling mode
- In the heating mode it will cause the operating pressures and temperatures to be high.

Refrigeration Theory Review

- Heat travels from hot to cold.
- The rate of heat transfer is affected by surface area and contact time.
  - Contact time is dictated by the fan CFM and Compressor speed.
  - The bigger the coil the better the heat transfer with less of a temperature differential (TD).
    - This also decreases the systems head pressure and increase the efficiency of the system

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10 SEER

2 Tons

20 SEER

Surface Area & Contact Time

10 SEER
30^0 TD

14 SEER
20^0 TD

20 SEER
10-15^0 TD

A larger Volume of the outdoor unit reduces the high side pressure
Larger coils allow more heat transfer with lower splits

Thermodynamics
Temperature Difference

- Refrigeration is based on the principle of thermodynamics.
- The laws of thermodynamics state:
    - Heat travels from hot to cold.
    - In order of heat to be transferred that there must be a difference in temperature.
    - The rate at which heat can be transferred depends on surface area, contact time and temperature difference.

Always remember that refrigeration is based on temperatures NOT pressures

A + B = C

Temperature Differences & Head Pressures

(A+B=C)

Saturation Temperature

80^0 Outdoor Air Temperature

Head Pressure

110^0 Saturation Temp

10 SEER
30^0 TD

365 psig

100^0 Saturation Temp

14 SEER
20^0 TD

317 psig

90^0-95^0 Saturation Temp

20 SEER
10-15^0 TD

275 psig

296 psig

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What Effect Does Lower Head Pressures Have On a System?

A typical evaporator – low side saturation temperature is 40^0 which has a corresponding pressure of 118 psig

10 SEER
30^0 TD

20+ SEER
10^0 TD

The difference between high side and low side is the compression ratio. High compression rations cause higher heat of compression.

Which Compressor Has to work harder ?
Which Uses More Energy and why?

How does air flow effect efficiency?

- 2 ton systems

Air Volume	CFM (L/M/H)	466 / 537 / 629

19.5 Seer

16.9 Seer

Air Volume	CFM (L/M/H)	547 / 636 / 759

The required air flow depends on the equipment and its efficiency!

Indoor Unit Rating at High Speed

- In order to get the correct capacity out of the indoor unit the fan must be on high speed.
- The coil and filter must be clean.
- Ducted units must be operating under the maximum ESP of the unit.
- How does the speed of the fan effect contact time?

CFM: L-293/ M-321/ H-364

CFM: L-300/ M-353/ H-424

CFM: L-459/ M-494/ H-547.

Each unit has a capacity of 12,000 Btu

Contact Time
Air Flow

- There are no more rules of thumb when it comes to air flow.
- The air flow has to be correct for the system you're working on.
- Too much air flow is a bad as not enough.
- It is recommended that the ESP of any ducted unit be tested during start up.
- Do you know what the system ESP was tested at?

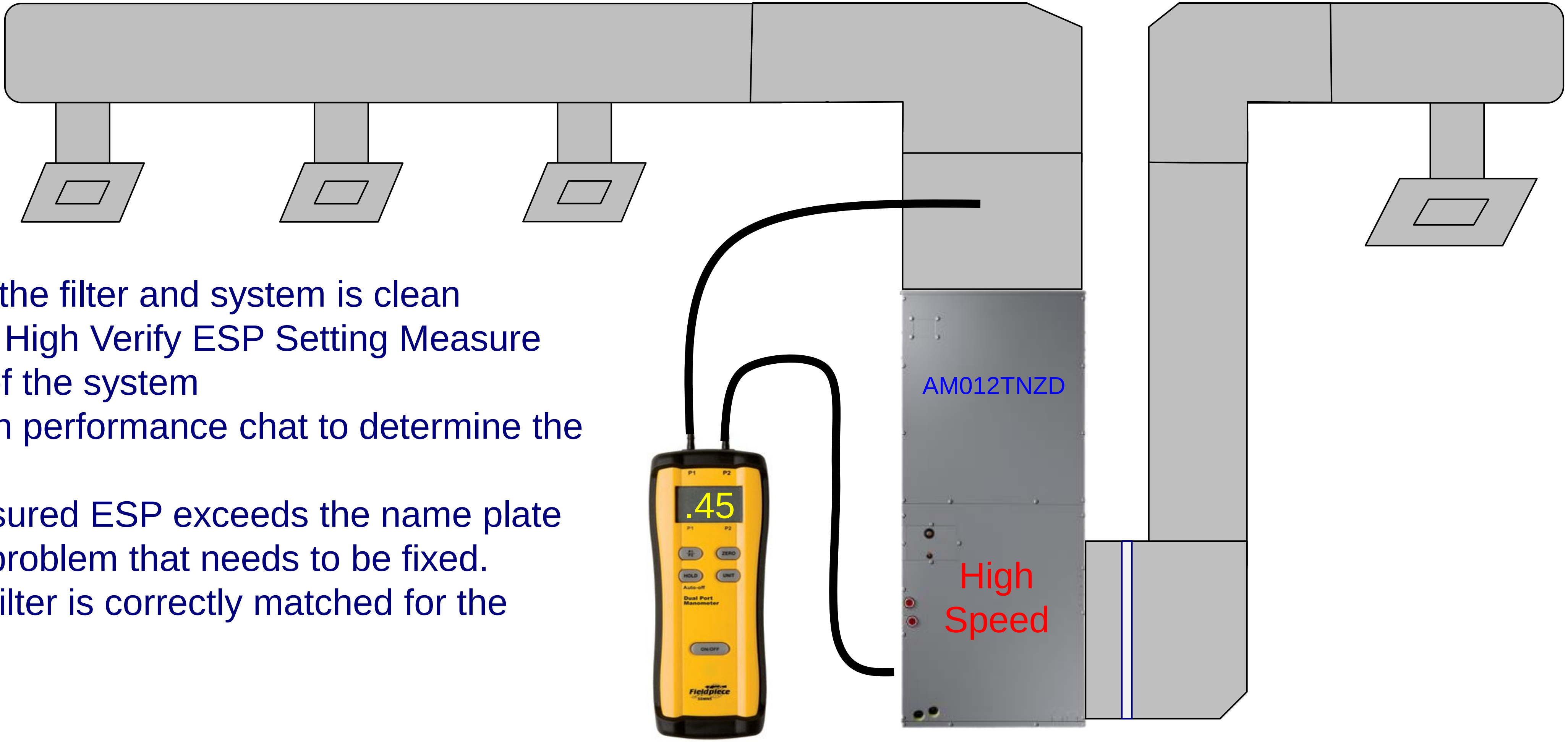
Specifications

Performance	Nominal Capacity (Btu/h)	Cooling	24,000
		Heating	27,000
Power	Voltage	ø / V / Hz	1 / 208-230 / 60
	Nominal Input Current*	Cooling (A)	1.12
	MCA*	Amps	0.9
	MOCP*	Amps	10
Fan	Type		Double-inlet, forward curve, centrifugal
	Motor	Type	Constant-torque (ECM)
		HP	1/3
		Output (W)	290
Airflow	CFM @ 0.4" ESP (UL)	L / M / H	450 / 560 / 706
External Static Pressure	Standard	"WC	0.4
	Min. / Max.	"WC	0.1 / 0.7

Always verify the proper air flow for each unit

Ducted Units
Verifying CFM

Verify that the filter and system is clean
Set Fan to High Verify ESP Setting Measure
The ESP of the system
Use the fan performance chat to determine the CFM.
If the measured ESP exceeds the name plate there is a problem that needs to be fixed.
Verify the filter is correctly matched for the system.



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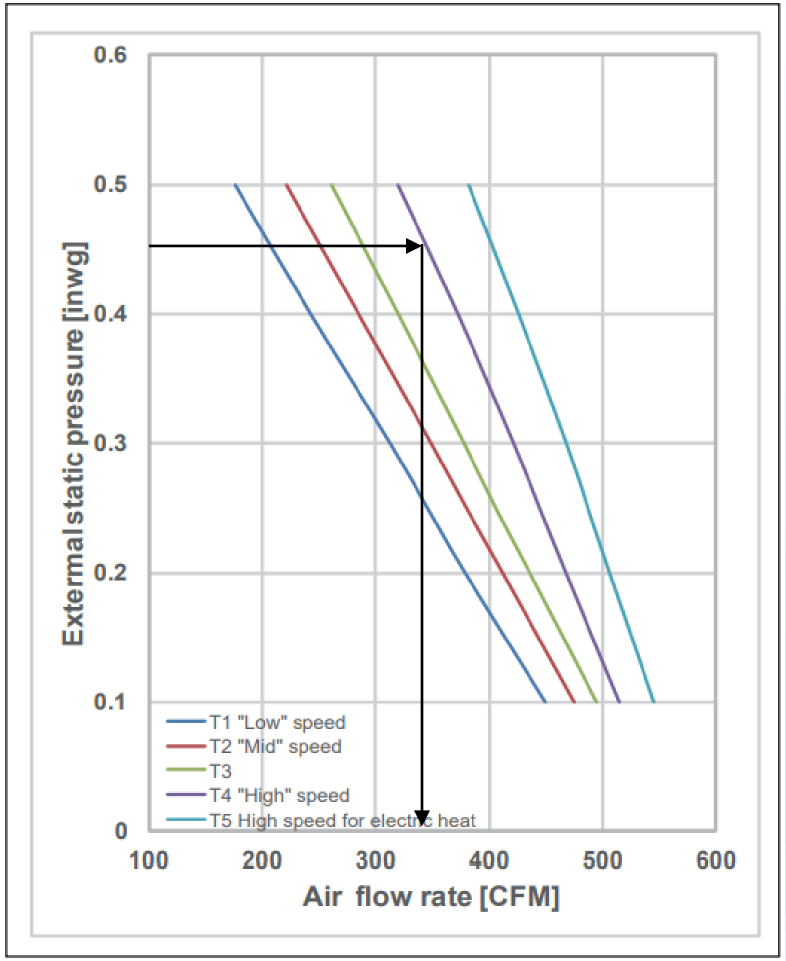
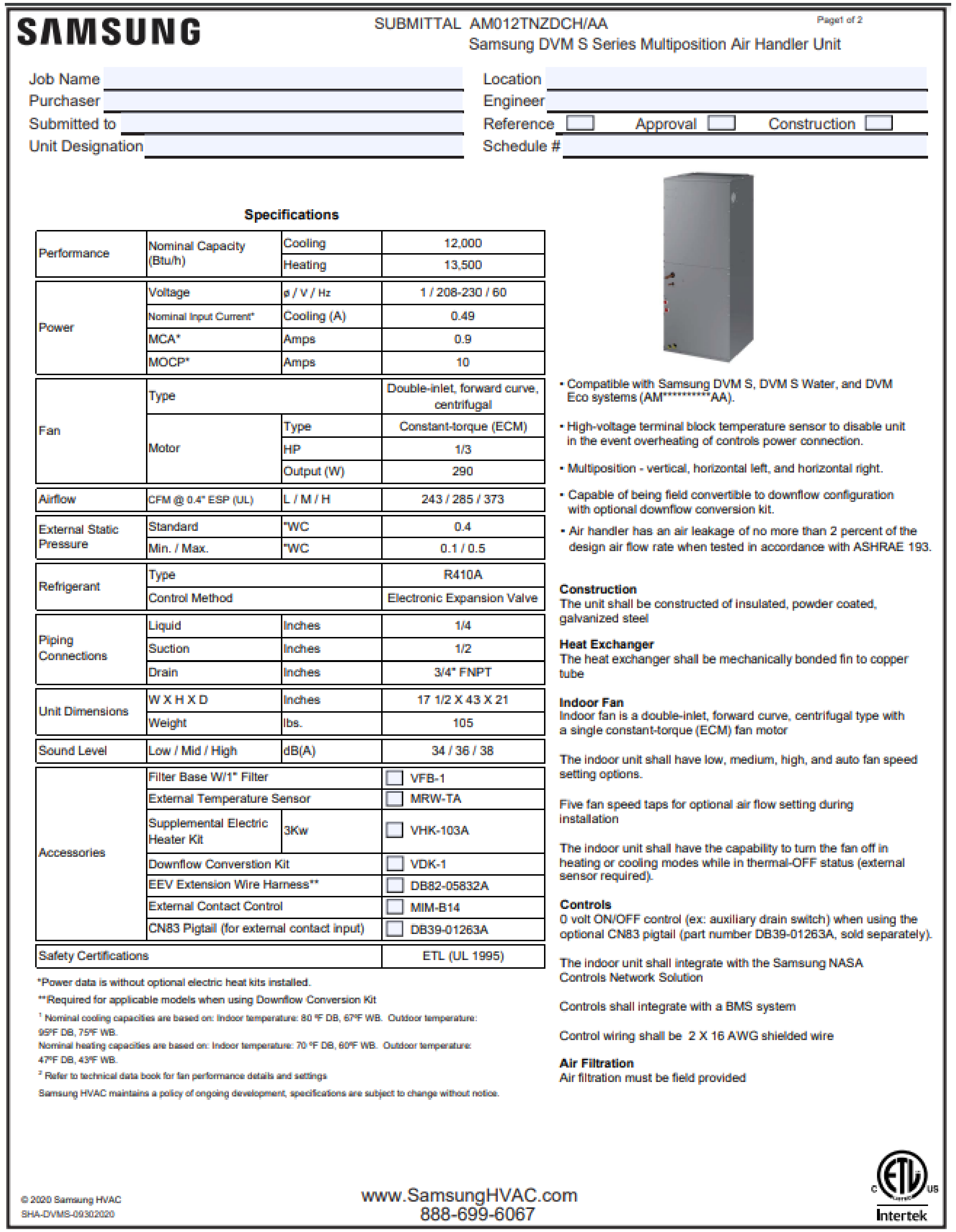
Air Handler CFM AM012

1) AM012TNZDCH/AA





SAMSUNG SUBMITTAL AM012TNZDCH/AA

Page1 of 2

Samsung DVM S Series Multiposition Air Handler Unit

Job Name ____ Location ____
Purchaser ____ Engineer ____
Submitted to ____ Reference ☐ Approval ☐ Construction ☐
Unit Designation ____ Schedule # ____

Specifications

Performance	Nominal Capacity (Btu/h)	Cooling	12,000
		Heating	13,500
Power	Voltage	ø / V / Hz	1 / 208-230 / 60
	Nominal Input Current*	Cooling (A)	0.49
	MCA*	Amps	0.9
	MOCP*	Amps	10
Fan	Type		Double-inlet, forward curve, centrifugal
	Motor	Type	Constant-torque (ECM)
		HP	1/3
		Output (W)	290
Airflow	CFM @ 0.4" ESP (UL)	L / M / H	243 / 285 / 373
External Static Pressure	Standard	"WC	0.4
	Min. / Max.	"WC	0.1 / 0.5
Refrigerant	Type		R410A
	Control Method		Electronic Expansion Valve
Piping Connections	Liquid	Inches	1/4
	Suction	Inches	1/2
	Drain	Inches	3/4" FNPT
Unit Dimensions	W X H X D	Inches	17 1/2 X 43 X 21
	Weight	lbs.	105
Sound Level	Low / Mid / High	dB(A)	34 / 36 / 38
Accessories	Filter Base W/1" Filter		☐ VFB-1
	External Temperature Sensor		☐ MRW-TA
	Supplemental Electric Heater Kit	3Kw	☐ VHK-103A
	Downflow Conversion Kit		☐ VDK-1
	EEV Extension Wire Harness**		☐ DB82-05832A
	External Contact Control		☐ MIM-B14
	CN83 Pigtail (for external contact input)		☐ DB39-01263A
Safety Certifications			ETL (UL 1995)

*Power data is without optional electric heat kits installed.

**Required for applicable models when using Downflow Conversion Kit

[1] Nominal cooling capacities are based on: Indoor temperature: 80 °F DB, 67°F WB. Outdoor temperature: 95°F DB, 75°F WB.
Nominal heating capacities are based on: Indoor temperature: 70 °F DB, 60°F WB. Outdoor temperature: 47°F DB, 43°F WB.

[2] Refer to technical data book for fan performance details and settings

Samsung HVAC maintains a policy of ongoing development, specifications are subject to change without notice.

- Compatible with Samsung DVM S, DVM S Water, and DVM Eco systems (AM**********AA).
- High-voltage terminal block temperature sensor to disable unit in the event overheating of controls power connection.
- Multiposition - vertical, horizontal left, and horizontal right.
- Capable of being field convertible to downflow configuration with optional downflow conversion kit.
- Air handler has an air leakage of no more than 2 percent of the design air flow rate when tested in accordance with ASHRAE 193.

Construction
The unit shall be constructed of insulated, powder coated, galvanized steel

Heat Exchanger
The heat exchanger shall be mechanically bonded fin to copper tube

Indoor Fan
Indoor fan is a double-inlet, forward curve, centrifugal type with a single constant-torque (ECM) fan motor

The indoor unit shall have low, medium, high, and auto fan speed setting options.

Five fan speed taps for optional air flow setting during installation

The indoor unit shall have the capability to turn the fan off in heating or cooling modes while in thermal-OFF status (external sensor required).

Controls
0 volt ON/OFF control (ex: auxiliary drain switch) when using the optional CN83 pigtail (part number DB39-01263A, sold separately).

The indoor unit shall integrate with the Samsung NASA Controls Network Solution

Controls shall integrate with a BMS system

Control wiring shall be 2 X 16 AWG shielded wire

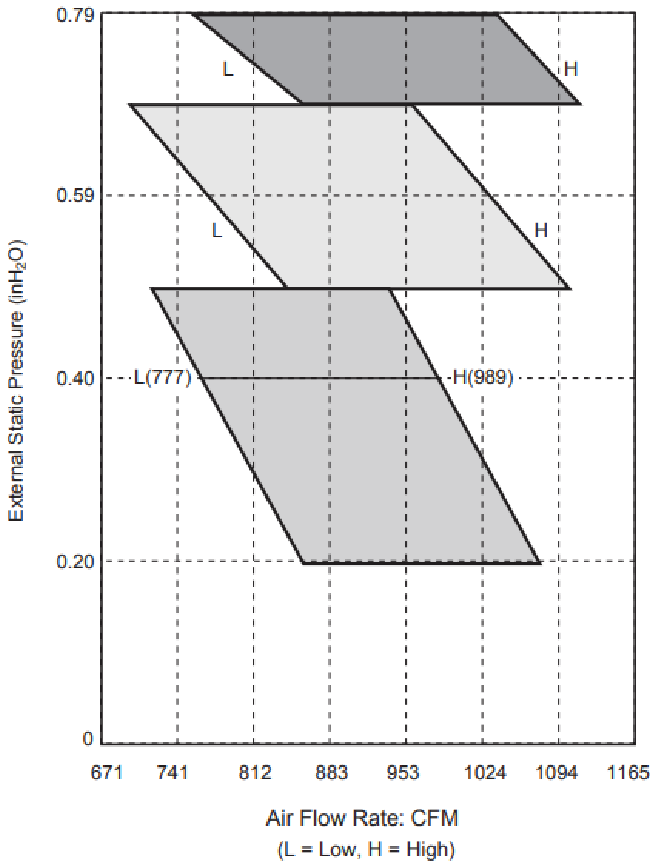
Air Filtration
Air filtration must be field provided

www.SamsungHVAC.com
888-699-6067

Intertek

Has the system been set up properly?

Fan Performance Data



External Static pressure (InH2O / mmAq)	Option code
0.2 / 5	010054-1355C6-206E6E-331110
0.4/ 10	010054-1355C6-206E6E-331110
0.6 /15	010054-13596E-206E6E-331110
0.8 / 20	010054-135AE4-206E6E-331110

AM036FNHDCH/AA

Airflow	CFM (UL)	H/M/L	989 / 883 / 777
	Total CFM Range [2]		706 - 1,116

External Static Pressure	Standard	"WC	0.39
	Min. / Max.	"WC	0.20 - 0.79

Product Option Code

Adjusting Static Pressure

0" - 0.04" ESP

Model number: AM012MNMDCH/AA
Description: DVM S, Duct S - 12000 Btu/h,
0" - 0.04" ESP (DEFAULT)
Option code: 010054-1E5072-202323-331102
Static pressure (inches): 0" - 0.04" (DEFAULT)

0.12" - 0.20“ ESP

Model number: AM012MNMDCH/AA
Description: DVM S, Duct S - 12000 Btu/h, 0.12" - 0.20" ESP
Option code: 010054-1E5449-202323-331102
Static pressure (inches): 0.12" - 0.20"

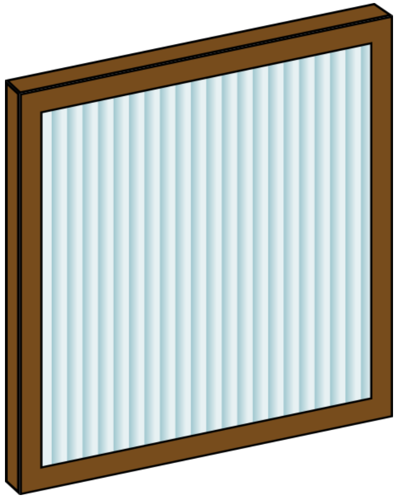
0.5" - 0.60“ ESP

Model number: AM012MNMDCH/AA
Description: DVM S, Duct S - 12000 Btu/h, 0.5" - 0.60" ESP
Option code: 010054-1E59CA-202323-331102
Static pressure (inches): 0.5" - 0.60"

Filter Selection

Know what’s in your system!

Air Flow: 1000 CFM
300 FPM



16 x 25 Air Filters @ 1,000 CFM

290	430	575	720	970*
0.09	0.15	0.22	0.31	0.46

Air Flow: 1400 CFM
500 FPM

Resistance : .2” wc

S-NET pro 2 - DVM S Water

Home | Trend Graph | Replay | Add-On | Help

8/4/2022 8:49:59 AM
Duration 00:11:57

8/4/2022 9:39:10 AM
Current Time '4/2022 9:01 AM

16x Fast Slow

Player

Indoor Unit Installation Data

Indoor Unit Installation Data

Address	Model	RMC	MCU ADDRES	MCU PORT	Location	Product Option	Installation Option	Installation Option2	Main Micom	Error History1	Error History2	Error History3	Error History4	Error History5
0	Slim 1Way	00	7	A	-	[0]17044-[1]180C8-[2]01616-[3]30010	[0]20010-[1]00000-[2]00000-[3]00000	[0]50000-[1]00000-[2]00000-[3]00000	DB91-01888C 180502	000	000	000	000	000
1	Slim 1Way	01	6	D	-	[0]17044-[1]180C8-[2]01616-[3]30010	[0]20010-[1]00000-[2]00000-[3]00000	[0]50000-[1]00000-[2]00000-[3]00000	DB91-01888C 180502	000	000	000	000	000
2	Slim 1Way	03	6	A	-	[0]17044-[1]180C8-[2]01616-[3]30010	[0]20010-[1]00000-[2]00000-[3]00000	[0]50000-[1]00000-[2]00000-[3]00000	DB91-01888C 180502	000	000	000	000	000
3	Duct	03	7	B	-	[0]10054-[1]E54BB-[2]04646-[3]31115	[0]20010-[1]20000-[2]00000-[3]00000	[0]50000-[1]00000-[2]40000-[3]00000	DB91-01889A 170309	-	-	-	-	-
4	Duct	00	6	C	-	[0]10054-[1]E54BB-[2]04646-[3]31115	[0]20010-[1]20000-[2]00000-[3]00000	[0]50000-[1]00000-[2]40000-[3]00000	DB91-01889A 170309	-	-	-	-	-
5	Slim 1Way	05	7	E	-	[0]17044-[1]180C8-[2]01616-[3]30010	[0]20010-[1]00000-[2]00000-[3]00000	[0]50000-[1]00000-[2]00000-[3]00000	DB91-01888C 180502	000	000	000	000	000
6	Duct	06	7	F	-	[0]10054-[1]E5438-[2]03535-[3]31104	[0]20010-[1]20000-[2]00000-[3]00000	[0]50000-[1]00000-[2]40000-[3]00000	DB91-01889A 170309	-	-	-	-	-
7	Slim 1Way	07	6	B	-	[0]17044-[1]180C8-[2]01616-[3]30010	[0]20010-[1]00000-[2]00000-[3]00000	[0]50000-[1]00000-[2]00000-[3]00000	DB91-01888C 180502	000	000	000	000	000
8	Slim 1Way	02	7	C	-	[0]17044-[1]180C8-[2]01616-[3]30010	[0]20010-[1]00000-[2]00000-[3]00000	[0]50000-[1]00000-[2]00000-[3]00000	DB91-01888C 180502	000	000	000	000	000

AM024MNHDCH/AA

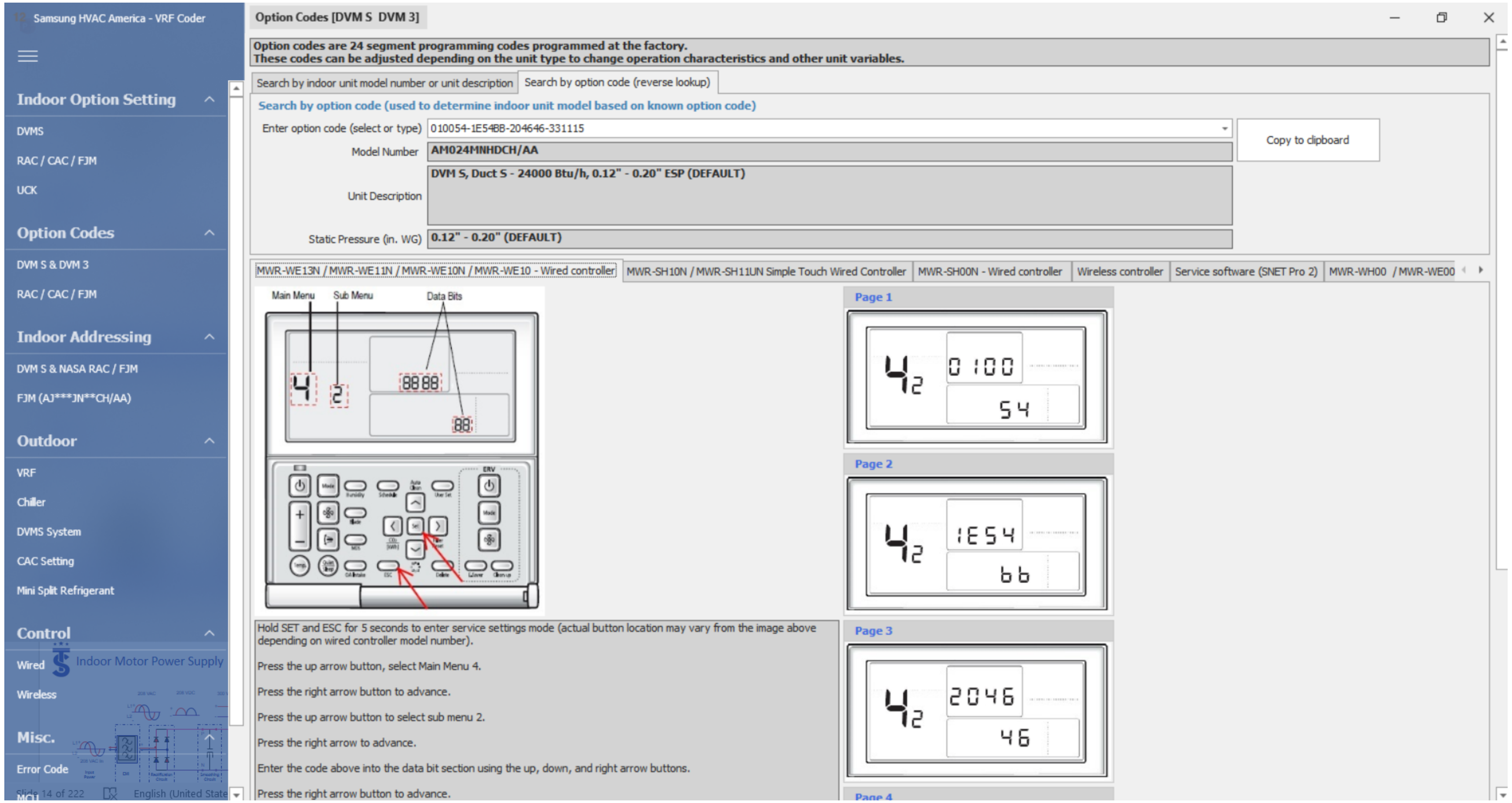
Fan options		ESP	Sound Pressure (dBA)		
		inAq	H	M	L
Default	010054-1E54BB-204646-331115	0.2	39	36	30
Option	010054-1E581F-204646-331115	0.3	41	36	31
	010054-1E5973-204646-331115	0.4	43	38	33
	010054-1E59C6-204646-331115	0.5	45	40	34
	010054-1E5D1A-204646-331115	0.6	47	42	36
	010054-1E5D6D-204646-331115	0.7	48	43	37
	010054-1E5EA0-204646-331115	0.8	48	44	39

This customer is complaining about the temperatures in these 2 zones.
Units are AM024MNHDCH
System had been tested and balanced
This is a commercial system with long duct runs
What does the product and installations codes tell us.

Outdoor Unit Data | Outdoor Unit Installation Data | Outdoor Unit Cycle Diagram | Indoor Unit Data | Indoor Unit Installation Data | Control for Unoccupied and Entering Room | MCU Unit Data

Version 1.13.0 Unit - Temp.: ℉ Power : Btu Pressure : psi 9/15/2022 4:59 PM COM -1

VRF Coder Reverse Look Up



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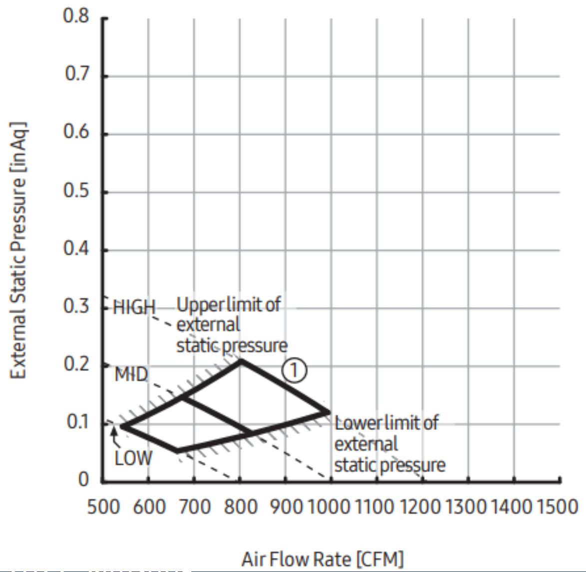
Product Option Code

Example

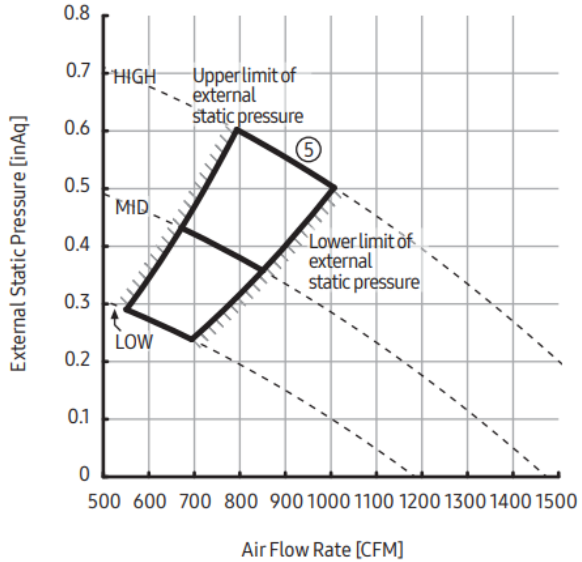
Model number: AM024MNHDCH
Description: DVM S, Duct S - 24,000 Btu/h,
Option code: 010054-1E54BB-204646-331115
Static pressure (inches): 0.12" - 0.20"

7) AM024MNHDCH

①	External Static Pressure(inAq)	Option Code
	0.12≤ SP ≤0.20	010054-1E54BB-204646-331115

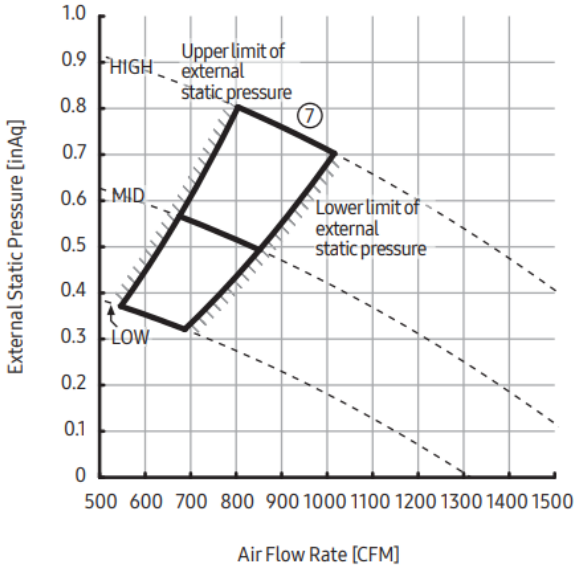


⑤	External Static Pressure(inAq)	Option Code
	0.50< SP ≤0.60	010054-1E5D1A-204646-331115



7) AM024MNHDCH

⑦	External Static Pressure(inAq)	Option Code
	0.69< SP ≤0.79	010054-1E5EA0-204646-331115



Air Flow Review

- Air flow is the most common overlooked issue when it comes to ducted units.
- Having the correct air flow is crucial for system performance.
- Each unit has it's design CFM
- Ducts must be designed to deliver the proper CFM
- If the measured ESP exceeds the nameplate the delivered CFM will be reduced.
- If you know the CFM you can determine the performance of any unit.

Verify the operation of this system

AM036TNZDCH/AA

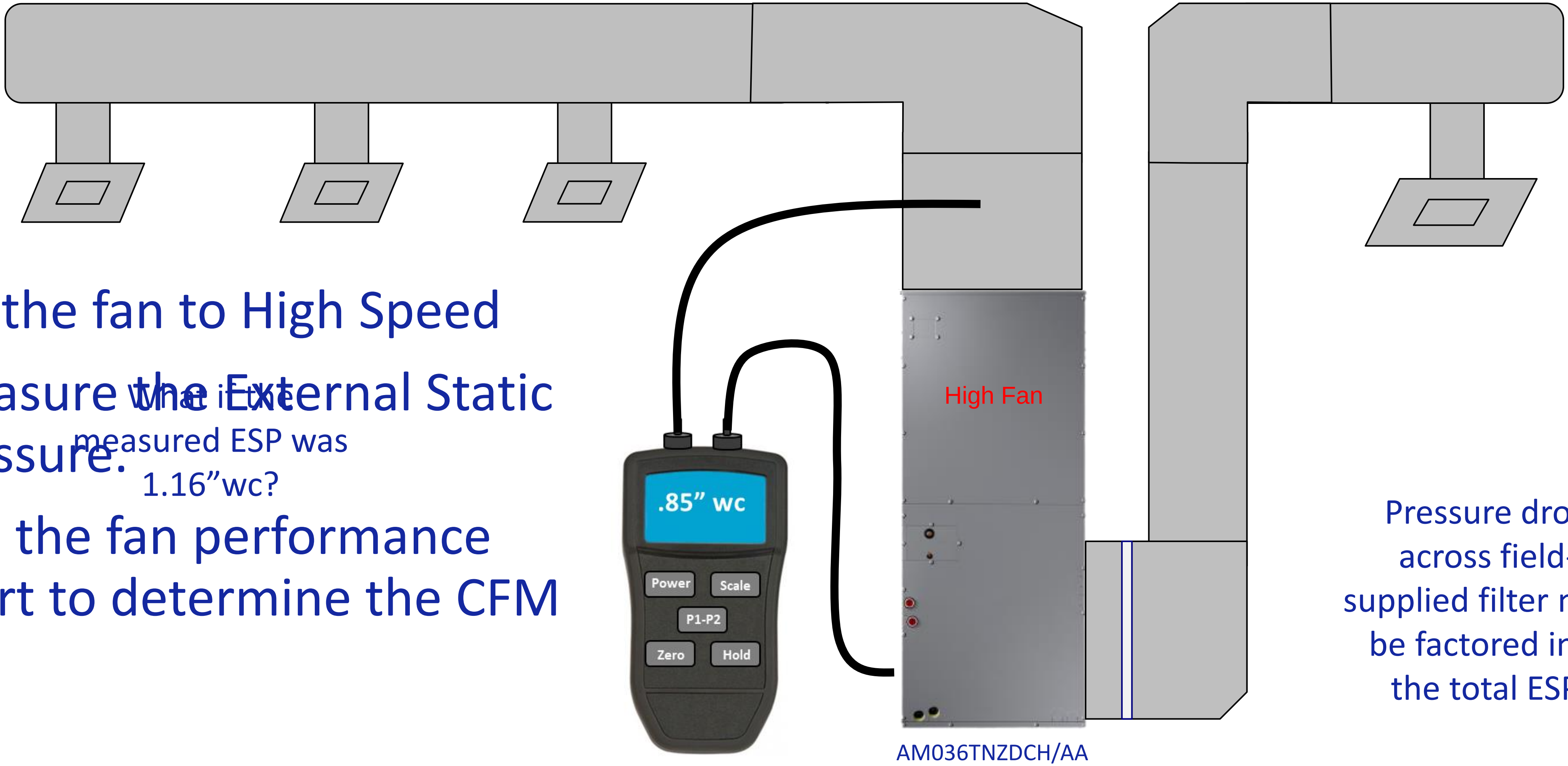
- On start up the system was put on high fan in the heating mode.
- The measured temperature rise in heat was 30 degrees.
- The measured TD in cooling mode was 20 degrees
- System passed the air balance test.
- The end user complains about lack of heating from time to time.
- What are some thing to look at

AM036TNZDCH
HP: 1/2
Default motor taps:
high=4, mid=3, low=1

Specifications

Performance	Nominal Capacity (Btu/h)	Cooling	36,000
		Heating	40,000
Airflow	CFM @ 0.4" ESP (UL)	L / M / H	758 / 924 / 1,053
External Static Pressure	Standard	"WC	0.4
	Min. / Max.	"WC	0.1 / 1.0

Does the system provide the correct airflow?



- Set the fan to High Speed
- Measure the External Static pressure.
- Use the fan performance chart to determine the CFM

What if the measured ESP was 1.16"wc?

Pressure drop across field-supplied filter must be factored into the total ESP.

Calculating CFM Example

- Set the fan to High Speed
- Fan tap in high is #4 (default)
- Measured External Static pressure is .85” w.c.
- The delivered CFM is approximately 889CFM
- Is this enough air flow (rated airflow is 1,053CFM)
- What can you do now that you know the CFM?
- You can determine the delivered capacity of the indoor unit in heating and cooling modes.

Motor Tap	ESP (inch)	CFM	RPM
2	0.1	989	670
	0.2	943	711
	0.25	940	737
	0.3	911	762
	0.4	878	810
	0.5	829	851
	0.6	791	881
	0.7	752	933
	0.8	682	1,010
	0.9	635	1,074
	1	568	1,122
1	0.1	888	615
	0.2	850	671
	0.25	826	691
	0.3	800	718
	0.4	762	771
	0.5	714	808
	0.6	663	879
	0.7	605	947
	0.8	524	1,016
	0.9	487	1,059
	1	431	1,105

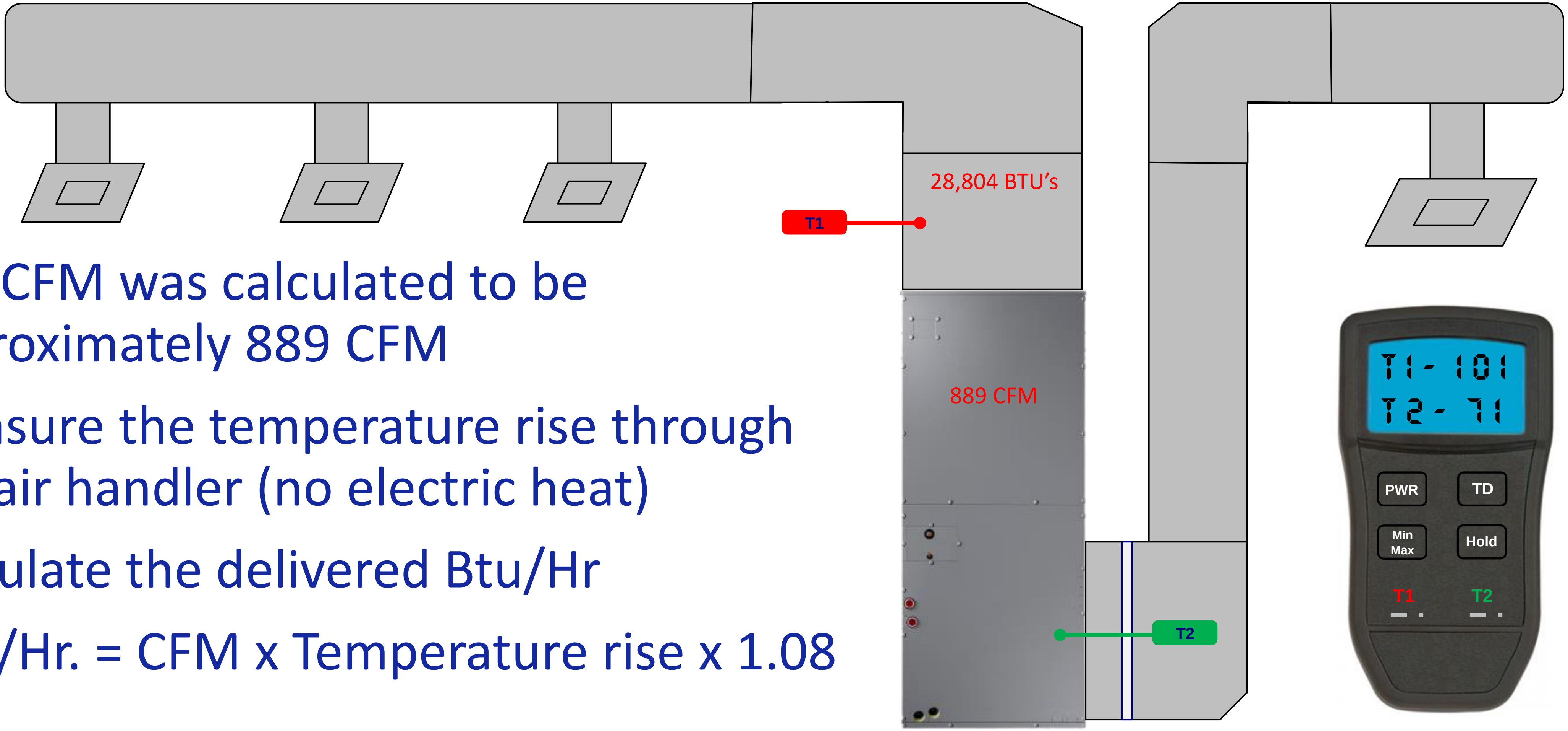
Motor Tap	ESP (inch)	CFM	RPM
4	0.1	1,165	748
	0.2	1,121	786
	0.25	1,110	809
	0.3	1,089	832
	0.4	1,053	877
	0.5	1,021	903
	0.6	982	946
	0.7	941	976
	0.8	906	1,016
	0.9	872	1,042
	1	816	1,108
3	0.1	1,039	684
	0.2	992	743
	0.25	980	752
	0.3	952	780
	0.4	921	828
	0.5	884	860
	0.6	841	903
	0.7	809	929
	0.8	767	984
	0.9	715	1,049
	1	643	1,132

Motor Tap	ESP (inch)	CFM	RPM
5	0.1	1,318	826
	0.2	1,281	864
	0.25	1,270	876
	0.3	1,246	896
	0.4	1,207	938
	0.5	1,183	978
	0.6	1,152	997
	0.7	1,120	1,032
	0.8	1,077	1,061
	0.9	1,043	1,097
	1	994	1,126

AM036TNZDCH
HP: 1/2
Default motor taps:
high=4, mid=3, low=1

Calculate delivered BTUH

- The CFM was calculated to be approximately 889 CFM
- Measure the temperature rise through the air handler (no electric heat)
- Calculate the delivered Btu/Hr
- BTU/Hr. = CFM x Temperature rise x 1.08



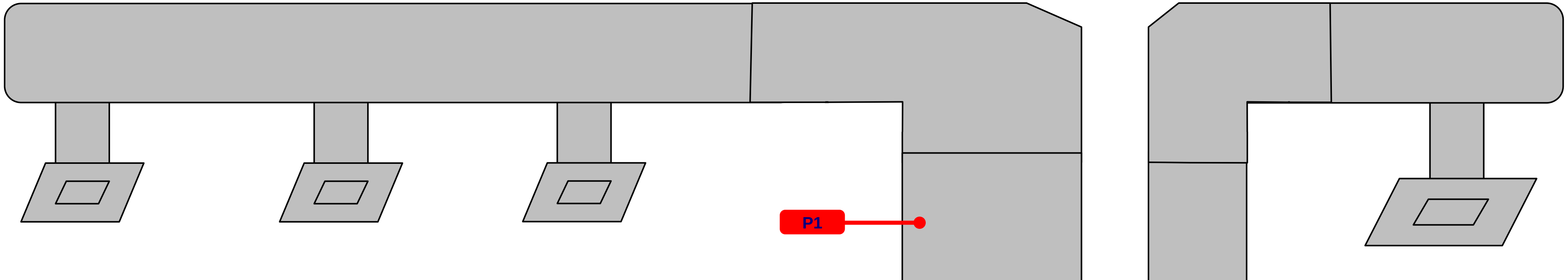
Conclusion

- When the system was tested the temperature rise was 30 degrees.
- It was stated that this is a “normal” temperature rise for a heat pump.
- If the temperature rise is correct, it must be moving the correct air flow.
- The balance report stated the CFM delivery was 1,053 CFM
- Should temperature rise be used as the only means of checking system performance???
- What is the correct temperature rise for this unit to deliver 40,000 Btu’s of heat at the rated air flow?

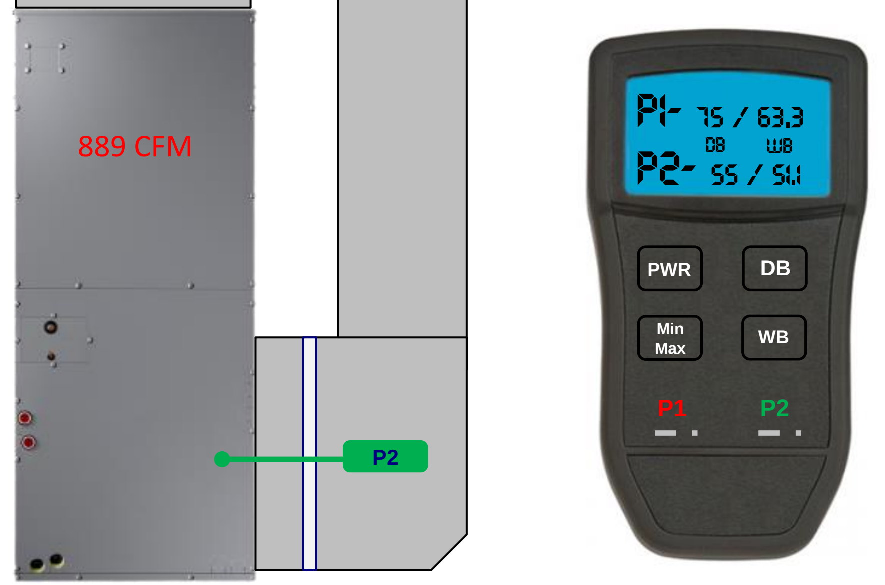
Cooling capacity in the cooling mode

- Cooling BTUs = CFM x Δh x 4.5
  - CFM = The fan CFM plotted on the fan tables, or measured at several points in the system
  - Δh = the enthalpy change through the system (we get this number by converting wet bulb readings.)
  - 4.5 = the cooling BTU multiplier at sea level, or .075-lbs. of air per CFM x 60 (minutes in an hour.) This factor will also change at higher altitudes. BTU multipliers will also change due to extreme air temperatures.

Calculate Cooling Capacity



- 889 CFM was previously calculated
- Using a reliable psychrometer measure Supply and Return dry bulb and wet bulb temperatures:
  - Return conditions are: 75 BD – 63.3 WB
  - Supply conditions are: 55 BD – 51.1 WB



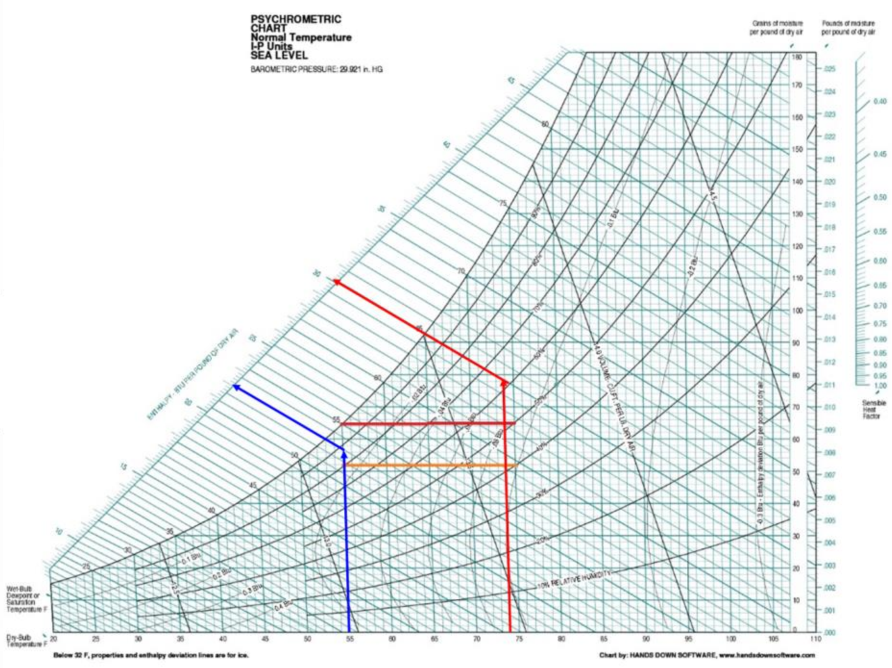
Calculate Enthalpy Differential

- Using a psychrometric calculator determine the enthalpy of the return and supply air then calculate the difference
  - Return Air: 75 BD – 63.3 WB – Enthalpy 29.2
  - Supply Air: 55 BD – 51.1 WB – Enthalpy 21.1
  - Enthalpy 8.1

Free online Psychrometric Calculator Print

IP SI altitude 0 ft b press 30.01 in.Hg

T (°F) dry bulb	rH (%) rel.humidity	TDP (°F) dew point	W (GPP) abs. humidity	h (btu/lb) enthalpy	Tw (°F) wet bulb	sp. vol (cft/lb)		
75	55	57.7	71	29.2	63.3	13.78	Calculate	Remove
53.1	90	50.4	54	21.1	51.1	13.14	Calculate	Remove



Calculating Cooling Capacity

- Cooling BTUs = CFM x Δh x 4.5
  - CFM = 889
  - Δh = 8.1
- Cooling BTUs = 889 x 8.1x 4.5
  - Cooling Capacity = 32,404 BTUH

AM036TNZDCH
HP: 1/2
Default motor taps:
high=4, mid=3, low=1

Specifications

Performance	Nominal Capacity (Btu/h)	Cooling	36,000
		Heating	40,000
Airflow	CFM @ 0.4" ESP (UL)	L / M / H	758 / 924 / 1,053
External Static Pressure	Standard	"WC	0.4
	Min. / Max.	"WC	0.1 / 1.0

Thermodynamics Contact Time

- The amount of time that the air is contact with the refrigerant is regulated by:
  - The size of the coil.
  - The fan
    - CFM of the Evaporator and Condenser
  - The compressor speed
    - Frequency of the compressor

Variable Speed Compressor

(Contact Time Refrigerant)

- The speed of the refrigerant is directly related to the frequency the compressor is running.
- The lower the Hz the slower the refrigerant moves through the system. (more Contact Time with the coil)

Maximum Cooling

Minimum

14 DVMS

10 DVMS 2

Maximum Heat

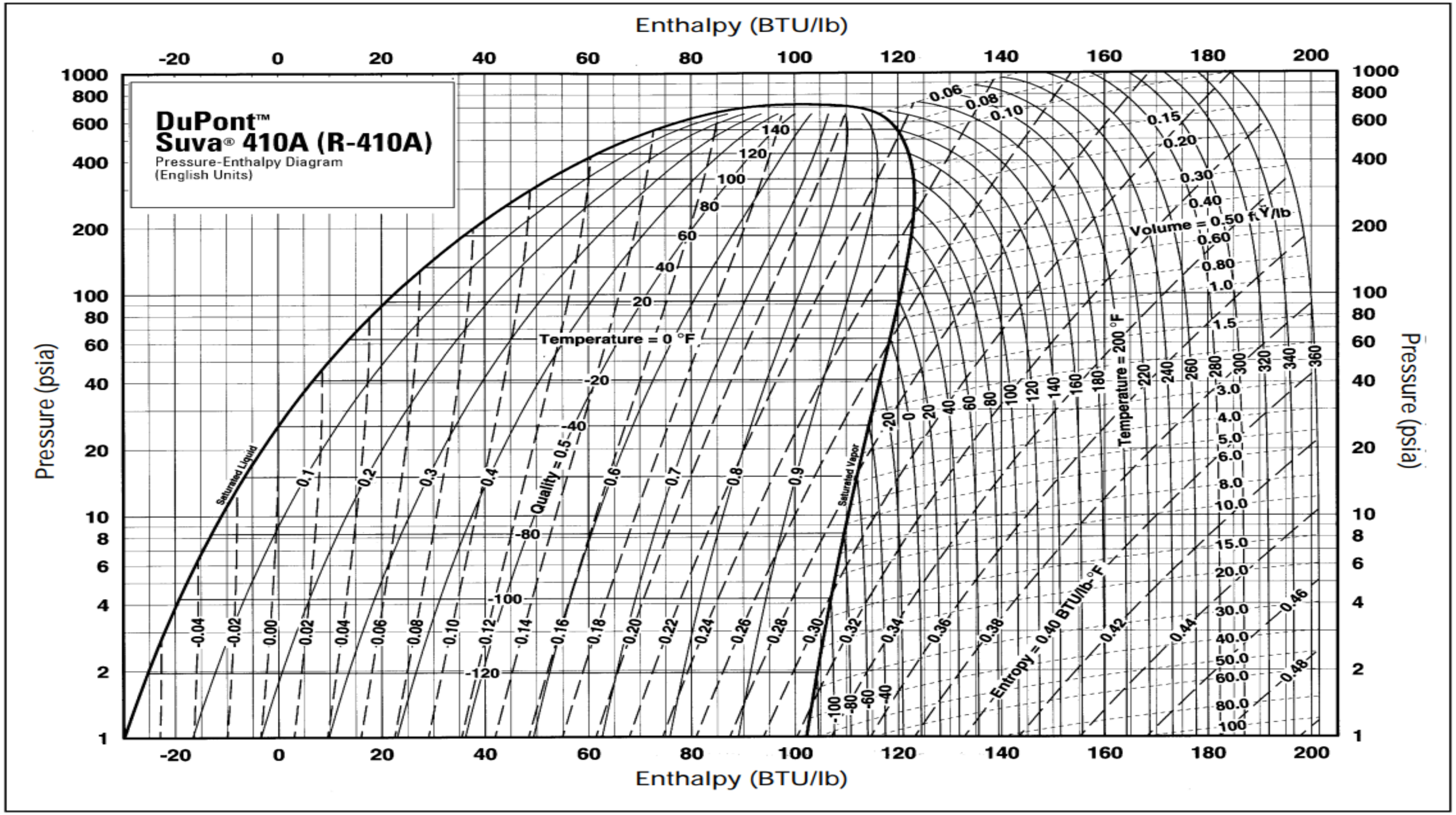
160 DVMS

180 DVMS 2

Beyond Basics:
Detailed System Operation

- The operation of a refrigeration system is predictable.
- P-H diagram give a graphical representation of the refrigerant as it travels through the refrigeration system.
- Engineers use these charts to map out and calculate system operating characteristics of a given system
- From these chart they can calculate how the system responds to changing conditions such as low pressure, low superheat and low sub-cooling.
- Information from these charts are used in VRF systems to control it's operation.
- Having a basic understanding of the PE diagram will aide in understating the operation of a VRF system as a whole.

PE Diagrams
Show The Why



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Ever Wonder?

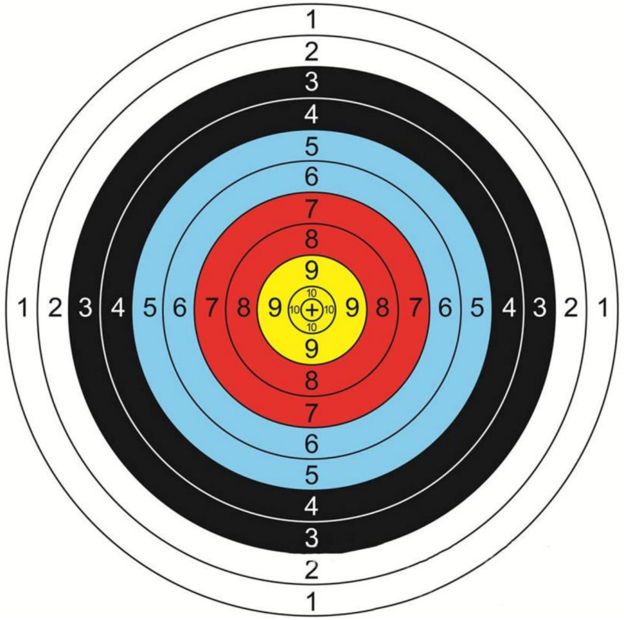
- Why the heating capacities are greater than the cooling capacities of a system?
- Where does the extra heat come from?
- Why system discharge temperature are lower in the cooling mode than the heating mode?
- How manufacturers produce target operational data for a system?

AM120BXVGJH/AA

Performance	US Ton (nominal)		10
	Capacity (Btu/h) [1]	Nominal / Rated Cooling	120,000 / 114,000
		Nominal / Rated Heating	135,000 / 129,000
	EER	Ducted / Mixed / Non-Ducted	12.00 / 12.45 / 12.90
	IEER	Ducted / Mixed / Non-Ducted	24.60 / 28.22 / 31.85
	High Heat COP	Ducted / Mixed / Non-Ducted	3.80 / 3.96 / 4.13

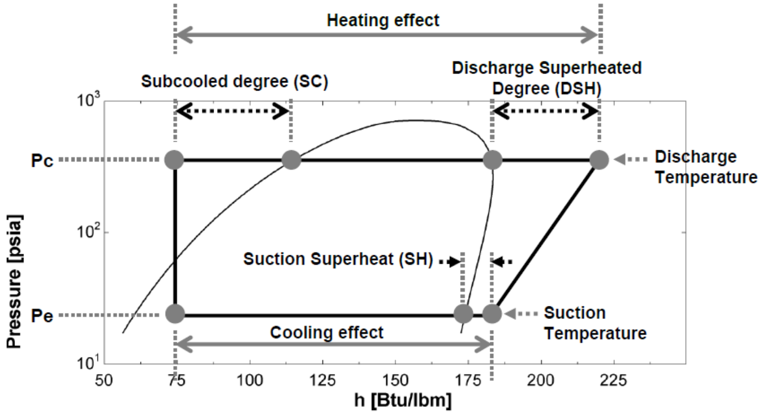
Putting It All Together

- The operating characteristics are determined during the testing process using PE diagrams.
- From these tables engineers can calculate the operating characteristics at any given condition for each mode.
- They use this information to establish targeted operating values of the system for the conditions and mode it is running at.
- The system components will adjust to varying conditions to maintain the "Targeted" operation.

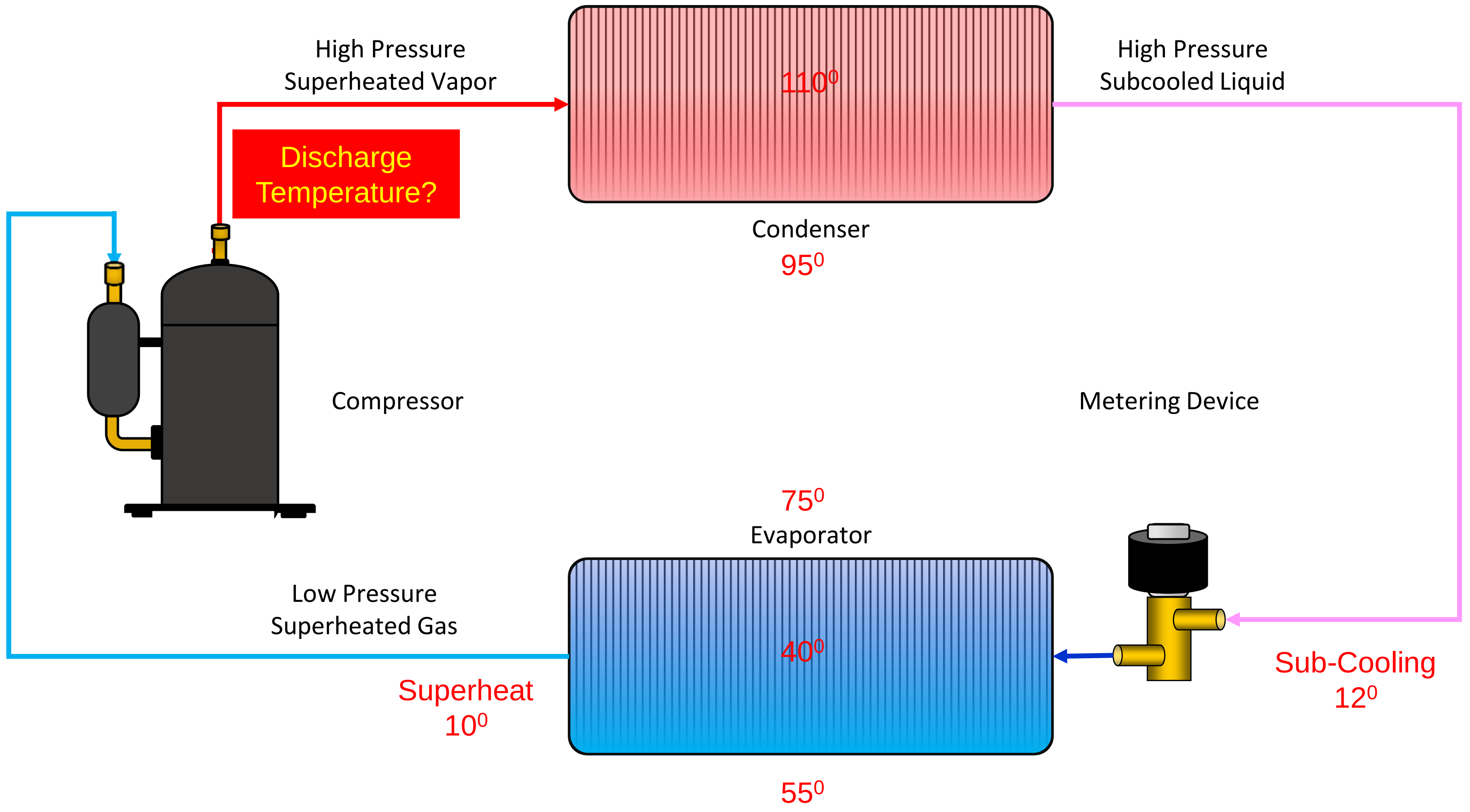


PE Diagrams

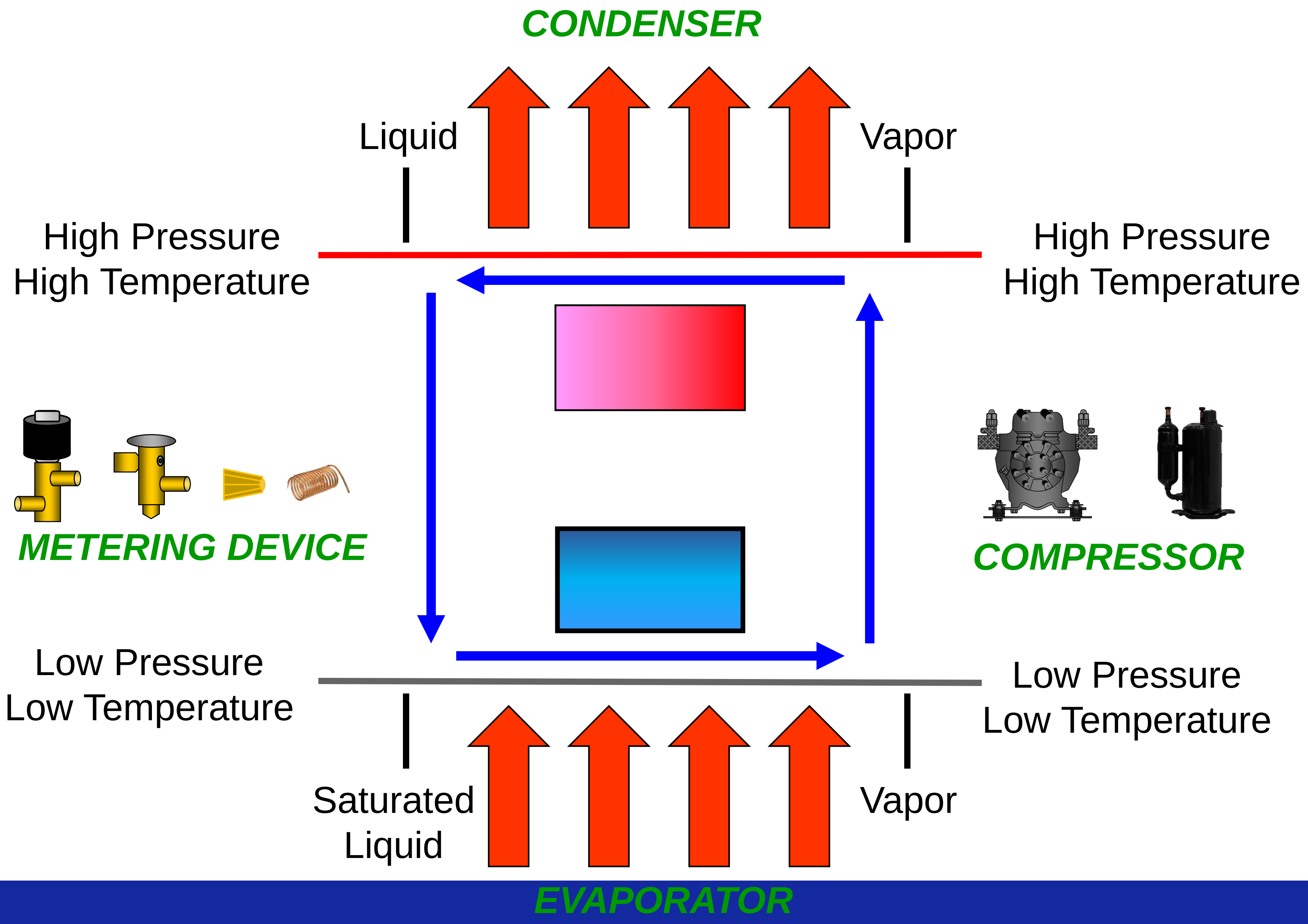
- Are charts that graphically describe the relationship between pressure and enthalpy (total heat content)of a select refrigerant.
- They are the engineering charts for a refrigeration system.
- They show the total heat transfer within a system
- PE diagrams show the operating characteristics of a system at various operating conditions.
- They are used to calculate:
  - Net Refrigeration Effect of the evaporator
  - Heat of Compression of the compressor
  - Total Heat of Rejection from the condenser
  - Discharge Temperature
  - The amount of refrigerant required for system operation



Unitary Air Conditioning Design



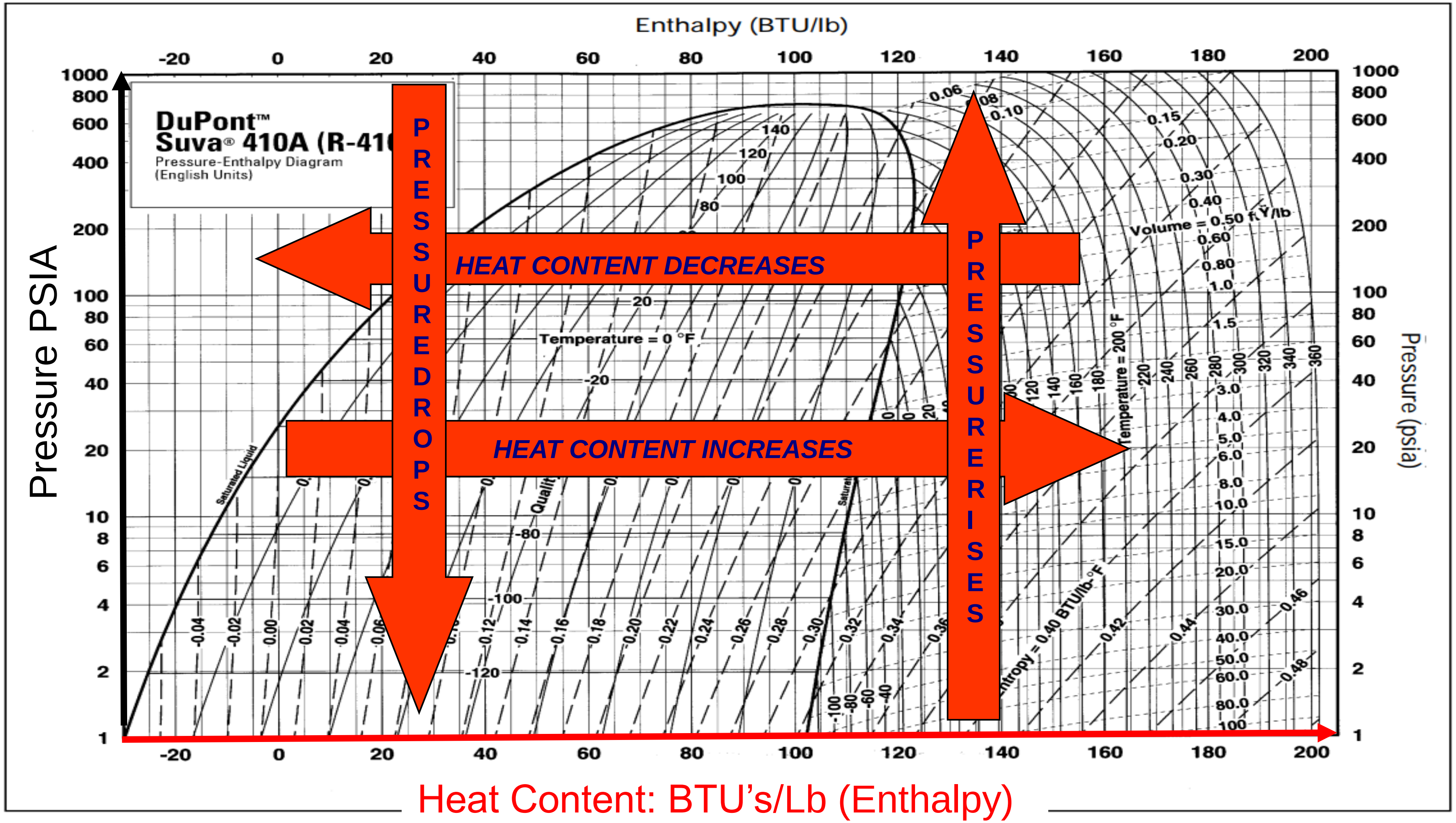
Refrigeration Cycle Review



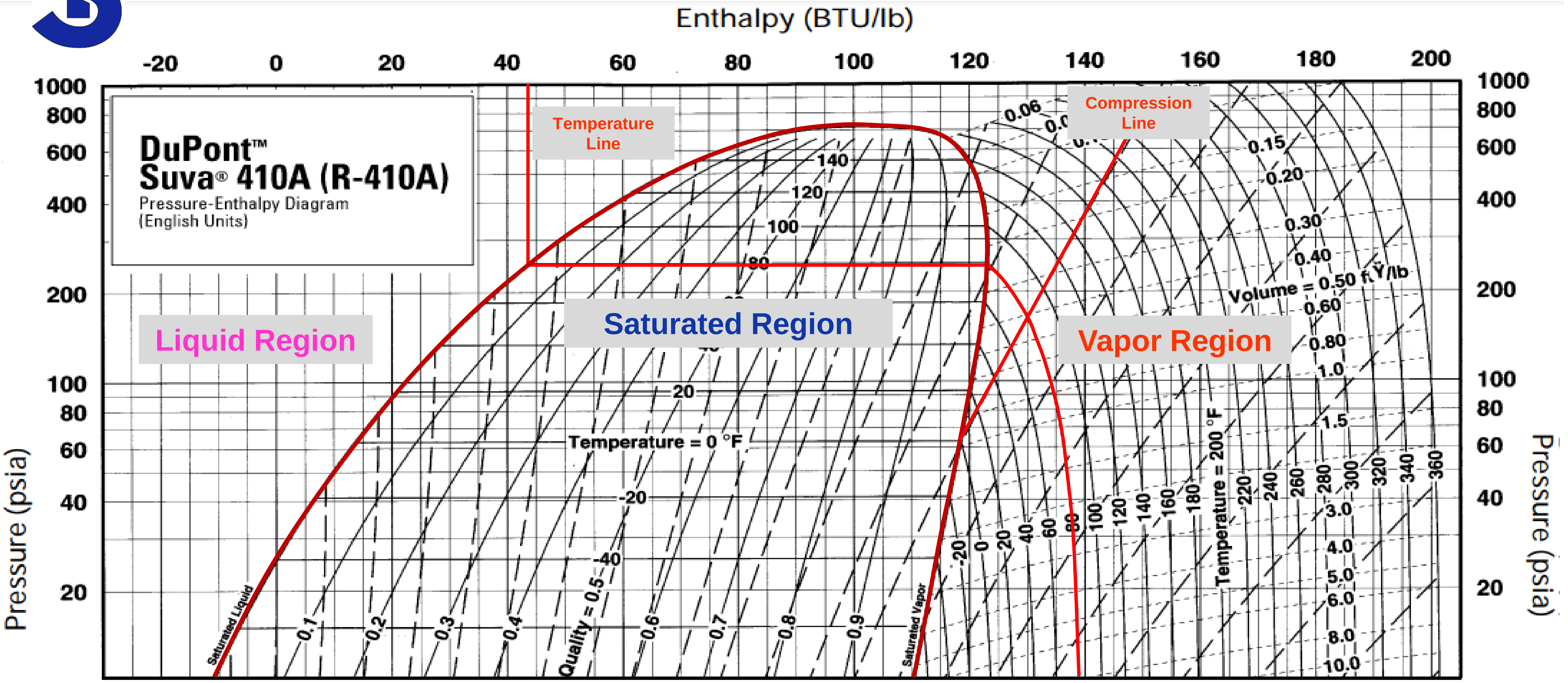
PE Basics

PSIA =

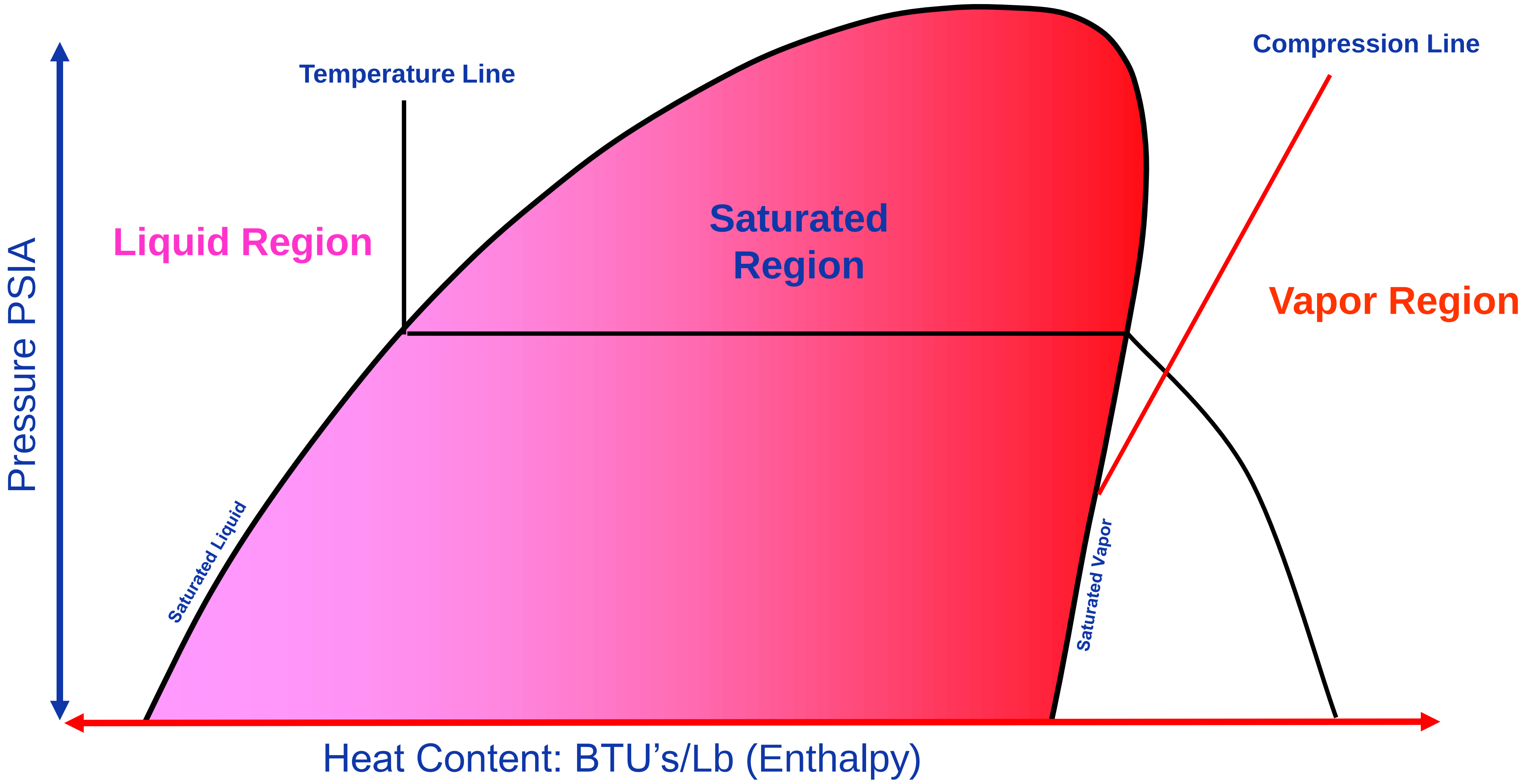
Gauge Pressure + 14.7



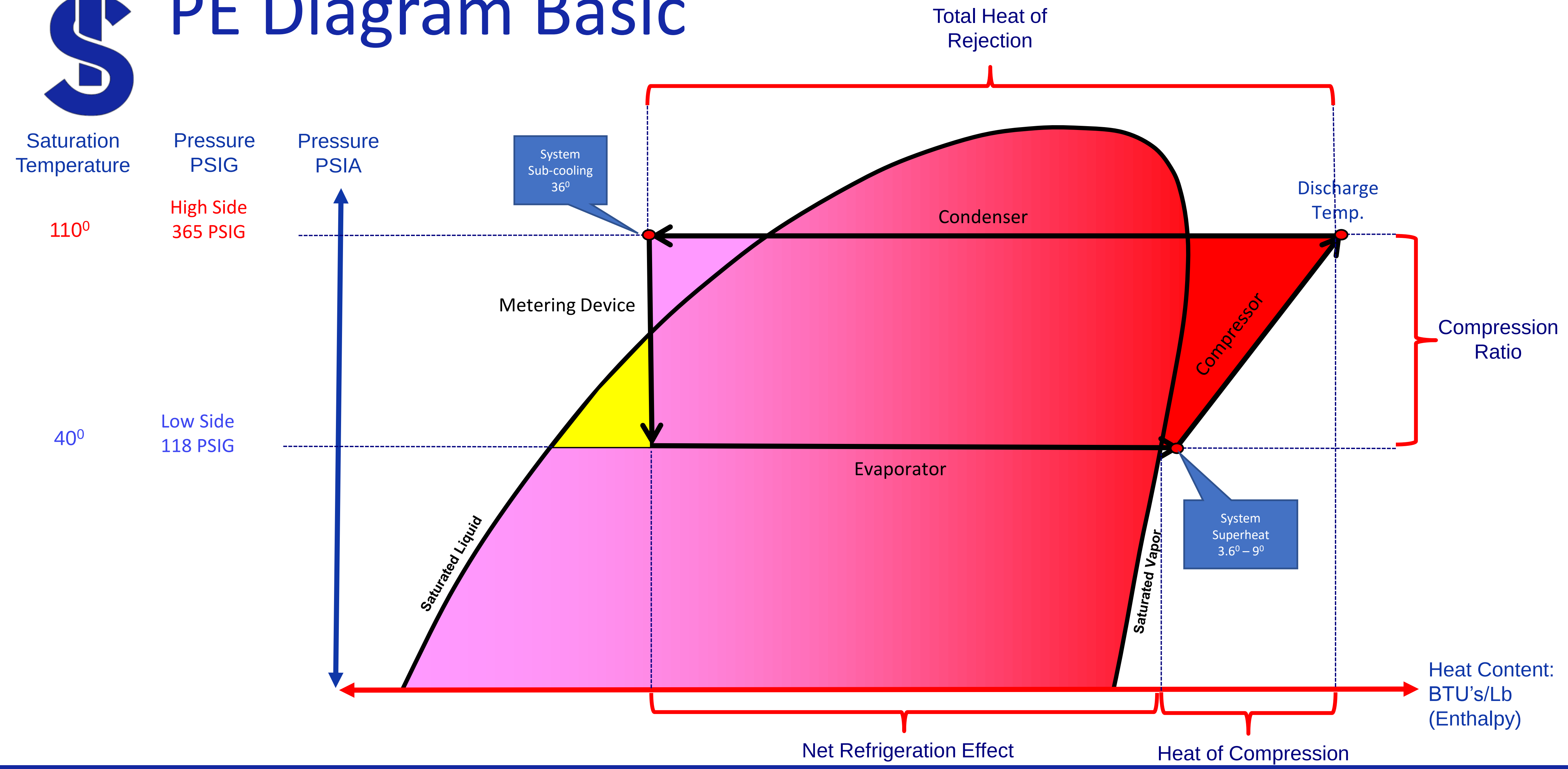
PE Diagram Basics



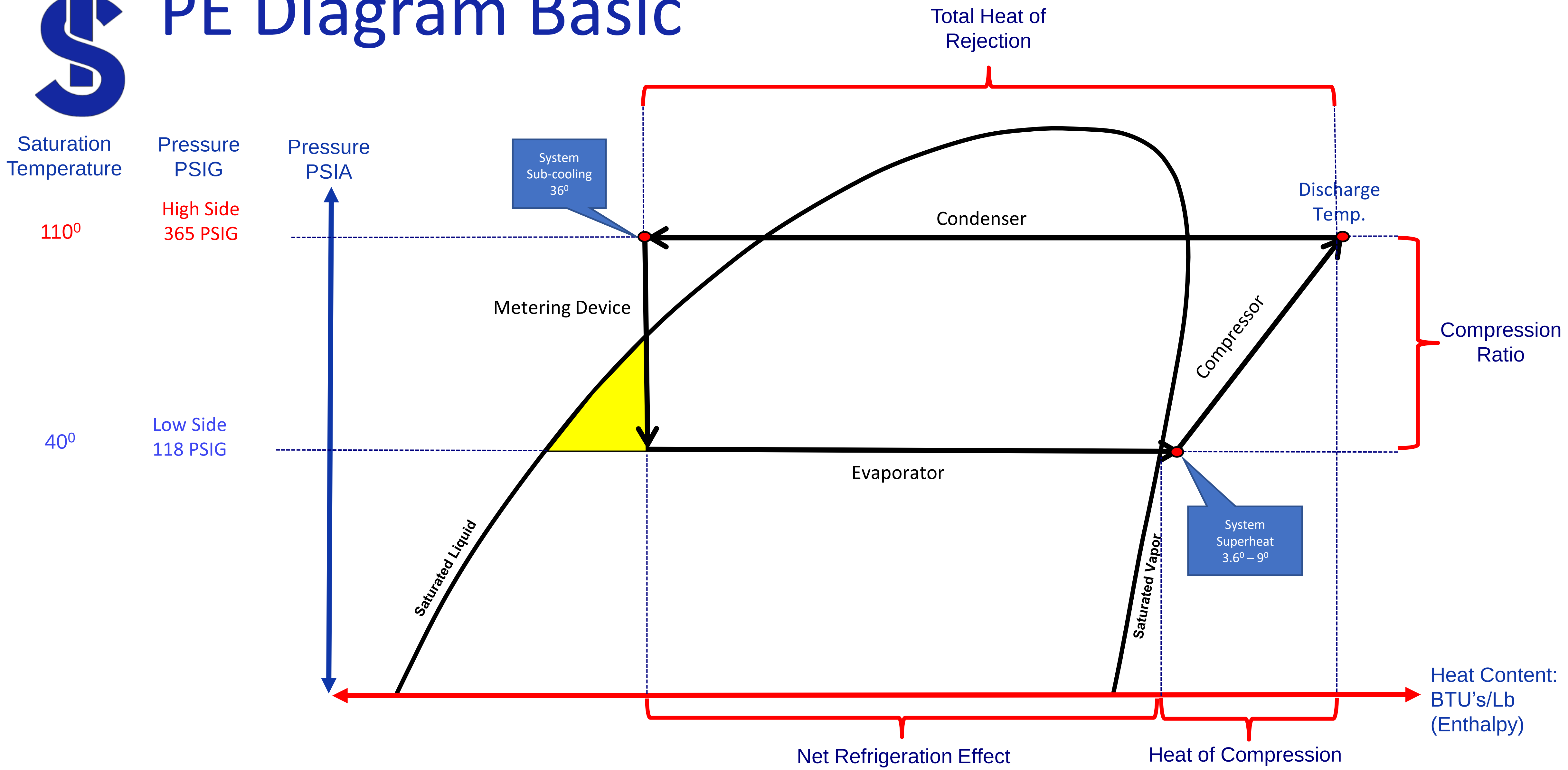
PE Diagrams


Compression Line
Temperature Line
Saturated Region
Liquid Region
Vapor Region
Pressure PSIA
Saturated Liquid
Saturated Vapor
Heat Content: BTU's/Lb (Enthalpy)

PE Diagram Basic



PE Diagram Basic



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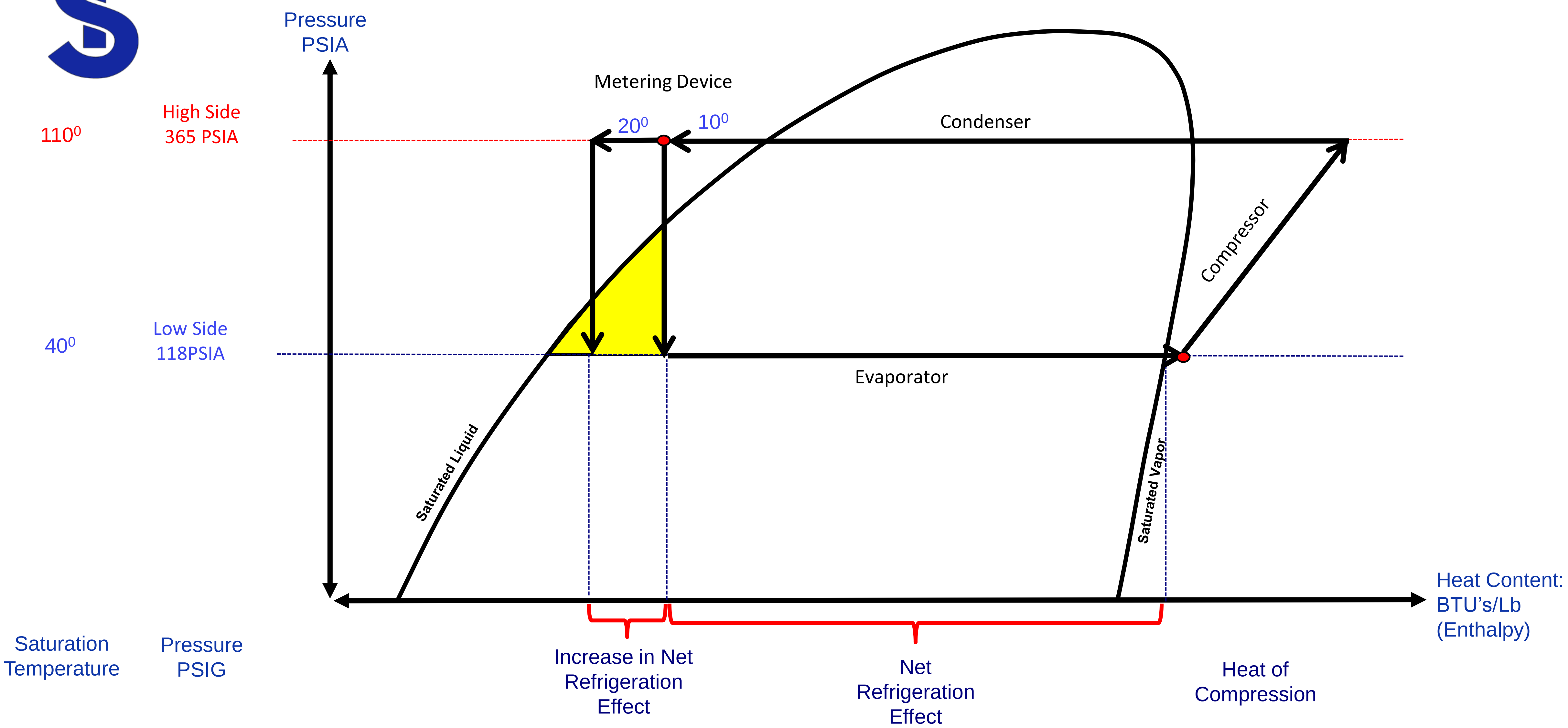
Remember this?

Which pot requires more heat to boil all the water?

Why?

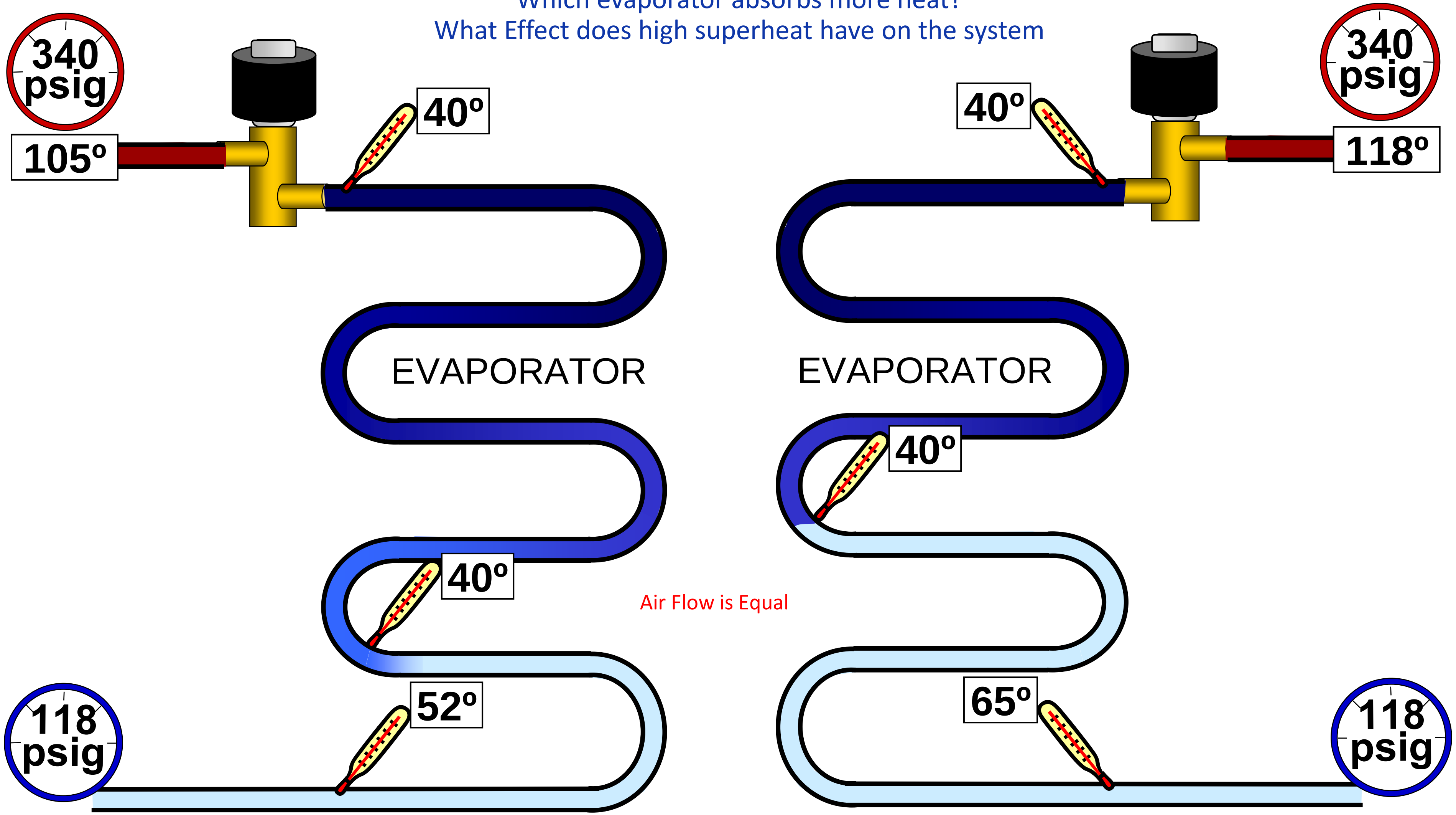
The more liquid refrigerant in the evaporator the more heat that can be removed from the space
This is why high efficiency units have lower evaporator superheats and high subcooling?

Example


Pressure
PSIA
Metering Device
High Side
365 PSIA
110⁰
20⁰
10⁰
Condenser
Compressor
Low Side
118PSIA
40⁰
Evaporator
Saturated Liquid
Saturated Vapor
Heat Content:
BTU's/Lb
(Enthalpy)
Saturation
Temperature
Pressure
PSIG
Increase in Net
Refrigeration
Effect
Net
Refrigeration
Effect
Heat of
Compression

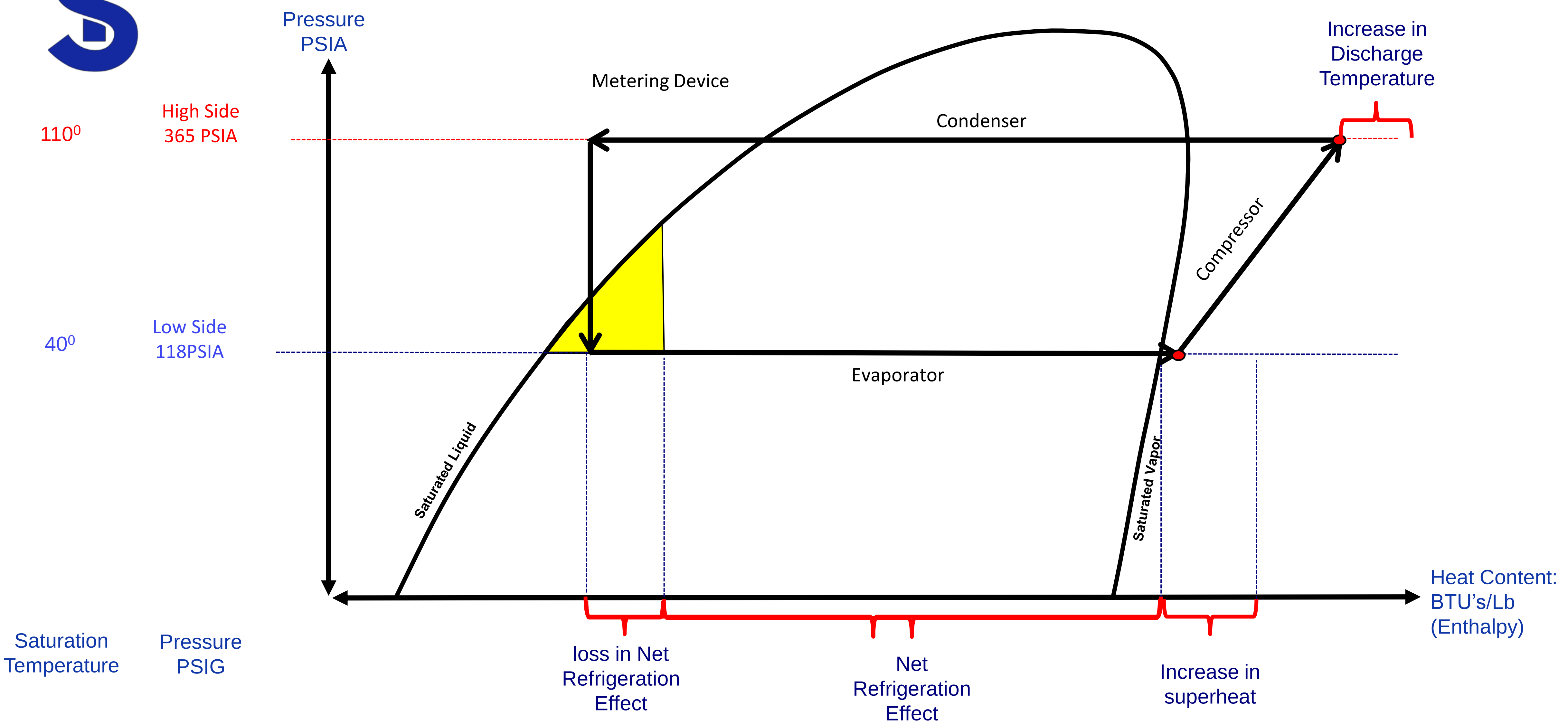
Remember this?

Which evaporator absorbs more heat?
What Effect does high superheat have on the system



Air Flow is Equal

Example: Less liquid in the Evaporator



Heat Pump Operation

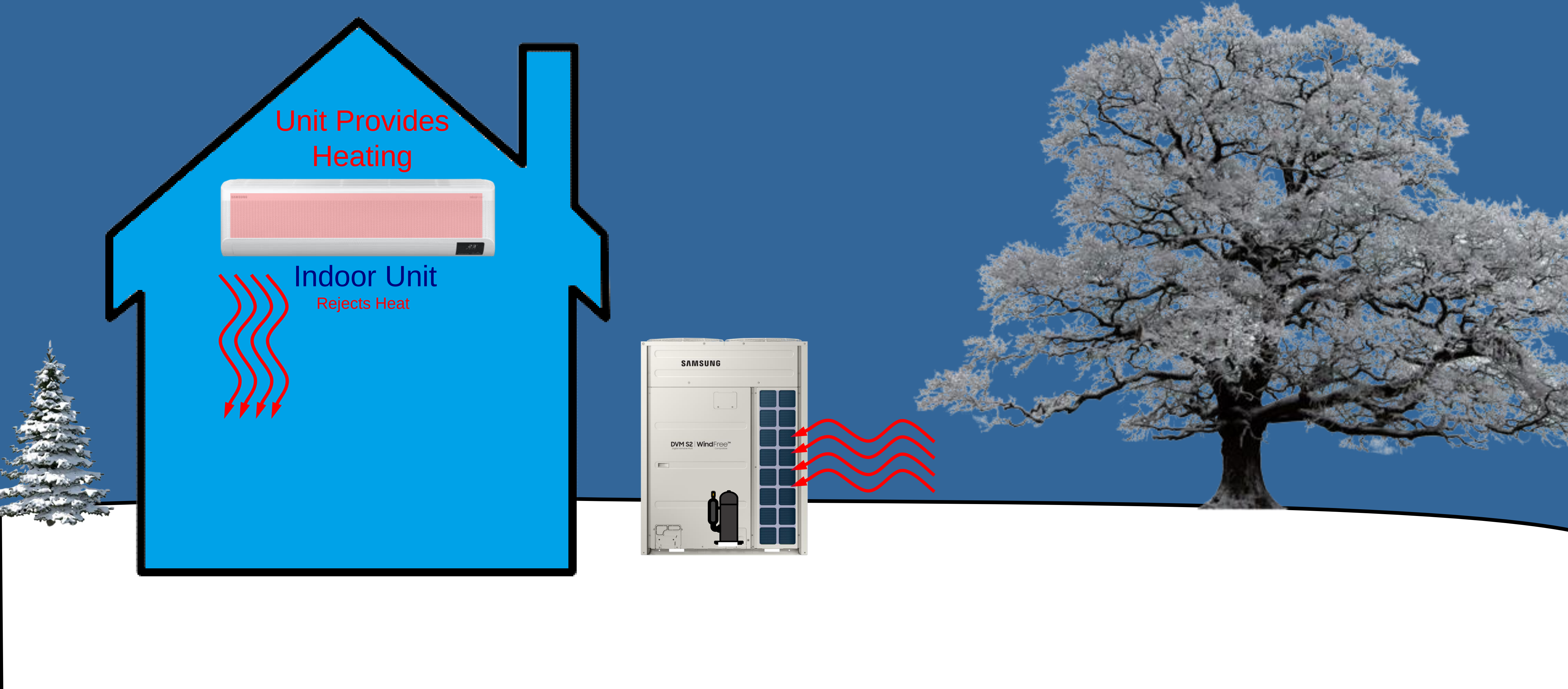
- What's the function of a heat pump?

Heating Mode

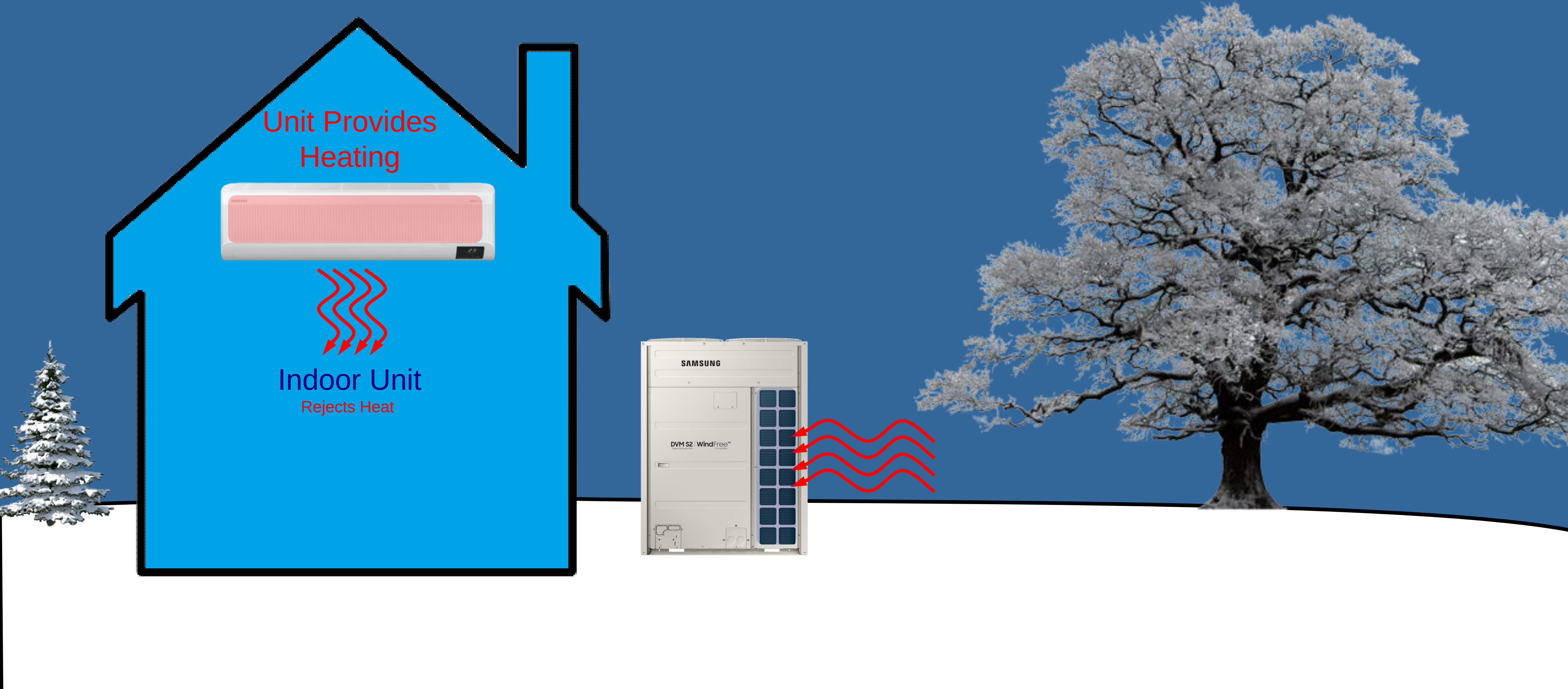
Cooling Mode

Winter Operation

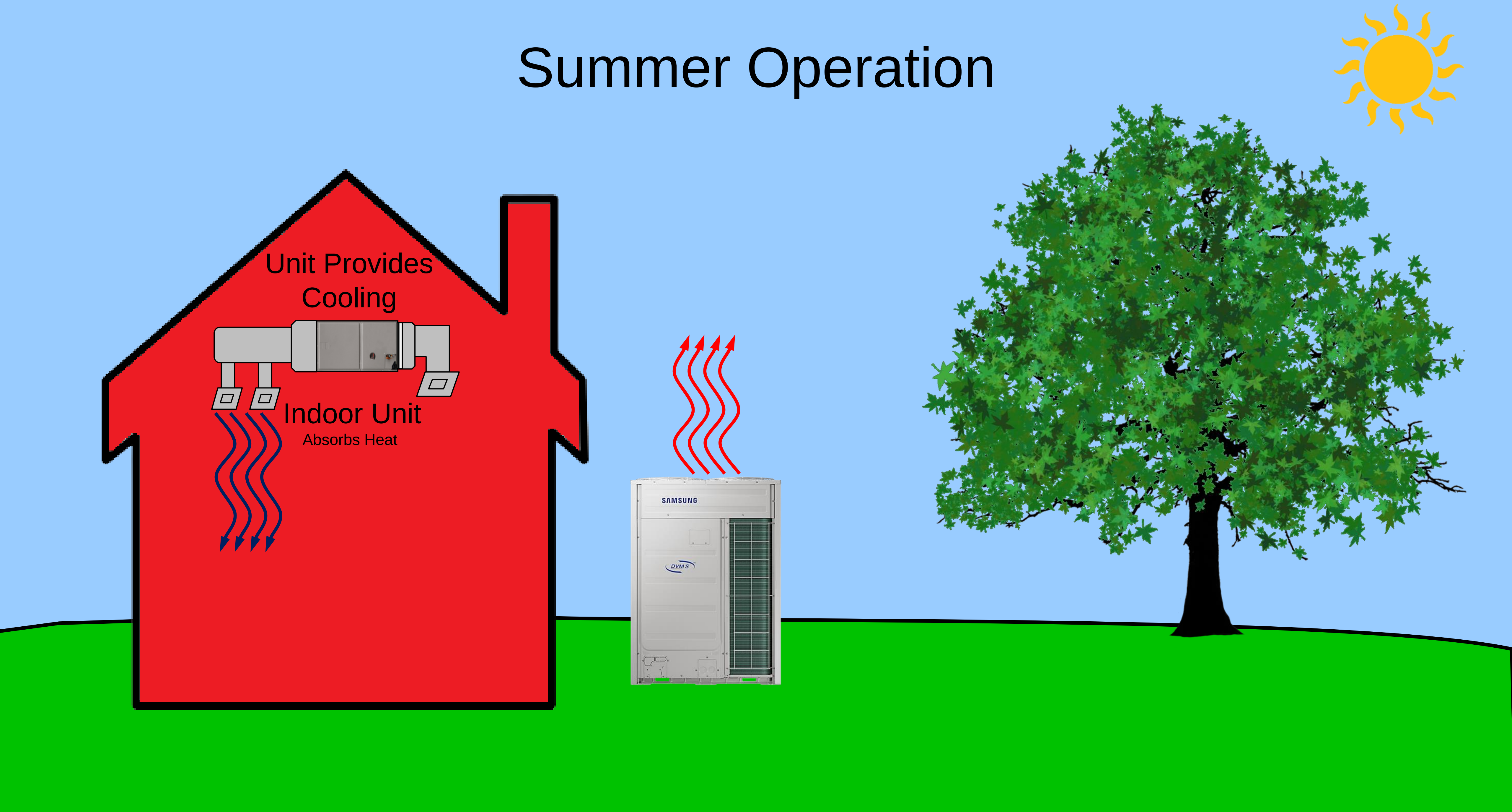
Winter Operation

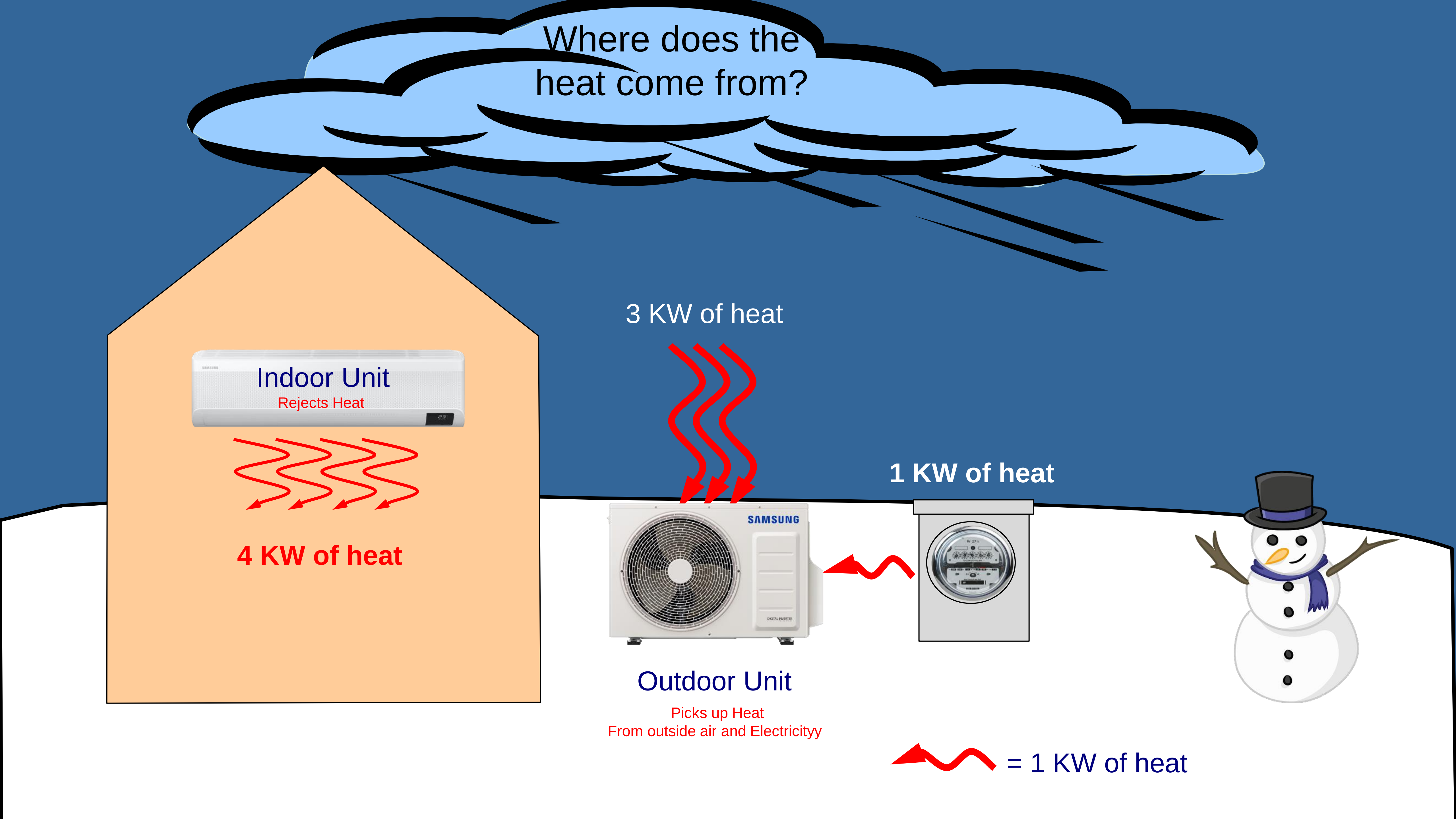


Winter Operation

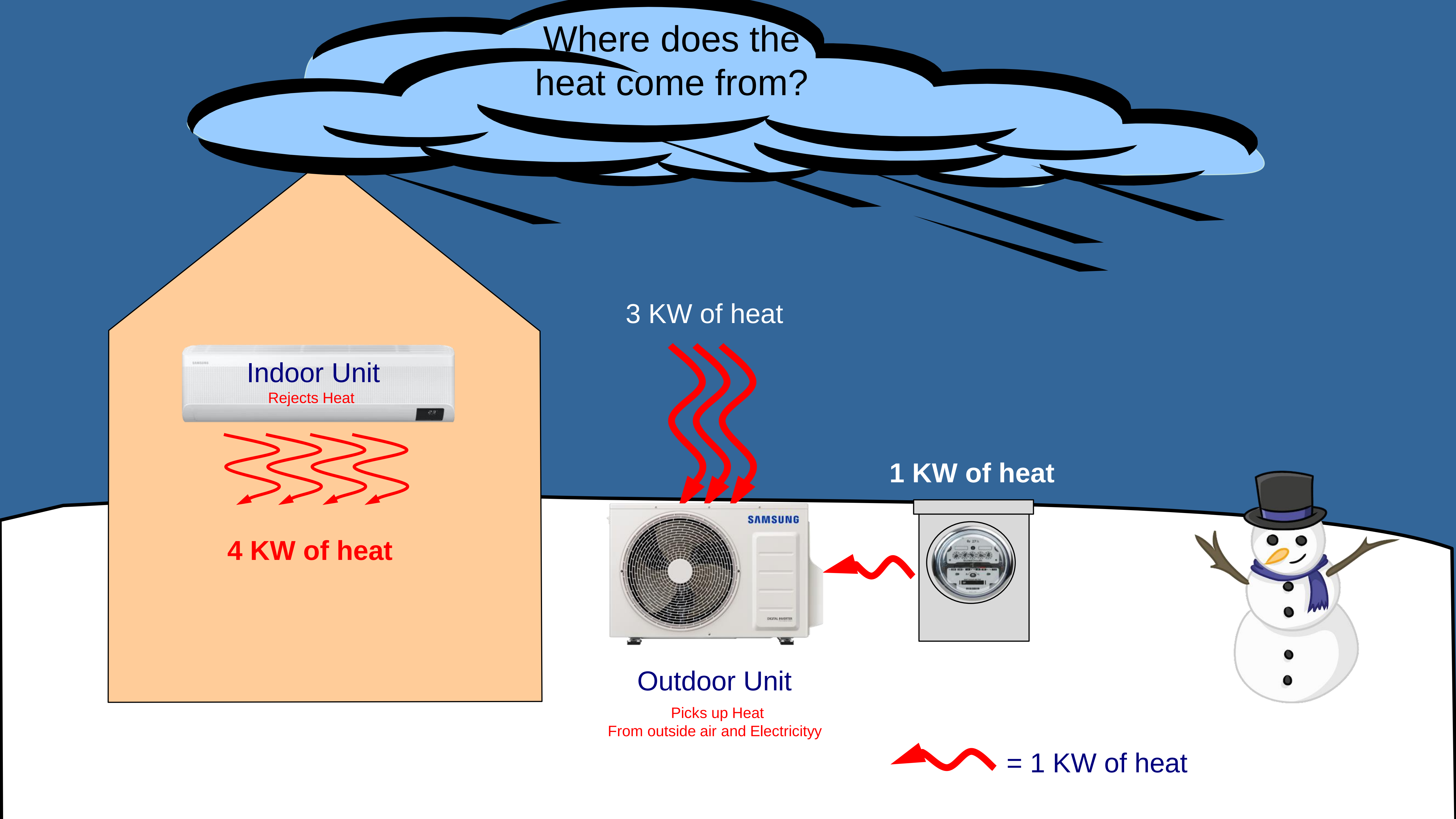


Summer Operation



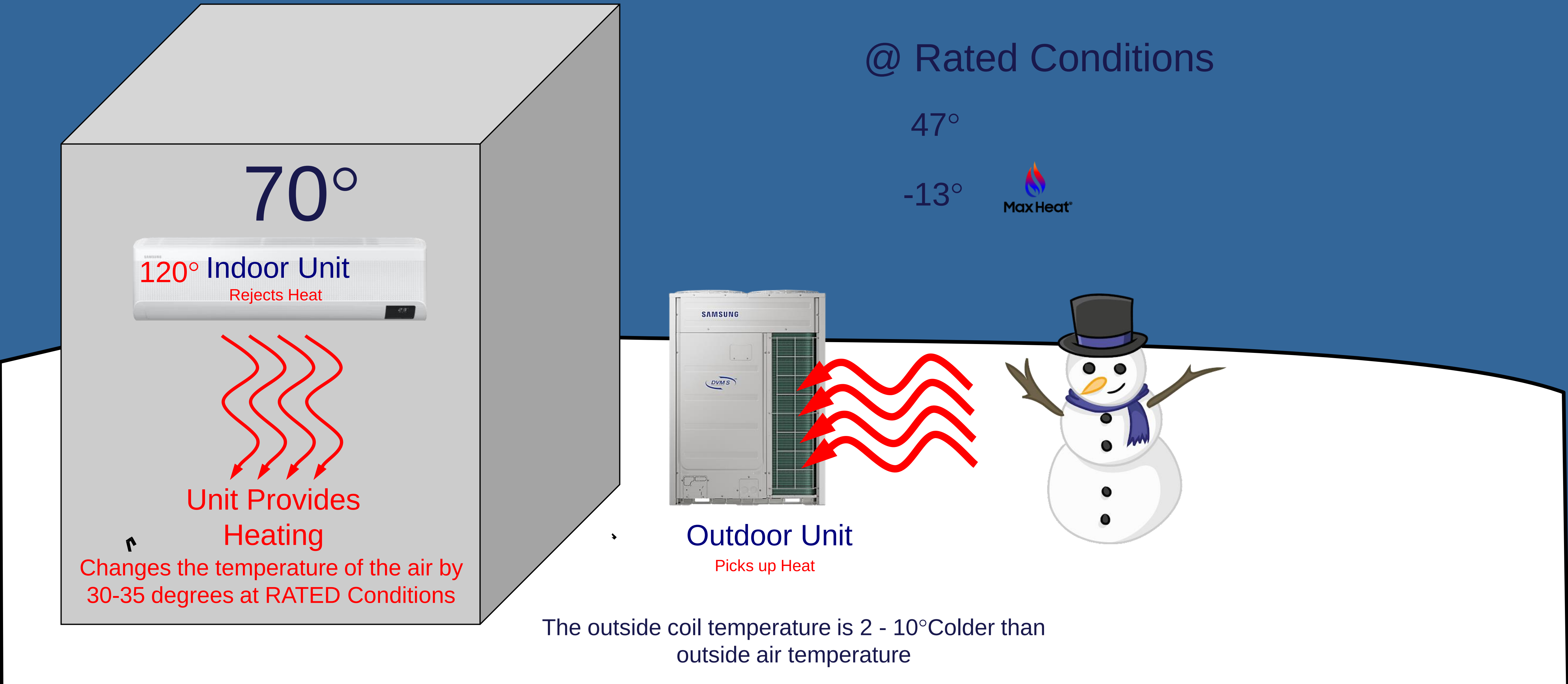


Where does the heat come from?
Indoor Unit
Rejects Heat
4 KW of heat
3 KW of heat
SAMSUNG
1 KW of heat
Outdoor Unit
Picks up Heat
From outside air and Electricityy
= 1 KW of heat



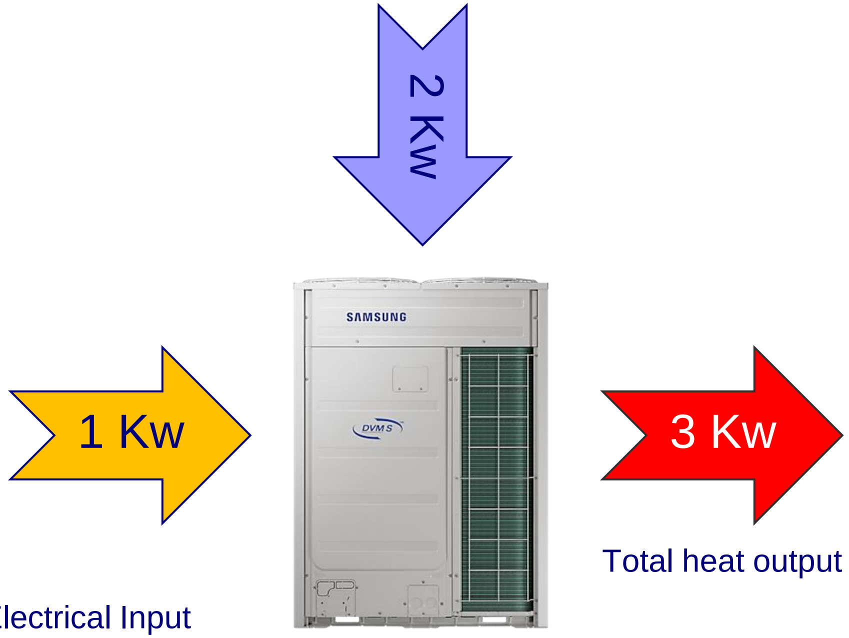
Where does the heat come from?
3 KW of heat
Indoor Unit
Rejects Heat
4 KW of heat
1 KW of heat
SAMSUNG
Outdoor Unit
Picks up Heat
From outside air and Electricityy
= 1 KW of heat

Winter Operation



Heat Pump Operation

Heat from outside air



Total heat output

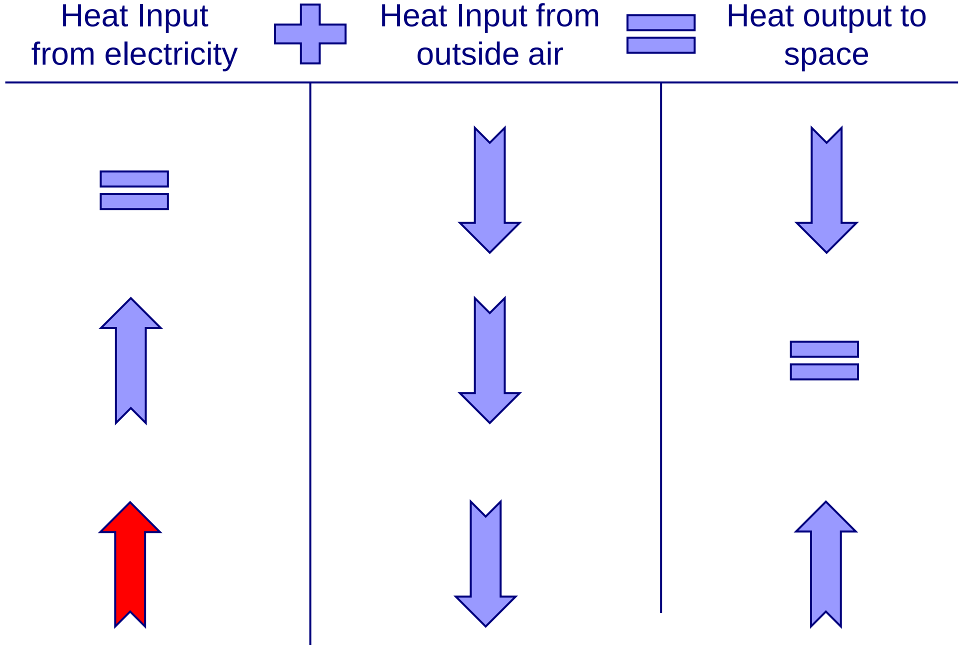
Electrical Input

Electrical BTU Input + Outdoor air BTU input = System 0utput

COP = 3:1

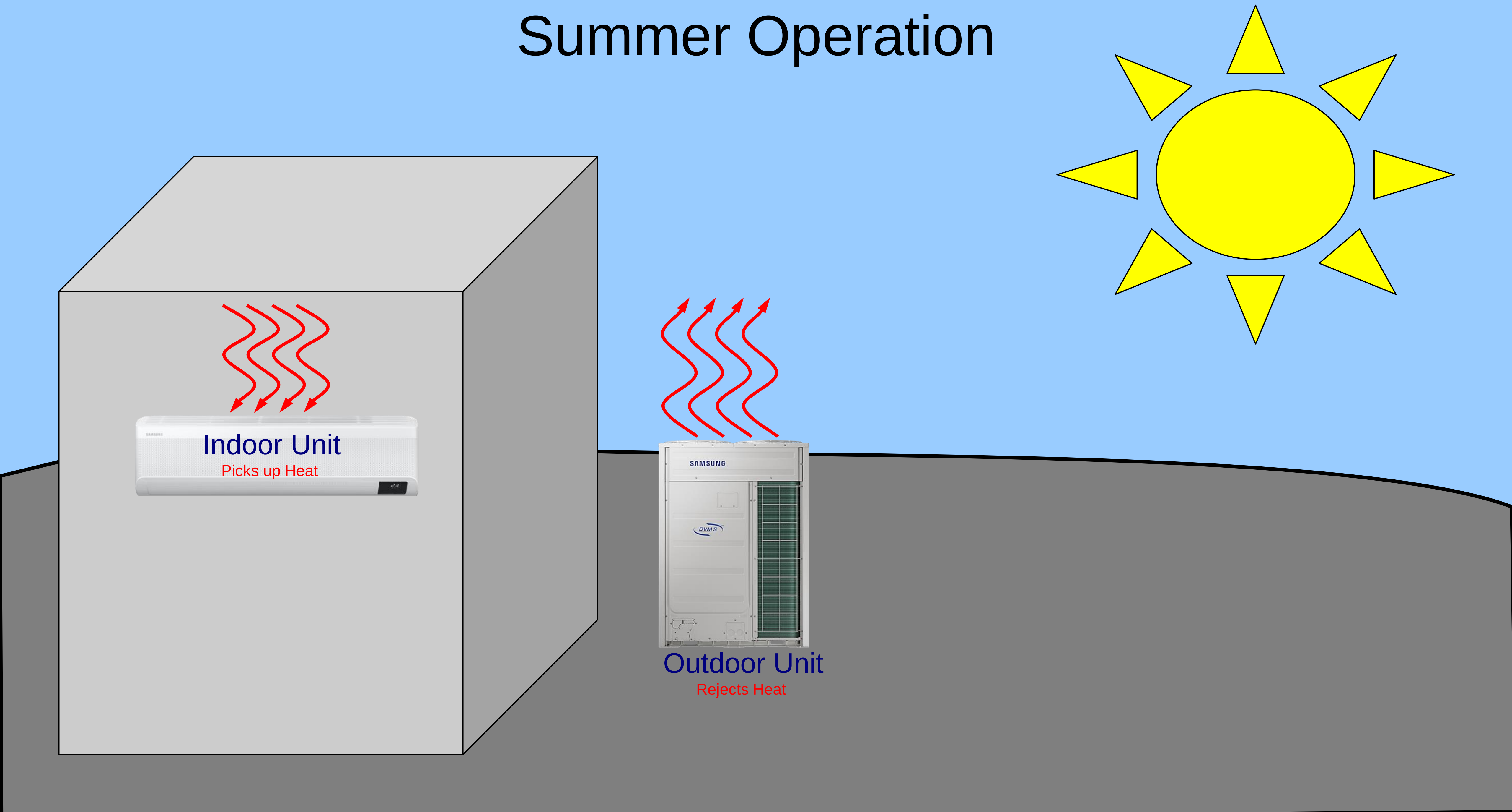
COP is the ratio of how much useful heat a heat pump will produce at a given energy input.

A + B = C



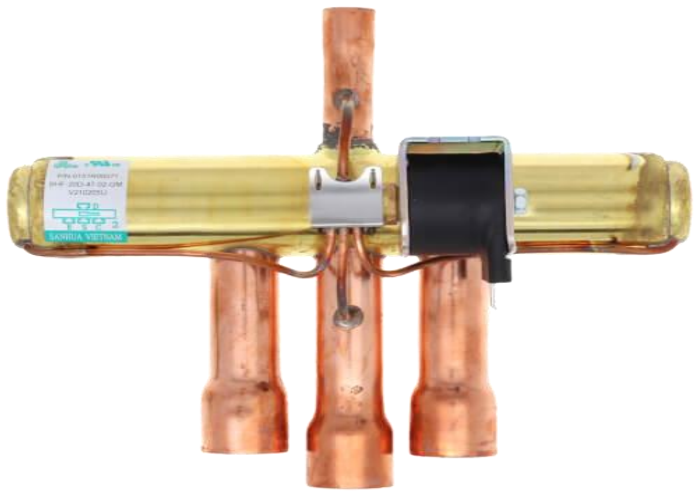
Where does the heat come from?

Summer Operation

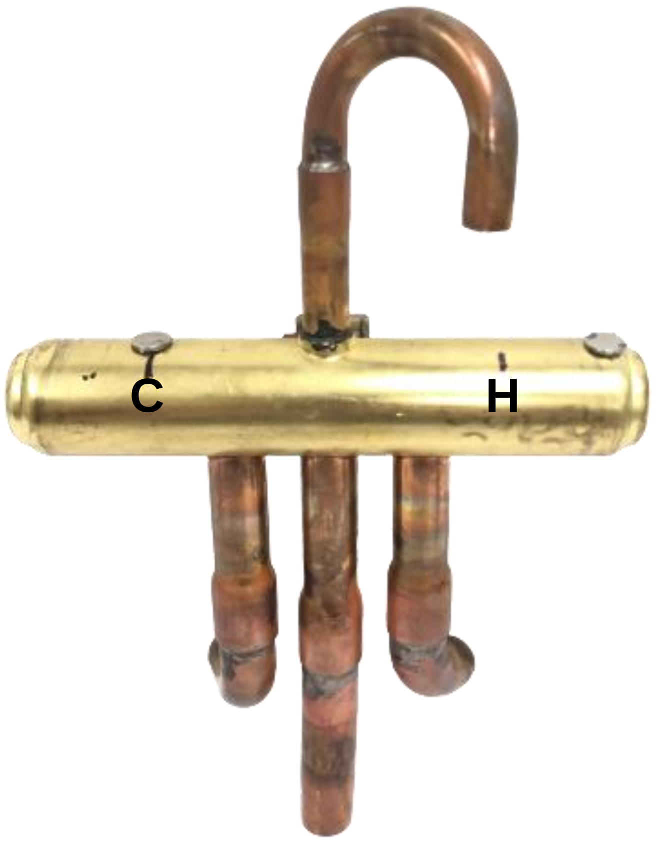


The Reversing Valve

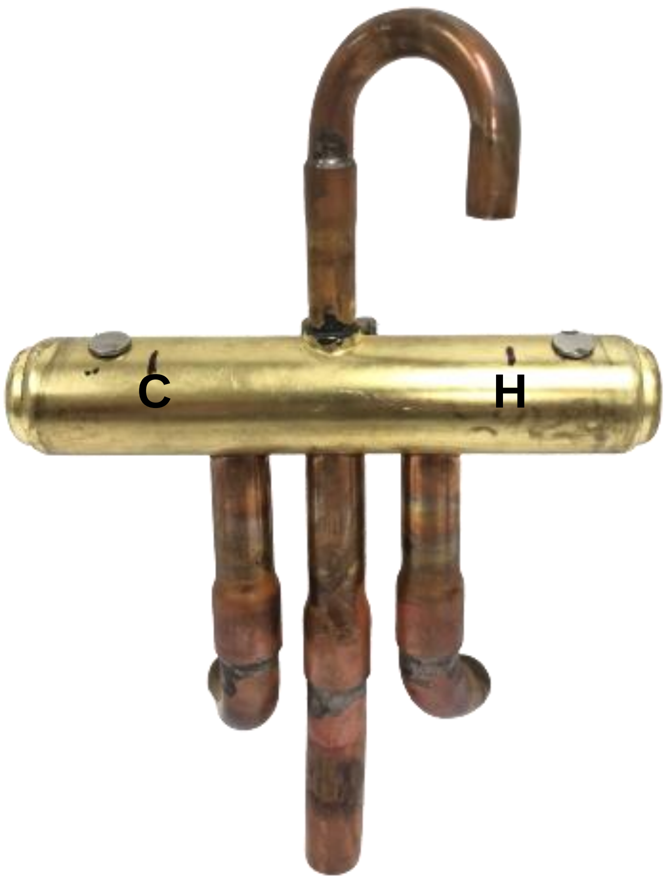
- In a heat pump the function of the indoor and outdoor coils change depending on the mode of operation.
- Directs hot gas from the compressor to the indoor coil in heating mode and to the outdoor coil in cooling mode.
- Its the difference between the high and low side pressure that causes the reversing valve to shift.
- The reversing valve is energized in the heating modes of operation.



Magnets Used as a Service Tool
Normal Position Deenergized



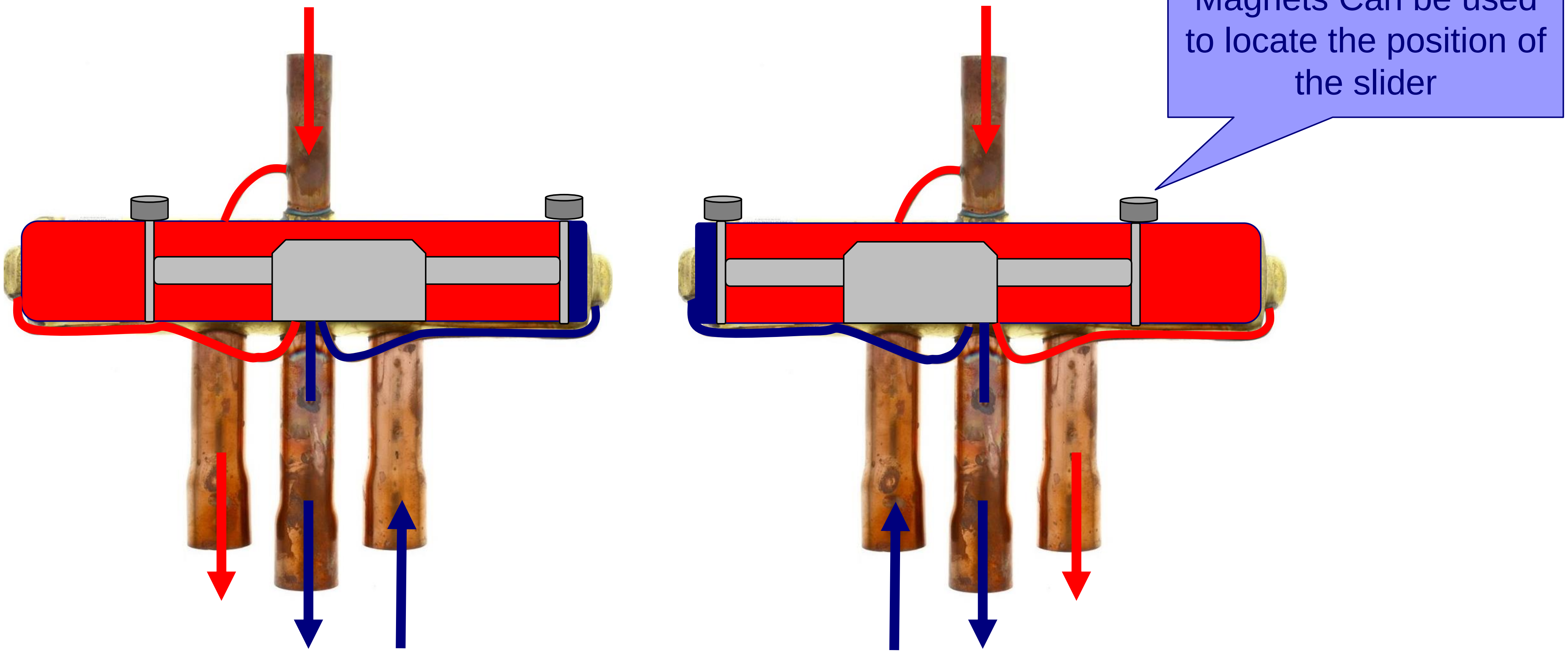
Cooling Position



Mid Seated

Heating Position

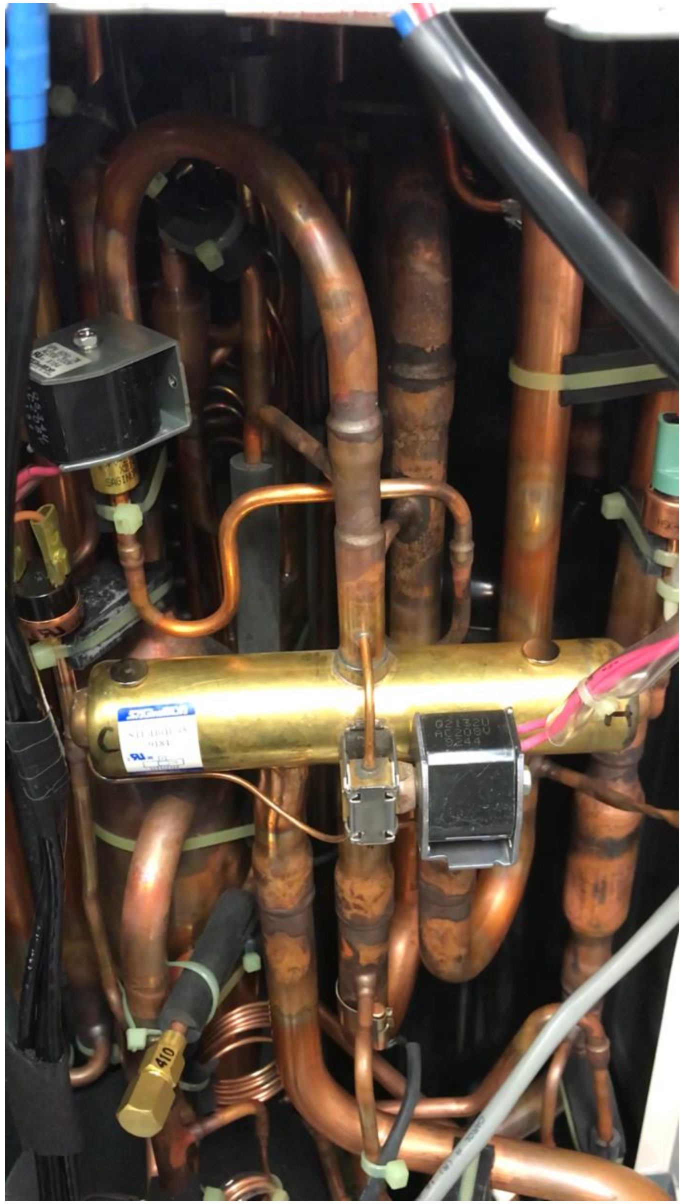
Reversing Valve Refrigerant Flow



A minimum PSIG differential between high and low side pressure is required to move the slider. I may shift with as low as 18 PSIG differential.

Reversing Valve Operation

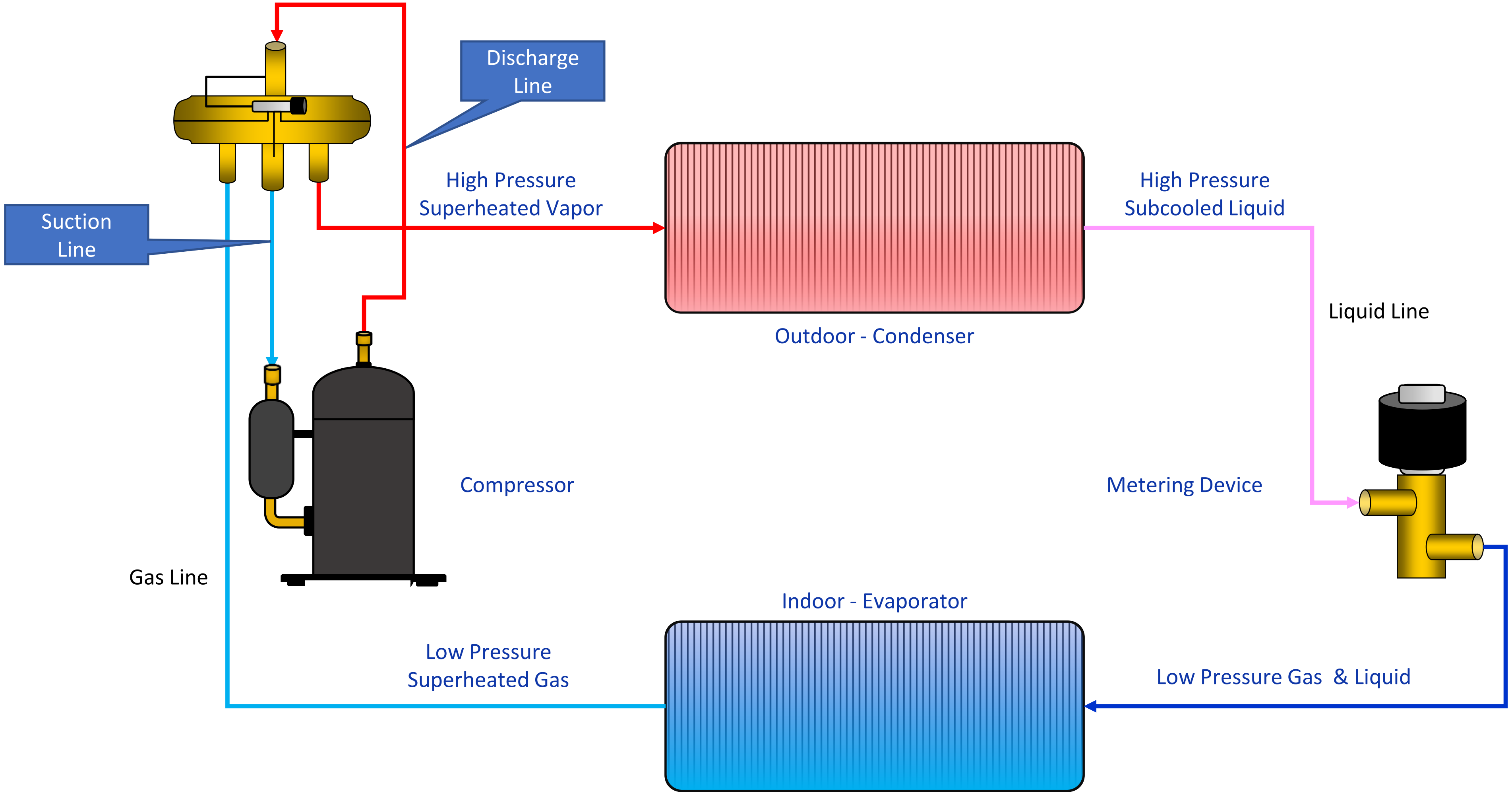
- On compressor start up you will see the magnets move to the left as the compressor start to make pressure.
- You will heat the compressor ramp up and the solenoid valve will be energized.
- Then the magnets move to the right to change to mode of operation.



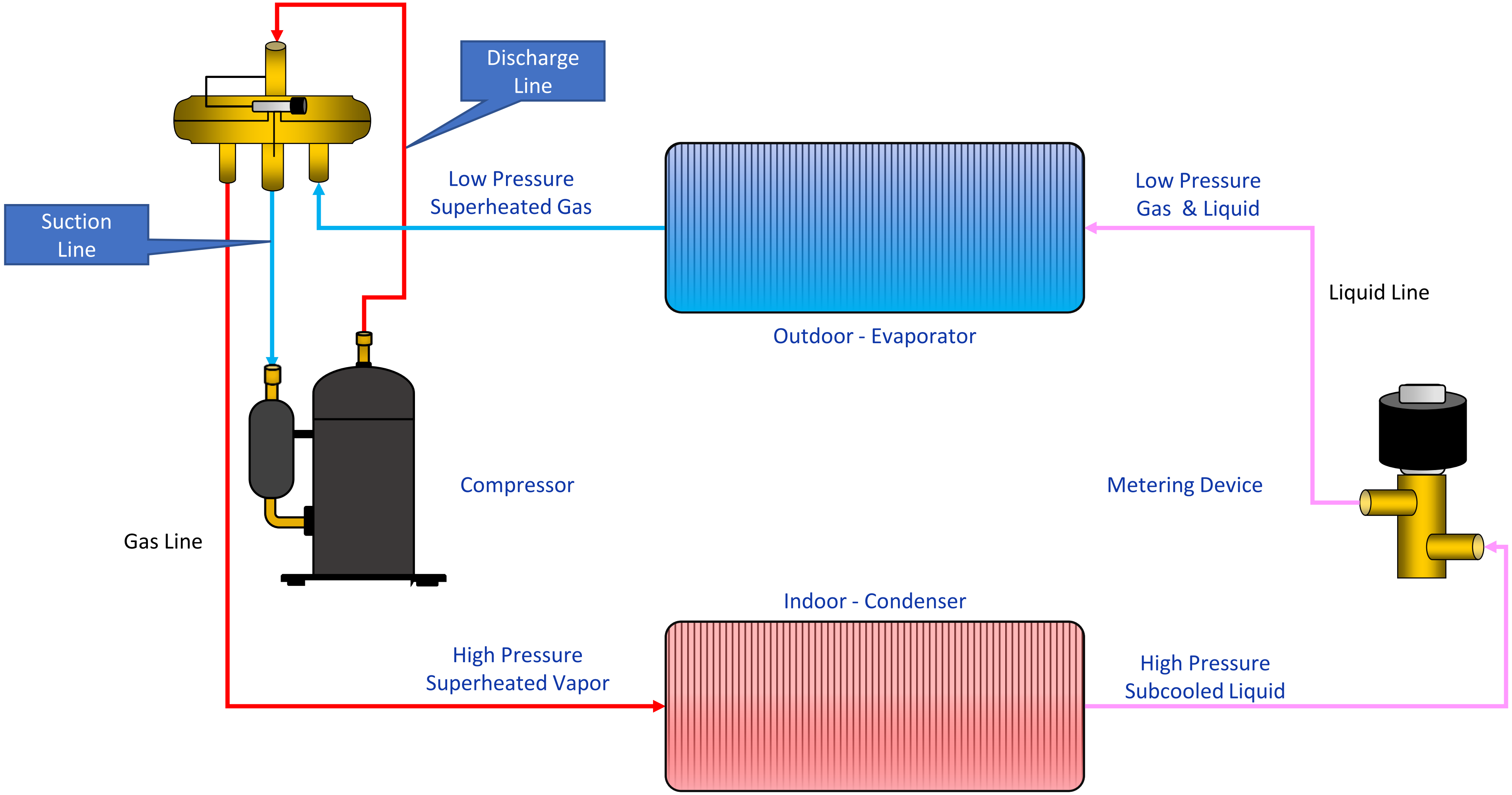
Heat Pump Piping systems

- When dealing with heat pump systems it is critical to understand the terminology used when it comes to the piping system.
- A basic heat pump refrigeration system will contain:
  - Discharge line
    - Connects the discharge of the compressor to the reversing valve
  - Gas line
    - Carries high pressure, high temperature superheated vapor to the Indoor coil in Heating Mode.
    - Carries high pressure, high temperature superheated vapor to the Outdoor coil in Cooling Mode.
  - Liquid line
    - Connects the outlet of the condenser to the inlet of the metering device.
    - Carries high pressure high temperature sub-cooled liquid to the Indoor Coil in Cooling Mode
    - Carries high pressure high temperature sub-cooled liquid to the Outdoor Coil in Heating Mode
  - Suction Line
    - Connects Reversing valve to the inlet of the compressor.

Basic Heat Pump Cooling Cycle



Basic Heat Pump Heating Cycle



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Heat Pump Design

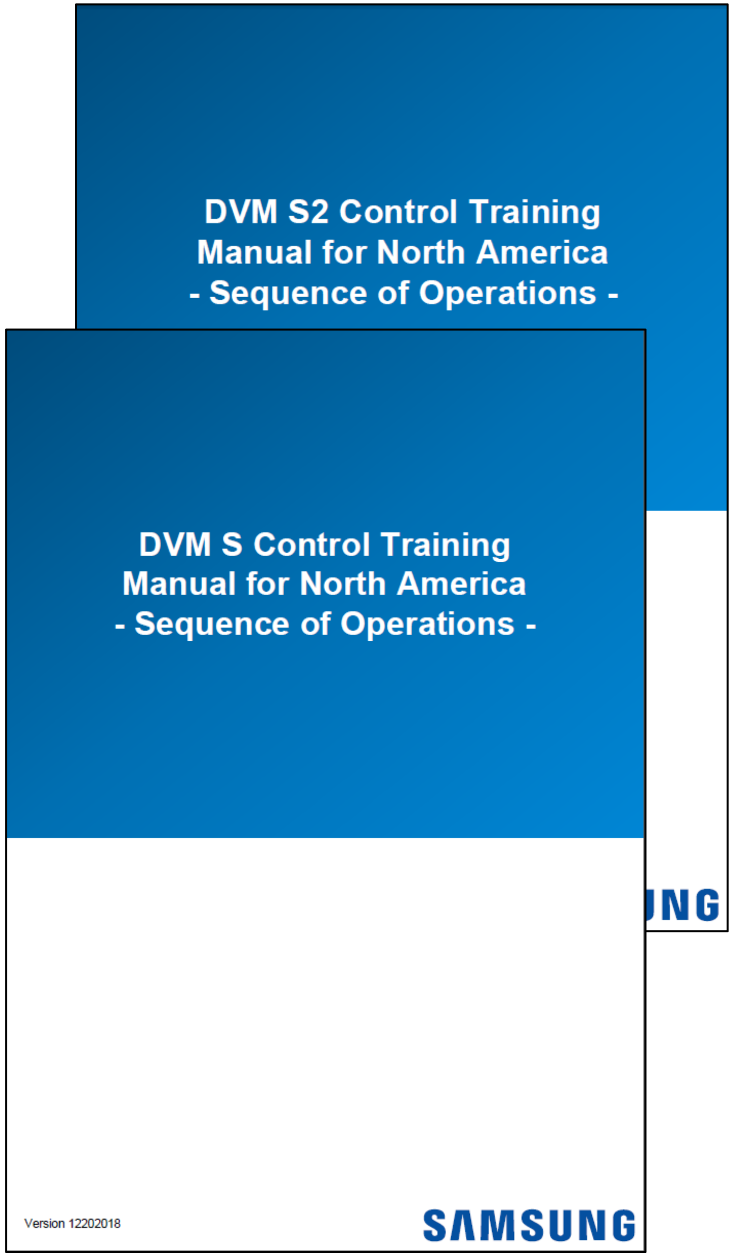
- All manufacturers are bound by the laws of thermodynamics.
- Heat pump are designed to perform a specific function under specific conditions.
- They must produce a specific output at a given condition “Rated Condition”.
- Knowing this information is key to understanding how the system operates and how to trouble shoot system issues.
- Remember that refrigeration is based on temperature differentials!

Thermodynamics Applied System Design

- Heat Pumps are designed to transfer heat from one place to another.
- Manufactures use design coil temperatures to achieve the correct amount of heat transfer in the heating and cooling modes.
- VRF Manufacturers test their equipment under the following conditions:
    - Nominal cooling capacities are based on: Indoor temperature: 80°F DB, 67°F WB. Outdoor temperature: 95°F DB, 75°F WB
    - Nominal heating capacities are based on: Indoor temperature: 70°F DB, 60°F WB. Outdoor temperature: 47°F DB, 43°F WB

Building a Foundation

- The sequence of operation manuals gives the "target" operational data in the heating and cooling modes.
- Knowing the operational data and design conditions we can start to construct the basic operating characteristics of the heating and cooling cycles.


DVM S2 Control Training
Manual for North America
- Sequence of Operations -

DVM S Control Training
Manual for North America
- Sequence of Operations -

Version 12202018

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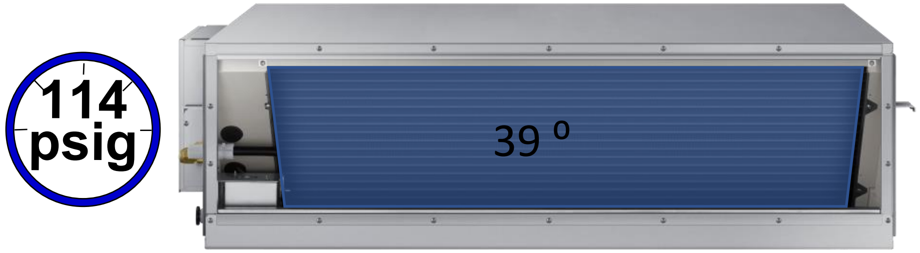
Air Conditioning Operation Design Conditions

Outdoor temperature: 95^{o}F DB

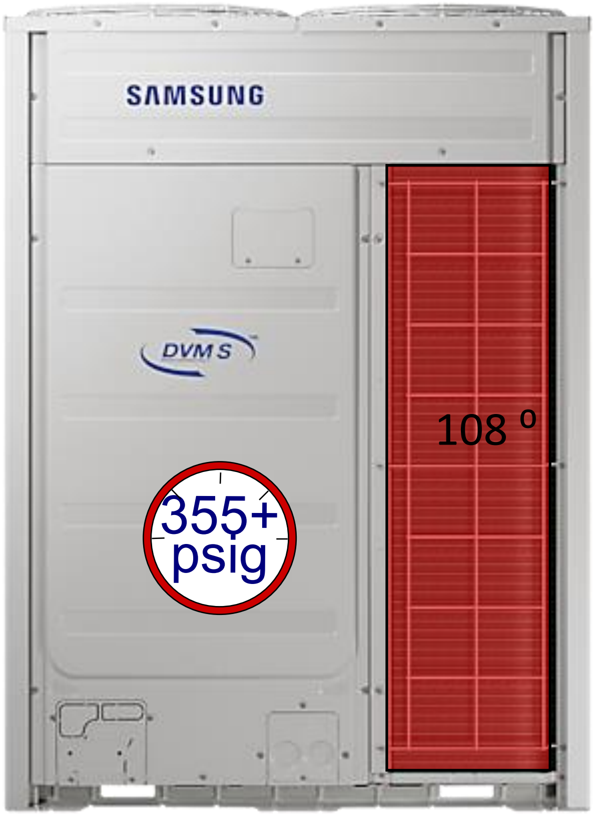
What are the corresponding pressures at each device?

Indoor temperature: 80 ^{o}F DB, 67^{o}F WB



ΔT: 20 º -25 º

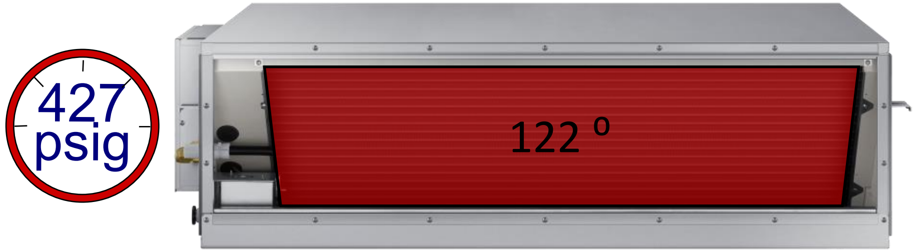
Average Evap. in
39º F - 50º



Heating Operation Design Conditions

What are the corresponding pressures at each device?

Indoor temperature: 70 °F DB, 60°F WB



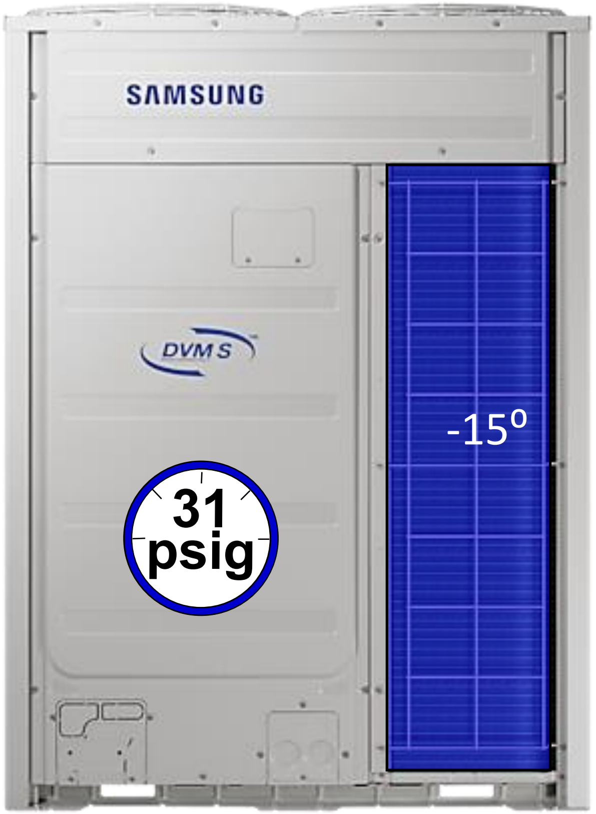
ΔT: 30 ° -35 °

At Design Conditions

Outdoor temperature:
47°F DB, 43°F WB

Max Heat®

Outdoor temperature:
-13°F DB, 43°F WB



Refrigerant Components

- Refrigerant components have a specific function and purpose within a refrigerant system.
- Their function does not change when applies in a VRF system
- In this section we will identify the components within the refrigeration system.

Basic components of a VRF Heat Pump System

Outdoor Unit

System Piping

Indoor Unit

Basic components of a VRF Heat Recovery System

Outdoor Unit

Main System Piping

MCU

IDU Piping

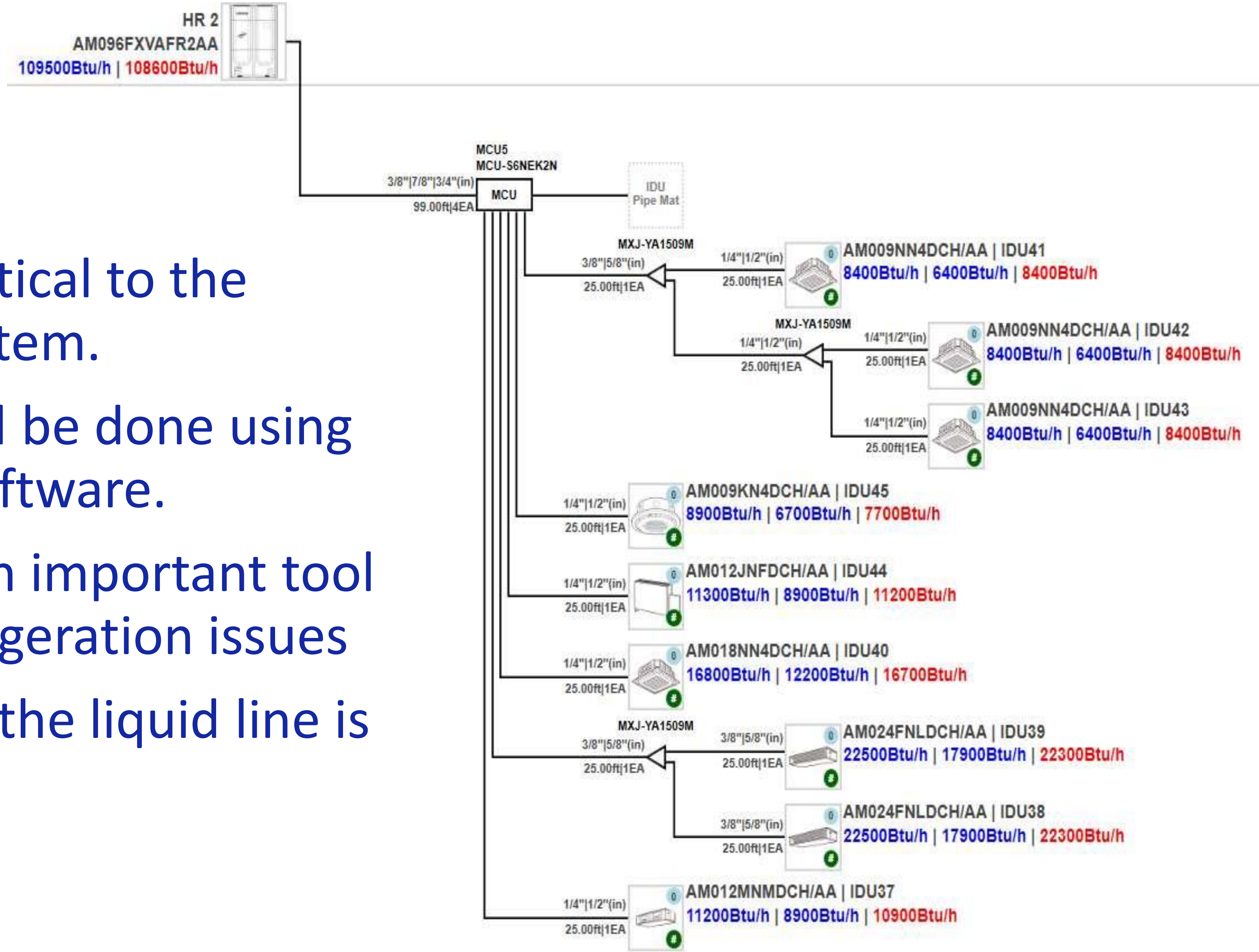
Indoor Unit

Piping systems

- The piping system is commonly overlooked when it comes to refrigeration system.
- A basic refrigeration system will contain:
  - Discharge line / Hot gas line
    - Connects the discharge of the compressor to the inlet of the condenser
    - Carries high pressure, high temperature superheated vapor
  - Liquid line
    - Connects the outlet of the condenser to the inlet of the metering device.
    - Carried high pressure high temperature sub-cooled liquid
  - Suction Line
    - Connects the outlet of the evaporator to the inlet of the compressor.
    - Carries low pressure low temperature superheated vapor.

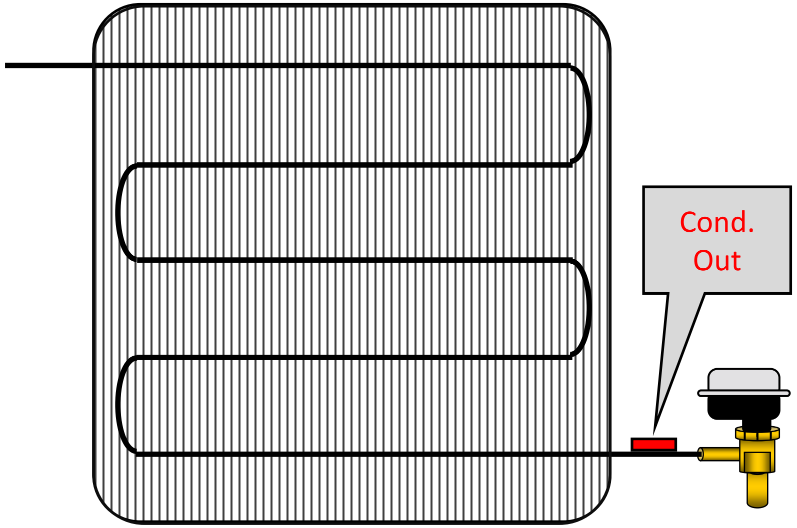
Piping

- The piping system is critical to the performance of the system.
- All piping design should be done using Samsung DVM S Pro Software.
- As built diagrams are an important tool in troubleshooting refrigeration issues
- What happens when if the liquid line is too small?



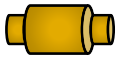
DVM S ODU

Components



ODU Coil

EEV



Strainer

Reversing Valve

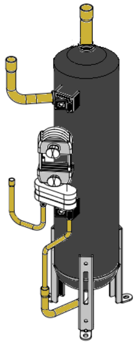
Thermistors

Heat Exchanger

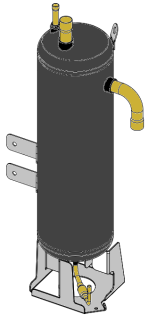
Pressure Transducer

EEV

Component pictures are for demonstration purposes only. Actual components may look slightly different



Oil Separator



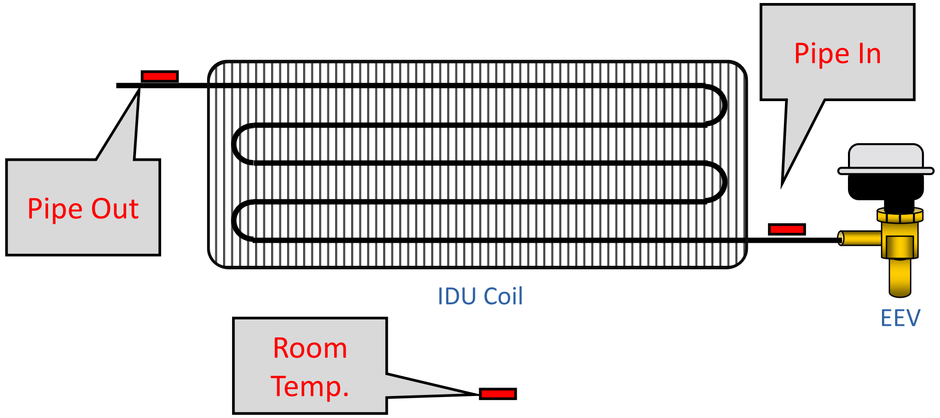
Accumulator

Compressor

Solenoid Valves

DVM S IDU

Refrigerant Components



EEV

Thermistors

Fan Motor

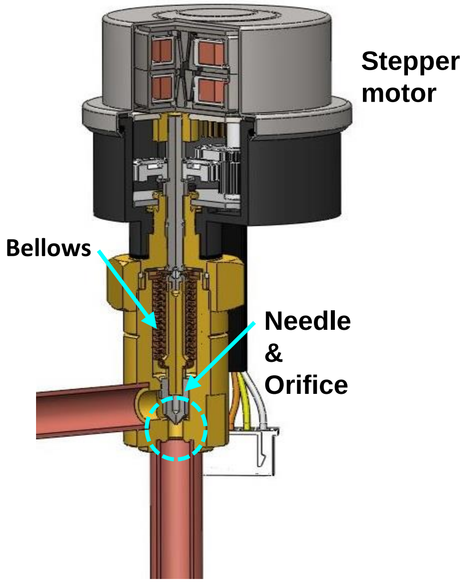
Fan Motor

Solenoid Valves

- Solenoid Valves control refrigerant flow through the system.
- There are several Solenoid valves in the system.
  - Main Cooling (HR systems only)
  - Hot Gas By-Pass
  - Hot Gas By-pass 2(HR systems only)
  - Accumulator Oil Return
  - EVI
  - EVI By-Pass
  - H
  - Heating solenoid In the MCU or HR Changer
  - Cooling solenoid In the MCU or HR Changer
  - Liquid By-Pass solenoid In the MCU or HR Changer

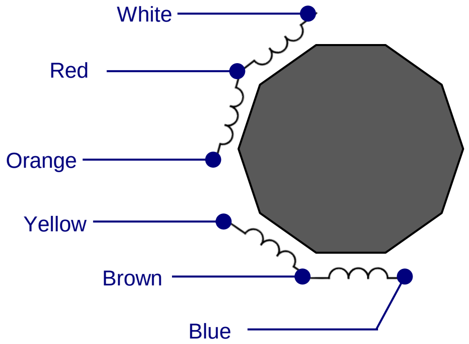
Electronic Expansion Valves
ODU EEV

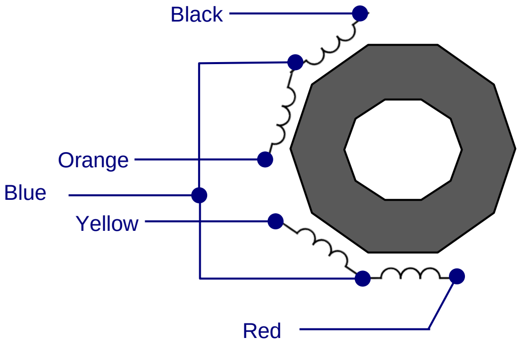
- EEV's are used as metering devices and flow control devices is commercial VFR systems.
- Its purpose is to **maintain a differential between to point** or to allow flow trough a device.
- The ODU has two EEVs in the system.
  - The Main EEV
  - The EVI EEV.
- The IDU Has 1 EEVs in the system.
  - EEV
- The MCU has Multiple EEV's
  - Port – Zone EEV
  - Sub-Cooler EEV
  - Liquid By-Pass (HR Changer)



EEV Coil Resistance

BLUE-BLACK	92Ω +/- 2Ω
BLUE-YELLOW	92Ω +/- 2Ω
BLUE-RED	92Ω +/- 2Ω
BLUE-ORANGE	92Ω +/- 2Ω
COM : Blue	DVMS EVI EEV



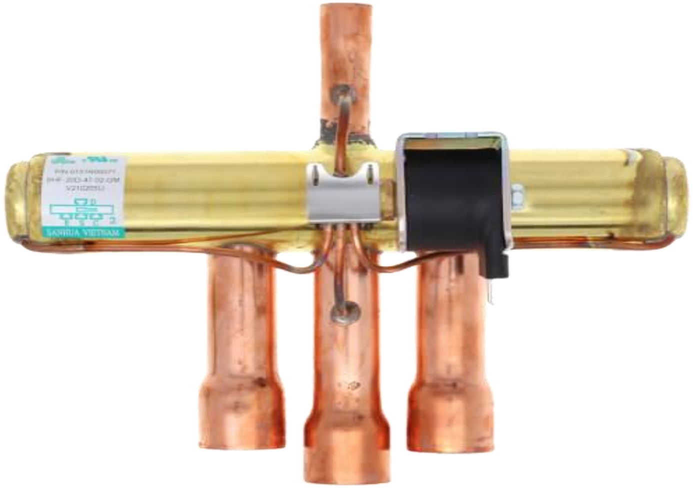


RED – WHITE	150±15Ω
RED – ORANGE	〃
BROWN-YELLOW	〃
BRWON-BLU	〃
COM : RED, BROWN	

BLUE-BLACK	46±2Ω
BLUE-YELLOW	〃
BLUE-RED	〃
BLUE-ORANGE	〃
COM : Blue	RAC CAC FJM

The Reversing Valve

- In a heat pump the function of the indoor and outdoor coils change depending on the mode of operation.
- Directs hot gas from the compressor to the indoor coil in heating mode and to the outdoor coil in cooling mode.
- Its the difference between the high and low side pressure that causes the reversing valve to shift.
- The reversing valve is energized in the heating modes of operation.
- DVM S2 Systems have 2 Reversing- 4 Way Valves.



The Compressor

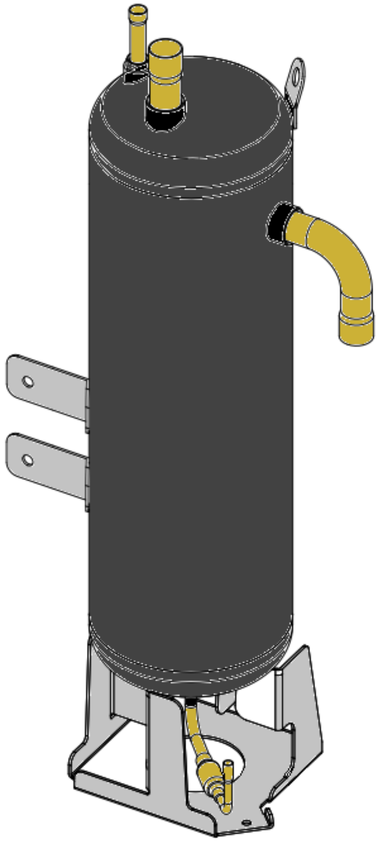
- The compressor operates to maintain:
- Suction Pressure in the Cooling Mode.
  - The compressor speeds up to lower suction pressure
  - The compressor slows down to increase suction pressure
- Head pressure in the heating mode.
  - The compressor speeds up to increase head pressure
  - The compressor slows down to lower pressure.

Cooling Mode

Heating Mode

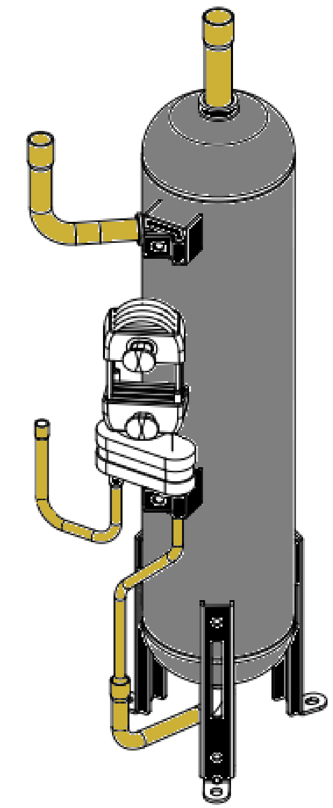
Accumulator

- Purpose:
  - Prevent liquid refrigerant from entering the compressor.
  - Allow oil that is captured in the accumulator to return to the compressor.



Oil Separator

- Purpose:
  - Remove oil from the refrigerant and return it back to the compressor.



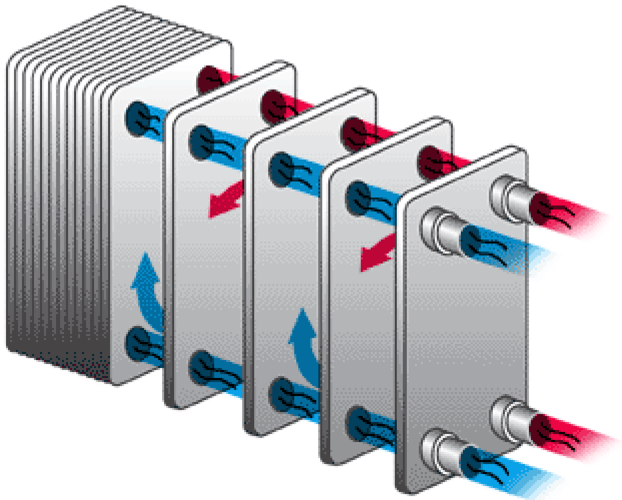
SAMSUNG

EVI - Heat Exchanger Control

- Provides sub-cooled liquid to the system in cooling mode
- Provides refrigerant to the vapor or flash injection circuit of the compressor in the heating mode.

Or

- Provides refrigerant to accumulator in the heating mode.

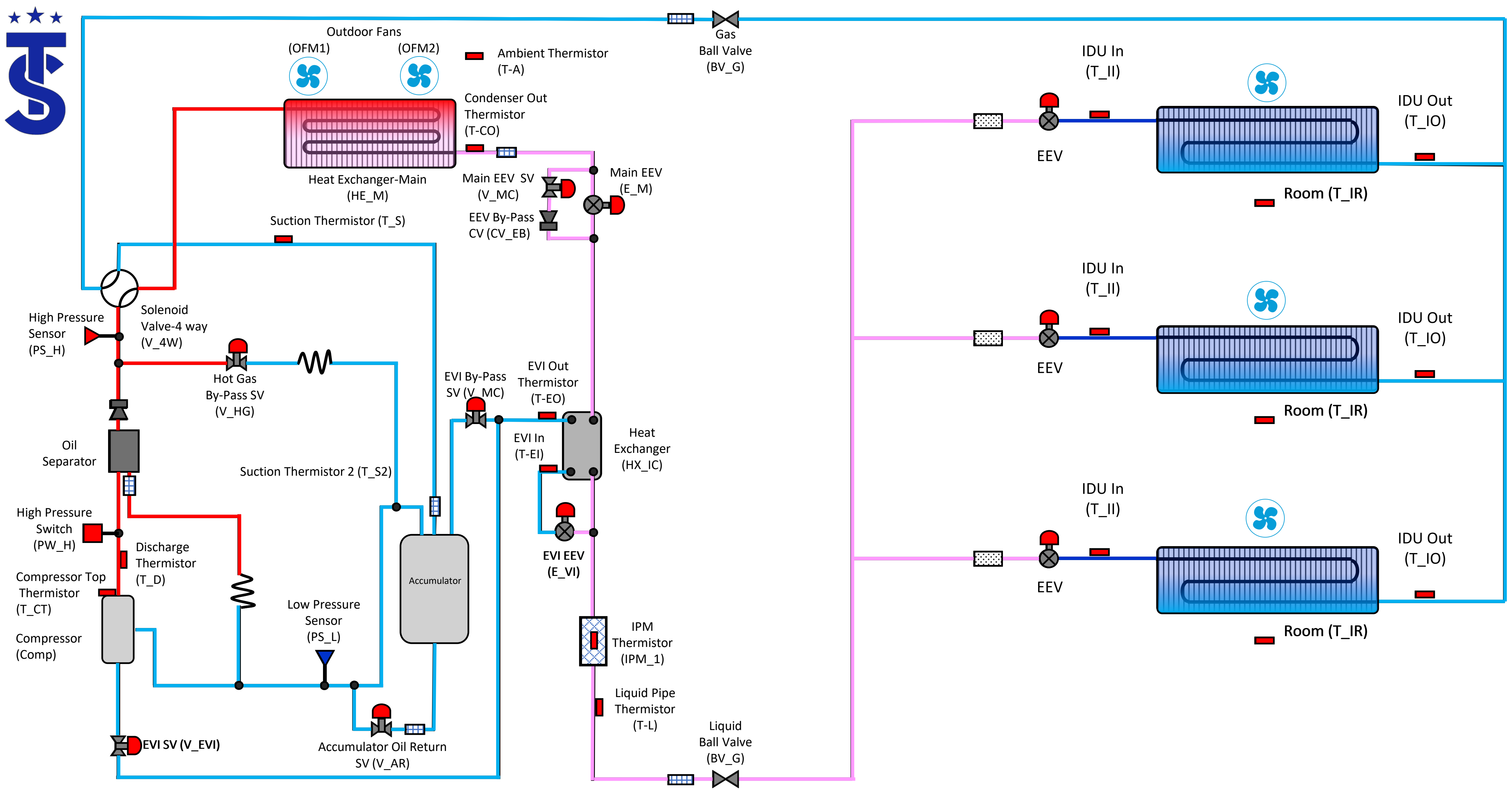


Air Flow

- System capacity is directly related to the air flow.
- To get the "Rated Capacity" the fan has to be at high speed.
- Low air flow has a negative effect on the system operation.
- Air flow must be verified prior to refrigerant cycle troubleshooting.

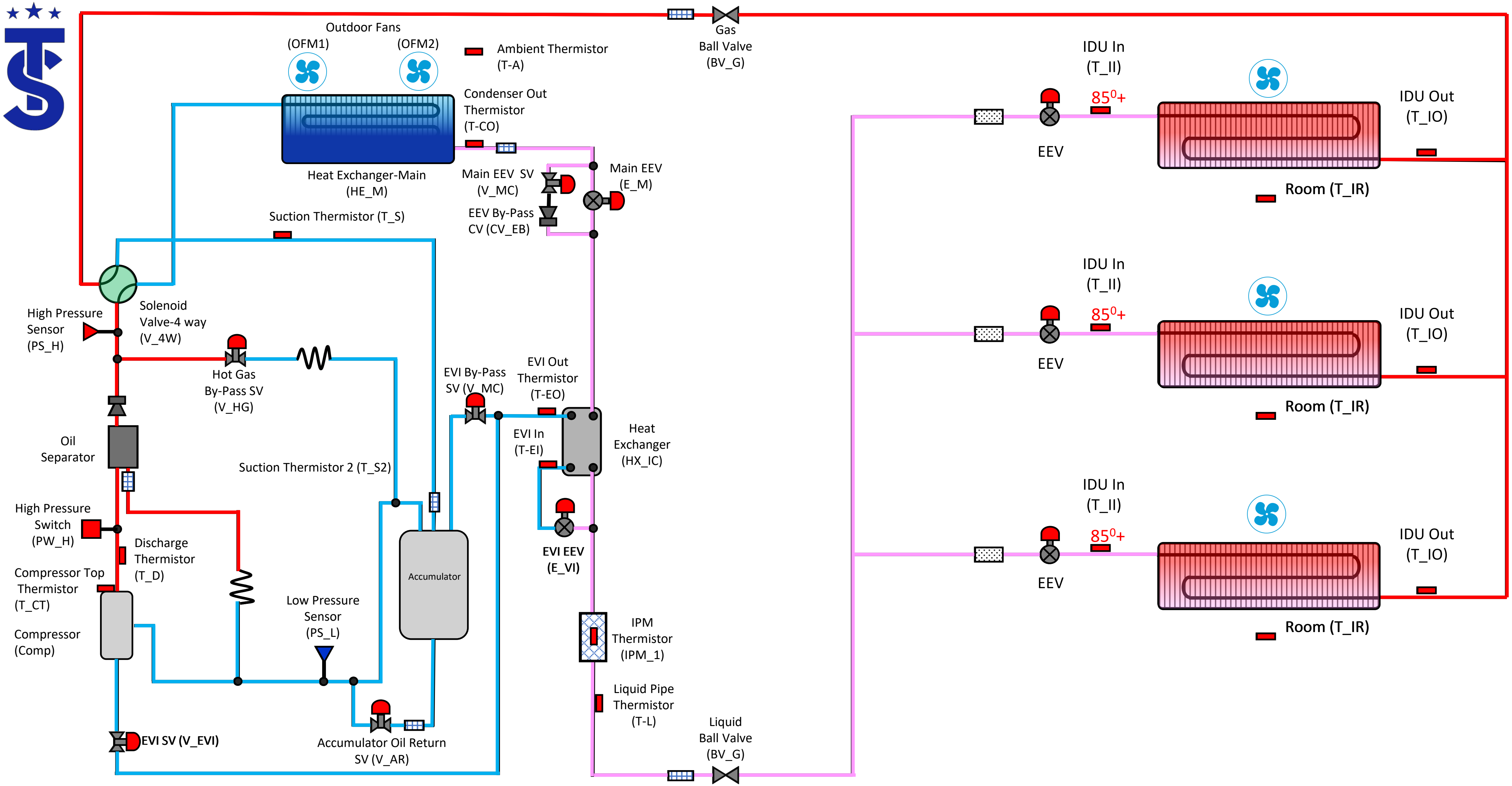
Putting It All Together

- Now that the function of each component has been reviewed let’s see how the combine to make the complete system



DVM S HP – Cooling Mode

SAMSUNG



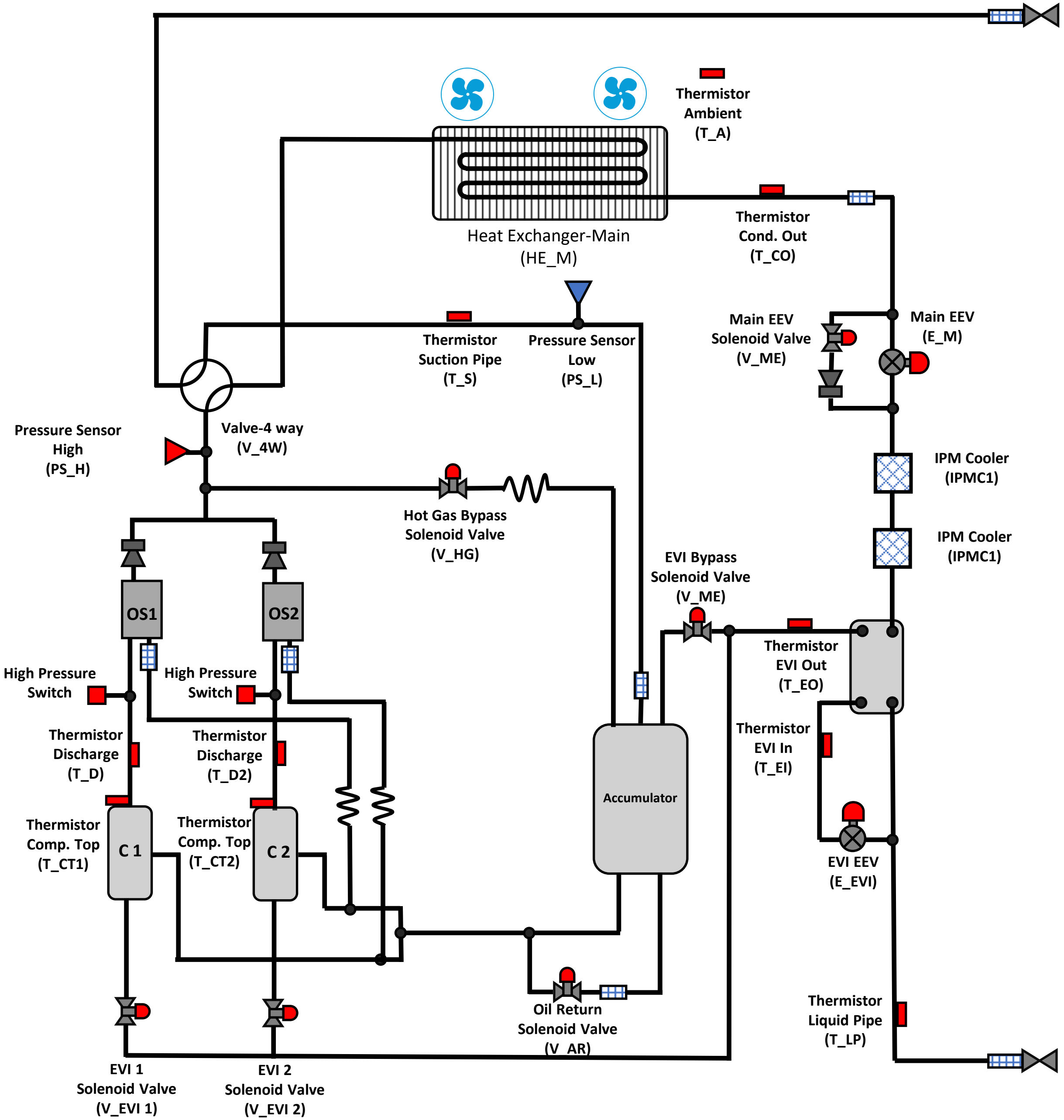
DVM S HP – Heating Mode

SAMSUNG

System Variations

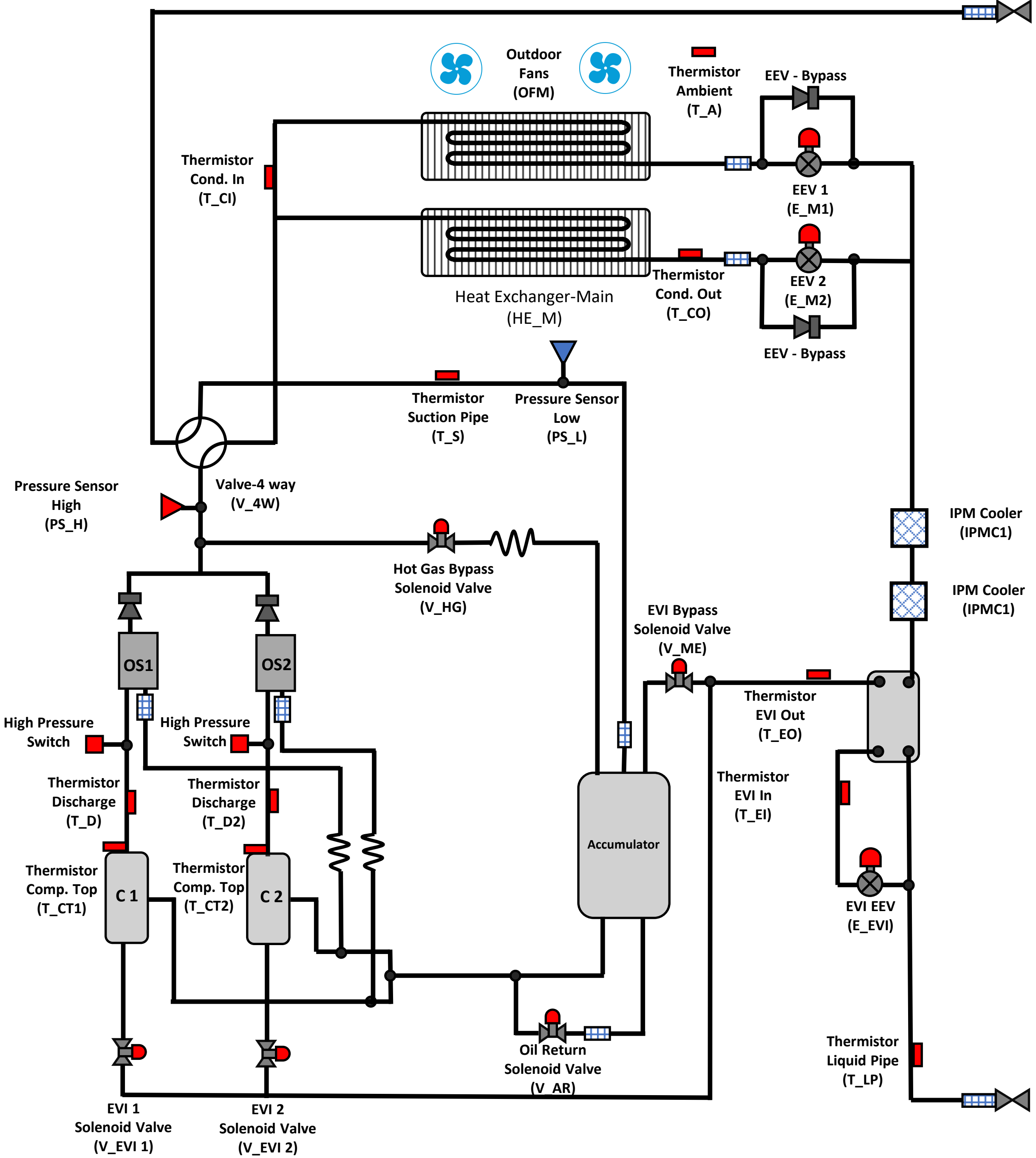
- There are different variations of the outdoor unit but the basic concept is still the same.
- Don’t let the added components worry you.

Dual Compressors



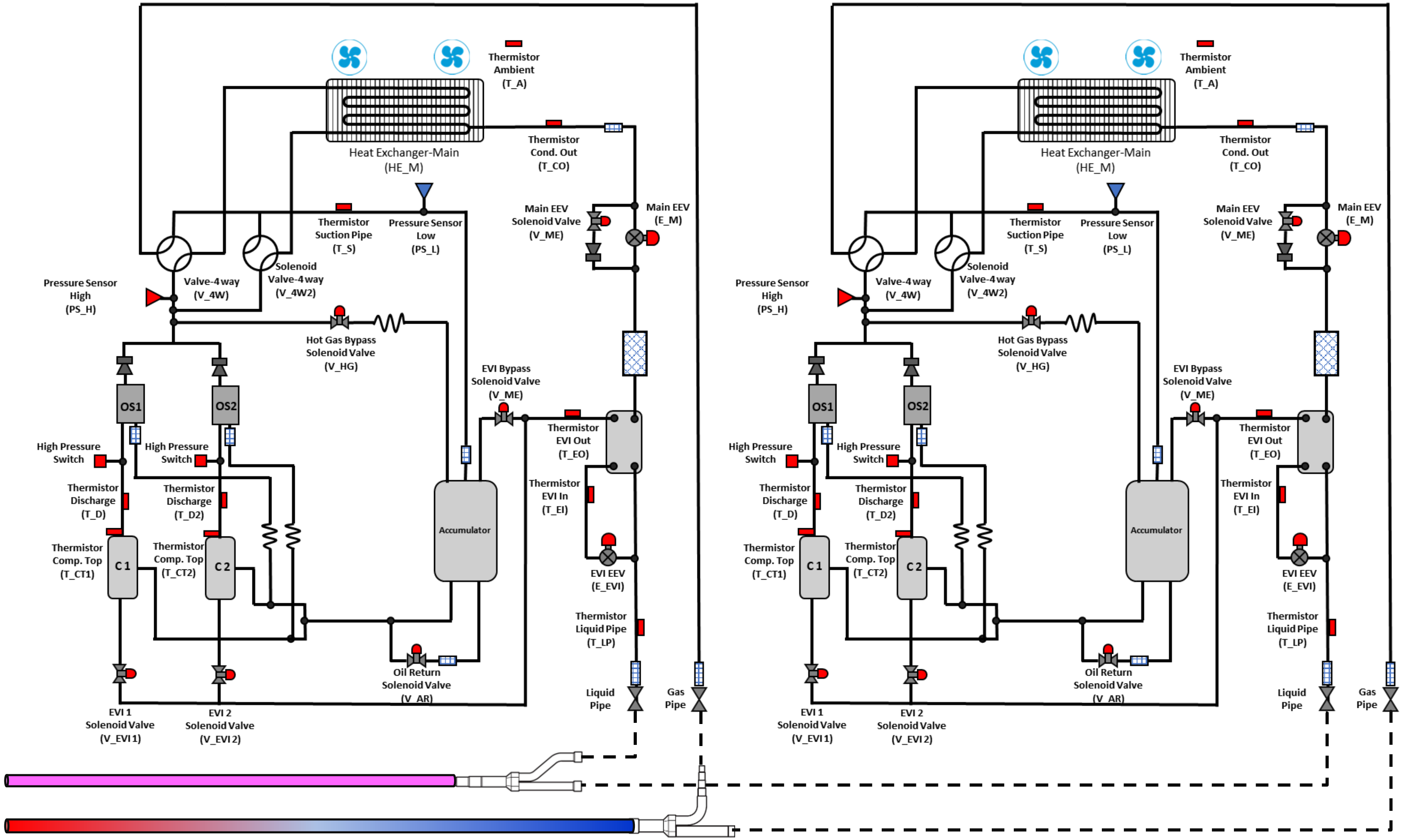
Dual Coils

AM216

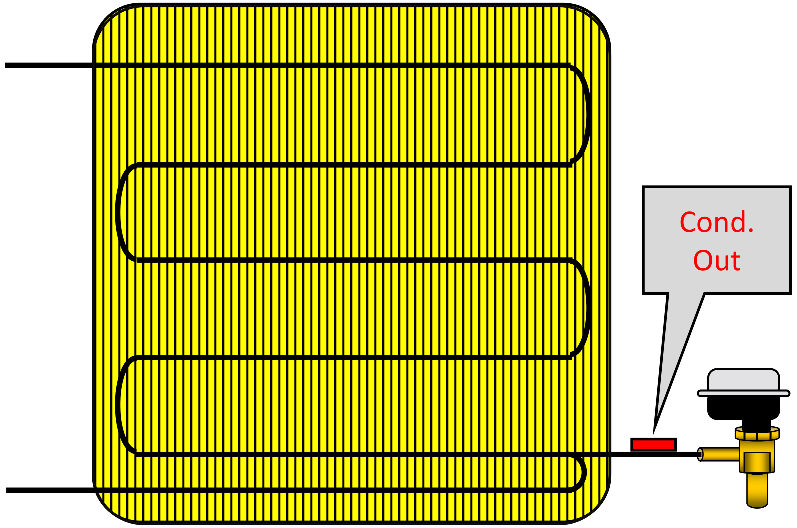


SAMSUNG

Dual Chassis

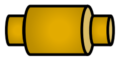


DVMS 2



ODU Coil

EEV



Strainer

Sub 4-way valve

4 way Valve

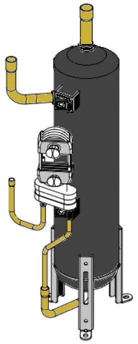
Thermistors

Heat Exchanger

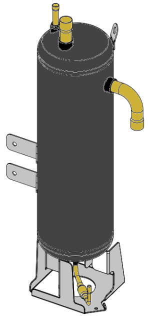
Pressure Transducer

EEV

Component pictures are for demonstration purposes only. Actual components may look slightly different



Oil Separator



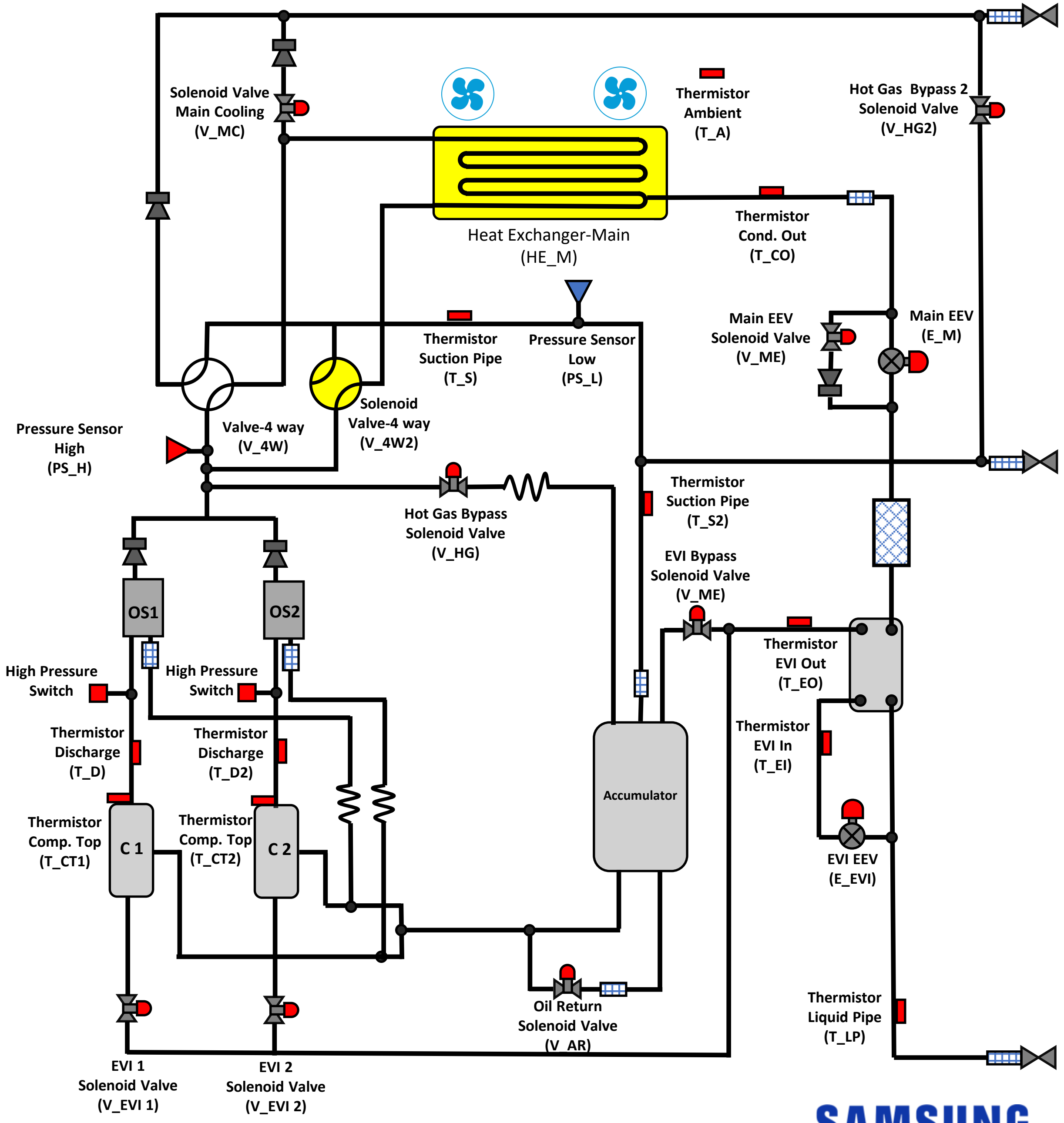
Accumulator

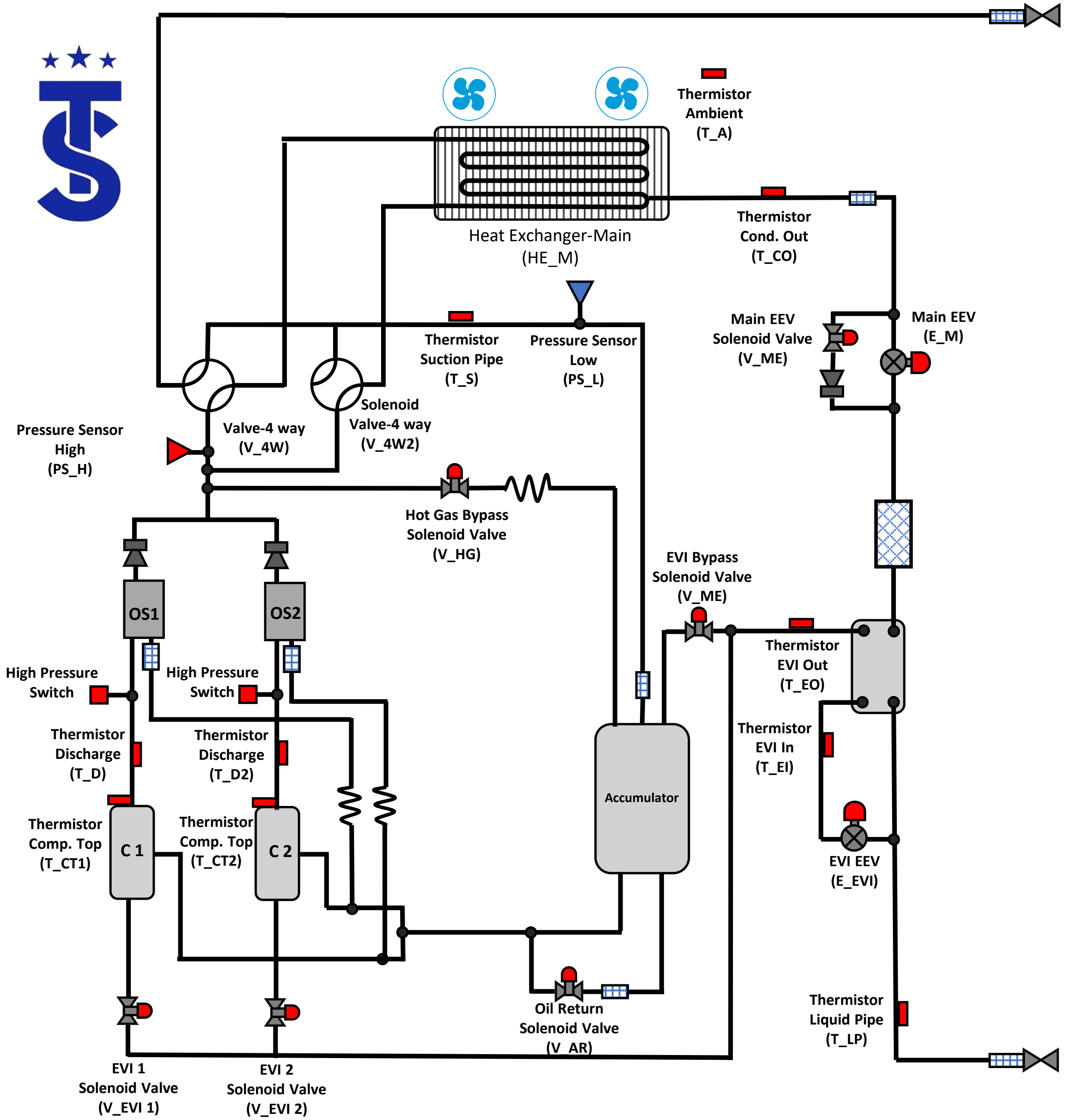
Compressor

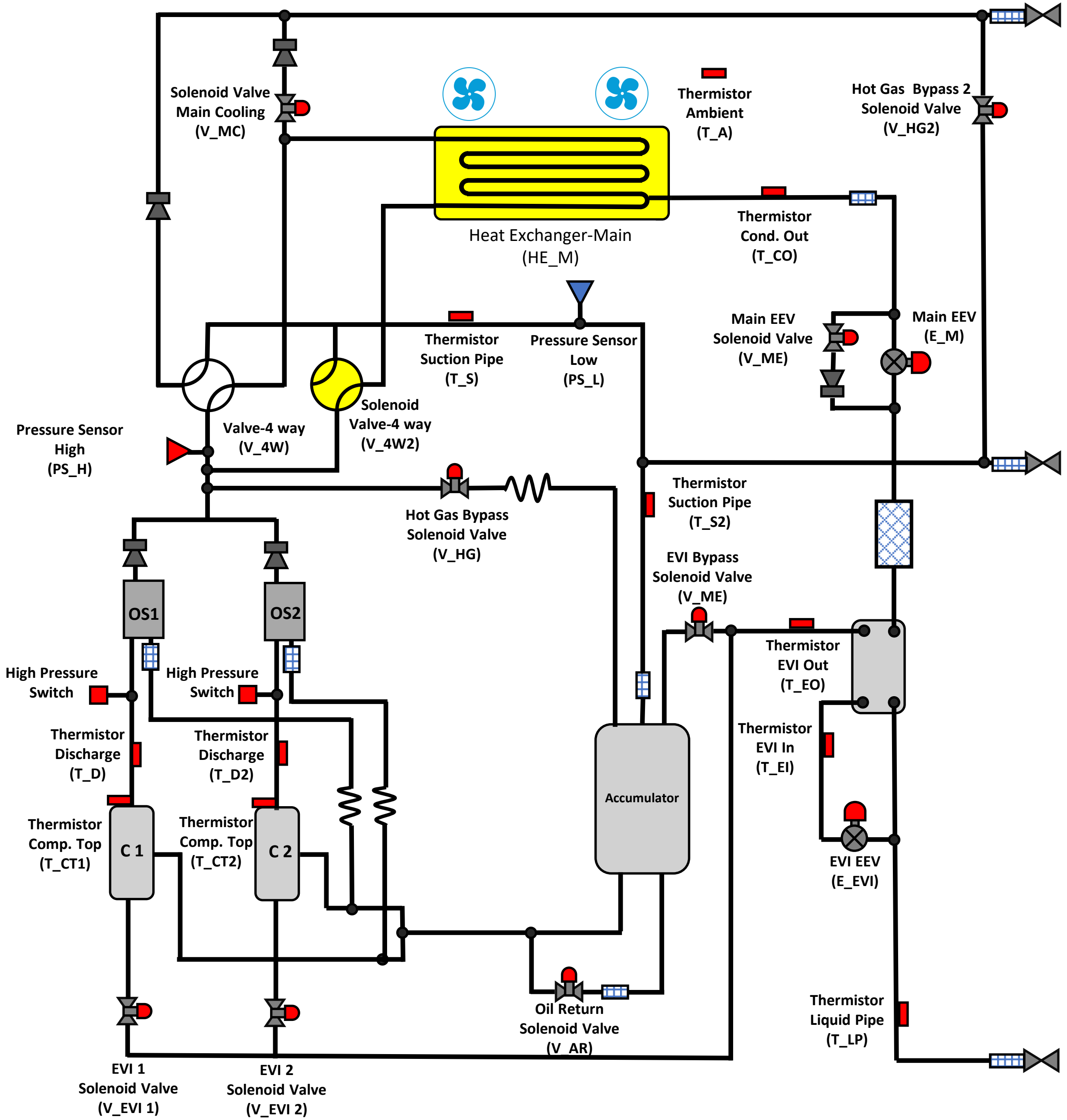
Solenoid Valves

DVM S2

- There are 2 primary differences between the DVM S and DVM S2 outdoor units.
- DVM S2 ODU's have:
  - A sub 4way valve
  - Split ODU Coil
- The function of the other component remain unchanged.



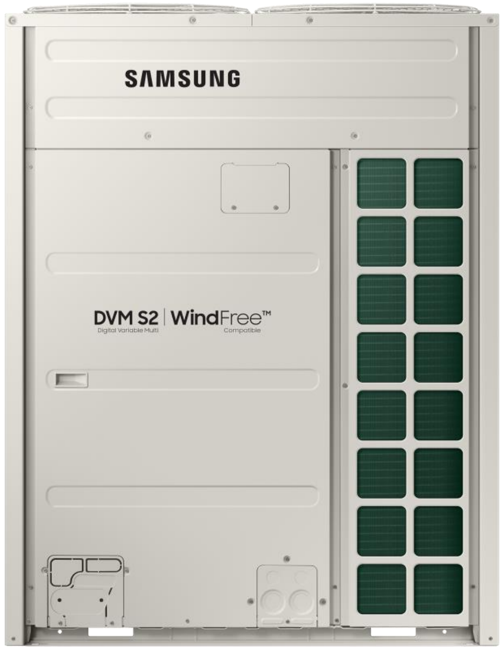

Thermistor
Ambient
(T_A)
Heat Exchanger-Main
(HE_M)
Thermistor
Cond. Out
(T_CO)
Thermistor
Suction Pipe
(T_S)
Pressure Sensor
Low
(PS_L)
Main EEV
Solenoid Valve
(V_ME)
Main EEV
(E_M)
Pressure Sensor
High
(PS_H)
Valve-4 way
(V_4W)
Solenoid
Valve-4 way
(V_4W2)
Hot Gas Bypass
Solenoid Valve
(V_HG)
EVI Bypass
Solenoid Valve
(V_ME)
OS1
OS2
High Pressure
Switch
High Pressure
Switch
Thermistor
EVI Out
(T_EO)
Thermistor
Discharge
(T_D)
Thermistor
Discharge
(T_D2)
Thermistor
EVI In
(T_EI)
Accumulator
Thermistor
Comp. Top
(T_CT1)
C 1
Thermistor
Comp. Top
(T_CT2)
C 2
EVI EEV
(E_EVI)
Thermistor
Liquid Pipe
(T_LP)
Oil Return
Solenoid Valve
(V_AR)
EVI 1
Solenoid Valve
(V_EVI 1)
EVI 2
Solenoid Valve
(V_EVI 2)


Solenoid Valve
Main Cooling
(V_MC)
Thermistor
Ambient
(T_A)
Hot Gas Bypass 2
Solenoid Valve
(V_HG2)
Heat Exchanger-Main
(HE_M)
Thermistor
Cond. Out
(T_CO)
Thermistor
Suction Pipe
(T_S)
Pressure Sensor
Low
(PS_L)
Main EEV
Solenoid Valve
(V_ME)
Main EEV
(E_M)
Pressure Sensor
High
(PS_H)
Valve-4 way
(V_4W)
Solenoid
Valve-4 way
(V_4W2)
Thermistor
Suction Pipe
(T_S2)
Hot Gas Bypass
Solenoid Valve
(V_HG)
EVI Bypass
Solenoid Valve
(V_ME)
OS1
OS2
High Pressure
Switch
High Pressure
Switch
Thermistor
EVI Out
(T_EO)
Thermistor
Discharge
(T_D)
Thermistor
Discharge
(T_D2)
Thermistor
EVI In
(T_EI)
Accumulator
Thermistor
Comp. Top
(T_CT1)
C 1
Thermistor
Comp. Top
(T_CT2)
C 2
EVI EEV
(E_EVI)
Thermistor
Liquid Pipe
(T_LP)
Oil Return
Solenoid Valve
(V_AR)
EVI 1
Solenoid Valve
(V_EVI 1)
EVI 2
Solenoid Valve
(V_EVI 2)

Heat recovery systems

Single Phase ECO

HR Changer



3 Phase DVM S2

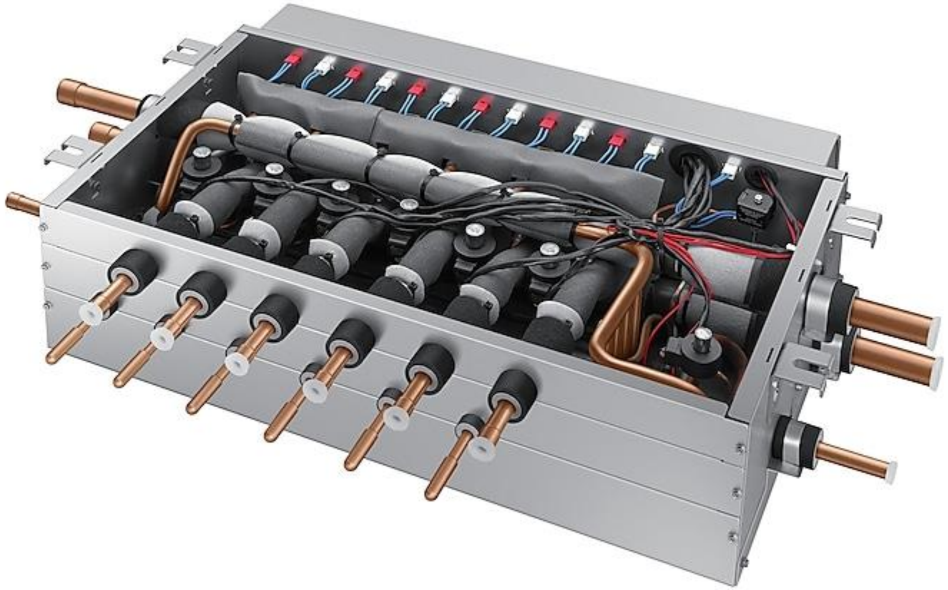
MCU

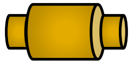
MCU and HR Changer

- The MCU and HR Changer (MCU-R4NEK0N) are used on heat recovery systems.
- They allow zones to operate between different modes of operation
- The HR Changer is used exclusively for the Eco single phase system
- Externally the appear the same.

MCU and HR Changer

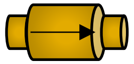
Refrigerant Components





Strainer

EEV's



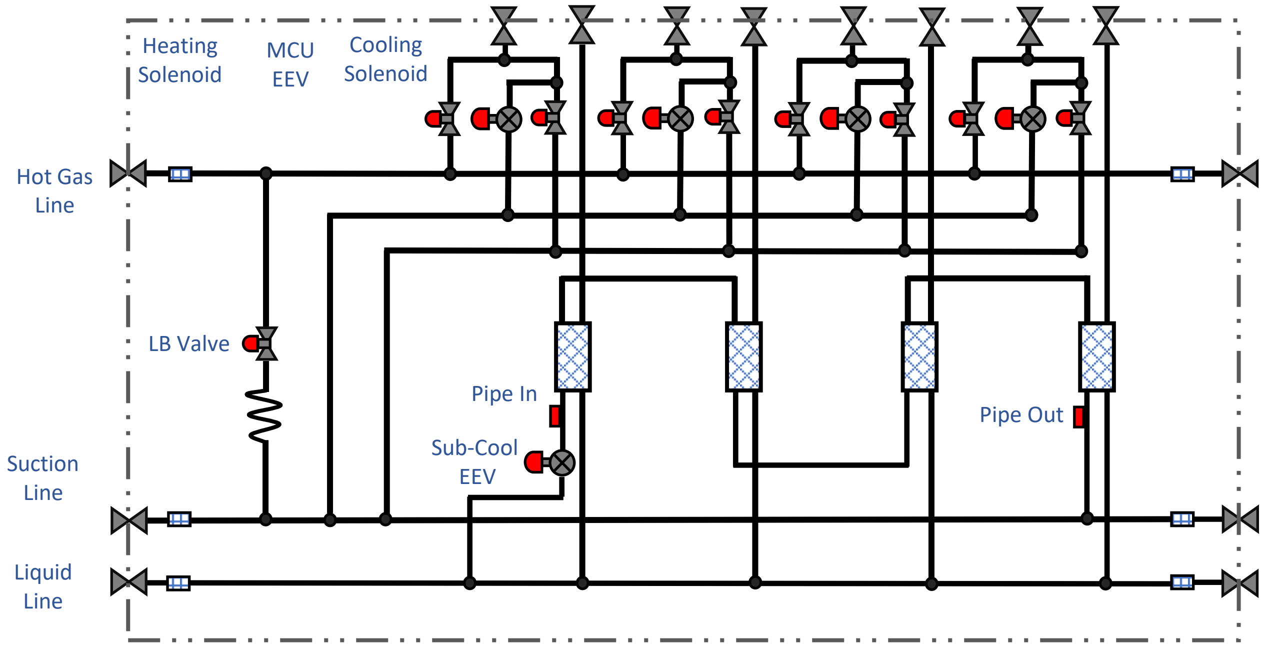
Check Valve

Thermistor

Solenoid

Heat Exchanger

MCU-Mode Change Unit
Refrigerant Components

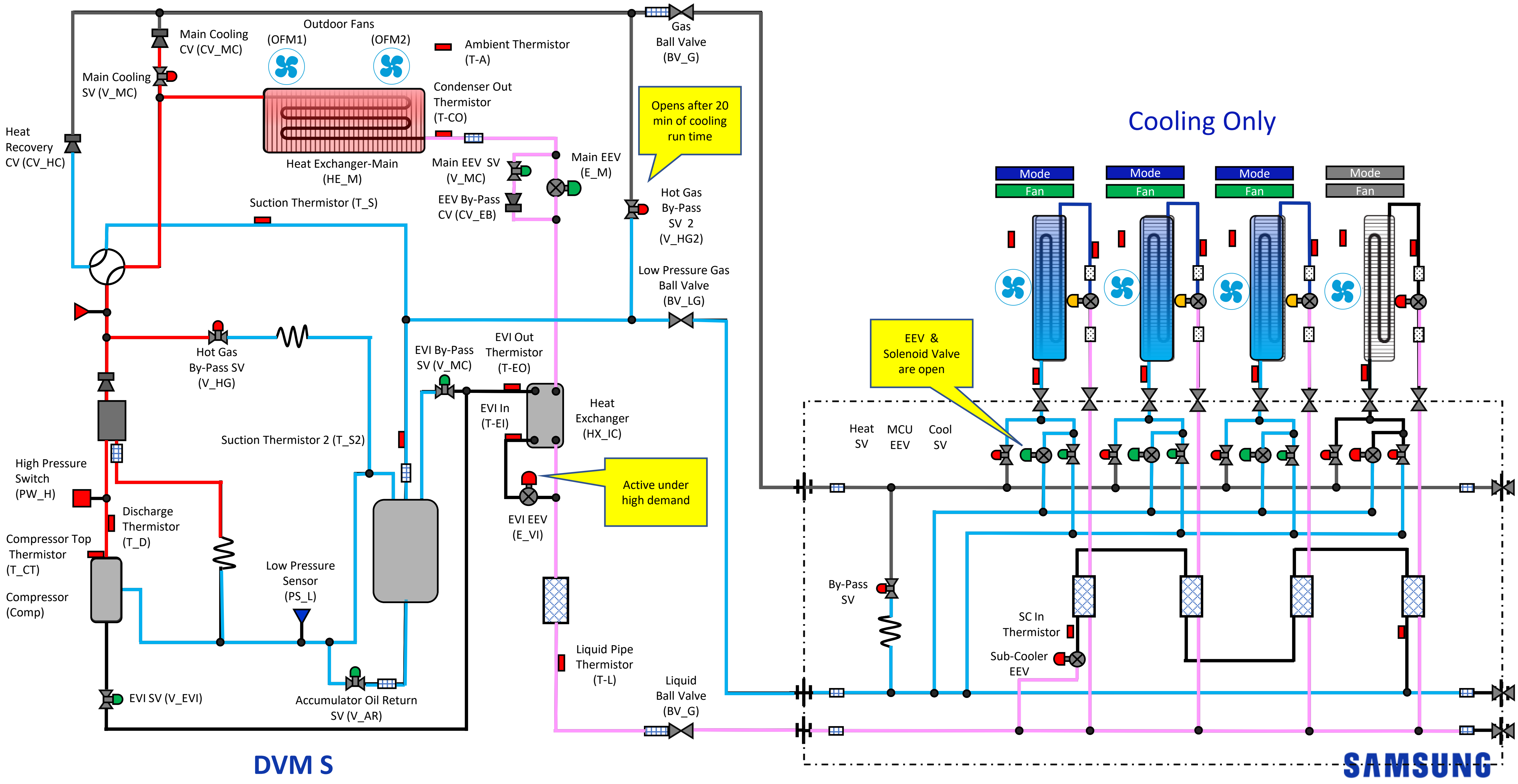


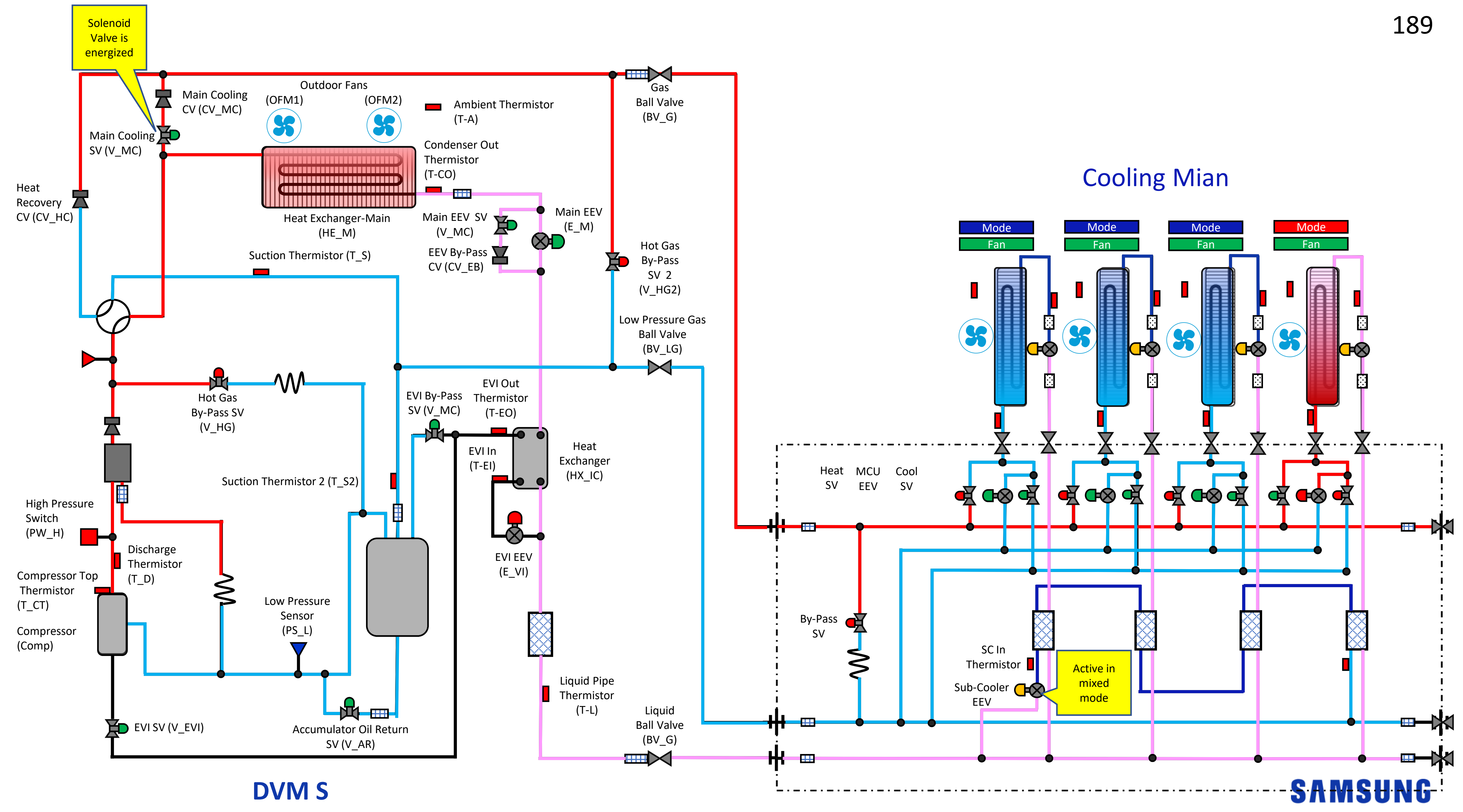
MCU

No	Cycle parts	Functions
1	Cooling solenoid valve	This valve opens during Indoor unit's cooling operation
2	Heating solenoid valve	This valve opens during Indoor unit's heating operation
3	MCU EEV	To reduce modulation noise while mode is changed. (Heating → Cooling)
4	Sub-cooler	To secure the sub-cooling of refrigerant for each indoor.
5	Sub-cooler EEV	To control super heat of sub-cooler
6	MCU liquid bypass valve	To prevent refrigerant accumulation in high pressure gas pipe

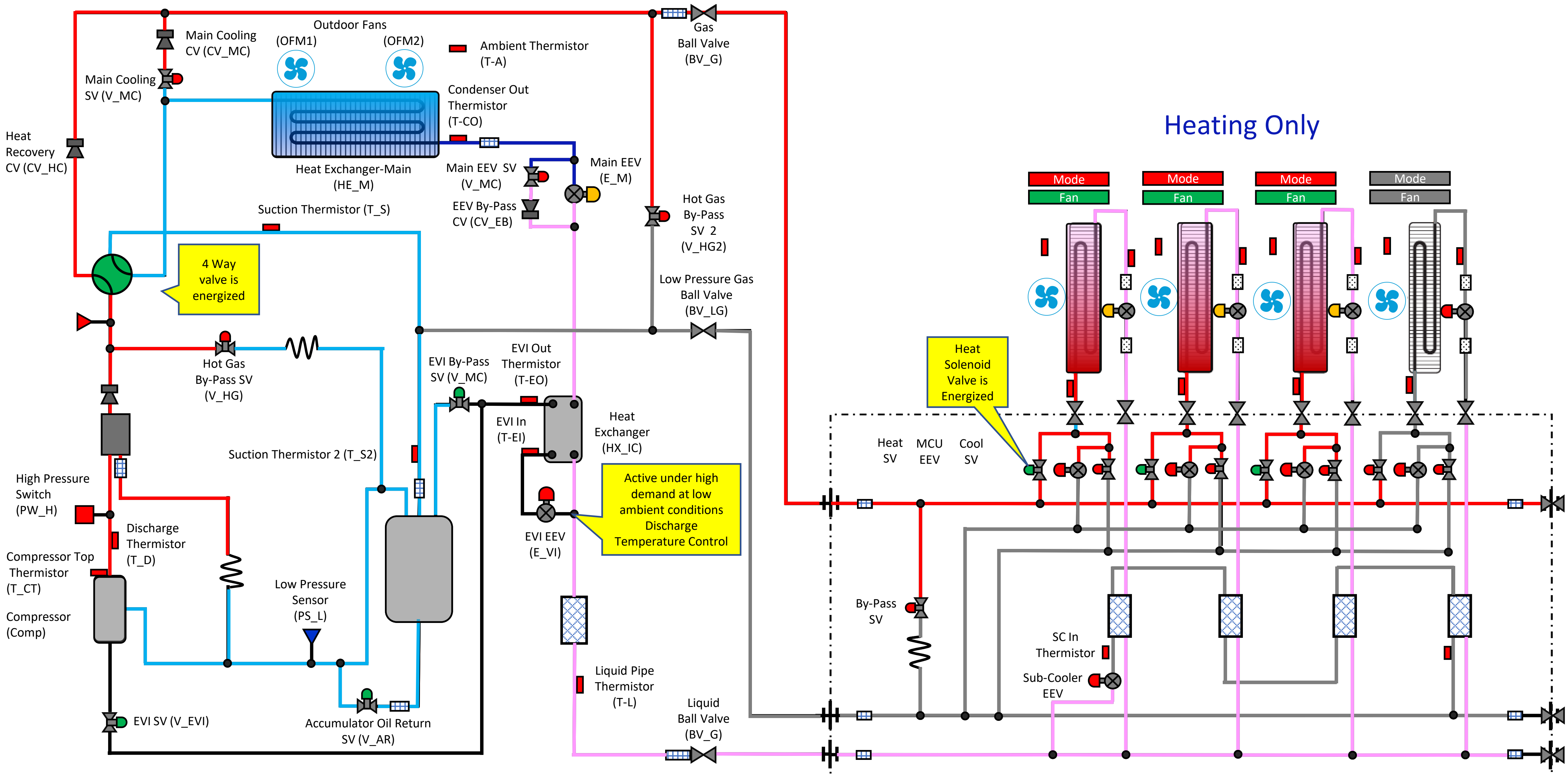
VRF Heat Recovery Modes of Operation

- Modes of Operation
  - Cooling
  - Cooling main
  - Heating
  - Heating Main


Main Cooling
CV (CV_MC)
Outdoor Fans
(OFM1)
(OFM2)
Ambient Thermistor
(T-A)
Gas
Ball Valve
(BV_G)
Main Cooling
SV (V_MC)
Condenser Out
Thermistor
(T-CO)
Opens after 20
min of cooling
run time
Heat
Recovery
CV (CV_HC)
Heat Exchanger-Main
(HE_M)
Main EEV SV
(V_MC)
Main EEV
(E_M)
Cooling Only
EEV By-Pass
CV (CV_EB)
Hot Gas
By-Pass
SV 2
(V_HG2)
Suction Thermistor (T_S)
Mode
Fan
Low Pressure Gas
Ball Valve
(BV_LG)
Hot Gas
By-Pass SV
(V_HG)
EVI Out
Thermistor
(T-EO)
EVI By-Pass
SV (V_MC)
EEV &
Solenoid Valve
are open
Heat
Exchanger
(HX_IC)
EVI In
(T-EI)
Suction Thermistor 2 (T_S2)
Heat
SV
MCU
EEV
Cool
SV
High Pressure
Switch
(PW_H)
Active under
high demand
Discharge
Thermistor
(T_D)
EVI EEV
(E_VI)
Compressor Top
Thermistor
(T_CT)
Low Pressure
Sensor
(PS_L)
By-Pass
SV
Compressor
(Comp)
SC In
Thermistor
Liquid Pipe
Thermistor
(T-L)
Sub-Cooler
EEV
Liquid
Ball Valve
(BV_G)
EVI SV (V_EVI)
Accumulator Oil Return
SV (V_AR)
DVM S
SAMSUNG

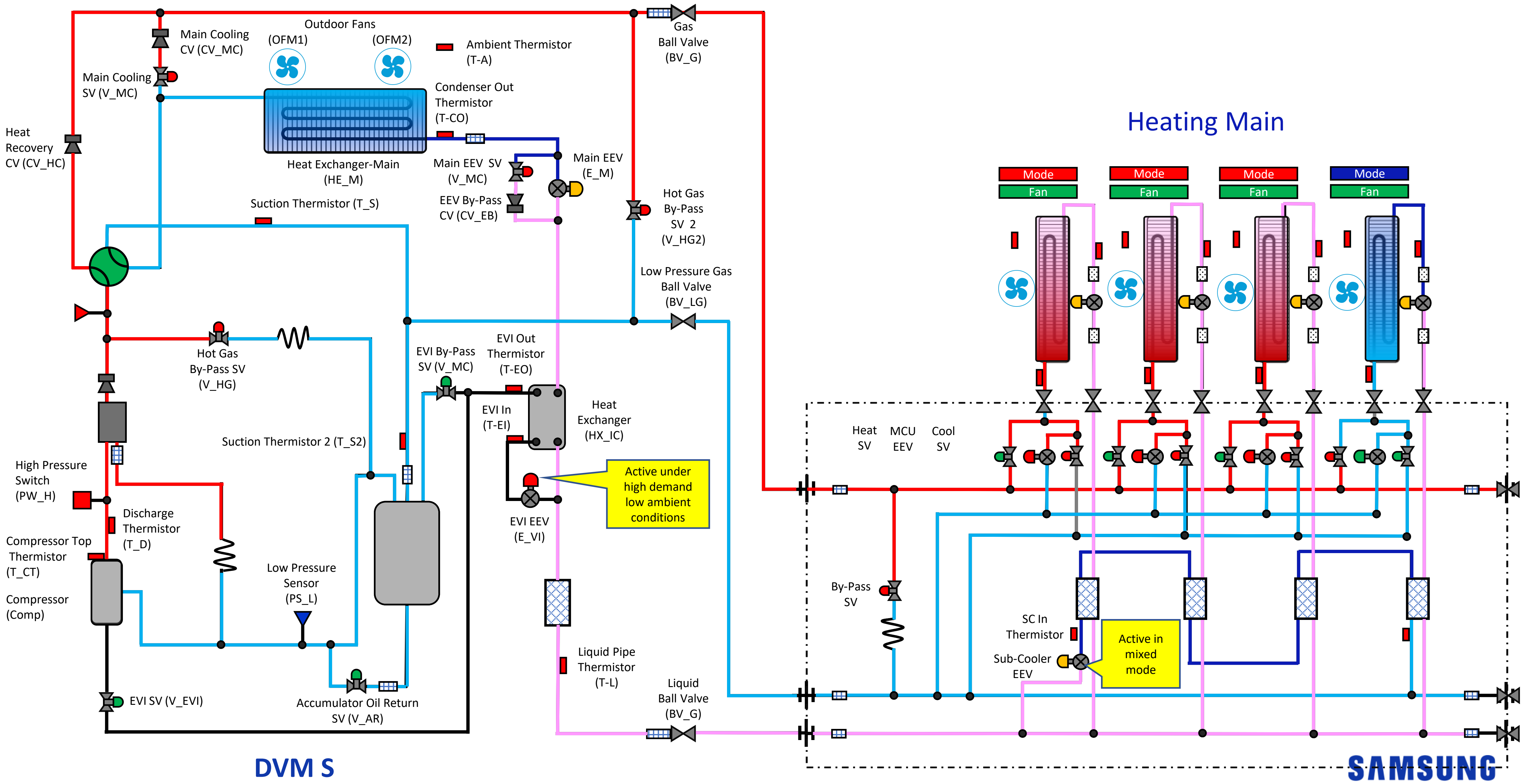


Solenoid Valve is energized
Main Cooling CV (CV_MC)
Outdoor Fans
(OFM1)
(OFM2)
Ambient Thermistor (T-A)
Gas Ball Valve (BV_G)
Main Cooling SV (V_MC)
Condenser Out Thermistor (T-CO)
Heat Recovery CV (CV_HC)
Heat Exchanger-Main (HE_M)
Main EEV SV (V_MC)
Main EEV (E_M)
EEV By-Pass CV (CV_EB)
Hot Gas By-Pass SV 2 (V_HG2)
Suction Thermistor (T_S)
Low Pressure Gas Ball Valve (BV_LG)
Hot Gas By-Pass SV (V_HG)
EVI By-Pass SV (V_MC)
EVI Out Thermistor (T-EO)
Heat Exchanger (HX_IC)
EVI In (T-EI)
Suction Thermistor 2 (T_S2)
High Pressure Switch (PW_H)
EVI EEV (E_VI)
Discharge Thermistor (T_D)
Compressor Top Thermistor (T_CT)
Low Pressure Sensor (PS_L)
Compressor (Comp)
Liquid Pipe Thermistor (T-L)
Liquid Ball Valve (BV_G)
EVI SV (V_EVI)
Accumulator Oil Return SV (V_AR)
DVM S
Cooling Mian
Mode
Fan
Mode
Fan
Mode
Fan
Mode
Fan
Heat SV
MCU EEV
Cool SV
By-Pass SV
SC In Thermistor
Sub-Cooler EEV
Active in mixed mode
SAMSUNG



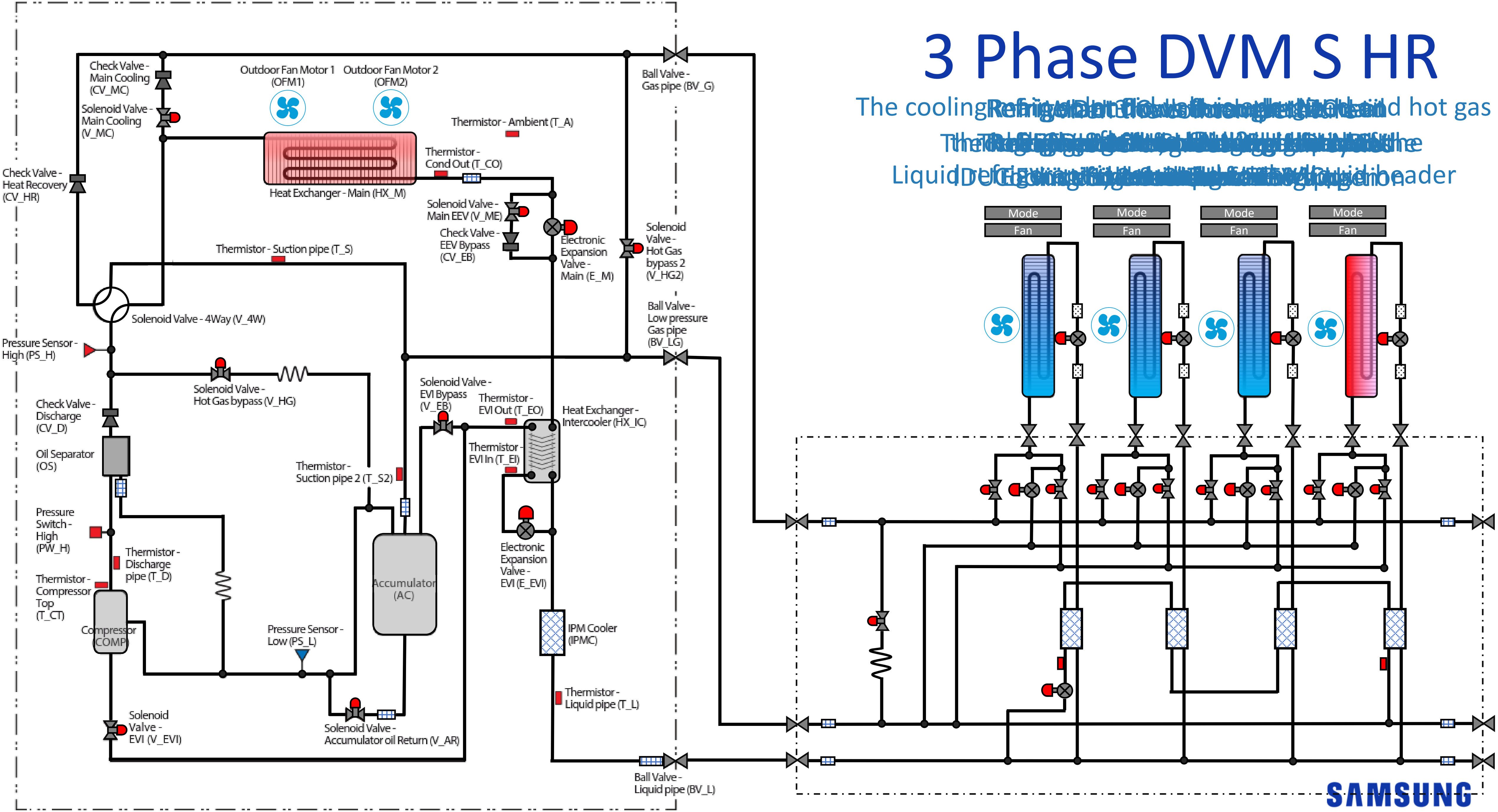
DVM S

SAMSUNG



Heating Main
Outdoor Fans
(OFM1)
(OFM2)
Ambient Thermistor
(T-A)
Main Cooling
CV (CV_MC)
Main Cooling
SV (V_MC)
Heat
Recovery
CV (CV_HC)
Condenser Out
Thermistor
(T-CO)
Heat Exchanger-Main
(HE_M)
Main EEV SV
(V_MC)
Main EEV
(E_M)
EEV By-Pass
CV (CV_EB)
Gas
Ball Valve
(BV_G)
Hot Gas
By-Pass
SV 2
(V_HG2)
Suction Thermistor (T_S)
Low Pressure Gas
Ball Valve
(BV_LG)
Hot Gas
By-Pass SV
(V_HG)
EVI By-Pass
SV (V_MC)
EVI Out
Thermistor
(T-EO)
EVI In
(T-EI)
Heat
Exchanger
(HX_IC)
Suction Thermistor 2 (T_S2)
High Pressure
Switch
(PW_H)
Active under
high demand
low ambient
conditions
EVI EEV
(E_VI)
Discharge
Thermistor
(T_D)
Compressor Top
Thermistor
(T_CT)
Compressor
(Comp)
Low Pressure
Sensor
(PS_L)
Liquid Pipe
Thermistor
(T-L)
Liquid
Ball Valve
(BV_G)
EVI SV (V_EVI)
Accumulator Oil Return
SV (V_AR)
Mode
Fan
Mode
Fan
Mode
Fan
Mode
Fan
Heat
SV
MCU
EEV
Cool
SV
By-Pass
SV
SC In
Thermistor
Sub-Cooler
EEV
Active in
mixed
mode
DVM S
SAMSUNG

3 Phase DVM S HR



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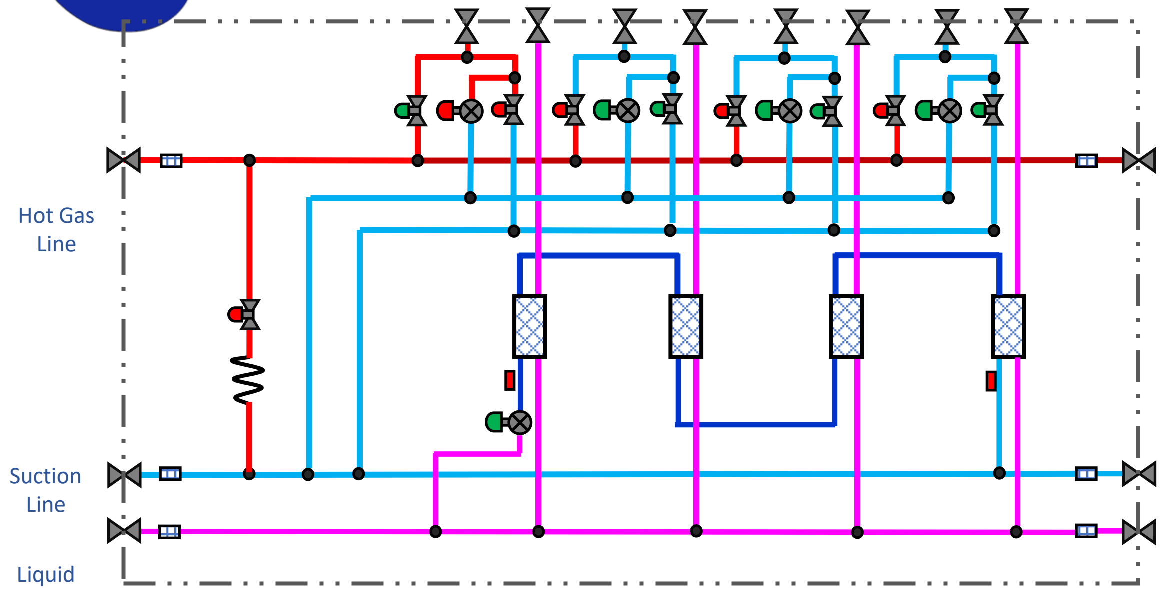
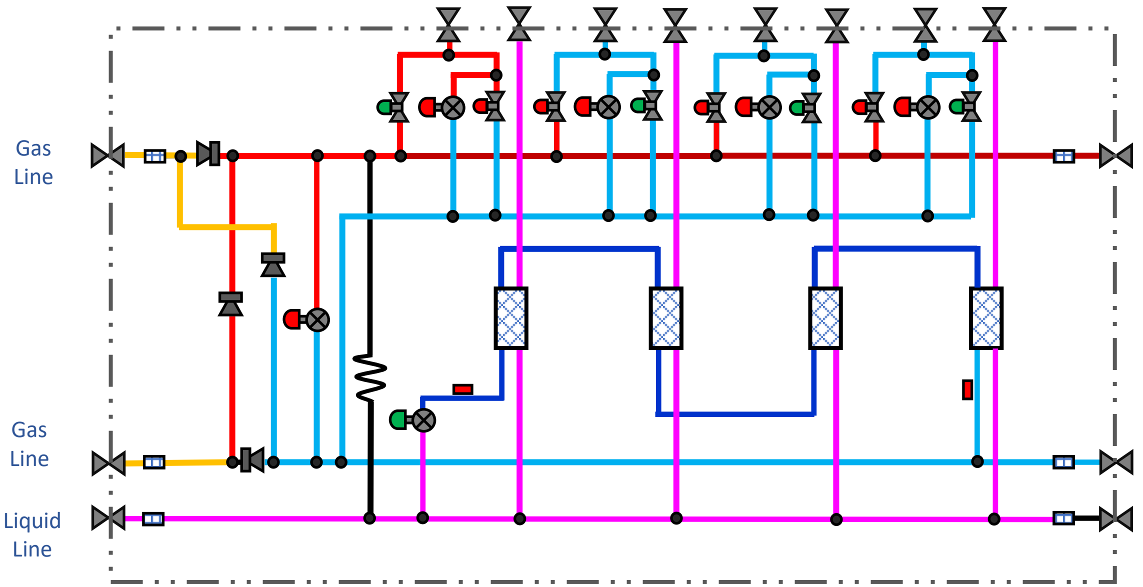
MCU vs HR Changer



MCU



HR Changer

No	Cycle parts	Functions
1	Cooling solenoid valve	This valve opens during Indoor unit's cooling operation
2	Heating solenoid valve	This valve opens during Indoor unit's heating operation
3	MCU EEV	To reduce modulation noise while mode is changed. (Heating → Cooling)
4	Sub-cooler	To secure the sub-cooling of refrigerant for each indoor.
5	Sub-cooler EEV	To control super heat of sub-cooler
6	MCU liquid bypass valve	To prevent refrigerant accumulation in high pressure gas pipe

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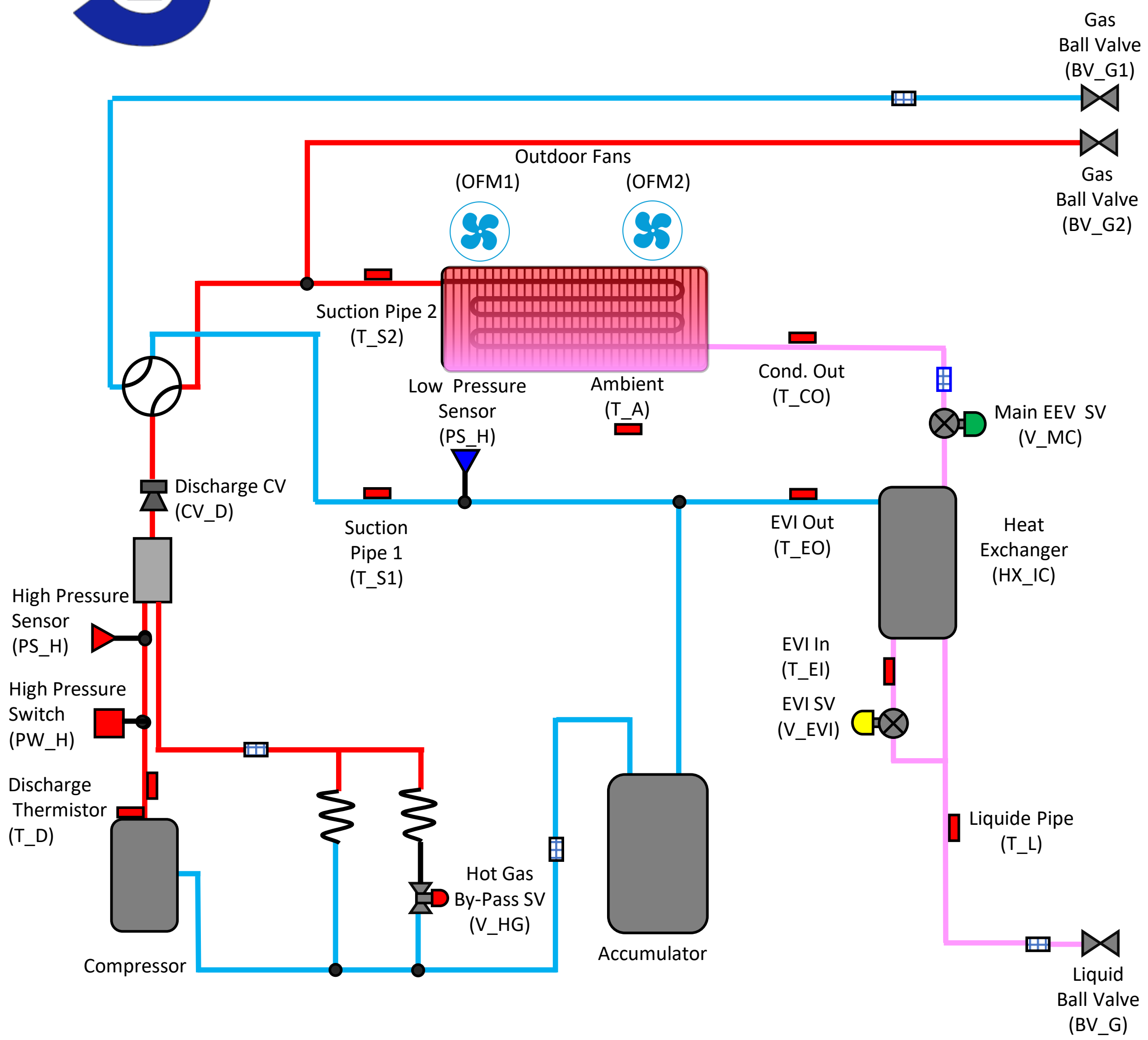
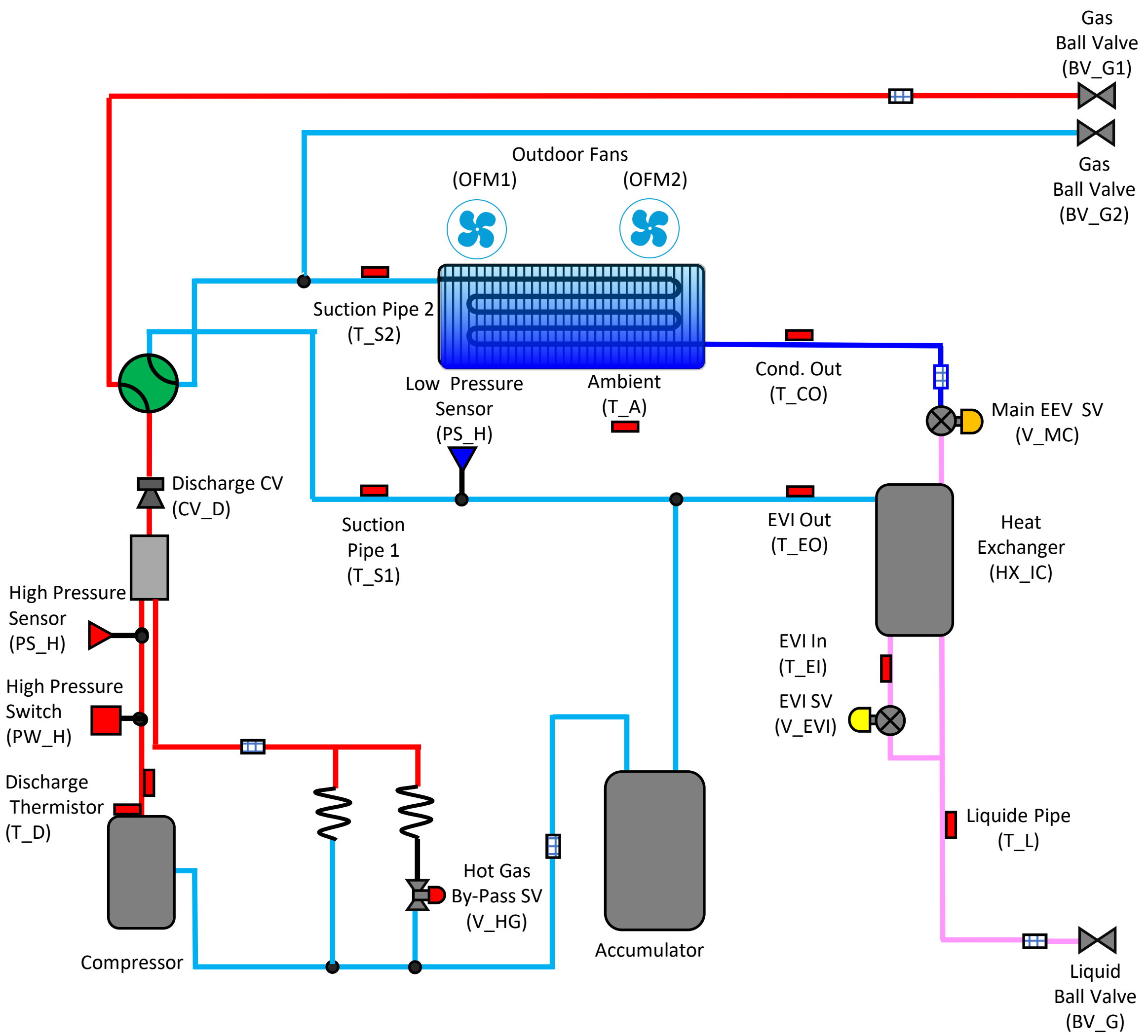
Why the need for an HR Changer

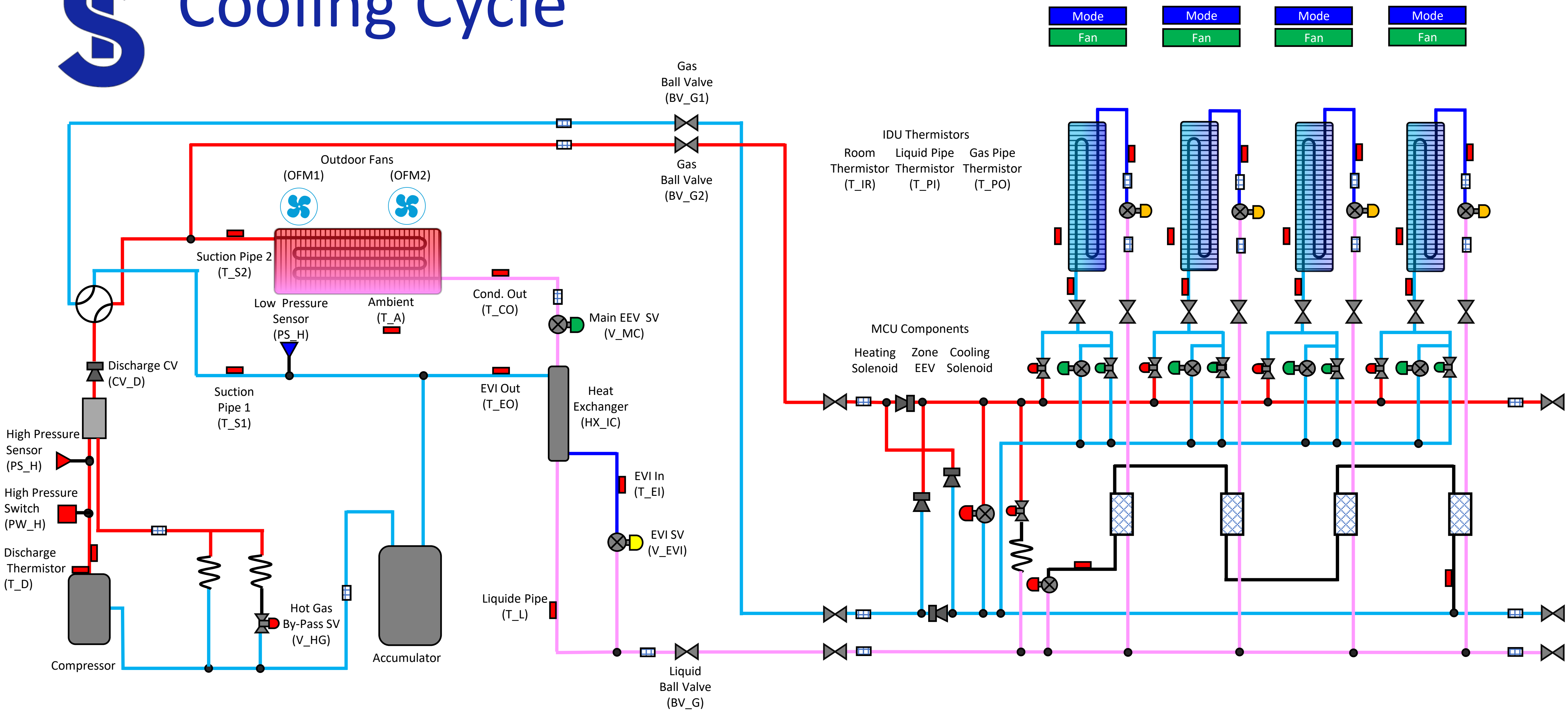
ECO HR

AM036TXMDCH/AA, AM048TXMDCH/AA, AM053TXMDCH/AA

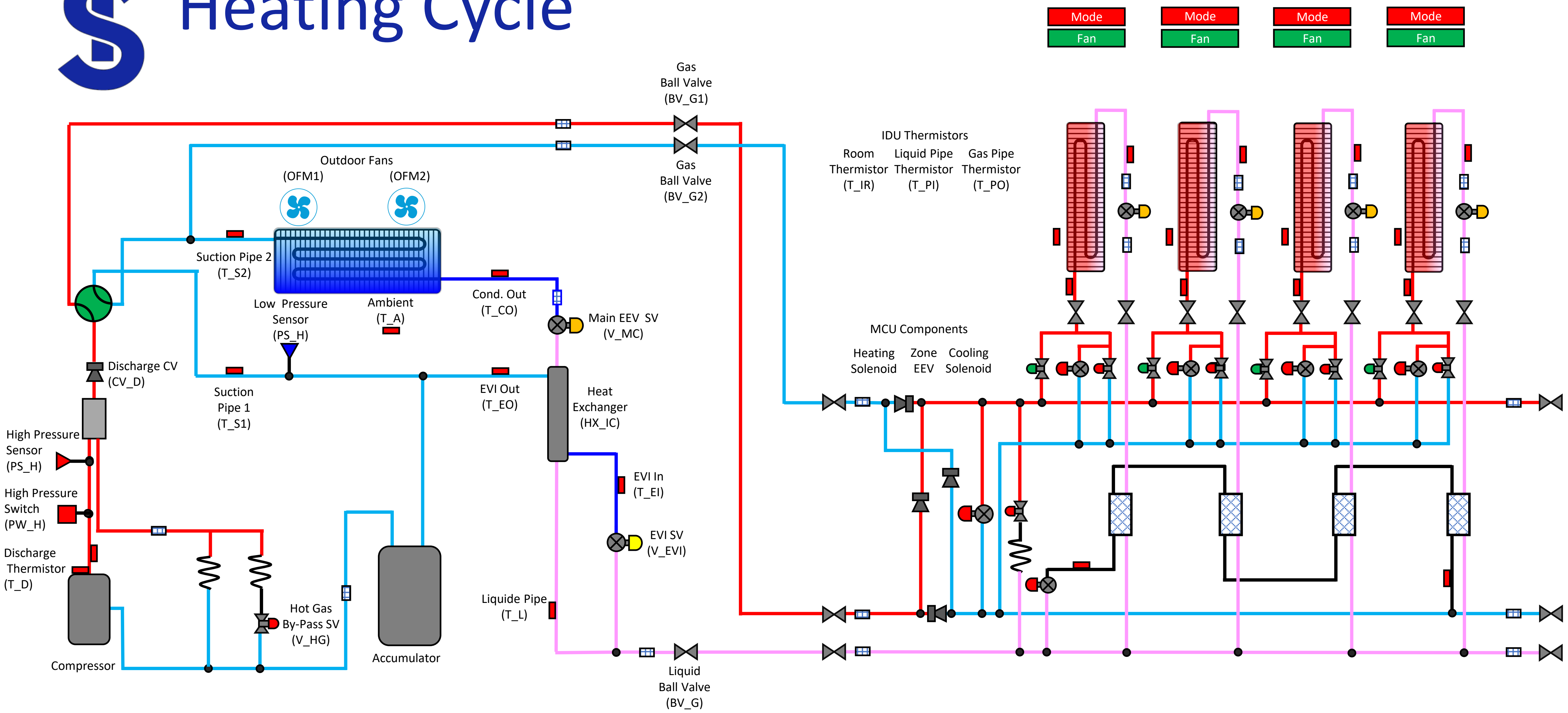




Cooling Cycle



Heating Cycle



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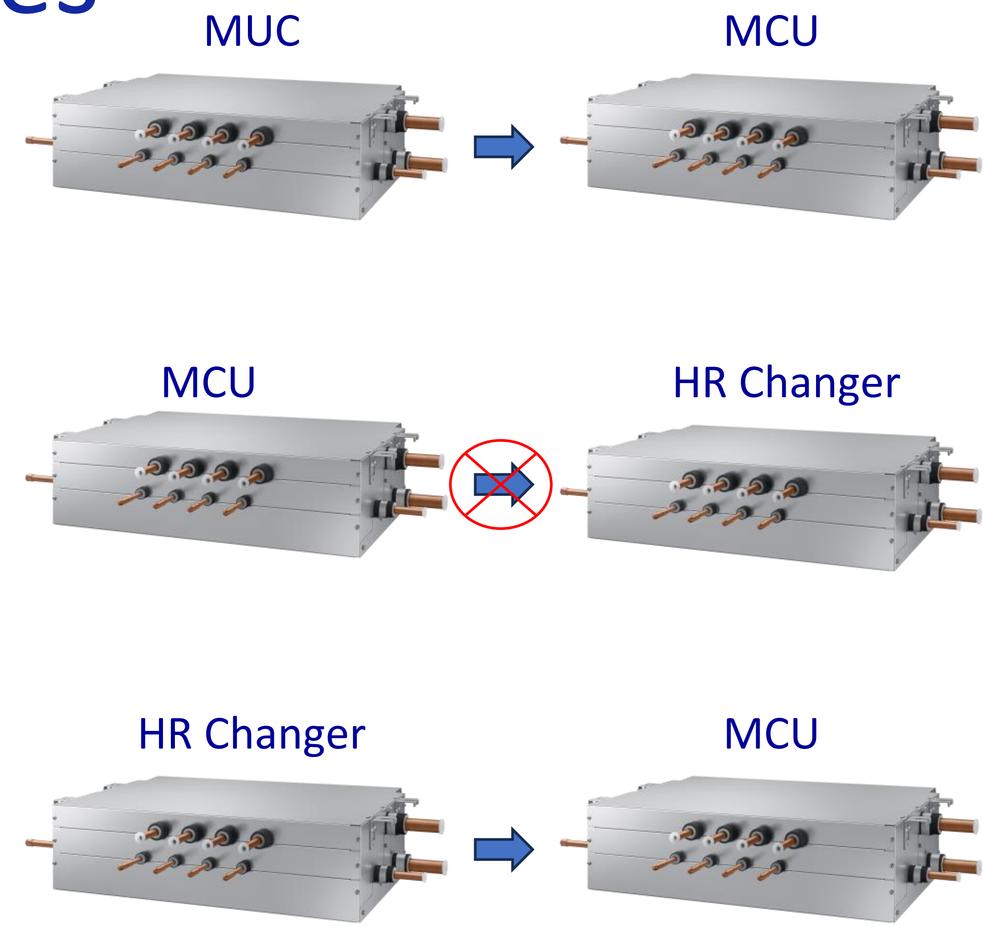
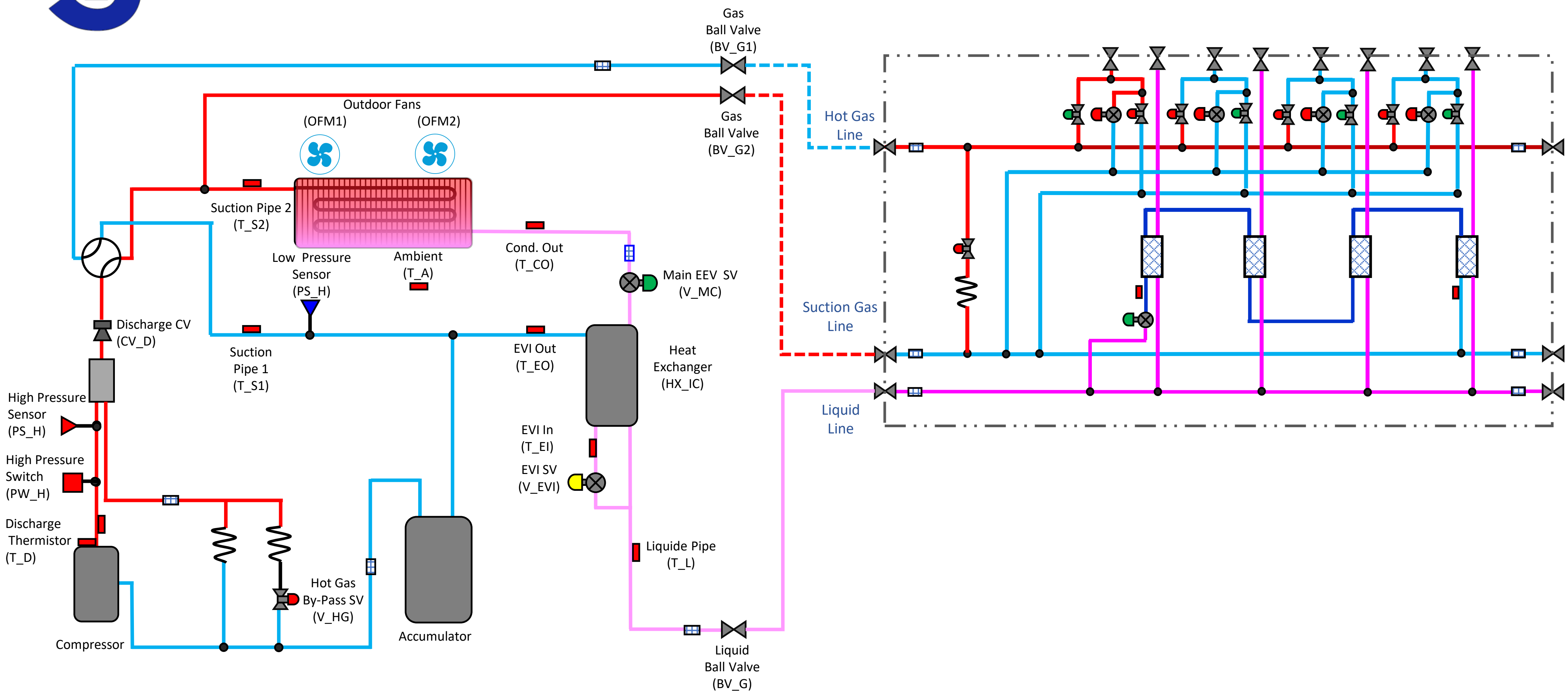
MCU and HR Changer Rules

- The MCU and HR Changer look a like, but they are very different
- The HR Changer is used exclusively for the Eco single phase system
- A MCU can be connected to a HR changer



MCU Connected to a Eco HR ODU



Section End

- This concludes the section on the function of the system components.
- In the neck section we will take a deeper dive into the system components and discuss their specific function and operating parameters.

Are the values you are seeing good or bad?

- This is a loaded question the answer depends on what you are working on.
- If an evaporator has 1 degree of superheat is that bad?
- How do you verify charge on a system?
- What do meters or gauges tell you?
- If you don’t know what normal is how can you determine if the system is operating correctly?

What to look for.

- Here is the caveat
  - The system are dynamic and constantly adjusting
  - We can not precisely identify these targets.
  - There for when troubleshooting we look for measurements to be with in a range, Not a specific value
- Sometimes it feels like you're hitting a moving target.
- We can use the controls sequence of operation to get an operating range for the heating and cooling cycles.

Fuzzy Logic
Automatic braking in Cruise Control

Distance Ft.

100 30

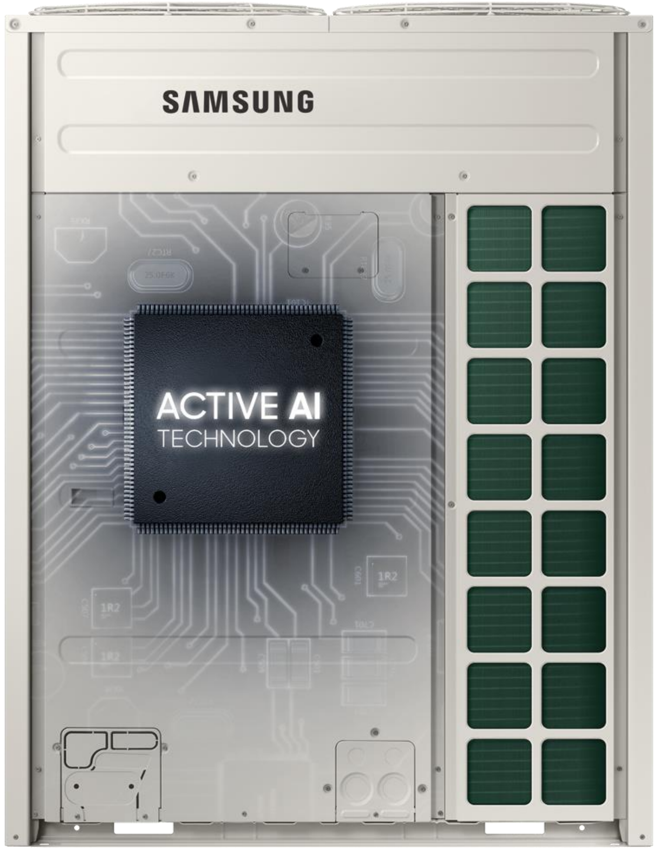
Brake Pressure

0 100

Breaking force is not directly proportional to distance.
50% reduction in distance does not relate to 50 brake pressure

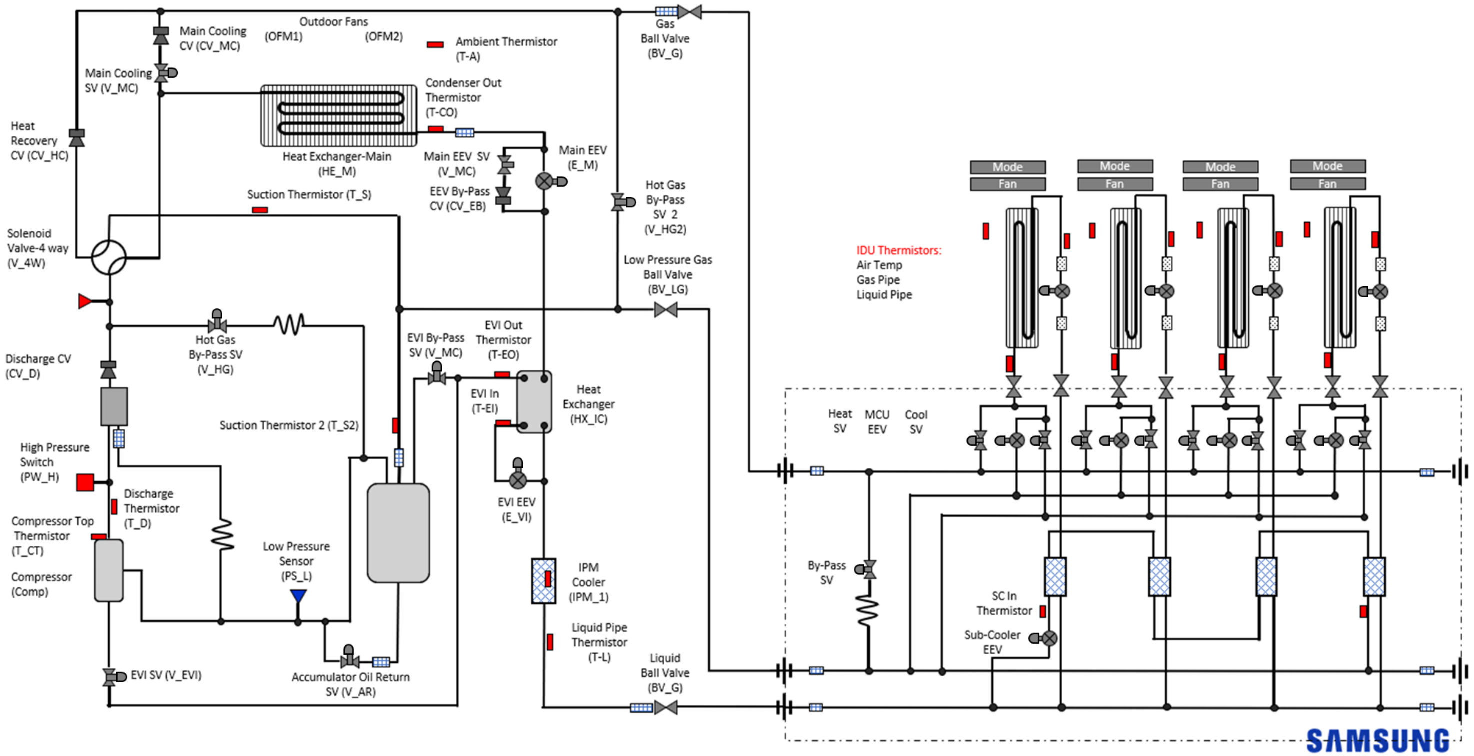
Outdoor Targets

- System operation is based on maintaining system targets.
- The system responds to changes within the system to maintain Individual component targets.
- The sequence of operation is for each generation is basically the same.
- The DVM S2 has some slight difference when it comes to the refrigeration circuit.



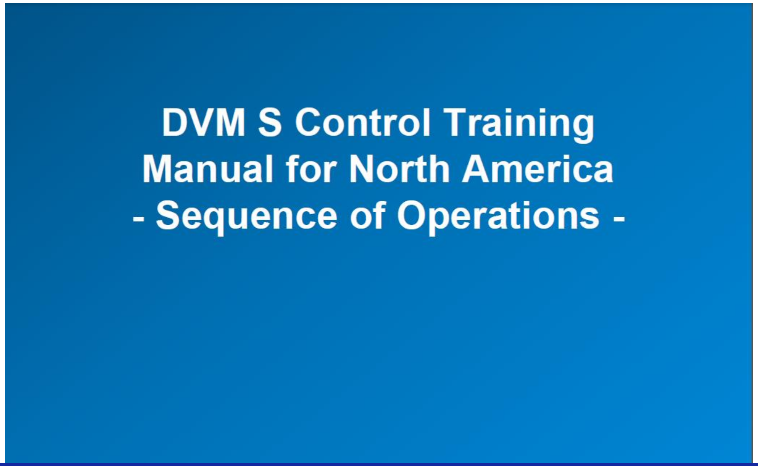
Component Operation

- This section will review the operating parameter for each component in the DVM S refrigeration system.
- It's important to know when and why each component Is activated.



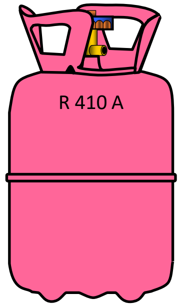
Target Operation

- The system operates to maintain target settings based on the equipment and mode of operation.
- Knowing the target operation is the first step in determining if the system is operating correctly.

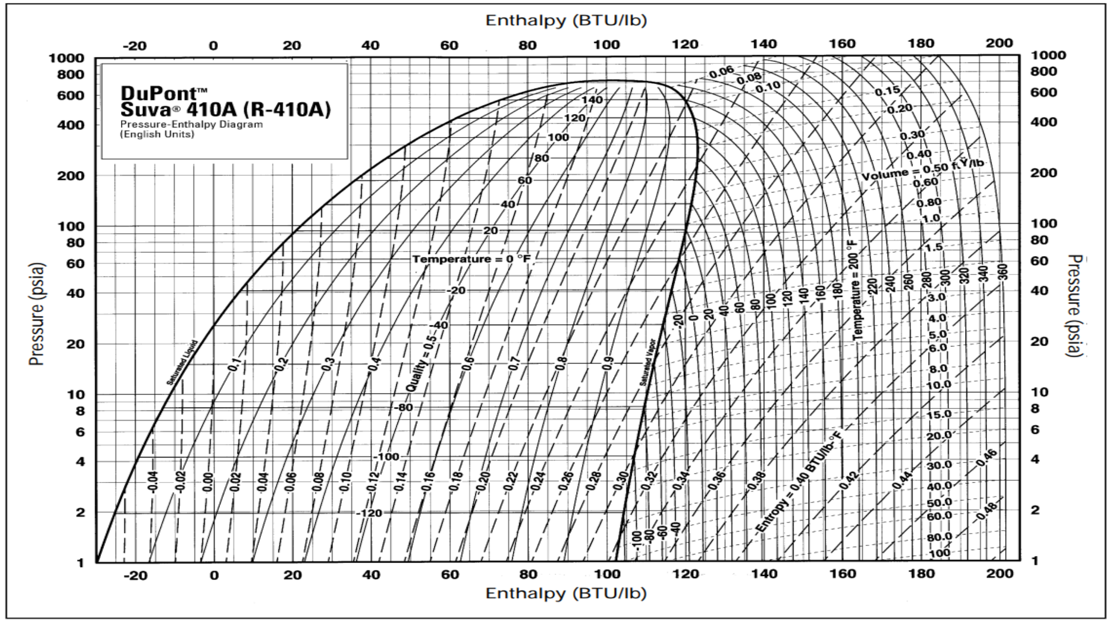


Where do the targets come from?

- System operation is predictable!
- The systems are tested at various conditions.
- The operating characteristics are determined during the testing process.
- The system uses this information to determine how the system should operate under a given set of conditions.

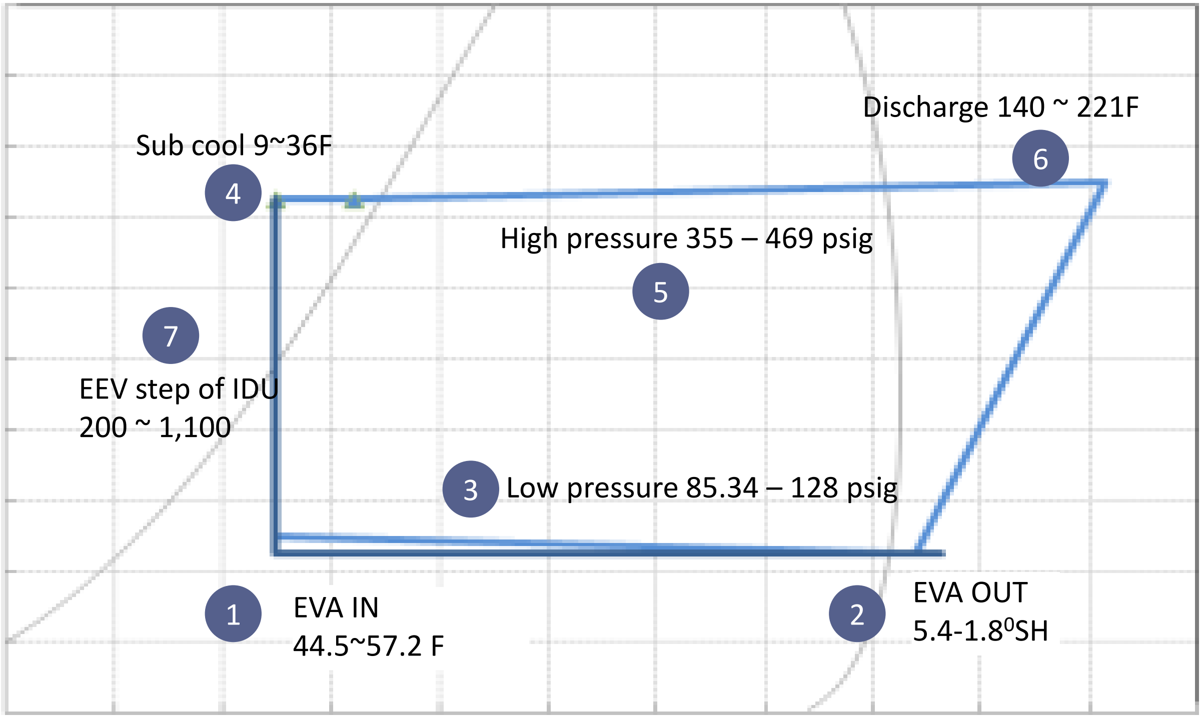


A given amount of refrigerant



At given conditions

Reference Cooling Cycle Data

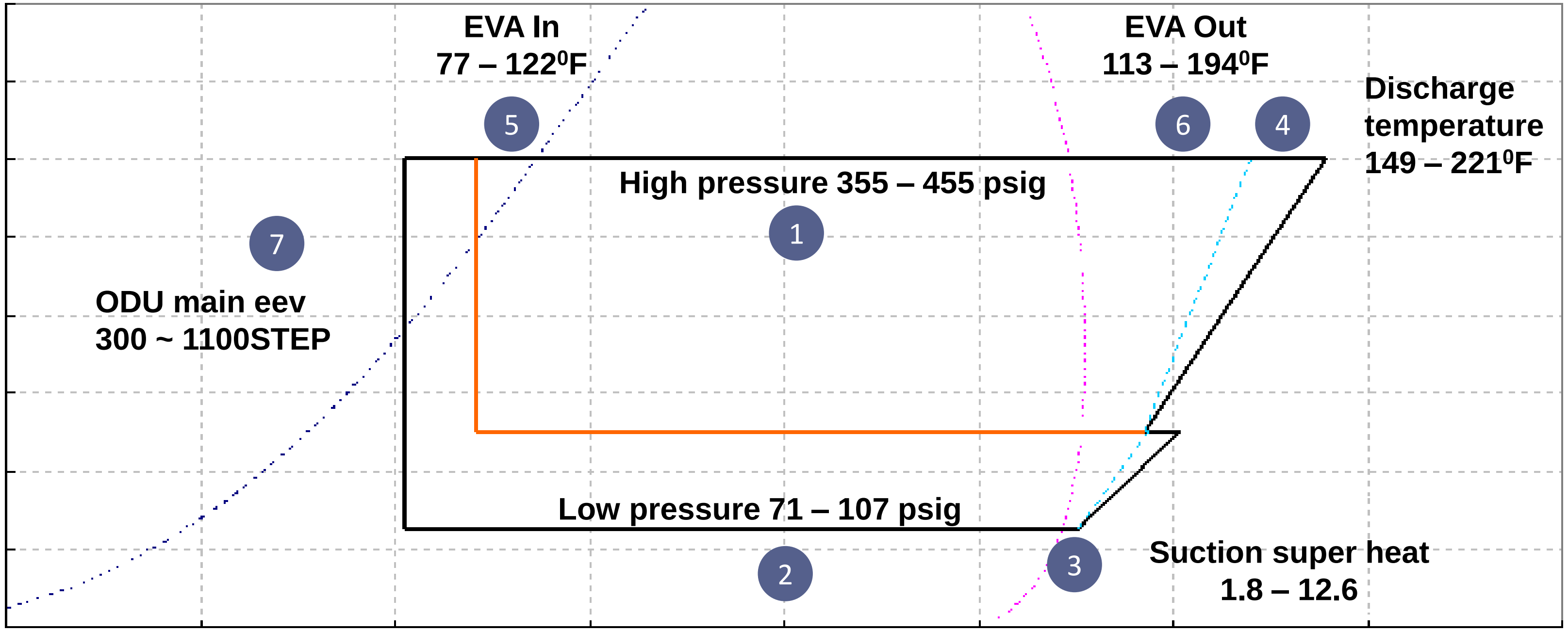


The figures presented in the graph are not absolute standards to judge the system.

If the surround conditions are out of the typical condition, the figures may be changed.

(outdoor temperature) 77^{0}F ~ 104^{0}F, room temperature 71^{0}F ~ 82^{0}F

Reference Heating Cycle Data



Figures presented in the graph is not absolute standards to judge abnormal of valves, sensors and refrigerant.

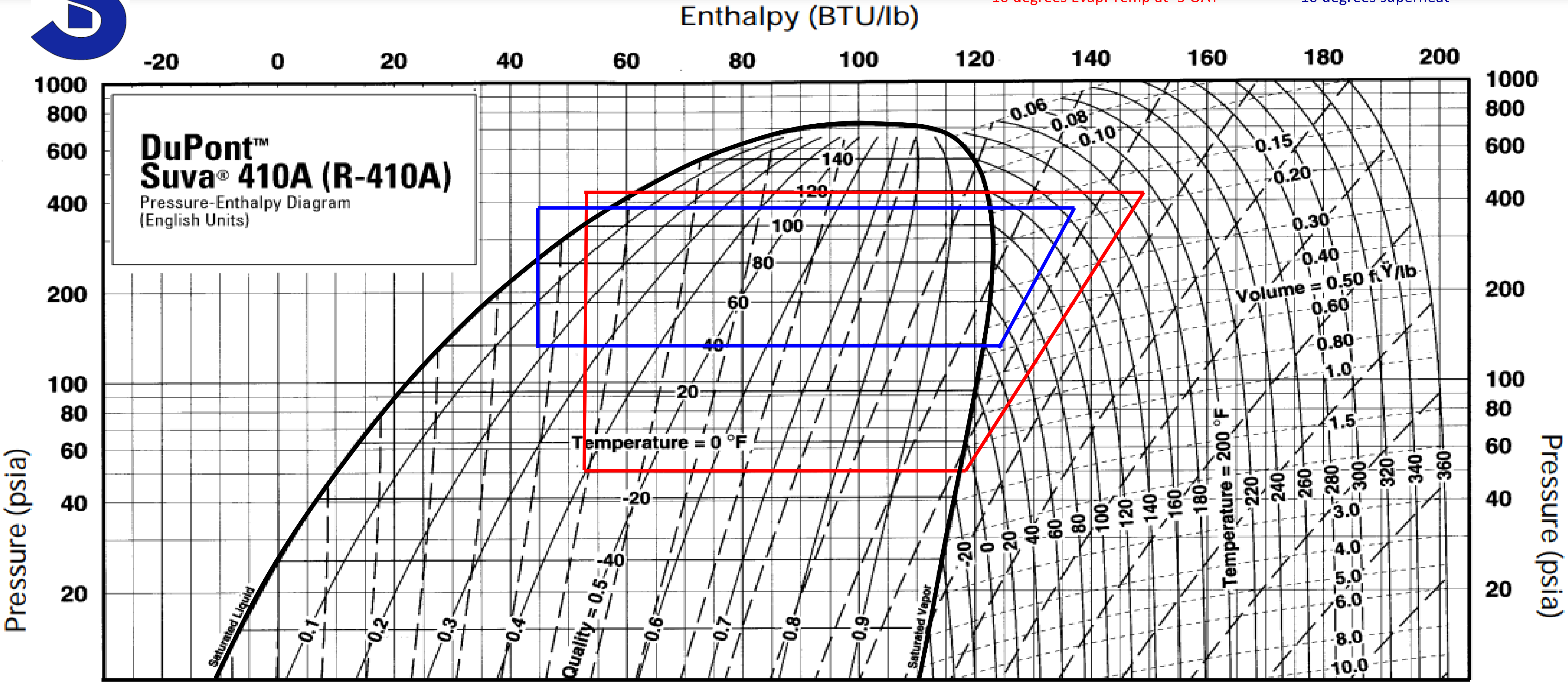
If the ambient temperature is outside of typical healing conditions (outdoor temperature 23 – 50 F, room temperature 64 – 77 F) the values can be changed.

Cooling Cycle vs Heat Cycle

Which Cycle uses more energy?

Heating Conditions:
- 120 High Side Saturation Temp
- 10 degrees sub-cooling
- 5 degrees superheat
- -10 degrees Evap. Temp at -5 OAT

Cooling Conditions:
- 40 degree Evap.
- 110 High Side Saturation Temp
- 30 degrees sub-cooling
- 10 degrees superheat



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DVM S ODU Target Pressures

How does the unit operate in each mode?

118 psig

356+ psig

? psig

427 psig

Based on OAT

AI Control

- Purpose
  - - The DVM S2 optimizes its cooling performance automatically, based on a learning and optimizing algorithm about the installation conditions and usage patterns.
- Concept
  - - Active AI Pressure Control intelligently adjusts the refrigerant condensing pressure and
  - Evaporating pressure, so it cools faster and reduces energy usage

DVM S2 AI Control

- The target low pressure in the cooling mode is determined by the Active AI control.
  - Next starting target low pressure : Determined by the Active AI
  - - After considering the average value of the eva_in temp., it varies
  - - Control interval : 15 min.
  - - Target pressure range :
  - → Heat pump : 85.3 ~ 128 psig
  - → Heat recovery : 49.8 ~ 128 psig

Target low pressure (psig)		
- 7.1 psig	Active AI	**+ 7.1 psig**

- The target high pressure in *cooling only* operation is varied by the Active AI control
  - Within the range of 241.8 psig (17 kg/cm2) - 355.6 psig (25 kg/cm2) .
  - (Main cooling target head pressure: 398.3 psig (28 kgf/cm2))

AI Control

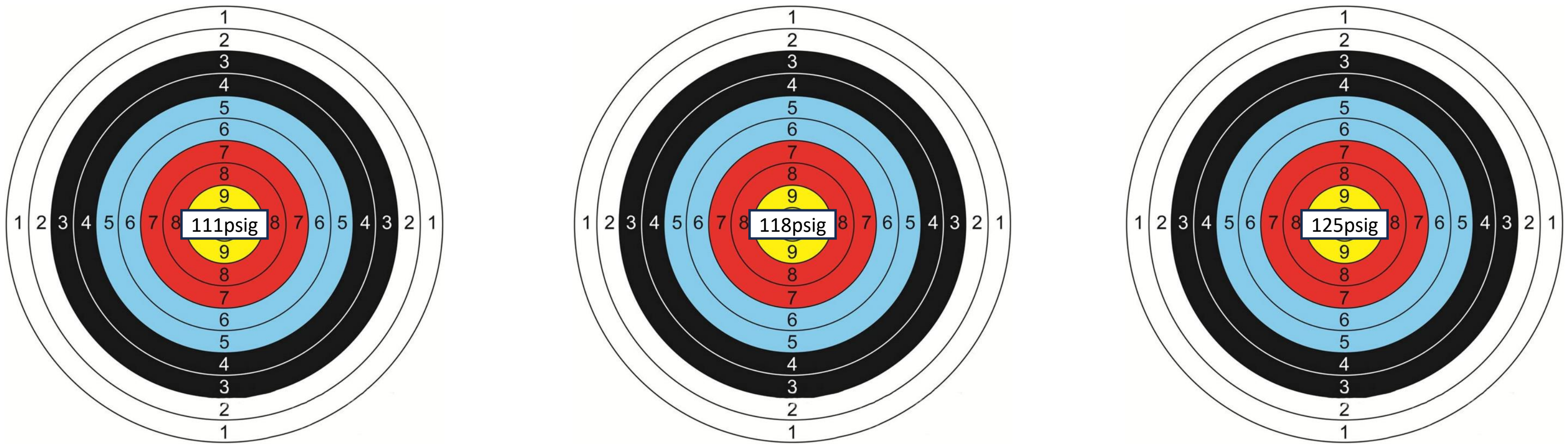
- By learning usage patterns from recent cooling operations and the surrounding conditions, DVM S2 proactively creates the optimal cooling environment to suit users' general requirements.
- For example:
  - (1) If a user frequently lowers the room temperature when turning on the air conditioner, the Active AI Pressure Control recognizes this pattern. So, when the air conditioner is turned on again, it automatically lowers the pressure of the inflow refrigerant by up to 33% and cools up to 20% faster*.
  - (2) However, if there's no need for fast cooling, it saves energy by adjusting the refrigerant pressure to be higher than normal AI

Next target low pressure control

- The target low pressure is determined by the Active AI control for next operation.

  Active AI calculates the target low pressure considering the previous environment and operation data.

- The target low pressure varies depending on the average Eva_in temp.



Initial Setting

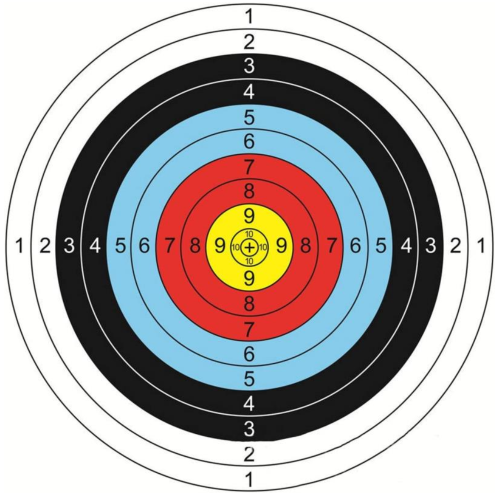
- Control interval : 15 min.
- Target pressure range

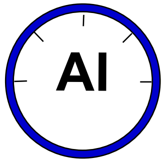
  . Heat pump : 6 ~ 9 kg/cm^2

  . Heat recovery : 3.5 ~ 9 kg/cm^2

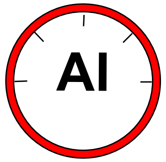
Basic ODU Target Operation DVM S 2

How does the unit operate in each mode?

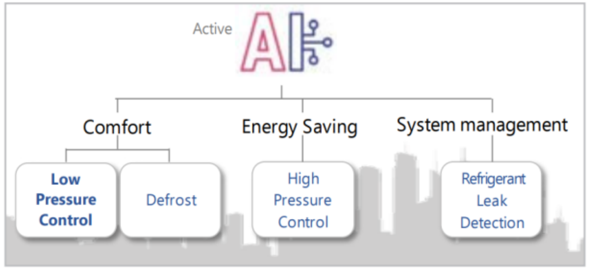




114 psig +/-
Active AI Control



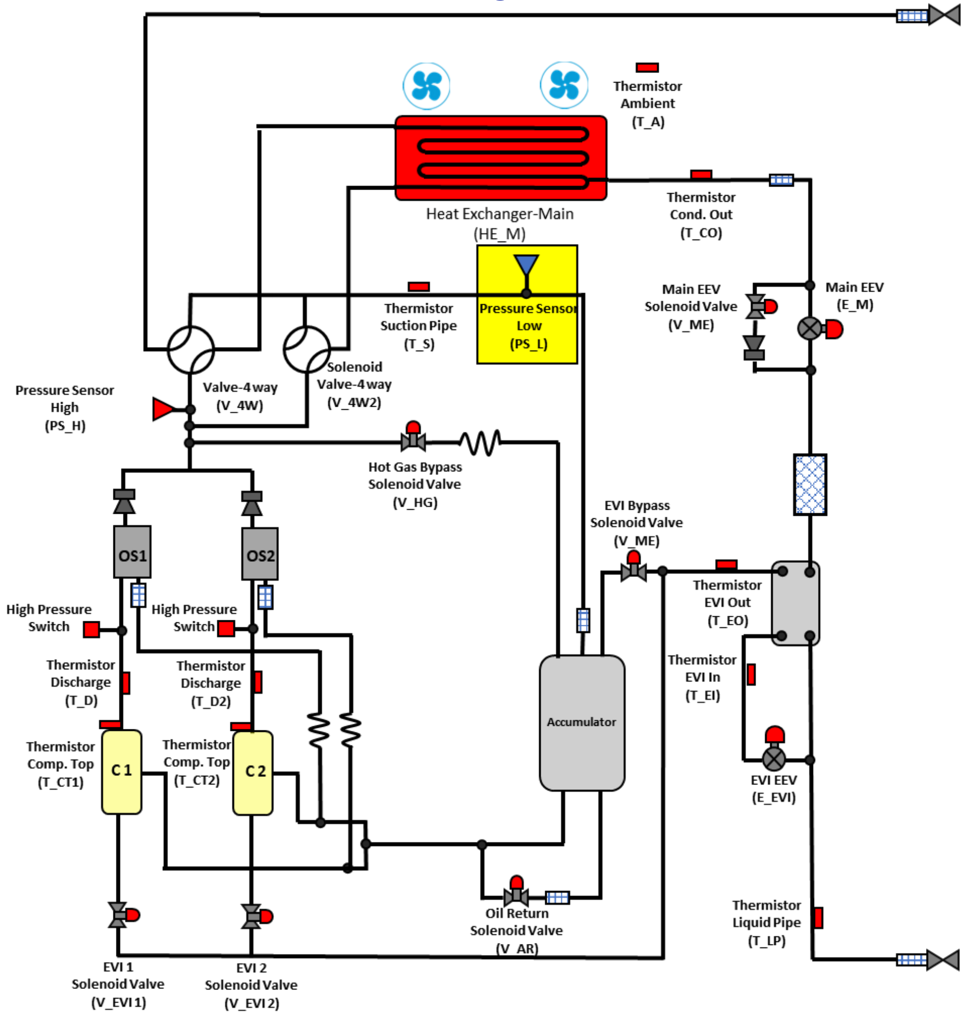
241.8 - 355.6 psig
Active AI Control



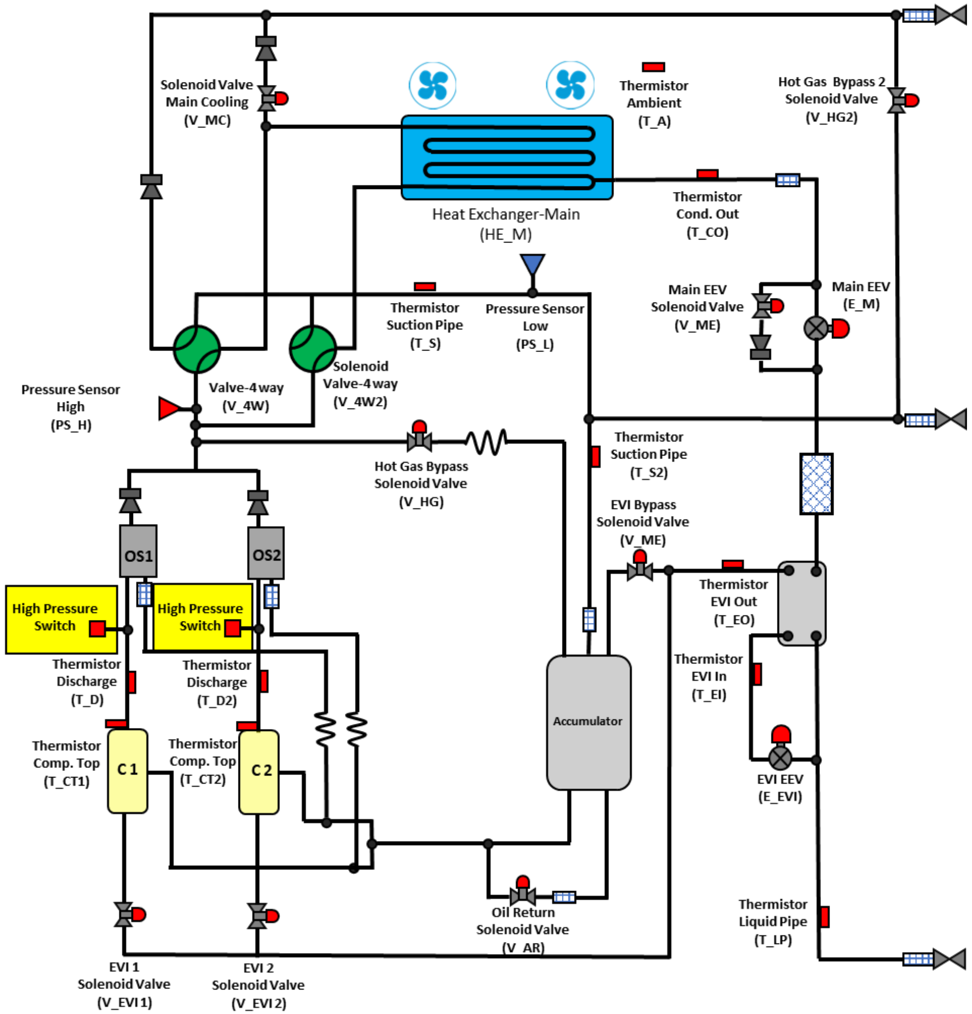
106.7 – 123.7

Compressor Operation

Cooling Mode

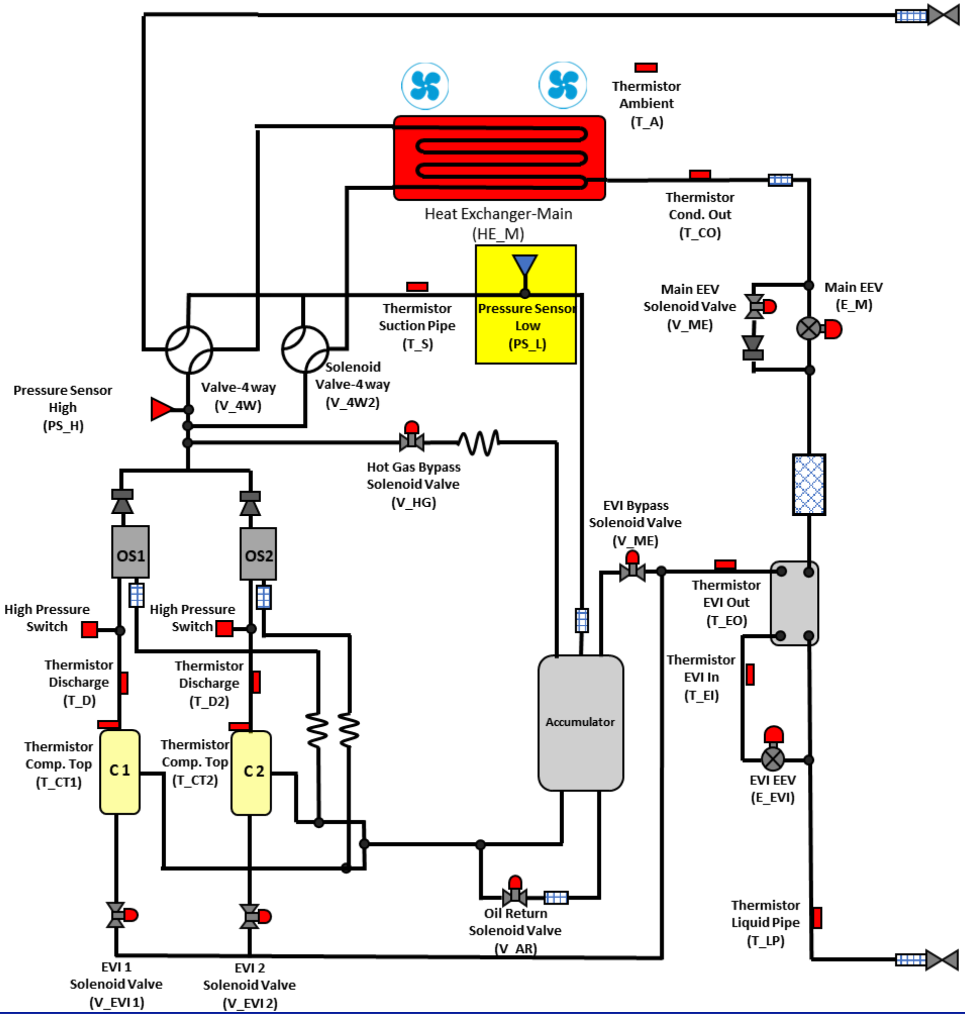


Heating Mode



Compressor Operation

Cooling Mode



✓ **Target low pressure control**

- avg. Evap-In temp. = setting → target low pressure maintain
- avg. Evap-In temp. < setting → target low pressure increase
- avg. Evap-In temp. > setting → target low pressure decrease

Target low pressure (psig)						
85	92	100	107	114	121	128

- Default target : 114 psig
- Setting for target avg. Evap-In = ODU option setting

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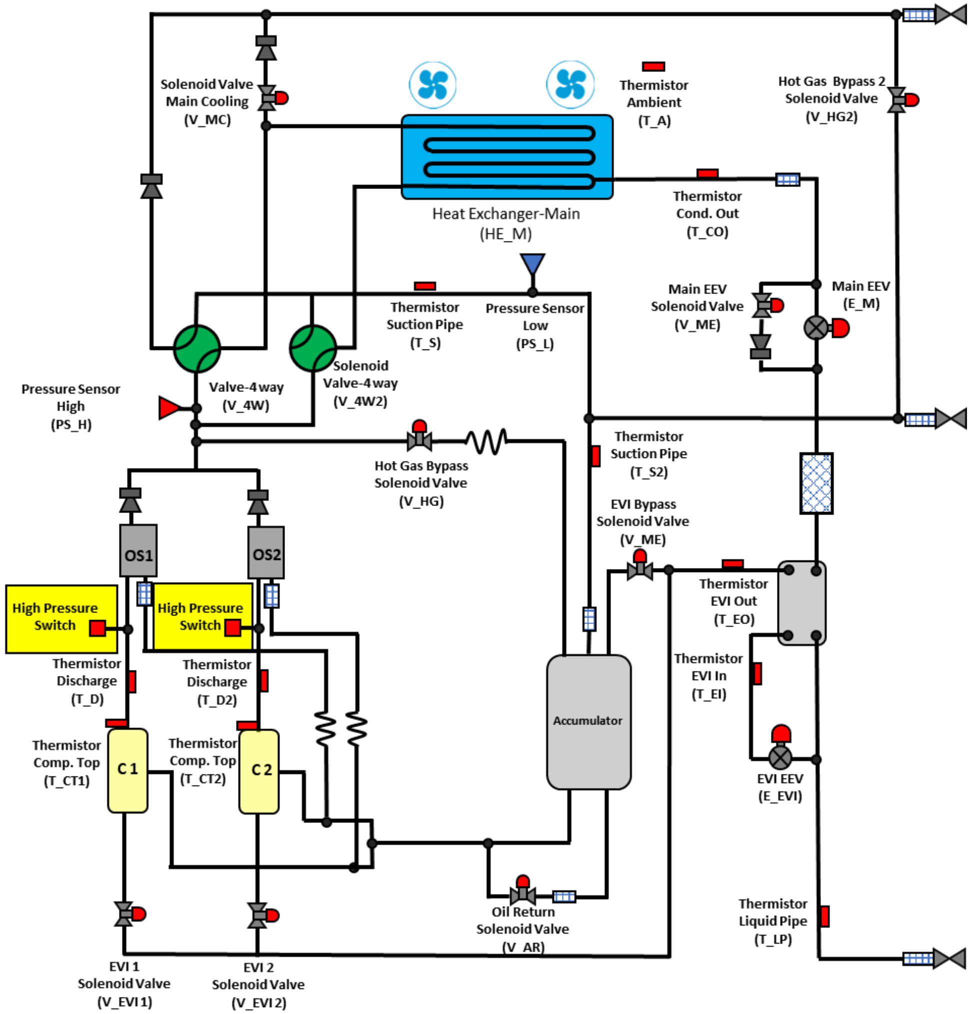
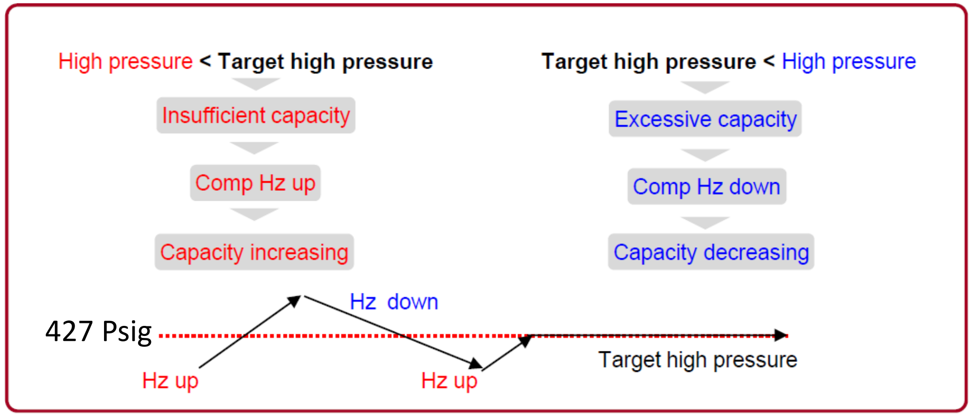
Compressor Operation

Heating Mode





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Compressor Operation

Cooling Mode

✓ **Target low pressure control**

- avg. Evap-In temp. = setting → target low pressure maintain
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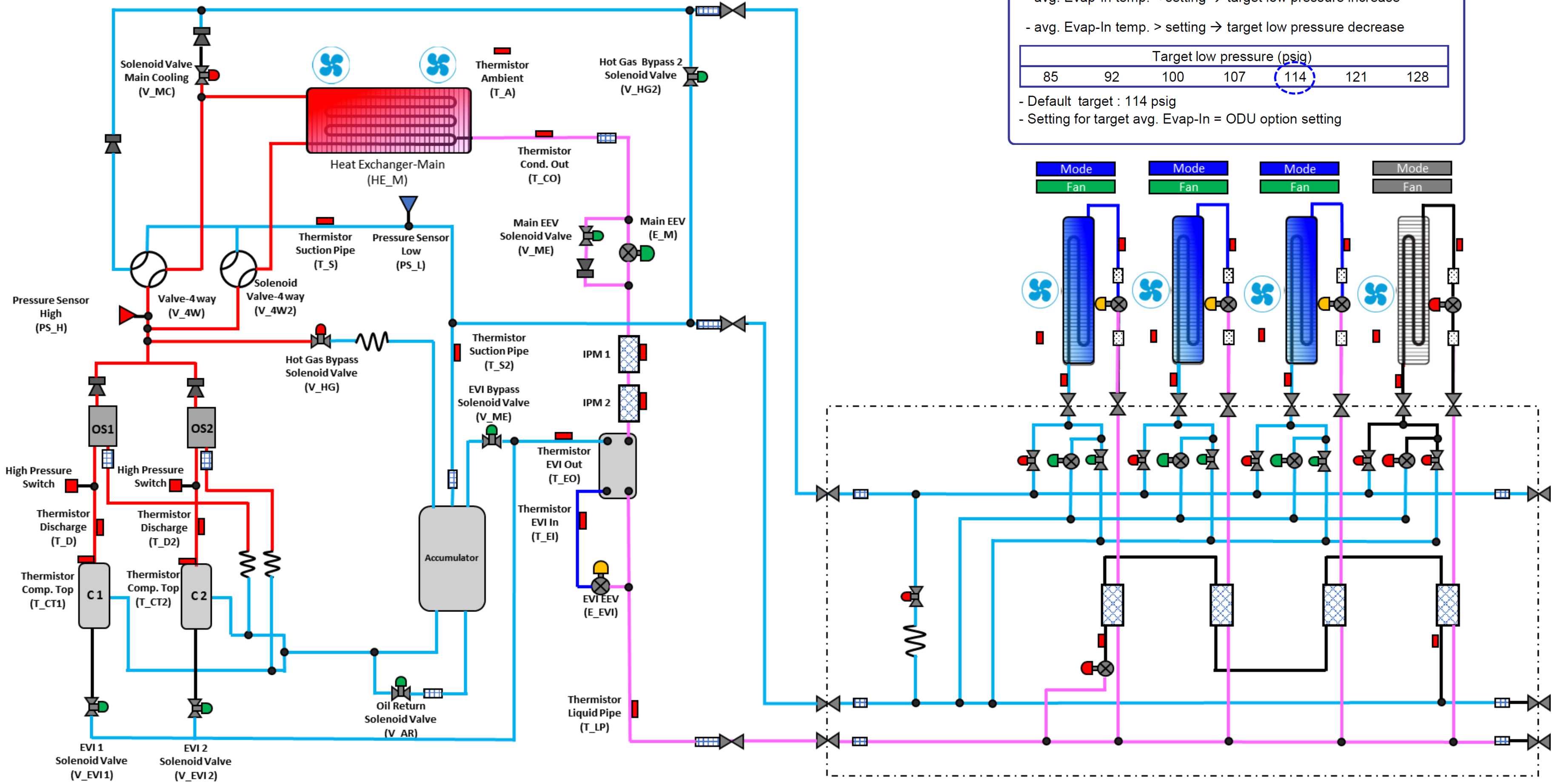

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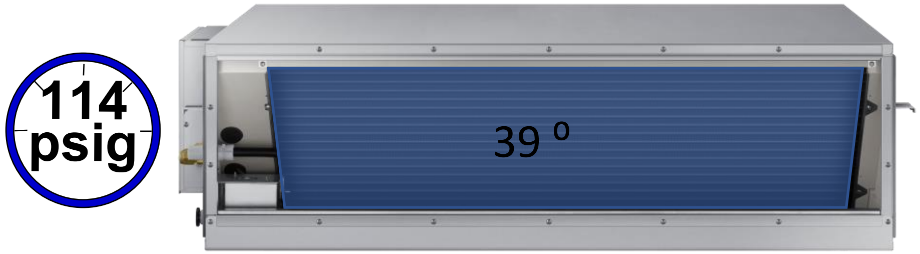
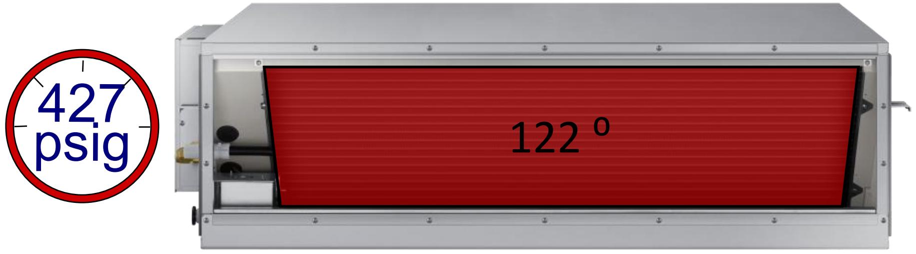
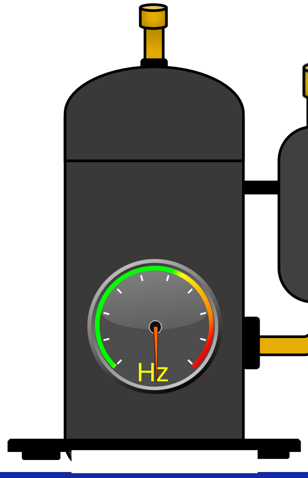
VRF Compressor Function.





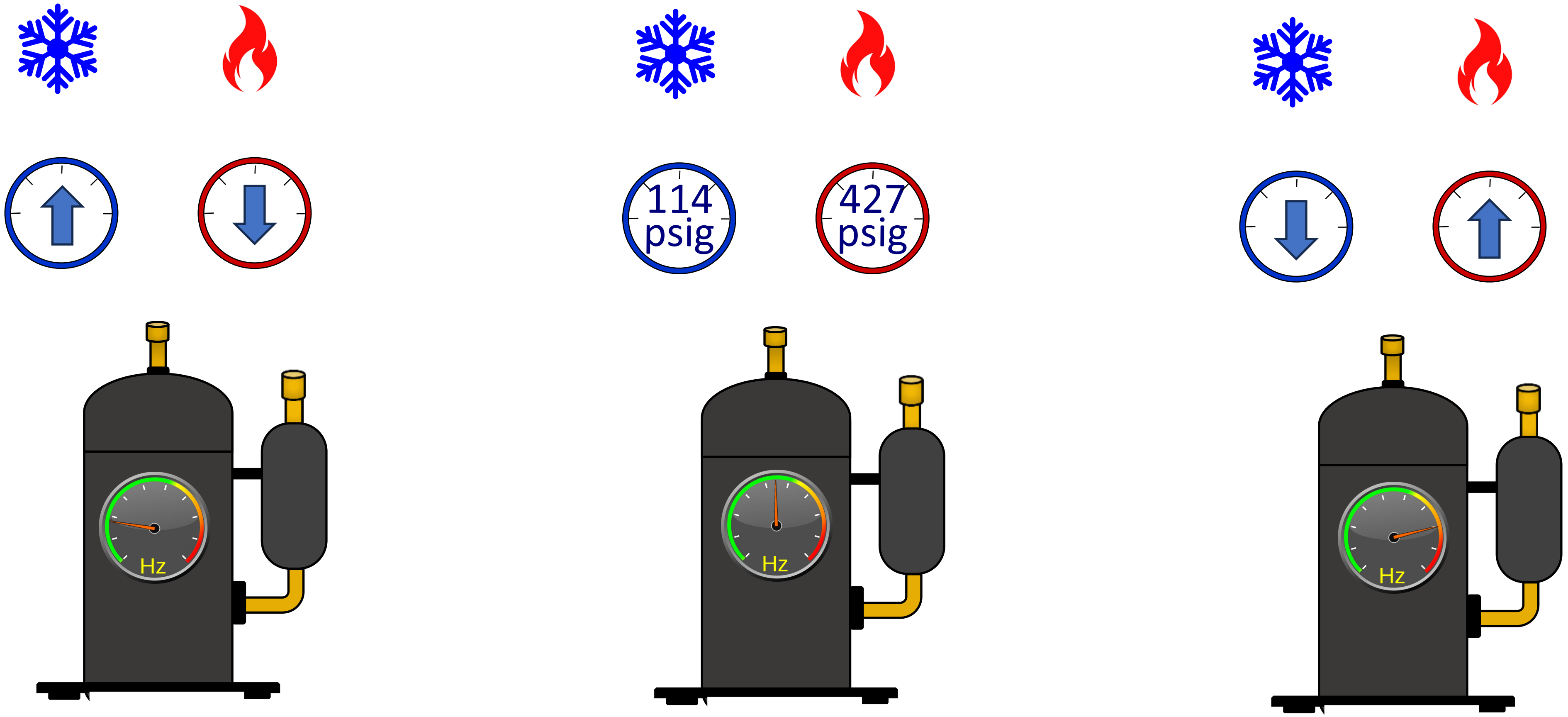


The compressors main function in the Cooling mode is to respond to changes in the system and maintain target suction pressure.

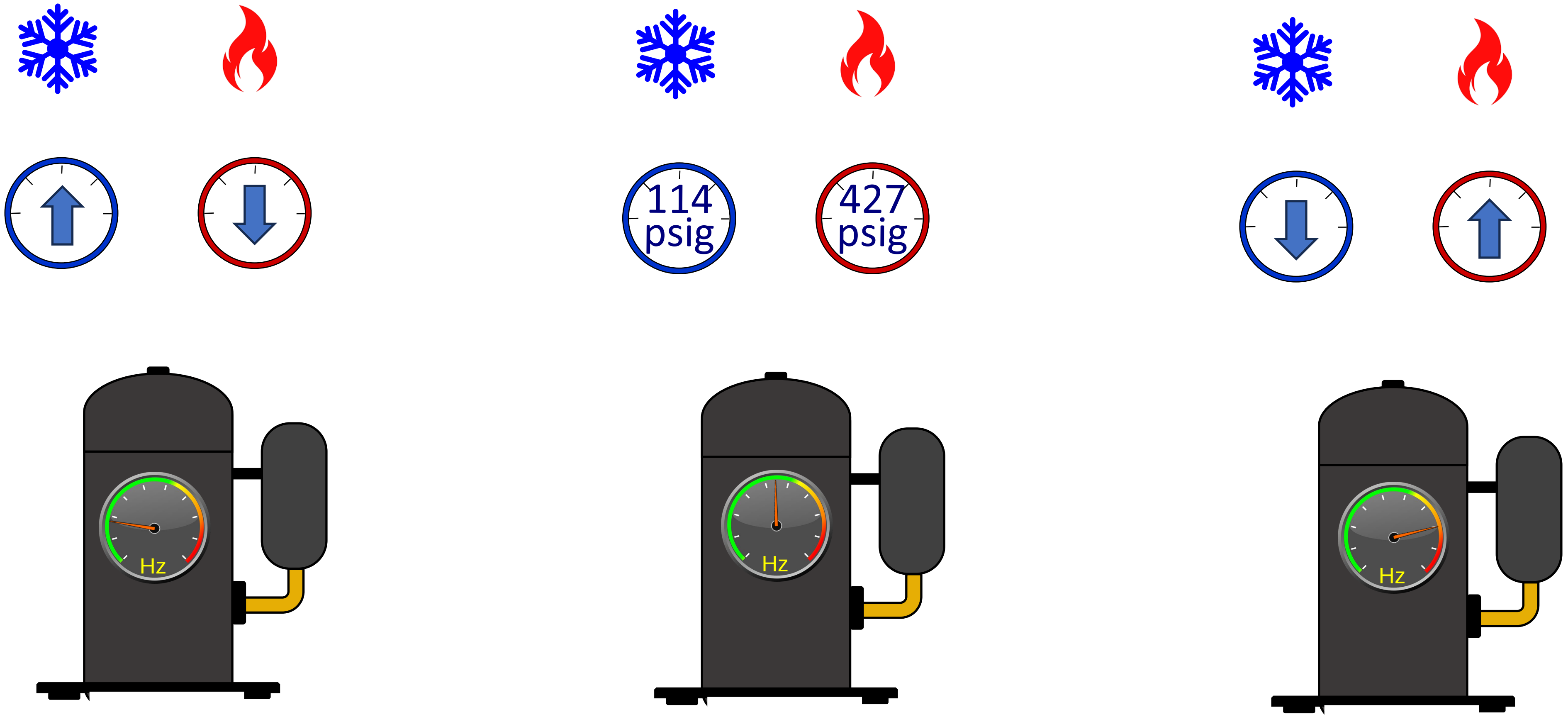
The system references average indoor coil saturation temperature

The compressors main function in the Heating mode is to respond to changes in the system and maintain target head pressure

The VRF
Speed Vs Pressure Relationship



The VRF
Speed Vs Pressure Relationship



The VRF Compressor

- The compressors main function in the Cooling mode is to respond to changes in the system and maintain target suction pressure.
- The compressors main function in the Heating mode is to respond to changes in the system and maintain target head pressure

Compressor Regulation

IN HEATING MODE, THE COMPRESSOR RUNS TO MAINTAIN A HEAD PRESSURE OF APPROXIMATELY 427 PSIG.

IN COOLING MODE, THE COMPRESSOR MODULATES TO MAINTAIN SUCTION PRESSURE OF APPROXIMATELY 114 PSIG

DVM S2 AI CONTROL ACTIVE IN COOLING ONLY MODE

THE ARE SEVERAL SAFETY CONDITIONS THAT WILL OVERRIDE COMPRESSOR OPERATION

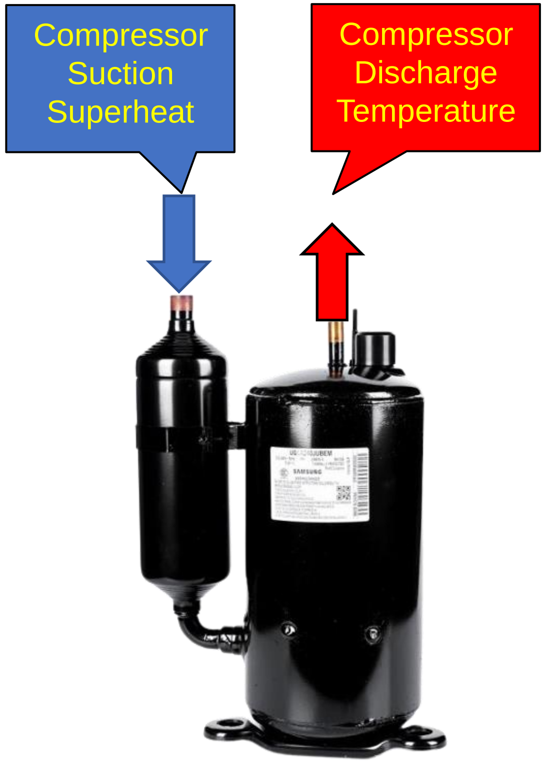
FLOOD BACK CONDITION

HIGH DISCHARGE TEMPERATURE

HIGH IMP TEMPERATURE

Discharge Temperature

- The compressor operation is controlled by target pressure and discharge temperature
- The discharge temperature is directly related to the compressor suction super heat.
- If the suction superheat is high the discharge temperature will be high.
- If the discharge superheat is low the discharge superheat will be low.
- In either scenario the compressor speed will be reduced.
- The discharge temperature is a vital measurement in the troubleshooting process.

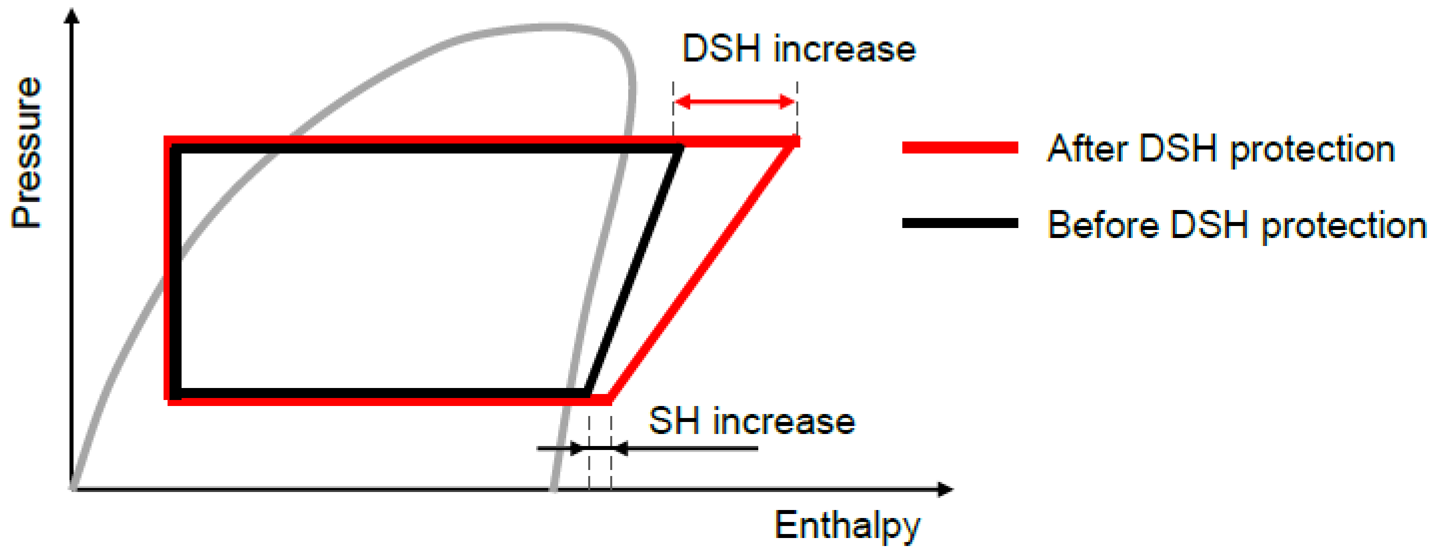


Superheat-Discharge Temperature Relationship

Concept

- The compressor Suction Super Heat (SH) is raised to increase the DSH.

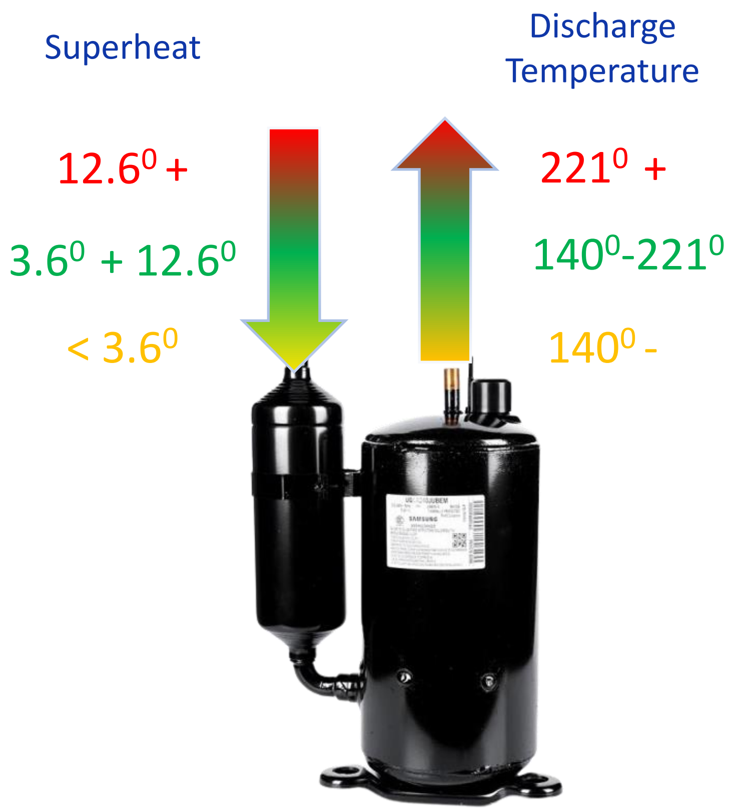
Ph-Diagram



- DSH = Compressor discharge temp. – High pressure saturation temp.

Discharge Temperature Range

- Normal 140^0-221^0
- 221^0 + May be low on Charge
- Below 140^0 system may be overcharged
- Discharge Superheat 54^0+ during trial heat operation



E438 Error

- EVI EEV cannot fully close
  - Discharge superheat less than 9degrees
  - Or
  - EVI in is Less than 32^0F
  - When
  - Compressor frequency is greater than 65Hz for 45 minutes,

Cycle error code

◆ Refrigerant flood back operation error. (E 438)

■ Detection condition
In case of both condition(1 & 2) are satisfied for 40min.

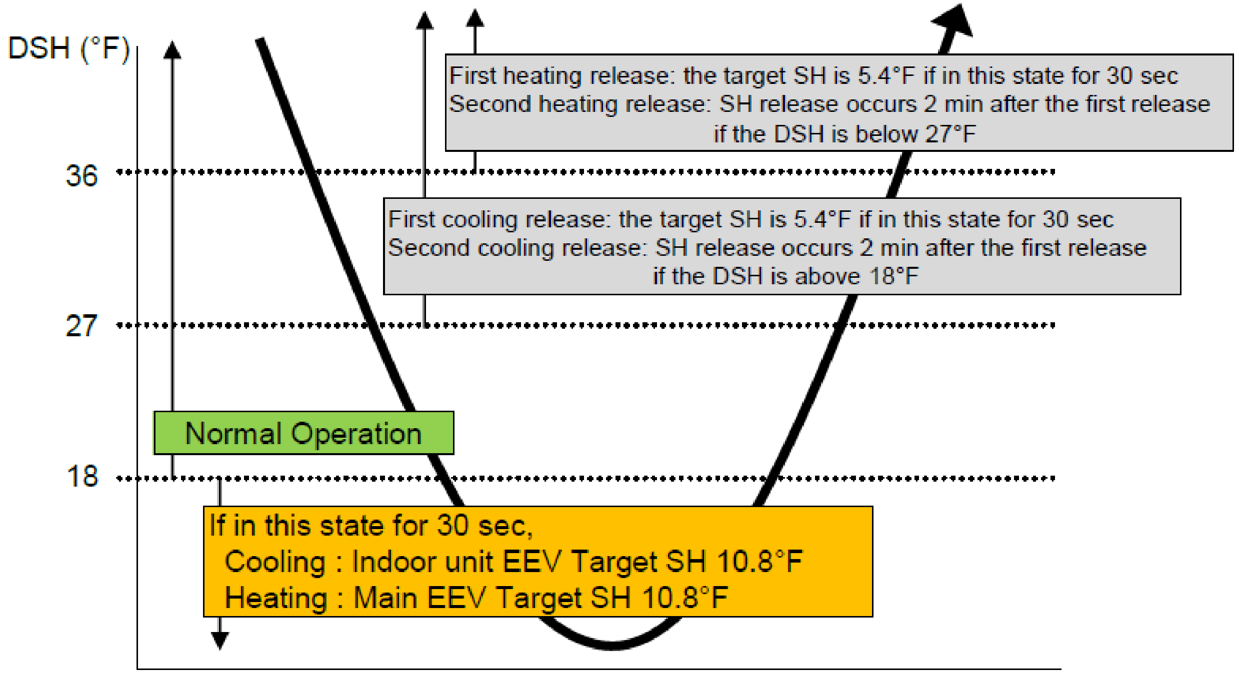
1) DSH < 5℃ &, 2) [(EVI out – EVI in temp. ≤ 0℃) or (SH < 5℃)]

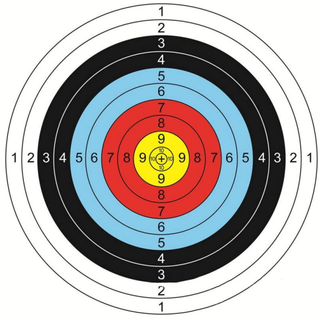
Flood back Discharge Temperature Protection

- If the compressor discharge superheat is below 18 degrees, the system is considered in a “Flood Back” condition
- The system will first readjust the target ***Evaporator*** (outdoor coil in heating mode and indoor coils in colling mode) superheats from 3.6 degrees to 5.4 degrees then to 10.4 degrees.
- If the superheat adjustment does not raise the discharge superheat compressor may slow down or the system will shut off.
- If the system is in a flood back condition the AR valve is closed.

Control specifications in detail

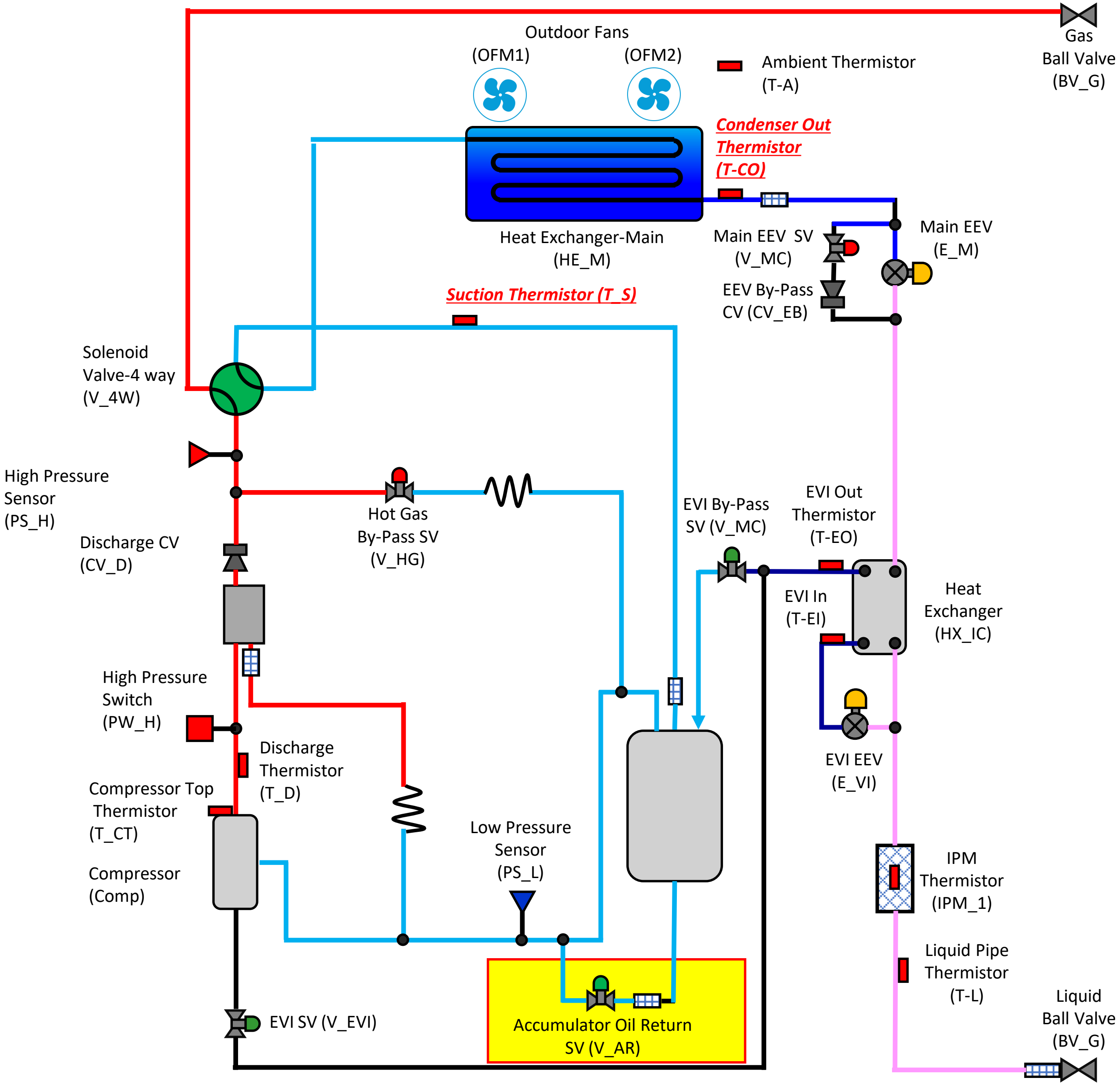
- The target superheat changes to 10.8°F if the DSH is 18°F or lower.





AR Valve

- During compressor operation, the valve is opened to return oil stagnating in the accumulator back to the compressor.
- It remains OFF (closed) under the following conditions during startup:
  - During Pump Down Start
  - If DSH < 18°F during Soft Start, remain in the OFF (closed) state
- Keep it in the OFF state when thermal off
- ARV OFF conditions during normal operation:
  - Turn off for 2 minutes if DSH < 18°F, but turn on immediately if DSH > 27°F
  - After turning off and on during normal operation, do not turn off (close) for the next 10 minutes from the time of turning on (open)
- Except for the cases mentioned above, always keep the ON (open) state during operation.



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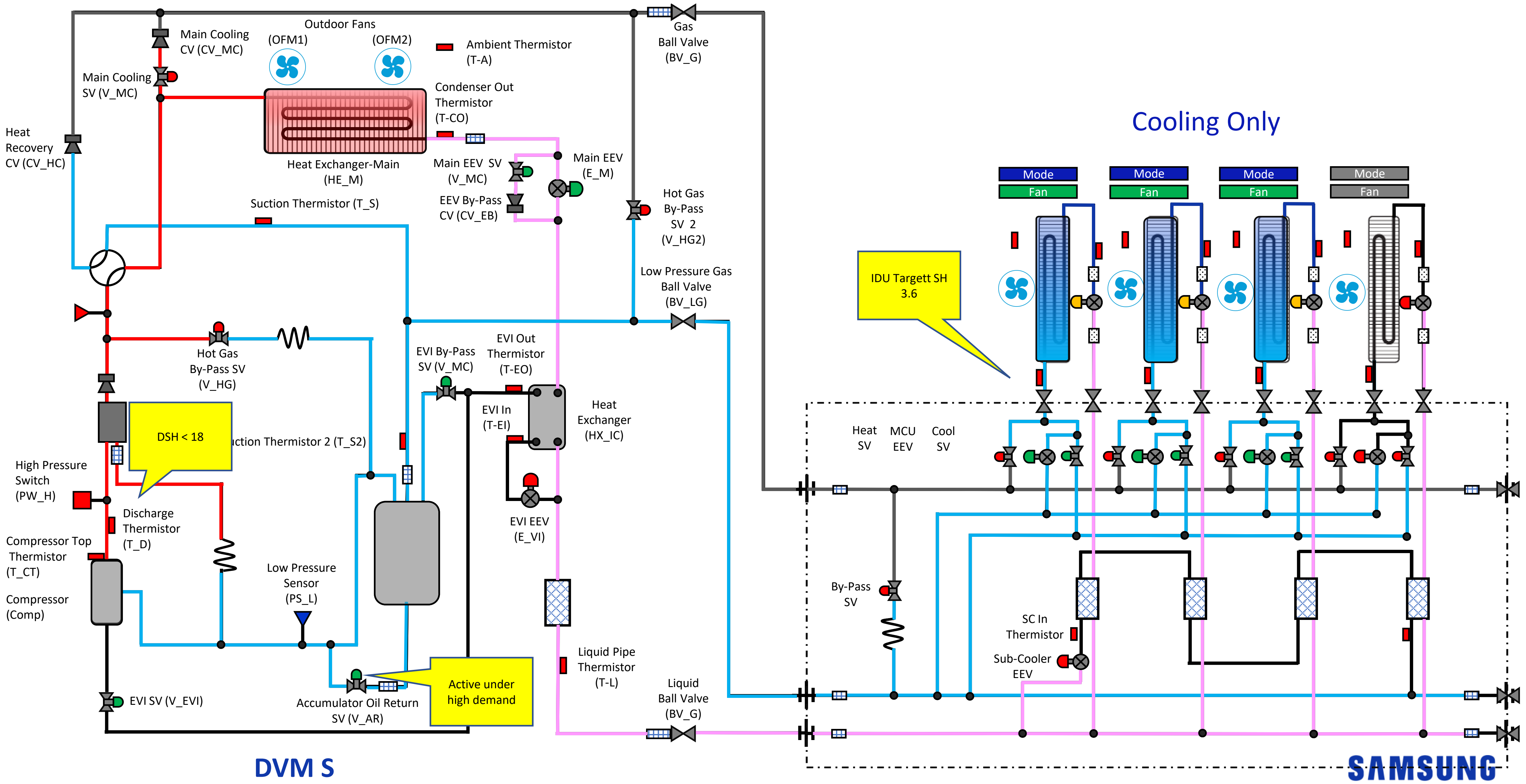
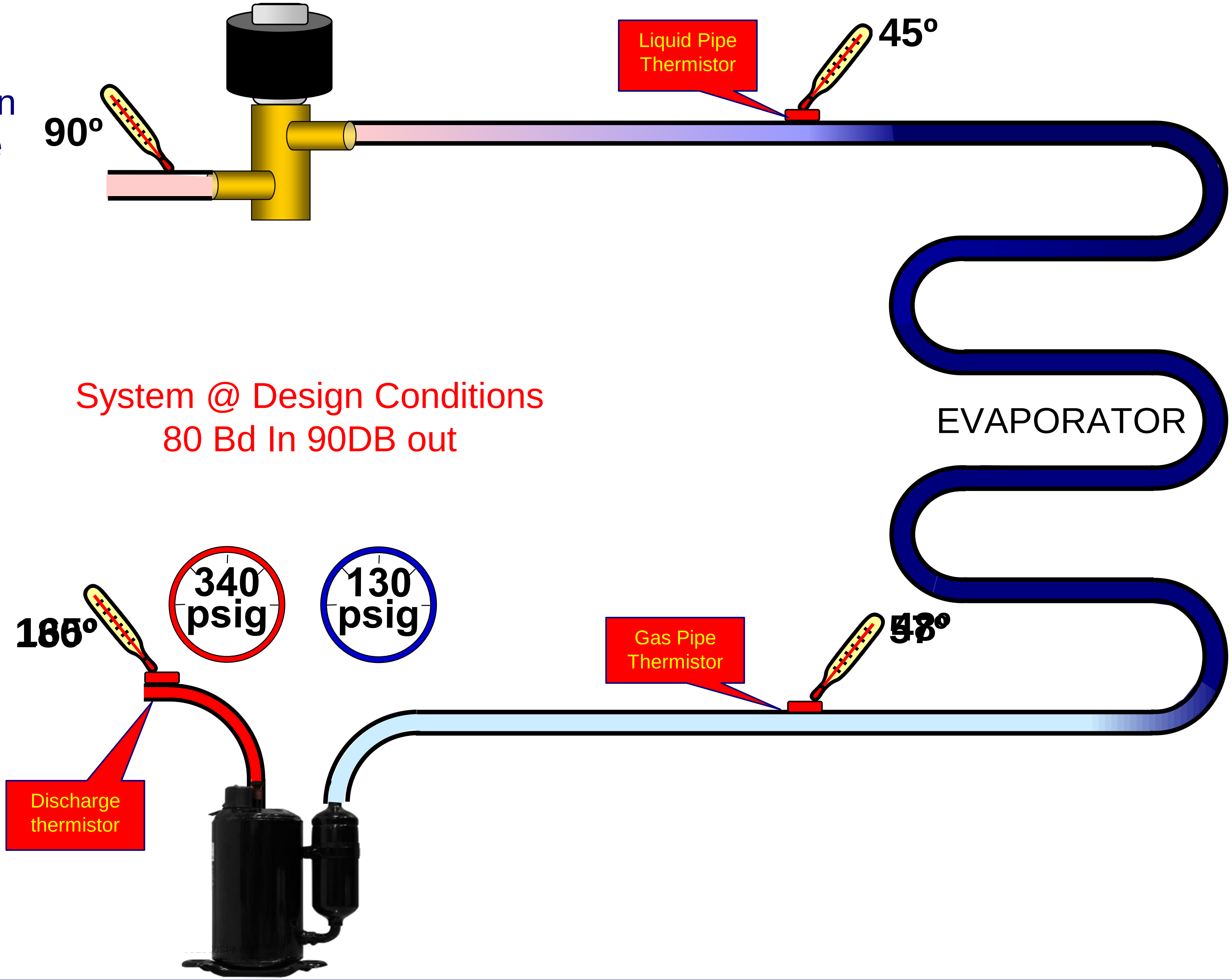


Outdoor Fans
(OFM1)
(OFM2)
Main Cooling CV (CV_MC)
Main Cooling SV (V_MC)
Ambient Thermistor (T-A)
Gas Ball Valve (BV_G)
Condenser Out Thermistor (T-CO)
Heat Recovery CV (CV_HC)
Heat Exchanger-Main (HE_M)
Main EEV SV (V_MC)
Main EEV (E_M)
EEV By-Pass CV (CV_EB)
Hot Gas By-Pass SV 2 (V_HG2)
Suction Thermistor (T_S)
Low Pressure Gas Ball Valve (BV_LG)
Cooling Only
Mode
Fan
IDU Targett SH 3.6
Hot Gas By-Pass SV (V_HG)
EVI By-Pass SV (V_MC)
EVI Out Thermistor (T-EO)
Heat Exchanger (HX_IC)
EVI In (T-EI)
DSH < 18
uction Thermistor 2 (T_S2)
High Pressure Switch (PW_H)
Discharge Thermistor (T_D)
Compressor Top Thermistor (T_CT)
Compressor (Comp)
EVI EEV (E_VI)
Low Pressure Sensor (PS_L)
Heat SV
MCU EEV
Cool SV
By-Pass SV
SC In Thermistor
Sub-Cooler EEV
Liquid Pipe Thermistor (T-L)
Liquid Ball Valve (BV_G)
EVI SV (V_EVI)
Accumulator Oil Return SV (V_AR)
Active under high demand
DVM S
SAMSUNG

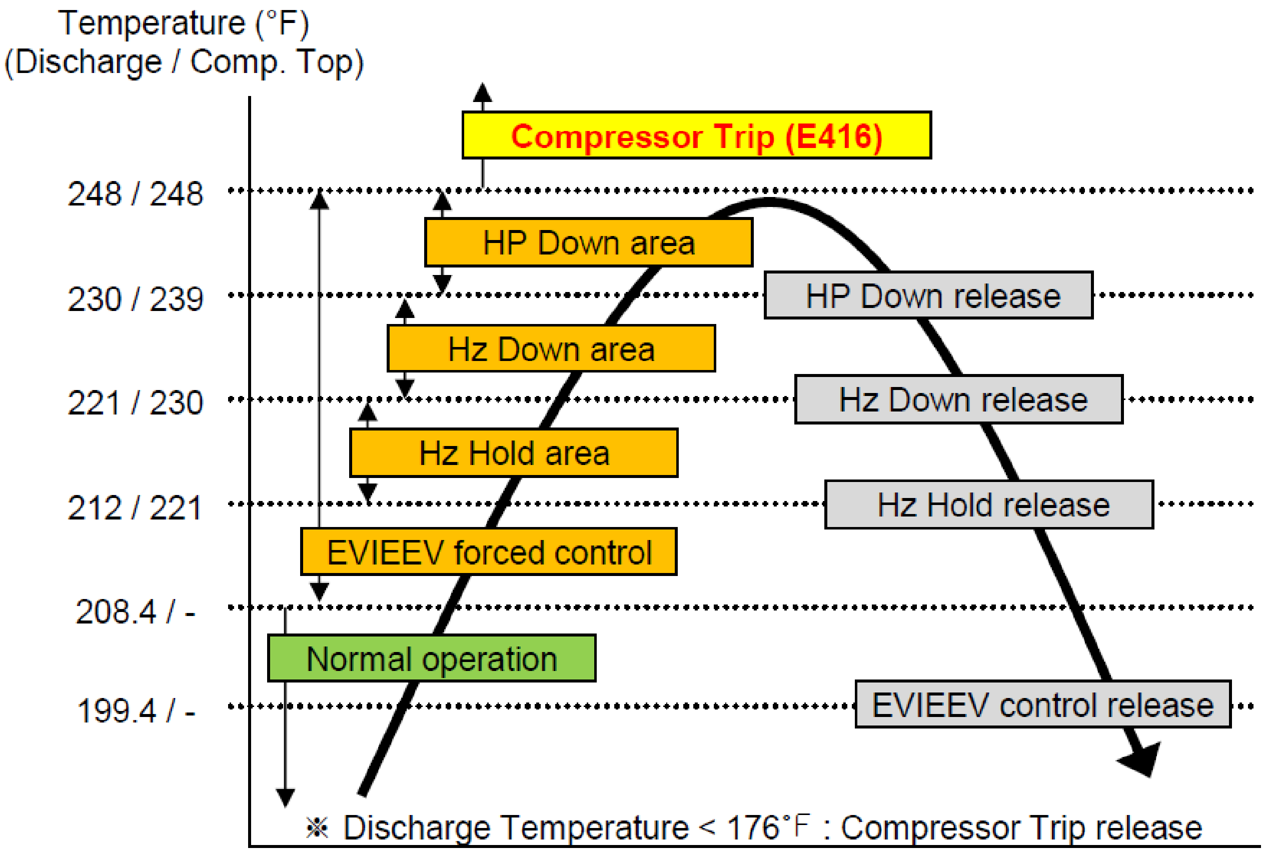
Effects of Superheat on Compressor discharge

- The system is designed to operate at given temperatures
- At design you will have the proper superheat and discharge temperatures
- If Super heat goes up what happens to discharge temperature?



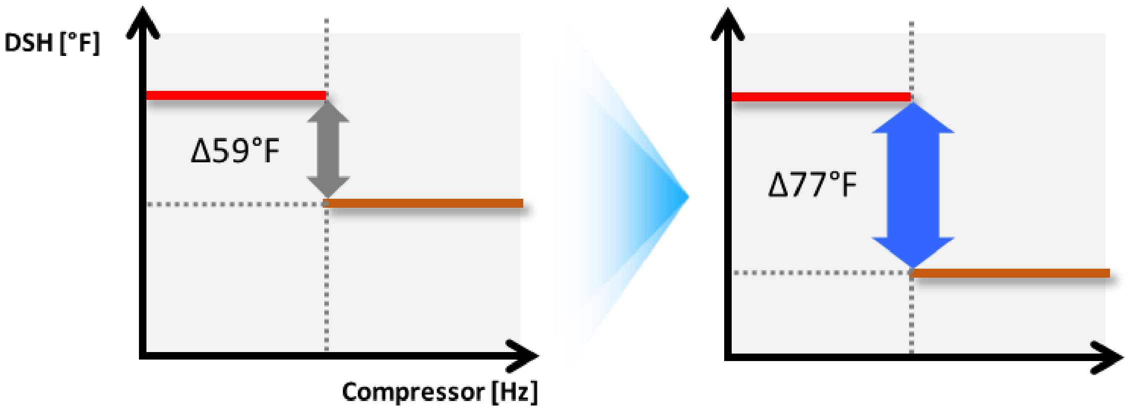
High Discharge Temperature Protection

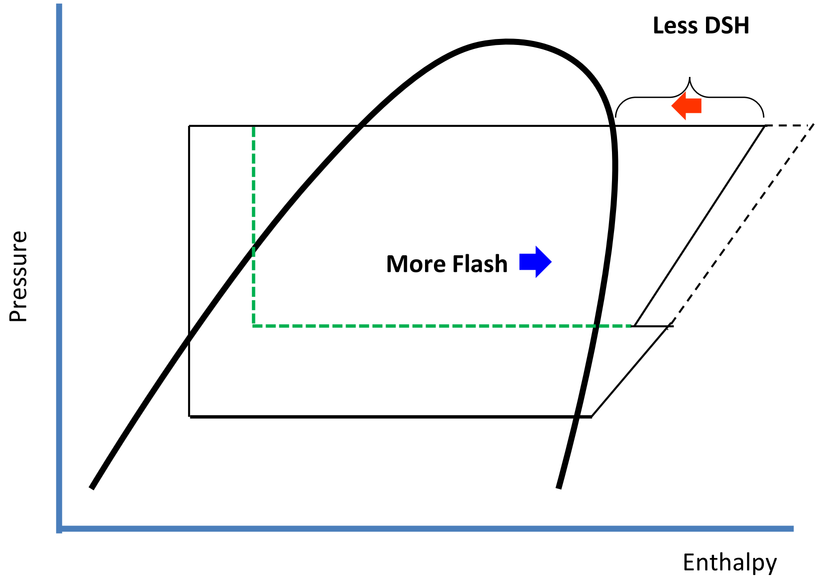
- The compressor frequency is controlled according to the discharge temp, as shown below.
- 1st step EVI EEV Open : EVI EEV is forcibl opened.
- 2nd step EVI EEV Open : The current frequency is fixed.
- (It is possible to lower the frequency if necessary)
- 3rd step Hz down : The frequency is forcibl lowered by 5Hz at specified intervals.
- 4th step HP down : The frequency is forcibl lowered by 1Hz every second.
- 5th step Comp trip : Compressors stop and an error is triggered (E416).



DVM S2
Upgraded Discharge Superheat Algorithm

- Upgraded Discharge Superheat (DSH) Algorithm automatically adjusts the degree of discharge superheat to heat to improve efficiency (68 ~ 113°F)
- Wider target DSH provides a more flexible response to various operating conditions
- Reducing the discharge superheat will reduce the amount of heat to be rejected from the refrigerant





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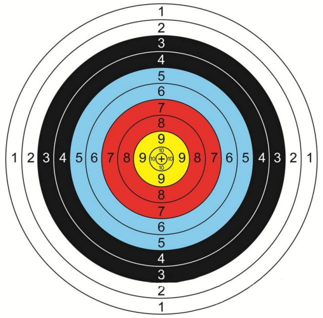
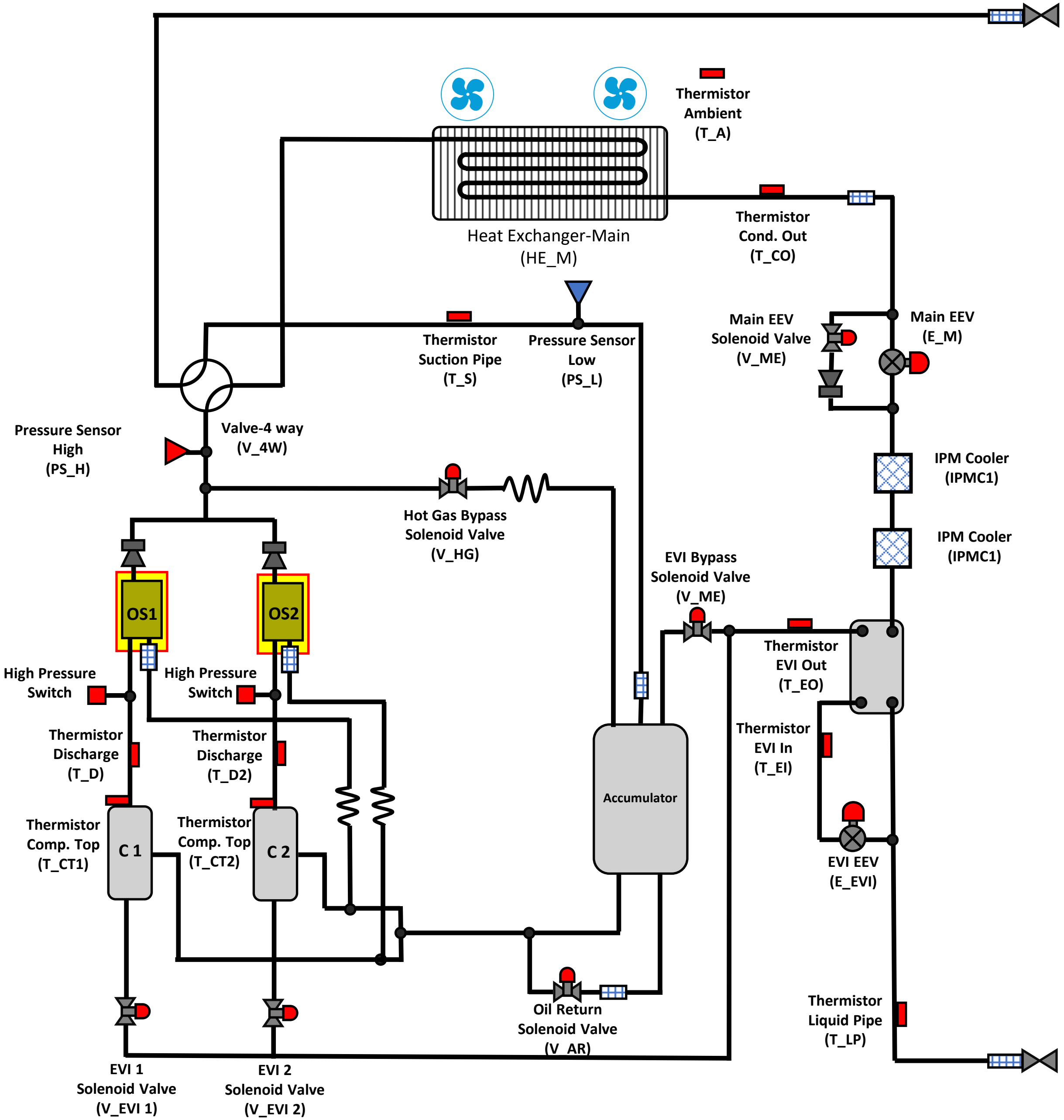
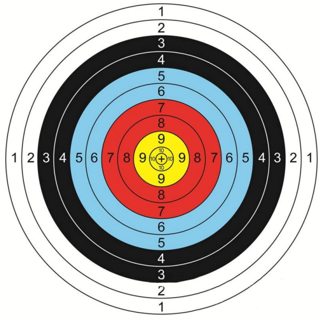


Oil Separator

- Removes entrained oil from the refrigerant and returns it to the suction side of the opposite compressor.

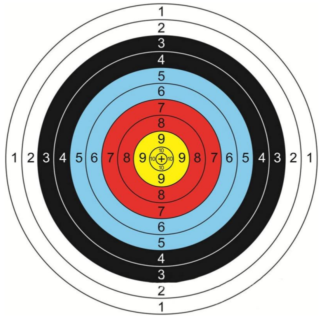
For Demonstration ONLY!





Oil Recovery Mode

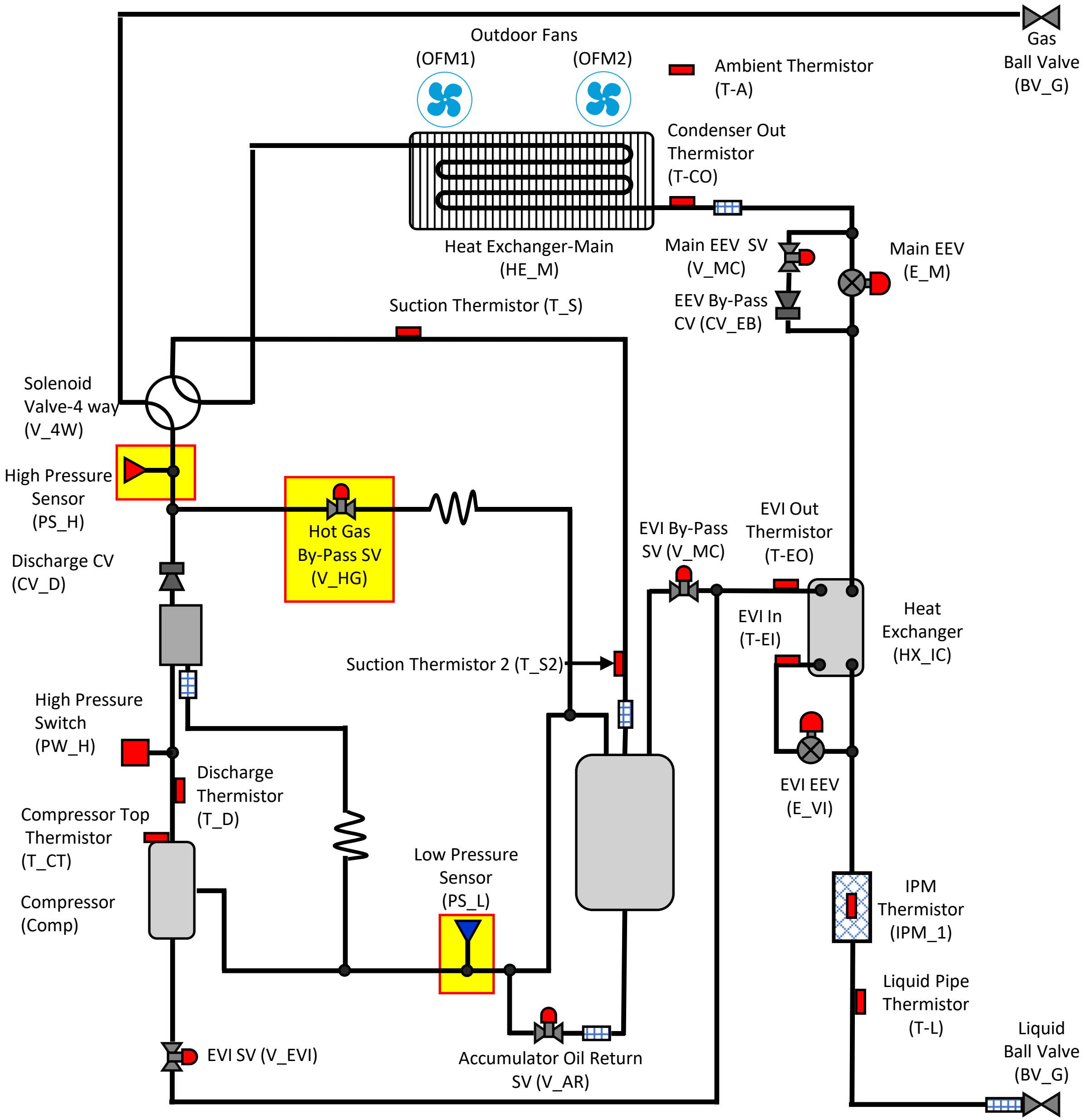
- Oil recovery operation shall be performed to prevent a compressor from being short of oil.
- Oil in pipes or indoor unit will be collected to outdoor unit by starting oil collecting operation.
- Oil collecting operation will begin 7 hours after outdoor unit starts to operate but time might be reduced depending on the operating condition (# of compressor in operation and operating frequency).
- The compressor capacity is raised to increase the amount of refrigerant circulation.
- This will return oil remaining in indoor units and piping.
- EEVs are opened in a controlled manner for active and inactive indoor units to facilitate oil return.

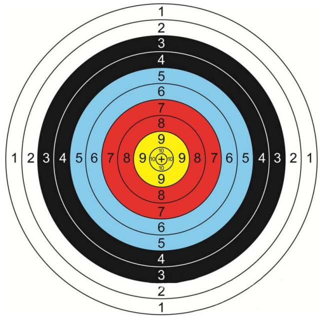


Hot Gas By-Pass

- Protects the system from high- and low-pressure conditions.
- Opens when
  - High Pressure exceeds 561 psig.
  - Low Pressure is below:
    - 45.5 cooling mode
    - 12.8 heating mode

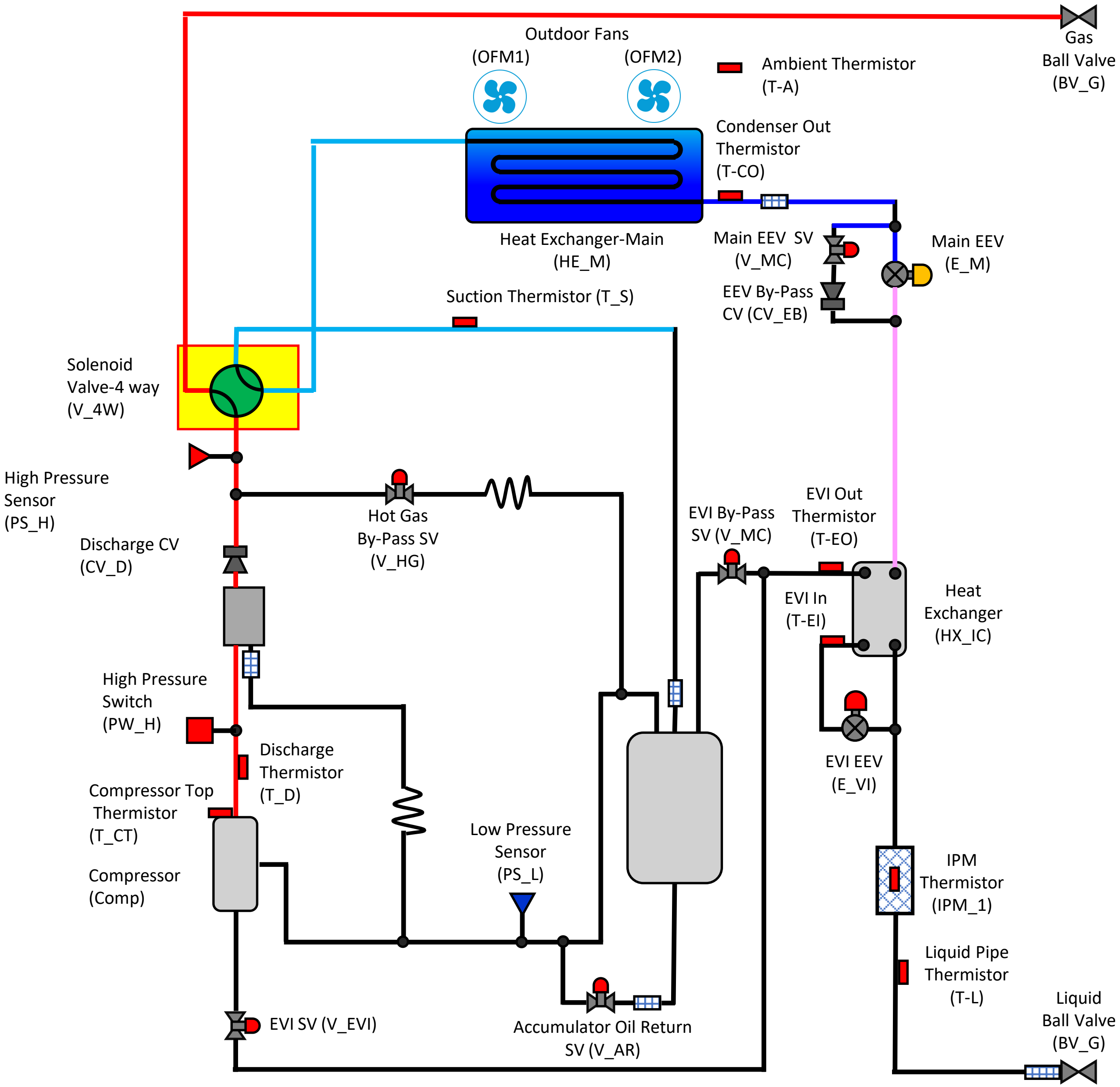
Valve control	Cooling	Heating
On (psi)	≤ 46	≤ 13
Off (psi)	≥ 57	≥ 24





Reversing Valve

- The reversing valve is energized in the heating mode

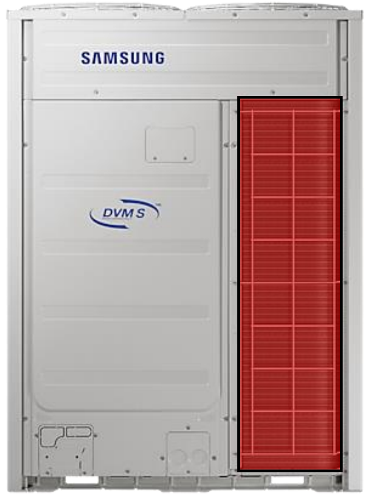


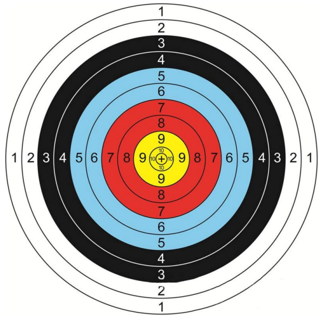
Heat Pump ODU Coil Detailed

- In the Heating Modes the outdoor coil becomes the evaporator.
- In the cooling mode the outdoor coil is the condenser.

Outdoor Unit Regulation Cooling Mode

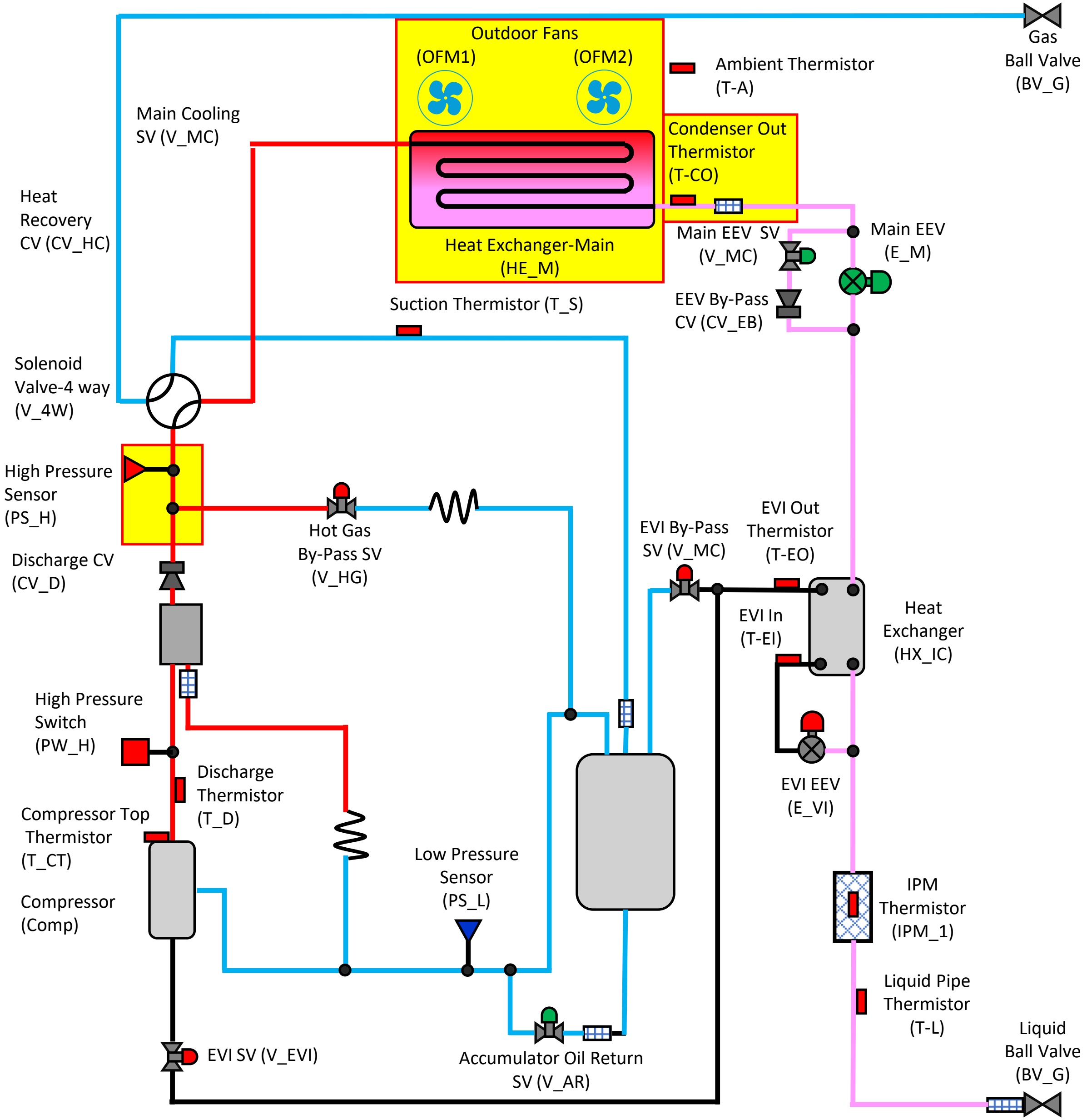
- The compressor is modulated to maintain an approximate suction pressure of 114 spig
- The DVM S condenser (outdoor coil) fans are cycled to maintain a head pressure of approximately 355 psig.
- The DVM S2 has Active AI Control.
- Condenser (outdoor coil) fans are cycled to maintain a head pressure of approximately 241.8 - 355.6 psig
- The Main EEV is commanded to full open (2,000 steps)
- The Main EEV solenoid is energized





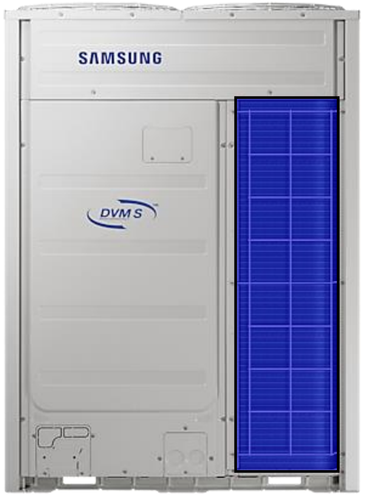
The Outdoor Coil Cooling Mode

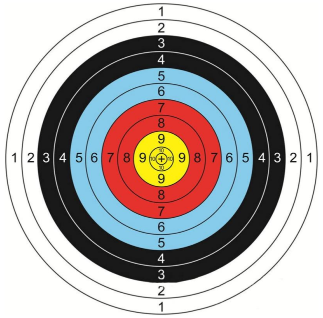
- In the cooling mode the outdoor coil is the condenser.
- The Condenser out temperature should be 5 – 30 degrees greater than ambient temperature.
- The condenser fan will modulate to maintain target head pressure measured at the high-pressure sensor.



Outdoor Unit Regulation Heating Mode

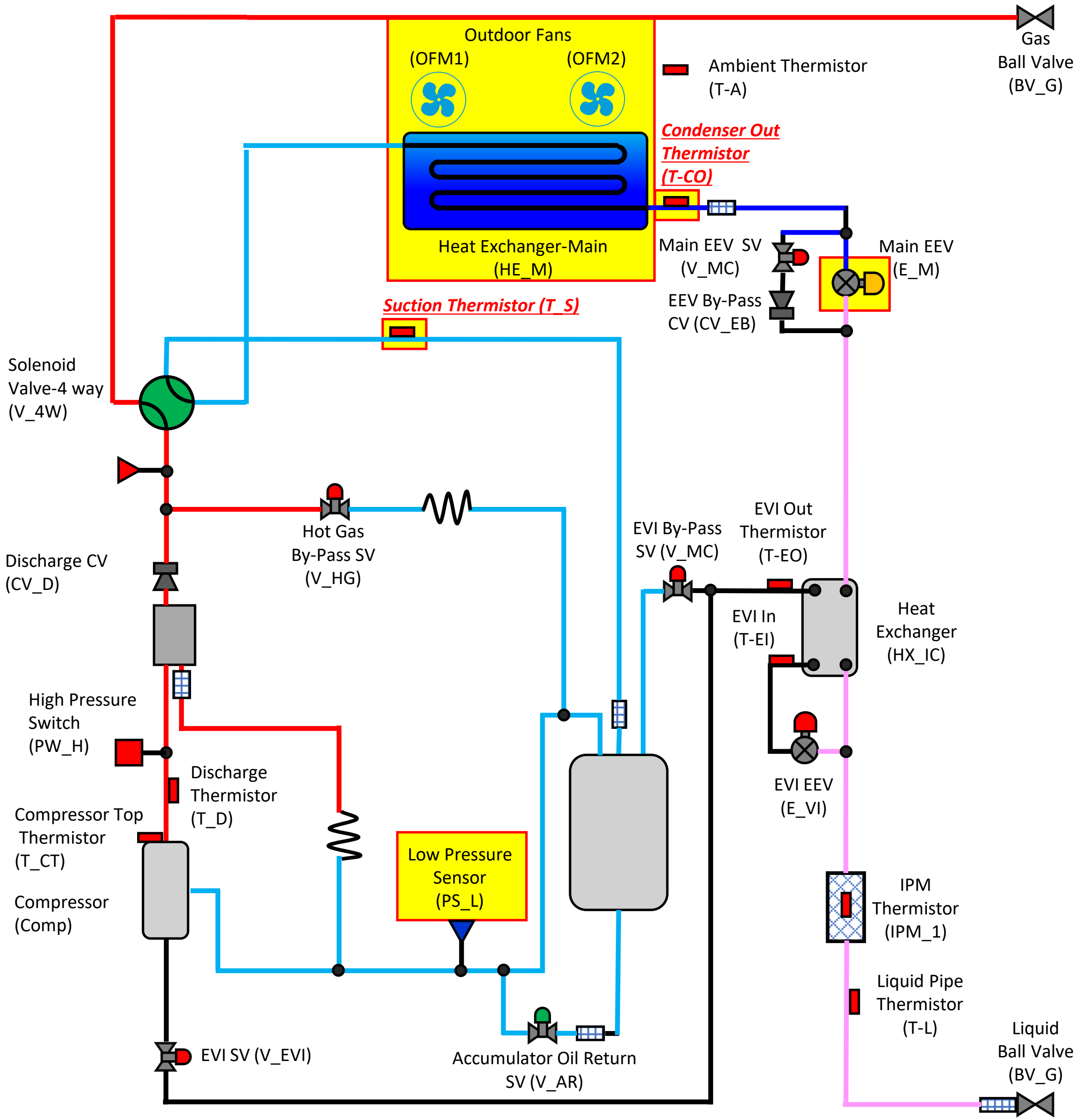
- In the heating mode the compressor will modulate to maintain a high side pressure of approximately 427 psig.
- This will cause the saturation temperature at the indoor coil to be approximately 120+ degrees
- The suction pressure is a by-product of the outside air temperature.
- The evaporator saturation temperature is “typically” 2 to 10 degrees lower than outside air temperature.

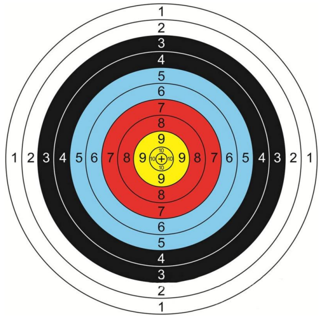




The Outdoor Coil Heating Mode

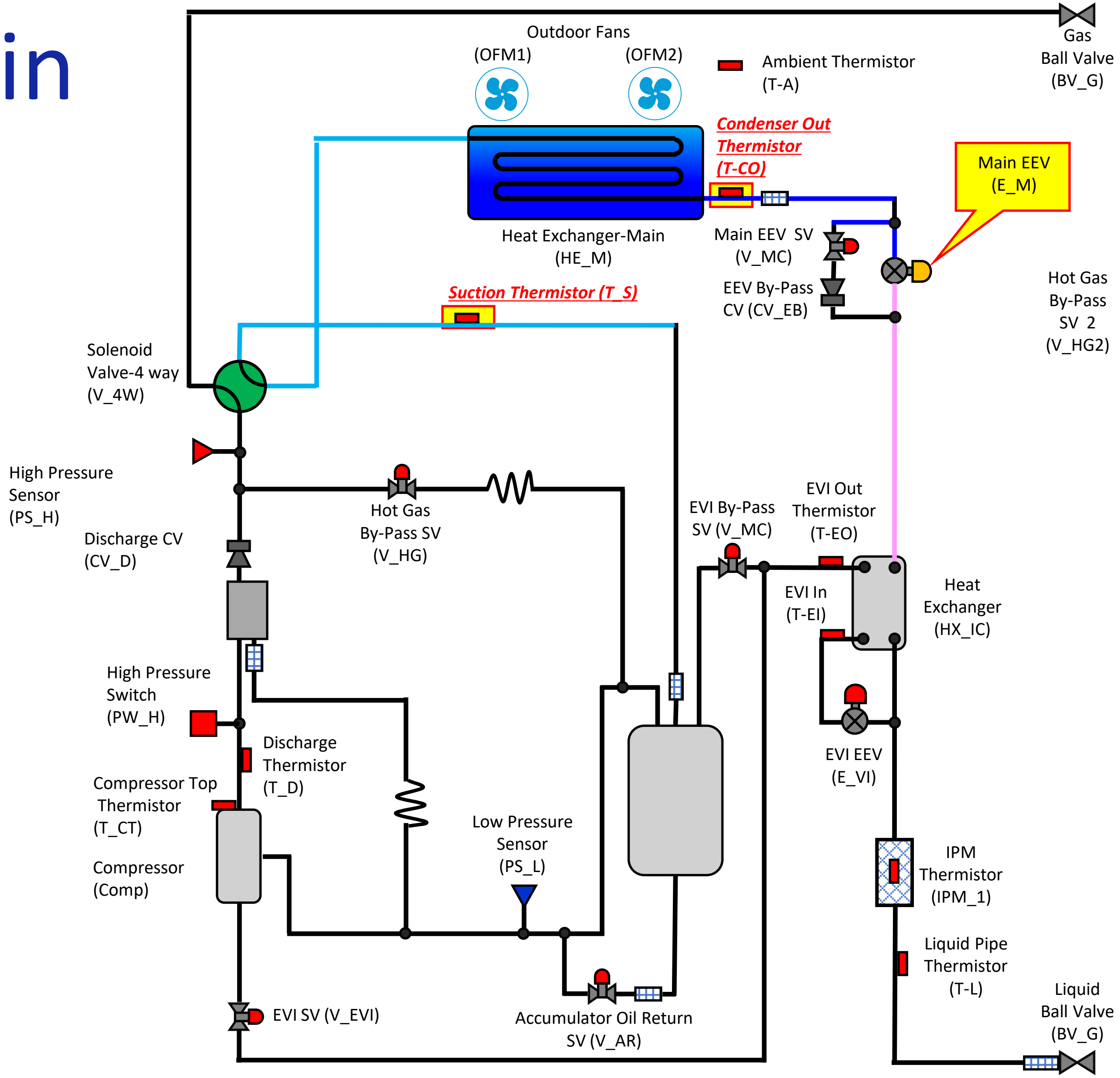
- In the heating mode the outdoor coil is the evaporator.
- The evaporators metering device is the Main EEV
- It modulates to maintain the desired differential (superheat) between the condenser out and suction thermistor.
- The low side pressure is determined by the outside air temperature





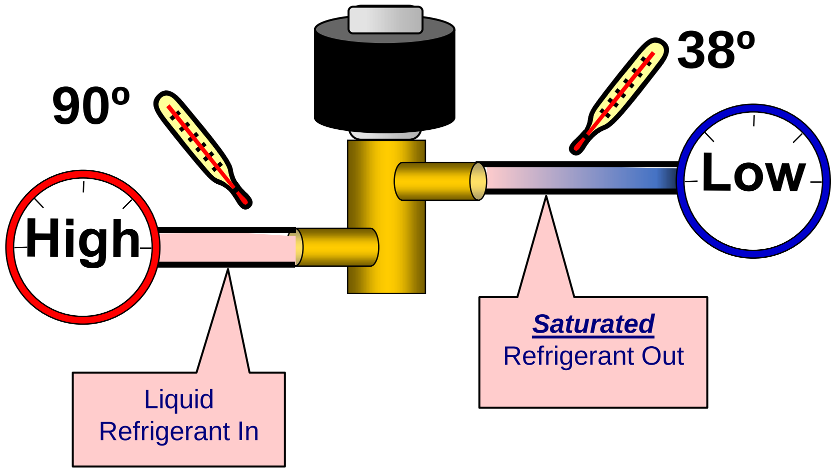
Main EEV Target in Heating Mode

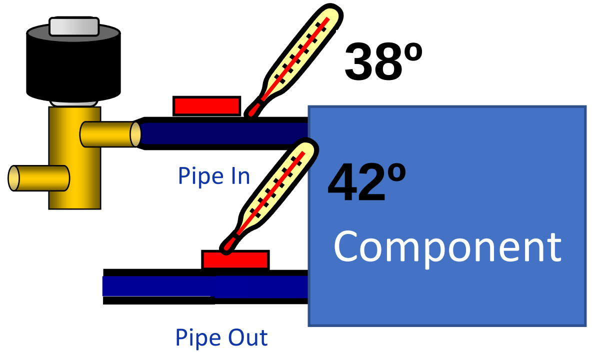
- In the heating mode the outdoor coil is the evaporator.
- The Main EEV is the metering device for the coil.
- The target evaporator superheat is:
- 3.6 degrees for a single compressor system
- 9 degrees for a dual compressor system.



Main EEV
ODU Evaporator Control

- It's purpose is to creates a pressure drop.
- It regulates refrigerant flow to maintain a design differential (superheat) through the component.
- If superheat is high the valve opens
- If superheat in low the valve closes
- If the discharge temperature is high in the heating mode the EEV may be driven open to help regulate discharge temperature





Outdoor Unit Control

Main EEV Control - Heating

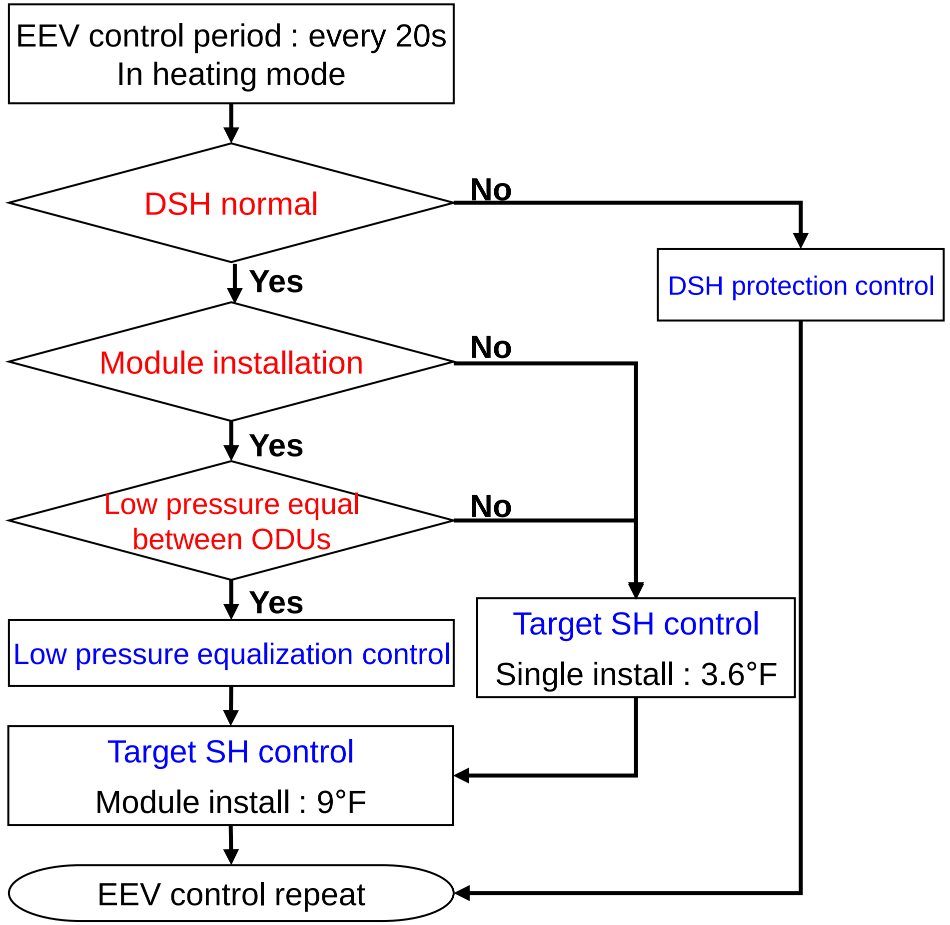
▶ Main EEV control

① Stop ODU : Main EEV close

② Cooling ODU : Main EEV Full open

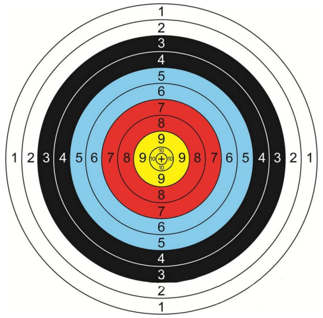
▶ Main EEV control for module install

1. Refrigerant distribution between ODUs
2. Then Super Heat control (SH)
3. If discharge temp is high → EEV step increases



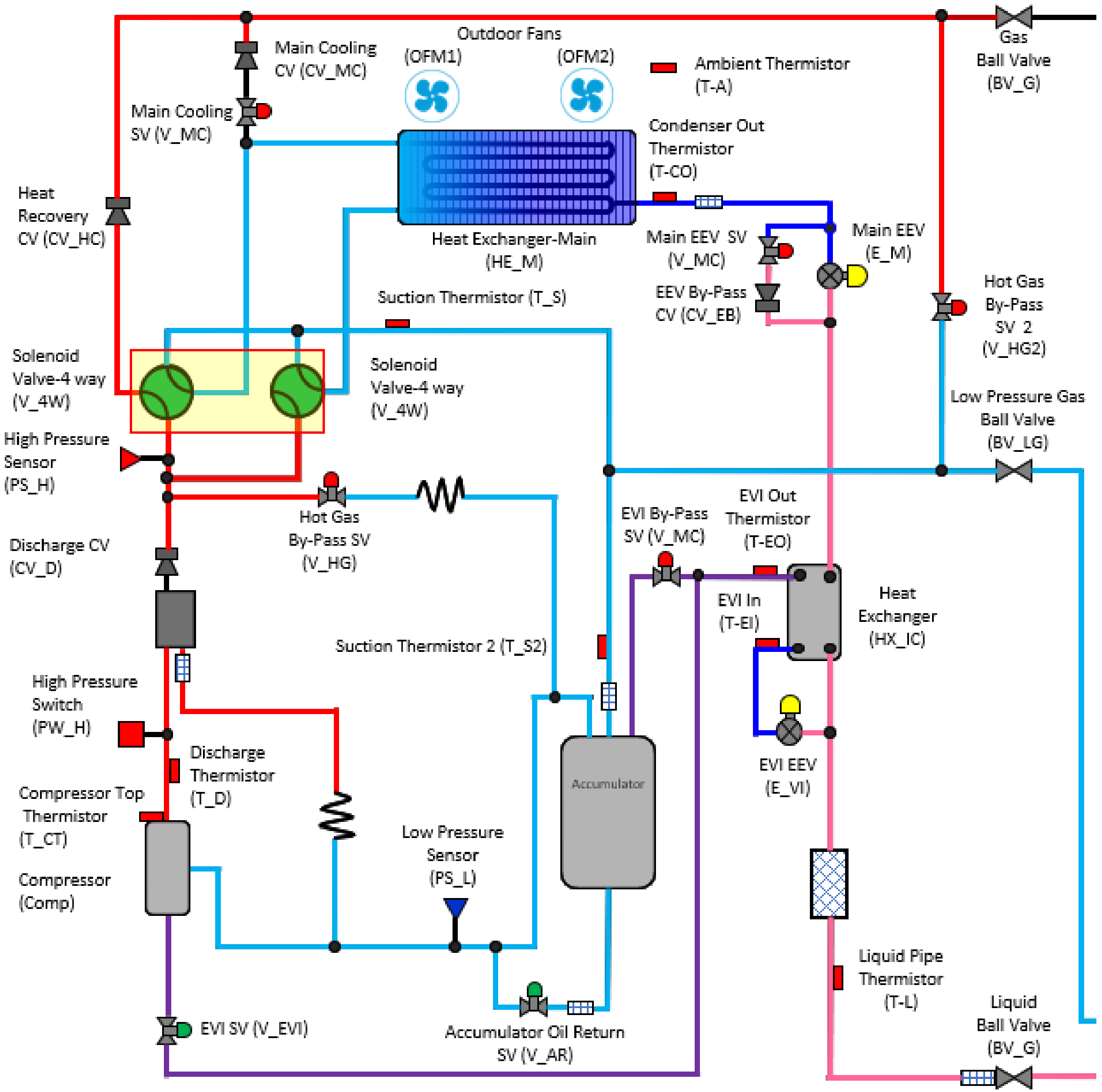
* SH = super heat
(Suction temp. – Low pressure saturated temp.)

* DSH = Discharge super heat
(discharge temp. – high pressure saturated temp.)



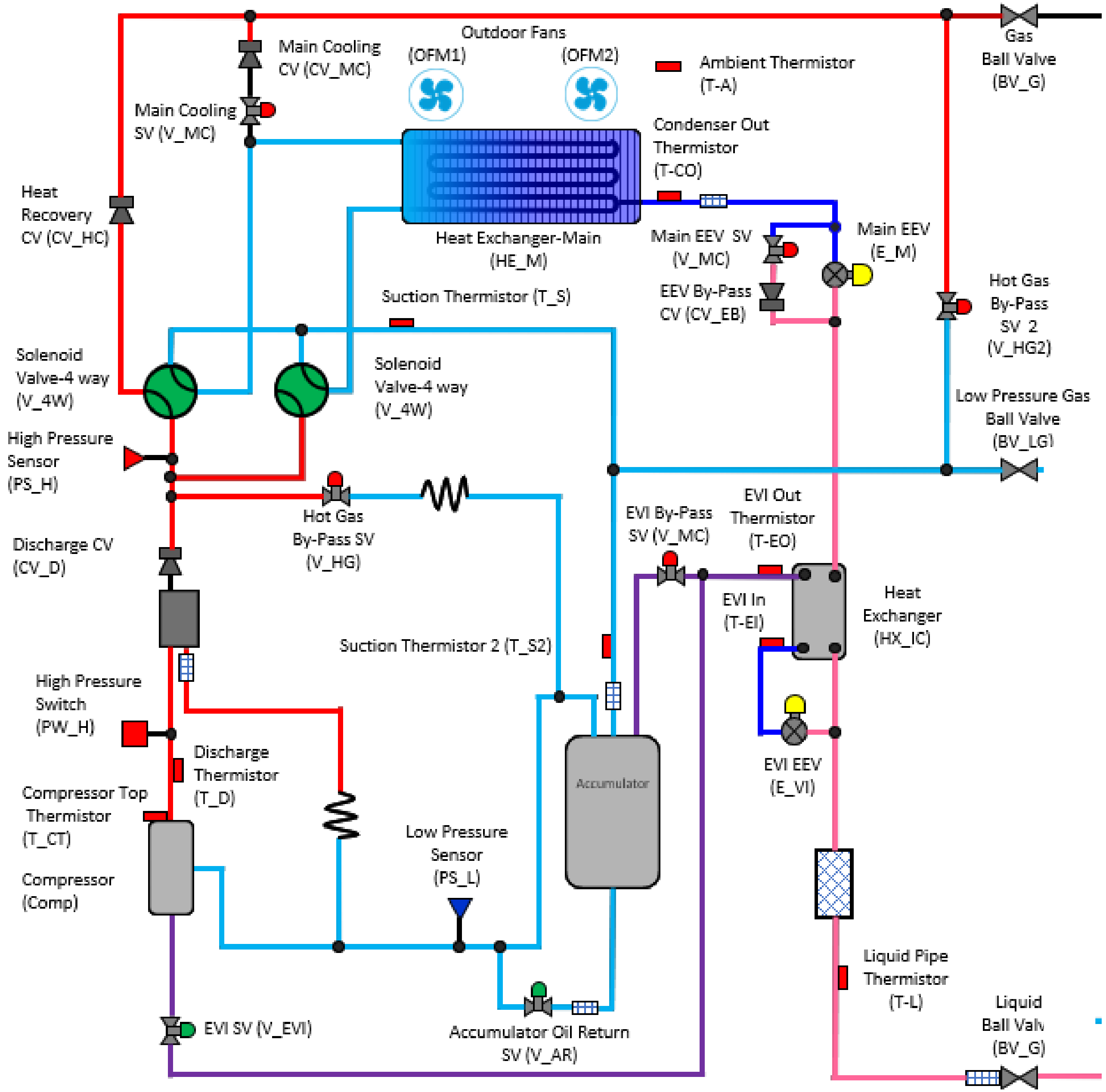
Reversing Valve DVM S2

- The DVM S2 has 2 reversing valves that allow split coil defrost.



Heating Cycle

- In the heating cycle the outdoor coil becomes the evaporator.
- In low ambient conditions ice will build up on the evaporator.
- When enough ice forms the it inhibits the heat transfer capability of the coil the unit needs to enter the defrost cycle



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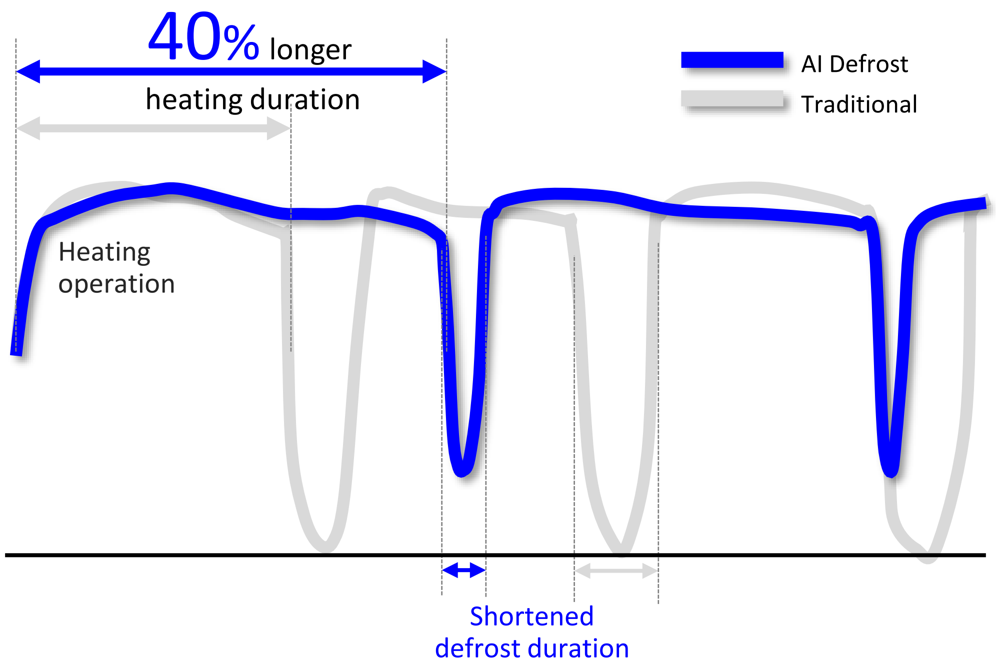
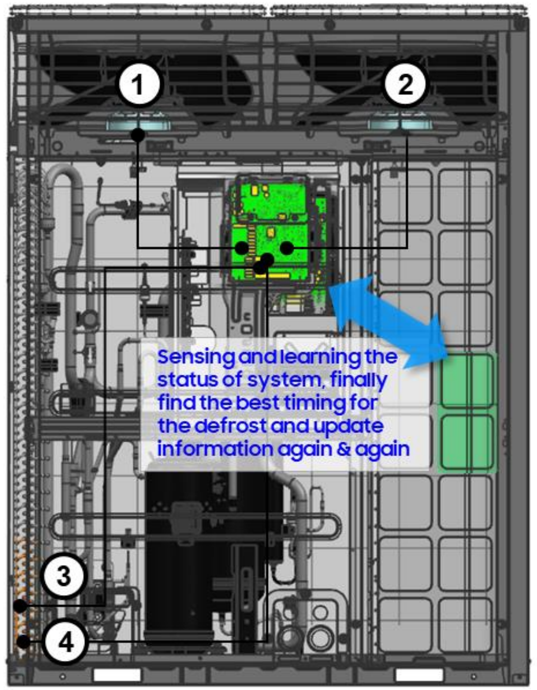
DVM S2
AI Defrost

- DVM S2 learns and trends fan motor current to detect ice formation on the condenser coil during heating operation. This also detects temperature in order to increase heating operation duration between defrost cycles to reduce overall defrost cycle duration.
- It carefully determines the exact status of the system with AI algorithms to provide optimal comfort during the heating season.



* Samsung internal test result. (Condition : Heating operation at -10℃, RH 40%)

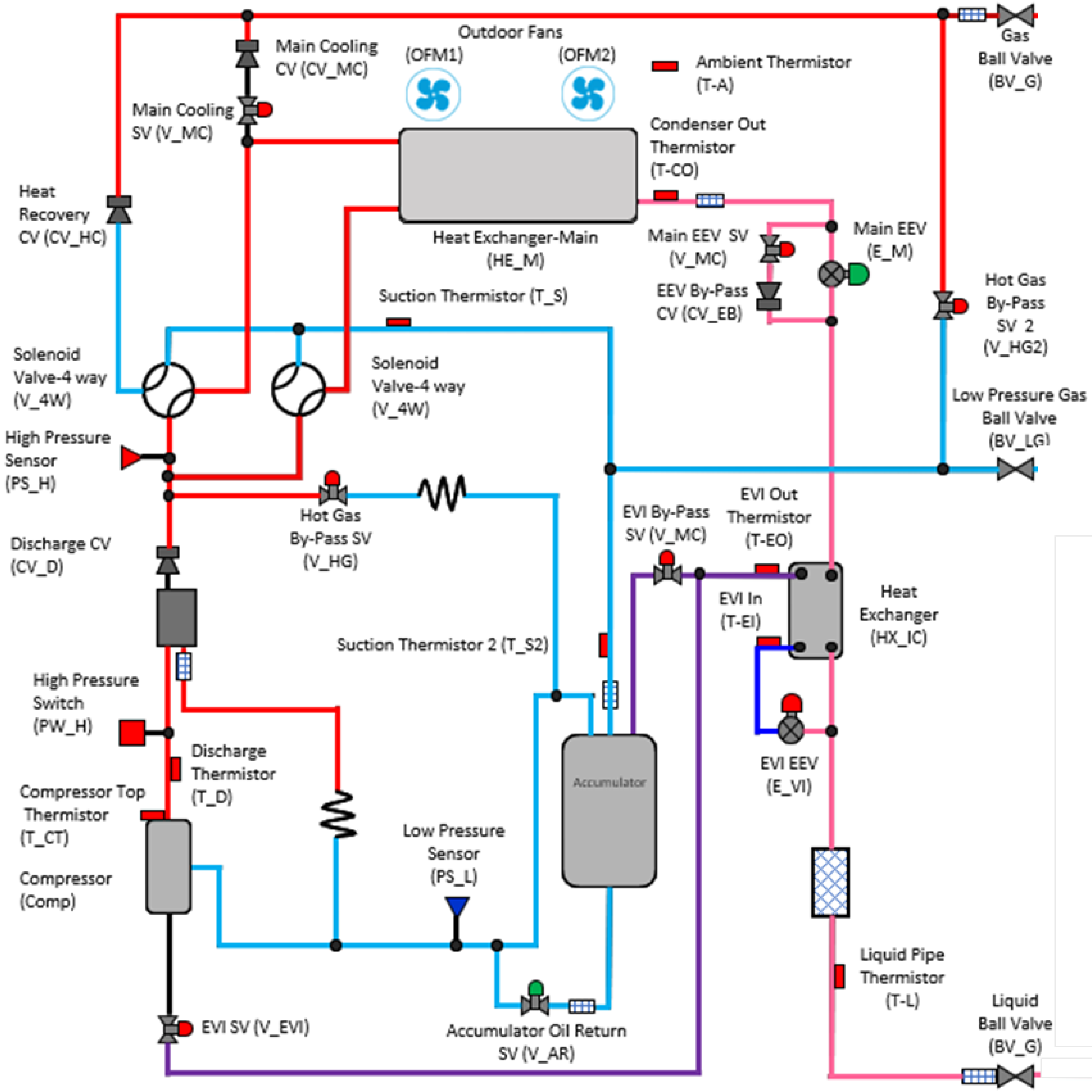


Enhanced Defrost Control

- Defrost operation uses the bottom path of the heat exchanger as hot gas bypass circuit for faster and more effective defrost cycles compared to previous models.
- Reduces the potential for frost build up on the bottom section of the coil.

Defrost Cycle

- In the defrost cycle the reversing valves are deenergized.
- The reversing valve will now direct hot gas to the outdoor coil instead of the indoor units
- The outdoor fan shut
- After the ice is melted the unit needs to return to the heating mode.



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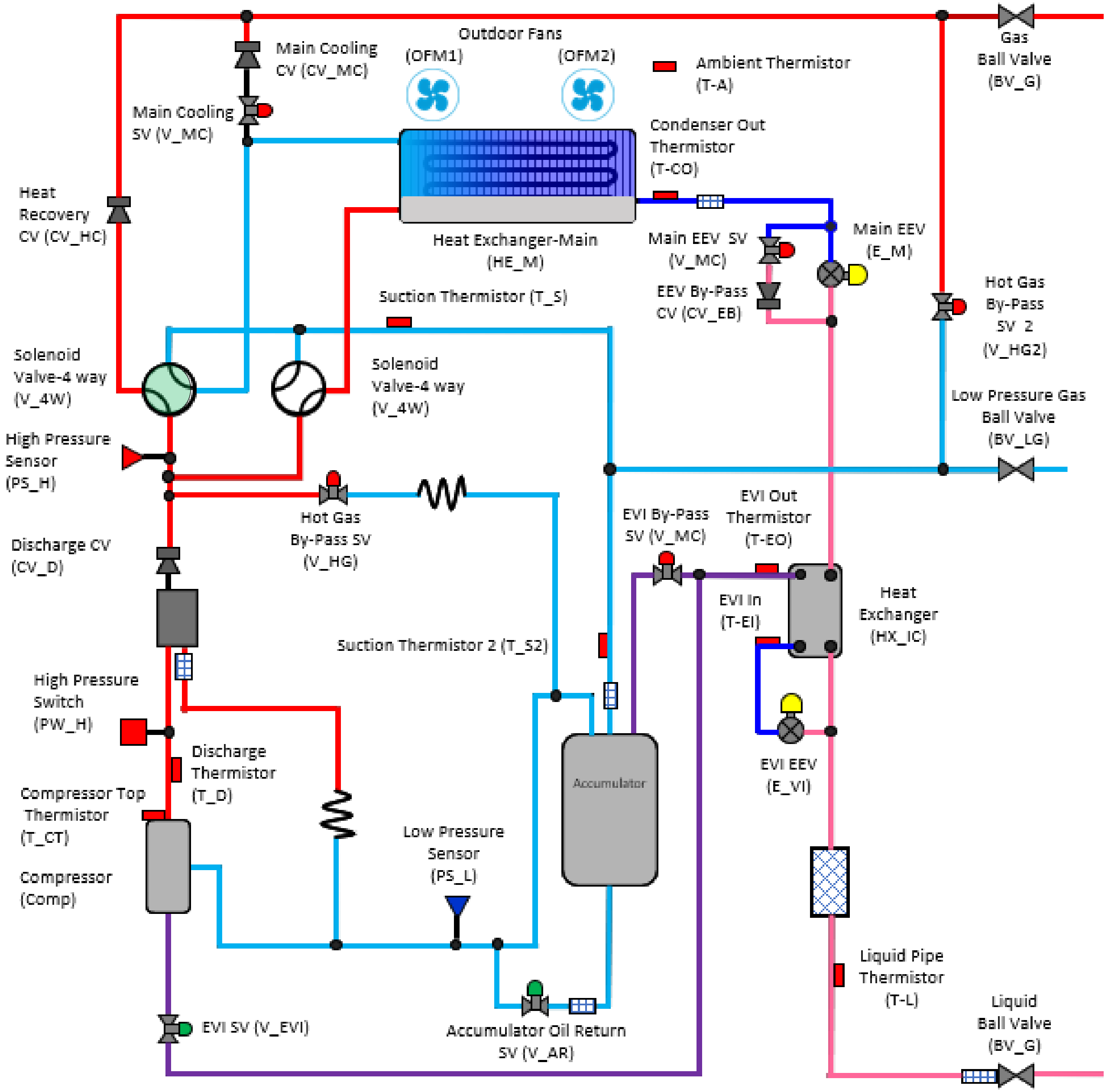
Enhanced Defrost

- The Sub 4 way valve continues to operate for up to 6 minutes after the main 4 way main returns to the heating mode.
- The sub 4 way valve is energized if there enough pressure differential between the high and low side pressures.



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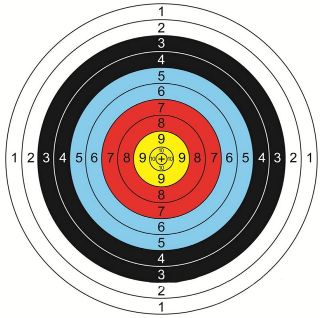
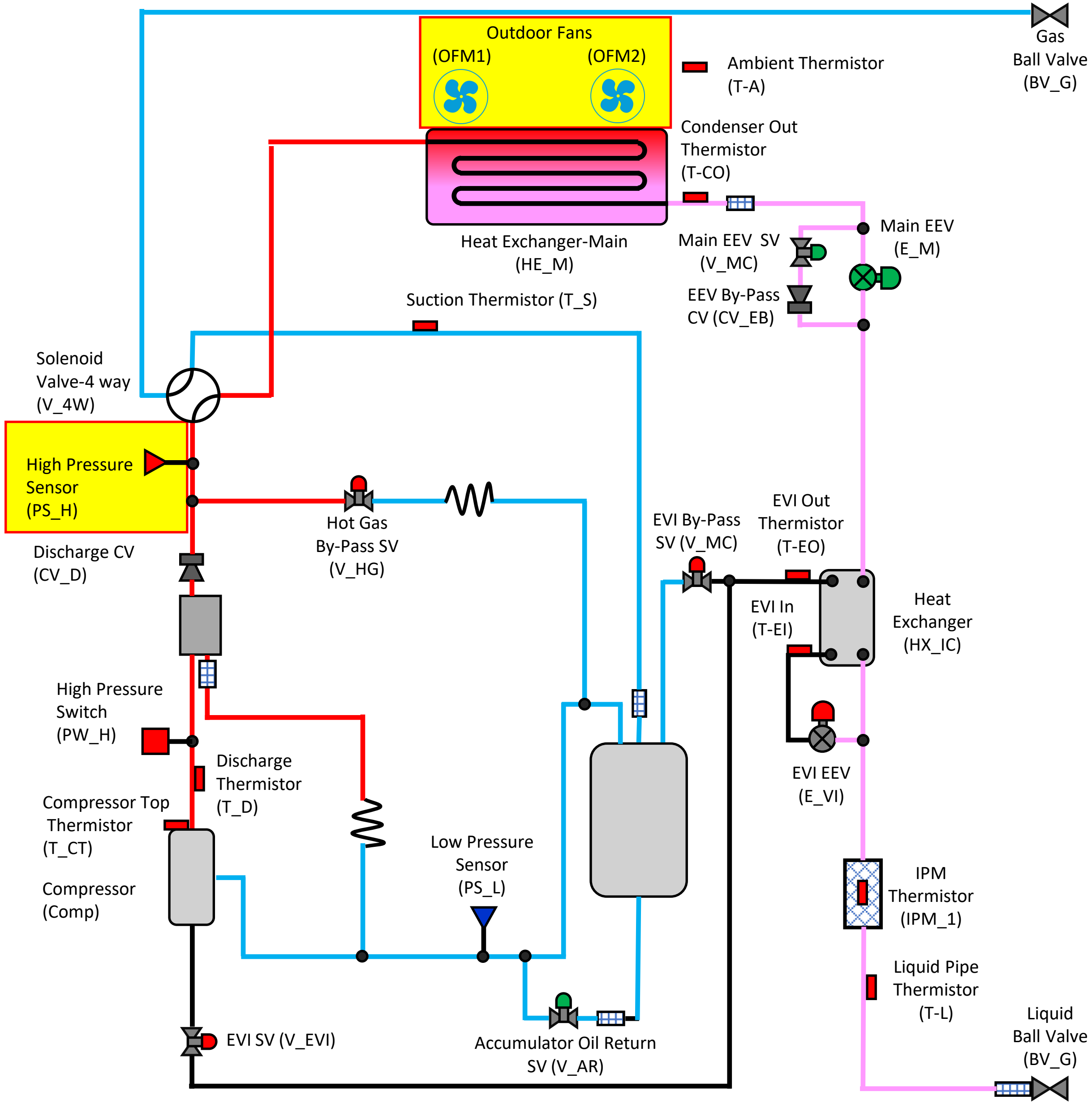


Outdoor Fans in Cooling Mode

- ❑The condenser fans will modulate to maintain head pressure measured at the high-pressure sensor in the cooling modes.
- ❑DVM S Cooling Only
  - Target is 356 psig
- ❑DVM S2 Cooling Only
  - Target 241.8 psig - 355.6 psig
- ❑Cooling Main
  - 356 psig – 425 psig



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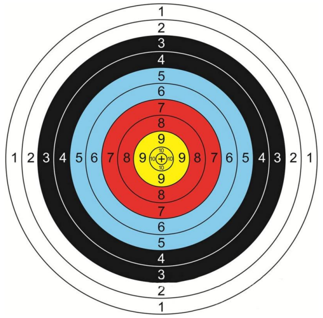
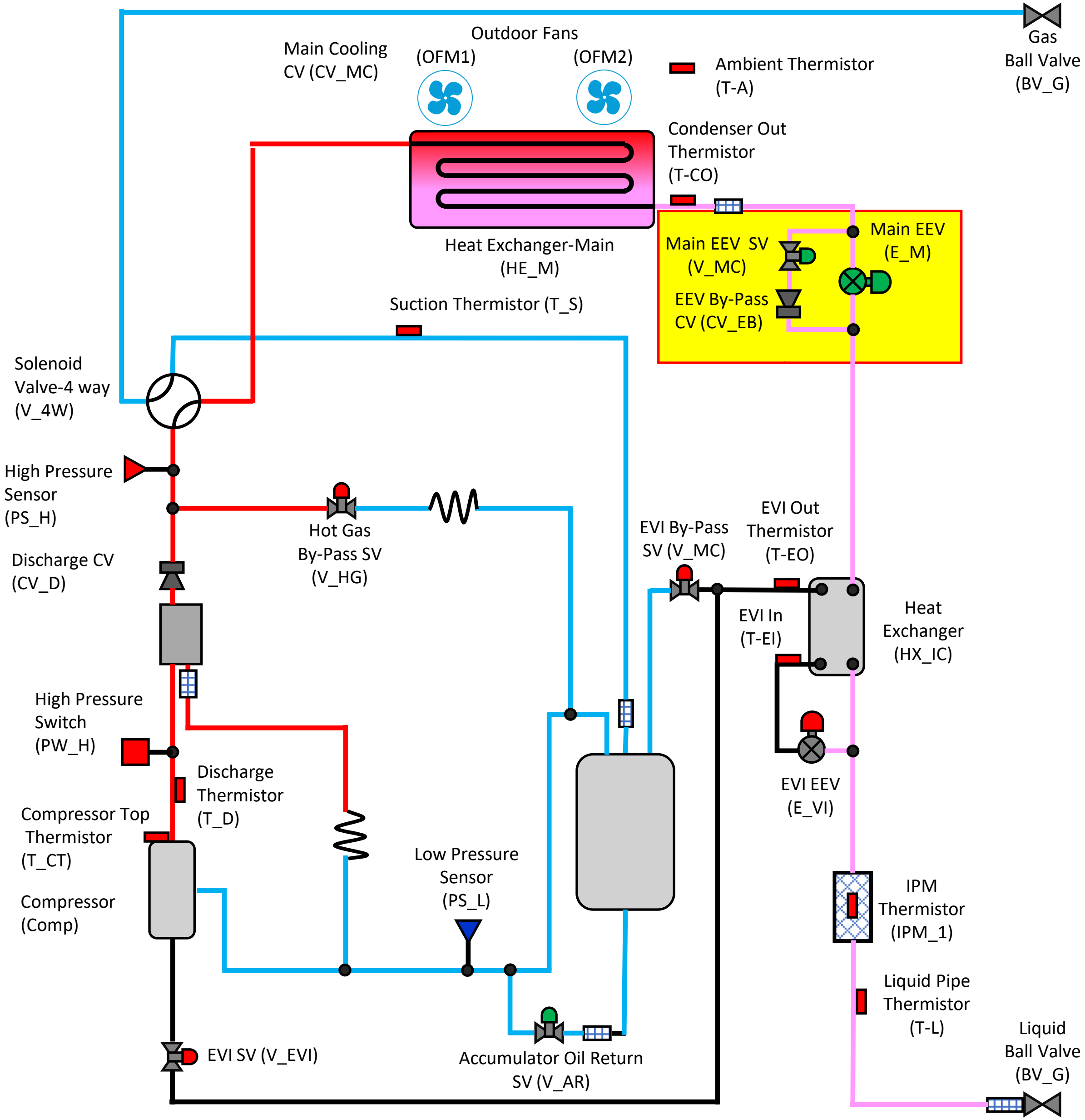


Main EEV & Solenoid Valve Cooling Modes

- The main EEV solenoid valve is energized in the cooling modes.
- Under “Typical” cooling conditions the EEV is commanded to full open.
- Under low loads conditions or low ambient conditions, it may modulate to help build head pressure.
- In Cooling Main the valve will be commanded to a set position



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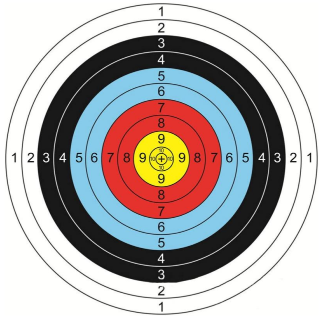
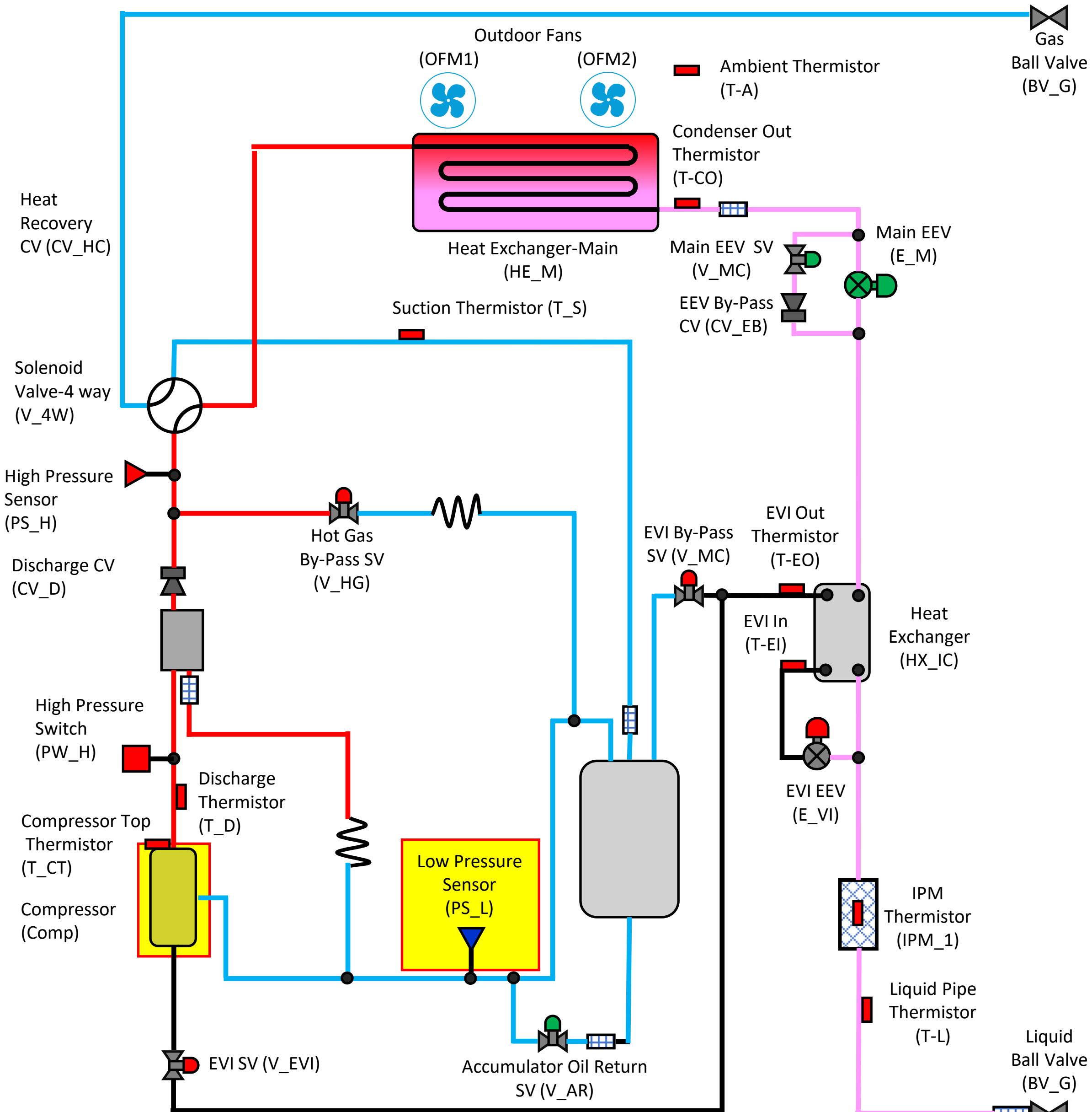
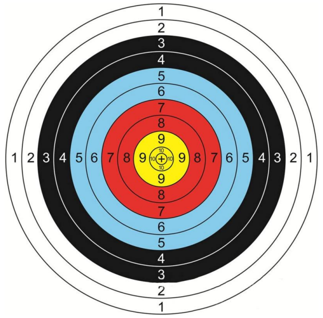


Low Psi Target in Cooling Mode

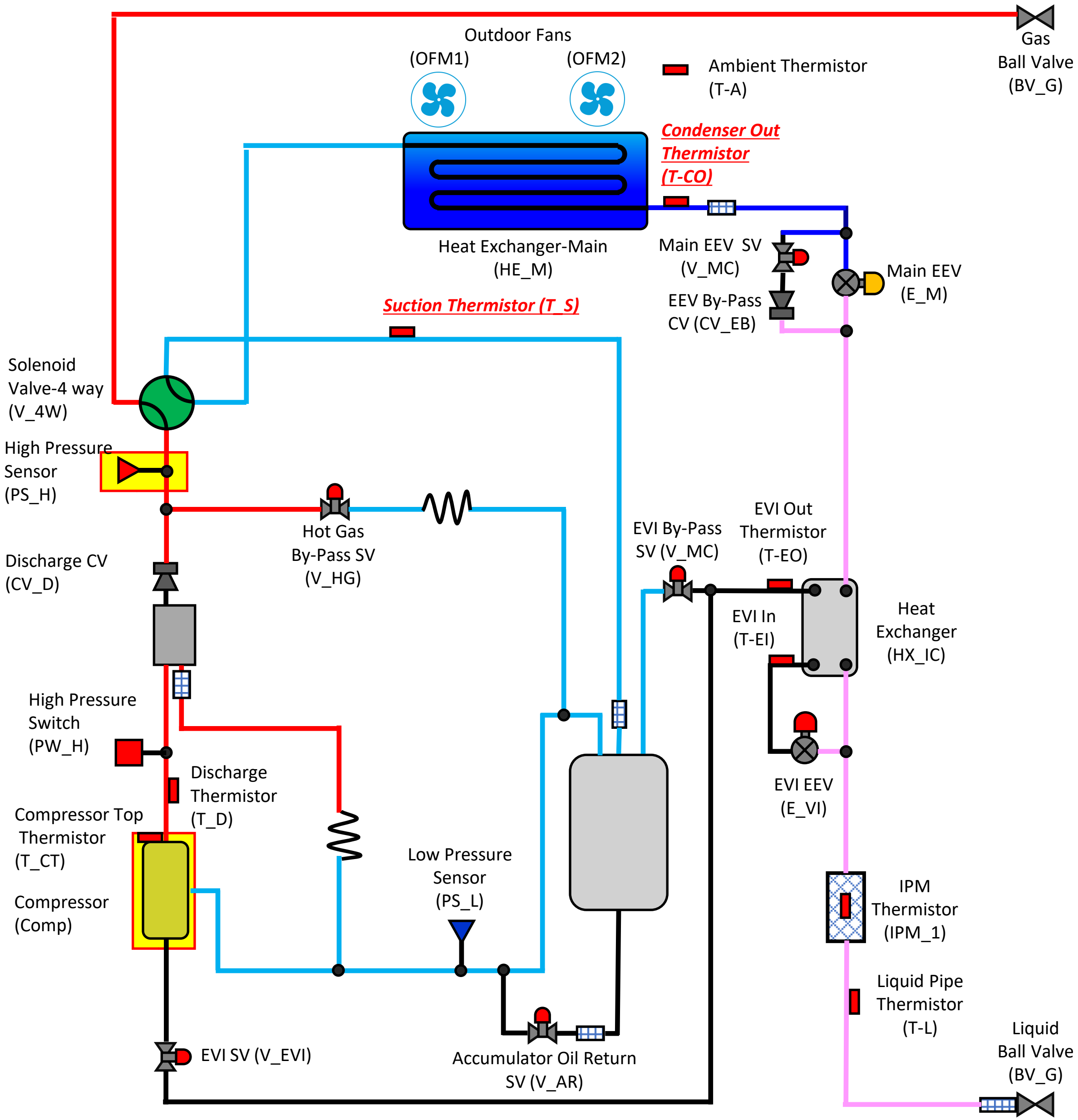
- The compressor will modulate in the cooling mode to maintain a suction pressure measured by the low-pressure sensor of 114 psig.
- Evap Saturation temperature may be higher than 39^0 due to pressure loss within the system





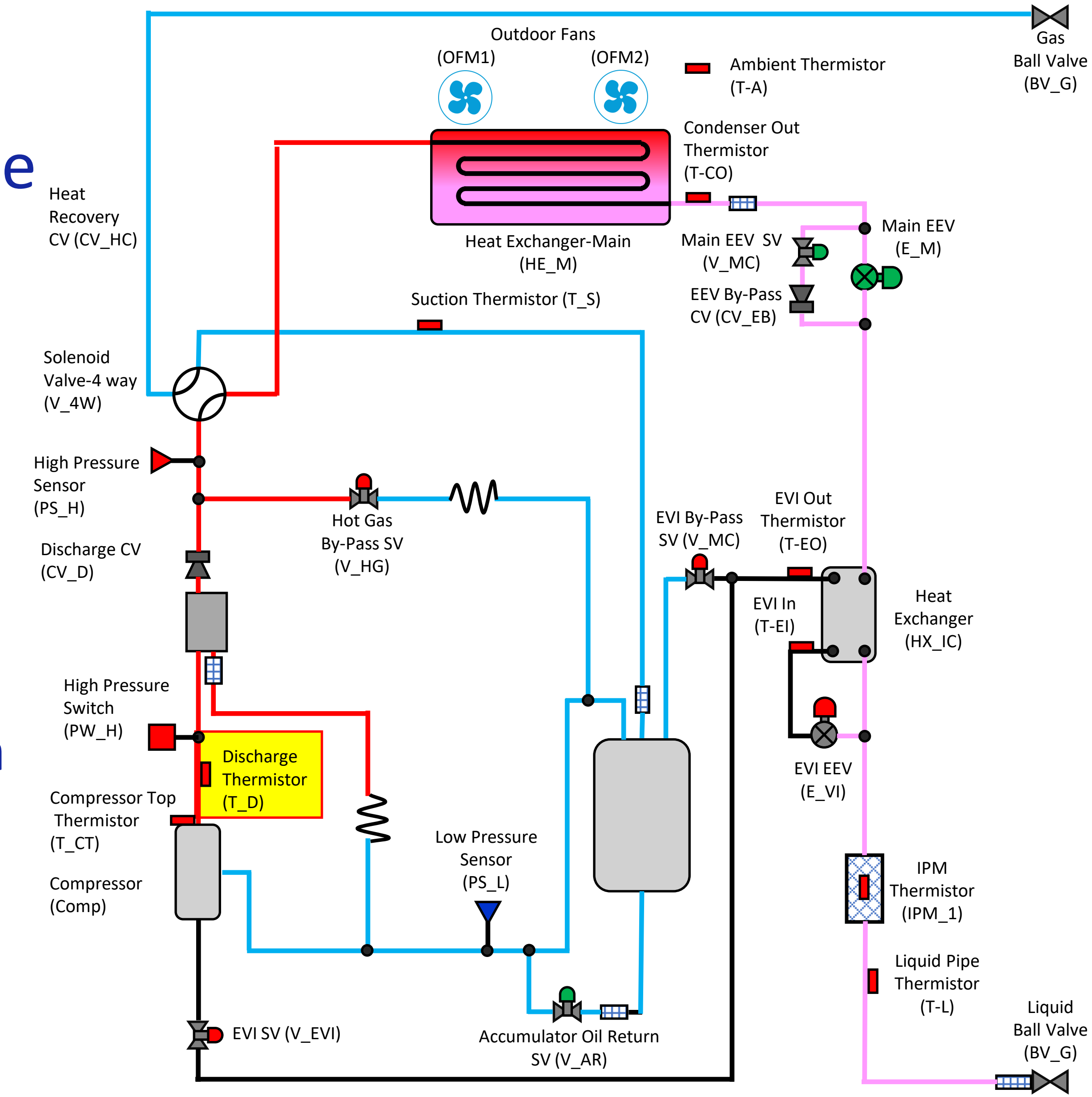
High Pressure Target in Heating Mode

- The compressor will modulate to maintain a target head pressure in the heating mode of 427 psig



Compressor Discharge Temperature Cooling Mode

- The compressor discharge thermistor is used to monitor and protect the compressor from flood back conditions and overheating conditions.
- The compressor discharge temperature is directly related to the compressor super heat. (low in low out)
- Typical discharge temperatures in the cooling mode range between 120-170 degrees.



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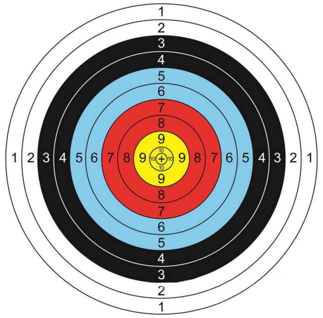
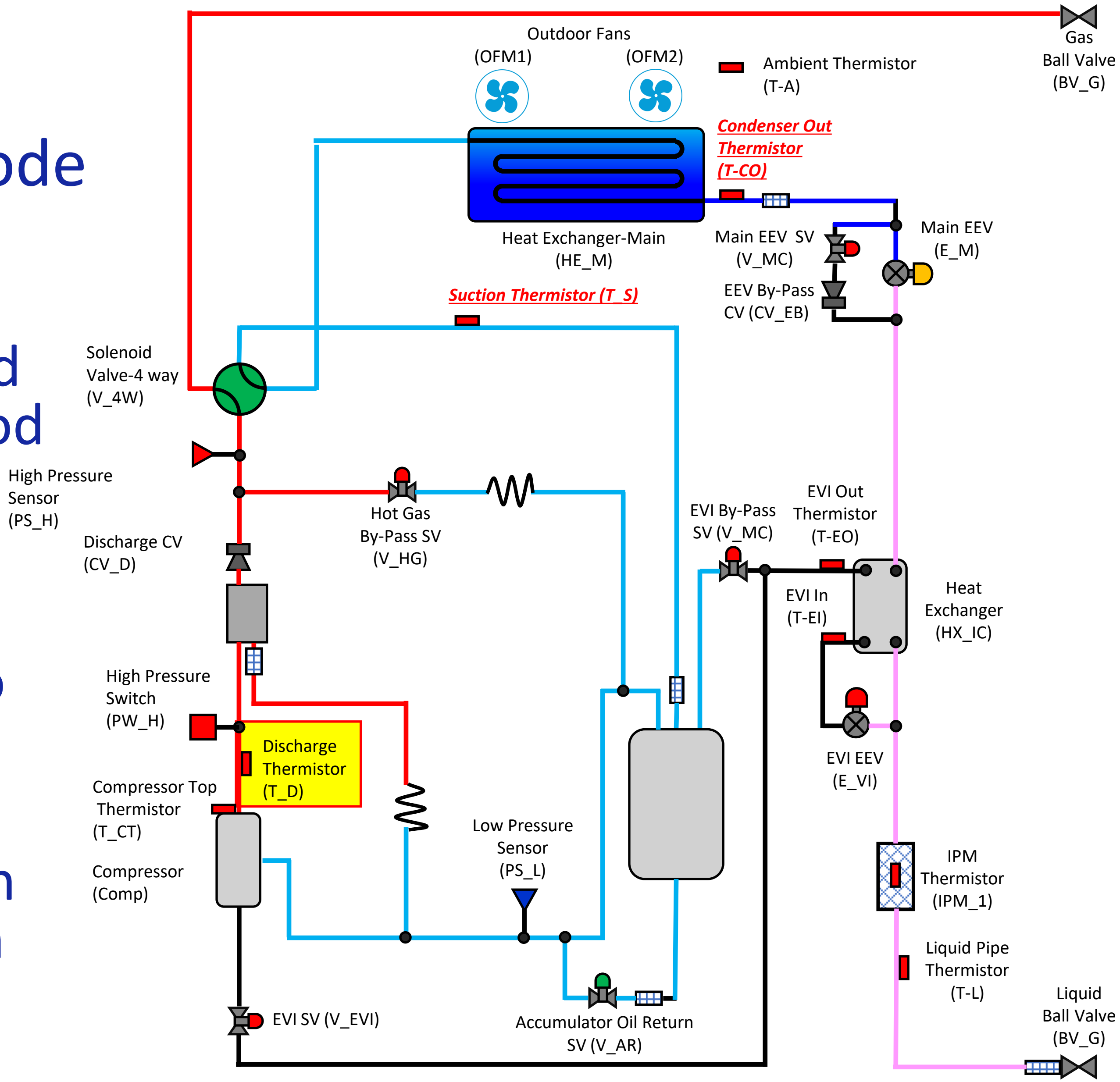
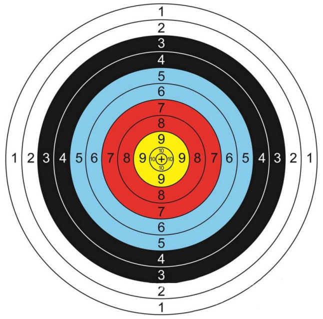


Compressor Discharge Temperature Heating Mode

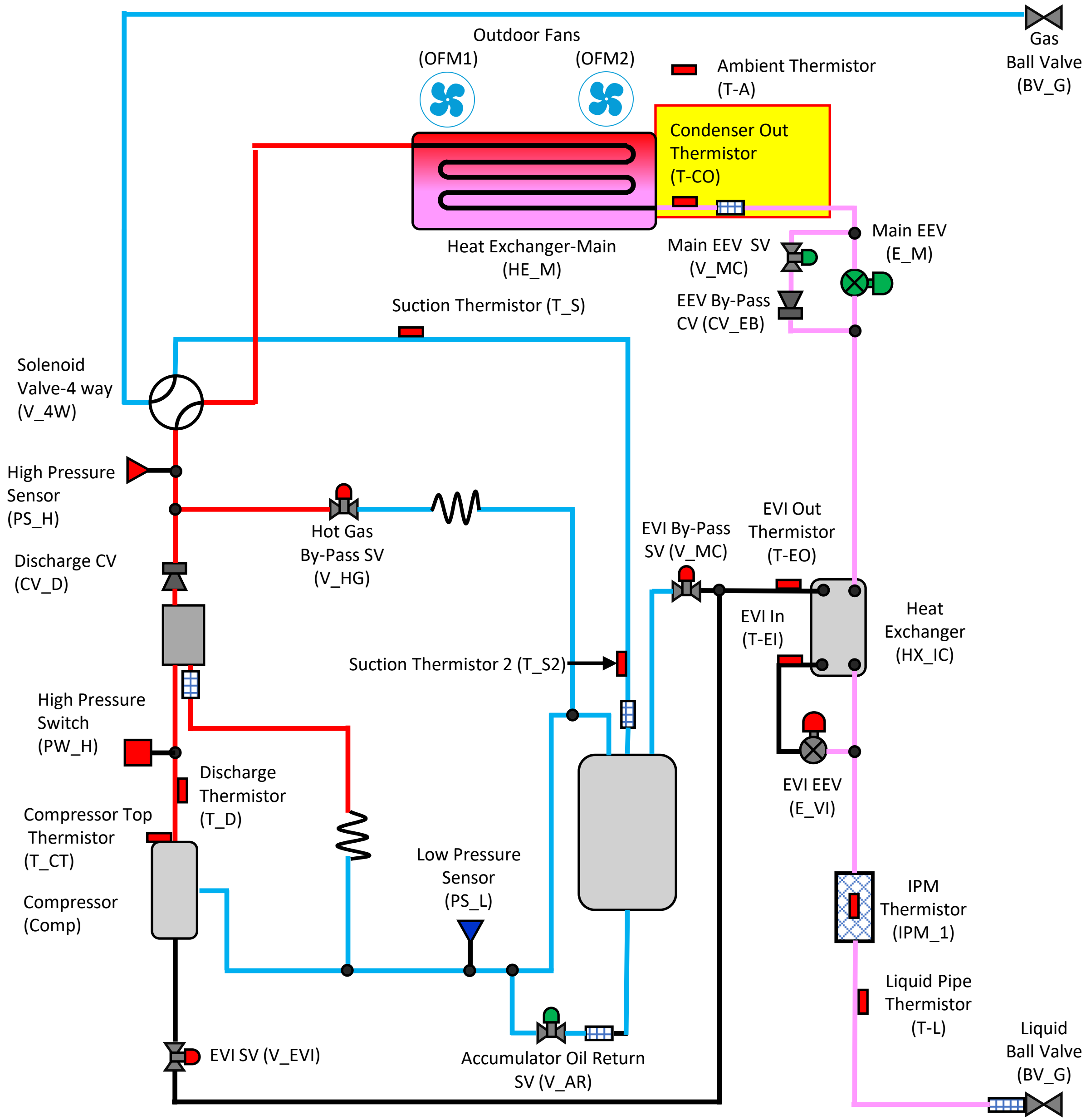
- The compressor discharge thermistor is used to monitor and protect the compressor from flood back conditions and overheating conditions.
- The compressor discharge temperature is directly related to the compressor super heat. (low in low out)
- Typical discharge temperatures in the cooling mode range between 140-210 degrees





Condenser Out Cooling Mode

- The condenser out temperature should be between 5-36 degrees greater than outside air temperature (T-A).
- Normal Sub-Cooling:
  - H.P saturated temp - Cond out temp = 3.6^0 - 14.4^0F



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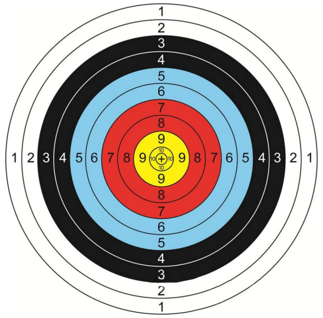
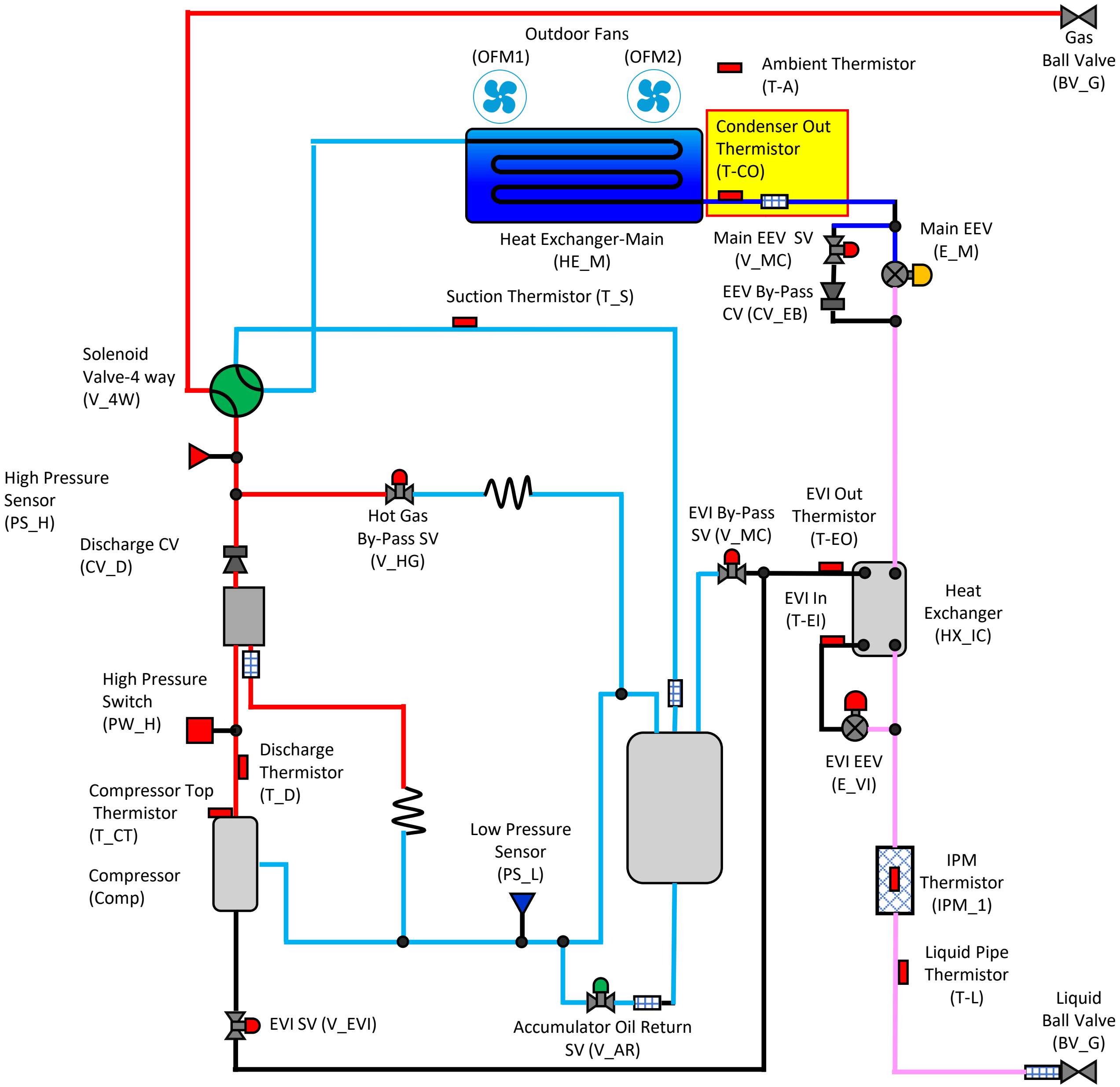
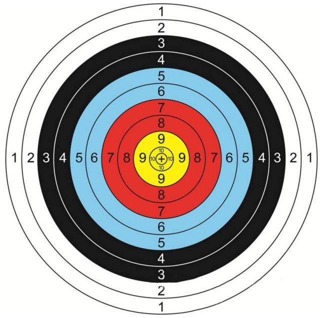


Condenser Out Heating Mode

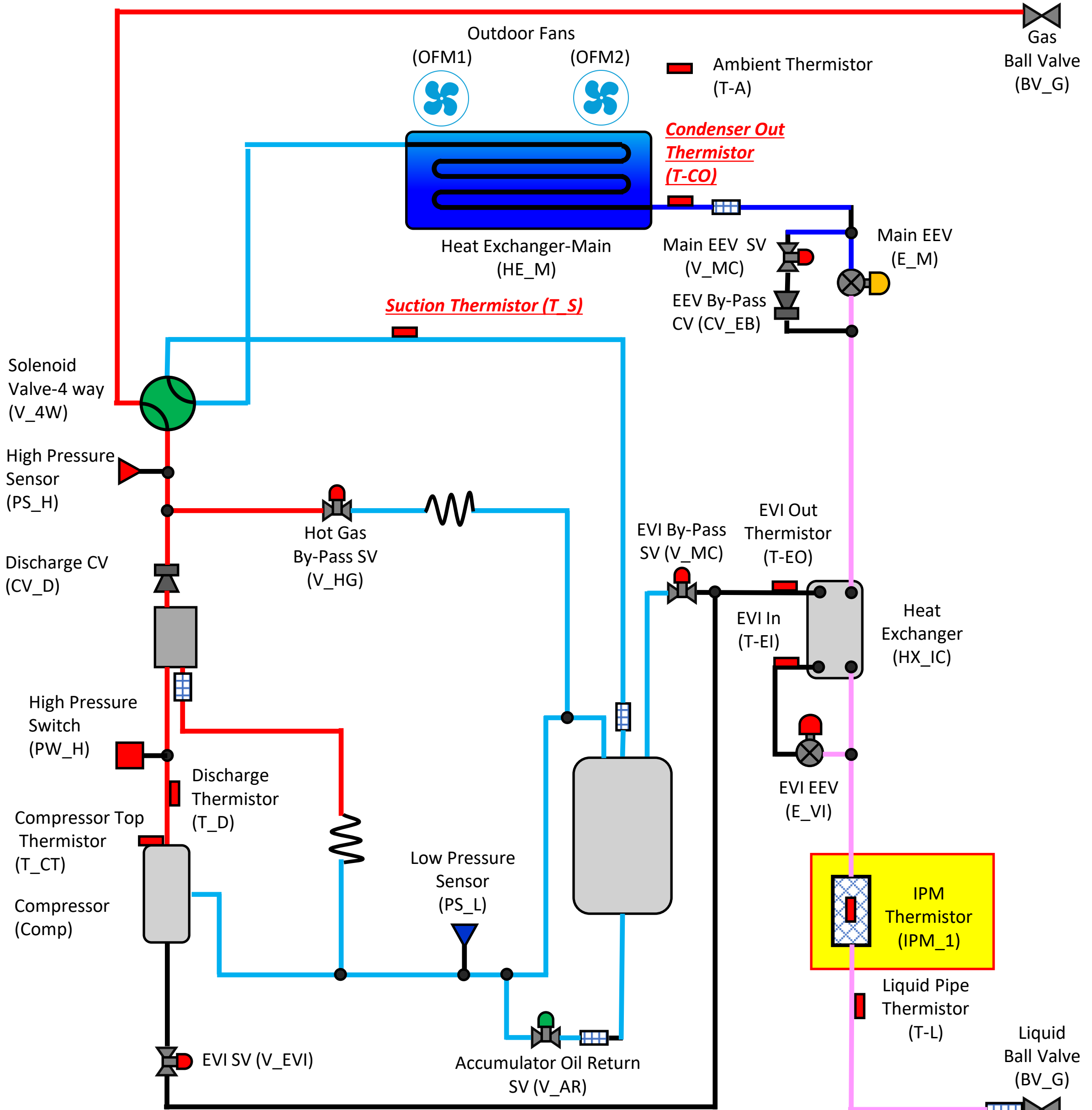
- Condenser out temperature should be at lest 2 degrees below outside air temperature .

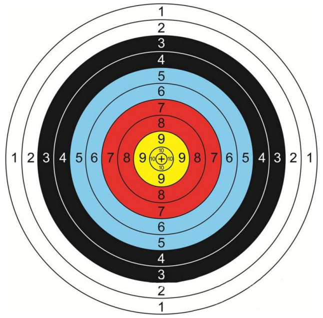




IPM Heat Sink

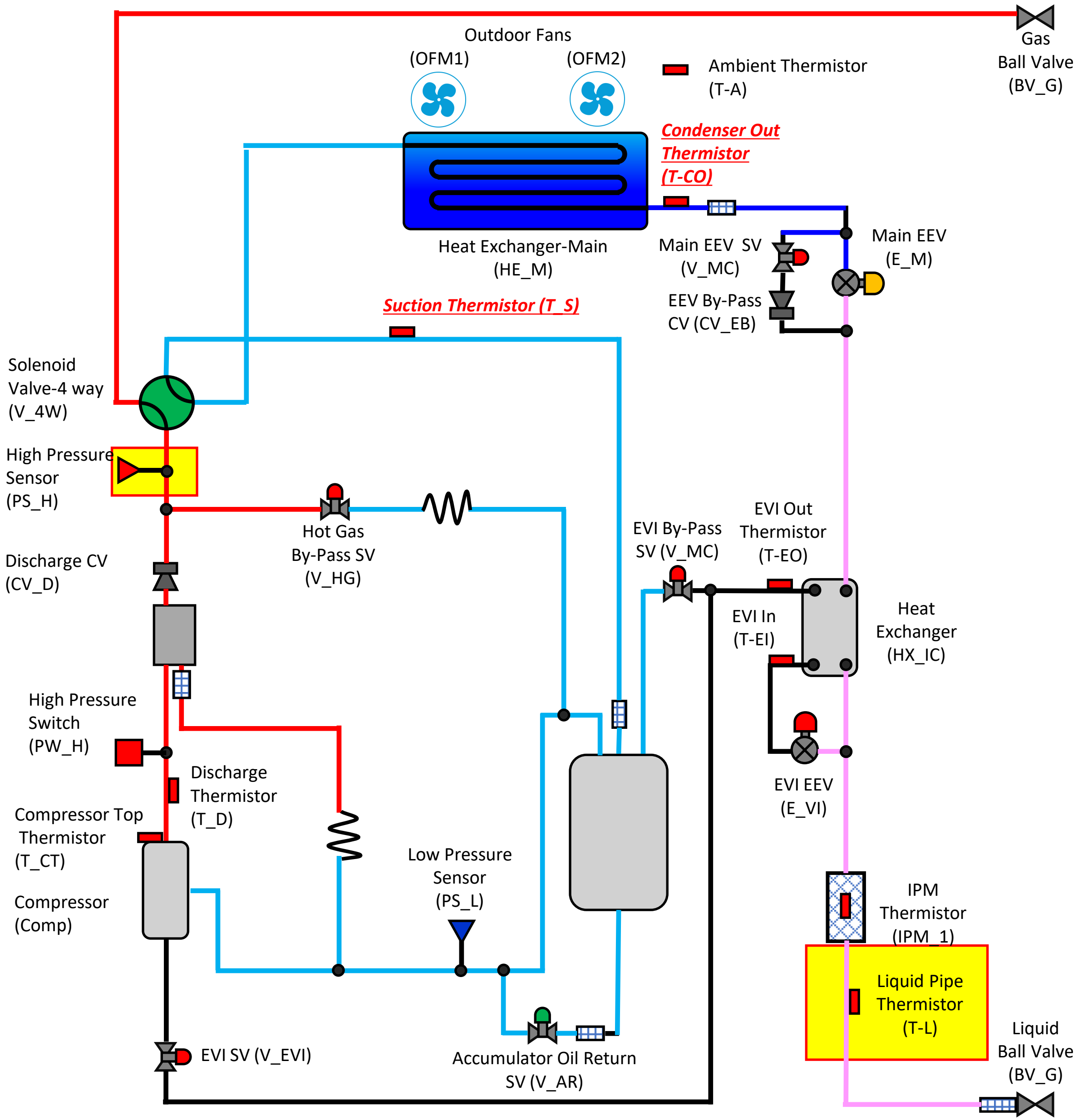
- The compressor IPM is cooled by the liquid refrigerant.
- Less than 194 degrees.
- If the IPM temperature exceeds 199.4^{0} F the system will modify operation to prevent overheating
- Compressor Hz will be lowered when temperature increases

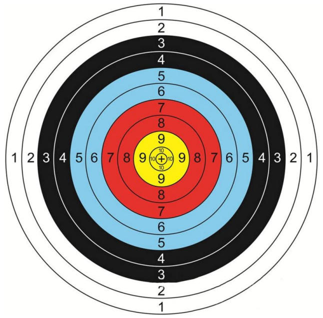




Liquid Line

- Sub-Cooling- Avg.
  - 9^0-36^0F
- Sub-cooling =
  - H.P saturated temp – Liquid tube temp

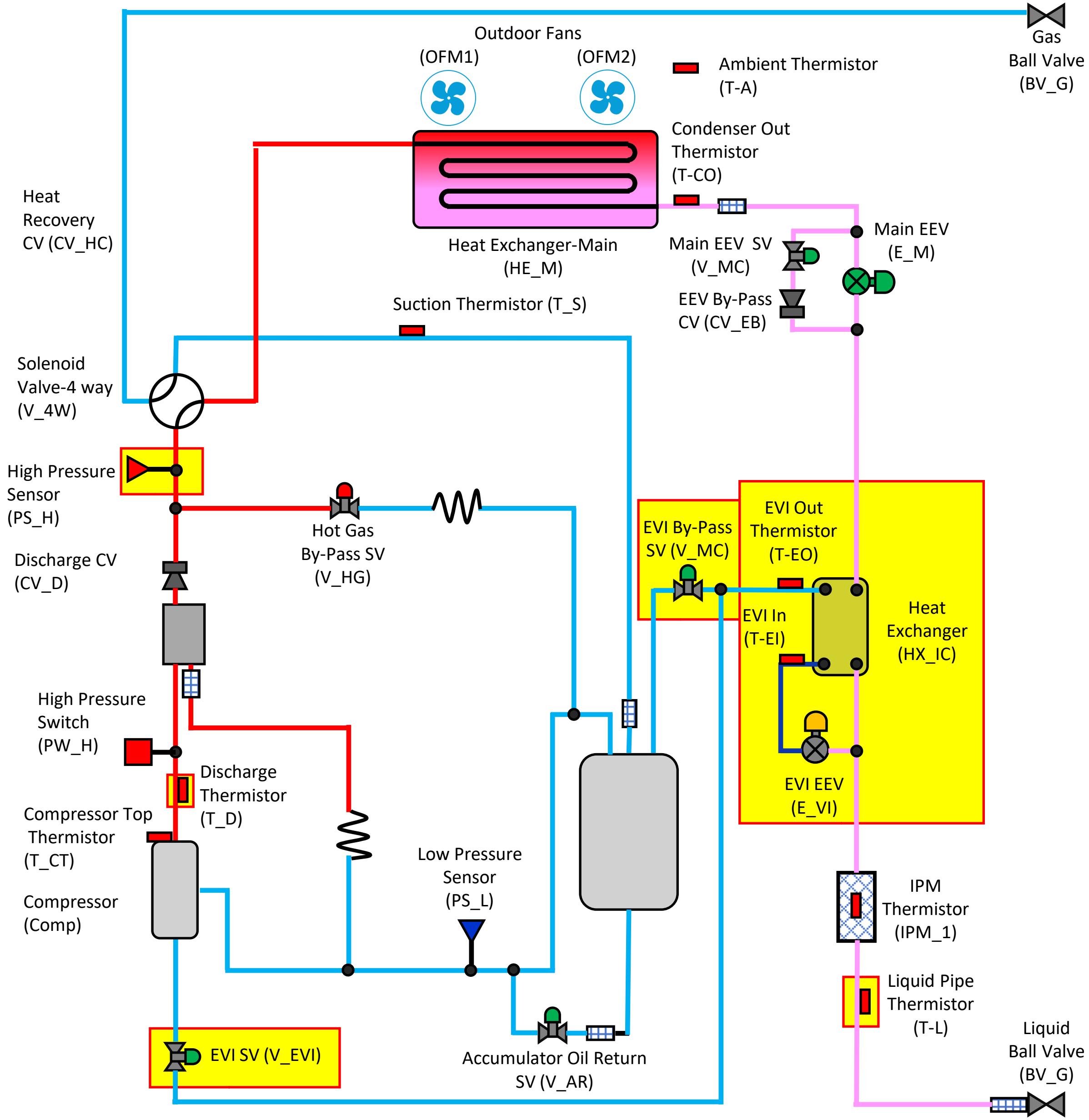




EVI Circuit
Cooling Mode
Normal Operation

- In the cooling mode the EVI circuit provides sub-cooled refrigerant to the indoor units.
- The target amount of sub-cooling measured at the liquid pipe thermistor is 9^0 -36^0
- The EVI EEV will modulate to maintain a 3.6^0 and 9^0 differential between EVI in and EVI out.

EVI Mode	Electronic Expansion Valve - EVI	Solenoid valve - EVI	Solenoid valve - EVI Bypass	*Condition
EVI Bypass mode	Open	Open	Open	Normal Operation
EVI On mode	Open	Open	Close	- All comp. in unit 70hz ↑ - Ambient Temp. < 96.8°F (cool) < 50°F (heat) - Control by unit
EVI Off mode	Close	Open	Open	- Safety start - Defrost



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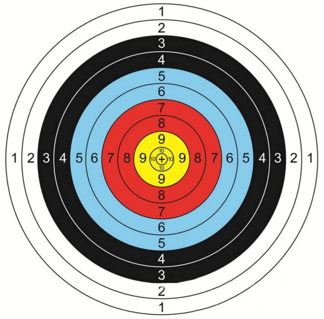
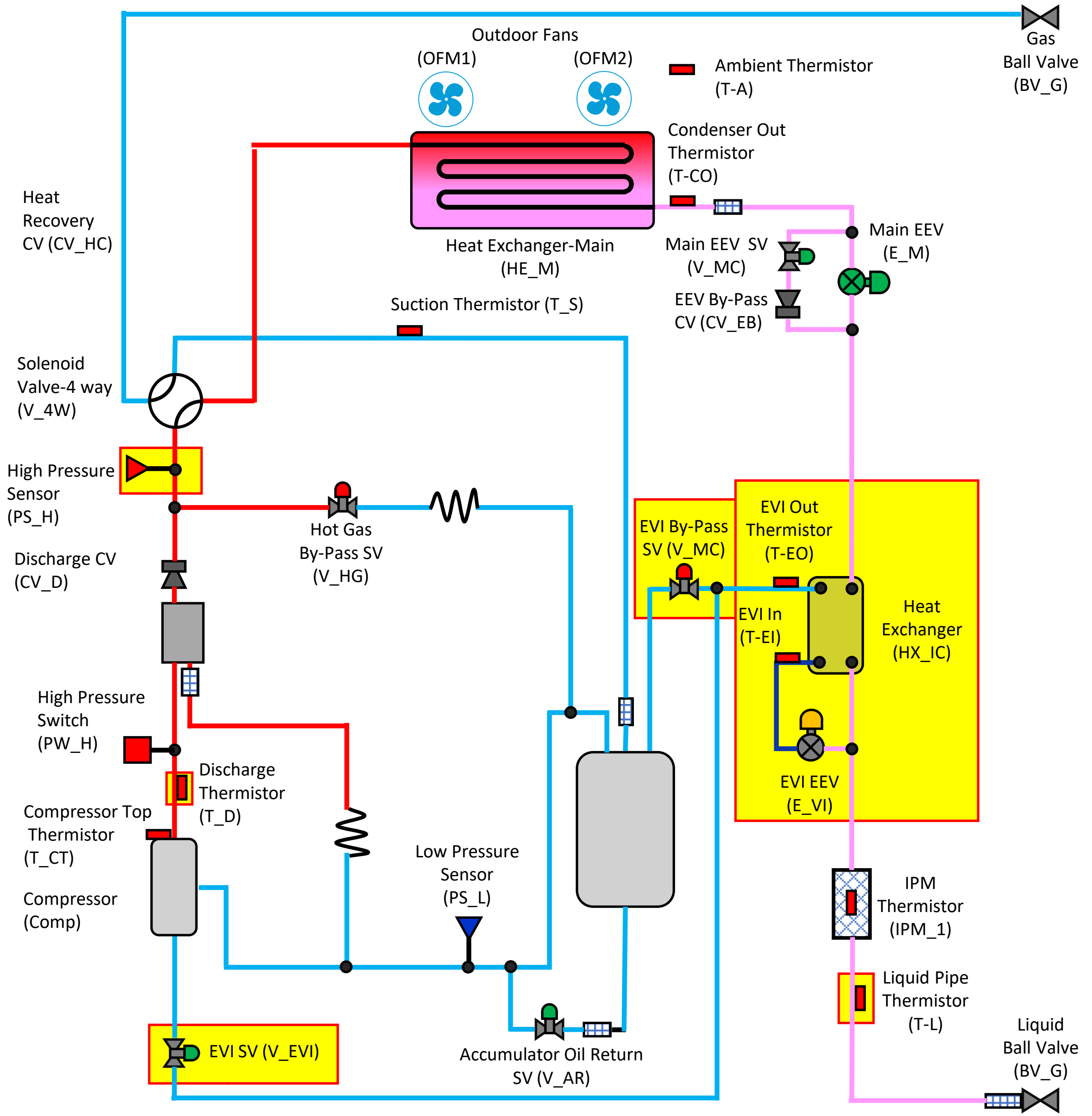
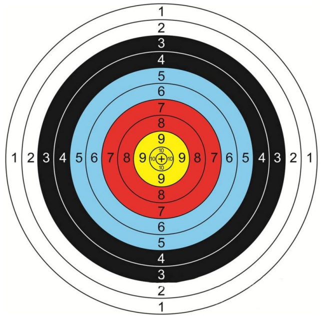


EVI Circuit
Cooling Mode
High Load Operation

- In high load condition the compressor needs to run at higher speeds.
- High speeds cause higher discharge temperatures.
- The EVI circuit is used for discharge temperature control.

EVI Mode	Electronic Expansion Valve - EVI	Solenoid valve - EVI	Solenoid valve - EVI Bypass	*Condition
EVI Bypass mode	Open	Open	Open	Normal Operation
EVI On mode	Open	Open	Close	- All comp. in unit 70hz ↑ - Ambient Temp. < 96.8°F (cool) < 50°F (heat) - Control by unit
EVI Off mode	Close	Open	Open	- Safety start - Defrost

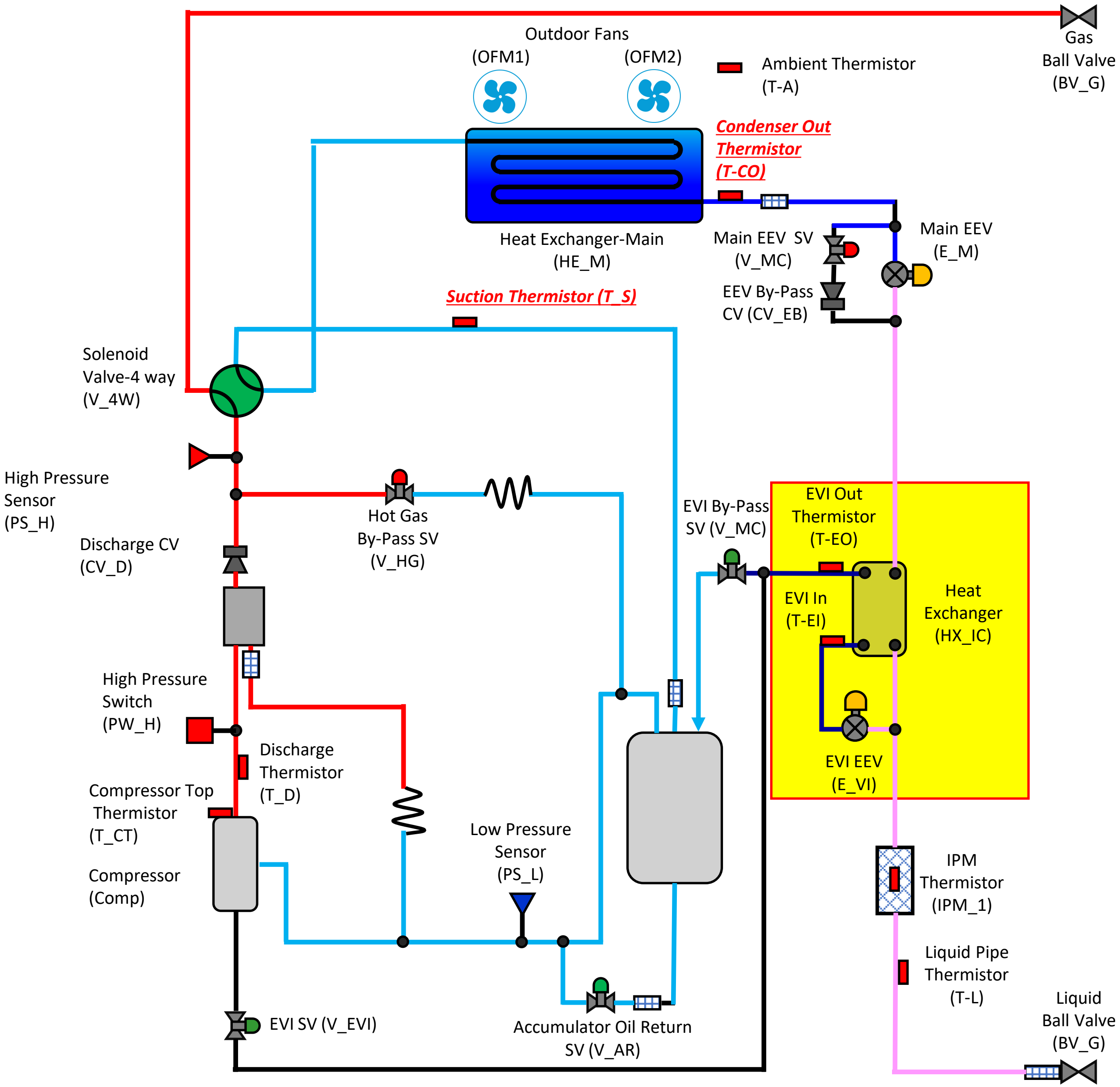




EVI Circuit Heating Mode

- Provides refrigerant to accumulator in heating mode for discharge temp. protection
- Heating mode operation
  - Open when
  - All compressors running at < 70 Hz
  - Target EVI Superheat:
    - (EVI out –EVI in) > 18^0
  - Discharge temp =194^0

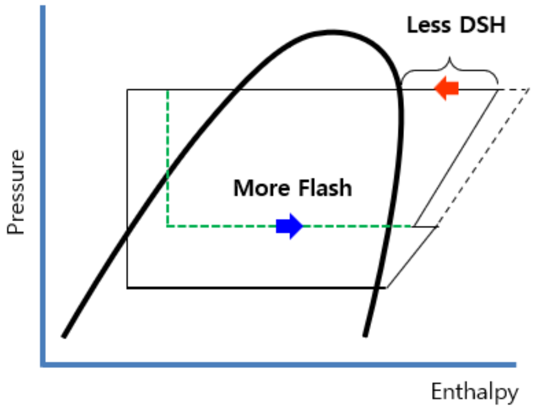
Heating
▷ SH control 1. EVI Out – EVI In temp. > 18°F (priority) 2. Tdis = 194°F

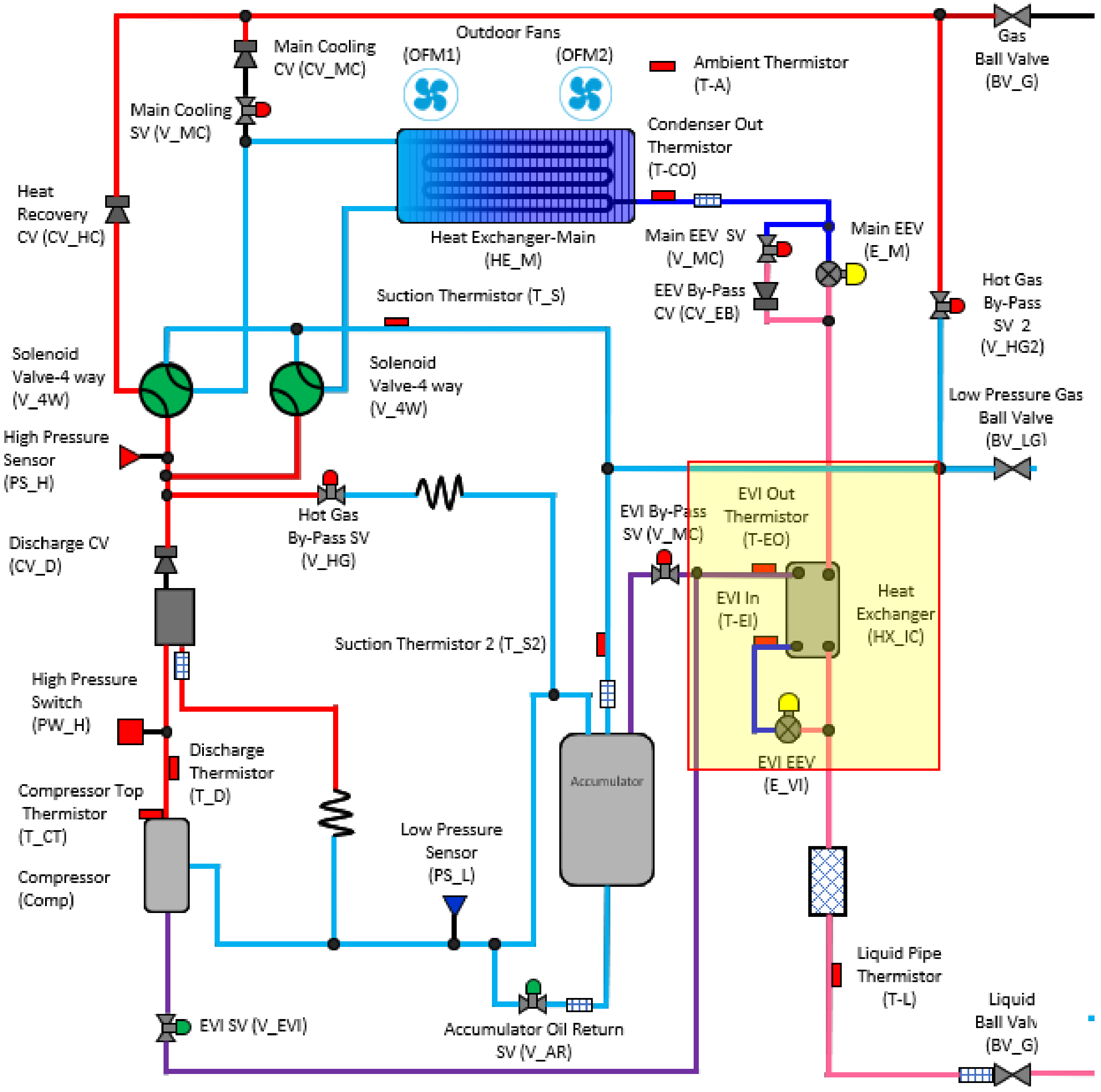


DVM S2 EVI Circuit Heating Mode

- Low ambient operation.
- EVI By-pass is closed.
- EVI EEV maintains discharge superheat within the range below.

EVI Mode	Electronic Expansion Valve - EVI	Solenoid valve - EVI	Solenoid valve - EVI Bypass	*Condition
EVI Bypass mode	Open	Open	Open	Normal Operation
EVI On mode	Open	Open	Close	- All compressors in unit 70hz ↑ - Ambient temp. < 96.8°F (cool) < 50°F (heat) * - Control by unit. EVI EEV: DSH control (68~113°F)
EVI Off mode	Close	Open	Open	- Safety start - Defrost





Understanding the need for Vapor- Flash Injection

IN HEATING MODE, THE COMPRESSOR RUNS TO MAINTAIN A HEAD PRESSURE OF APPROXIMATELY 427 PSIG.

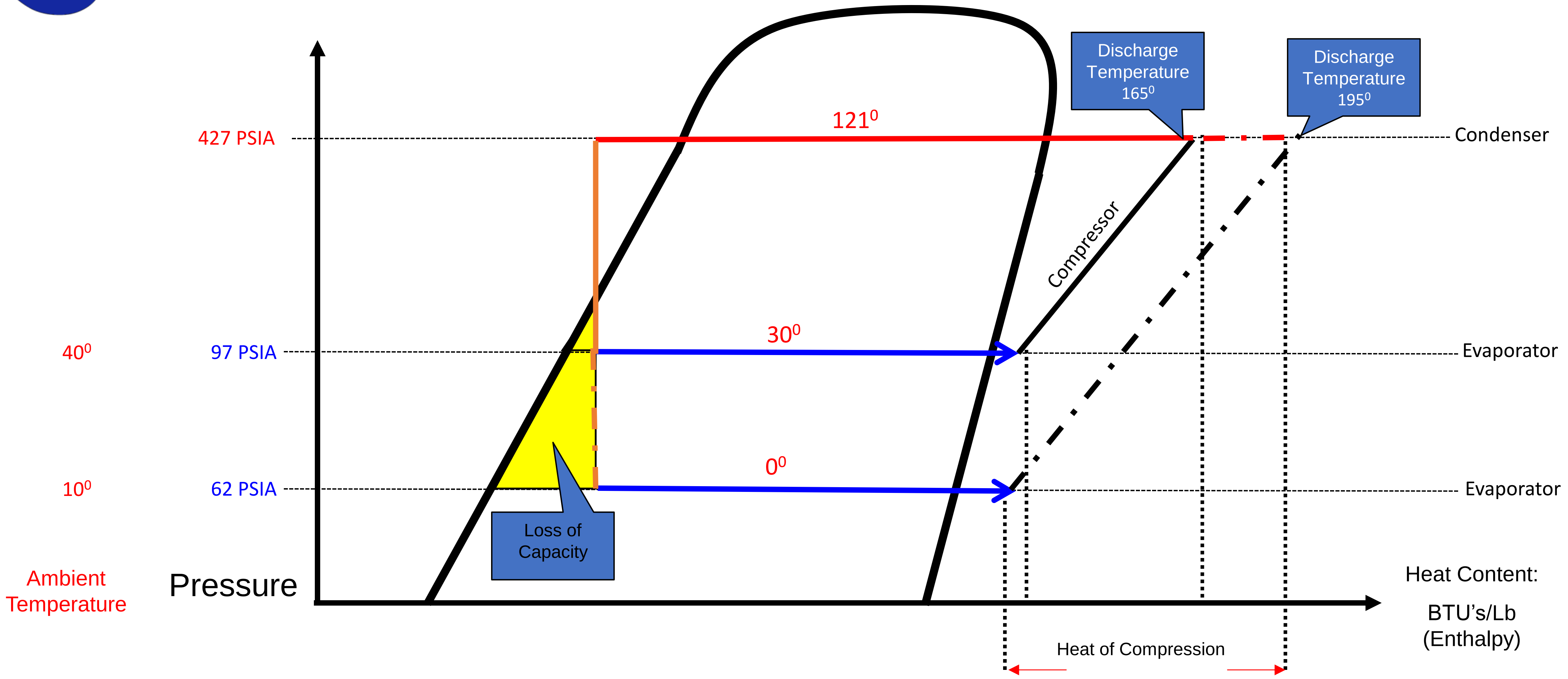
THE LOW SIDE PRESSURE IS DETERMINED BY THE OUTSIDE AIR TEMPERATURE.

THE LOWER THE AMBIENT TEMPERATURE THE LOWER THE SUCTION PRESSURE

WHEN SUCTION PRESSURE IS LOWERED THE COMPRESSION RATIO INCREASES

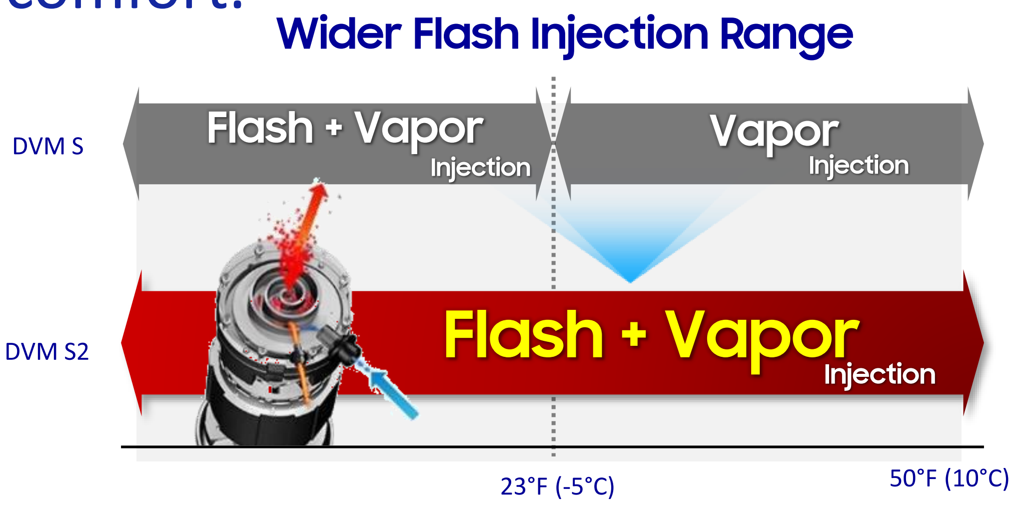
THE HIGHER COMPRESSION RATIOS CAUSE HIGHER DISCHARGE TEMPERATURES

Understanding the Vapor Injection Circuit

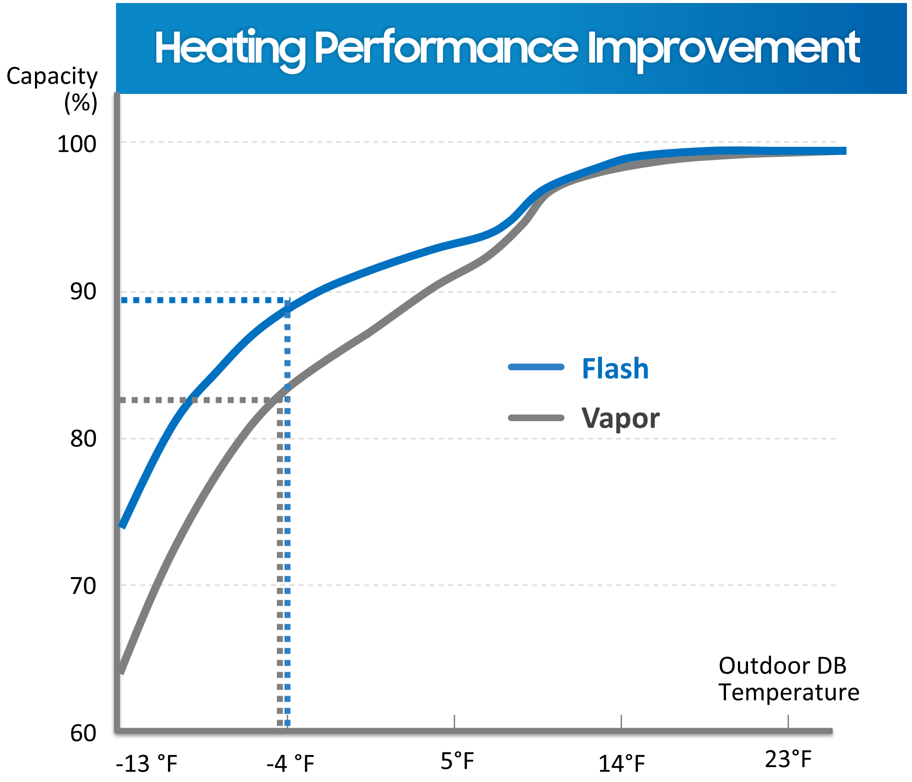


Flash + Vapor / Vapor Injection

- Samsung continues to develop specialized technologies for low temperature heating performance to provide customers with optimal comfort.



Flash injection technology increases the flow of refrigerant so the compressor continues to work reliably and improves heating performance at a lower temperature.

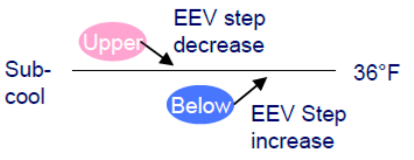
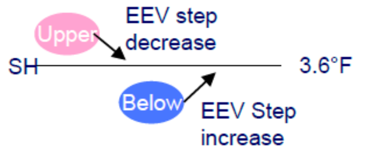


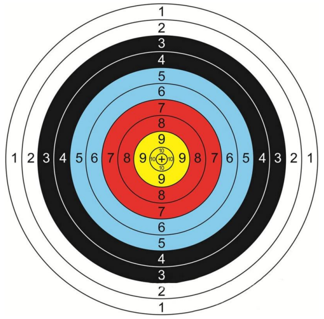
EVI Circuit

- **Purpose**
  - Control the EVI EEV to secure a subcooling degree in cooling mode.
  - Control the EVI EEV to optimize the performance in heating mode.
- **Concept**
  - In case you need additional refrigerant flow in the compressor during low temperature heating, or if you can operate compressor effectively by increasing the flow rate through the injection, it controls the EVI EEV in the EVI ON mode status.
  - In addition, in case of cooling, it controls EVI EEV when needs super cooling in order to deal with the installation conditions such as the long pipe installation or high head.
  - In EVI ON mode, optimizes the performance improvement through the injection by controlling the SH as 3.6°F and 9°F to EVI EEV.
  - When there is need of sub cooling in cooling mode, it controls the degree of sub cooling as 36°F.
  - The EVI EEV is regulated through PD control to secure the target subcooling degree in cooling mode.
  - The EVI EEV step is regulated through PD control to enable operation according to the target EVI SH (superheat) degree in heating mode.
  - The EVI EEV step is regulated to adjust the compressor discharge temperature (Td) to obtain the target Td if additional performance is required in the low temperature section while in heating mode.
- **Control specifications in detail**
  - Control the EVI EEV to secure a subcooling degree in cooling mode.

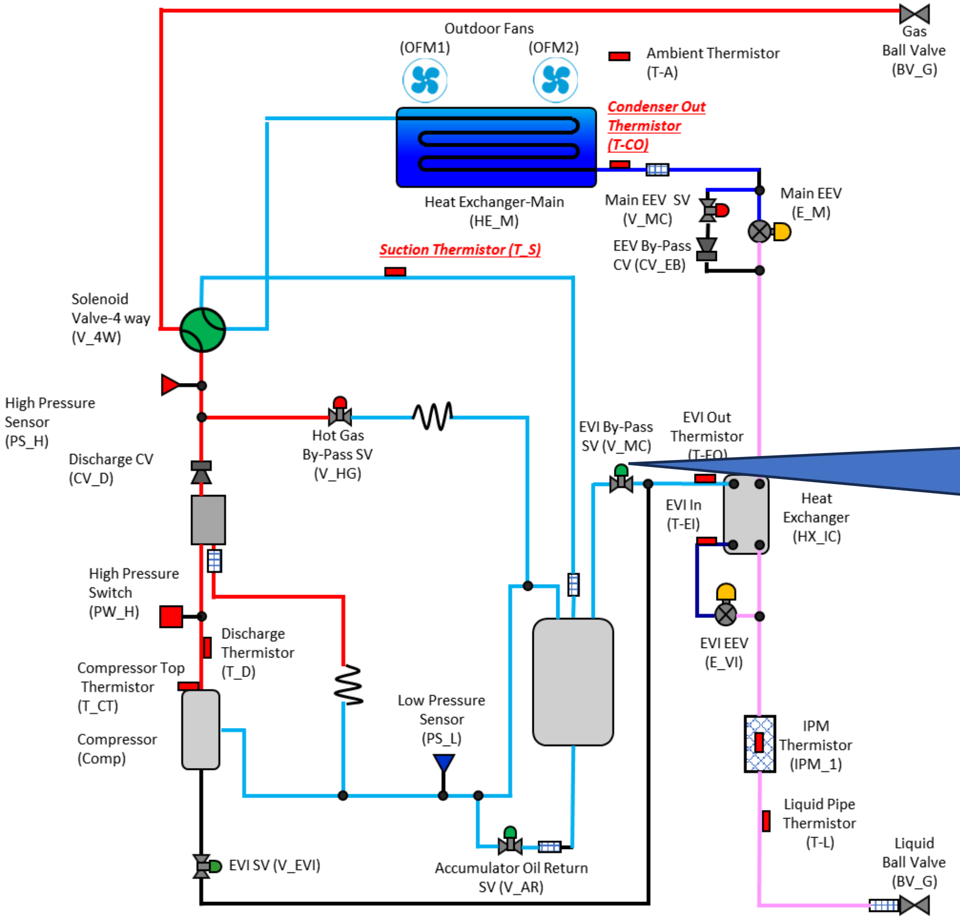
Flash & Vapor Injection

EVI Mode	Electronic Expansion Valve - EVI	Solenoid valve - EVI	Solenoid valve - EVI Bypass	*Condition
EVI Bypass mode	Open	Open	Open	Normal Operation
EVI On mode	Open	Open	Close	- All comp. in unit 70hz ↑ - Ambient Temp. < 96.8°F (cool) < 50°F (heat) - Control by unit
EVI Off mode	Close	Open	Open	- Safety start - Defrost

EVI mode	Cooling	Heating
EVI Bypass mode	▷ Sub-cooling control : 36°F (High pressure Saturated temp. – Liquid tube temp.)	▷ SH control 1. EVI Out – EVI In temp. > 18°F (priority) 2. Tdis = 194°F
EVI On mode	 	▷SH (EVI out temp. – EVI in temp.) control : 3.6°F 
Remark	※ If compressor discharge temperature is abnormally high, open EVI EEV to decrease the discharge temperature	



EVI Circuit Operation



Vapor-Flash Injection

All Comp at 70HZ or above

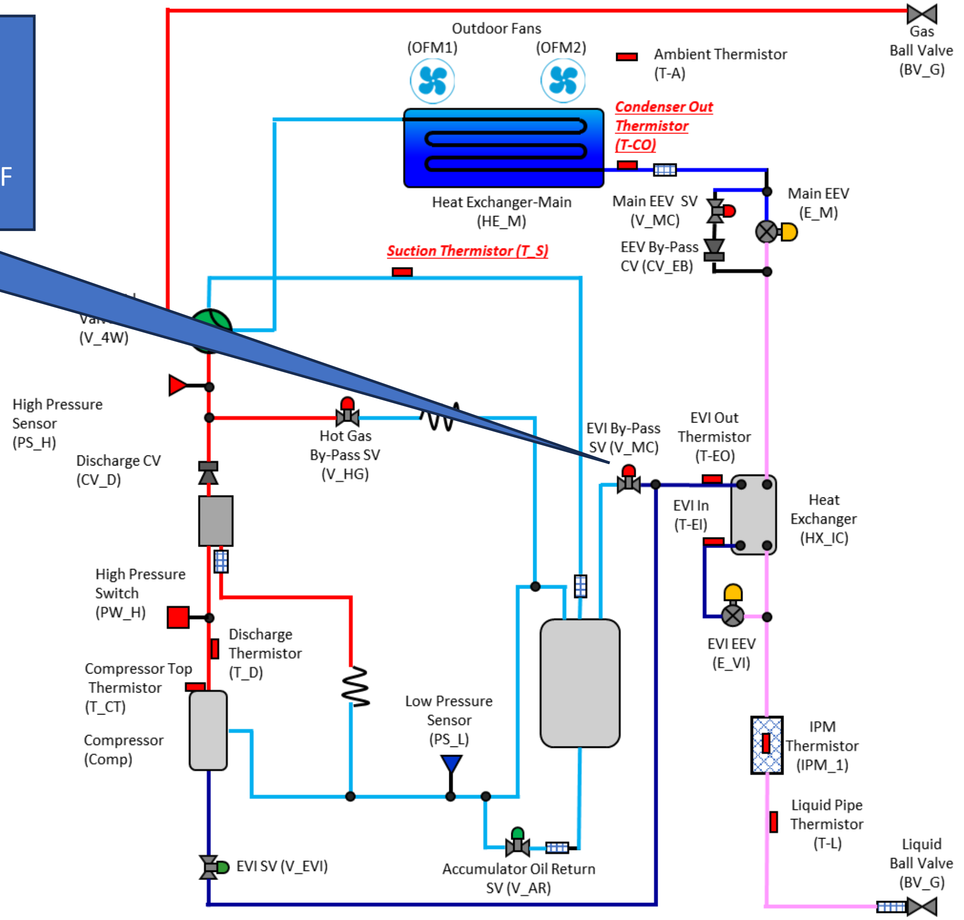
Ambient Temperature less than 50

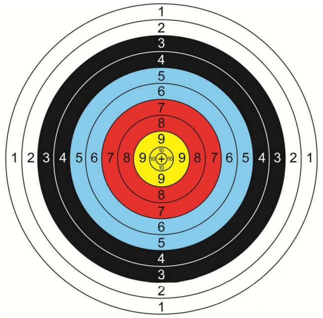
SH (EVI out temp. – EVI intemp.) control : 3.6°F

By-Pass Mode

SH control

1. EVI Out – EVI In temp. > 18°F (priority)
2. Tdis = 194°F



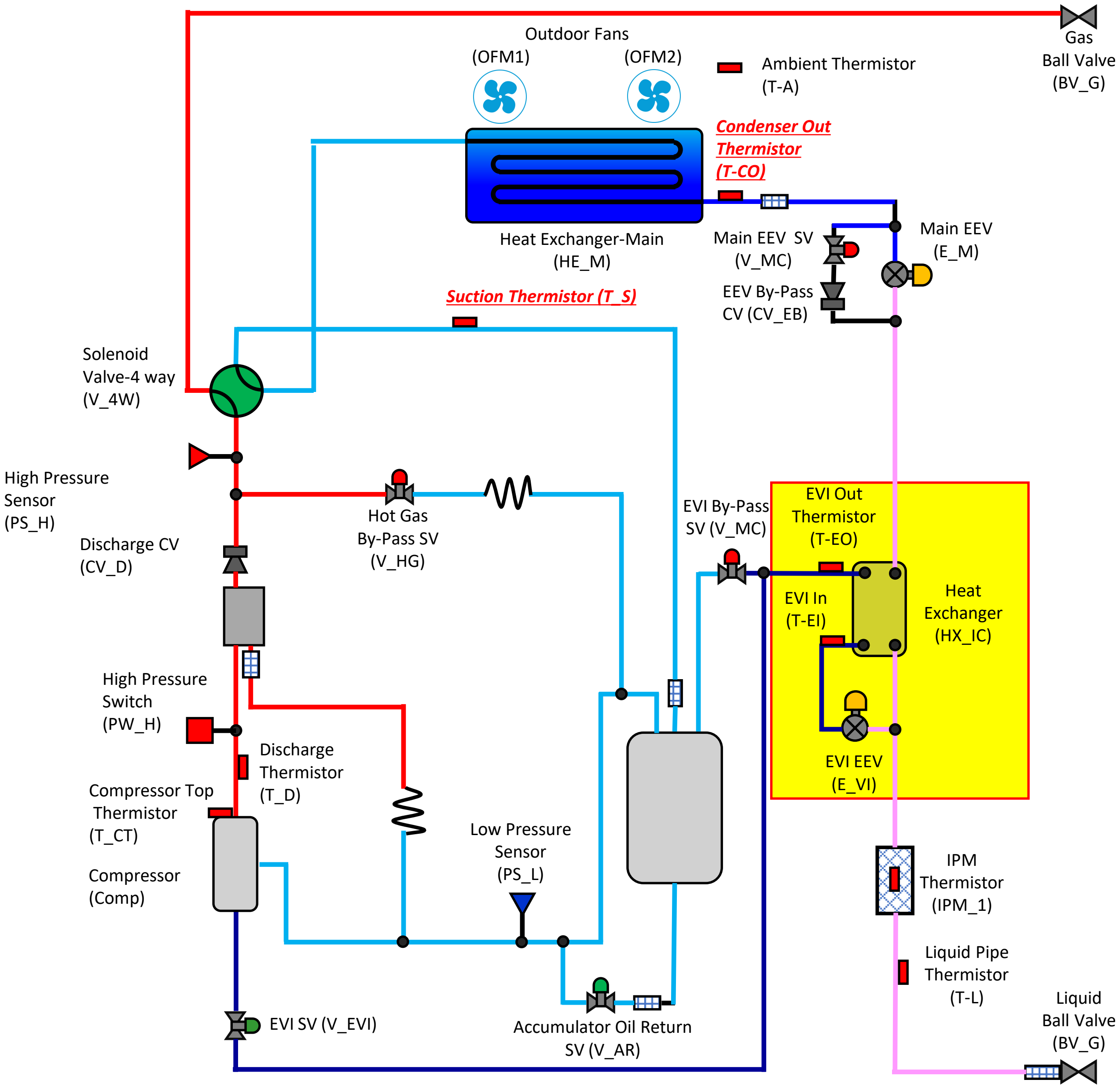


EVI Circuit Heating Mode

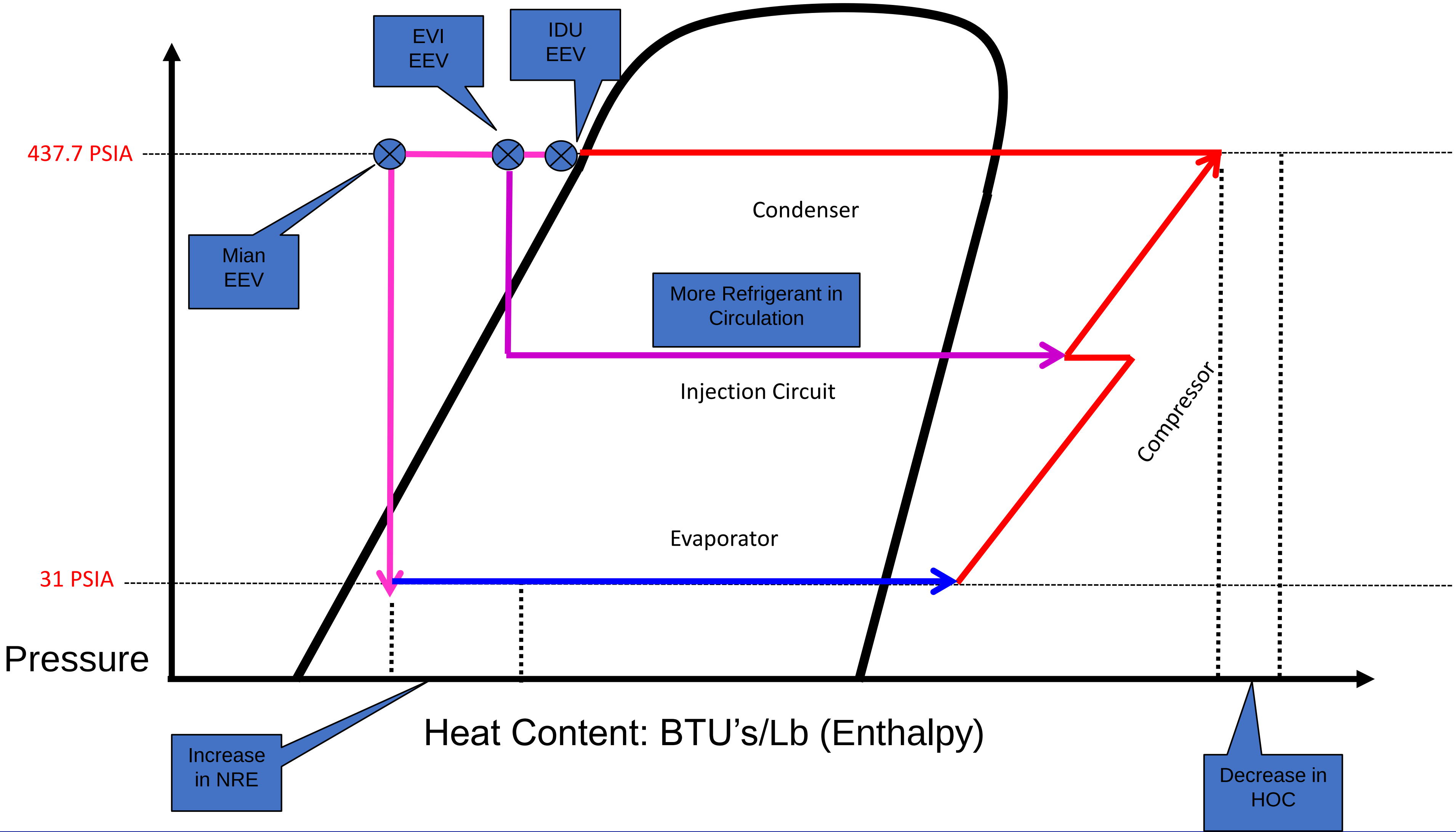
On (Open): *T_a > 100.4°F or 50Hz↓
Off (Close): T_a < 96.8°F↓ and 70Hz↑

- Provides refrigerant to the vapor injection circuit of the compressor in the heating mode.
- Heating mode operation
    - Ambient temperature below 50degrees
    - All compressors running at 70 Hz +
    - Target EVI Superheat = 3.6 degrees (single compressor)
    - 9 degrees superheat (multi compressors

EVI Mode	Electronic Expansion Valve - EVI	Solenoid valve - EVI	Solenoid valve - EVI Bypass	*Condition
EVI Bypass mode	Open	Open	Open	Normal Operation
EVI On mode	Open	Open	Close	- All comp. in unit 70hz ↑ - Ambient Temp. < 96.8°F (cool) < 50°F (heat) - Control by unit
EVI Off mode	Close	Open	Open	- Safety start - Defrost



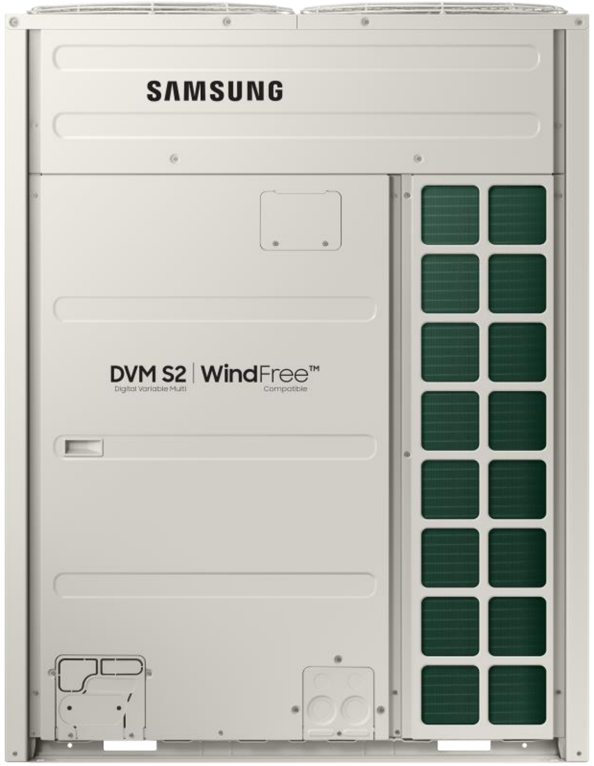
Vapor Injection Circuit



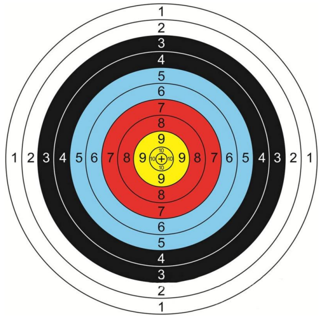
Heat recovery systems

Single Phase ECO

3 Phase DVM S

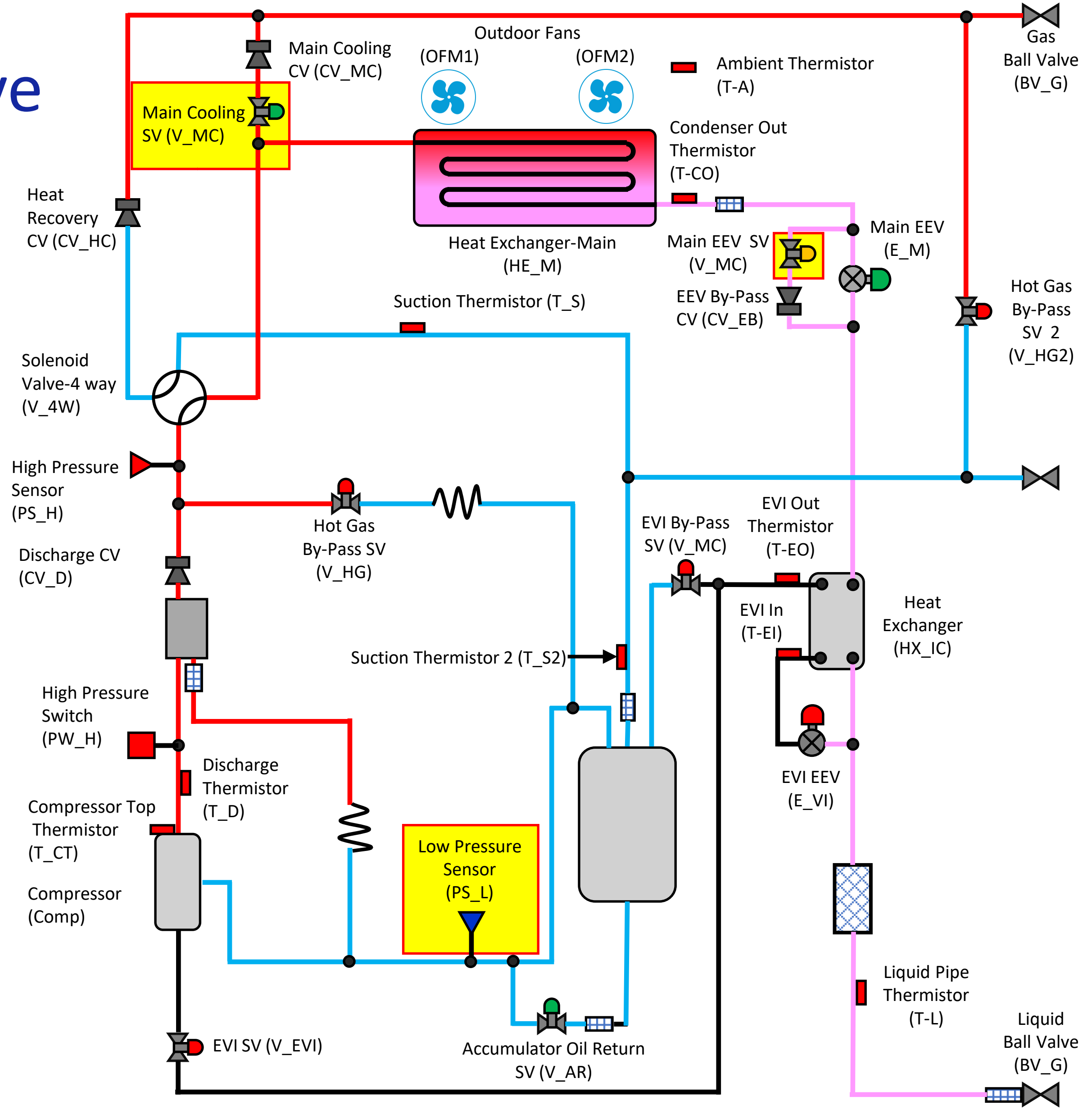


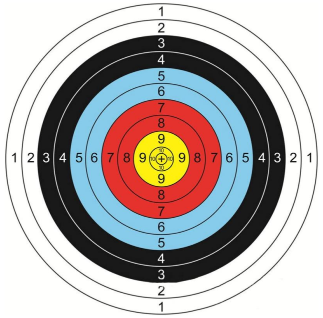
3 Phase DVM S2



Cooling Main Solenoid Valve
Cooling Main Mode

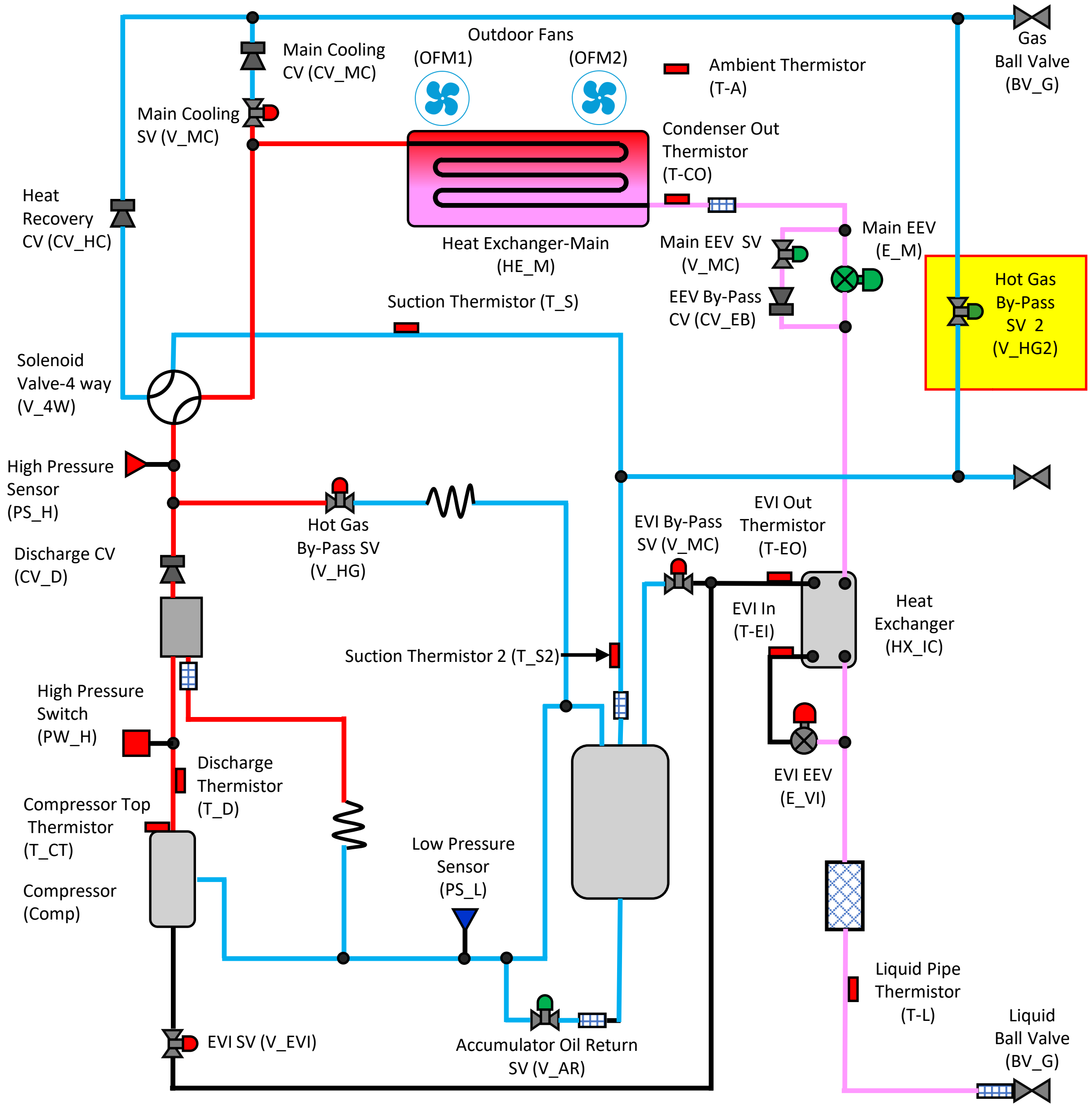
- The Cooling main solenoid is used in a heat recovery system.
- It is energized when the system is in mixed mode of operation and the outdoor unit is in the cooling mode (Cooling Main Operation)
- When energized it sends hot gas to the MCU or HR Changer depending on model of ODU.
- This allows the units calling for heat to get hot gas.





Hot Gas By-Pass 2 Cooling Only Mode

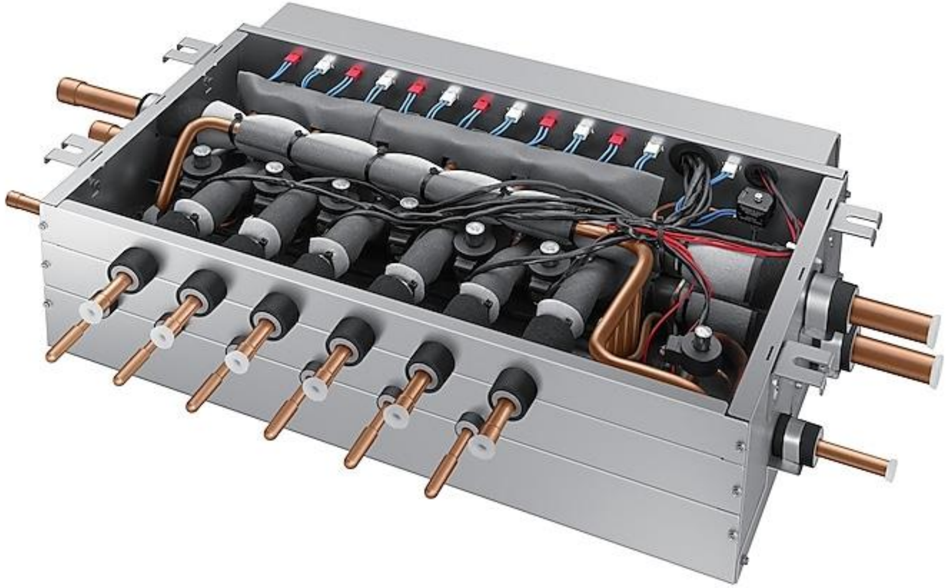
- The valve is energized in the cooling only mode after the system has operated for 20 minutes.
- This helps prevent refrigerant logging out in the system



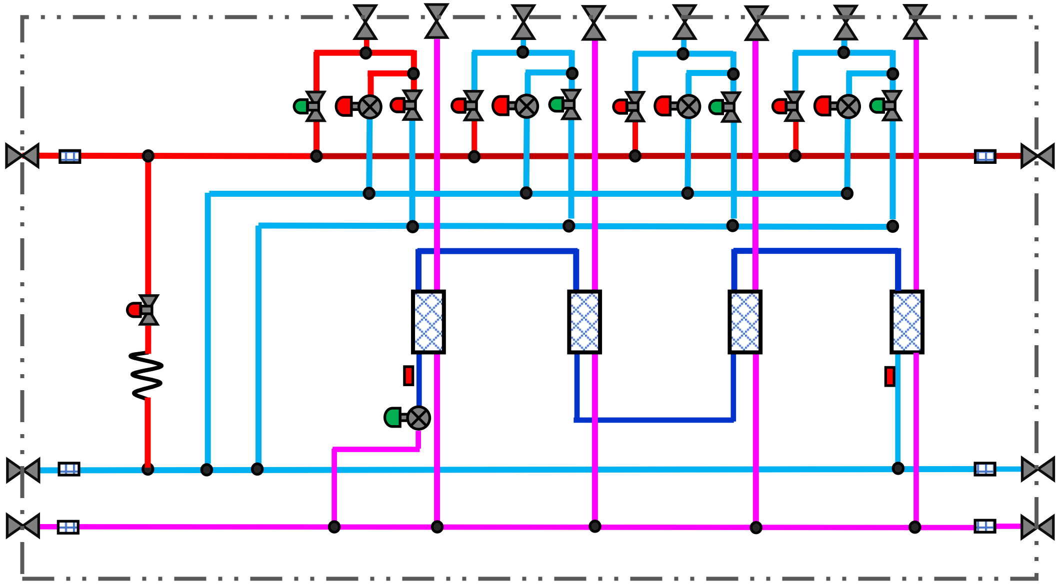
MCU and HR Changer

Refrigerant Components

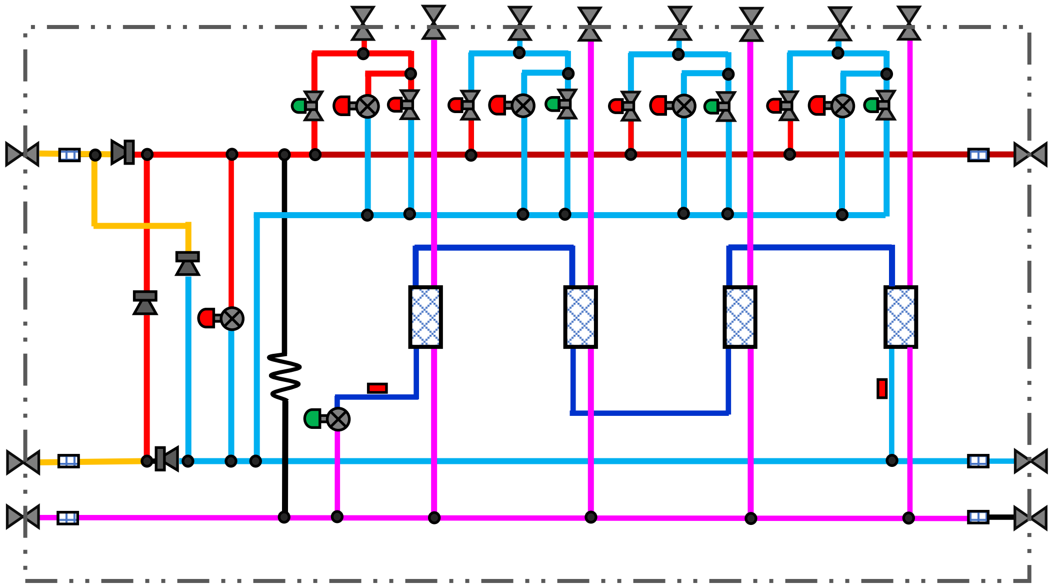
- MUC’s and HR Changer’s primary components and function are essentially the same.
- In this section we will discus the operation of the components within the MUC and HR Changer



MCU vs HR Changer Core Components



MCU



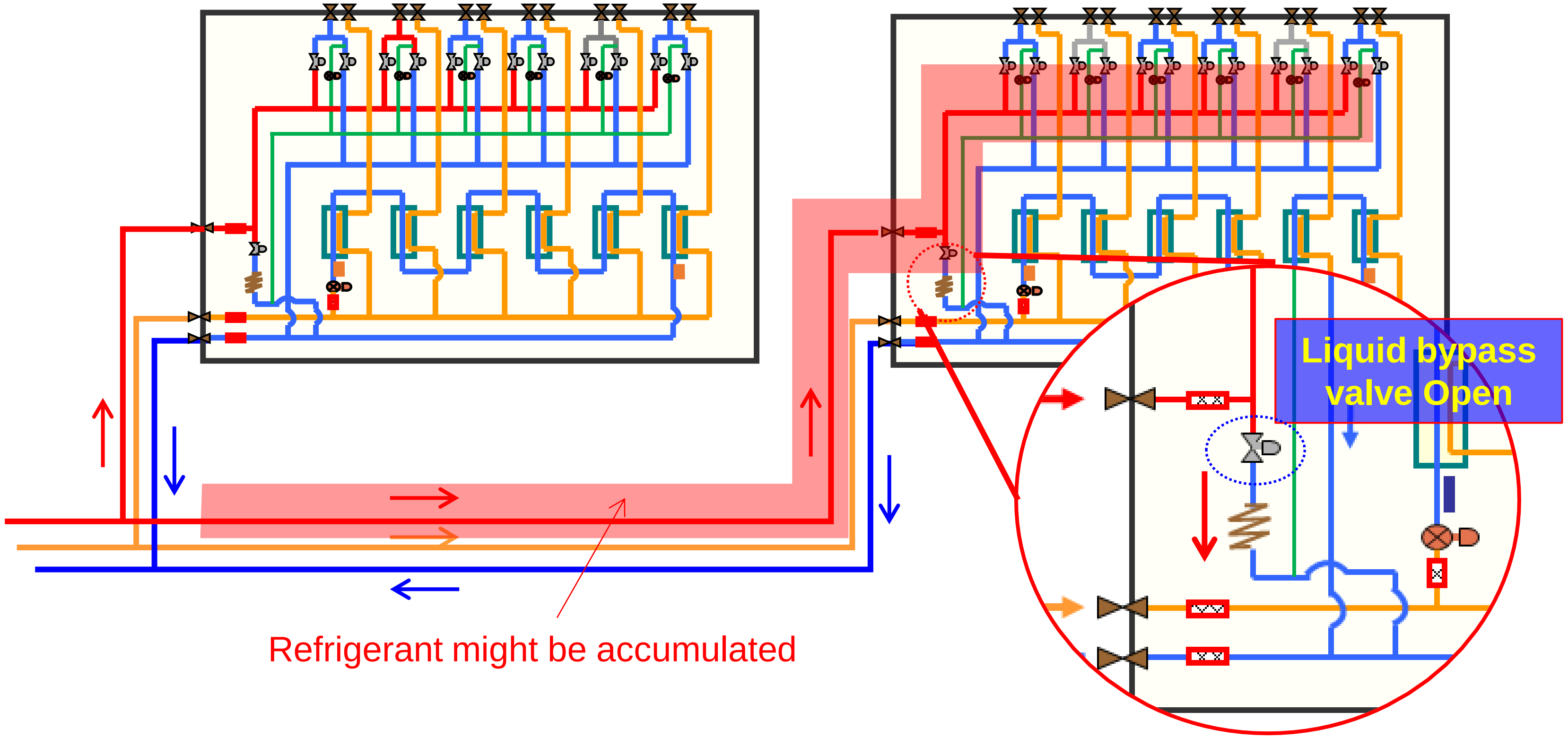
HR Changer

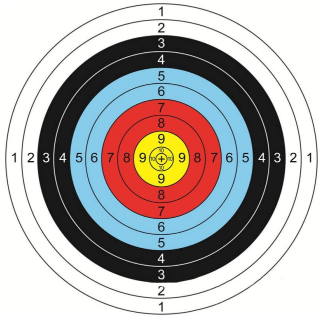
No	Cycle parts	Functions
1	Cooling solenoid valve	This valve opens during Indoor unit's cooling operation
2	Heating solenoid valve	This valve opens during Indoor unit's heating operation
3	MCU EEV	To reduce modulation noise while mode is changed. (Heating → Cooling)
4	Sub-cooler	To secure the sub-cooling of refrigerant for each indoor.
5	Sub-cooler EEV	To control super heat of sub-cooler
6	MCU liquid bypass valve	To prevent refrigerant accumulation in high pressure gas pipe

MCU Liquid By-Pass

- To prevent refrigerant accumulation in high pressure gas pipe

Ex) Main cooling condition

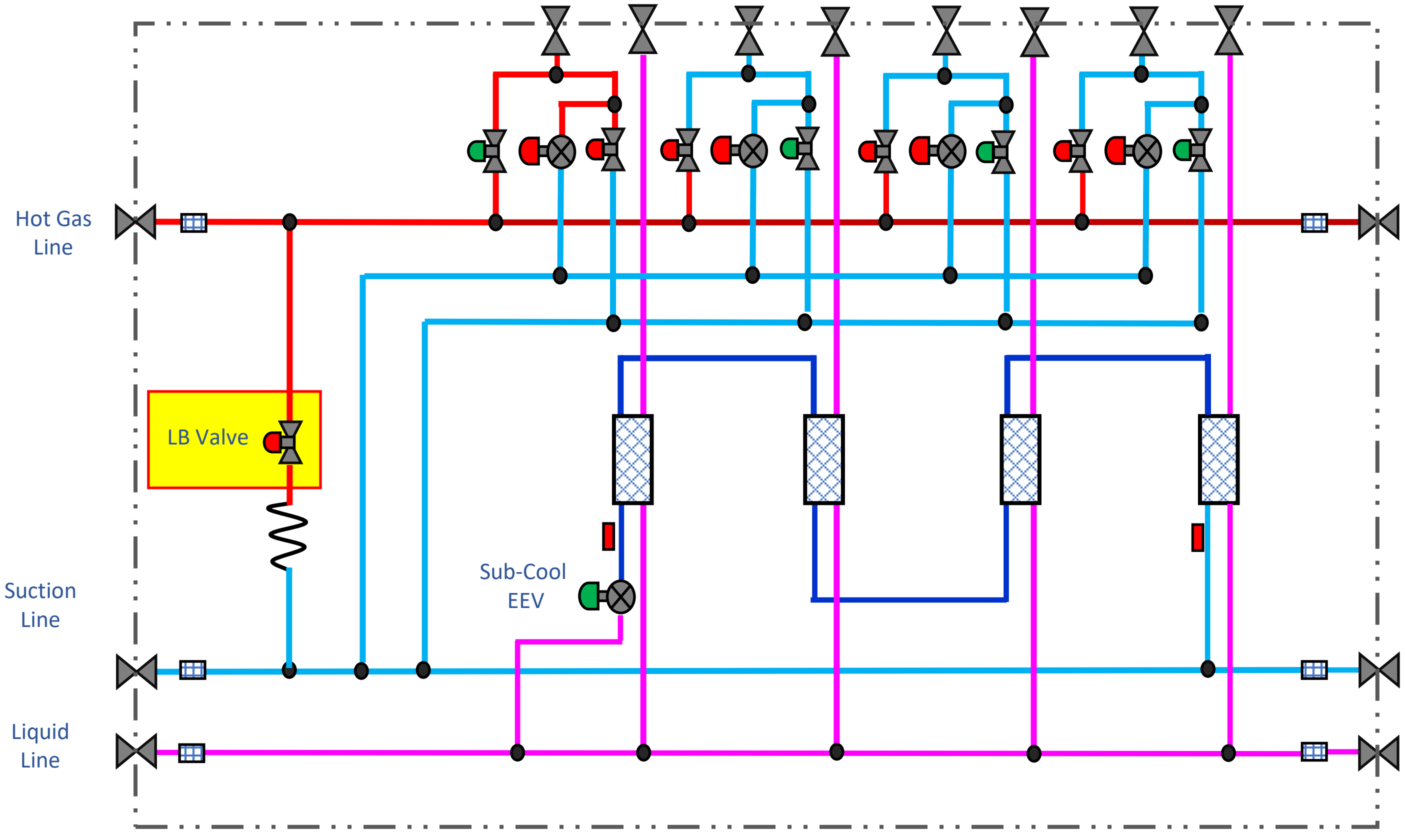




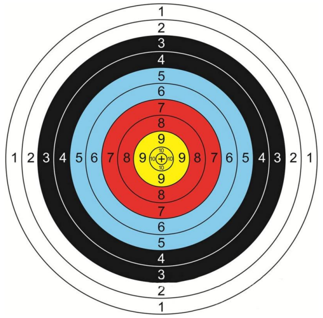
Liquid By-Pass Solenoid

- Energized in Cooling Only
- Prevents refrigerant accumulation in hot gas line

- main heating and main cooling not cooling or heating only

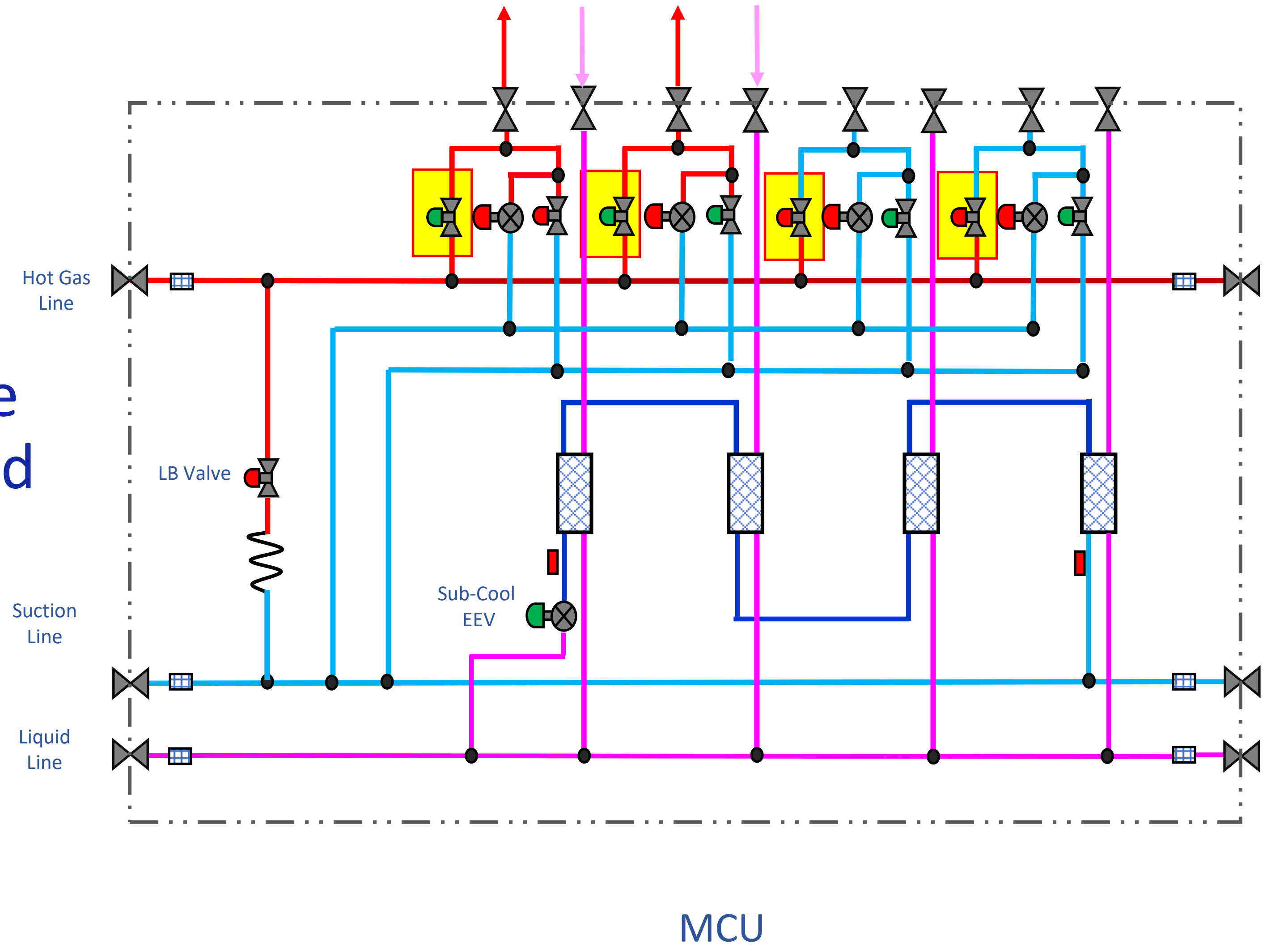


MCU



Heating Solenoid

- Energized when the IDU is calling for heat
- When energized the refrigerant flow is from the hot gas header to the liquid header



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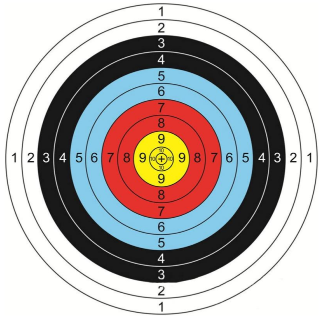
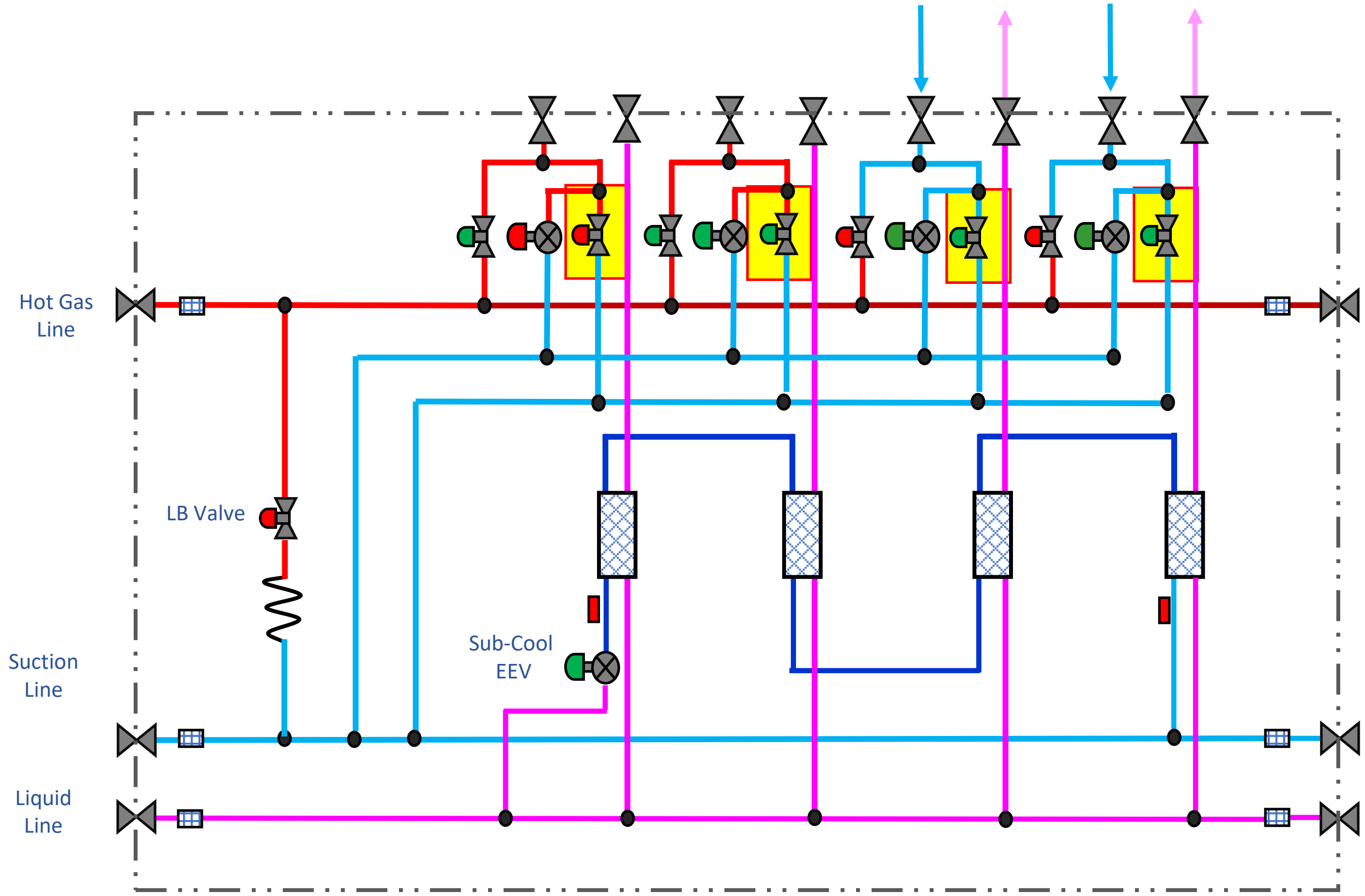


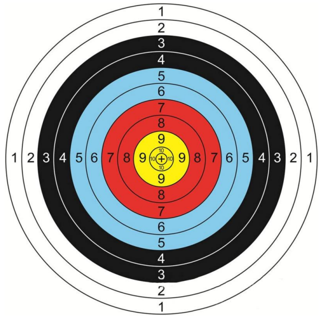
Cooling Solenoid

- Energized when the IDU is calling for Cooling or Dry Modes
- When energized the refrigerant flow is from the liquid header to the suction header



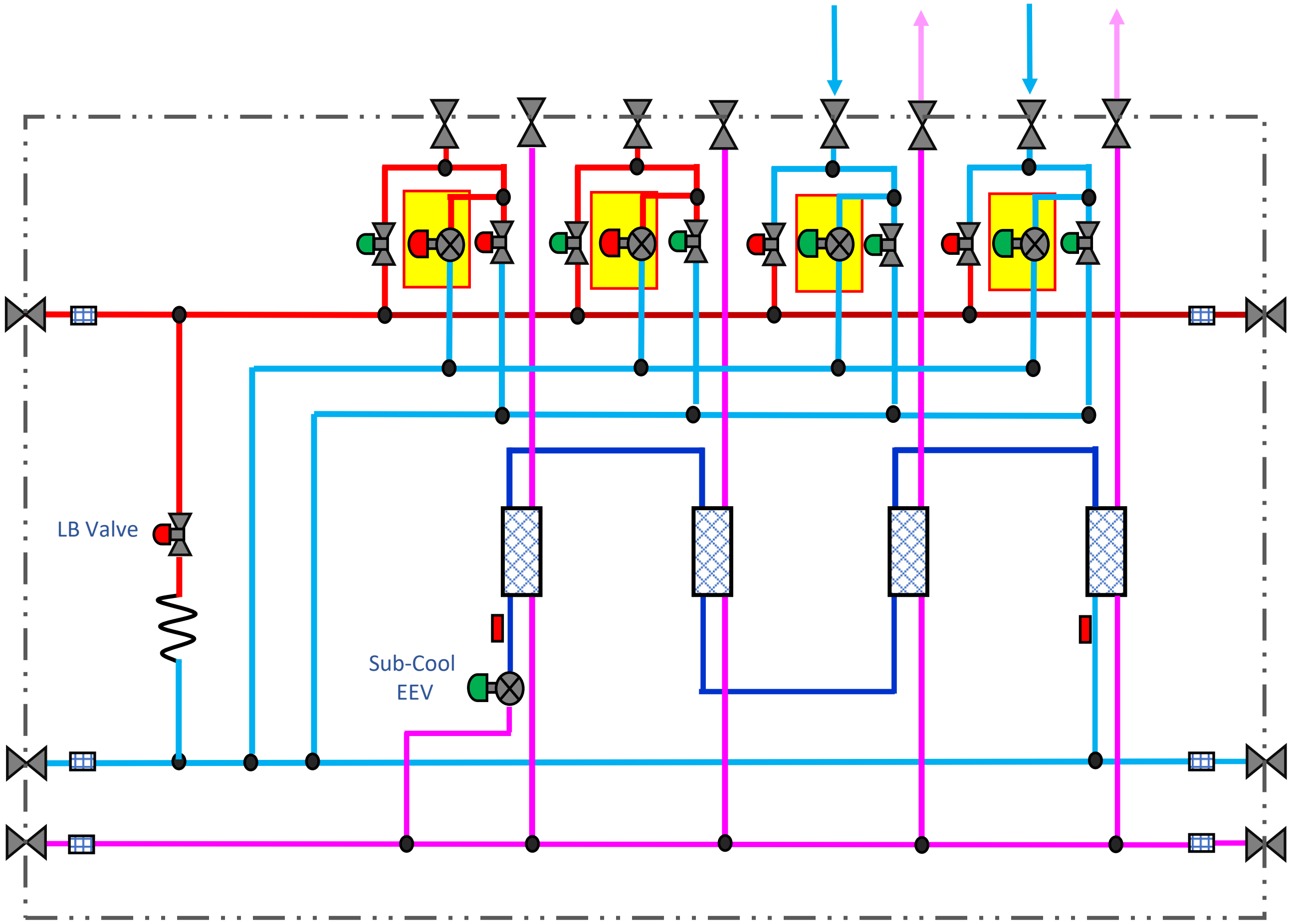
MCU

SAMSUNG



MCU EEV

- EEV is opened gradually during mode change(Heating to cooling) to prevent noise during change over.
- In cooling mode, it is commanded to 480 Steps (full open)
- When IDU is stopped : the EEV gradually opens



MCU

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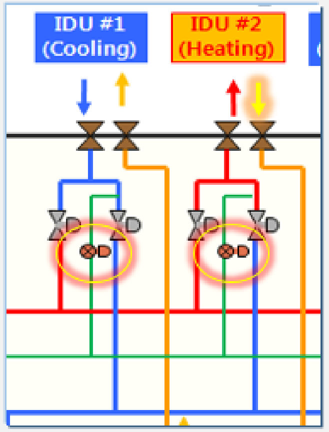
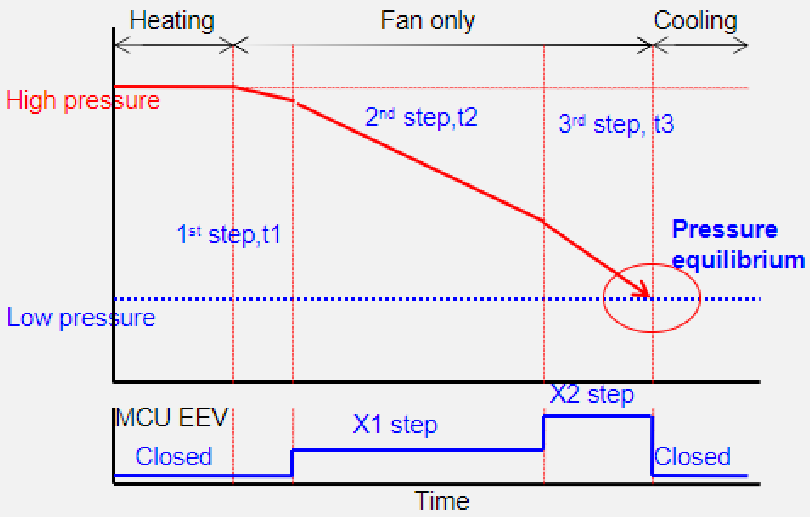
MCU
K4 Buttons to test solenoids

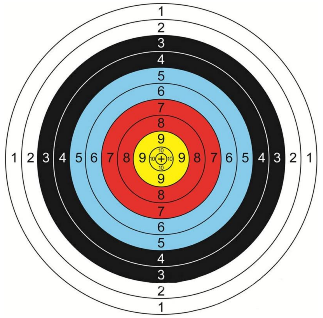
K4 Switch (Solenoid Valve Manual Control)

- According to the push time of K4 Switch, A_C, A_H, …, F_C, F_H, Liquid bypass solenoid valve opens in order.
- In Solenoid Valve Manual Control mode, valve operates by K4 Push time irrespective of indoor operation mode.
- In Solenoid Valve Manual Control mode, push K1 Switch makes DATA DISPLAY MODE to start and valves will operate following indoor operation mode.

K4 (Push time)	Display Contents	Display segment			
		1	2	3	4
1	A_C sol valve ON, other sol valve Off	P	A	1	0
2	A_H sol valve ON, other sol valve Off	P	A	0	1
3	B_C sol valve ON, other sol valve Off	P	B	1	0
4	B_H sol valve ON, other sol valve Off	P	B	0	1
5	C_C sol valve ON, other sol valve Off	P	C	1	0
6	C_H sol valve ON, other sol valve Off	P	C	0	1
7	D_C sol valve ON, other sol valve Off	P	D	1	0
8	D_H sol valve ON, other sol valve Off	P	D	0	1
9	E_C sol valve ON, other sol valve Off	P	E	1	0
10	E_H sol valve ON, other sol valve Off	P	E	0	1
11	F_C sol valve ON, other sol valve Off	P	F	1	0
12	F_H sol valve ON, other sol valve Off	P	F	0	1
13	Liquid b/p sol valve ON, other sol valve Off	P	S	1	0
14	sol valve Manual Control MODE end	P	Communication DATA Display		

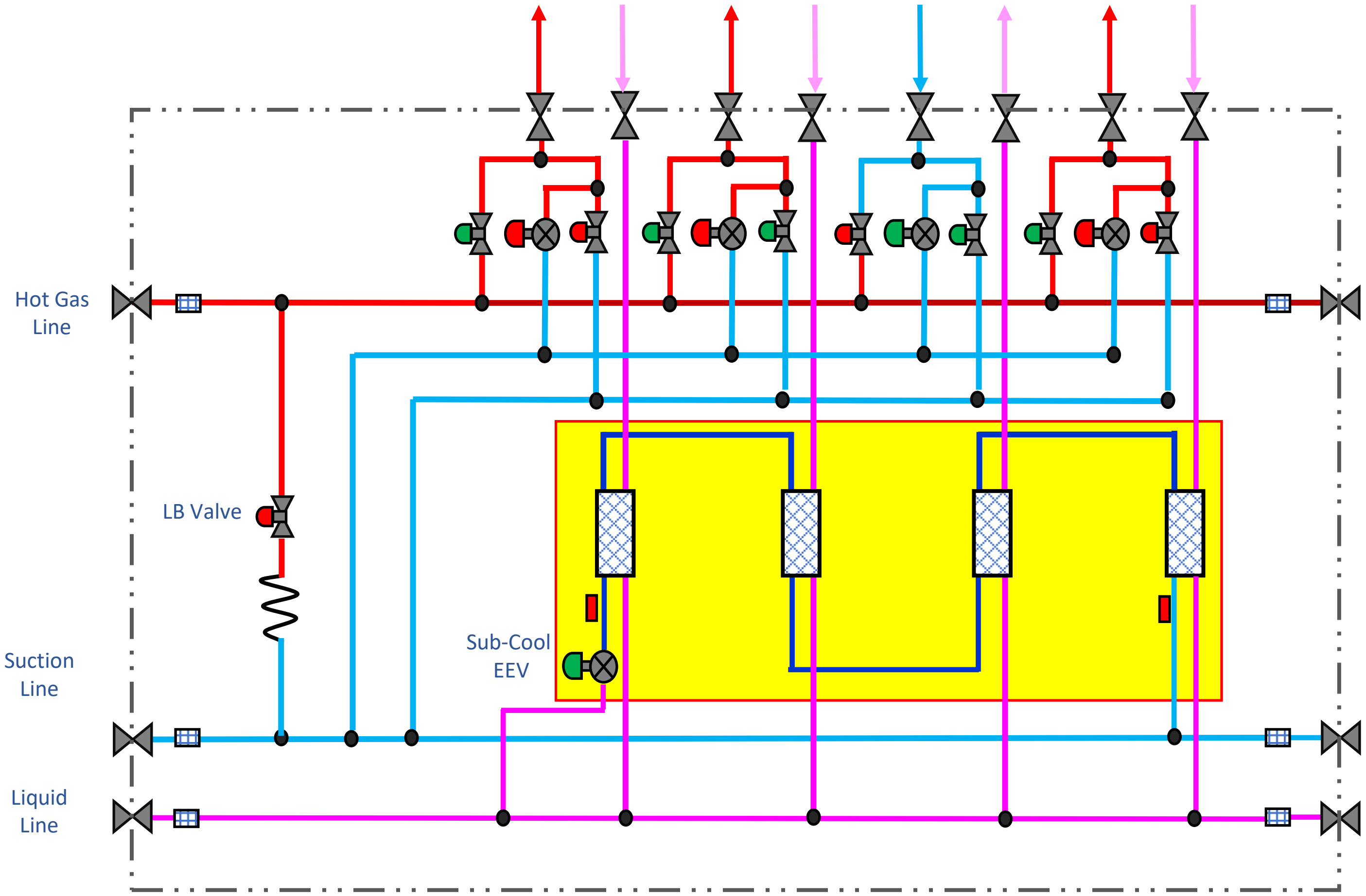
MCU EEV Operation

<table>
<tr><th>MCU EEV</th><th>Operation</th><th>effect</th></tr>
<tr><td rowspan="2">
</td>
<td>EEV is opened gradually during mode change(Heating → cooling)

</td>
<td>Pressure equilibrium without noise</td></tr>
<tr><td>When IDU is in cooling mode : Full open
When IDU is in heating mode : Full close
When IDU is stop : Full close
When all IDU is stop : open the EEV gradually, keep full open</td>
<td>Reduce pressure drop or Return refrigerant in nonworking IDU without noise</td></tr>
</table>



Sub-Cooling Circuit

- Active in Heating modes including Cooling Main.
- Provides sub-cooled refrigerant for IDU in cooling mode.
- Refrigerant from heating IDU doesn't enter to cooling IDU directly but though Sub cooler
  - Secure sufficient sub cooling
  - No boiling noise in cooling IDU
  - higher performance
- EEV step is controlled by the difference between pipe in and pipe out temperatures to maintain a 5.4^0 Superheat



MCU

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Indoor Unit Targets

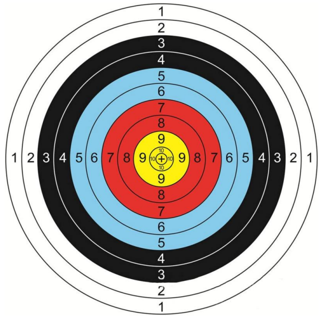
- System operation is based on maintaining system targets.
- The system responds to changes within the system to maintain Individual component targets.
- This section will review individual components target operation.

The Indoor coil

- In the cooling mode it is the evaporator of the system and functions like any other evaporator.
- In the heating mode it is the condenser and functions like any other condenser.
- The EEV (metering device) is in the indoor units.
  - In the cooling mode it maintains superheat across the coil.
  - In heating mode modulates to control sub-cooling.

Indoor EEV Control

- The indoor EEV in Cooling mode modulates to maintain a differential across the evaporator (super heat) between the pipe in and pipe out thermistor
- The indoor EEV in the heating mode modulates to maintain a differential between pip in temperature and the high side saturation temperature



Indoor EEV Control

① Cooling operation

- SH control : 2℃ / 3.6^{0}F (SH = Eva out temp. – Eva in temp.)

Eva in temp. = 7℃ / 44.6^{0}F

Eva out temp. = 10℃ / 50^{0}F

SH = 3℃ / 5.4^{0}F

EEV Step = (+) Step

Eva in temp. = 9℃/ 48.2^{0}F

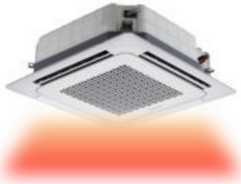
Eva out temp. = 10℃/ 50^{0}F

SH = 1℃/ 1.8^{0}F

EEV Step = (-) Step

② Heating operation

- SC control : Target = average SC

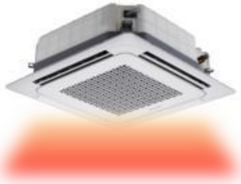


Eva in temp. = 46℃/ 114.8^{0}F

Eva in average temp. = 48℃/ 118.4^{0}F

△T = -2℃/ 3.6^{0}F

EEV Step = (+) Step



Eva in temp. = 50℃/ 122^{0}F

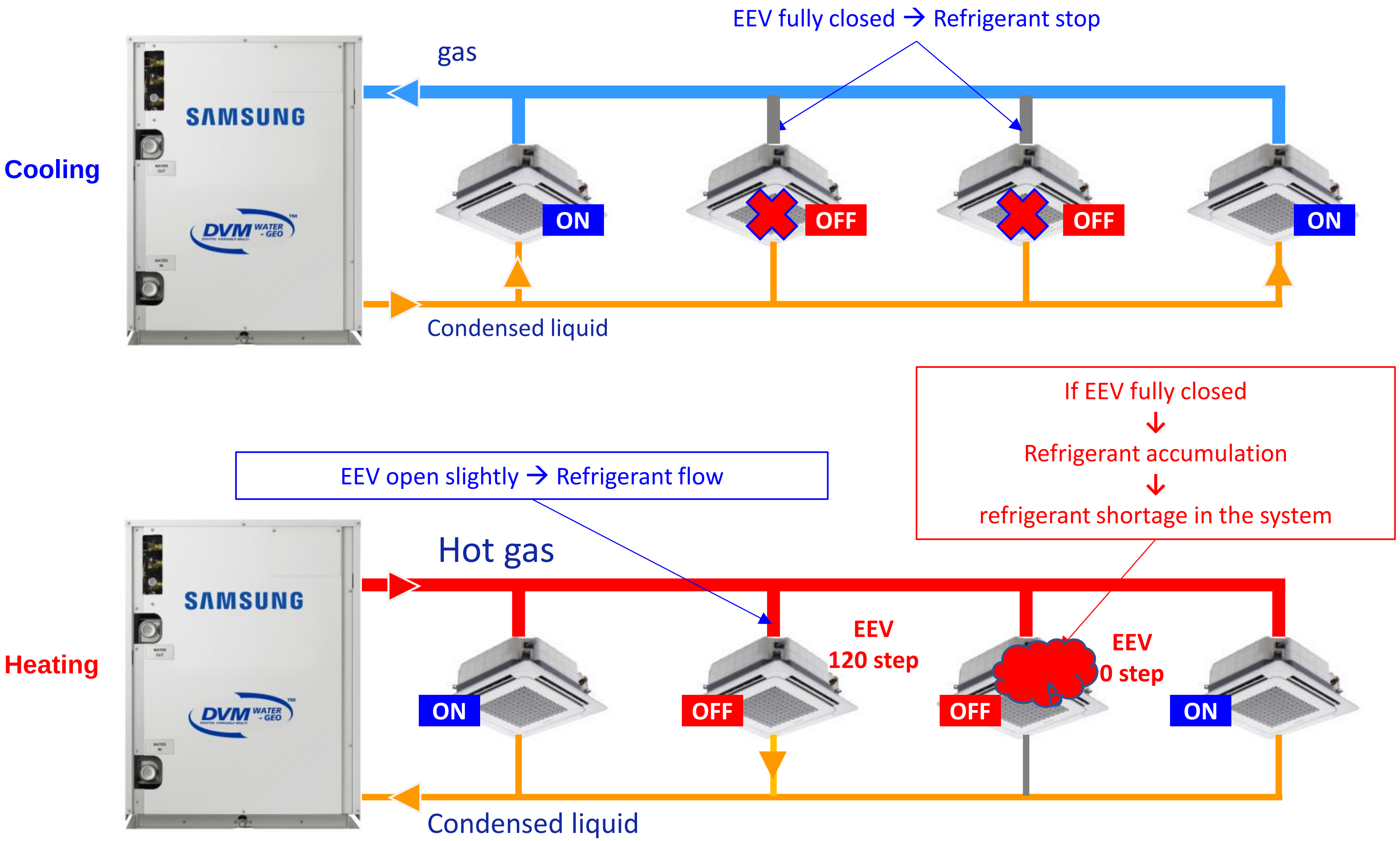
Eva in average temp. = 48℃ / 118.4^{0}F

△T = 2℃/ 3.6^{0}F

EEV Step = (-) Step

※ In special condition, EEV control can be changed.

Electronic Expansion Valve Control



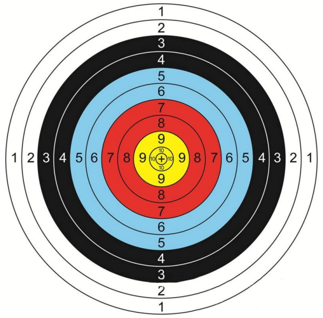
Indoor Unit Control Air Conditioning

- In the air conditioning mode, the compressor modulates to maintain and average evaporator saturation temperature of 41^0-44.6^0 degrees or 114 psig
- The average coil temperature is calculated by using the pipe in thermistor of the indoor coils
- The main EEV modulates to maintain 0-7 degrees of superheat across the evaporator depending on the system make up.
- Fan operation is set by the controller (Auto or fixed speed)

Input unit	Seg1	Seg2	Seg3	Seg4	Temperature(°F)
Main	0	1	0	0	44.6 – 48.2
			0	**1**	**41 – 44.6 (default)**
			0	2	48.2 – 51.8
			0	3	50 – 53.6
			0	4	51.8 – 55.4
			0	5	53.6 – 57.2
			0	6	55.4 - 59

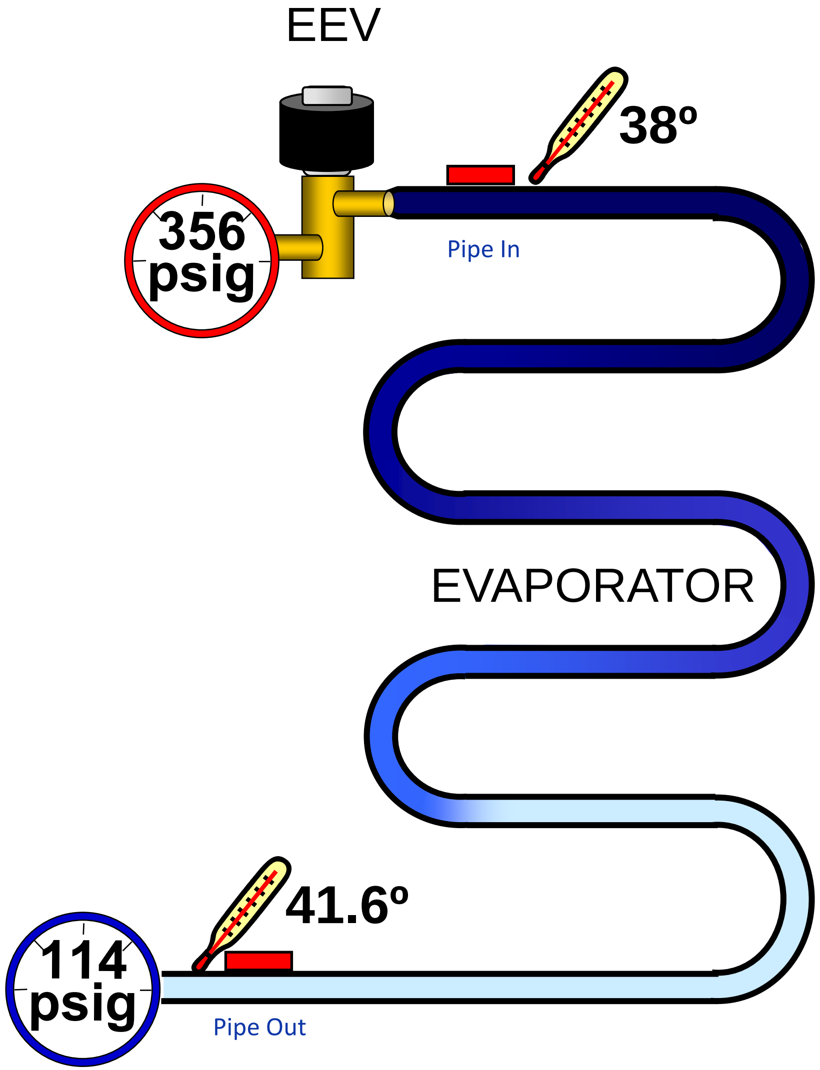
Indoor Unit Control Air Conditioning

- In the air conditioning mode, the compressor modulates to maintain and average evaporator saturation temperature of 38 degrees or 114 psig
- The average coil temperature is calculated by using the pipe in thermistor of the indoor coils
- The main EEV modulates to maintain 0-7 degrees of superheat across the evaporator depending on the system make up.
- Fan operation is set by the controller (Auto or fixed speed)



IDU Target Cooling Mode

- The EEV in the cooling mode will modulate to maintain a 3.6-degree differential between the pipe in (Low Side Saturation Temperature) and pipe (Suction Pipe Temperature)
- If load increases the valve will open to maintain that differential.
- If the load drops, the valve will close to maintain that differential.
- What are some thing that cause the load to change?



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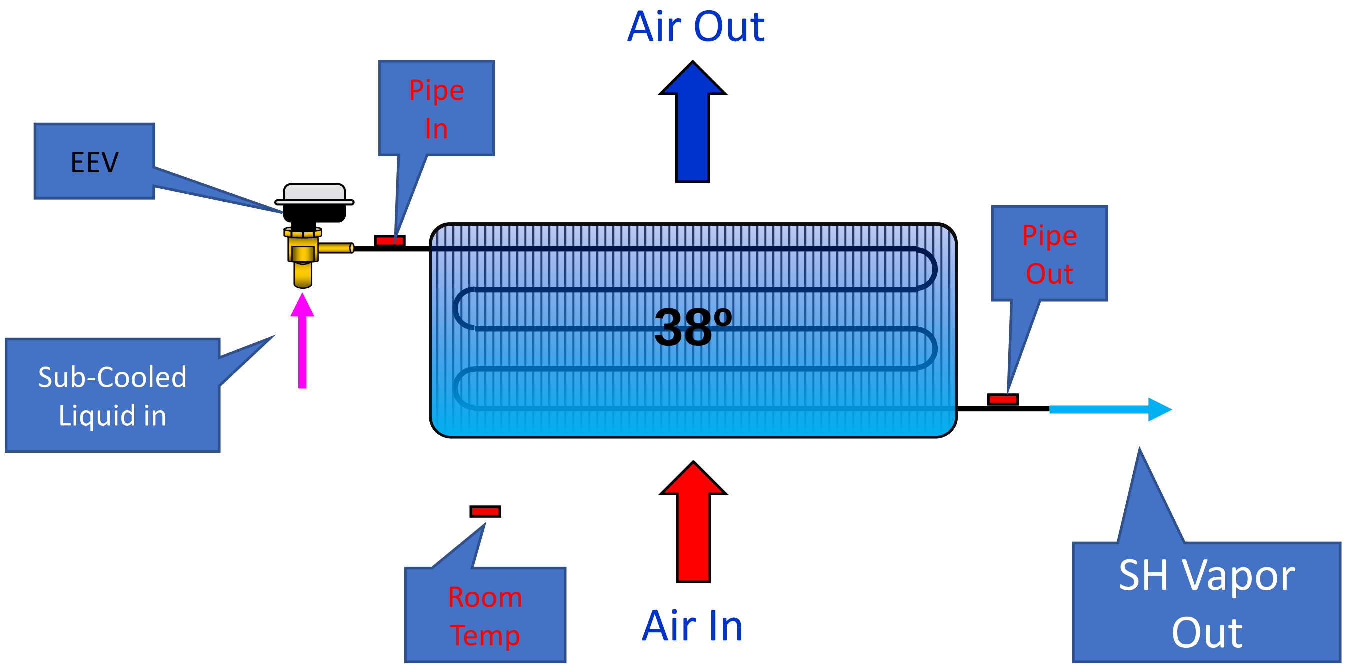
Indoor Unit in Cooling Mode

Coil operates as an evaporator



The EEV maintains a deferential between pipe in and pipe out temperatures
If the temperature difference is high the EEV opens. If it is low the EEV closes

Indoor Unit Control in Heating

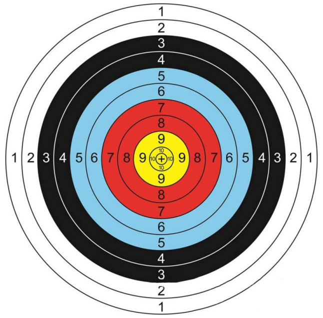
- In the heating mode the compressor will modulate to maintain a high side pressure of approximately 427 psig.
- This will cause the saturation temperature at the indoor coil to be approximately 121 degrees
- The design room air temperature across the indoor coil in the heating mode is 70 degrees.
- This should produce 100+ degrees to be discharged for the coil.
- Fan operation is delayed until the pipe in temperature reaches 85degrees
- The EEV modulates to maintain Sub-Cooling.

- ODU option setting (Capacity correction for heating)

	Seg1	Seg2	Seg3	Seg4	pressure(psig)
Main	0	2	**0**	**0**	**427 (default)**
			0	1	356
			0	2	370
			0	3~7	384 ~ 455
			0	8	469

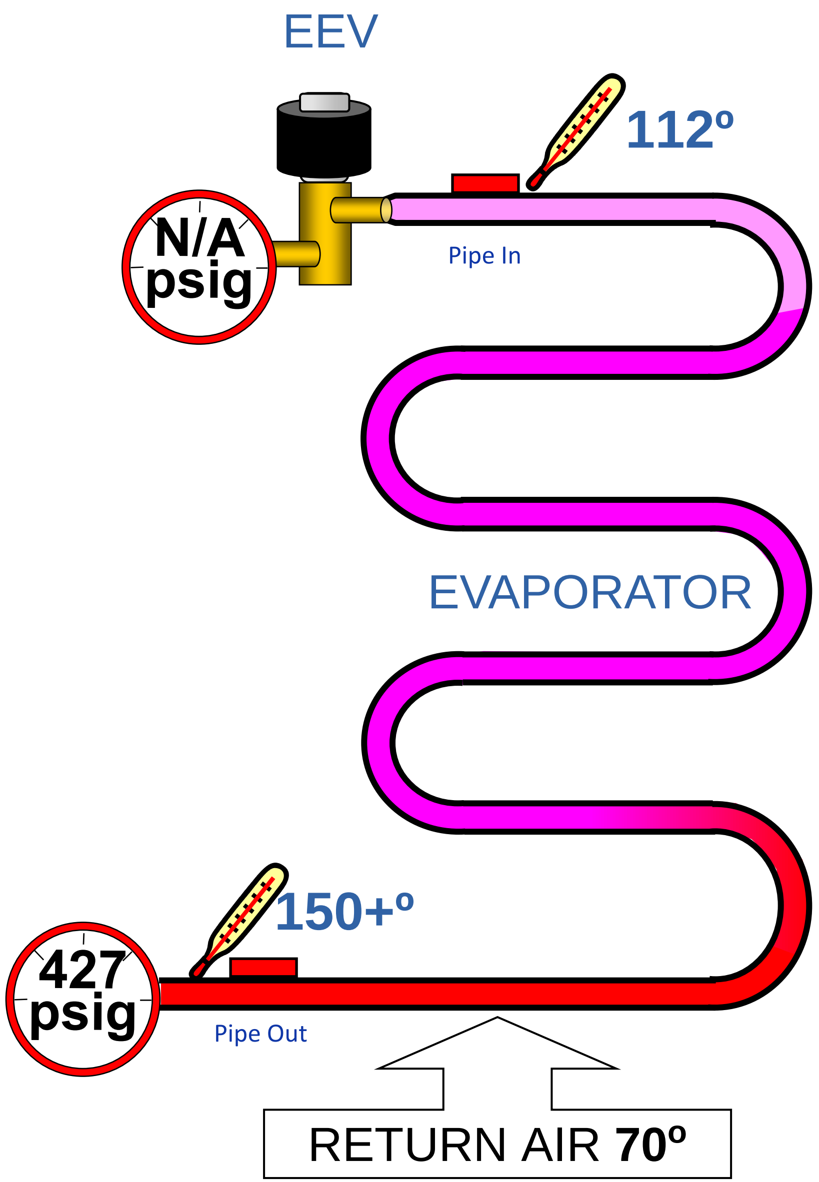
Indoor Unit Control in Heating

- In the heating mode the compressor will modulate to maintain a high side pressure of approximately 427 psig.
- This will cause the saturation temperature at the indoor coil to be approximately 121 degrees
- The design room air temperature across the indoor coil in the heating mode is 70 degrees.
- This should produce 100+ degrees to be discharged for the coil.
- Fan operation is delayed until the pipe in temperature reaches 85degrees
- The EEV modulates to maintain Sub-Cooling.



IDU Target Heat Mode

- The EEV in the heating mode will modulate to maintain a 9-degree differential between the pipe in (Liquid Temperature) and pipe (High Side Saturation Temperature)
- If the load increases the valve will open to maintain that differential.
- If the load drops, the valve will close to maintain that differential.
- What effect does low space temperatures have on the operation of the system?



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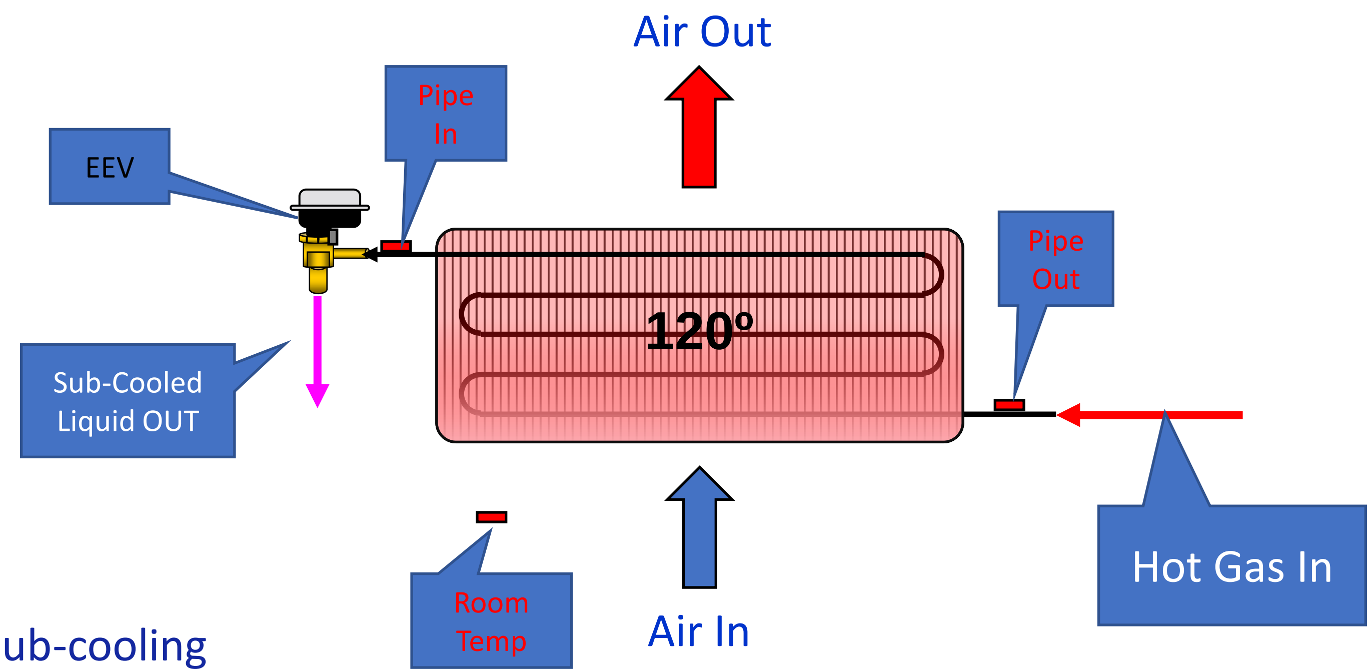
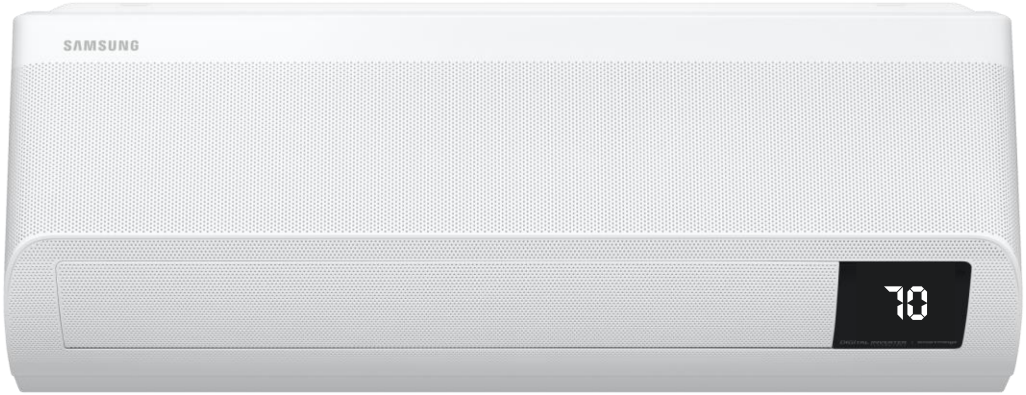
Indoor Unit in Heating Mode

Coil Operates as a Condenser



The EEV maintains a desired sub-cooling between pipe out and high side saturation temperatures of the ODU
If sub-cooling is low the EEV closes if the Sub-cooling is high the EEV opens

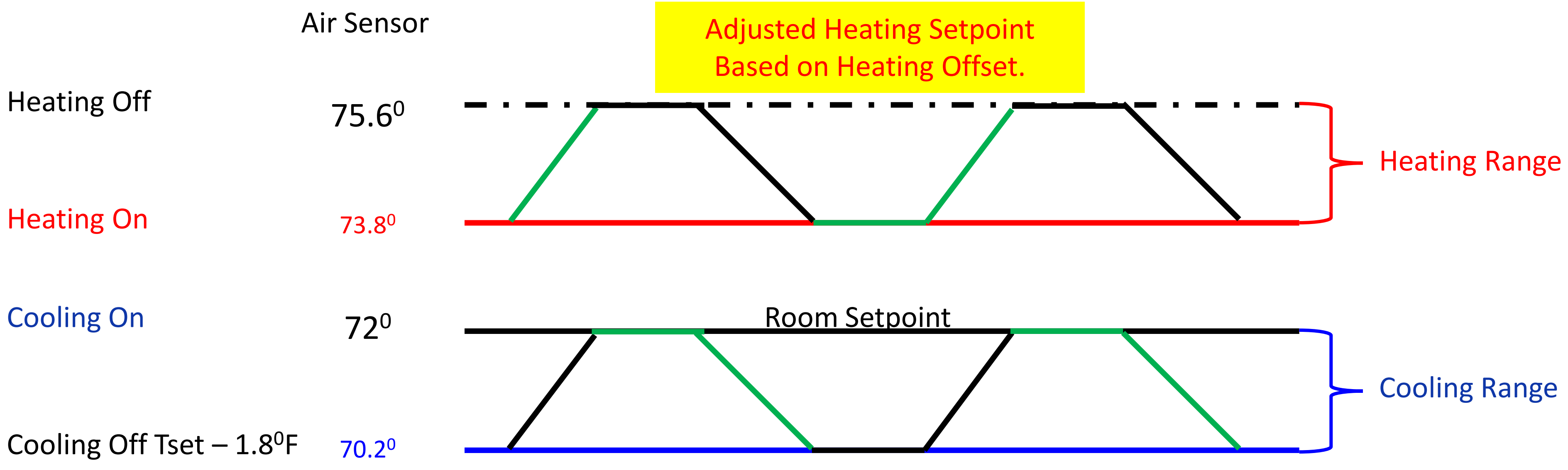
Indoor Fan Operation in Auto Mode



$\lvert Ts - Ti \rvert \leq 1.8°F$	$1.8°F < \lvert Ts - Ti \rvert \leq 5.4°F$	$5.4°F < \lvert Ts - Ti \rvert$
Low	Mid	High

The fan speed is adjusted based on the difference between the set temperature(Ts) and indoor temperature (Ti)

Thermal On & Off



Cooling mode
Thermo on: Room temp. ≥ Set temp
Thermo off: Room temp. ≤ Set temp. -1.8°F

Heating Mode
Thermo on: Room temp. ≤ Set temp. -1.8°F + △T*
Thermo off: Room temp. ≥ Set temp. + △T

Factory default (product option code SEG6)
4way type : △T* = 9°F
others : △T* = 3.6°F

Segment 6 of the 01-Product Code.

- Here are examples of the heat offset settings for different units
- The chart that explains the heat offset and the heating off calculations will vary depending on the type of equipment.

Address	Model	RMC	MCU ADDRES	MCU PORT	Location	Product Option	Installation Option	Installation Option2
0	RAC	02	-	-	Monkey Room	[0]11A35-[1]7C0F7-[2]71A20-[3]7140F	[0]20A10-[1]00000-[2]00101-[3]00346	[0]00000-[1]00000-[2]00000-[3]00000
1	RAC	00	-	-	Living Room	[0]11A35-[1]7C217-[2]72323-[3]7140F	[0]20A10-[1]00000-[2]00101-[3]00346	[0]00000-[1]00000-[2]00000-[3]00000
2	RAC	01	-	-	Master Bedroom	[0]11A35-[1]7C0F7-[2]71A20-[3]7140F	[0]20A10-[1]00000-[2]00101-[3]00346	[0]00000-[1]00000-[2]00000-[3]00000

Indoor Unit Installation Data

						Indoor Unit Installation Data		
Address	Model	RMC	MCU ADDRES	MCU PORT	Location	Product Option	Installation Option	Installation Option2
0	Duct	00	-	-	������� �������	[0]1E26C-[1]05020-[2]76470-[3]70005	[0]20610-[1]00100-[2]00000-[3]00000	[0]5FFFF-[1]FFFFF-[2]FFFFF-[3]FFFFF

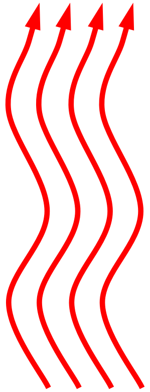
Hot air rise and in some cases the temperature at the unit will be higher than living space.
Because of this there is an automatic Heat Compensation offset built into the systems.

Segment 6 of the 01product code - Heat Compensation Values = 5

△T = 5.4°F

OFF: 79.1^0

ON: 75.4^0



Set Point
70^0

RAC IDU 1:1 or with a FJM

Heat Mode Control Algorithm:

Thermo on:

Room temp. ≤ Set temp. + △T

Thermo off:

Room temp. ≥ Set temp. + 3.6°F + △T

Example:

Heating mode Thermo on:

Room temp. ≤ 70 + 5.4 = 75.4

Thermo off:

Room temp. ≥ 70 + 3.6 + 5.4 = 79.1

01 SEG6	△T ⁰C	Note
0	-2 / 0	
1	-1 / 0	
2	0 / 0	
3	1 / 0	
4	2 / 0	
5	3 / 0	Heating △T / Cooling △T
6	4 / 0	
7	5 / 0	
8	3 / 1	
9	3 / 2	
A	3 / 3	
B	3 / -1	

Segment 6 option chart from an older presentation

Hot air rise and in some cases the temperature at the unit will be higher than living space.
Because of this there is an automatic Heat Compensation offset built into the systems.

OFF: 79.1^0

ON: 75.4^0

Segment 6 of the 01product code - Heat Compensation Values = 5

△T = 5.4°F

RAC IDU 1:1 or with a FJM

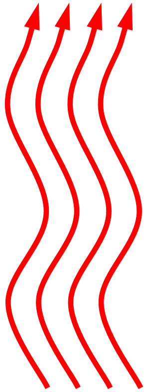
Heat Mode Control Algorithm:

Thermo on:

Room temp. ≤ Set temp. + △T

Thermo off:

Room temp. ≥ Set temp. + 3.6°F + △T



Set Point
70^0

Example:

Heating mode Thermo on:

Room temp. ≤ 70 + 5.4 = 75.4

Thermo off:

Room temp. ≥ 70 + 3.6 + 5.4 = 79.1

Product option code(01) SEG6		
Value	△T	△X
0	-2 ℃ / -3.6 ^{0}F	0 ℃
1	-1 ℃ / -1.8 ^{0}F	
2	0 ℃ / ^{0}F	
3	1 ℃ / 1.8 ^{0}F	
4	2℃ / 3.6 ^{0}F	
5	3 ℃ / 5.4 ^{0}F	
6	4 ℃/ 7.2^{0}F	
7	5 ℃ / 9^{0}F	
8	-2 ℃ / -3.6 ^{0}F	1 ℃ / 1.8 ^{0}F
9	-1 ℃ / -1.8 ^{0}F	
A	0 ℃ / ^{0}F	
B	1 ℃ / 1.8 ^{0}F	
C	2℃ / 3.6 ^{0}F	
D	3 ℃ / 5.4 ^{0}F	
E	4 ℃/ 7.2^{0}F	
F	5 ℃ / 9^{0}F	

Segment 6 option chart from an newer presentation

Removing the Heat-Offset

Segment 4 of the 02 Installation Code

Installation Option: 02xxxx-1xxxxx-2xxxxx-3xxxxx

Segment 1	Segment 2	Segment 3	Segment 4	Segment 5	Segment 6
Page	Option Type	Reserved	**Use of External Room Temperature Sensor / Minimizing Fan Operation when Thermostat is OFF**	Use of Central Control	Compensation of Fan RPM
Segment 7	**Segment 8**	**Segment 9**	**Segment 10**	**Segment 11**	**Segment 12**
Page	Use of Drain Pump	Reserved	Reserved	Reserved	Dew Removal Operation in Wind-Free Mode
Segment 13	**Segment 14**	**Segment 15**	**Segment 16**	**Segment 17**	**Segment 18**
Page	Use of External Control	Setting the Output of External Control	Reserved	Buzzer Control	Hours of Filter Usage
Segment 19	**Segment 20**	**Segment 21**	**Segment 22**	**Segment 23**	**Segment 24**
Page	Individual Control with Remote Control	Heating Setting Compensation Offset	The Lowest Limit of Outdoor Temperature for Heating Operation	Reserved	Up/Down Louver Swing Speed

Segment 4: External Room Temperature Sensor / Minimizing Fan Operation In Thermal-OFF

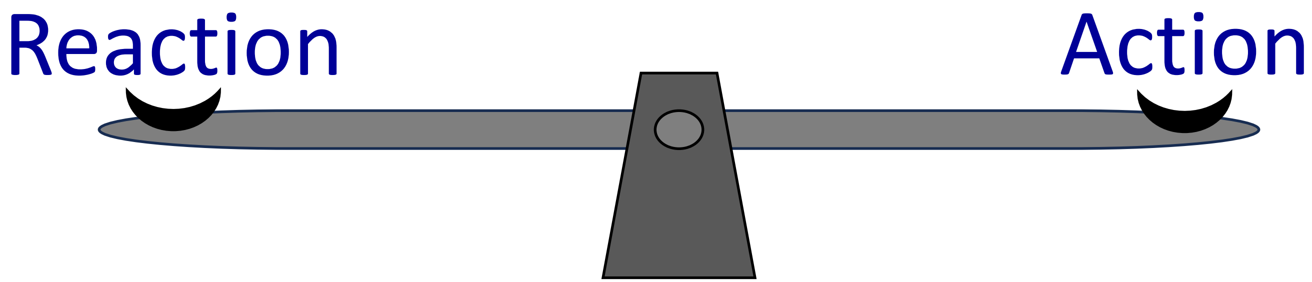
Value	External Room Temp Sensor	Minimizing fan operation when Thermal-OFF	
		Cooling Thermal-OFF	Heating Thermal-OFF
0 (Default)	Disuse	Fan operation: **ON**	EVA-IN temp ≥ 68°F: **Ultra Low** EVA-IN temp < 68°F: **OFF**
1	Use	Fan operation: **ON**	EVA-IN temp ≥ 68°F: **Ultra Low** EVA-IN temp < 68°F : **OFF**
2	Disuse	Fan operation: **ON**	EVA-IN temp ≥ 68°F: **Ultra Low for 20 seconds every 5 minutes** EVA-IN temp < 68°F: **OFF**
3	Use	Fan operation **ON**	EVA-IN temp ≥ 68°F: **Ultra Low for 20 seconds every 5 minutes** EVA-IN Temp < 68°F: **OFF**
4	Disuse	Fan operation: **OFF**	EVA-IN Temp ≥ 68°F: **Ultra Low** EVA-IN Temp < 68°F: **OFF**
5	Use	Fan operation: **OFF**	EVA-IN Temp ≥ 68°F: **Ultra Low** EVA-IN Temp < 68°F: **OFF**
6	Disuse	Fan operation: **OFF**	EVA-IN Temp ≥ 68°F: **Low Low for 20 seconds every 5 minutes** EVA-IN Temp < 68°F: **OFF**
7	Use	Fan operation: **OFF**	EVA-IN Temp ≥ 68°F: **Low Low for 20 seconds every 5 minutes** EVA-IN Temp < 68°F : **OFF**
8	Disuse	Fan operation: **ON at Ultra Low speed**	EVA-IN Temp ≥ 68°F: **Low Low** EVA-IN Temp < 68°F: **OFF**
9	Use	Fan operation: **ON at Ultra Low speed**	EVA-IN Temp ≥ 68°F: **Low Low** EVA-IN Temp < 68°F: **OFF**
A	Disuse	Fan operation: **ON at Ultra Low speed**	EVA-IN Temp ≥ 68°F: **Ultra Low for 20 seconds every 5 minutes** EVA-IN Temp < 68°F : **OFF**
B	Use	Fan operation: **ON at Ultra Low speed**	EVA-IN Temp ≥ 68°F: **Ultra Low for 20 seconds every 5 minutes** EVA-IN Temp < 68°F: **OFF**

NOTE

- EVA-IN = Evaporator Inlet Temperature.
- External temperature sensing can be achieved using a wired controller with the internal sensor enabled or MRW-TA External Temperature Sensor.
- "Use" of External Room Temp Sensor options are only selected when the MRW-TA is installed.

VRF System Operation

- The operation of a VRF system is a constant balancing act between changes in the system and the operation of components.
- For every change within the system the is a reaction by the components to maintain their target operation.
- One change within the system can cause a domino effect of component reactions.
- Its important to understand the cause and effects that take place within the VRF system to understand the system operation.



System Operation

- The operation of a Variable system is a constant balancing act between changes in the system and the operation of components.
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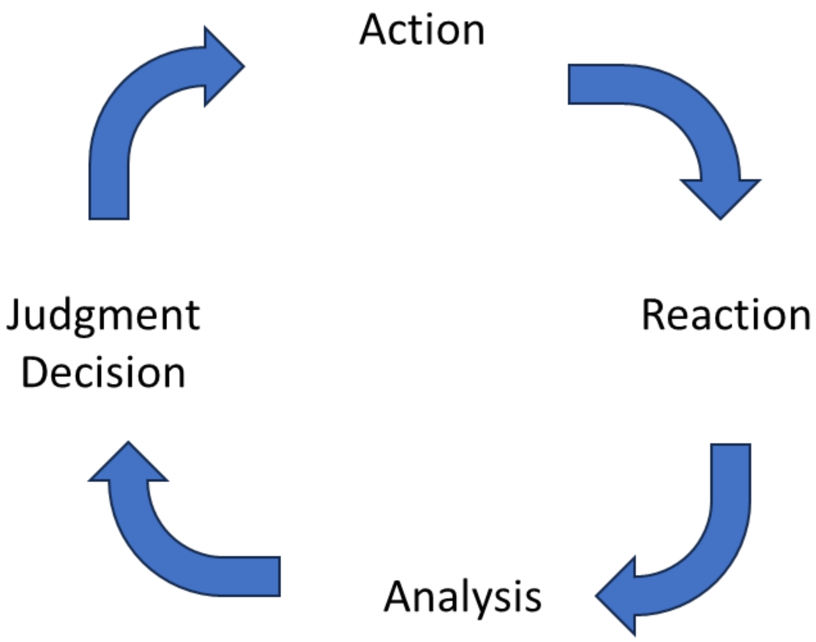
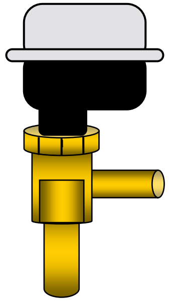


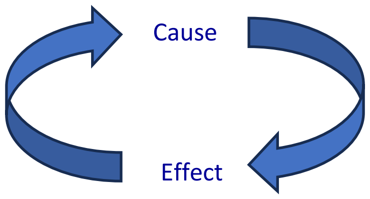
Interrelationship between components

- What effect dose the operation of one component have on the others?
- How does the system react to changes in the system?
- What is the domino effect of actions within the system that occur when there is a change within the system
- For example, what happens effect does being under charged have on the system operation?

Under Charge

- In the cooling mode the suction pressure will drop due to a shortage of gas in the system.
- The compressor will slow down to try to maintain target suction pressure.
- The IDU superheat will be high and the EEV's will be driven open more than normal.
- Evaporator super heat will be high
- The discharge temperature will be high
- There will be a reduced amount of Sub-Cooling.
- There will be a lack of cooling performance

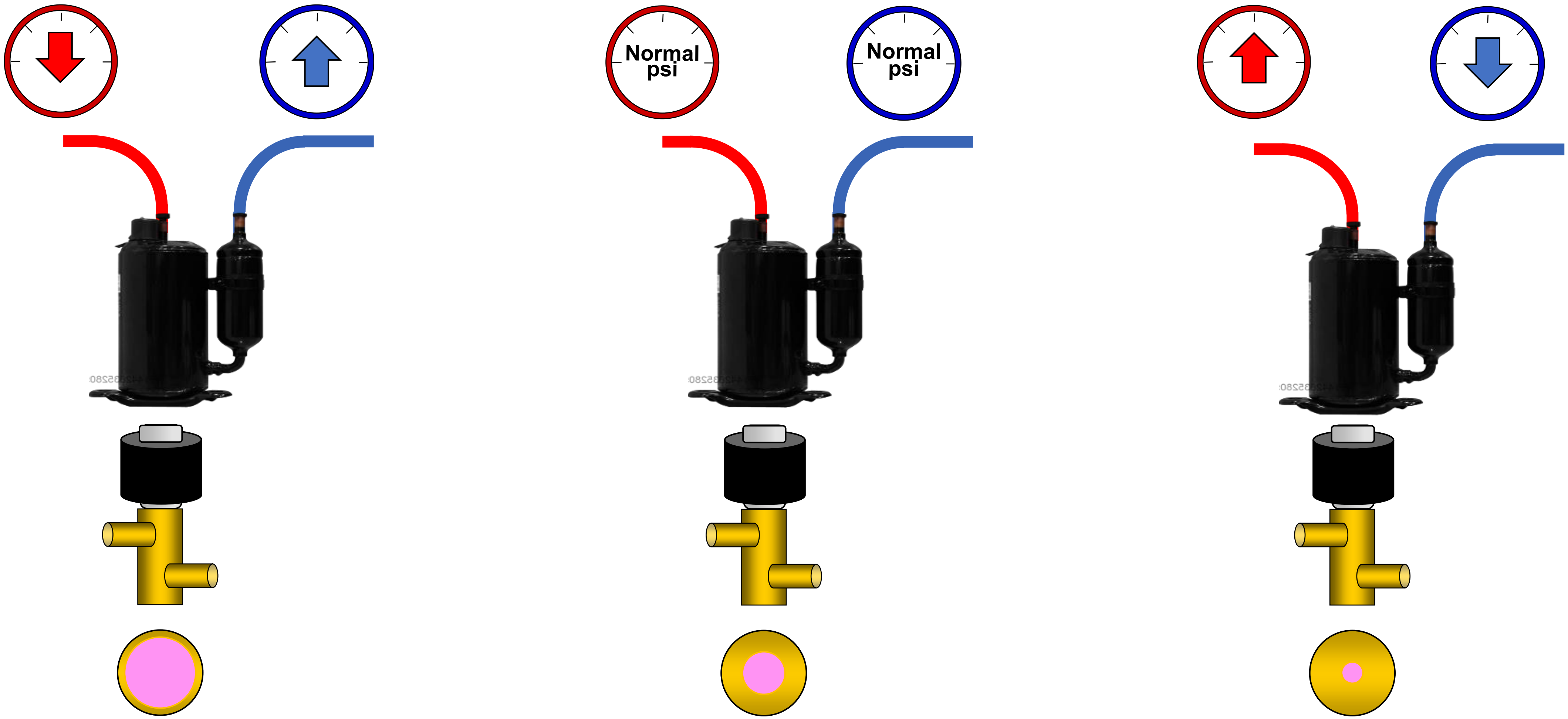


Over Charge

- In the heating mode the head pressure will be high due to a excess refrigerant in the system.
- The compressor will achieve target head pressure at lower than need Hz.
- The discharge temperature will be low
- The IDU superheat will be lower - normal and the EEV's will be driven closed to their lower limit.
- There will be a large amount of Sub-Cooling.
- There will be a lack of heating performance because of the reduced compressor speed.

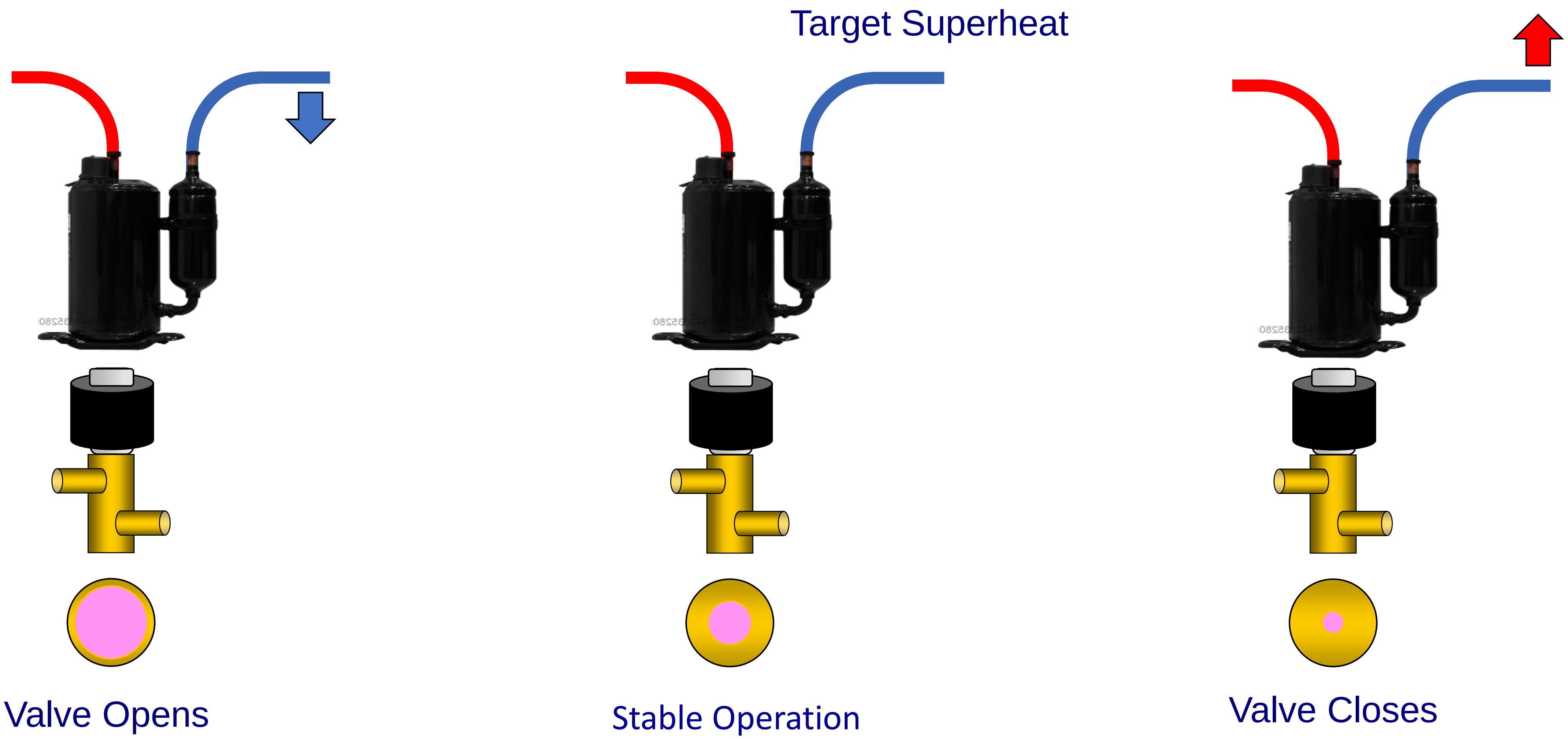
LEV Opening Vs Pressure Relationships

(With no other changes in the system)



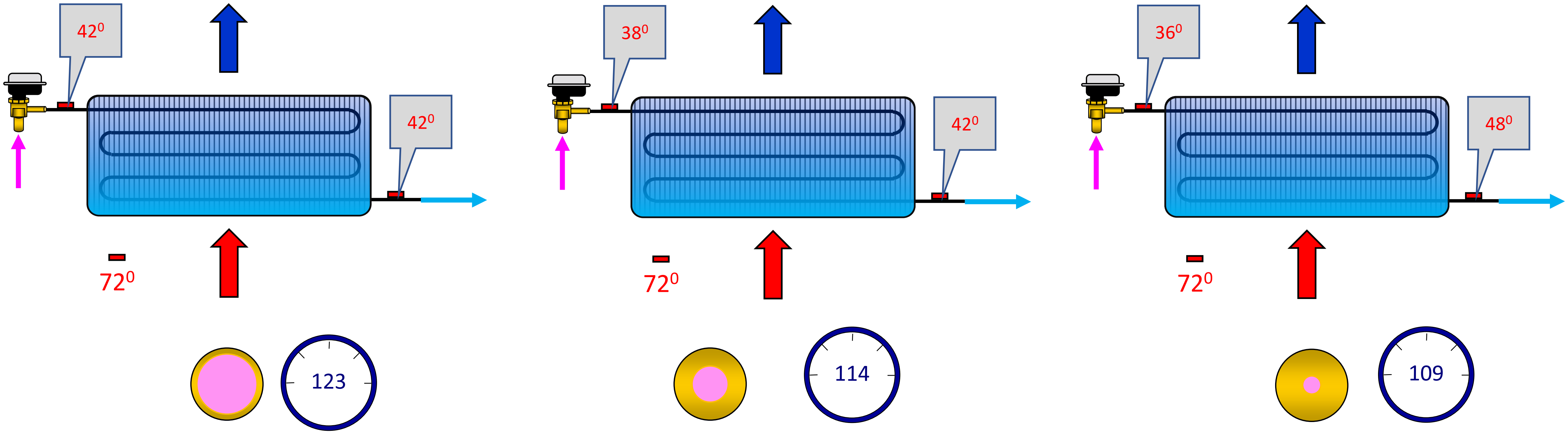
LEV Opening Vs Superheat

(With no other changes in the system)



LEV opening Effects on Super Heat

(With no other changes in the system)



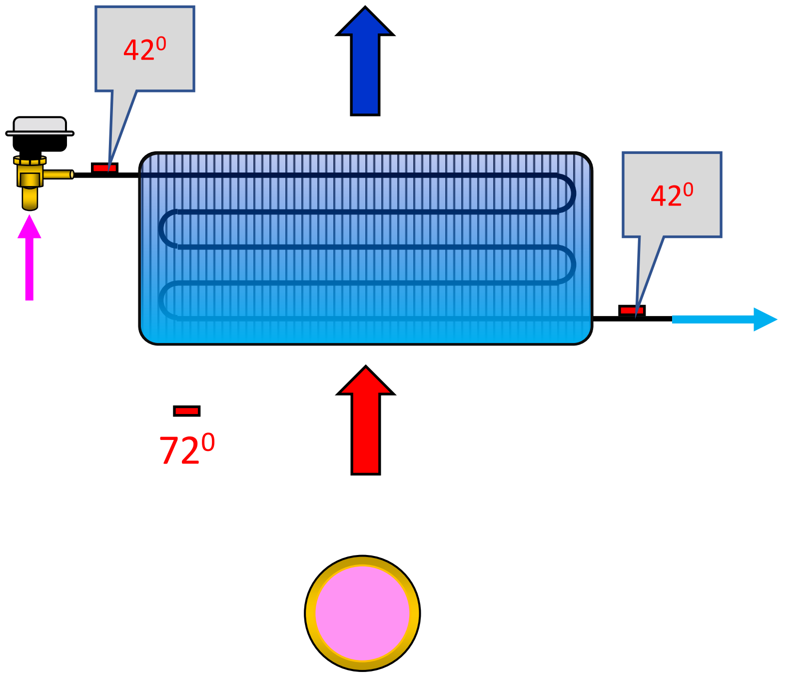
Open Valve to lower Temp

Stable Operation

Close Valve to increase Temp

LEV opening Effects on Super Heat

(With no other changes in the system)



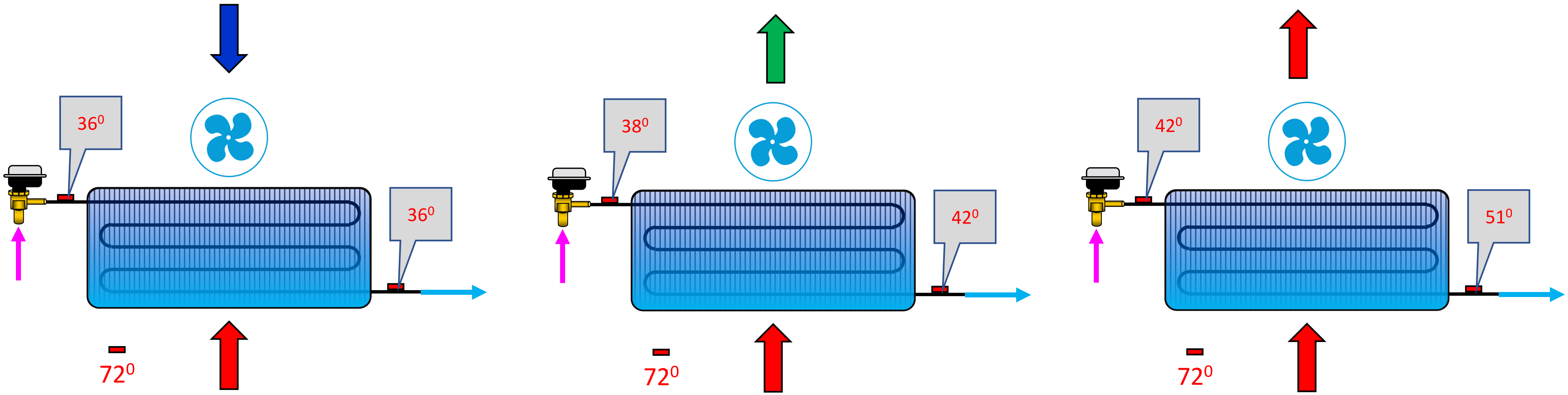
Open Valve to lower Temp

Stable Operation

Close Valve to increase Temp

Air flow effects on superheat heat

(With no other changes in the system)

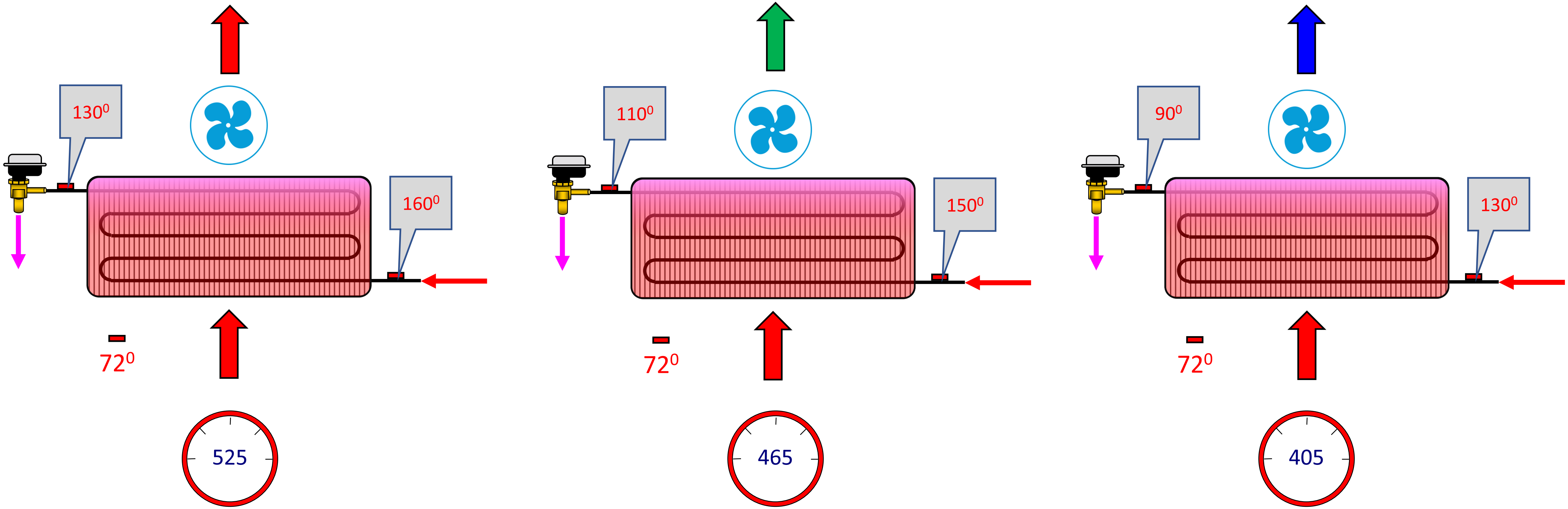


Fan cycles to low speed

Stable Operation
Medium Fan

Fan cycles to high

Air flow effects on heat PSI

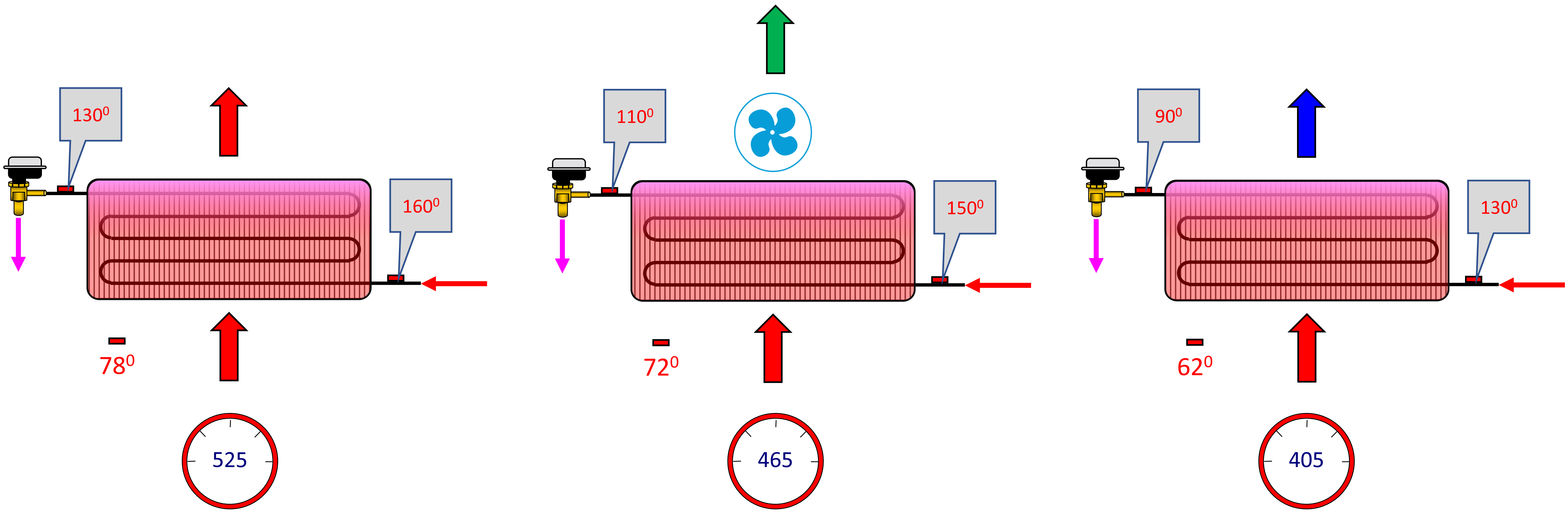


Low Air Flow

Correct Air Flow
High Fan

High Air Flow

Space Temperature effects on heat PSI

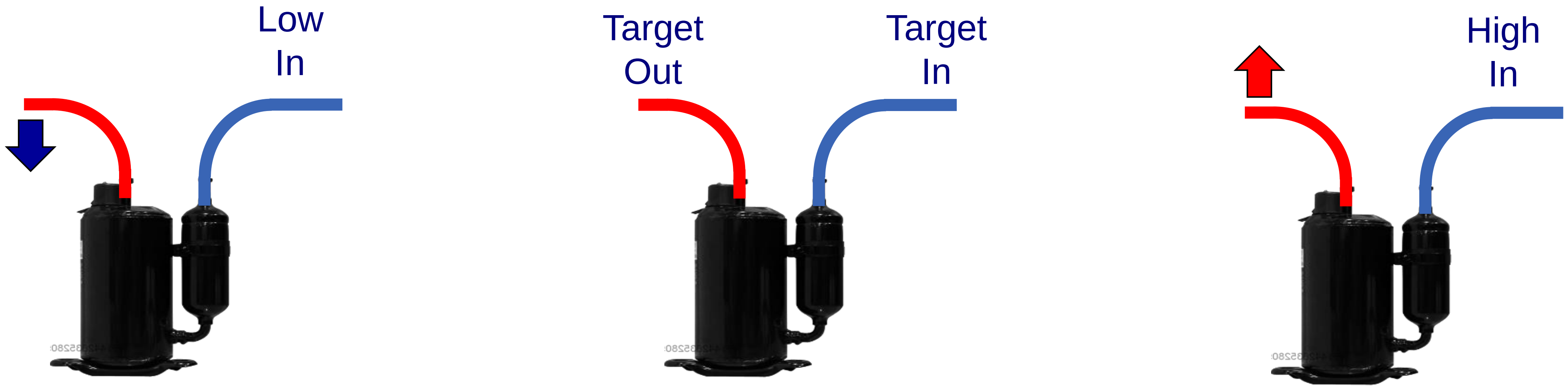


Correct Air Flow
High Fan

Discharge Temperature Control

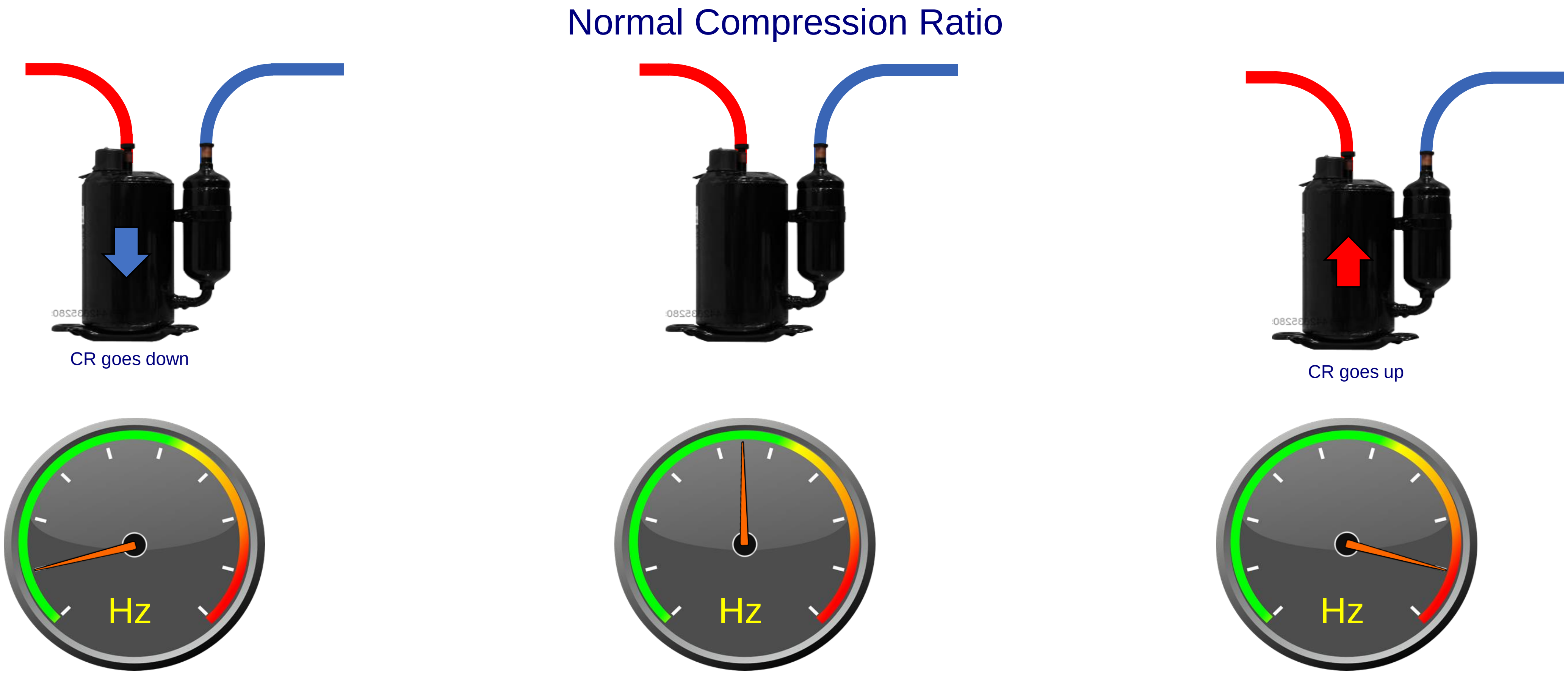
- The discharge temperature is directly related to the suction super heat and the compression ratio.
- If the discharge temperature is low and the system wants to increase the discharge temperature then it will slow down the compressor allowing the refrigerant to sit in the evaporator longer to absorb more heat
- It the system wants to lower the discharge temperature it will increase the speed of the compressor.
- Discharger temperature control is a key factor in the operation of a VRF system.

Compressor Temperature In Vs Out



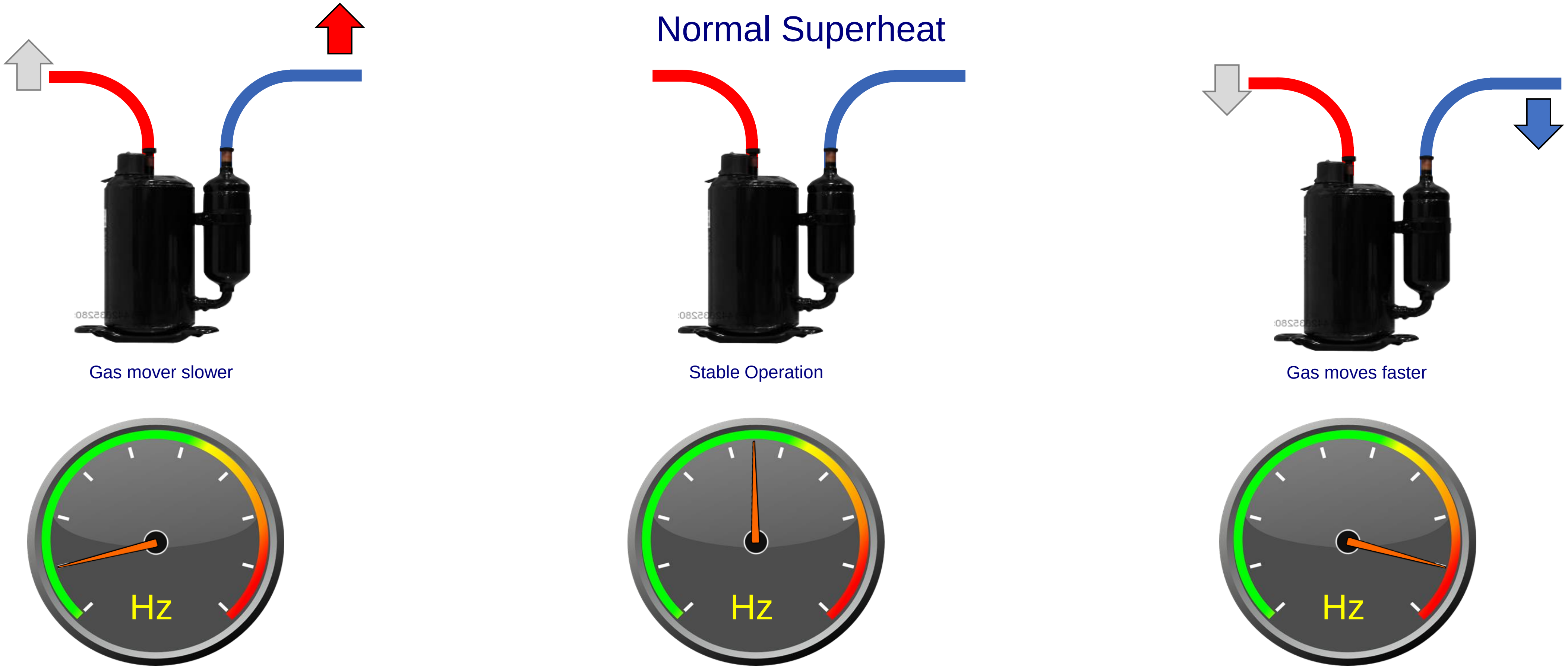
The Compressor
Speed Vs Compression Ration

(With no other changes in the system)

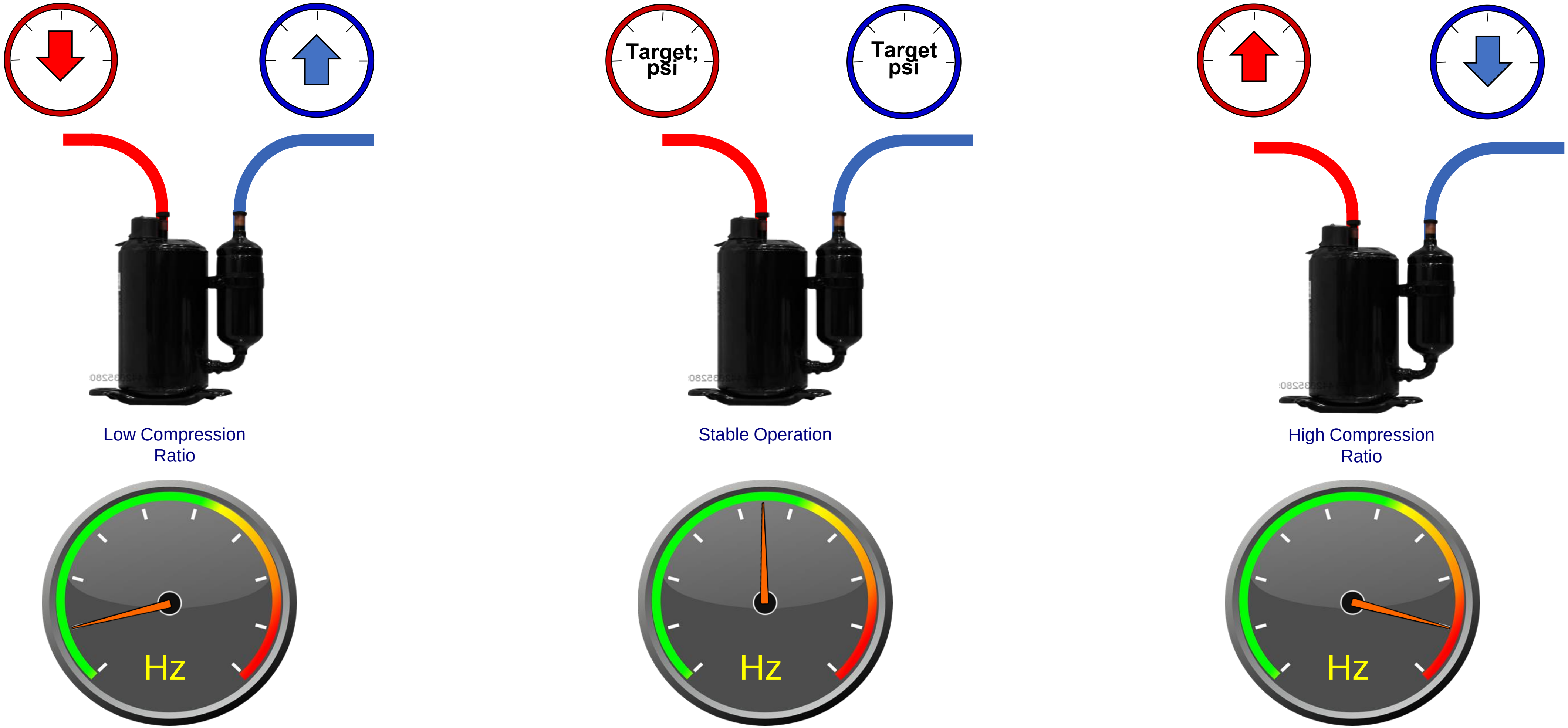


The Compressor
Speed Vs Velocity of the refrigerant (SH)

(With no other changes in the system)

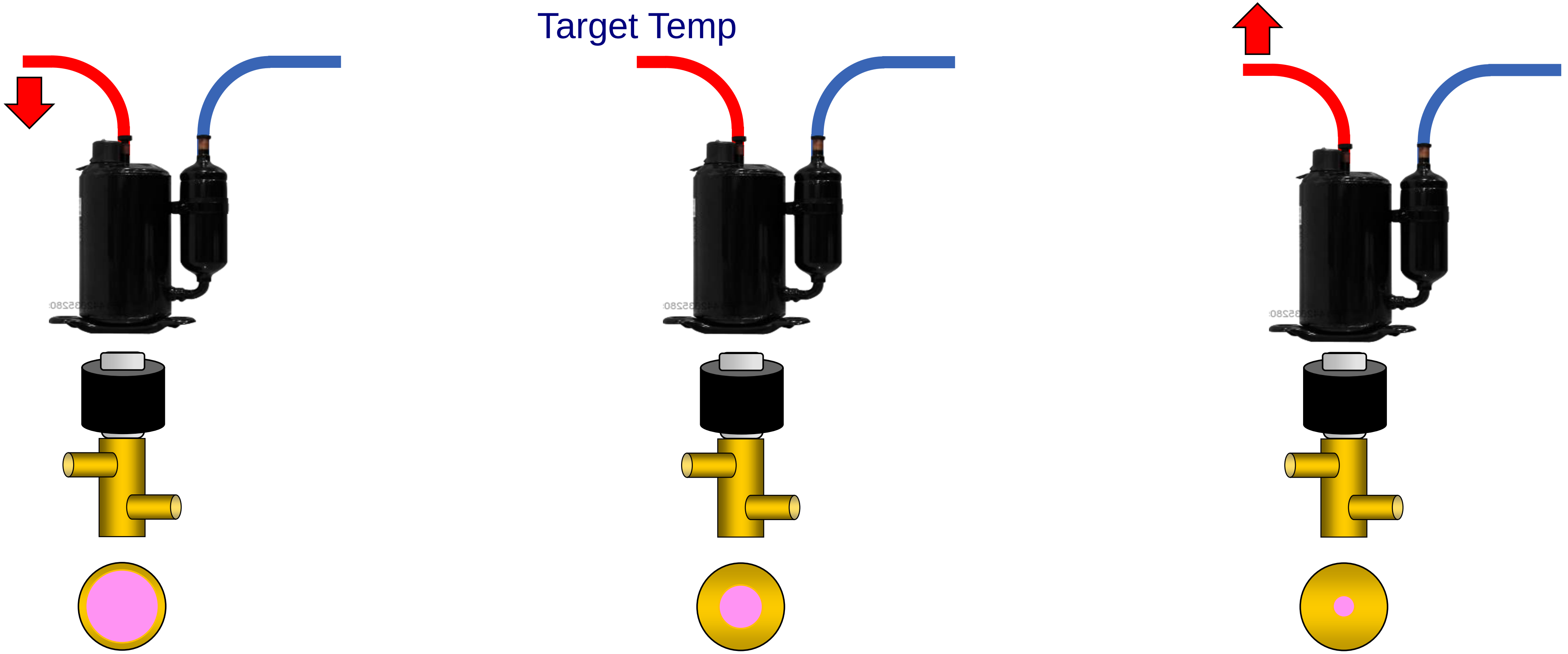


The Compressor
Speed Vs Pressure Relationships
(With no other changes in the system)



LEV Opening Vs Discharge Temperature Relationships

(With no other changes in the system)



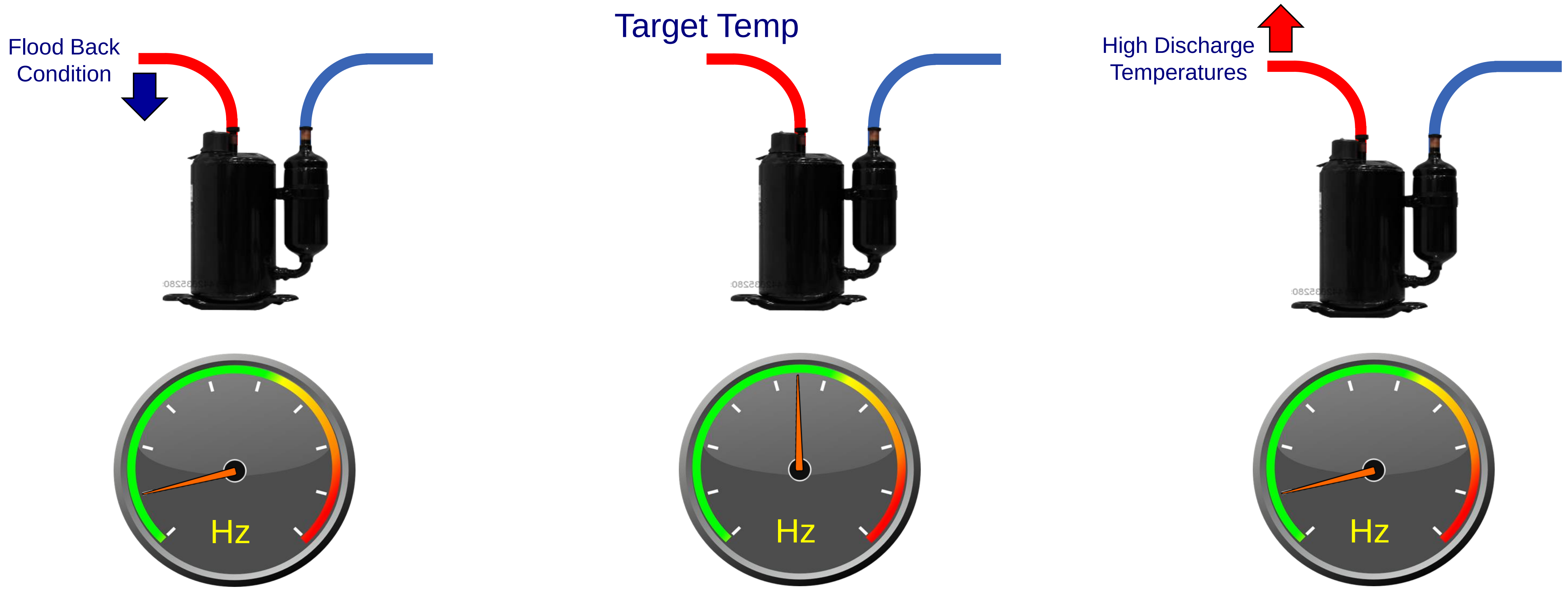
Open Valve to lower Temp

Stable Operation

Close Valve to increase Temp

Discharge Temperature protections Vs Compressor Frequency

(With no other changes in the system)



Compressor Slow Down

Stable Operation

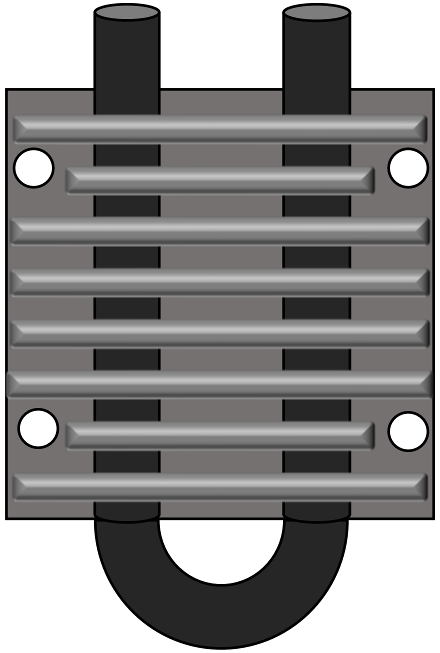
Compressor Slows Down

The Compressor
Speed Vs IPM Heat Sink Temperature

(With no other changes in the system)

High IPM
Temperature

Stable Operation



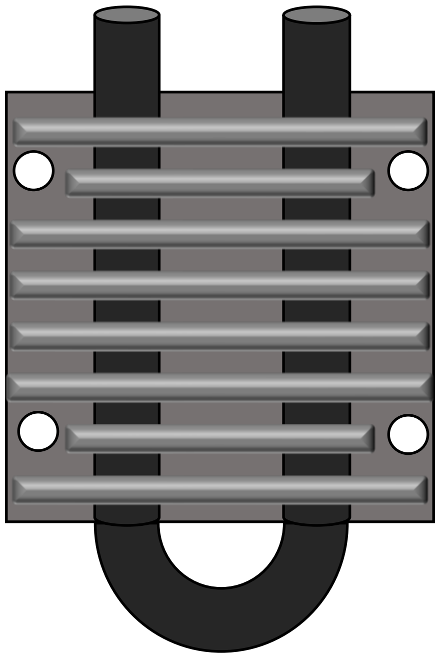
When the IPM temperature reaches 194^0 and above the compressor speed is reduced

The Compressor
Speed Vs IPM Heat Sink Temperature

(With no other changes in the system)

Stable Operation

IPM Protection



When the IPM temperature reaches 194^0 and above the compressor speed is reduced

Putting It All Together

- Let’s apply what we used to explain the system operation form a call

System Start Up

18 K

24K

30 K

6 Ton

Mode

Fan Speed

Set Point

Room Temp

Calculated Capacity

On start up the system checks the demand of the IDU's and calculates a desired capacity based on size, air flow and deviation from set point.
The ODU calculates the total capacity needed to determine its operating parameters.

ODU frequency set to match total IDU capacity

System Start Up Calculations

6 Ton

34,000 BTU's Required




	18 K	18 K	18 K	18 K
Mode	Cool	Cool	Cool	Off
Fan Speed	Low	High	Medium	Off
Set Point	70	70	72	74
Room Temp	72	76	73	74
Calculated Capacity	6,000	18,000	10,000	0

IDU EEV set to match IDU capacity or predetermined starting position

ODU frequency set to match total IDU capacity

System Start Up Calculations
What If……

6 Ton

? BTU’s Required



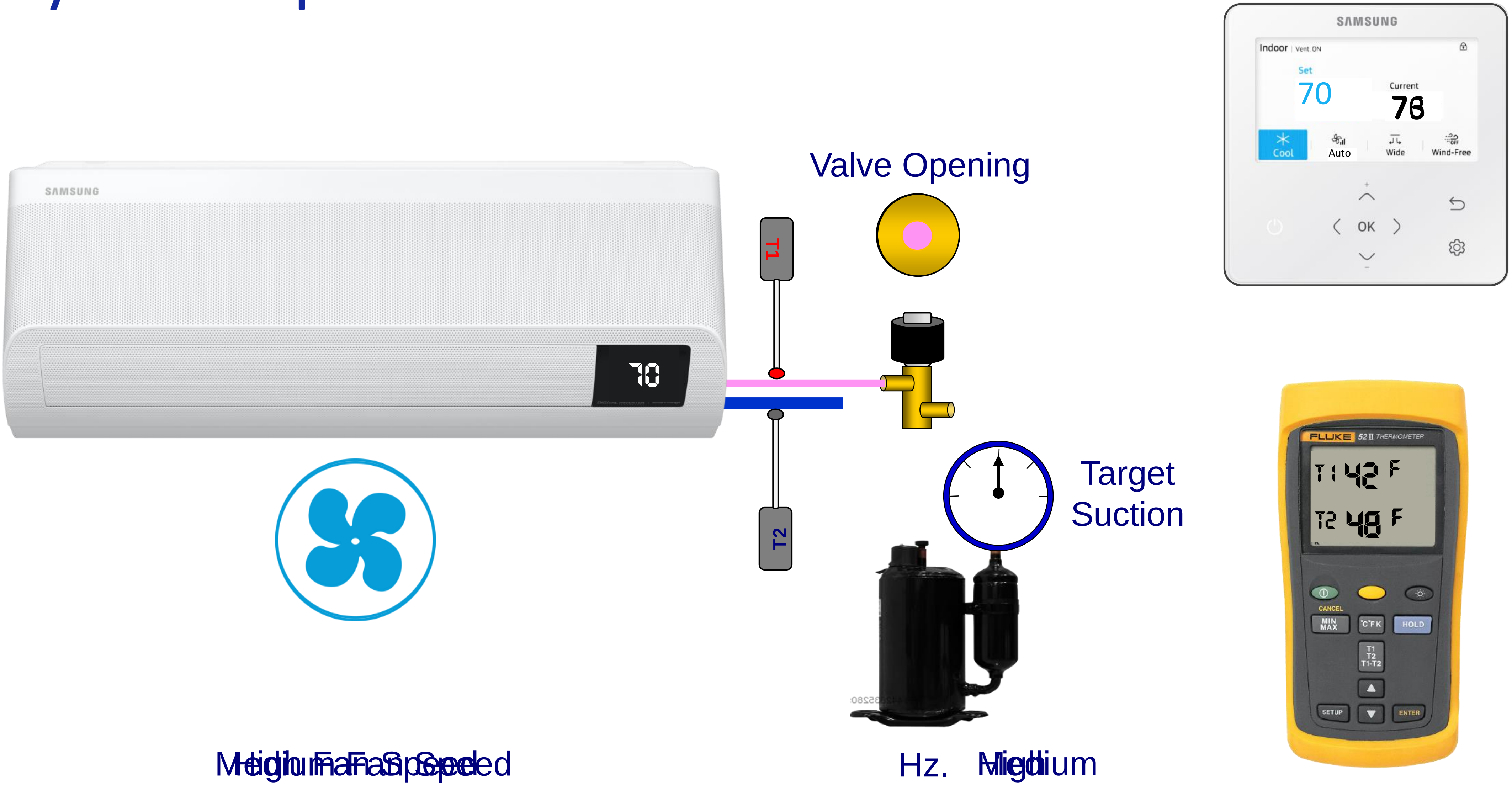


	18 K	18 K	18 K	18 K	18 K
Mode	Cool	Cool	Cool	Cool	Cool
Fan Speed	High	High	High	High	High
Set Point	68	68	68	68	68
Room Temp	80	78	78	81	80
Calculated Capacity	18,000	18,000	18,000	18,000	18,000

IDU EEV set to match IDU capacity

ODU frequency set to match total IDU capacity

Putting it all together
System Operation



Verifying Operation

- How can you verify the heating and cooling output of a unit?
- How does this apply A + B =C

Review

- Given the following system determine the fan speed settings for heat and cooling .
  - 70,000 BTU Input furnace?
  - 2 tons air 20 SEER air conditioning?
- Heat Speed.
- Cooling Speed
- More information
  - 95% 70,000 BTU Input furnace with a design TD of 55 degrees
  - 2 tons air high efficiency air conditioning requiring 350 cfm/ton?
- The CFM requirement for A/C is 700 CFM. The heating CFM can be determined on the following page

Calculating CFM

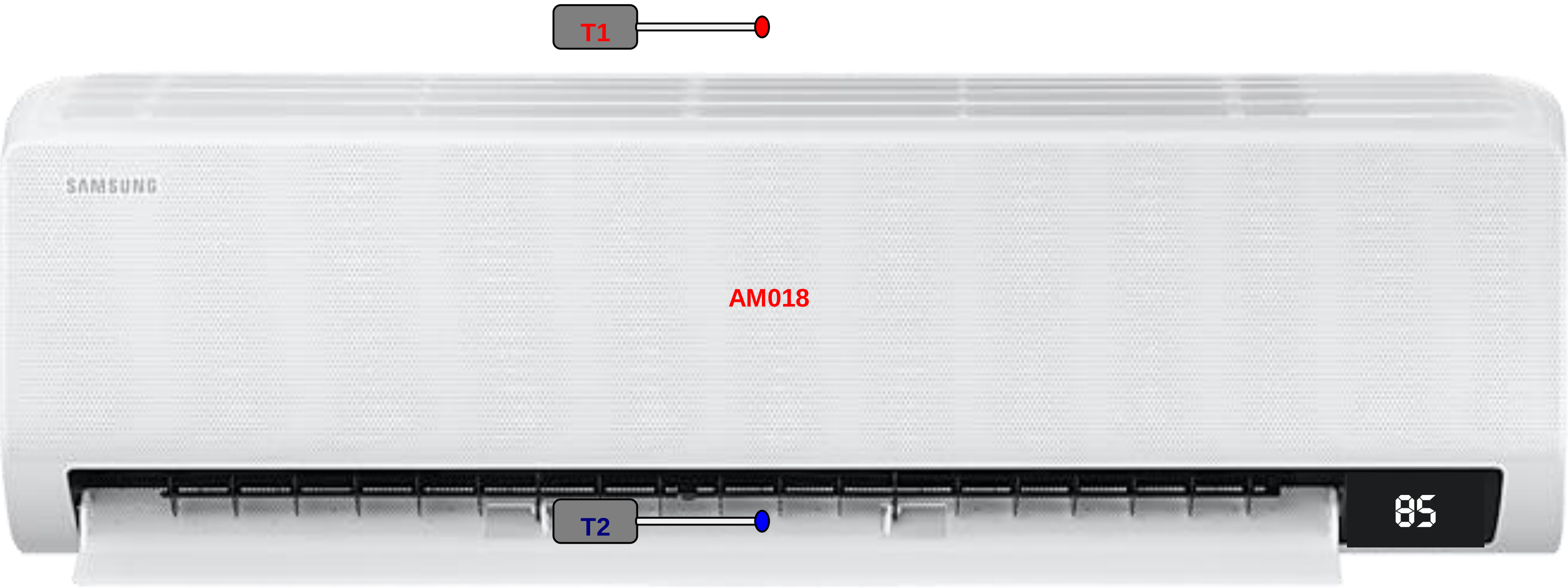
Gas Furnace

- When installing a gas burning piece of equipment you must calculate the required CFM needed for proper operation.
- To calculate the required CFM you need to know the systems Btu Output and the design TD.
- The formula is:
  - **CFM = BTU Output / TD x 1.08** (the specific heat of a cu’ of air).
- Calculate the CFM requirement for a 95% 70,000 BTU Input furnace with a design TD of 55 degrees?
- CFM = 66,500 / 55 x 1.08
- CFM = 1,103

Conclusion

- The heating system requires more air than the air conditioning system.
- To get the correct performance out of the air conditioning system it requires 700 CFM not 800.
- Why is this important?
- Does it really make a difference?
- Do your contractors know this?

Is this system operating properly?



IDU Capacity

- The IDU capacity is directly related to the air flow setting, refrigerant temperature and space conditions.
- If the air flow setting is low the output capacity is low
- If the air flow setting is high the output capacity will be high.
- On start up the system looks at air flow setting and deviation from set point and calculates the desired BTU out put.

Air Flow Setting

- Given the following system determine the fan speed settings for heat and cooling .
  - 70,000 BTU Input furnace?
  - 2 tons air high efficiency air conditioning?
- Heat Speed.
- Cooling Speed
- More information
- 95% 70,000 BTU Input furnace with a design TD of 55 degrees
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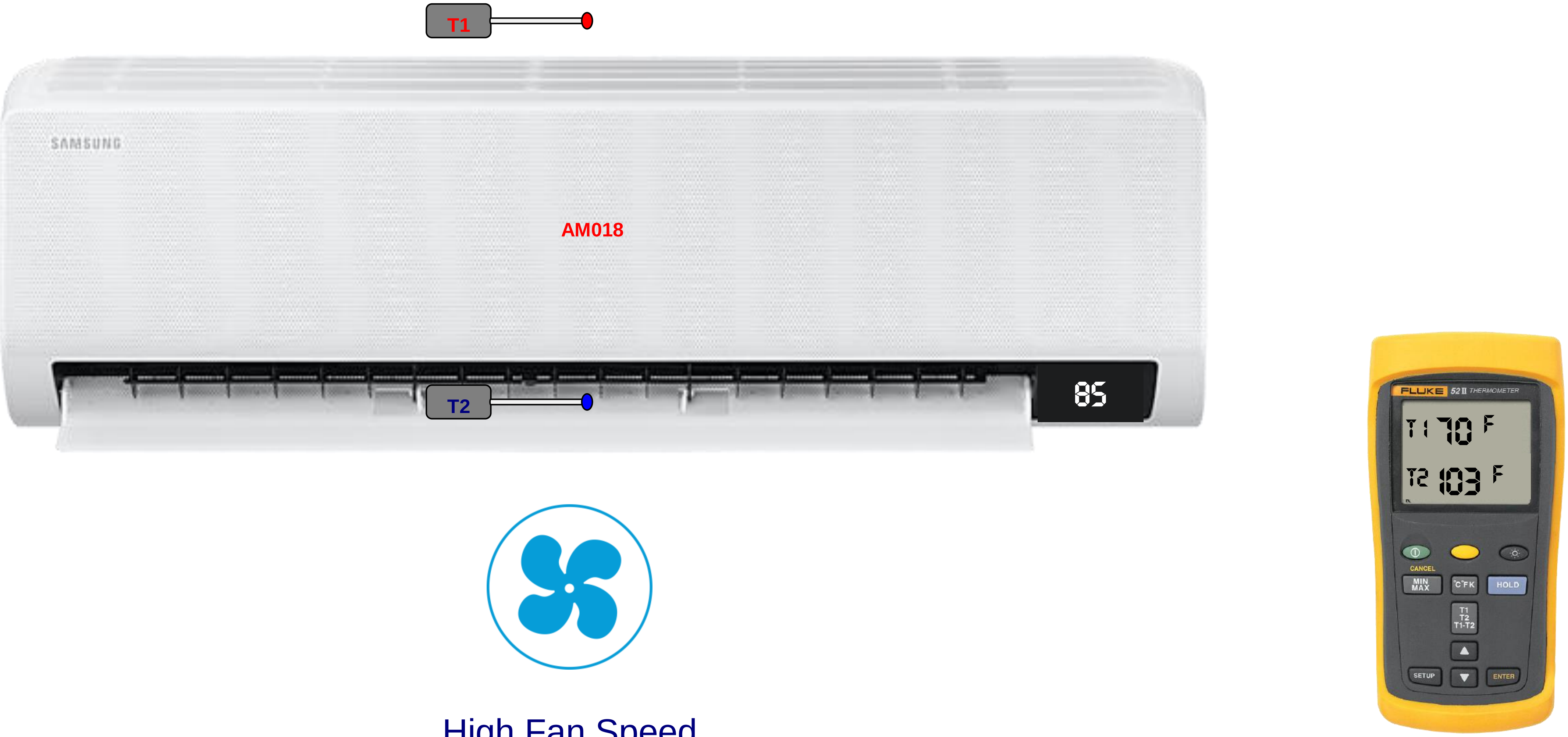
- CFM = 66,500 / 55 x 1.08
- CFM = 1,103

Calculating the IDU Heat Output

A+B=C

- **Heating BTUs = CFM x Δt x (Sensible Heat Factor) 1.08**
  - CFM = the fan CFM plotted on the fan tables, or measured at several points in the system
  - Δt = the dry bulb temperature change through the system
  - 1.08 = the heating BTU multiplier at sea level, or .075-lbs. of air per CFM x .24 (the specific heat of air) x 60 (minutes in an hour.) This factor will vary at higher altitudes and temperatures

Indoor Unit Testing in Heat Mode



High Fan Speed

Indoor unit CFM

- Finding the indoor CFM for a ductless unit is quite simple.
- If the coil and fan wheel is clean
- And there are no issues with the fan motor the CFM for the unit can be found using the specification sheet.

SAMSUNG SUBMITTAL AM018TNVDCH/AA

Samsung DVM S Series, Wind-Free™ Wall-Mounted Unit

Job Name
Purchaser
Submitted to
Unit Designation
Location
Engineer
Reference ☐ Approval ☐ Construction ☐
Schedule #

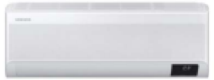
Specifications

Performance	Nominal Capacity[1]	Cooling (Btu/h)	18,000
		Heating (Btu/h)	20,000
Power	Voltage (ø/V/Hz)		1 / 208-230 / 60
	Nominal Running Current (A)		0.35
	MCA		0.44
	MOP		15
Fan	Type		Crossflow
	Motor	Type	BLDC (1)
		Output (W)	27
Airflow	CFM (UL)	L/M/H	424 / 487 / 555
Refrigerant	Type		R410A
	Control Method		Electronic Expansion Valve
Piping Connections	Liquid (flare)	Inches	1/4
	Suction (flare)	Inches	1/2
	Drain	Inches	ID 11/16 Hose
Unit Dimensions	W X H X D	Inches	41 1/2 X 11 3/4 X 8 7/16
	Weight	lbs.	26.5
Sound Level	L / M / H	dB(A)	34 / 37 / 40
Accessories	External Contact Control		☐ MIM-B14
	Condensate Pump	Aspen Mini Orange	☐ ASP-MO-UNIV 110-250
		Blue Diamond	☐ BD-BLUE230
	External Temperature Sensor		☐ MRW-TA
Safety Certifications			ETL (UL 1995)

[1] Nominal cooling capacities are based on: Indoor temperature: 80°F DB, 67°F WB. Outdoor temperature: 95°F DB, 75°F WB
Nominal heating capacities are based on: Indoor temperature: 70°F DB, 60°F WB. Outdoor temperature: 47°F DB, 43°F WB

*The Wind-Free™ unit delivers an air current that is under 0.15 m/s while in Wind-Free™ mode. Air velocity that is below 0.15 m/s is considered "still air" as defined by ASHRAE (American Society of Heating, Refrigerating, and Air-Conditioning Engineers).

Samsung HVAC maintains a policy of ongoing development, specifications are subject to change without notice.



- Compatible with Samsung DVM S, DVM S Water, and DVM S Eco systems (AM**********AA).
- Electro-static, washable, pleated filters as standard (included with unit)
- 7-segment digital display on front of unit to display temperature and unit status
- High-voltage terminal block temperature sensor to disable unit in the event overheating of power connection.
- Wind-Free™ function will close the supply air outlet louver while in cooling mode to gently disperse cool air into the space without blowing directly onto occupants. The Wind-Free™ feature is optional and can be enabled using central or local control options (MWR-SH11UN, MWR-WE13UN, MWR-WG00*N, and AR-EH0*U only).

Construction
HIPS chassis certified to UL94 V0 with galvanized steel mounting bracket

Heat Exchanger
The heat exchanger shall be mechanically bonded fin to copper tube

Indoor Fan
Indoor fan is a single, antibacterial, crossflow type

Three fan speed settings and auto setting

Automatic (motorized) vertical swing (up/down) and horizontal swing (left/right) louvers

Controls
The unit shall be operated via a wireless or wired remote control with DDC type signal

The unit shall integrate with the Samsung NASA Controls Network Solution

Controls shall integrate with a BMS system

Control wiring shall be 2 X 16 AWG shielded wire

SHA-DVMS-05052020

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888-699-6067

Intertek

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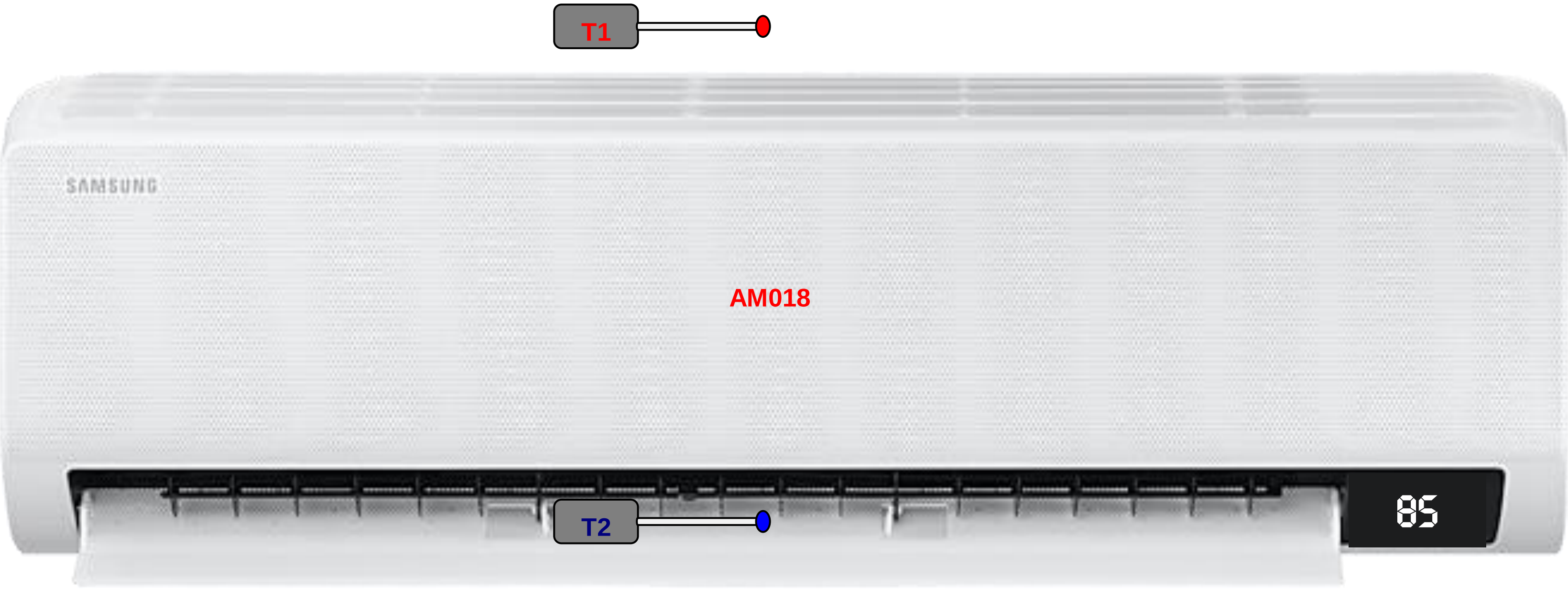
Calculate the Indoor Units Capacity

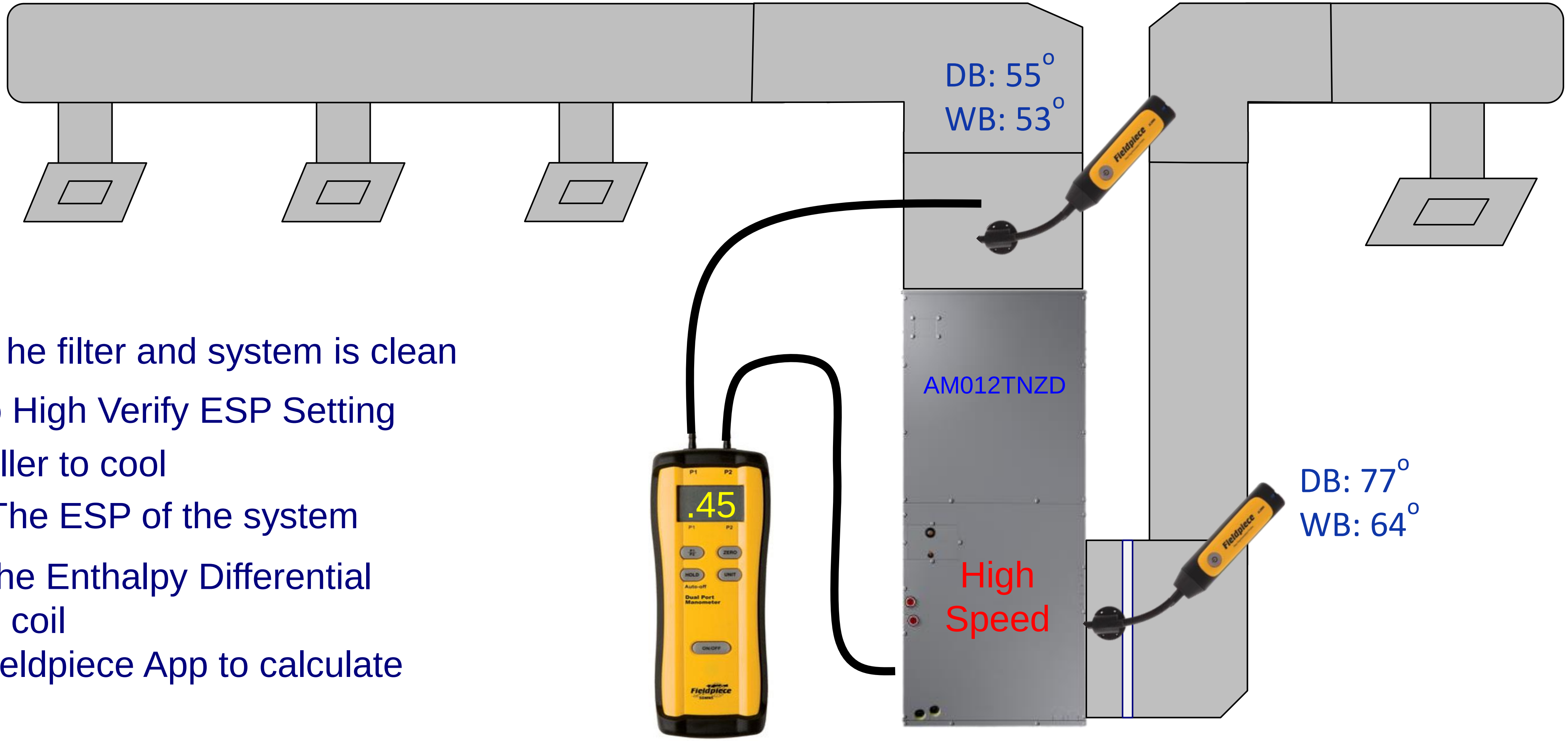


555 CFM

Cooling Capacity Calculations

- Cooling BTUs = CFM x Δh x 4.5
- CFM = The fan CFM plotted on the fan tables, or measured at several points in the system
- Δh = the enthalpy change through the system (we get this number by converting wet bulb readings.)
- 4.5 = the cooling BTU multiplier at sea level, or .075-lbs. of air per CFM x 60 (minutes in an hour.) This factor will also change at higher altitudes. BTU multipliers will also change due to extreme air temperatures

Air Handler CFM System Measurements



Verify that he filter and system is clean

Set Fan to High Verify ESP Setting

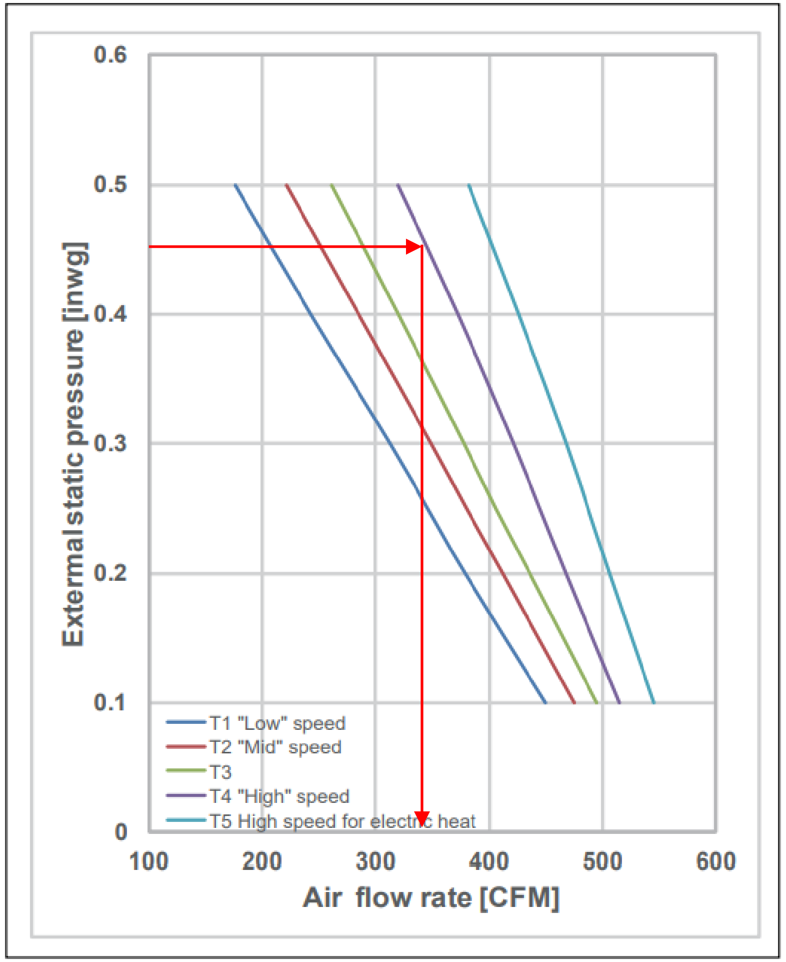
Set controller to cool

Measure The ESP of the system

Measure the Enthalpy Differential across the coil

Use the Fieldpiece App to calculate BTU's

Air Handler CFM AM012



SAMSUNG SUBMITTAL AM012TNZDCH/AA

Samsung DVM S Series Multiposition Air Handler Unit

Job Name		Location	
Purchaser		Engineer	
Submitted to		Reference ☐ Approval ☐ Construction ☐	
Unit Designation		Schedule #	

Specifications

Performance	Nominal Capacity (Btu/h)	Cooling	12,000
		Heating	13,500
Power	Voltage	ø / V / Hz	1 / 208-230 / 60
	Nominal Input Current*	Cooling (A)	0.49
	MCA*	Amps	0.9
	MOCP*	Amps	10
Fan	Type		Double-inlet, forward curve, centrifugal
	Motor	Type	Constant-torque (ECM)
		HP	1/3
		Output (W)	290
Airflow	CFM @ 0.4" ESP (UL)	L / M / H	243 / 285 / 373
External Static Pressure	Standard	"WC	0.4
	Min. / Max.	"WC	0.1 / 0.5
Refrigerant	Type		R410A
	Control Method		Electronic Expansion Valve
Piping Connections	Liquid	Inches	1/4
	Suction	Inches	1/2
	Drain	Inches	3/4" FNPT
Unit Dimensions	W X H X D	Inches	17 1/2 X 43 X 21
	Weight	lbs.	105
Sound Level	Low / Mid / High	dB(A)	34 / 36 / 38
Accessories	Filter Base W/1" Filter		☐ VFB-1
	External Temperature Sensor		☐ MRW-TA
	Supplemental Electric Heater Kit	3Kw	☐ VHK-103A
	Downflow Conversion Kit		☐ VDK-1
	EEV Extension Wire Harness**		☐ DB82-05832A
	External Contact Control		☐ MIM-B14
	CN83 Pigtail (for external contact input)		☐ DB39-01263A
Safety Certifications			ETL (UL 1995)

*Power data is without optional electric heat kits installed.

**Required for applicable models when using Downflow Conversion Kit

[1] Nominal cooling capacities are based on: Indoor temperature: 80 °F DB, 67°F WB. Outdoor temperature: 95°F DB, 75°F WB.
Nominal heating capacities are based on: Indoor temperature: 70 °F DB, 60°F WB. Outdoor temperature: 47°F DB, 43°F WB.

[2] Refer to technical data book for fan performance details and settings

Samsung HVAC maintains a policy of ongoing development, specifications are subject to change without notice.

- Compatible with Samsung DVM S, DVM S Water, and DVM Eco systems (AM**********AA).
- High-voltage terminal block temperature sensor to disable unit in the event overheating of controls power connection.
- Multiposition - vertical, horizontal left, and horizontal right.
- Capable of being field convertible to downflow configuration with optional downflow conversion kit.
- Air handler has an air leakage of no more than 2 percent of the design air flow rate when tested in accordance with ASHRAE 193.

Construction
The unit shall be constructed of insulated, powder coated, galvanized steel

Heat Exchanger
The heat exchanger shall be mechanically bonded fin to copper tube

Indoor Fan
Indoor fan is a double-inlet, forward curve, centrifugal type with a single constant-torque (ECM) fan motor

The indoor unit shall have low, medium, high, and auto fan speed setting options.

Five fan speed taps for optional air flow setting during installation

The indoor unit shall have the capability to turn the fan off in heating or cooling modes while in thermal-OFF status (external sensor required).

Controls
0 volt ON/OFF control (ex: auxiliary drain switch) when using the optional CN83 pigtail (part number DB39-01263A, sold separately).

The indoor unit shall integrate with the Samsung NASA Controls Network Solution

Controls shall integrate with a BMS system

Control wiring shall be 2 X 16 AWG shielded wire

Air Filtration
Air filtration must be field provided

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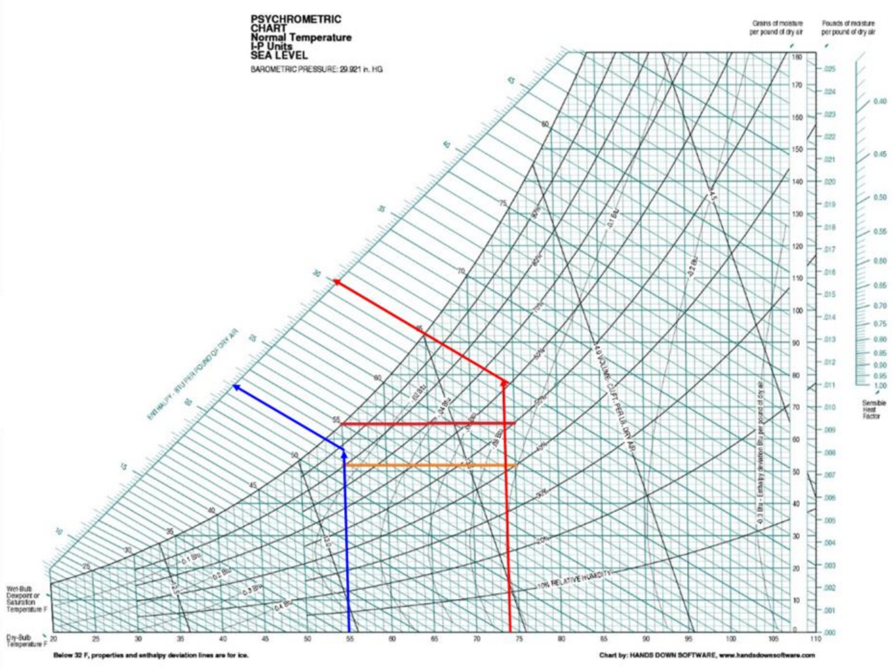
Calculate Enthalpy Differential

- Using a psychrometric calculator determine the enthalpy of the return and supply air then calculate the difference
  - Return Air: 75 BD – 63.3 WB – Enthalpy 29.2
  - Supply Air: 55 BD – 51.1 WB – Enthalpy 21.1
  - Enthalpy 8.1

Free online Psychrometric Calculator Print

IP SI altitude 0 ft b press 30.01 in.Hg

T (°F) dry bulb	rH (%) rel.humidity	TDP (°F) dew point	W (GPP) abs. humidity	h (btu/lb) enthalpy	Tw (°F) wet bulb	sp. vol (cft/lb)		
75	55	57.7	71	29.2	63.3	13.78	Calculate	Remove
53.1	90	50.4	54	21.1	51.1	13.14	Calculate	Remove



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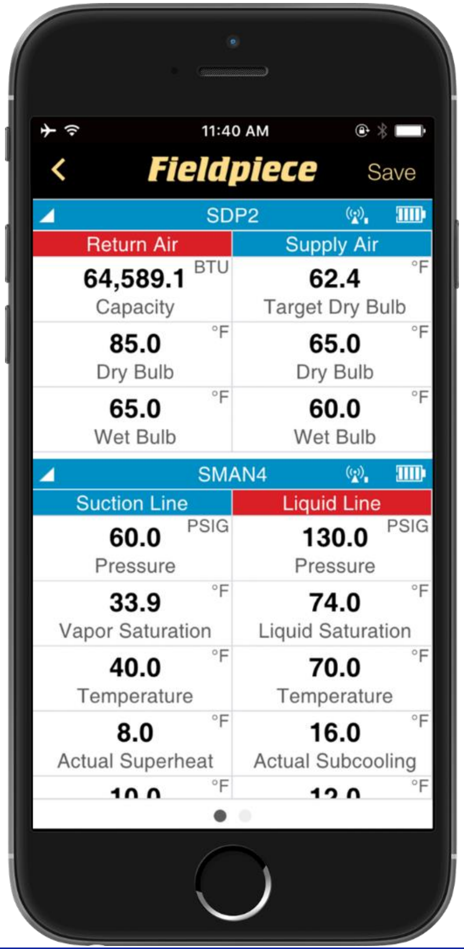
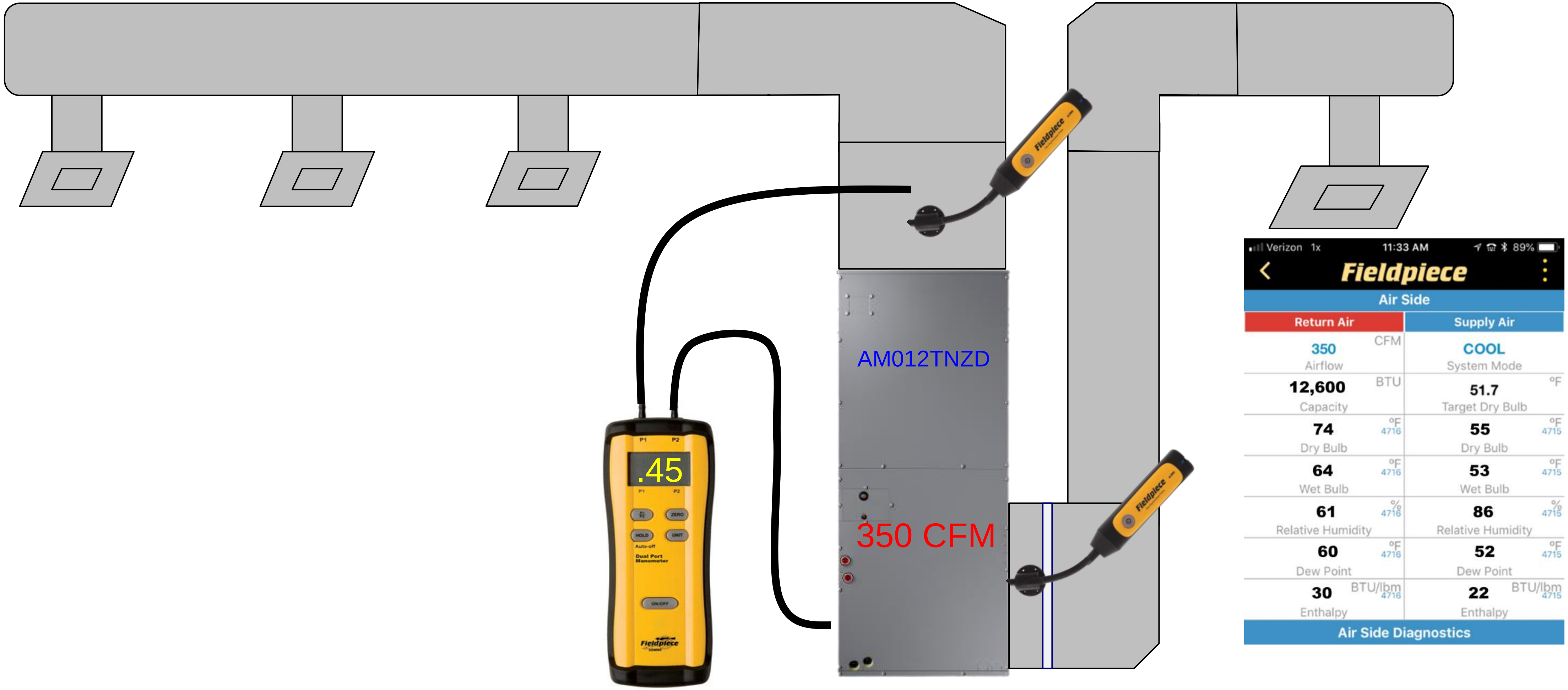
Cooling Capacity Formula

- BTUH = CFM x 4.5 X Delta Enthalpy
- As an **example**, measurements inside the main ducts of the air handler:
  - Before the evaporator (return): DB: 75^{o} F, WB: 63.3^{o} F, Enthalpy = 29.2 BTU/lb.
  - After evaporator (supply): DB: 55^{o} F, WB: 51.1^{o} F, Enthalpy = 21.1 BTU/lb.
- **System Enthalpy: 29.2** – 21.1= 8.1 BTU/lb**.**
- **System CFM =** 350 CFM
- **BTUH =** 350 x 4.5 x 8.1
- **BTUH =** 12,600 BTUH or 1 ton

Make Life Simple



Air Handler CFM
System Measurements



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System Startup Basic Operating Data

During Start up observe the following conditions

- Compressor discharge temperature
- Condenser outlet temperature
- High and Low pressure
- Compressor current
- Indoor unit EEV (cool)
- Indoor unit EEV (heat)
- Startup report

Test Modes
Cooling / Heating

1) Outdoors:

- Compressor: following normal operation logic but Max Hz is limited based on EERPOM data.
- Example: Compressor in Cooling –
  - Normal Operation 140hz
  - Test Cooling 80hz
- Example: Compressor in Heating –
  - Heating – Normal Operation 160hz
  - Test Heating 90hz
- These values can vary based on ODU.
- EEV: following normal operation
- Fan motor: following normal operation

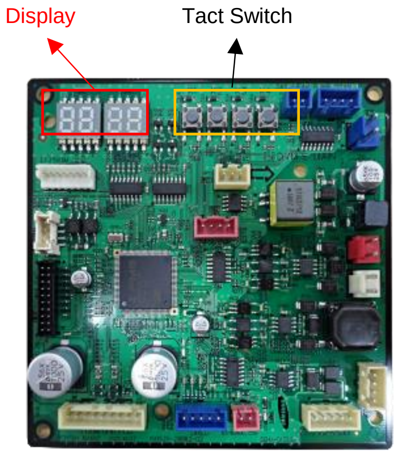
2) Indoor units:

- EEV: following normal operation
- Fan motor: high speed
- Set temperature: 37^0 F for colling / 40 degree(celsius) 104^0 F for heating
- Demand capacity: 100%

3) Protection logic:

- follows normal operation

Gathering Information.



S-NET pro 2 - DVM S Water

Home | Trend Graph | Replay | Add-On | Help

8/4/2022 8:49:59 AM
Duration 00:11:57

8/4/2022 9:39:10 AM
Current Time '4/2022 9:01 AM

16x Fast Slow

Player

Indoor Unit Installation Data

Indoor Unit Installation Data

Address	Model	RMC	MCU ADDRES	MCU PORT	Location	Product Option	Installation Option	Installation Option2	Main Micom	Error History1	Error History2	Error History3	Error History4	Error History5
0	Slim 1Way	00	7	A	-	[0]17044-[1]180C8-[2]01616-[3]30010	[0]20010-[1]00000-[2]00000-[3]00000	[0]50000-[1]00000-[2]00000-[3]00000	DB91-01888C 180502	000	000	000	000	000
1	Slim 1Way	01	6	D	-	[0]17044-[1]180C8-[2]01616-[3]30010	[0]20010-[1]00000-[2]00000-[3]00000	[0]50000-[1]00000-[2]00000-[3]00000	DB91-01888C 180502	000	000	000	000	000
2	Slim 1Way	03	6	A	-	[0]17044-[1]180C8-[2]01616-[3]30010	[0]20010-[1]00000-[2]00000-[3]00000	[0]50000-[1]00000-[2]00000-[3]00000	DB91-01888C 180502	000	000	000	000	000
3	Duct	03	7	B	-	[0]10054-[1]E54BB-[2]04646-[3]31115	[0]20010-[1]20000-[2]00000-[3]00000	[0]50000-[1]00000-[2]40000-[3]00000	DB91-01889A 170309	-	-	-	-	-
4	Duct	00	6	C	-	[0]10054-[1]E54BB-[2]04646-[3]31115	[0]20010-[1]20000-[2]00000-[3]00000	[0]50000-[1]00000-[2]40000-[3]00000	DB91-01889A 170309	-	-	-	-	-
5	Slim 1Way	05	7	E	-	[0]17044-[1]180C8-[2]01616-[3]30010	[0]20010-[1]00000-[2]00000-[3]00000	[0]50000-[1]00000-[2]00000-[3]00000	DB91-01888C 180502	000	000	000	000	000
6	Duct	06	7	F	-	[0]10054-[1]E5438-[2]03535-[3]31104	[0]20010-[1]20000-[2]00000-[3]00000	[0]50000-[1]00000-[2]40000-[3]00000	DB91-01889A 170309	-	-	-	-	-
7	Slim 1Way	07	6	B	-	[0]17044-[1]180C8-[2]01616-[3]30010	[0]20010-[1]00000-[2]00000-[3]00000	[0]50000-[1]00000-[2]00000-[3]00000	DB91-01888C 180502	000	000	000	000	000
8	Slim 1Way	02	7	C	-	[0]17044-[1]180C8-[2]01616-[3]30010	[0]20010-[1]00000-[2]00000-[3]00000	[0]50000-[1]00000-[2]00000-[3]00000	DB91-01888C 180502	000	000	000	000	000

AM024MNHDCH/AA

Fan options		ESP	Sound Pressure (dBA)		
		inAq	H	M	L
Default	010054-1E54BB-204646-331115	0.2	39	36	30
Option	010054-1E581F-204646-331115	0.3	41	36	31
	010054-1E5973-204646-331115	0.4	43	38	33
	010054-1E59C6-204646-331115	0.5	45	40	34
	010054-1E5D1A-204646-331115	0.6	47	42	36
	010054-1E5D6D-204646-331115	0.7	48	43	37
	010054-1E5EA0-204646-331115	0.8	48	44	39

This customer is complaining about the temperatures in these 2 zones.
Units are AM024MNHDCH
System had been tested and balanced
This is a commercial system with long duct runs
What does the product and installations codes tell us.

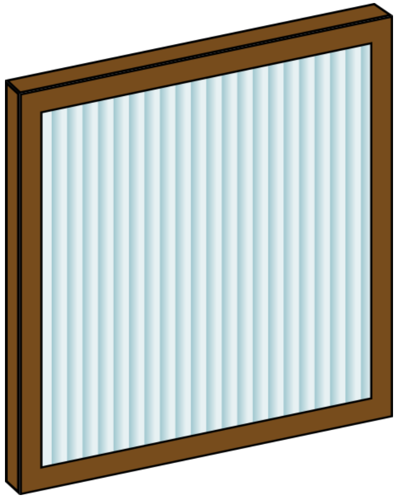
Outdoor Unit Data | Outdoor Unit Installation Data | Outdoor Unit Cycle Diagram | Indoor Unit Data | Indoor Unit Installation Data | Control for Unoccupied and Entering Room | MCU Unit Data

Version 1.13.0 Unit - Temp.: ℉ Power : Btu Pressure : psi 9/15/2022 4:59 PM COM -1

Filter Selection

Know what's in your system!

Air Flow: 1000 CFM
300 FPM



16 x 25 Air Filters @ 1,000 CFM

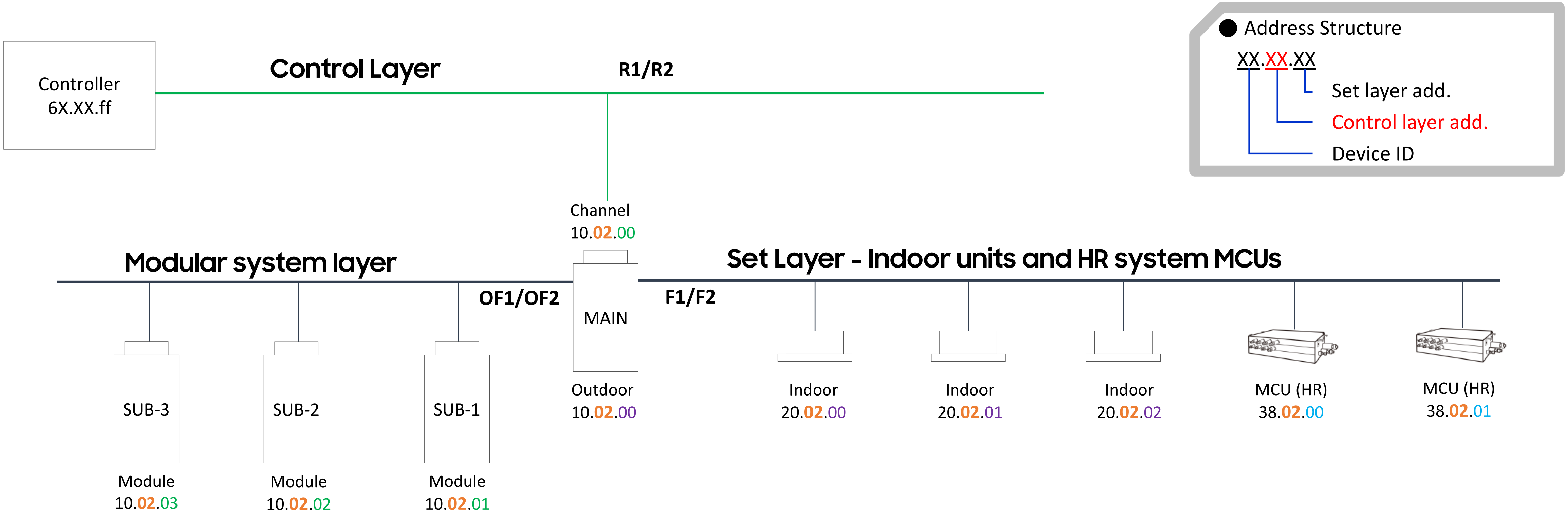
290	430	575	720	970*
0.09	0.15	0.22	0.31	0.46

Air Flow: 1400 CFM
500 FPM

Resistance : .2" wc

Samsung Equipment Structure

- Indoor units and Mode Control Units (MCU) are controlled and monitored through the MAIN ODU
- For modular systems, the second, third, and fourth (not USA) outdoor units are also seen through the MAIN ODU



STS Training Series

DVM S Refrigeration Circuits and Teoubleshooting

Overview

- Use the operational data for each mode to troubleshoot each exercise.
- State your diagnosis and how you came to your decision.
- Explain how the system responds to the changes in the system.

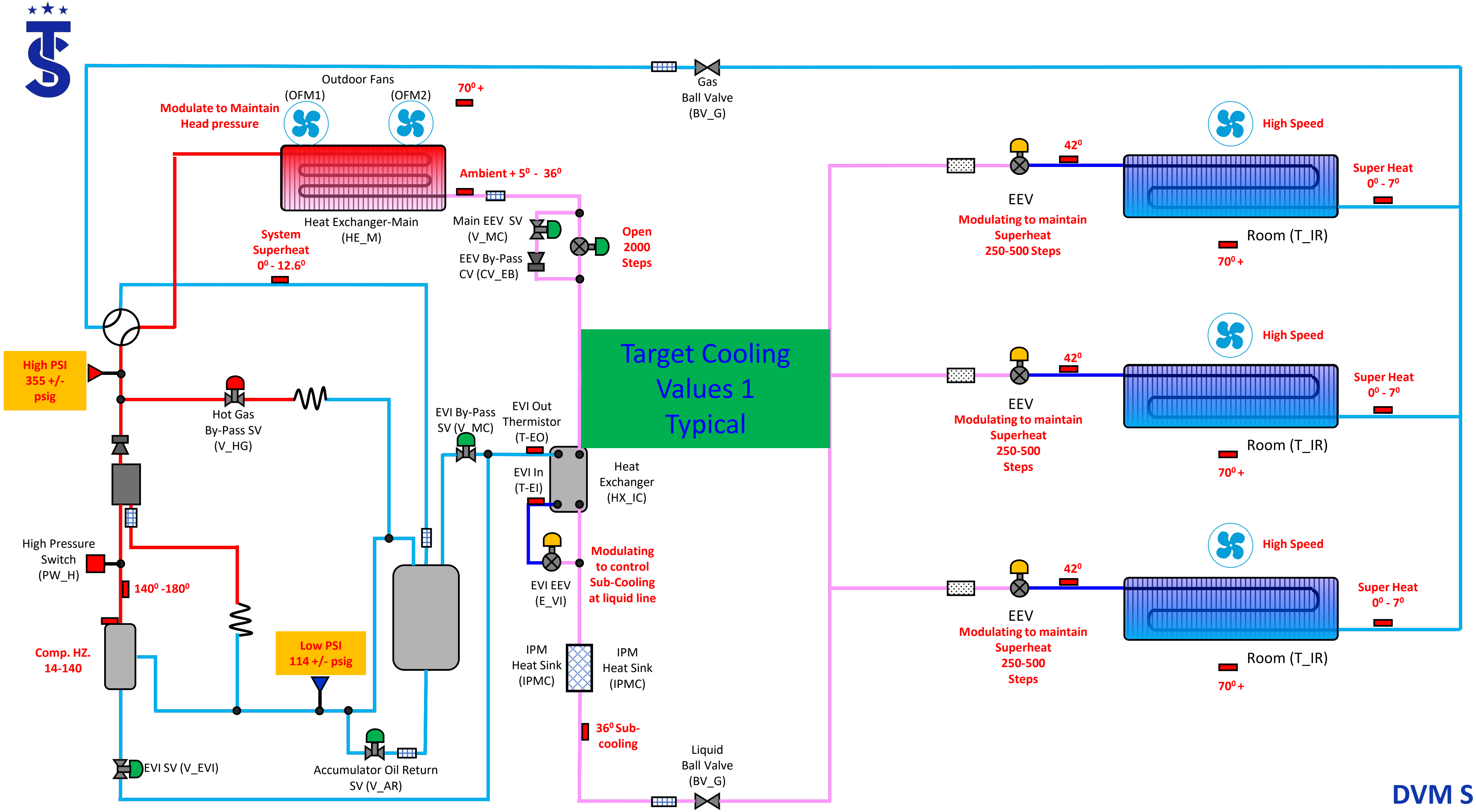
Test Mode Operation

- Outdoor Unit Operation
  - Compressor hertz (These values can vary based on ODU)
    - Normal Cooling Max 140 hz
    - Test Mode Cooling operation 80 hz
    - Normal Heating Max 160 hz.
    - Test Mode Heating operation 90hz
  - In test mode the all-other controls operate as normal to maintain target operation.
- Indoor units
  - Fan motor: high speed
  - Set Point Temperature (100% demand)
    - 3 degree(Celsius) for colling 36^0 F
    - 40 degree(Celsius) for heating 104^0 F
  - EEV operates as normal to maintain target operation
- Protection logic
  - Follows normal operation

Check point of cooling trial operation

※ The data is not absolute one, only can be used to check the abnormal status

Item	Normal	Check point
EVA IN Temp.	44.6^0 - 57.2^0	▷ Check **target Eva in temperature** of option setting(cooling capacity correction) The default : EVA IN 44.6^0 -57.2^0, Super heat = EVA OUT - EVA IN = 1.8^0 – 7.2^0 (Normal) ▷ **Refrigerant shortage check** With all IDU operation, If more than 50% of the indoor unit's **SH is over 10.6** & EEV step of those unit is **over 900 step**. (In case of EEV RAC : 1300 step) ▷ **Refrigerant over charged check** With all IDU operation, as the room temp. is over 77^0, If the majority of IDU of **SH is 0 -1.8^0** & these EEV step is **less then 200step**. (In case of EEV RAC : 300 step) ▷ If only few indoor unit's EEV step is the Max. step, check the pipe length from the first branch pipe to indoor unit (limited 45m) also ensure that if there is any kind of flow resistance
EVA OUT Temp.	44.6^0 – 57.2^0	
EEV Step of Indoor Unit	250 ~ 900	
Low Pressure psig	85 - 135	▷ The low pressure varies from 85 – 135 according to the piping length & target low pressure. When the ambient temperature is high or the indoor load is big, the low pressure may be higher. ▷ High pressure will rise when the ambient temperature rises (Max. 512 psig) When the ambient temperature decreases, high pressure may be decreased as well
High Pressure psig	355 - 469	
Compressor Discharge Temp.	140^0 - 221^0	▷ High pressure is over 355 psig but discharge temp. is less than 140^0 (DSH less than 36^0) Please check the EVI EEV, PHE leakage, indoor EEV opening error, Indoor fan operating error.
Condensing Temp.	95^0 - 131^0	▷ Condensing temperature will be changed according to the high pressure
Liquid Tube Temp.	77^0 - 122^0	▷ Condensing temp. – liquid tube temp.= Sub-cooling (9^0 - 36^0)



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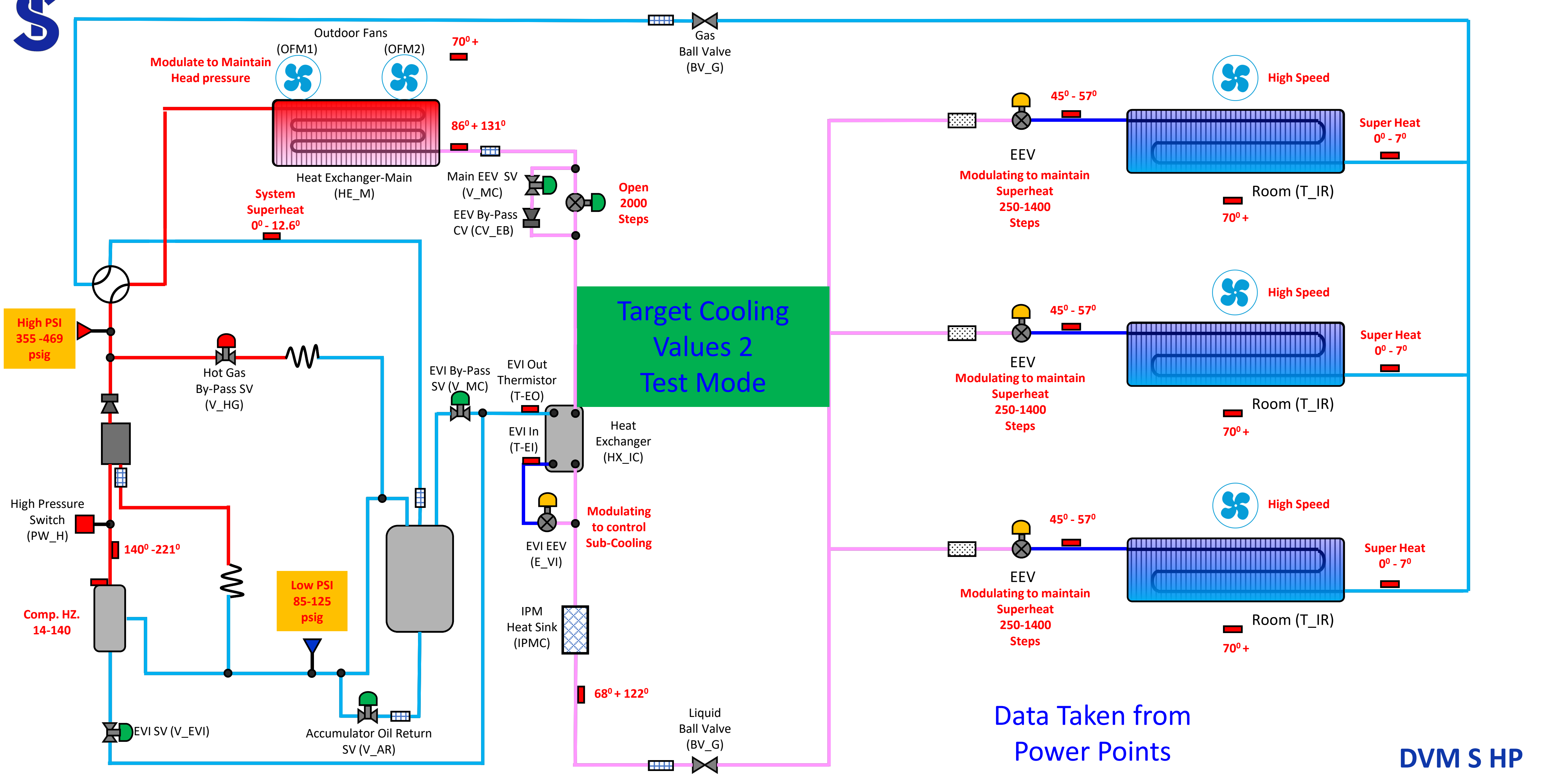

Outdoor Fans
(OFM1)
(OFM2)
Modulate to Maintain Head pressure
70^0 +
Gas Ball Valve (BV_G)
Heat Exchanger-Main (HE_M)
86^0 + 131^0
Main EEV SV (V_MC)
EEV By-Pass CV (CV_EB)
Open 2000 Steps
System Superheat 0^0 - 12.6^0
High PSI 355 -469 psig
Hot Gas By-Pass SV (V_HG)
EVI By-Pass SV (V_MC)
EVI Out Thermistor (T-EO)
EVI In (T-EI)
Heat Exchanger (HX_IC)
Target Cooling Values 2 Test Mode
High Pressure Switch (PW_H)
140^0 -221^0
Comp. HZ. 14-140
Low PSI 85-125 psig
EVI EEV (E_VI)
Modulating to control Sub-Cooling
IPM Heat Sink (IPMC)
68^0 + 122^0
EVI SV (V_EVI)
Accumulator Oil Return SV (V_AR)
Liquid Ball Valve (BV_G)
High Speed
45^0 - 57^0
EEV
Modulating to maintain Superheat 250-1400 Steps
Super Heat 0^0 - 7^0
Room (T_IR)
70^0 +
High Speed
45^0 - 57^0
EEV
Modulating to maintain Superheat 250-1400 Steps
Super Heat 0^0 - 7^0
Room (T_IR)
70^0 +
High Speed
45^0 - 57^0
EEV
Modulating to maintain Superheat 250-1400 Steps
Super Heat 0^0 - 7^0
Room (T_IR)
70^0 +
Data Taken from Power Points
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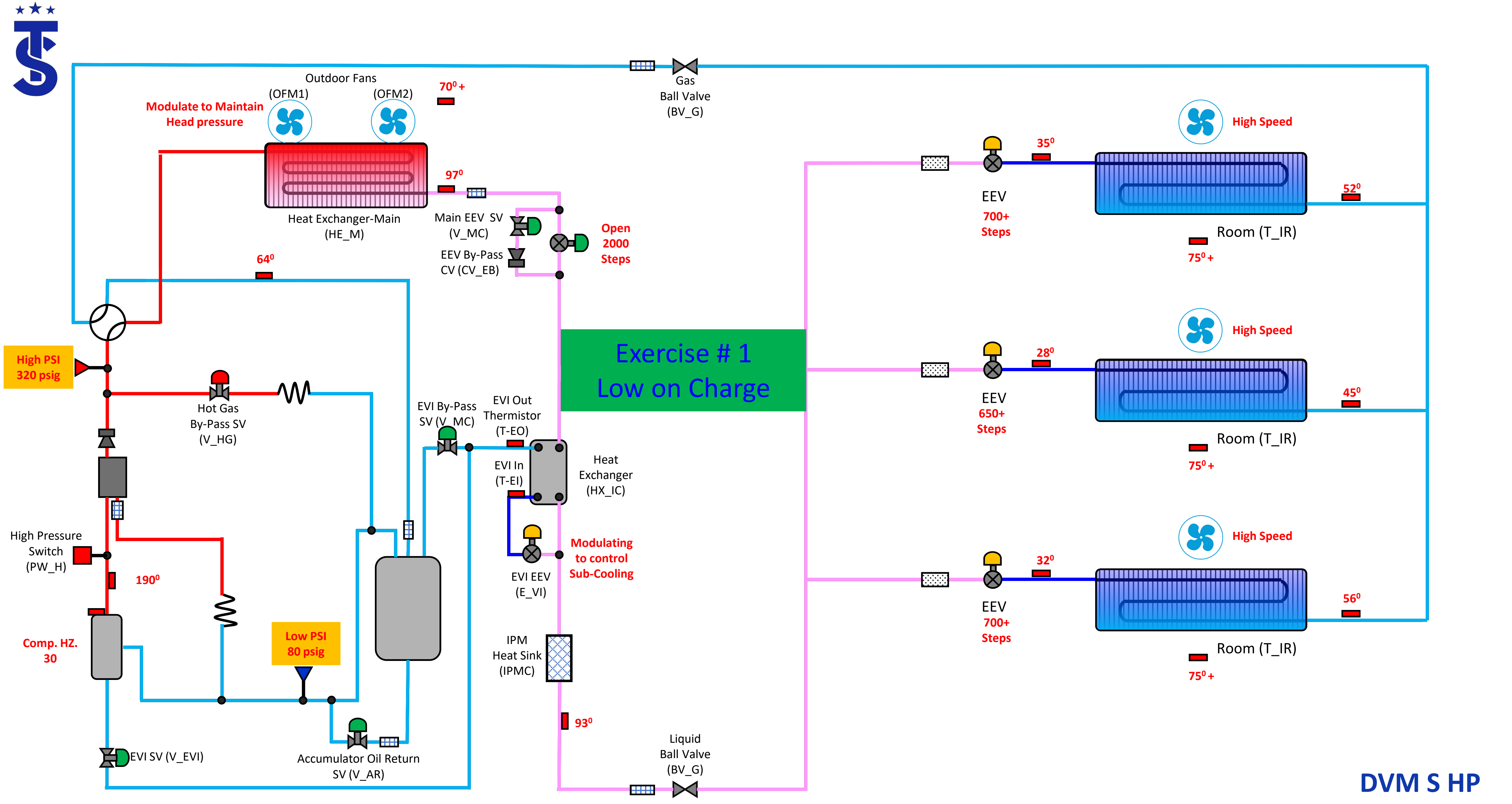
Exercise # 1
Operational Characteristics

- ❑ Inadequate cooling
- ❑ System Cools when the load is low but not when its high
- ❑ Compressor does not seem to run at a high enough speed for the load

We can use this section to put in operational characteristics of the system and any other data pertinent to the faut, like the compressor is in protection mode or system is throwing a fault. Thoughts



Outdoor Fans
(OFM1)
(OFM2)
Modulate to Maintain
Head pressure
70⁰ +
Gas
Ball Valve
(BV_G)
High Speed
35⁰
97⁰
Heat Exchanger-Main
(HE_M)
Main EEV SV
(V_MC)
EEV By-Pass
CV (CV_EB)
Open
2000
Steps
EEV
700+
Steps
52⁰
Room (T_IR)
75⁰ +
64⁰
High Speed
28⁰
High PSI
320 psig
Exercise # 1
Low on Charge
Hot Gas
By-Pass SV
(V_HG)
EVI By-Pass
SV (V_MC)
EVI Out
Thermistor
(T-EO)
EEV
650+
Steps
45⁰
Room (T_IR)
75⁰ +
Heat
Exchanger
(HX_IC)
EVI In
(T-EI)
High Pressure
Switch
(PW_H)
190⁰
Modulating
to control
Sub-Cooling
EVI EEV
(E_VI)
High Speed
32⁰
EEV
700+
Steps
56⁰
Room (T_IR)
75⁰ +
Comp. HZ.
30
Low PSI
80 psig
IPM
Heat Sink
(IPMC)
93⁰
Liquid
Ball Valve
(BV_G)
EVI SV (V_EVI)
Accumulator Oil Return
SV (V_AR)
DVM S HP

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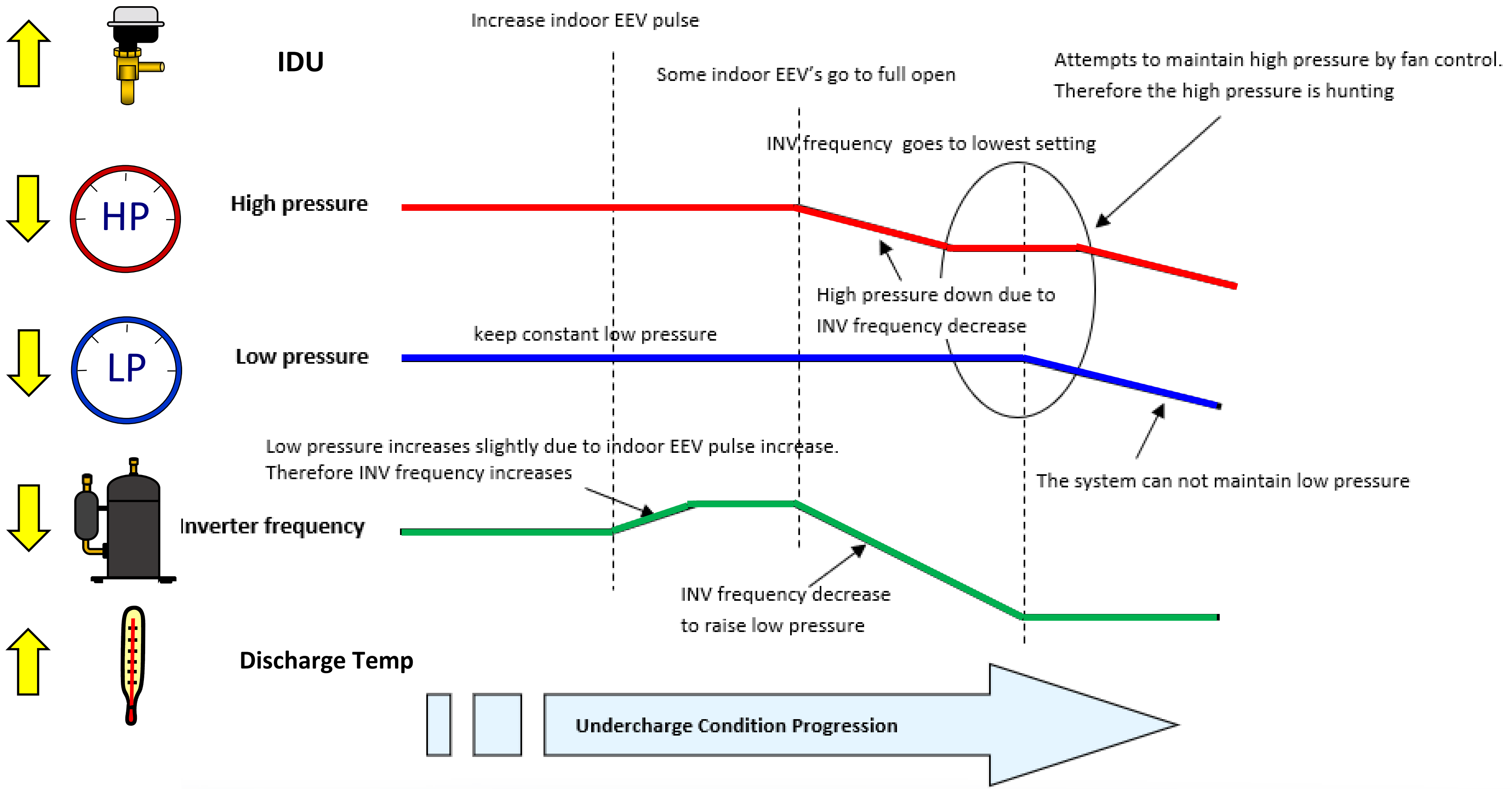
Under Charge Cooling

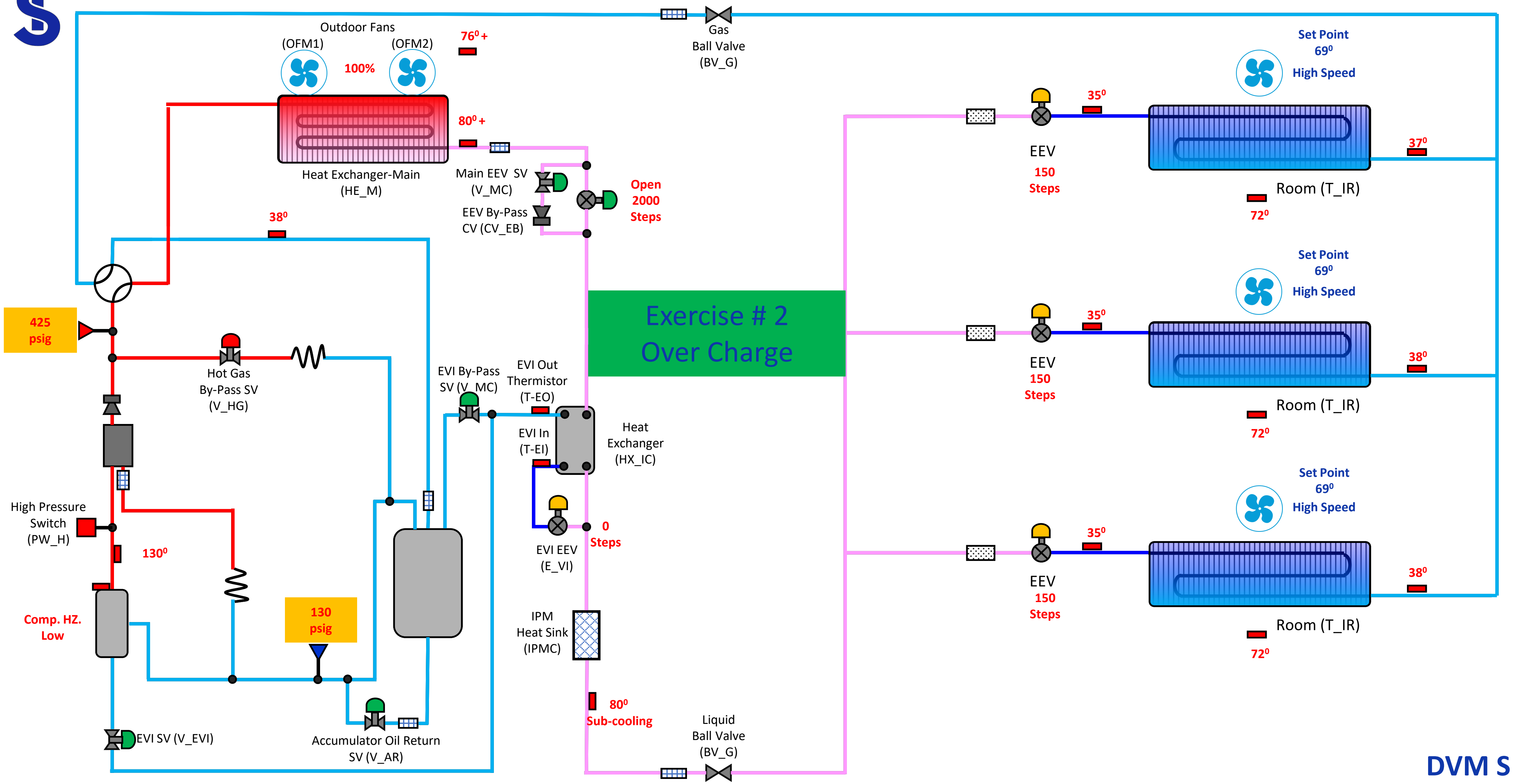


Discharge temperature is over 221º F,

Exercise # 2
Operational Characteristics

- ❑ Inadequate cooling
- ❑ System Cools when the load is low but not when its high
- ❑ Compressor does not seem to run at a high enough speed for the load

We can use this section to put in operational characteristics of the system and any other data pertinent to the faut, like the compressor is in protection mode or system is throwing a fault. Thoughts



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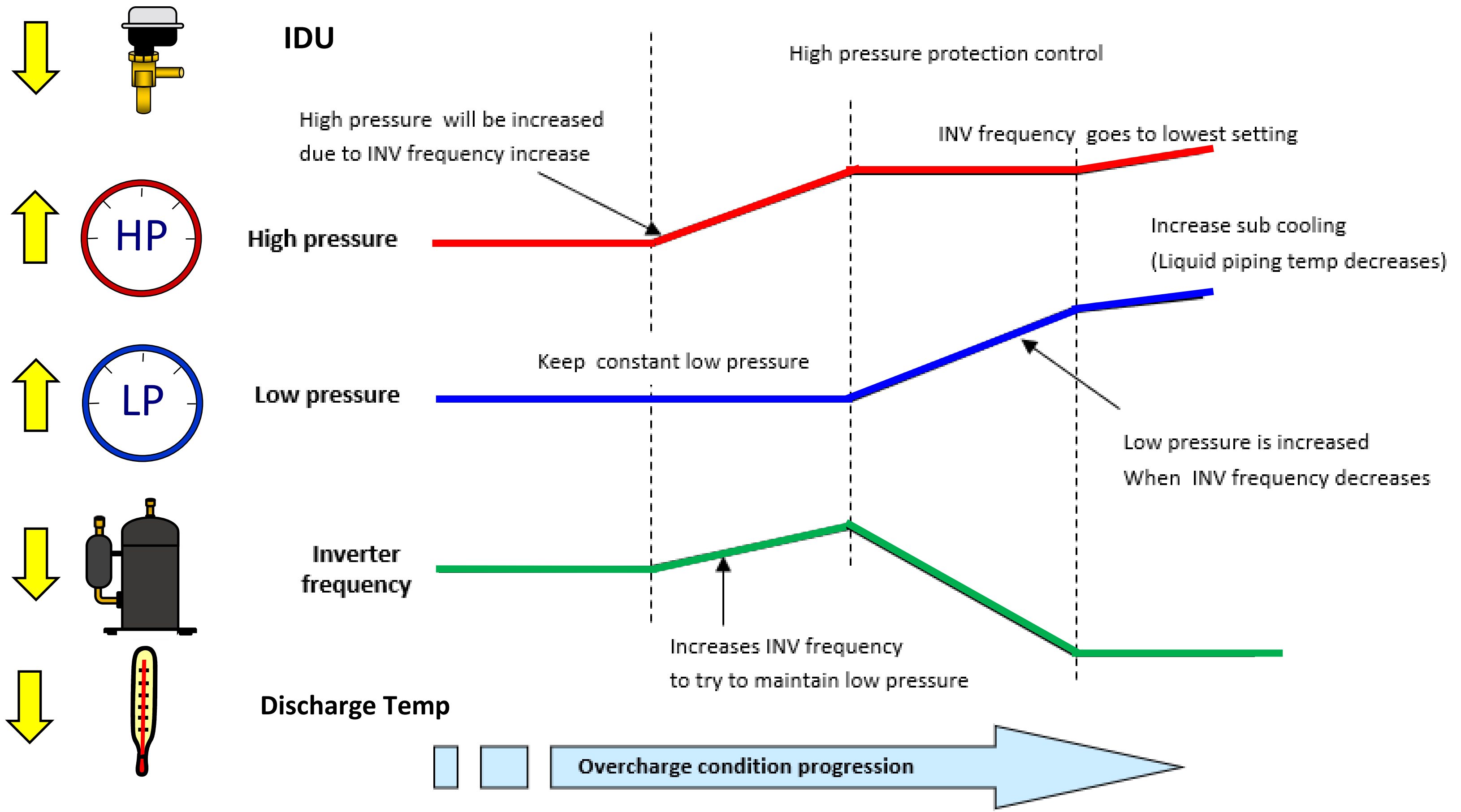
Over Charge Cooling

- **System will run in normal operation until it is above regulated capacity**

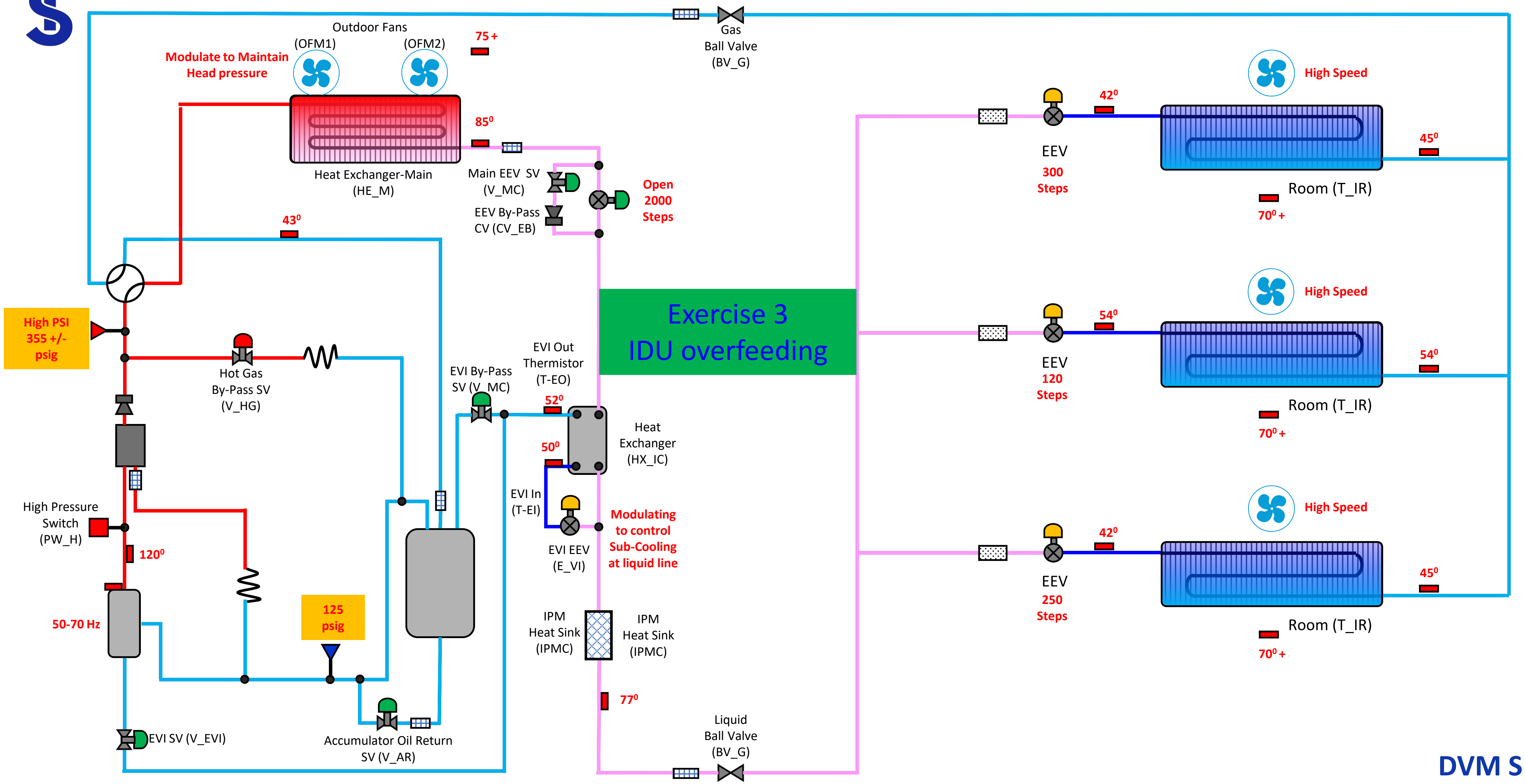
Discharge temperature is below 140º F,



Exercise # 3
Operational Characteristics

- ❑ Zone 2 is not cooling
- ❑ Zones 1 & 3 have diminished cooling capacity

We can use this section to put in operational characteristics of the system and any other data pertinent to the faut, like the compressor is in protection mode or system is throwing a fault. Thoughts



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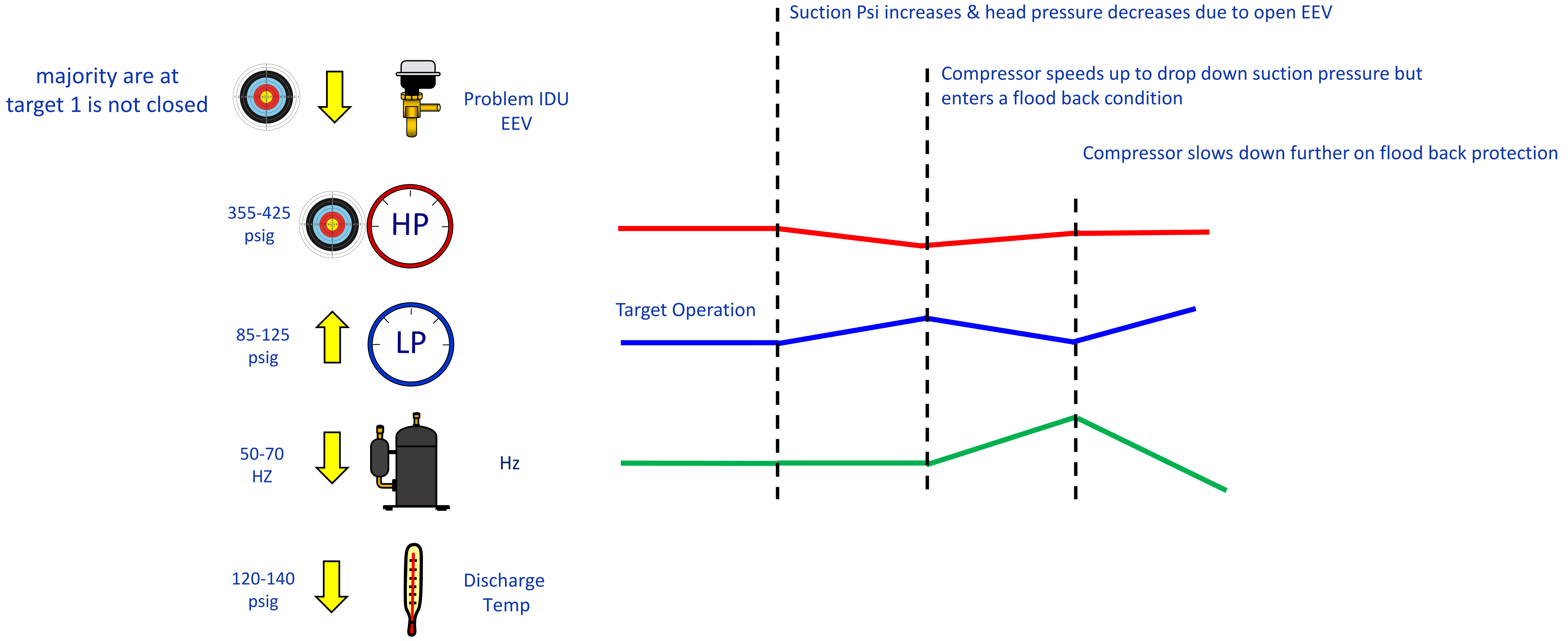
Overfeeding IDU



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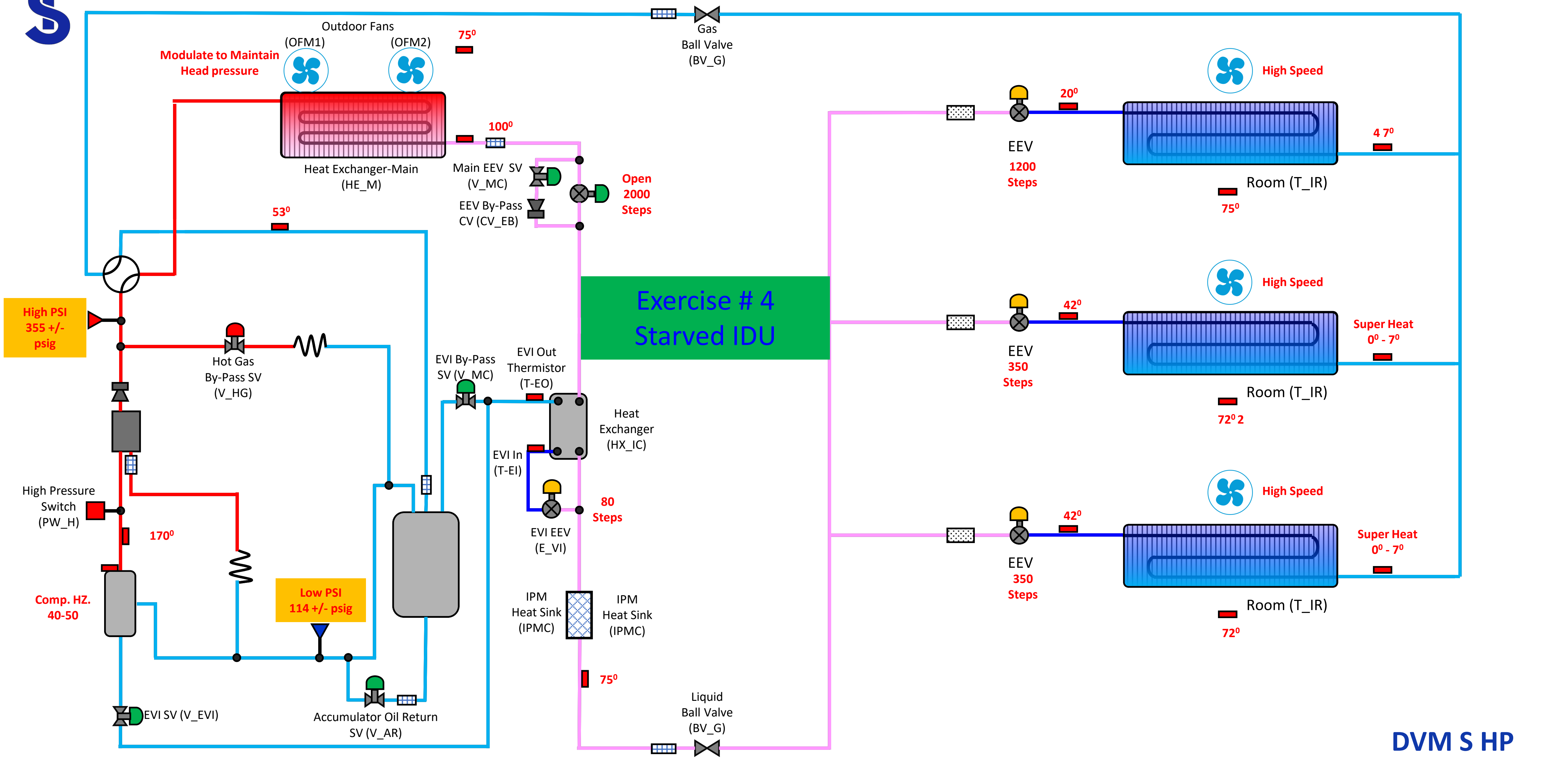
Exercise # 4
Operational Characteristics

- ❑ Zone 1 is not cooling
- ❑ Other zones are maintaining space temperature

We can use this section to put in operational characteristics of the system and any other data pertinent to the faut, like the compressor is in protection mode or system is throwing a fault. Thoughts


Outdoor Fans
(OFM1)
(OFM2)
Modulate to Maintain Head pressure
75⁰
Gas Ball Valve (BV_G)
High Speed
20⁰
100⁰
Heat Exchanger-Main (HE_M)
Main EEV SV (V_MC)
EEV By-Pass CV (CV_EB)
Open 2000 Steps
EEV
1200 Steps
4 7⁰
Room (T_IR)
75⁰
53⁰
High Speed
42⁰
Exercise # 4
Starved IDU
High PSI 355 +/- psig
Hot Gas By-Pass SV (V_HG)
EVI By-Pass SV (V_MC)
EVI Out Thermistor (T-EO)
EEV
350 Steps
Super Heat 0⁰ - 7⁰
Room (T_IR)
72⁰ 2
Heat Exchanger (HX_IC)
EVI In (T-EI)
High Pressure Switch (PW_H)
170⁰
80 Steps
EVI EEV (E_VI)
High Speed
42⁰
EEV
350 Steps
Super Heat 0⁰ - 7⁰
Comp. HZ. 40-50
Low PSI 114 +/- psig
IPM Heat Sink (IPMC)
IPM Heat Sink (IPMC)
Room (T_IR)
72⁰
75⁰
EVI SV (V_EVI)
Accumulator Oil Return SV (V_AR)
Liquid Ball Valve (BV_G)
DVM S HP

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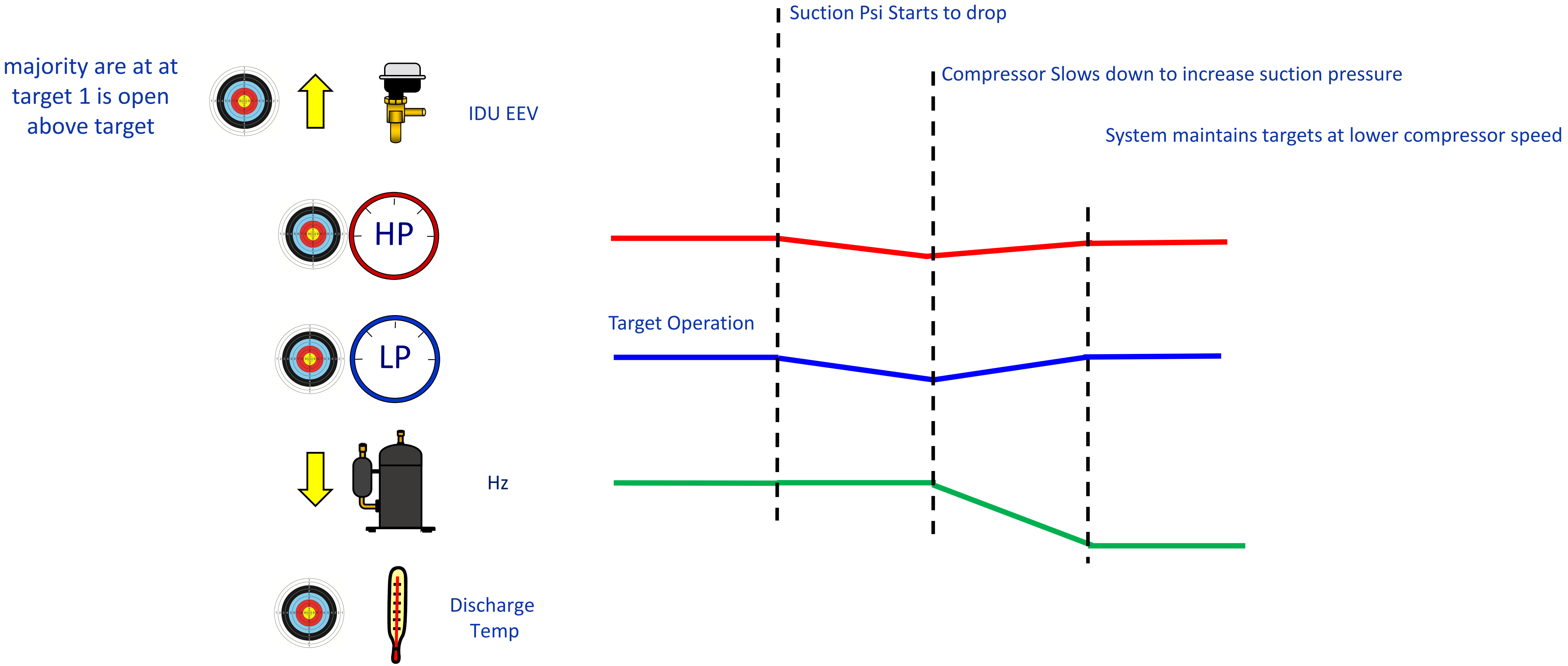
Starved IDU

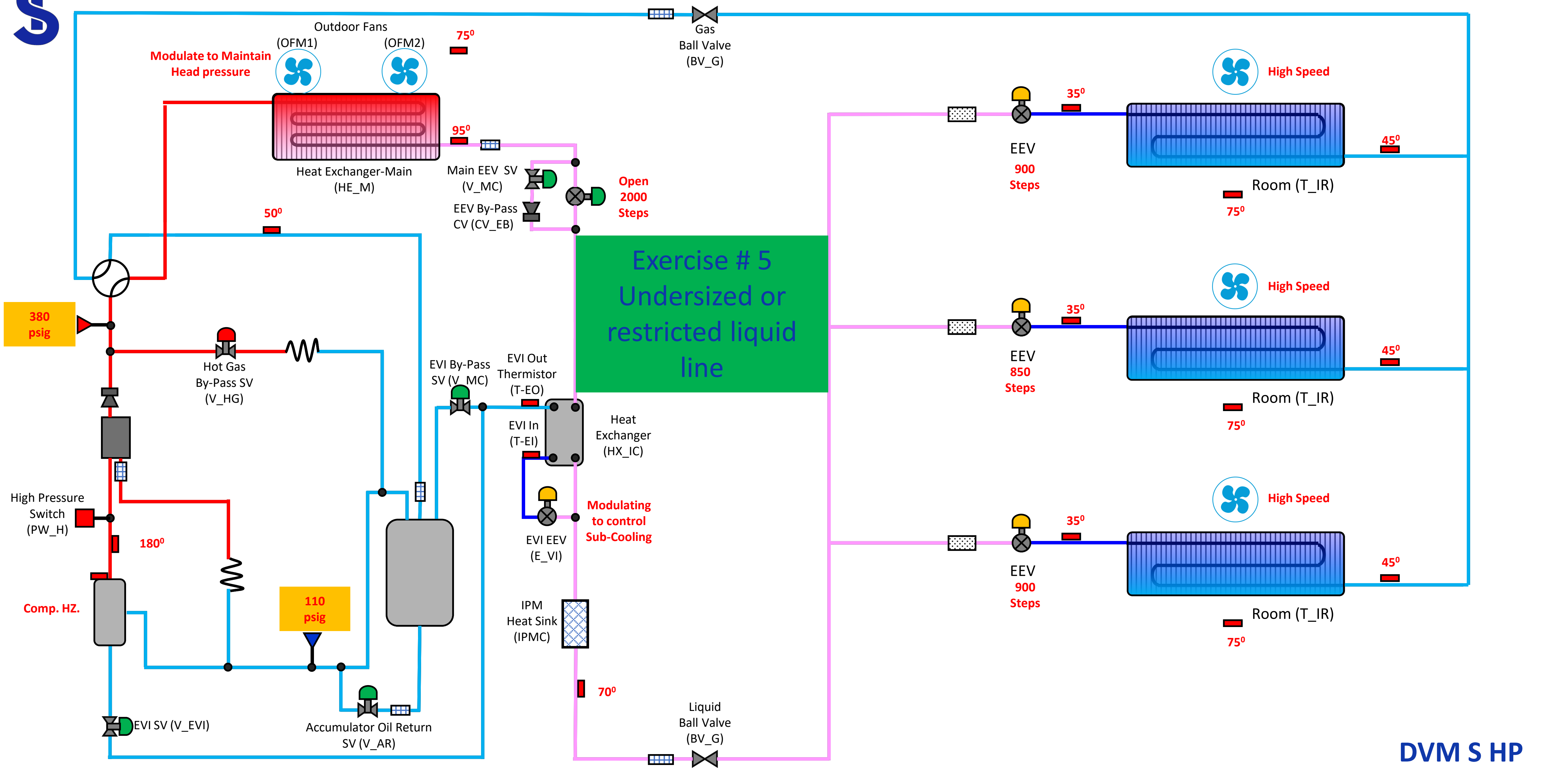


majority are at at target 1 is open above target
IDU EEV
HP
LP
Hz
Discharge Temp
Suction Psi Starts to drop
Compressor Slows down to increase suction pressure
System maintains targets at lower compressor speed
Target Operation

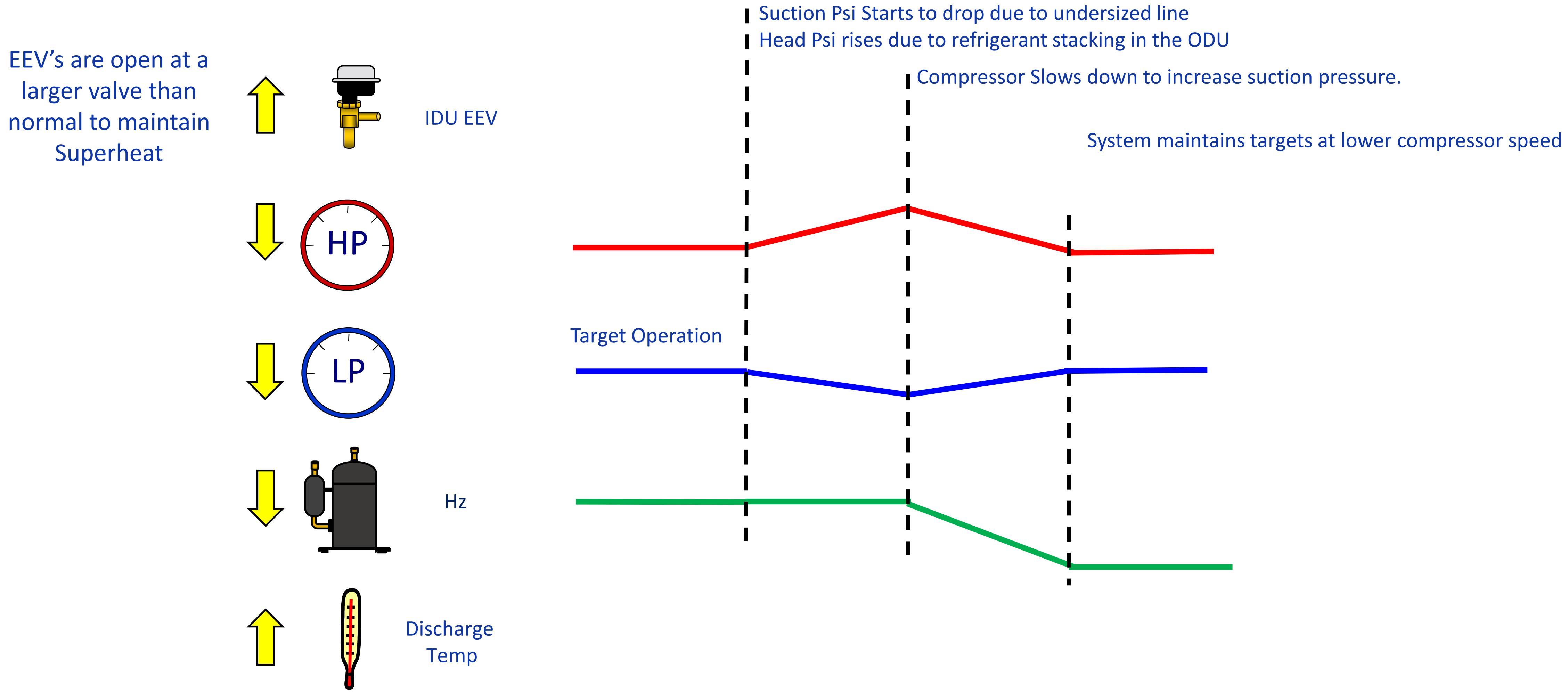
Exercise # 5
Operational Characteristics

- ❑ Inadequate cooling for all zones
- ❑ System Cools when the load is low but not when its high

We can use this section to put in operational characteristics of the system and any other data pertinent to the faut, like the compressor is in protection mode or system is throwing a fault. Thoughts


Exercise # 5
Undersized or restricted liquid line
Outdoor Fans
(OFM1)
(OFM2)
Modulate to Maintain Head pressure
75⁰
95⁰
Heat Exchanger-Main
(HE_M)
Gas
Ball Valve
(BV_G)
Main EEV SV
(V_MC)
EEV By-Pass
CV (CV_EB)
Open
2000
Steps
50⁰
380
psig
Hot Gas
By-Pass SV
(V_HG)
High Pressure
Switch
(PW_H)
180⁰
Comp. HZ.
110
psig
EVI SV (V_EVI)
Accumulator Oil Return
SV (V_AR)
EVI By-Pass
SV (V_MC)
EVI Out
Thermistor
(T-EO)
EVI In
(T-EI)
Heat
Exchanger
(HX_IC)
EVI EEV
(E_VI)
Modulating
to control
Sub-Cooling
IPM
Heat Sink
(IPMC)
70⁰
Liquid
Ball Valve
(BV_G)
High Speed
35⁰
EEV
900
Steps
45⁰
Room (T_IR)
75⁰
High Speed
35⁰
EEV
850
Steps
45⁰
Room (T_IR)
75⁰
High Speed
35⁰
EEV
900
Steps
45⁰
Room (T_IR)
75⁰
DVM S HP

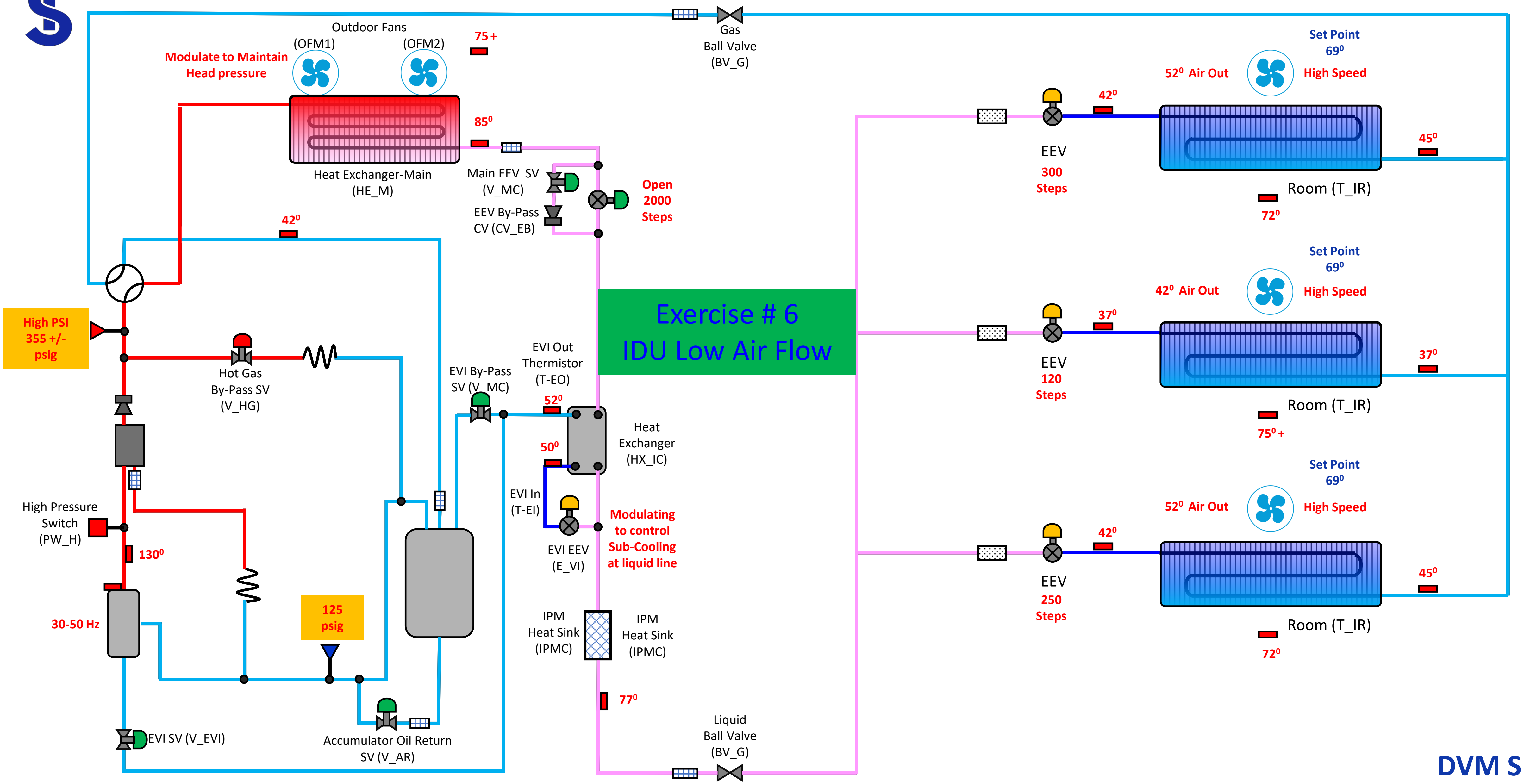
Undersized Liquid line



Exercise # 6
Operational Characteristics

- ❑ Inadequate cooling zone 2
- ❑ Cold air is coming from the unit but the space temperature will not drop

We can use this section to put in operational characteristics of the system and any other data pertinent to the faut, like the compressor is in protection mode or system is throwing a fault. Thoughts



DVM S HP

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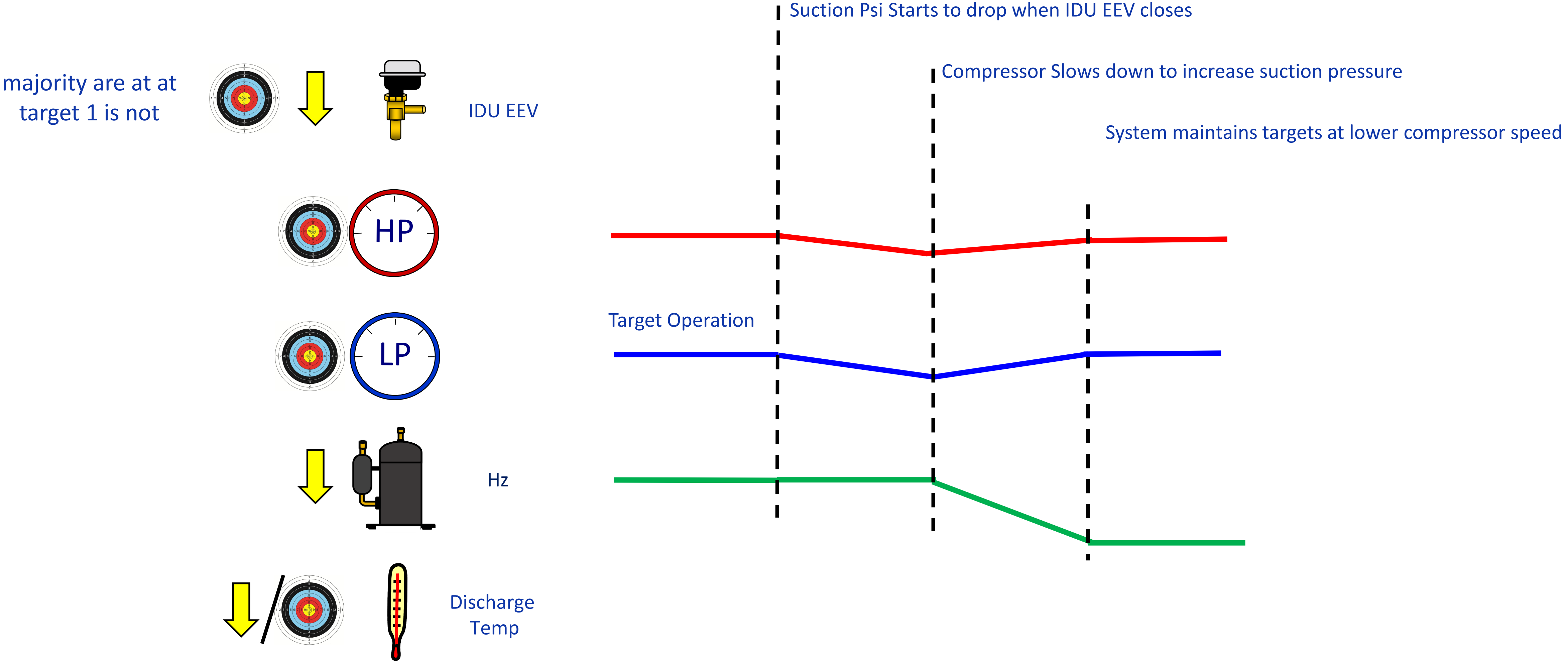
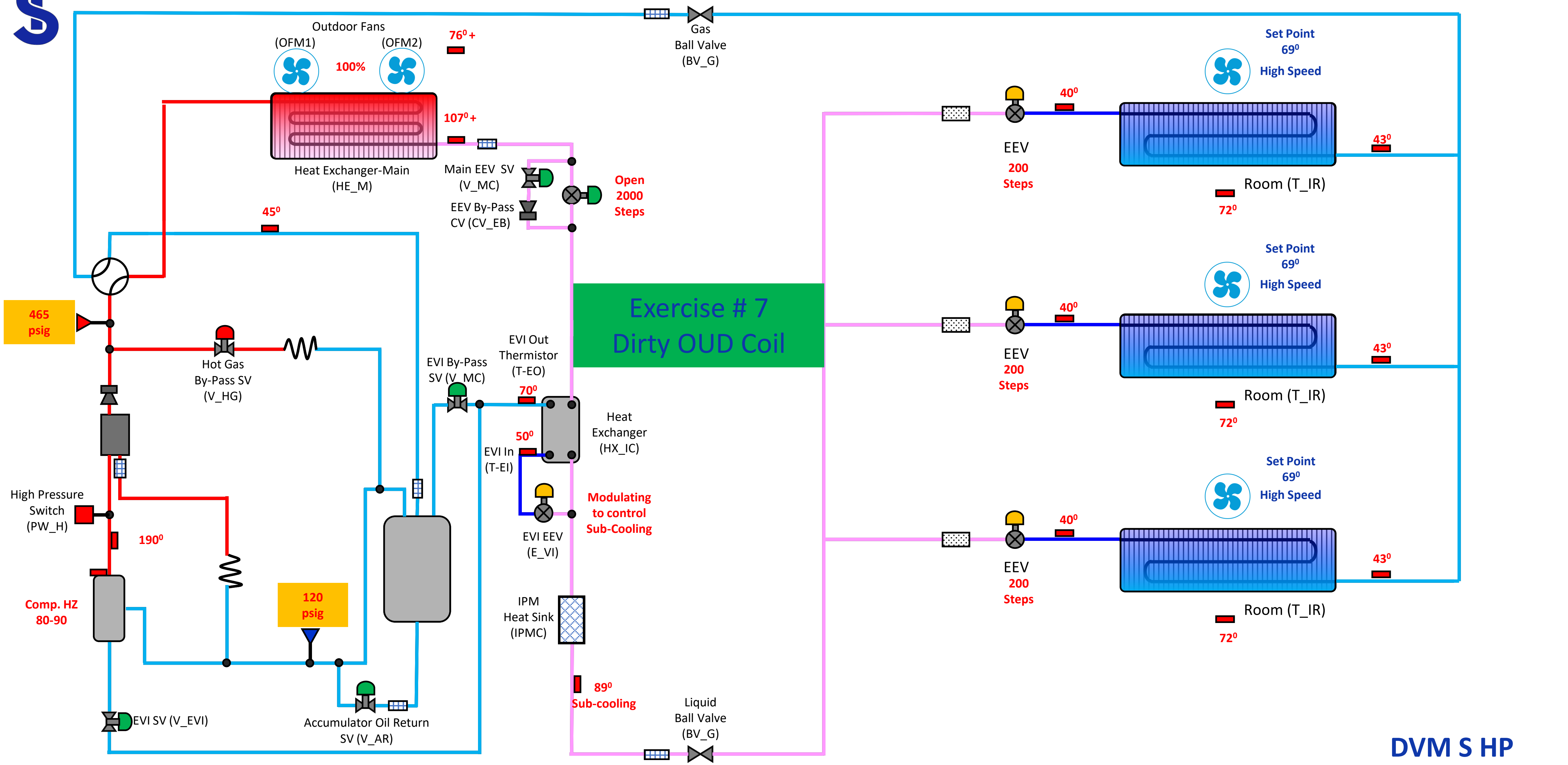
Low Air Flow IDU



Exercise # 7
Operational Characteristics

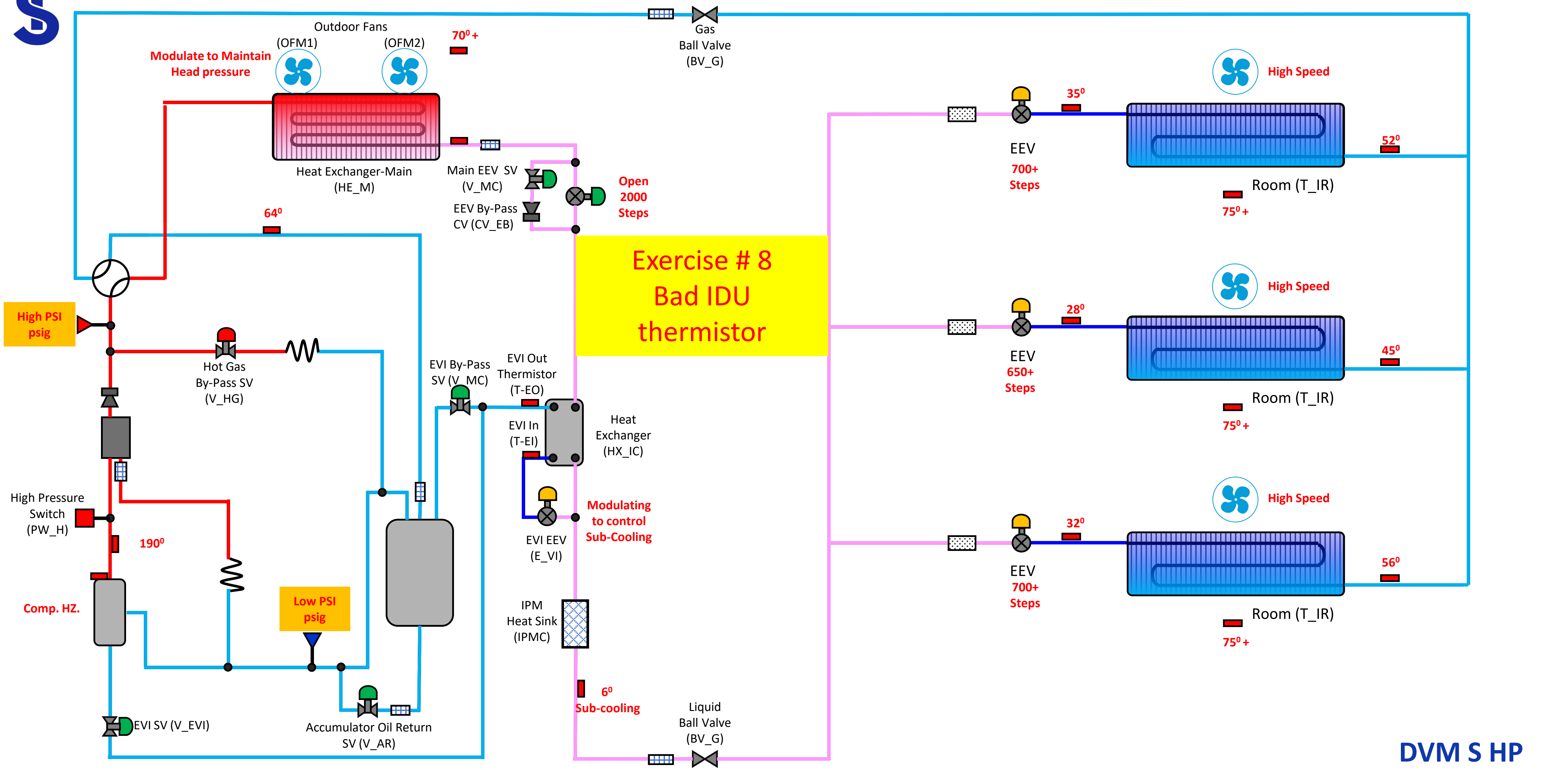
- ❑ Inadequate cooling
- ❑ System Cools when the load is low but not when its high
- ❑ Compressor making noise

We can use this section to put in operational characteristics of the system and any other data pertinent to the faut, like the compressor is in protection mode or system is throwing a fault. Thoughts



Outdoor Fans
(OFM1)
(OFM2)
100%
76⁰+
107⁰+
Heat Exchanger-Main
(HE_M)
Gas
Ball Valve
(BV_G)
Main EEV SV
(V_MC)
EEV By-Pass
CV (CV_EB)
Open
2000
Steps
45⁰
465
psig
Hot Gas
By-Pass SV
(V_HG)
High Pressure
Switch
(PW_H)
190⁰
Comp. HZ
80-90
120
psig
EVI By-Pass
SV (V_MC)
EVI Out
Thermistor
(T-EO)
70⁰
50⁰
EVI In
(T-EI)
Heat
Exchanger
(HX_IC)
Modulating
to control
Sub-Cooling
EVI EEV
(E_VI)
IPM
Heat Sink
(IPMC)
89⁰
Sub-cooling
Liquid
Ball Valve
(BV_G)
EVI SV (V_EVI)
Accumulator Oil Return
SV (V_AR)
Exercise # 7
Dirty OUD Coil
Set Point
69⁰
High Speed
40⁰
43⁰
EEV
200
Steps
Room (T_IR)
72⁰
Set Point
69⁰
High Speed
40⁰
43⁰
EEV
200
Steps
Room (T_IR)
72⁰
Set Point
69⁰
High Speed
40⁰
43⁰
EEV
200
Steps
Room (T_IR)
72⁰
DVM S HP

SAMSUNG

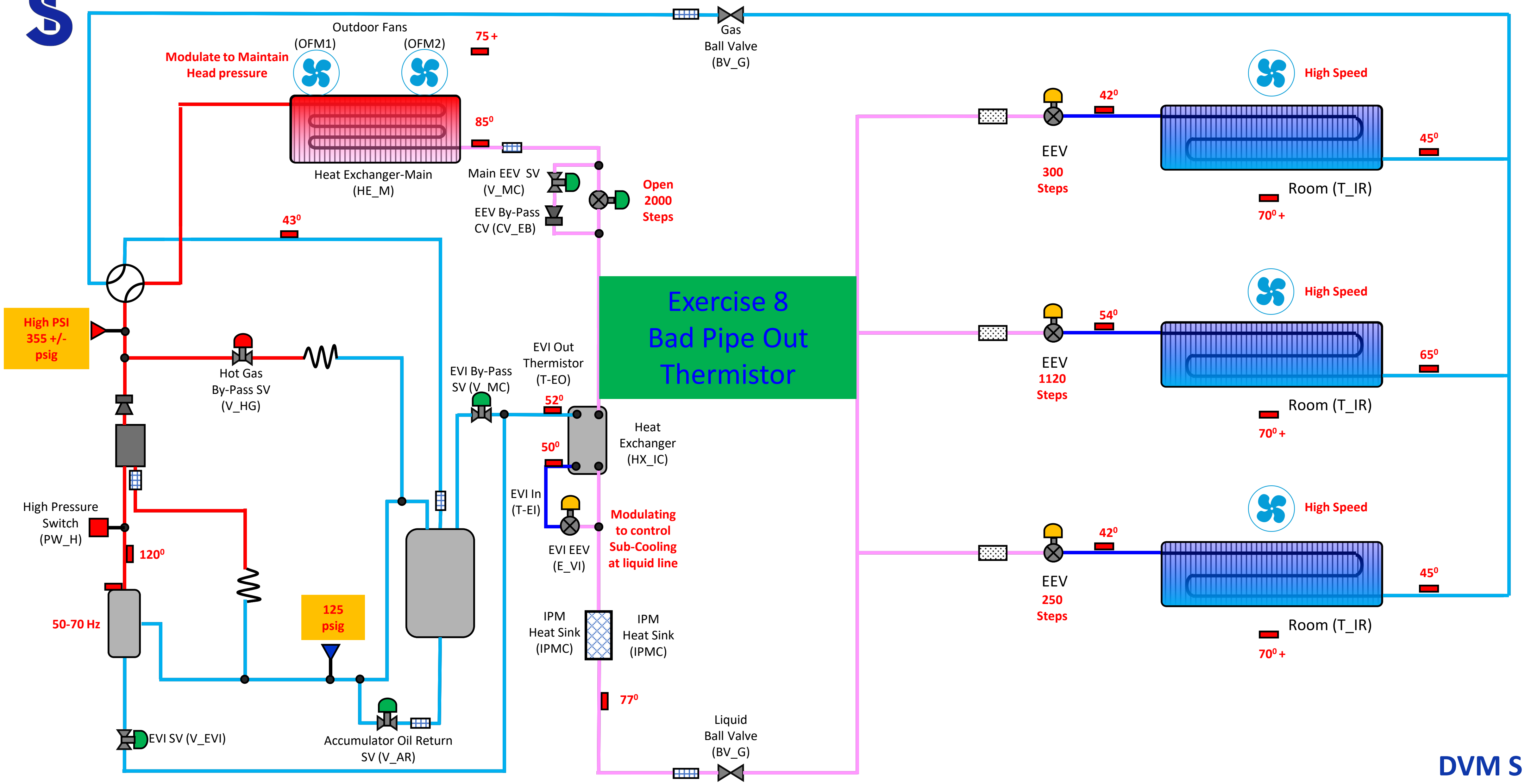


Outdoor Fans
(OFM1)
(OFM2)
Modulate to Maintain
Head pressure
70⁰ +
Gas
Ball Valve
(BV_G)
High Speed
35⁰
EEV
700+
Steps
52⁰
Room (T_IR)
75⁰ +
Heat Exchanger-Main
(HE_M)
Main EEV SV
(V_MC)
EEV By-Pass
CV (CV_EB)
Open
2000
Steps
64⁰
Exercise # 8
Bad IDU
thermistor
High PSI
psig
Hot Gas
By-Pass SV
(V_HG)
High Speed
28⁰
EEV
650+
Steps
45⁰
Room (T_IR)
75⁰ +
EVI By-Pass
SV (V_MC)
EVI Out
Thermistor
(T-EO)
EVI In
(T-EI)
Heat
Exchanger
(HX_IC)
High Pressure
Switch
(PW_H)
190⁰
Modulating
to control
Sub-Cooling
EVI EEV
(E_VI)
High Speed
32⁰
EEV
700+
Steps
56⁰
Room (T_IR)
75⁰ +
Comp. HZ.
Low PSI
psig
IPM
Heat Sink
(IPMC)
6⁰
Sub-cooling
Liquid
Ball Valve
(BV_G)
EVI SV (V_EVI)
Accumulator Oil Return
SV (V_AR)
DVM S HP

Exercise # 8
Operational Characteristics

- ❑ Zone 2 is not cooling
- ❑ Zones 1 & 3 have diminished cooling capacity

We can use this section to put in operational characteristics of the system and any other data pertinent to the faut, like the compressor is in protection mode or system is throwing a fault. Thoughts



DVM S HP

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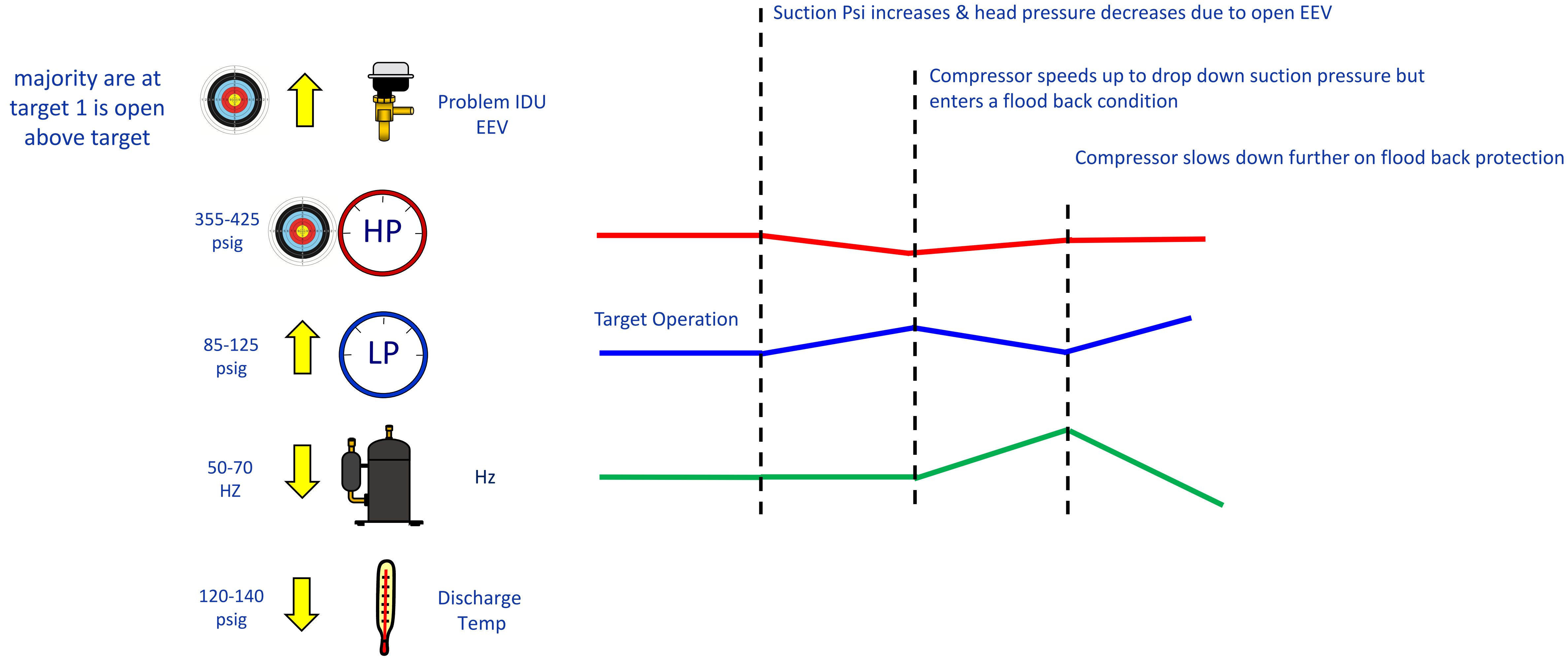
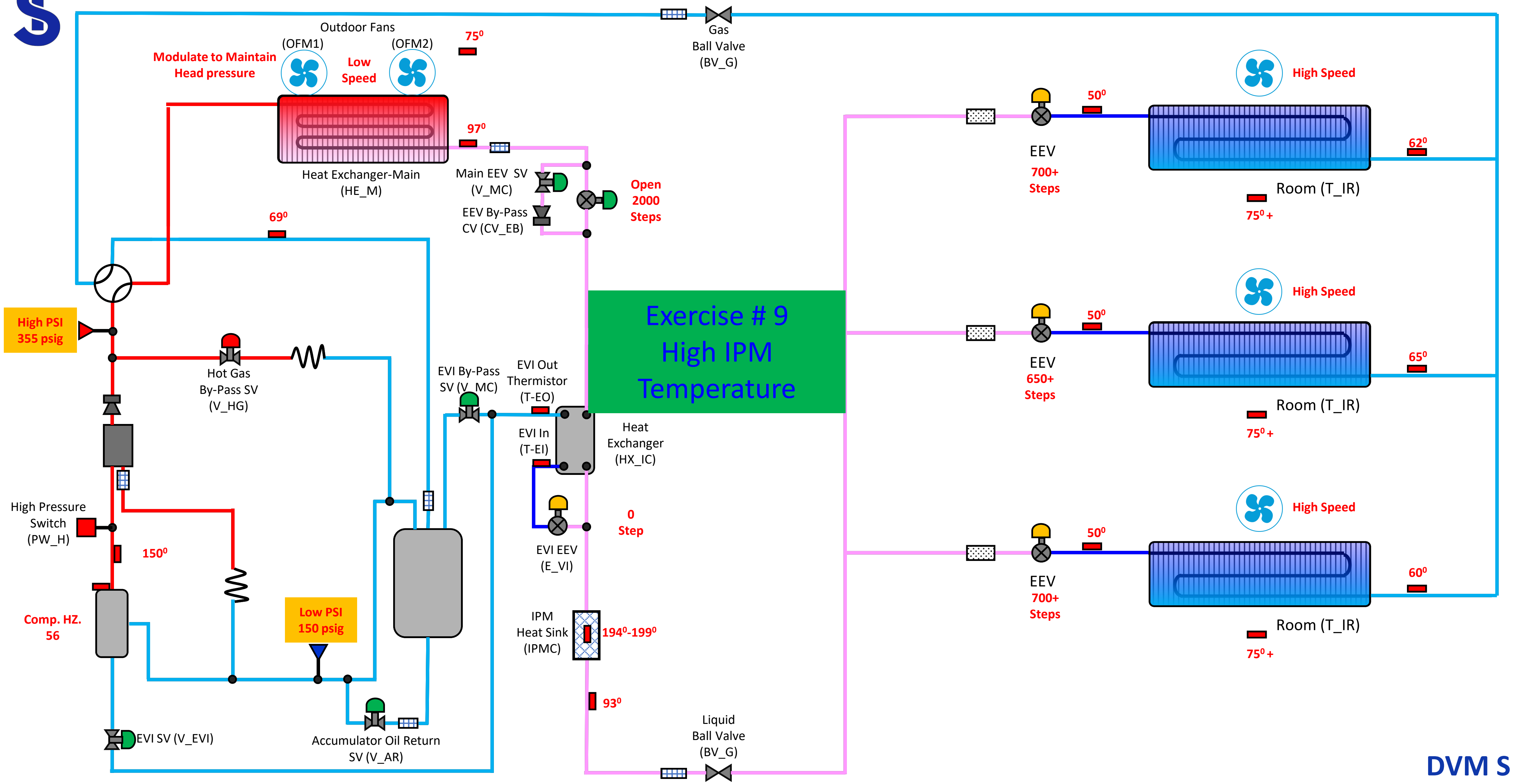
Pipe Thermistor IDU



Exercise # 9
Operational Characteristics

- ❑ In adequate cooling
- ❑ Contractor had a bad compressor
- ❑ They replaced the compressor and IPM
- ❑ They verified the charge
- ❑ The compressor will not ramp up

We can use this section to put in operational characteristics of the system and any other data pertinent to the faut, like the compressor is in protection mode or system is throwing a fault. Thoughts



Exercise # 9 High IPM Temperature

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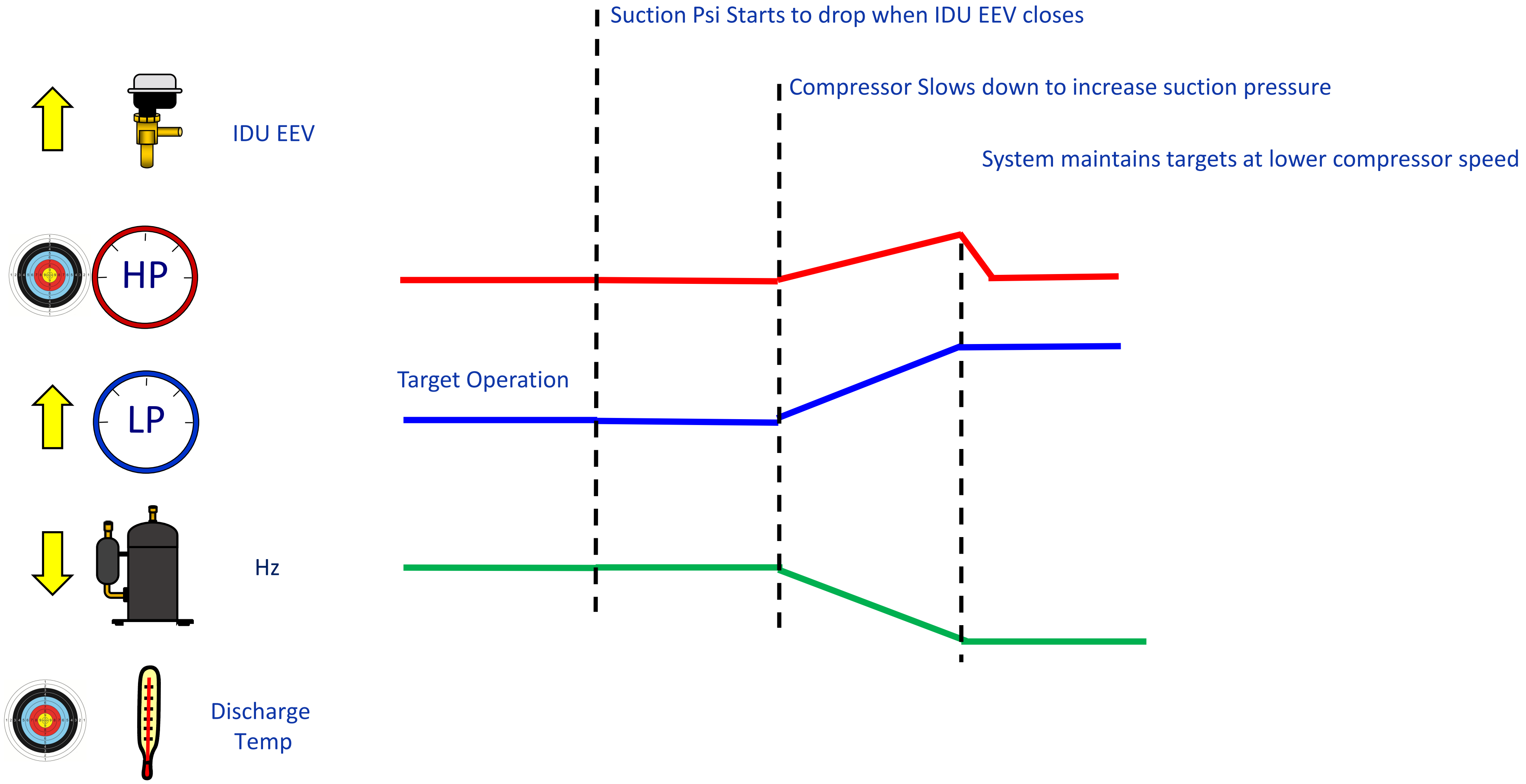
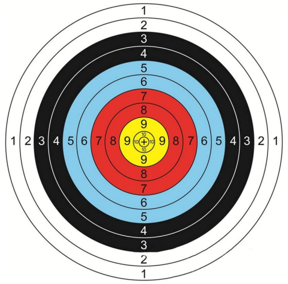
IPM Overheating

All EEV’s at a high opening setting



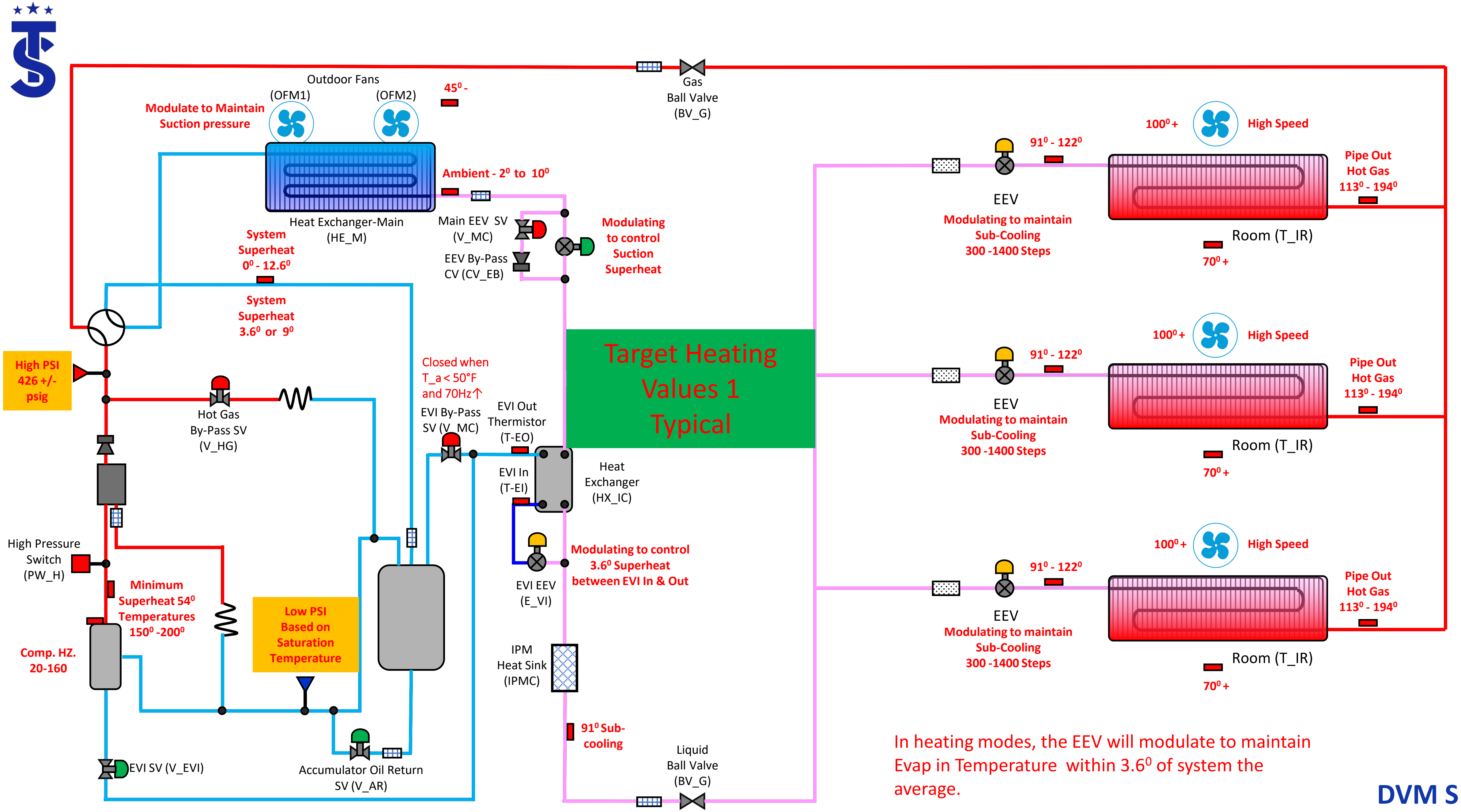
Heating Exercises



Check point of heat trial operation

※ The data is not absolute one, only can be used to check the abnormal status

Item	Normal	Check point
EVA out Temp.	113^{0} - 194^{0}	▷ Normal condition of the 'Evap In' temperature - 23.4^{0} higher than room temp (14.4^{0} for duct model) - not less than 30.6^{0} from high pressure saturated temp (39.6^{0} for duct model) ▷ If all 'Evap In temp' is lower than normal(above) or the high pressure is lower than 355 psig, - Check whether the ambient temp is lower than 23^{0} / Load calculation for heating. - It can be caused by 'Refrigerant overcharged' or 'Wrong load design'. ▷ If only few indoor unit's Evap In temp. is lower than normal, check the pipe length from the first Y-joint to the indoor unit (limited 45m) also ensure that if there is any kind of flow resistance.
EVA in Temp.	91^{0} - 122^{0}	
EEV Step of Indoor Unit	Initial 960 (RAC 1440) Higher floor ↓ Lower floor ↑ ODU closer ↓ ODU farther↑	
High Pressure psig	355 - 425 psig	▷ High pressure can be lower than normal in below case 1. Ambient temp 32 ↓ , 2. heating capacity shortage by wrong design, 3. Indoor temp 68 ↓ ▷ Low pressure will be a reference value because can be changed easily by ambient temperature
Low Pressure psig	71 – 106.6 psig	
Compressor Discharge Temp.	149^{0} - 221^{0}	▷ If all indoor units are operating and both 1 and 2 is satisfied, it may be normal operation. 1. Highest DSH(discharge temp - high pressure saturated temp) ≥ 36^{0} 2. Max. discharge temp ≤ 221^{0} * Highest DSH<36: Ref. overcharged, EEV or liquid bypass valve or EVI EEV check * Max. dis temp>221: Ref. shortage, EEV malfunction, Liquid service valve close
Suction super heat	1.8 – 12.6	▷ Check EEV working if suction super heat is not satisfied. - If main eev step is max. & super heat is high, then we can suspect refrigerant shortage
Main EEV step	300~1400	



In heating modes, the EEV will modulate to maintain Evap in Temperature within 3.6^0 of system the average.

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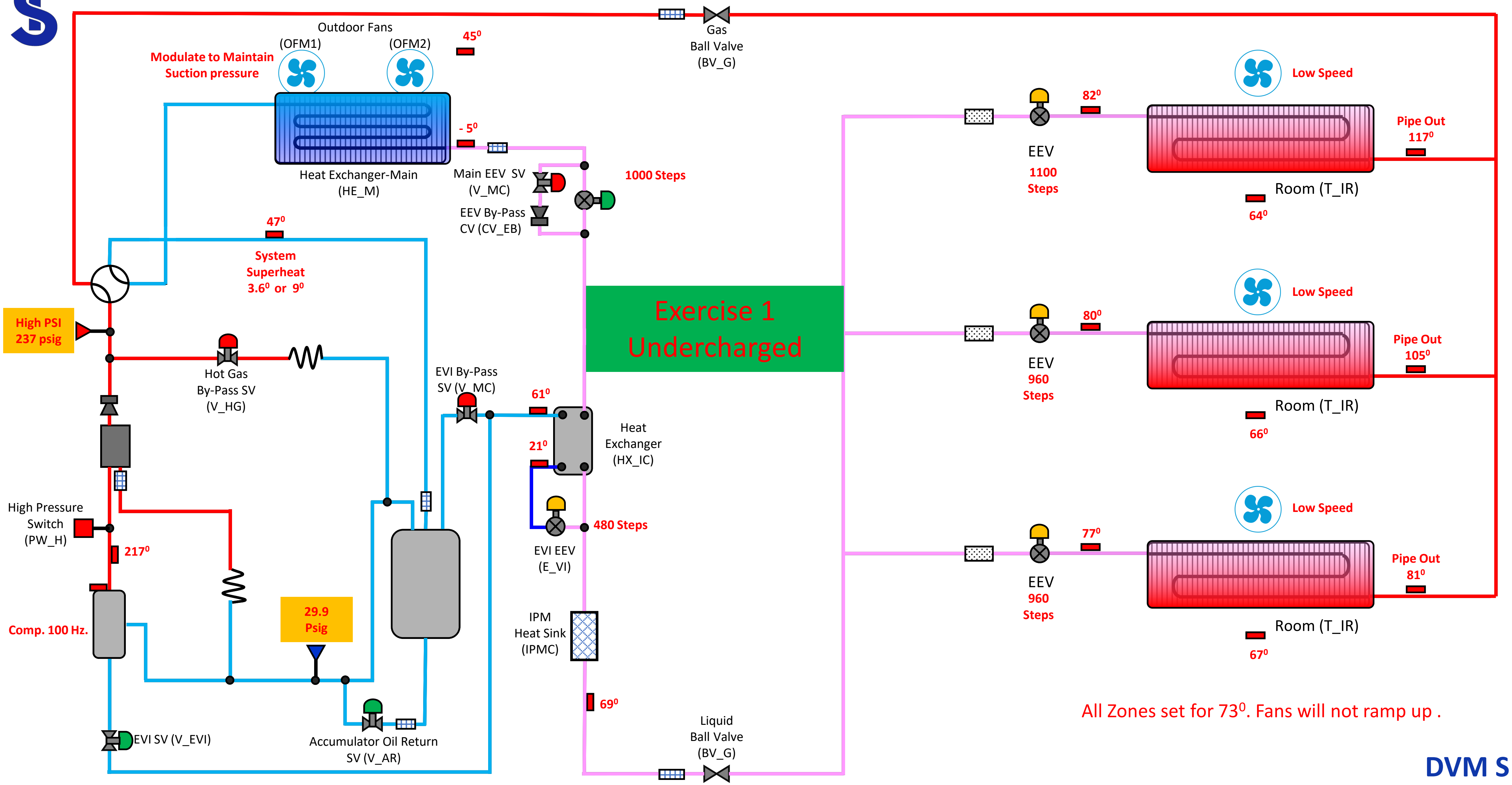
Exercise Heating #1
Operational Characteristics

- ❑ In adequate heating at full load
- ❑ Heating performance diminishes as outside air temperature drops.

We can use this section to put in operational characteristics of the system and any other data pertinent to the faut, like the compressor is in protection mode or system is throwing a fault. Thoughts



DVM S HP

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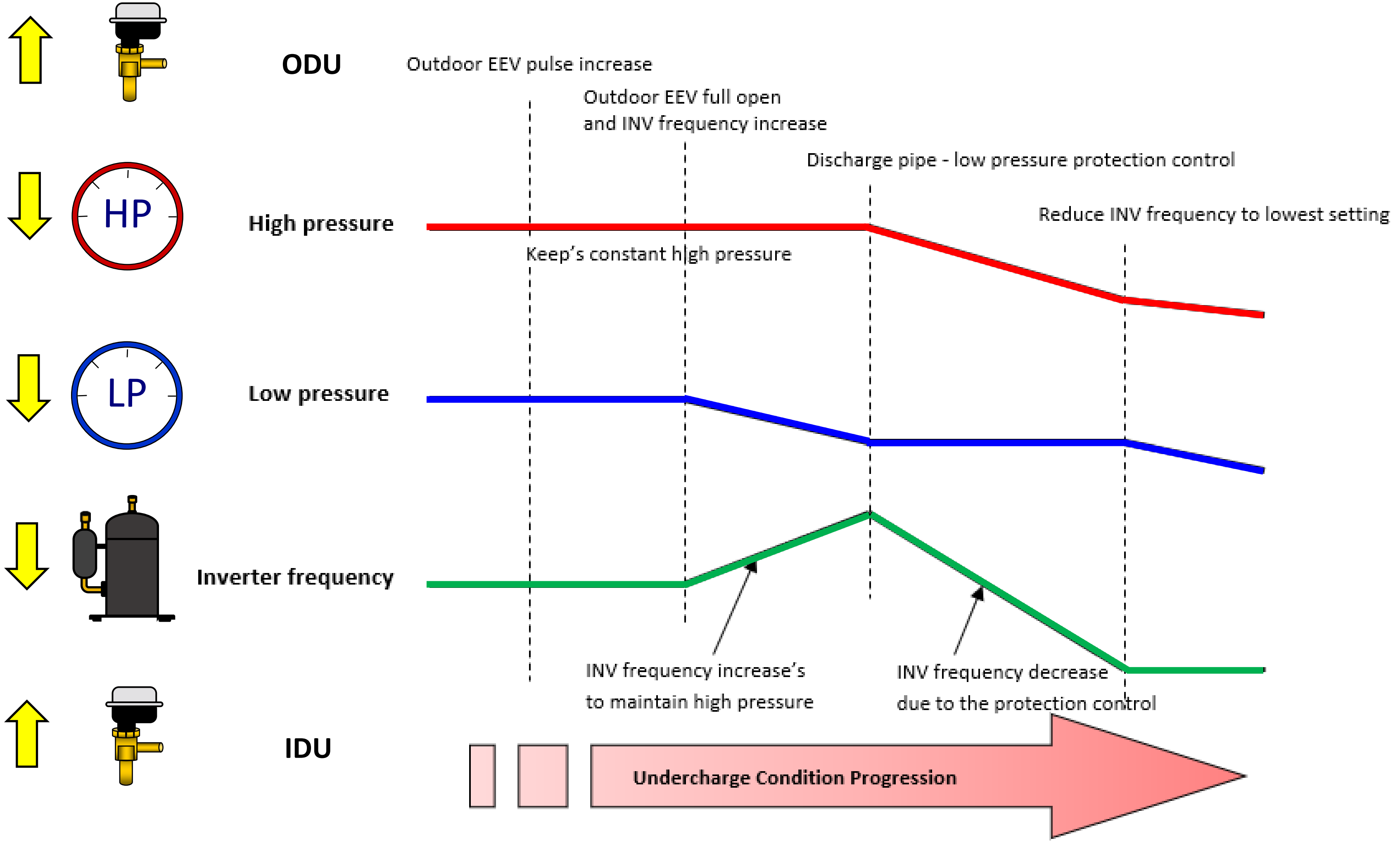
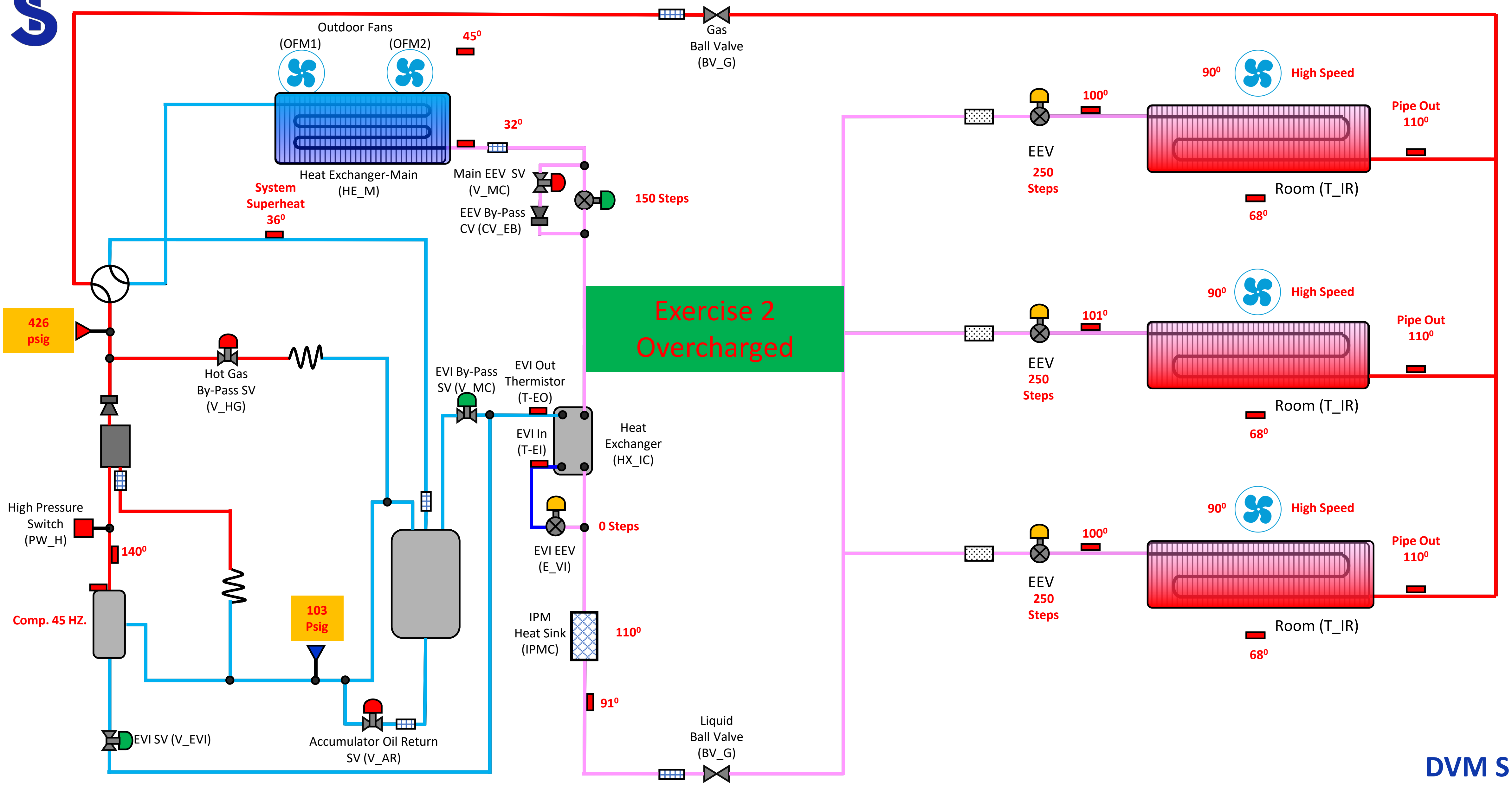
Undercharge Heating



Exercise Heating #2
Operational Characteristics

- ❑ In adequate heating
- ❑ Compressor will not ramp up

We can use this section to put in operational characteristics of the system and any other data pertinent to the faut, like the compressor is in protection mode or system is throwing a fault. Thoughts



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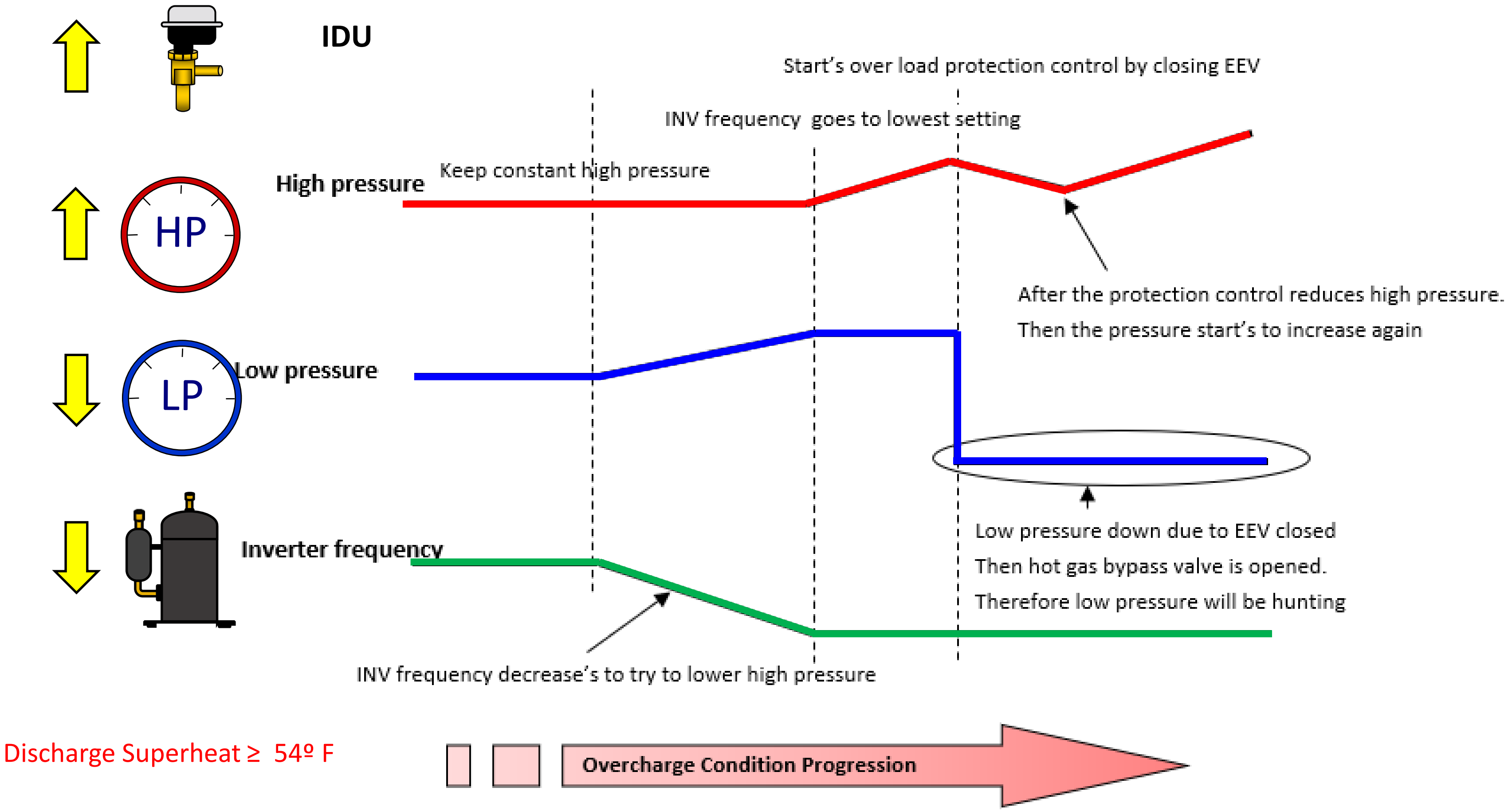
Overcharge Heating

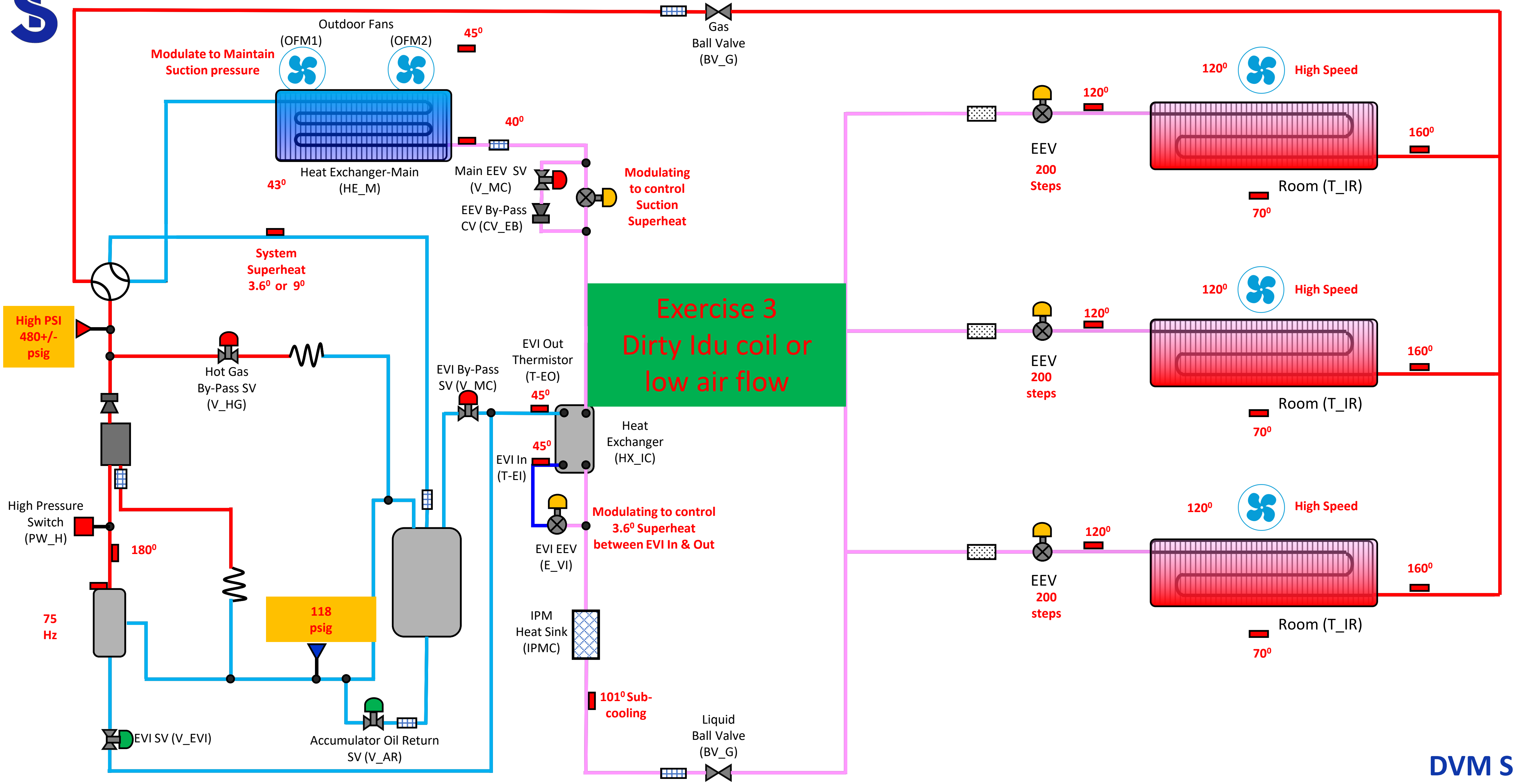
- **System will run in normal operation until it is above regulated capacity**



Exercise Heating #3
Operational Characteristics

- ❑ In adequate heating
- ❑ Compressor will not ramp up
- ❑ High TD's across IDU's
- ❑ Space temperature hot hitting setpoint

We can use this section to put in operational characteristics of the system and any other data pertinent to the faut, like the compressor is in protection mode or system is throwing a fault. Thoughts



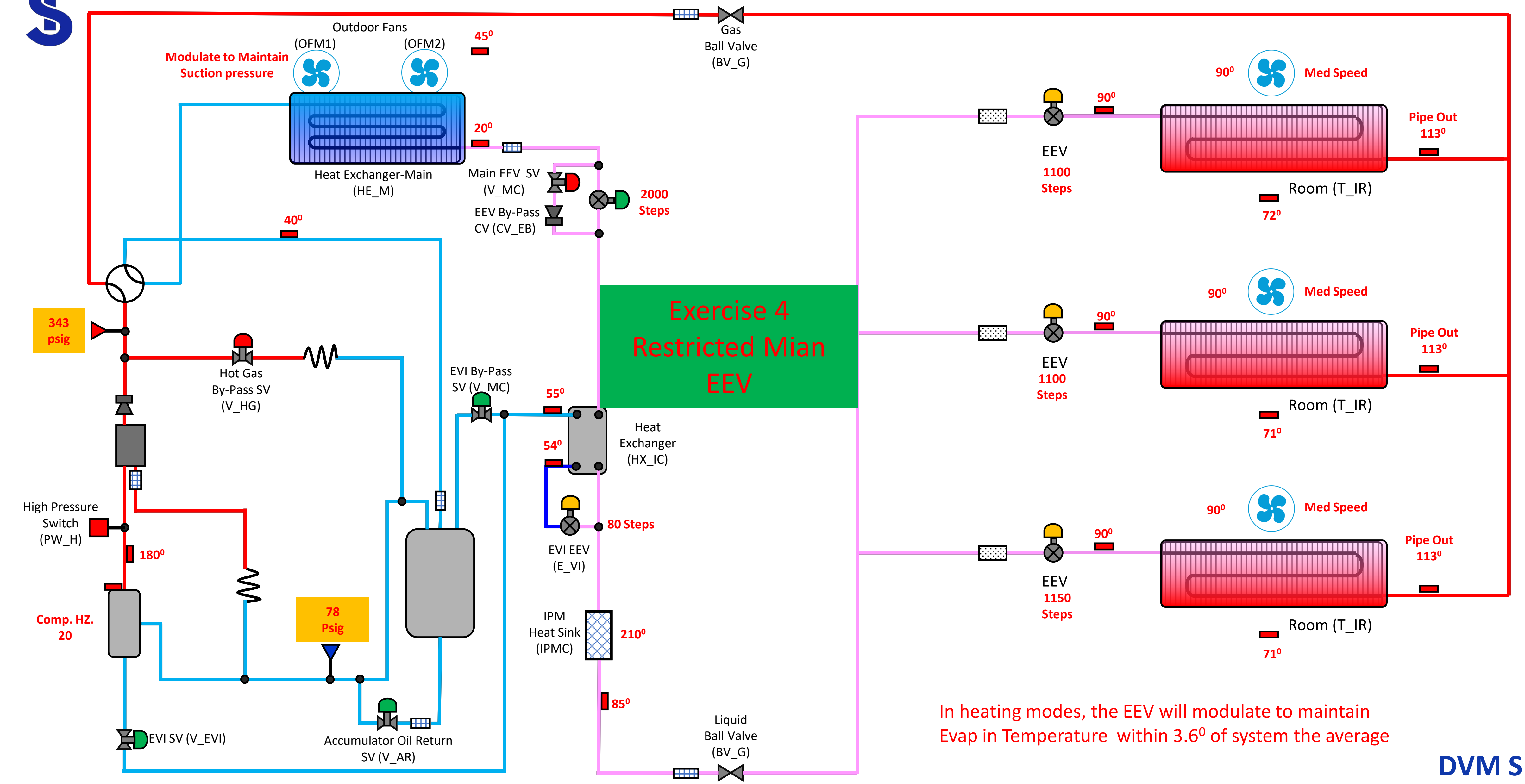
Outdoor Fans
(OFM1)
(OFM2)
Modulate to Maintain
Suction pressure
45^0
Gas
Ball Valve
(BV_G)
Heat Exchanger-Main
(HE_M)
40^0
43^0
Main EEV SV
(V_MC)
EEV By-Pass
CV (CV_EB)
Modulating
to control
Suction
Superheat
System
Superheat
3.6^0 or 9^0
High PSI
480+/-
psig
Hot Gas
By-Pass SV
(V_HG)
Exercise 3
Dirty Idu coil or
low air flow
EVI By-Pass
SV (V_MC)
EVI Out
Thermistor
(T-EO)
45^0
Heat
Exchanger
(HX_IC)
45^0
EVI In
(T-EI)
Modulating to control
3.6^0 Superheat
between EVI In & Out
EVI EEV
(E_VI)
High Pressure
Switch
(PW_H)
180^0
75
Hz
118
psig
IPM
Heat Sink
(IPMC)
101^0 Sub-
cooling
Liquid
Ball Valve
(BV_G)
EVI SV (V_EVI)
Accumulator Oil Return
SV (V_AR)
120^0
High Speed
120^0
EEV
200
Steps
160^0
Room (T_IR)
70^0
120^0
High Speed
120^0
EEV
200
steps
160^0
Room (T_IR)
70^0
120^0
High Speed
120^0
EEV
200
steps
160^0
Room (T_IR)
70^0

DVM S HP

Exercise Heating #4
Operational Characteristics

- ❑ In adequate heating
- ❑ Compressor will not ramp up

We can use this section to put in operational characteristics of the system and any other data pertinent to the faut, like the compressor is in protection mode or system is throwing a fault. Thoughts



In heating modes, the EEV will modulate to maintain Evap in Temperature within 3.6⁰ of system the average

DVM S HP

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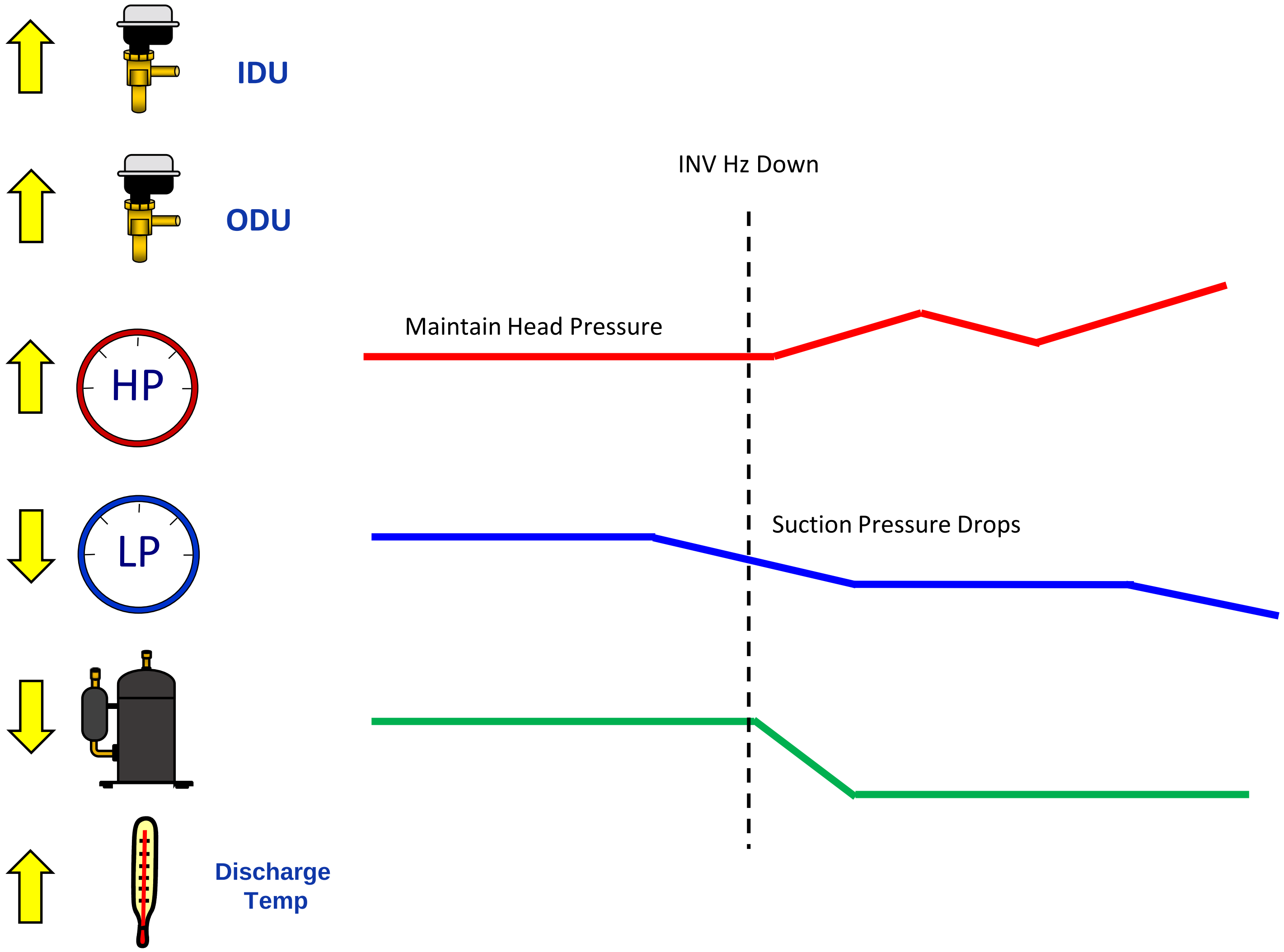
Main EEV Restricted in Heat



Explanation:

When the main EEV in the ODU is not feeding enough refrigerant the OD coil super heat rises. The Main EEV opens to lower the super heat. Suction pressure drops. Head pressure may rise or stay stable.

The compressor superheat rises do to the rise in coil superheat.

The compressors slow down due to high head pressure and Discharge temperature.

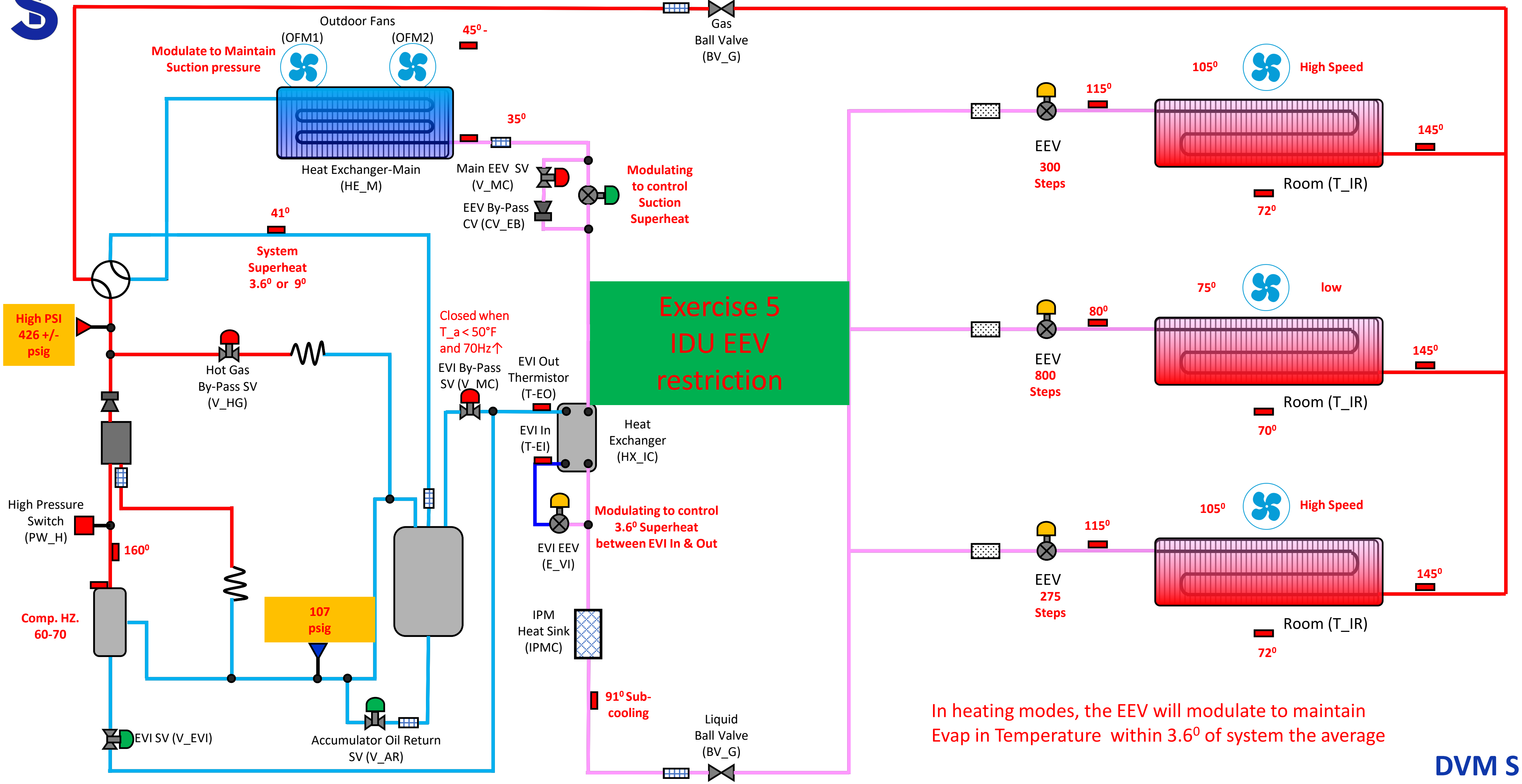
The system can maintain the high side pressure at low RPMS because refrigerant is being stacked I n the indoor units.

The indoor EEV's open because there is a high temperature difference between pipe in and out due to the low velocity of the refrigerant.

Exercise Heating #5
Operational Characteristics

- ❑ One zone not producing heat
- ❑ All other zones are working as required

We can use this section to put in operational characteristics of the system and any other data pertinent to the faut, like the compressor is in protection mode or system is throwing a fault. Thoughts



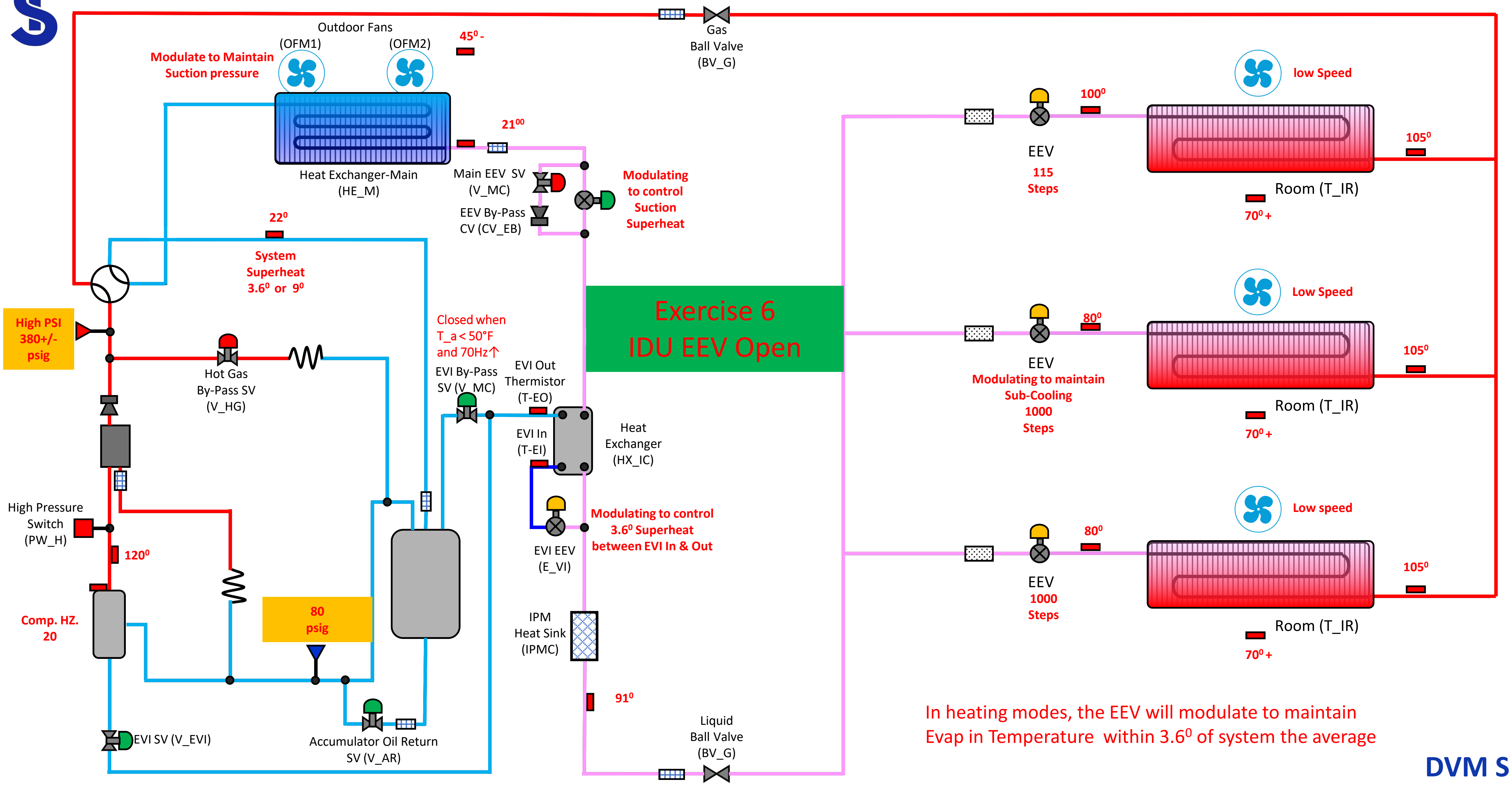
In heating modes, the EEV will modulate to maintain
Evap in Temperature within 3.6^0 of system the average

DVM S HP

Exercise Heating #6
Operational Characteristics

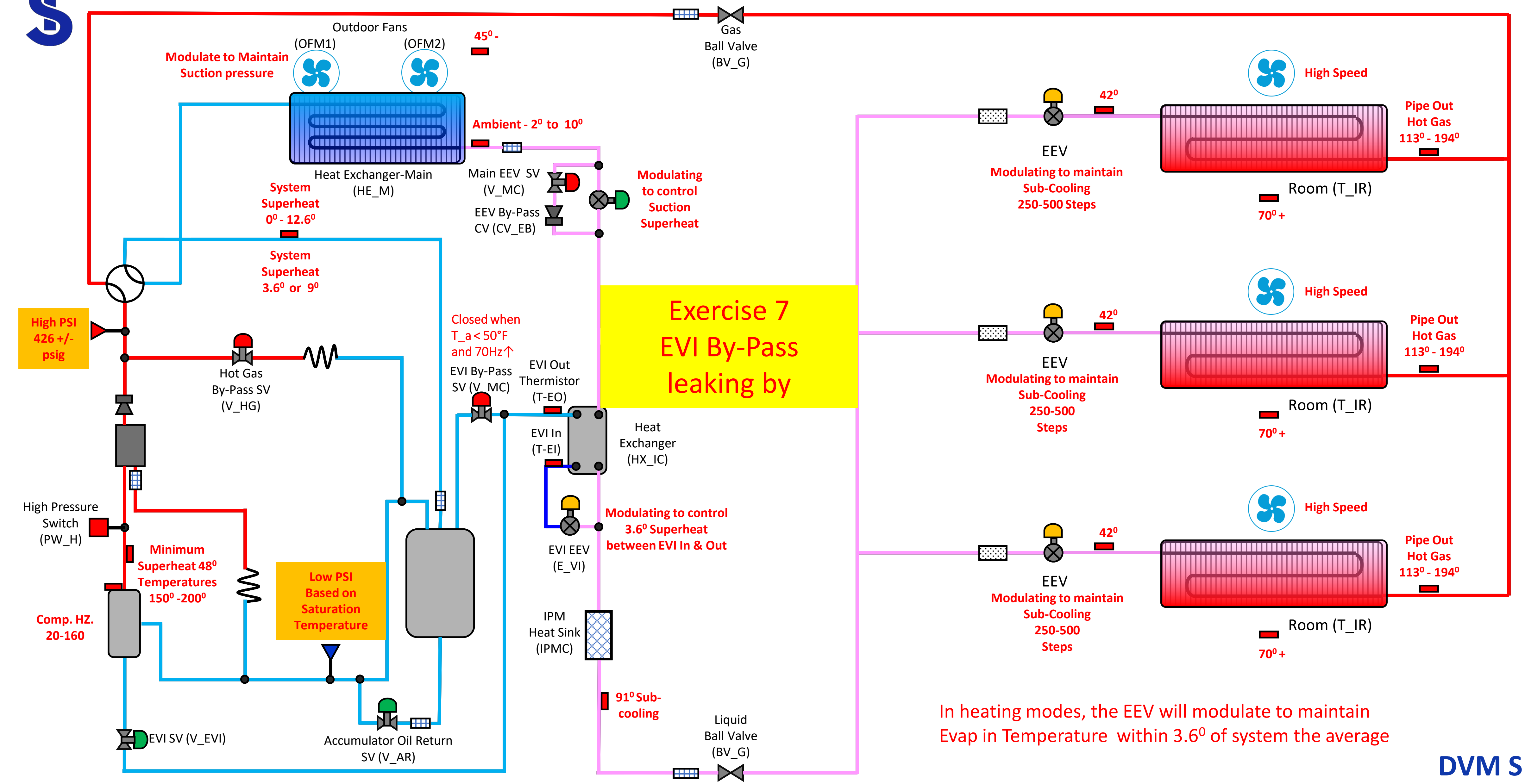
- ❑ One zone not producing heat
- ❑ All other zones are working as required

We can use this section to put in operational characteristics of the system and any other data pertinent to the faut, like the compressor is in protection mode or system is throwing a fault. Thoughts



In heating modes, the EEV will modulate to maintain Evap in Temperature within 3.6⁰ of system the average

DVM S HP



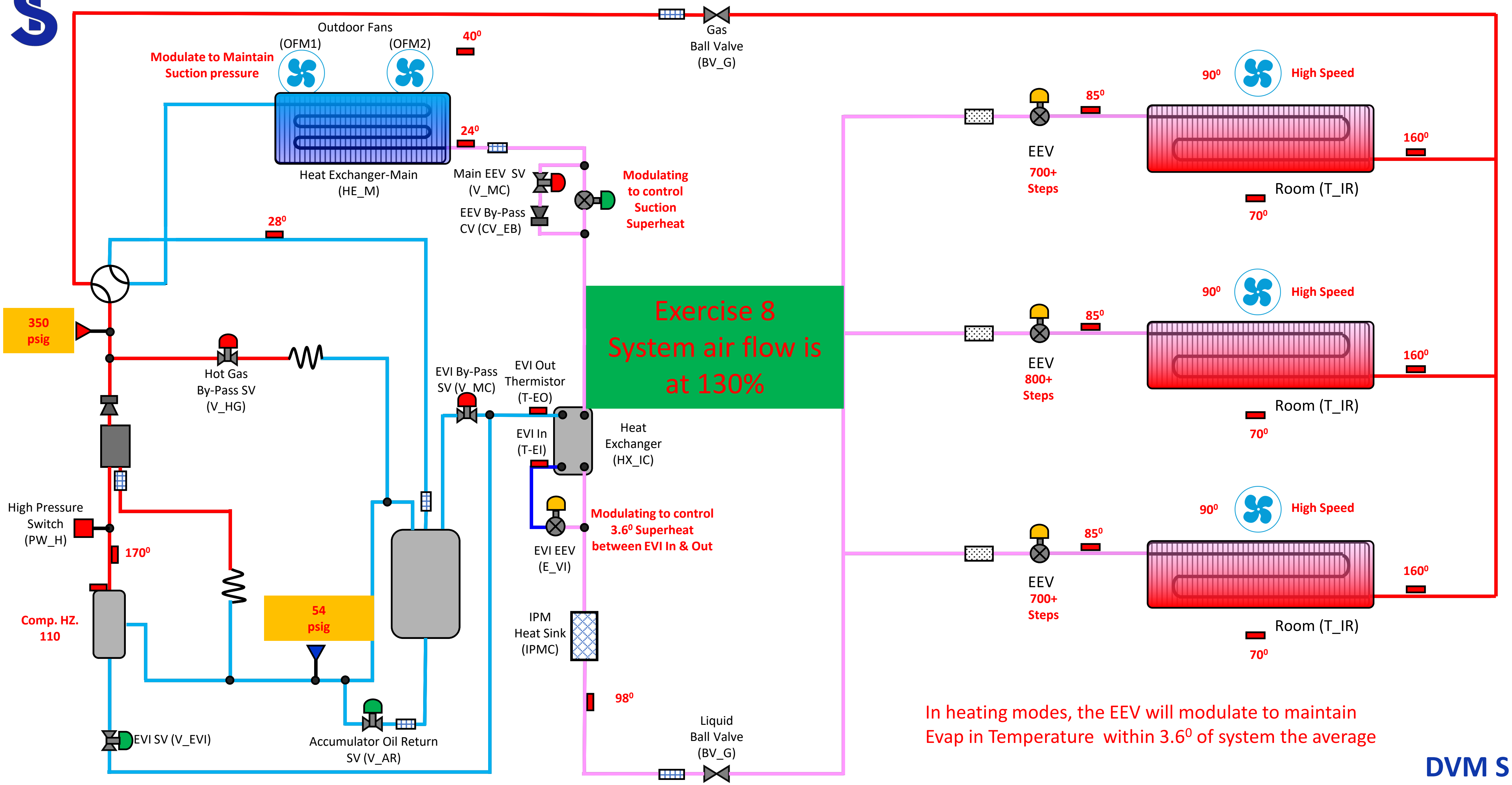
In heating modes, the EEV will modulate to maintain Evap in Temperature within 3.6⁰ of system the average

DVM S HP

Exercise
Operational Characteristics

- ❑ Erratic heating
- ❑ System heats under partial load an performance dies off as it get colder out and the interior load increases
- ❑ When all IDU's are calling with high fan speed the IDU Td's are small, and spaces will not hit setpoint

We can use this section to put in operational characteristics of the system and any other data pertinent to the faut, like the compressor is in protection mode or system is throwing a fault. Thoughts



In heating modes, the EEV will modulate to maintain Evap in Temperature within 3.6^0 of system the average

DVM S HP

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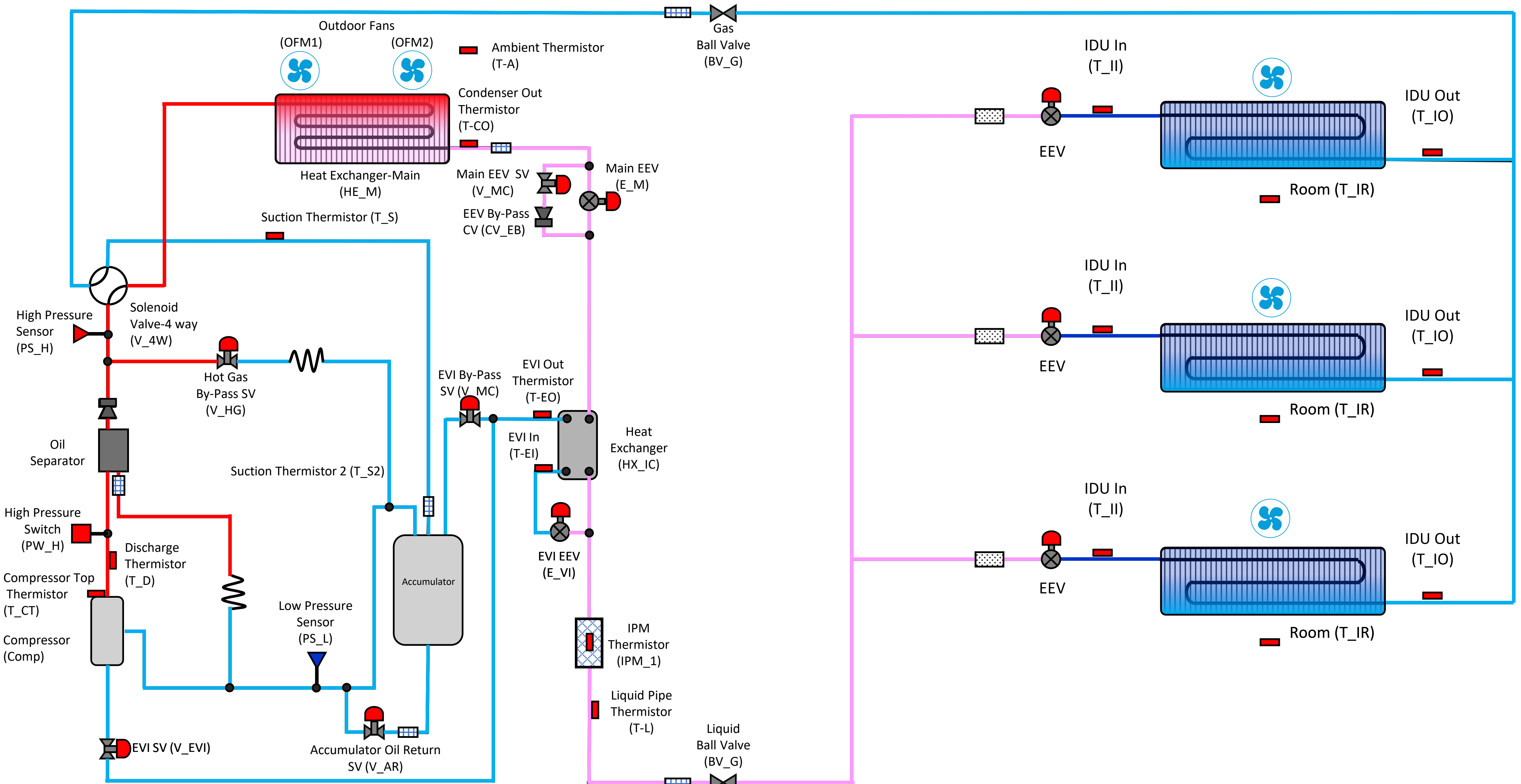
DVMS Heat Pump Piping Diagrams

- We will review the piping diagrams and refrigerant flow for a DVSM S Heat Pump in heating and cooling modes.
- Remember that the basic function of the components d not change.



DVM S HP – Cooling Mode

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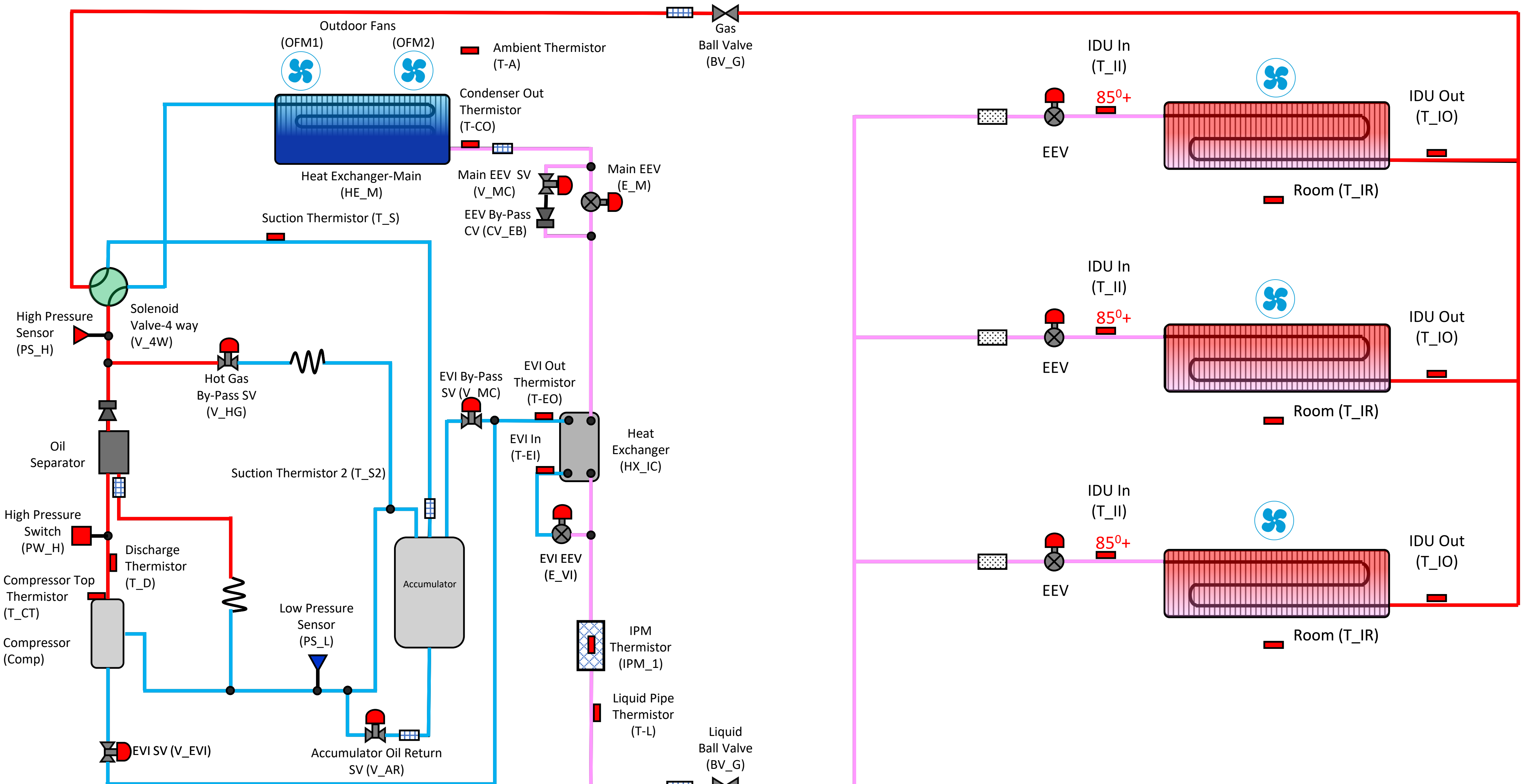

DVM S HP – Heating Mode

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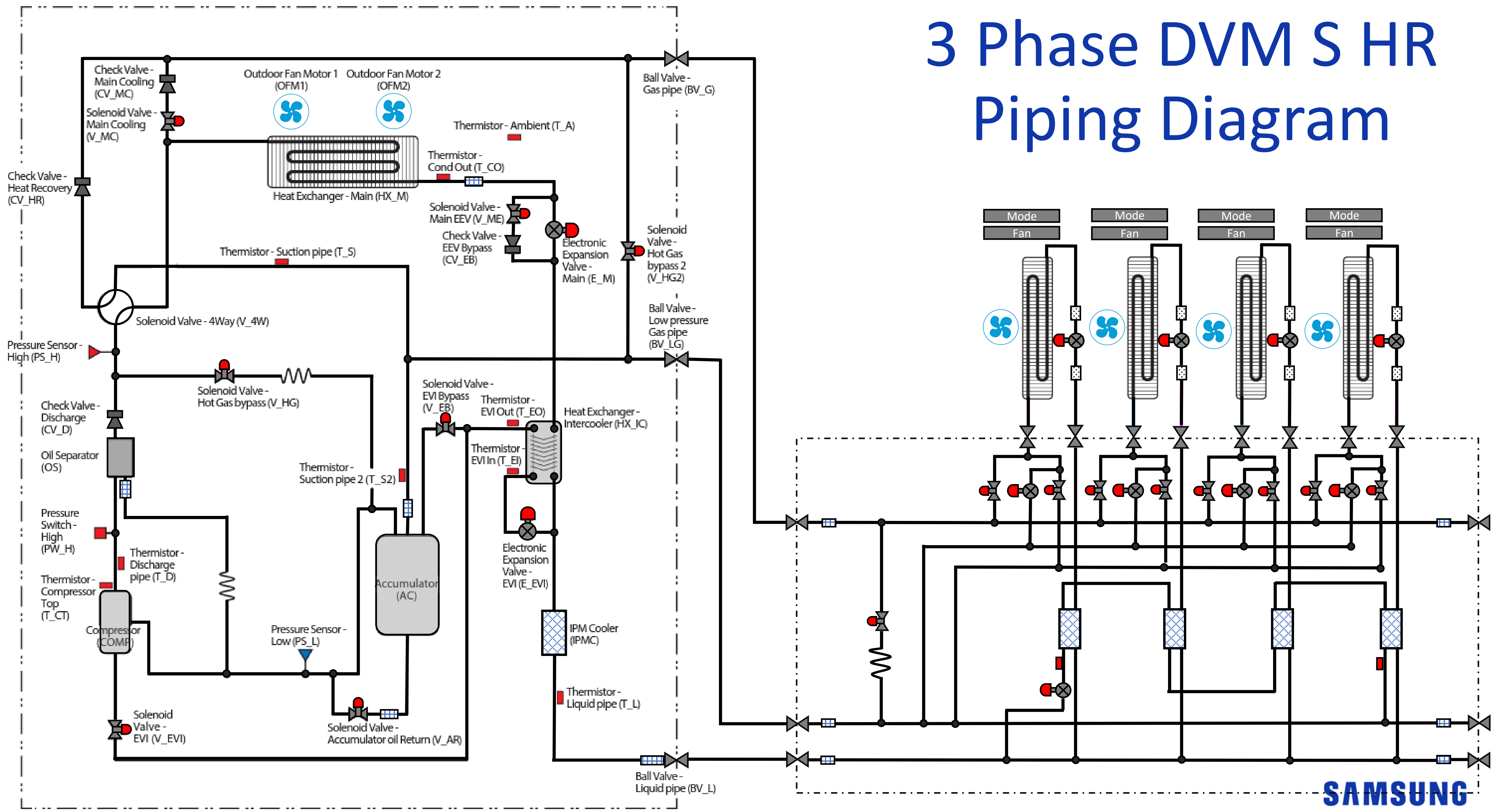
3 Phase DVM S HR Piping Diagram



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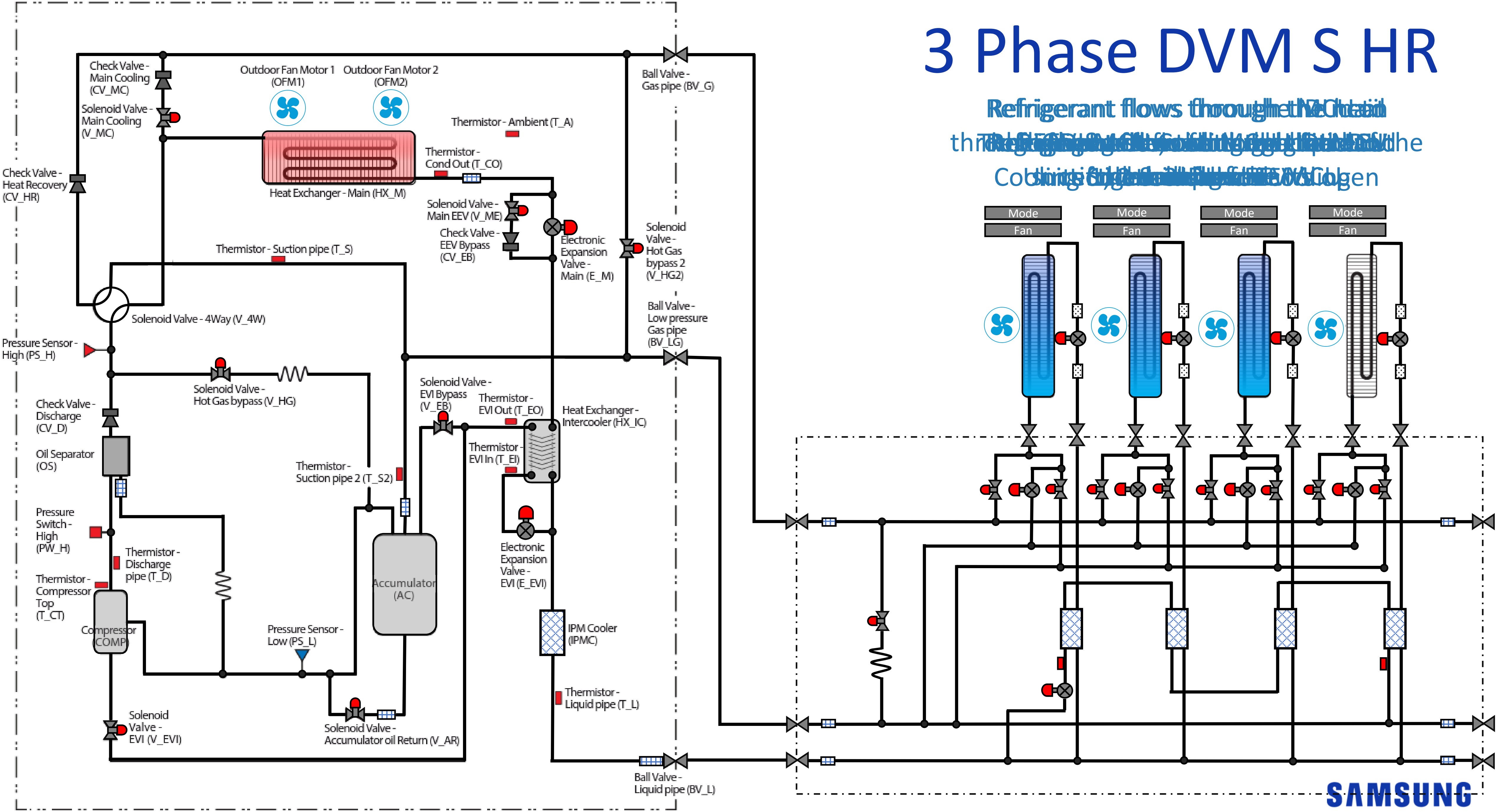
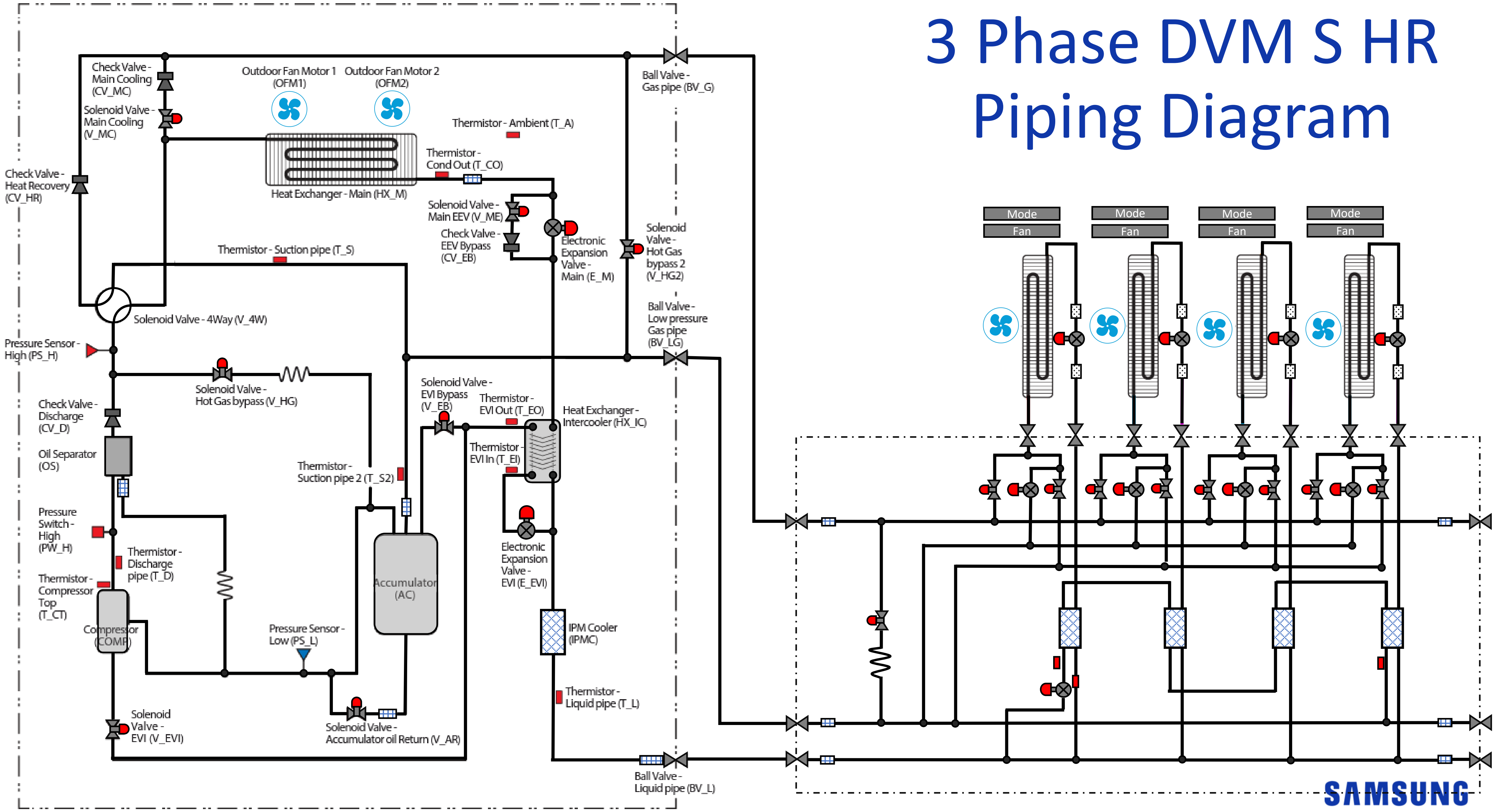
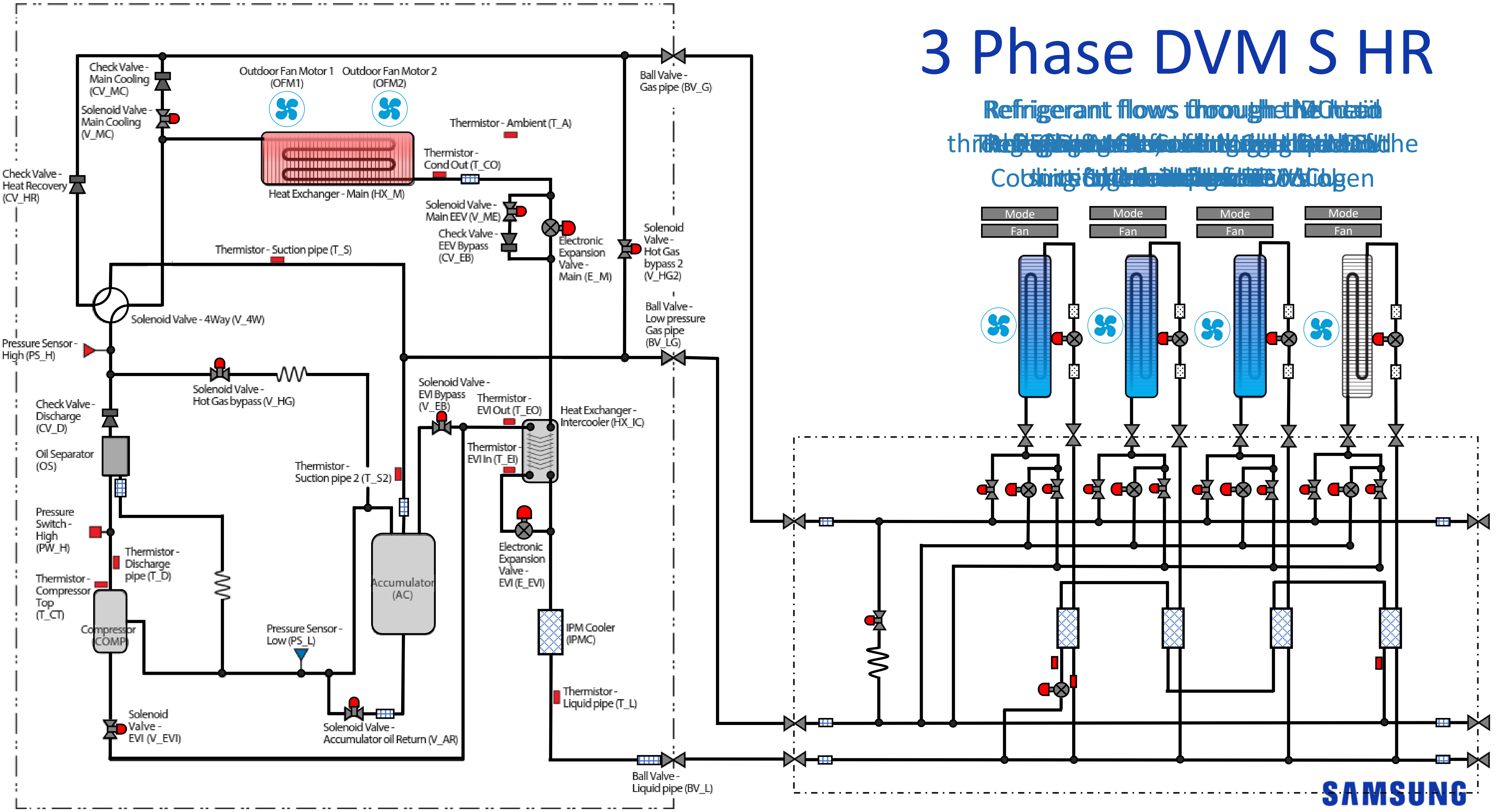
3 Phase DVM S HR



3 Phase DVM S HR Piping Diagram



3 Phase DVM S HR



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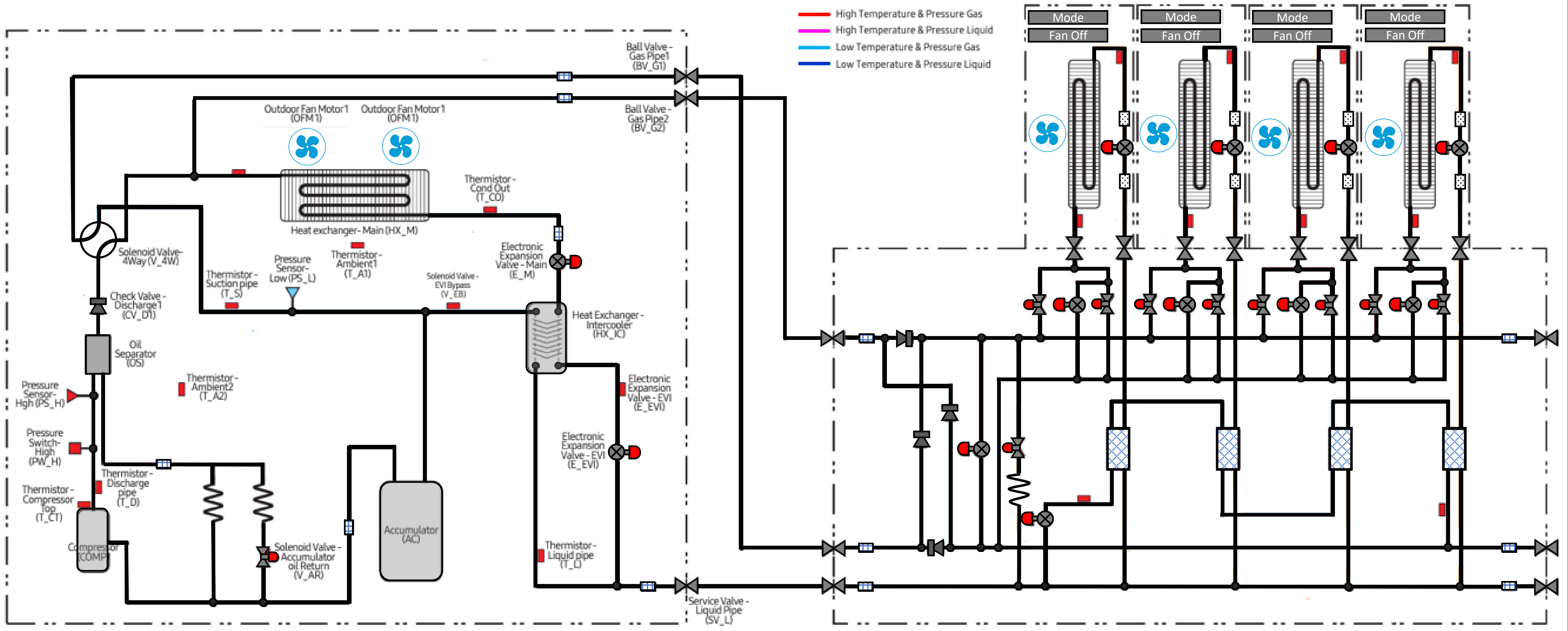
Single Phase ECO HP & HR

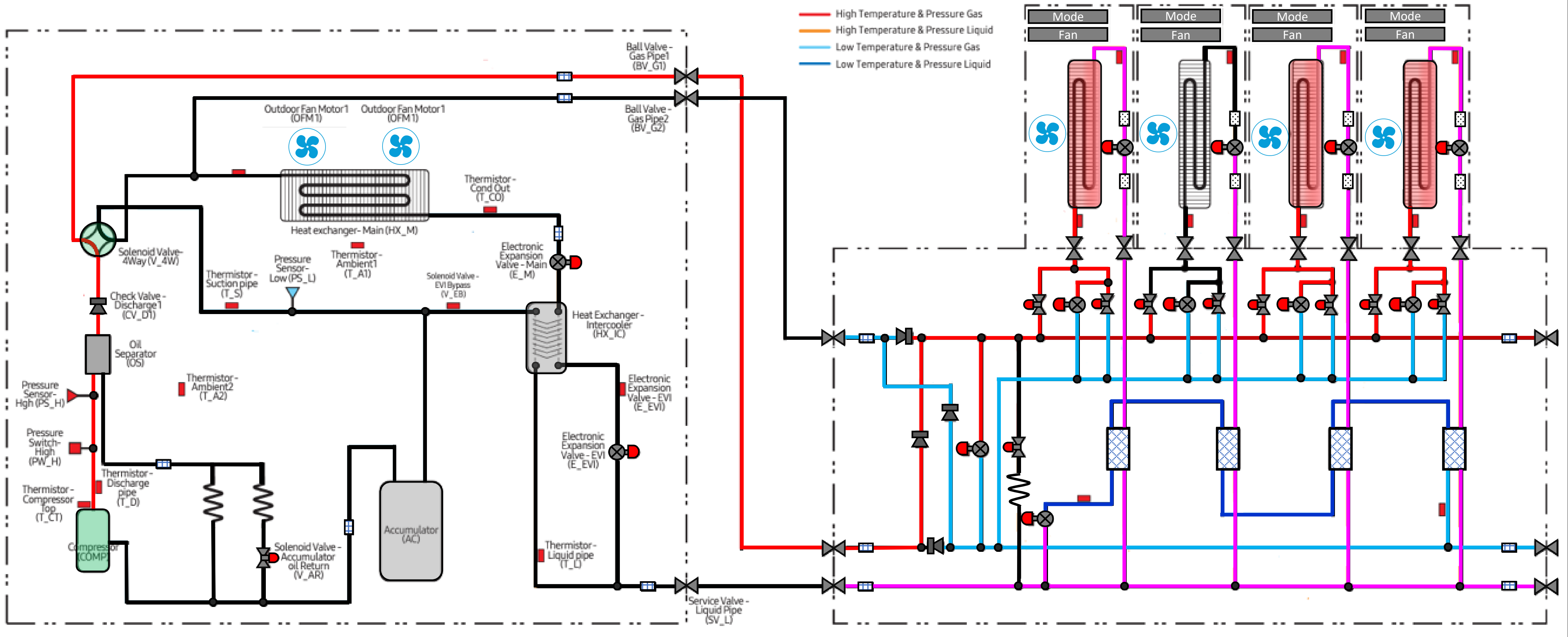
- ❑ Must use the Heat Changer box with the single-phase product.
- ❑ The piping at the indoor unit changes function based on the mode of operation

Eco HR Piping Diagram

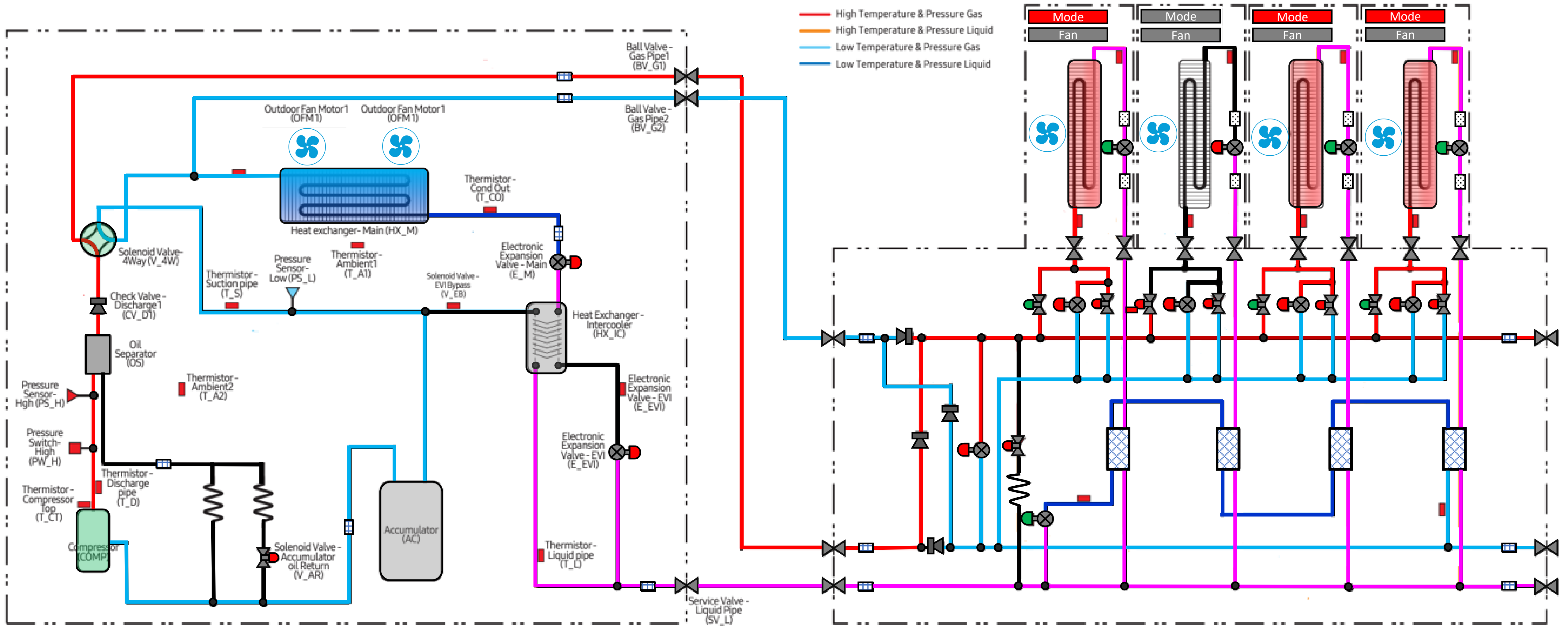


Single Phase Eco

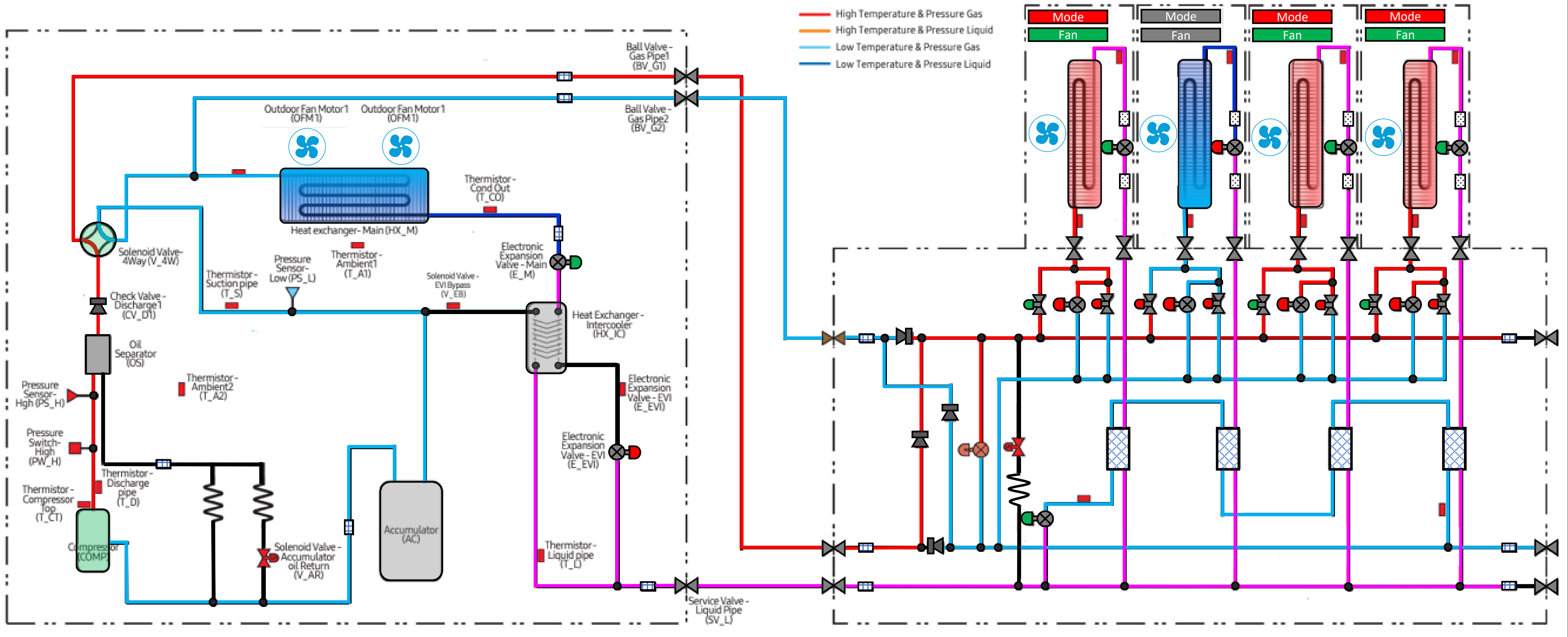
Call for Heat



Refrigerant flow to the ODU

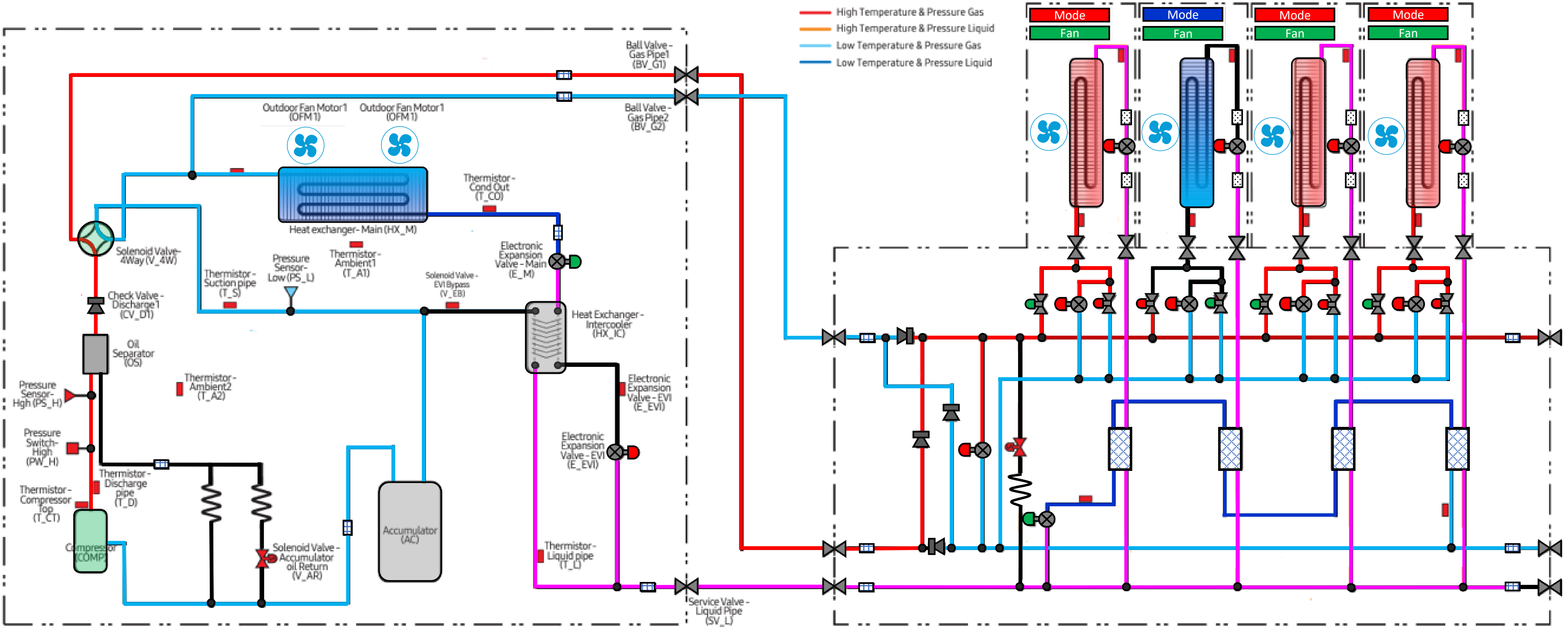


Call For Air Conditioning

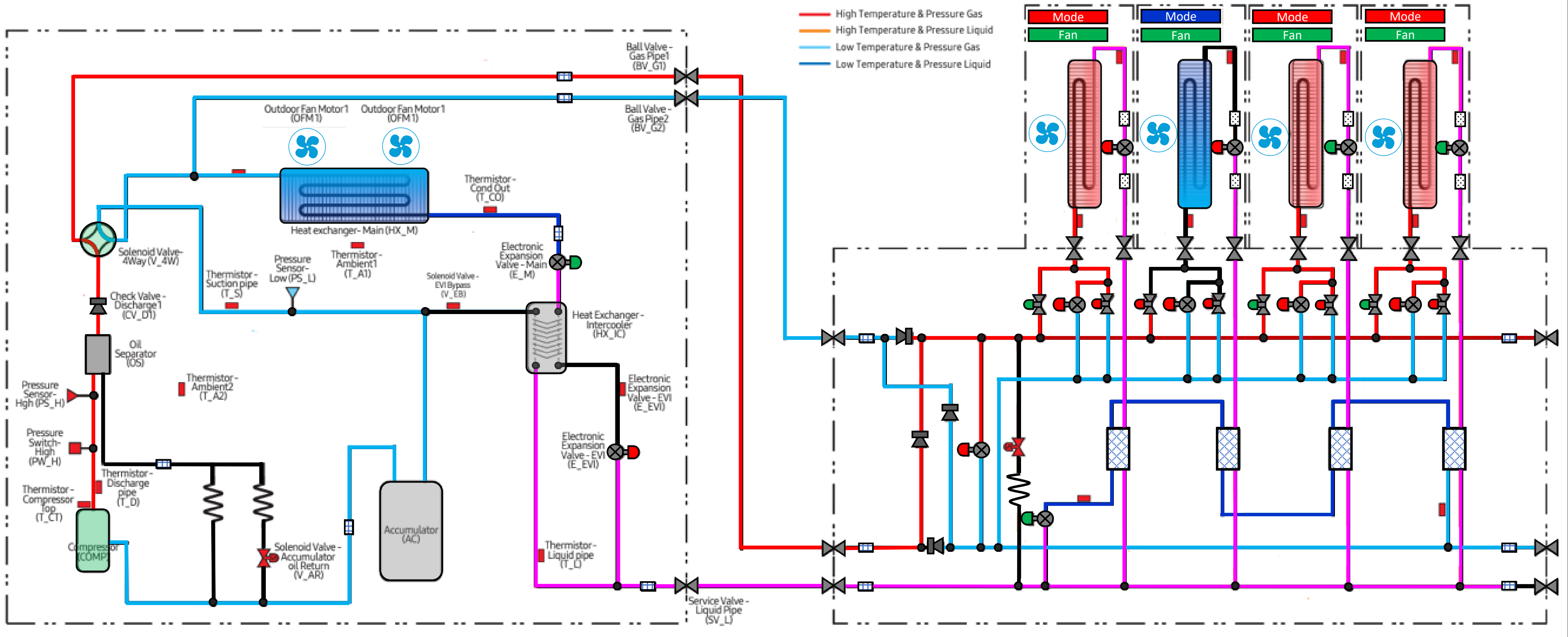


Stable Operation

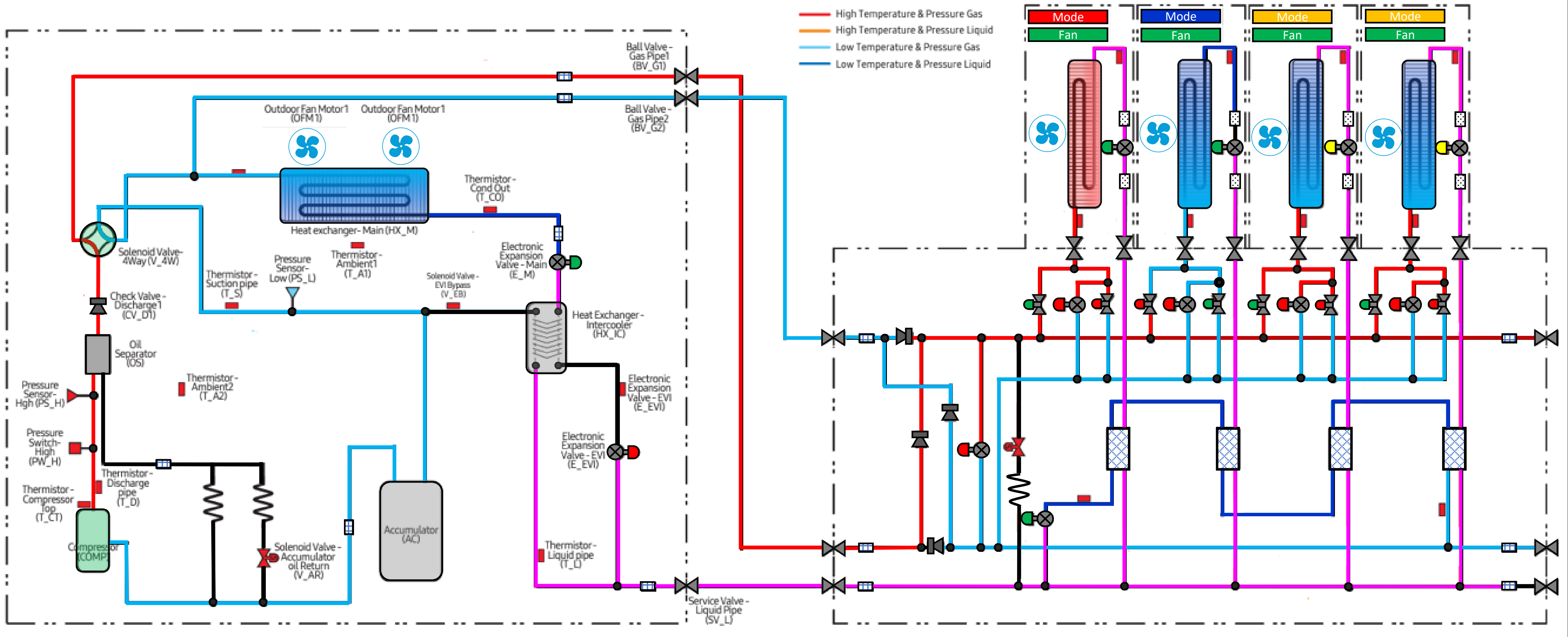
Heating Main



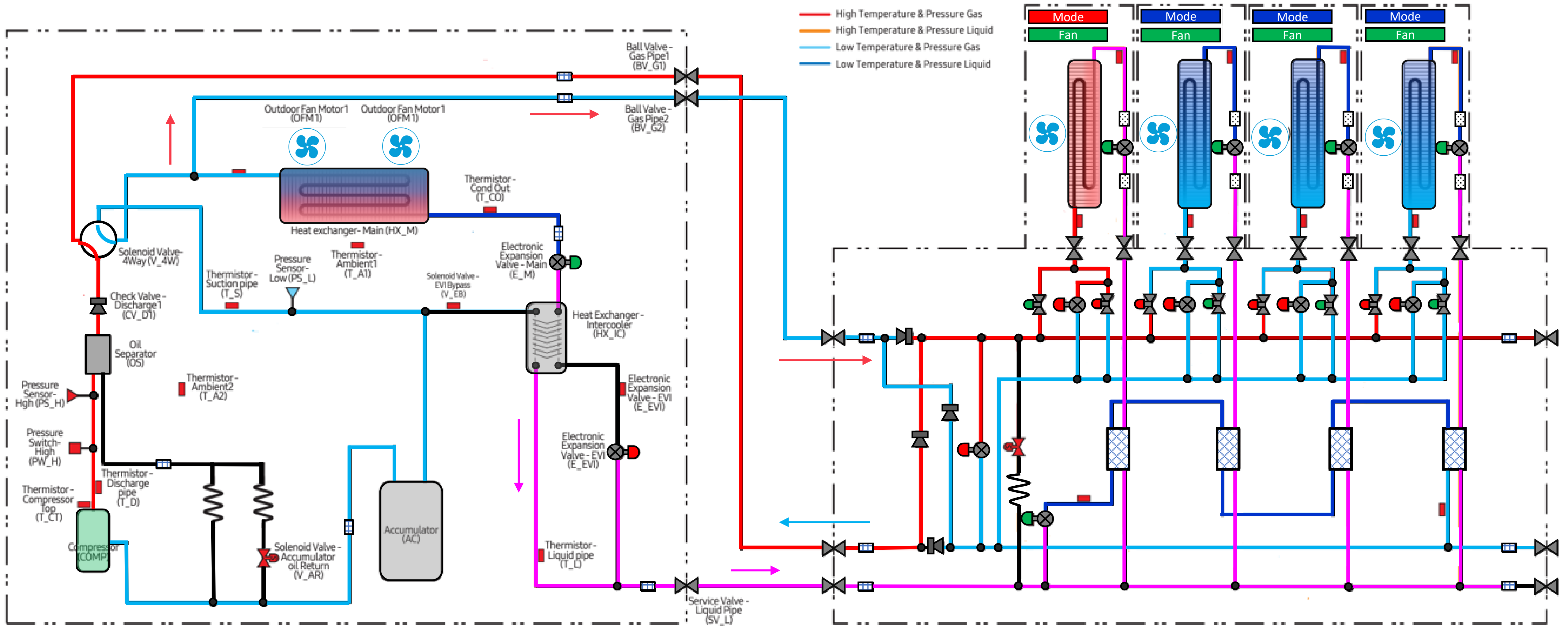
Unit 3 & 4 cycle satisfy



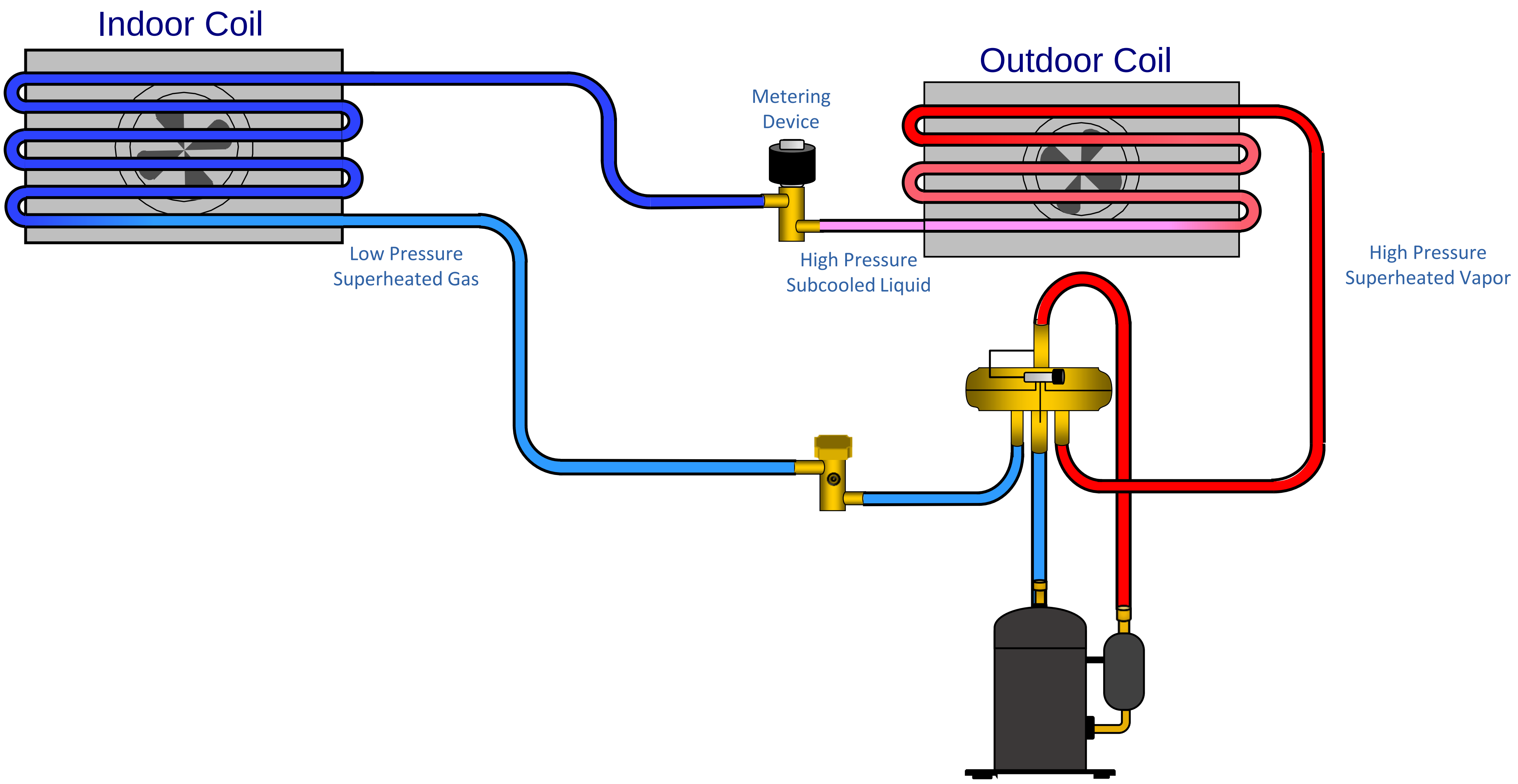
Unit 3 & 4 Switch to A/C



Outdoor unit switches to Air Conditioning



States of refrigerant within a system



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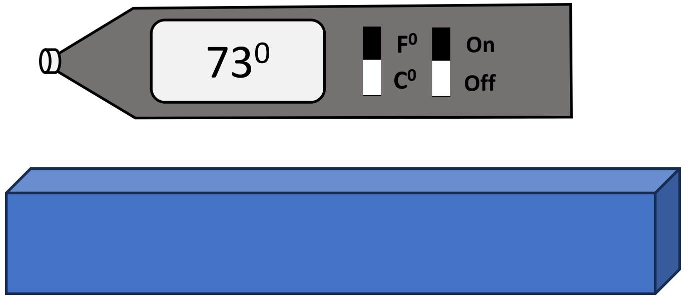
On
Off
F 0
C 0


On
Off

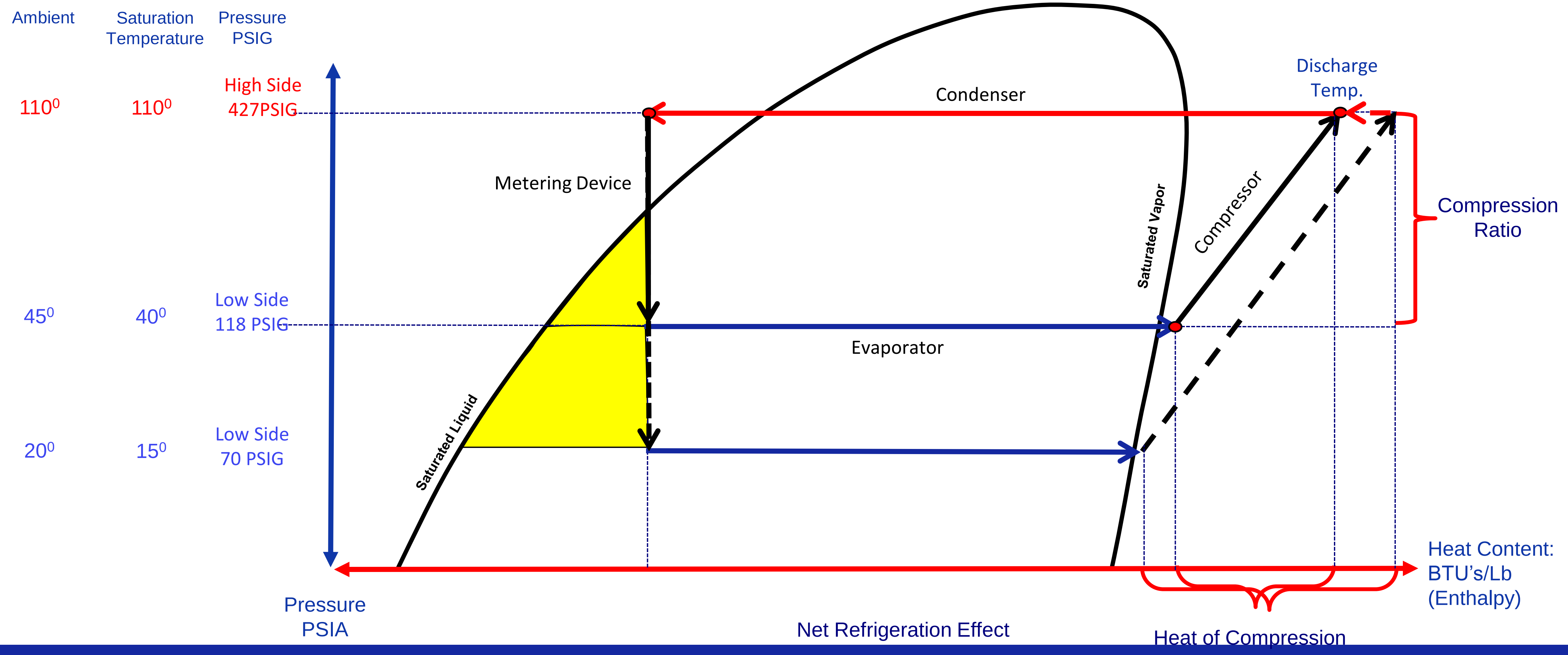

On
Off
On
Off
testo 115i


tenths
77.2
PT-100 Digital Thermometer


73⁰
F⁰
C⁰
On
Off

Increase in Compression Ratio

How can we increase the capacity of the evaporator at low ambient conditions



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Vapor Injection Circuit